THESE LINEAR ALGEBRA NOTES ARE EASY

(No, the title is lying)
(RUN AWAY!!!)
(ABANDON HOPE ALL YE WHO READ THESE!)

BY YILONG YANG

A SLAVE TO HIS WIFE AND TWO KIDS,
ITS MATH AND MADNESS! MADNESS, I TELL YOU!

Picture above by Richard Schwartz

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Linear Algebra in $\mathbb{R}^n$

Chapter 1

Introduction

1.1 A philosophical discourse on the philosophy of math and learning

Math is not an object-oriented subject. Rather, it is more relation-oriented. When we learn that $1 + 1 = 2$, the symbols 1 and 2 here do not refer to some genuin objects somewhere in the universe. Rather, the equation is suppose to points out a universal relation about quantities in this world. If you have one apple and another apple, then you have two apples. If you have one banana and another banana, then you have two bananas. You can easily find infinite examplifications of such a relation. As math students, we see this kind of quantity relation, and we write down the equation $1 + 1 = 2$ to capture this relation.

Similarly, linear algebra uses matrices. But what are matrices? They are not actual objects. They are relations of some kind. The object-oriented view versus relation-oriented view will lead to very different teaching practice and learning practice.

Suppose I am teaching my children (say, 3 year old) the concept of cars. From an object-oriented view, I would do the following: I show them a million pictures, and say "This is a car" or "This is not a car". Then after enough practices, they are taught the concept of cars. Perhaps your education before college is exactly like this. Do the same problem enough times, your exam scores will then improve, right?

However, from a relational point of view, I would do things differently. I would say the following: "A car has a steering wheel, and you can turn it." "A car has a gas paddle to run faster, and a break to go slower." "Yesterday we went to the park by a car." "This red car is our car, and that blue car is our neighbor's car." "The policeman use a police car to catch bad guys." "Let us sing a song about cars." "Let us play with this toy car." I bring out the relation of cars with everything else in my children's life. Then with enough connections, their knowledge of car is now solidified in the web of its relations with regards to everything else. In particular, they now understand the concept of cars, even though they have not seen millions of pictures of cars.

Now suppose a person A learned about cars in the object-oriented way. And a person B learned about cars in the relation-oriented way. If I give them exams ("which of the following picture is a car?"), both should perform adequately well. Maybe A would perform even better, because of the extra practices. However, which one is more likely to enjoy cars when they grow up? Which one is going to have a better intuition about car mechanics? Which one is more likely to be a better driver? I would guess that B is going to be better in these senses.

Think about learning a foreign language. You can memorize words and grammar rules to your heart's content, and you might still freeze when you try to speak the foreign language out loud. Instead, if you go live in an environment speaking that language, then you'll be able to speak it easily in no time, because now you have connected this language with various aspects of your life.

So I hope dearly that in your own studies, try to focus NOT on the WHAT, but on the HOW COME; NOT on the concepts, but on the CONNECTIONS between concepts. Everyone can search for the definition of some concepts online, but only YOU can connect the concept with YOUR own prior knowledge, in ways comfortable to YOU. These web of relations is what you learn.

Mere recitations will make you STOP learning, while making connections is how you START learning.

1.2 What is linear algebra

1.2.1 Linear combinations

In my personal view, the study of Linear Algebra is essentially the study of ***linear combinations***. But what is this? Is this something we can eat? Is it something we wear to keep us warm during winter?

Succintly put, I think a linear combination is a phenomenon in our daily lives. Think about the following examples:

Example 1.2.1.

1. For breakfast I like to eat apples, bananas and cherries. Today I ate 2 apples, 3 bananas and 4 cherries. Then I can say that I ate a linear combination of apples, bananas and cherries, with coefficients $2, 3, 4$. I can also write that I ate
$$(2 \times \text{apple} + 3 \times \text{banana} + 4 \times \text{cherry}).$$

2. Each of your class must use linear algebra at least once, to calculate your grade for the class. Your grade in a linear algebra class might be determined as the following: homework takes 20 percent, midterm takes 30 percent, and the final exam takes 50 percent. Then I can say that your grade is a linear combination of your homework, midterm and final with coefficients $0.2, 0.3, 0.5$. I can also write that your final grade is
$$(0.2 \times \text{Homework} + 0.3 \times \text{Midterm} + 0.5 \times \text{Final}).$$

3. Say I traveled to the USA. At the airport, I need to change yuans into dollars. Say I exchanged 12 yuan for 2 dollar. Then what's the change in my wallet? Well, it is a linear combination of yuans and dollars, with coefficients -12 and 2. I can also write that the change in my wallet is
$$(-12 \times \text{yuan} + 2 \times \text{dollar}).$$

4. My household is made of myself, my wife, two sons and no daughter. You can say that my household is a linear combination of me, wife, sons and daughters, with coefficients $1, 1, 2, 0$ respectively.
$$(1 \times \text{me} + 1 \times \text{wife} + 2 \times \text{son} + 0 \times \text{daughter}).$$
You can also say that my household is a linear combination of me, wife and sons, which is also correct. However, it would be WRONG to say that my house hold is a linear combination of me, sons and pets, because no matter how you combine them, you will always miss out my wife, which is an important part of my household. (THE most important part. My wife might read this.)

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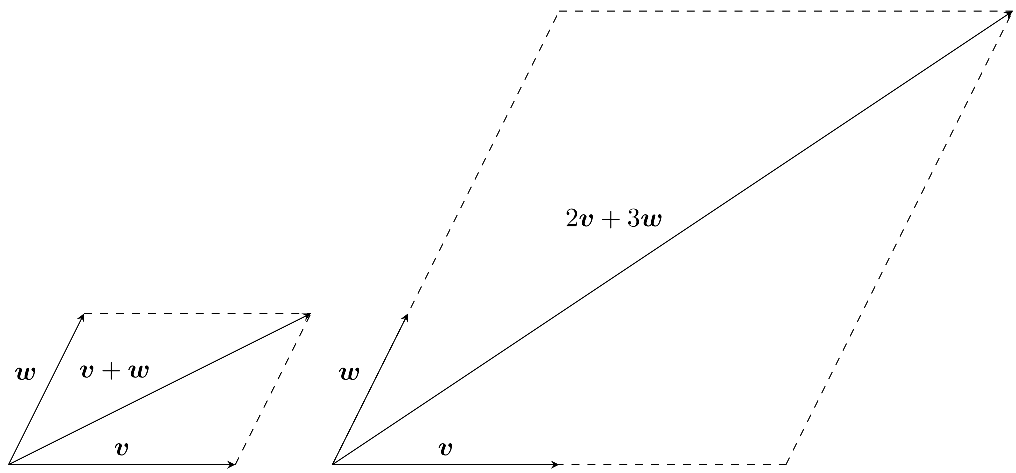
Remark 1.2.2. *As you can see from the examples above, the coefficients (or* ***scalars****, since they "scale" the corresponding item to some multiples of them.) here can be integers, non-integers, positive numbers, negative numbers, and even zero. All real numbers are good.*

Later we shall also introduce complex coefficients. But for the most part of this class, you can stay comfortably in the world of real numbers.

Here are some more abstract, mathematical structures, on which we can also do linear combinations.

Example 1.2.3.

1. In high school, we define "vectors" as some sort of arrows. We say that it is something with direction and magnitude. For example, if $\boldsymbol{v}, \boldsymbol{w}$ are two arrows, then we can perform "arrow additions" like $\boldsymbol{v}+\boldsymbol{w}$, or even linear combinations like $2\boldsymbol{v} + 3\boldsymbol{w}$, by drawing the corresponding parallelogram and find the diagonal arrow.



2. Given two functions $f(x), g(x)$, defined from real numbers to real numbers, we can do their linear combinations like $2f(x) + 3g(x)$, which will still be a function.

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Now, as long as we are doing linear combinations, here is a fun problem to comtemplate. It will also serve as a major indication of how one should learn mathematics in general.

1.2.2 Linear maps

So, intuitively at least, linear combination simply means we are combining things by multiplying each with a number, and add them together. As you can see, I'm teaching you nothing new so far. I'm merely pointing out a structure with which you are already familiar, since they are everywhere in your life.

But what's the point of having structures? Well, structures are meant to be related to each other. Consider the following examples:

Example 1.2.4.

1. Say I go shopping and buy apples, bananas and cherries, and the prices for each is 3 yuan, 1 yuan and 2 yuan respectively.

   I intend to buy the following:

$$(2 \times \text{apples} + 3 \times \text{bananas} + 4 \times \text{cherries}).$$

   When I check out, the linear combination of fruits will transform into the same linear combination of prices of each fruit, which is what I must pay:

$$(2 \times \text{apple price} + 3 \times \text{banana price} + 4 \times \text{cherry price}) = 17\text{yuan}.$$

In this example, the inputs are linear combinations of fruits, and the outputs are real numbers. In mathematical words, we have a function "CheckOut" that sends linear combination of fruits to their total price. In this case, we have

$$\text{CheckOut}(2 \times \text{apples} + 3 \times \text{bananas} + 4 \times \text{cherries}) = 17\text{yuan}.$$

You can easily see that, for any real number x, y, and any linear combination of fruits $\boldsymbol{a}, \boldsymbol{b}$, we should always have

$$\text{CheckOut}(x\boldsymbol{a} + y\boldsymbol{b}) = x\text{CheckOut}(\boldsymbol{a}) + y\text{CheckOut}(\boldsymbol{b}).$$

In particular, even though we are changing fruits into numbers, the structure of linear combination is in fact preserved.

2. I put a bunch of chickens and rabbits into a barn. Each chicken has 1 head, 2 legs. Each rabbit has 1 head and 4 legs. Then if I put a linear combination $3 \times \text{chickens} + 5 \times \text{rabbits}$ into the barn, then in my barn, I would have the linear combination

$$3(1 \times \text{head} + 2 \times \text{leg}) + 5(1 \times \text{head} + 4 \times \text{legs}) = 8 \times \text{head} + 26 \times \text{legs}.$$

In this example, the inputs are linear combinations of animals, and the outputs are linear combinations. In mathematical words, we have a function "Count", that sends linear combinations of animals to linear combinations of body parts.

If $\boldsymbol{a}, \boldsymbol{b}$ are two linear combination of animals, then $\text{Count}(x\boldsymbol{a} + y\boldsymbol{b}) = x\text{Count}(\boldsymbol{a}) + y\text{Count}(\boldsymbol{b})$.

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As you can see, at the center of this is a kind of functions with a special property. These are called ***linear maps***. In mathematical words, a linear map is a function f that preserves the structure of linear combinations, i.e., if $\boldsymbol{a}, \boldsymbol{b}$ are possible inputs, and we combine them into a new input $x\boldsymbol{a}+y\boldsymbol{b}$, then we always have $f(x\boldsymbol{a} + y\boldsymbol{b}) = xf(\boldsymbol{a}) + yf(\boldsymbol{b})$.

(Some would insiste that "functions" should only send numbers to numbers, whereas "map" can send anything to anything. So I shall use map to be safe from critisisms.)

In summary, linear algebra is the study of linear combinations, and functions that preserves them (i.e., linear maps).

1.2.3 Linear algebra

Mathematicians are super lazy. If you want to talk about $(2 \times \text{apples} + 3 \times \text{bananas} + 4 \times \text{cherries})$, why not just write $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$?

There are several advantages to this type of writing. For one, calculations are now much more intuitvie and simple. Compare the following two process. The first is the verbose way:

$$\begin{aligned}
&(2 \times \text{apples} + 3 \times \text{bananas} + 4 \times \text{cherries}) + (2 \times \text{apples} + 1 \times \text{bananas} + 2 \times \text{cherries}) \\
=&(2 \times \text{apples} + 2 \times \text{apples}) + (3 \times \text{bananas} + 1 \times \text{bananas}) + (4 \times \text{cherries} + 2 \times \text{cherries}) \\
=&(2 + 2) \times \text{apples} + (3 + 1) \times \text{bananas} + (4 + 2) \times \text{cherries} \\
=&4 \times \text{apples} + 4 \times \text{bananas} + 6 \times \text{cherries}.
\end{aligned}$$

This process used the law of commutativity and associativity for addition, the law of distribution for multiplication, and contains many steps.

The second is the shortened way:

$$\begin{bmatrix}2\\3\\4\end{bmatrix} + \begin{bmatrix}2\\1\\2\end{bmatrix} = \begin{bmatrix}2+2\\3+1\\4+2\end{bmatrix} = \begin{bmatrix}4\\4\\6\end{bmatrix}.$$

We simply add numbers in corresponding locations and call it a day.

Note that the ORDER of these real numbers are vital. If I have $\begin{bmatrix}3\\2\\4\end{bmatrix}$, then it will NOT be $(2 \times \text{apples} + 3 \times \text{bananas} + 4 \times \text{cherries})$. Instead, it is $(3 \times \text{apples} + 2 \times \text{bananas} + 4 \times \text{cherries})$.

Definition 1.2.5. *1. We define $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ or simply $\mathbb{R}^n$ to be the set of ordered list of n elements of $\mathbb{R}$, i.e., $\begin{bmatrix}a_1\\\vdots\\a_n\end{bmatrix}$ such that all these "coordinates" are in $\mathbb{R}$. (Traditionally people also write $(a_1, ..., a_n)$ to denote this ordered list of elements. But in our class, we uses the vertical notation. Don't ask why, it is just a tradition for linear algebra, probably for aethetic reasons that you shall see later, when we multiply matrices to vectors.)*

*2. For elements of $\mathbb{R}^n$, which we shall call **real vectors**, we define vector addition as $\begin{bmatrix}a_1\\\vdots\\a_n\end{bmatrix} + \begin{bmatrix}b_1\\\vdots\\b_n\end{bmatrix} = \begin{bmatrix}a_1+b_1\\\vdots\\a_n+b_n\end{bmatrix}$.*

3. For any real number $k \in \mathbb{R}$ and a real vector $\begin{bmatrix}a_1\\\vdots\\a_n\end{bmatrix} \in \mathbb{R}^n$, we define scalar multiplication as $k\begin{bmatrix}a_1\\\vdots\\a_n\end{bmatrix} = \begin{bmatrix}ka_1\\\vdots\\ka_n\end{bmatrix}$.

4. We say the dimension of $\mathbb{R}^n$ is the number n.

In our class, if we refer to $\mathbb{R}^n$, we shall always use the vector addition and scalar multiplication defined above.

Example 1.2.6.
When given a plane, say we are doing some high school geometry problem, then we can draw a coordinate chart. By doing so, we have declared that each point on the plane is now essentially a pair of real numbers. What is this? This is $\mathbb{R}^2$.

So in a sense, we can think of $\mathbb{R}^2$ as a plane with a coordinate chart. Similarly, one can think of $\mathbb{R}^3$ as the space with a coordinate chart. In this fashion, one can see that, the so called "n-dimensional space" is simply $\mathbb{R}^n$ if we draw a coordinate chart. We may not see higher dimensional space, but we can certainly calculate. Just treat it as $\mathbb{R}^n$ and we can happily compute away whatever we come across. ☺

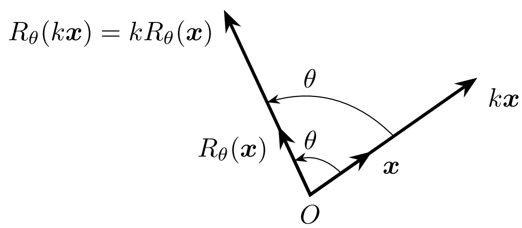
What about linear maps? Recall that a function is a linear map if it sends linear combinations of many inputs to the SAME linear combination of respective outputs, i.e., $f(\sum a_i \boldsymbol{v}_i) = \sum a_i f(\boldsymbol{v}_i)$.

Definition 1.2.7. *A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is said to be a **linear map** if $f(\sum a_i \boldsymbol{v}_i) = \sum a_i f(\boldsymbol{v}_i)$ for all $a_i \in \mathbb{R}, \boldsymbol{v}_i \in \mathbb{R}^n$.*

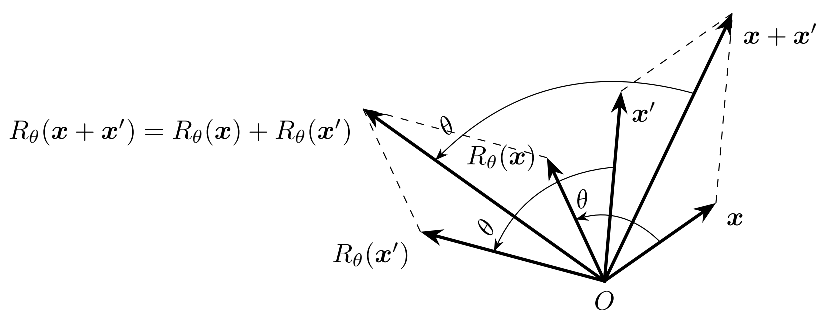
Here the notation "$f : X \to Y$" means all possible inputs of f come from the set X, and all possible outputs of f are inside the set Y. In particular, $f : \mathbb{R}^n \to \mathbb{R}^m$ means the inputs all have n coordinates, while the outputs all have m coordinates.

Example 1.2.8. To show that a map f is linear, it is enough to check that $f(k\boldsymbol{v}) = kf(\boldsymbol{v})$ and $f(\boldsymbol{v}+\boldsymbol{w}) = f(\boldsymbol{v}) + f(\boldsymbol{w})$. These will obviously allow you to show that f respect all linear combinations.

Consider a map $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$. For each arrow vector $\boldsymbol{v} \in \mathbb{R}^2$, R_θ simply rotate it by θ counter clockwise. Then this map is linear, as can be seen in the graphs below.



(c) Scalar multiplication



(d) Vector addition

Figure 1.2.1: Rotation is linear

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Keep in mind of a very important property of linear maps. Since the output is always proportional to the input, a linear map should always send no input to no output.

Proposition 1.2.9. *Given a linear map $f : \mathbb{R}^n \to \mathbb{R}^m$, we must have $f(\mathbf{0}) = \mathbf{0}$.*

Here $\mathbf{0}$ refers to the zero vector $\mathbf{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$, or the ***origin*** of the space $\mathbb{R}^n$ or $\mathbb{R}^m$. In the formula $f(\mathbf{0}) = \mathbf{0}$, the input $\mathbf{0}$ is the origin of $\mathbb{R}^n$, while the output $\mathbf{0}$ is the origin of $\mathbb{R}^m$, so technically they might not be the same zero vector. (They might have different number of coordinates, even though all coordinates are zero.)

Proof. We just calculate this

$$f(\mathbf{0}) = f(\mathbf{0}+\mathbf{0}) = f(\mathbf{0}) + f(\mathbf{0}).$$

Now substract both sides by $f(\mathbf{0})$, we see that $\mathbf{0} = f(\mathbf{0})$. □

Example 1.2.10. In particular, a translation map (shift all inputs by the same vector), say $T : \mathbb{R}^2 \to \mathbb{R}^2$ where $T(\boldsymbol{v}) = \boldsymbol{v} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, is NOT linear. This is because $T(\mathbf{0})$ is no longer $\mathbf{0}$.

This map is in fact an ***affine maps***, which is basically a linear map but you can then add a constant vector afterwards. So functions $f : \mathbb{R} \to \mathbb{R}$ such as $f(x) = 2x$ are linear, and functions such as $f(x) = 2x+1$

are affine. We do not do affine maps in this class. (But affine maps can still be studied in similar manners.) ☺

Finally, what is a matrix? It is a way to record linear maps.

Example 1.2.11 (Nutrition table is a linear map)**.** Suppose we want to study our fiber and sugar intake, and we want to consider eating apples and bananas and cherries.

The fiber and sugar in an apple is a vector, say $\begin{bmatrix}1\\2\end{bmatrix}$, which means one unit of fiber and two units of sugar. The fiber and sugar in a banana is also a vector, say $\begin{bmatrix}3\\4\end{bmatrix}$. Finally, for cherry, let's say the vector is $\begin{bmatrix}5\\6\end{bmatrix}$.

Say I eat a bunch of apples, bananas and cherries. What is the total amount of nutrition I have eaten? This "counting nutrition" process C is a linear map. It would send a fruit combination $\begin{bmatrix}x\\y\\z\end{bmatrix}$ to the corresponding fiber-sugar combination, which should be

$$C(\begin{bmatrix}x\\y\\z\end{bmatrix}) = x\begin{bmatrix}1\\2\end{bmatrix} + y\begin{bmatrix}3\\4\end{bmatrix} + z\begin{bmatrix}5\\6\end{bmatrix}.$$

In short, the input coordinates are coefficients that we use to combine some given vectors. The process C would send these coefficients to the result of the corresponding linear combination. As you can see, the process C is entirely described by the three vectors $\begin{bmatrix}1\\2\end{bmatrix}$, $\begin{bmatrix}3\\4\end{bmatrix}$ and $\begin{bmatrix}5\\6\end{bmatrix}$. As long as we know these vectors, then we know how to compute the output of any input.

So we simply write $C = \begin{bmatrix}1 & 3 & 5\\2 & 4 & 6\end{bmatrix}$. Welcome to your first matrix. ☺

Example 1.2.12. Suppose $f : \mathbb{R}^3 \to \mathbb{R}^2$ is some unknown linear map, whose matrix is $\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}$. Then to find $f(\begin{bmatrix}2\\2\\3\end{bmatrix})$, we have

$$\begin{aligned} f(\begin{bmatrix}2\\2\\3\end{bmatrix}) &= 2\begin{bmatrix}1\\4\end{bmatrix} + 2\begin{bmatrix}2\\5\end{bmatrix} + 3\begin{bmatrix}3\\6\end{bmatrix} \\ &= \begin{bmatrix}15\\36\end{bmatrix}. \end{aligned}$$

You can imagine that the input is a fruit combination. The output is some nutrition combination. The process is linear, and the computation is done by combining relevant vectors. This is how a linear map and a matrix relate to each other. ☺

Definition 1.2.13. *1. An $m \times n$ **real matrix** (plural of matrix is matrices) is a rectangular array of real numbers with m rows and n columns, like $\begin{bmatrix}a_{11} & \dots & a_{1n}\\ \vdots & \ddots & \vdots\\ a_{m1} & \dots & a_{mn}\end{bmatrix}$. We usually denote these matrices (plural form of "matrix") with capital letters like A, B, C, D, M, X, Y. For example, we might say $A = \begin{bmatrix}a_{11} & \dots & a_{1n}\\ \vdots & \ddots & \vdots\\ a_{m1} & \dots & a_{mn}\end{bmatrix}$.*

2. *We call the number in the i-th row and j-th column of this matrix the (i,j) entry of this matrix.*

3. *If we write matrix A in terms of its column vectors $A = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$, then we have the matrix-vector multiplication rule*

$$\begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n.$$

Or in short,

$$\begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \sum x_i\boldsymbol{a}_i.$$

4. *In this way, an $m \times n$ real matrix is a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$.*

Example 1.2.14. Here is an extra remark in its beauty. Consider the following calculation.

$$\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + 3y \\ 4x + 5y \end{bmatrix}.$$

If we write everything in an organized way, the coeffients on the output side are exactly in the formation of the array $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$. In this sense, we can say that $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ means exactly to use the matrix as coeffients in corresponding locations, and we simply insert x, y and addition symbols into the correct position and get $\begin{bmatrix} 2x + 3y \\ 4x + 5y \end{bmatrix}$. ☺

Example 1.2.15. Here are some computational examples, if you need some.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+0 \\ 1+1+0 \\ 1+1+1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 2+0+0+0 \\ -2+4+0+0 \\ 0-4+6+0 \\ 0+0-6+8 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}.$$

Can you imagine some generalized patterns for the above examples? ☺

Example 1.2.16. Here is another computational observation. For a matrix A and a vector $\boldsymbol{v}$, then $A\boldsymbol{v}$ is another vector. However, to compute the first coordinate of $A\boldsymbol{v}$, we only need the first row of A.

You can see this clearly from the following example.

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} a + 2b + 3c \\ d + 2e + 3f \end{bmatrix}.$$

As a result, when we compute, we can also feel free to compute one coordinate at a time. Recall that $A\boldsymbol{v}$ means we linearly combine columns of A according to coefficients of $\boldsymbol{v}$. Then consequently, if we linearly combine entries in the i-th row of A according to coefficients of $\boldsymbol{v}$, we will get the i-th coordinate of $A\boldsymbol{v}$.

For example, suppose $A = \begin{bmatrix} 2 & 1 & 3 \\ * & * & * \end{bmatrix}$, where $*$ is something we don't know. Then we have

$$A \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ * & * & * \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \times 4 + 1 \times 5 + 3 \times 6 \\ * \end{bmatrix} = \begin{bmatrix} 31 \\ * \end{bmatrix}.$$

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As we can see here, all matrices are linear maps. By doing matrix-vector multiplications, they send vectors to vectors in a linear way. On the other hand, are all linear maps matrices? They are indeed.

To see this, first let us have a definition/notation in place.

Definition 1.2.17. *In $\mathbb{R}^n$, we say the standard basis vectors are the following:*

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

We also write them as $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$. Here $\boldsymbol{e}_i$ is a vector with 1 in the i-th coordinate, and 0 in all other coordinates.

Example 1.2.18. For example, the standard basis vectors in $\mathbb{R}^3$ are the following:

$$\boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \boldsymbol{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

These vectors are super useful. For example, imagine that $\mathbb{R}^3$ is the apple-banana-cherry space. Then $\boldsymbol{e}_1$ represents a single apple. $\boldsymbol{e}_2$ represents a single banana. And $\boldsymbol{e}_3$ represents a single cherry. These are the most fundamental objects in the apple-banana-cherry space. Hence the name "standard basis". ☺

Now, the standard basis is very useful in the analysis of a linear map.

Example 1.2.19. What is, say, $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$? We can easily see that

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Suppose $f : \mathbb{R}^3 \to \mathbb{R}^2$ is some unknown linear map. Then

$$f(\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}) = 2f(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}) + 3f(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}) + 4f(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}).$$

As you can see, the process of f is a linear combination of the three vectors $f(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}), f(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}), f(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix})$, with the input coordinates as coefficients. So its matrix is simply

$$f = \begin{bmatrix} f(\begin{bmatrix}1\\0\\0\end{bmatrix}), f(\begin{bmatrix}0\\1\\0\end{bmatrix}), f(\begin{bmatrix}0\\0\\1\end{bmatrix}) \end{bmatrix}.$$

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Let us see this in action.

Example 1.2.20. Consider the counting map L in the chicken-rabbit problem. It sends chicken-rabbit combinations to head-leg combinations, and it is linear. What is its matrix?

$L\begin{bmatrix}1\\0\end{bmatrix}$ is the head-leg combination for one chicken and no rabbit. So $L\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix}$ by the anatomy of a chicken.... Similarly $L\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1\\4\end{bmatrix}$. These are the columns. So we have

$$L = \begin{bmatrix}1 & 1\\2 & 4\end{bmatrix}.$$

Does this work? Let us check it. If we have x chickens and y rabbits, then

$$L\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}x+y\\2x+4y\end{bmatrix}.$$

This is exactly the number of heads and legs we expect! ☺

Example 1.2.21. On a plane, each vector is given by two coordinates. (Note that in our class, we usually write vectors vertically.) Given a vector $\boldsymbol{v} = \begin{bmatrix}x\\y\end{bmatrix}$, we can rotate it counterclockwise by θ. What is the coordinate of the result in terms of x, y, θ?

On the face, this appears to be a rather annoying problem. Sure, in the traditional sense, you can draw graphs, draw auxilliary lines, analyze triangles, use trigonometries, and eventually find the answer. But here we shall present another proof, which is both straightforward and elementary, if we start with exploiting the linear structures. THIS IS HOW WE SHOULD THINK IN THIS CLASS!

Consider the rotation as a map R_θ sending vectors to vectors. Then this is a linear map!

Then in particular, we should have the following computation:

$$R_\theta(\begin{bmatrix}x\\y\end{bmatrix}) = R_\theta(x\begin{bmatrix}1\\0\end{bmatrix} + y\begin{bmatrix}0\\1\end{bmatrix}) = xR_\theta(\begin{bmatrix}1\\0\end{bmatrix}) + yR_\theta(\begin{bmatrix}0\\1\end{bmatrix}).$$

In particular, we will figure out the rotation formula as soon as we find out the value of $R_\theta(\begin{bmatrix}1\\0\end{bmatrix})$ and $R_\theta(\begin{bmatrix}0\\1\end{bmatrix})$! In fact, it is easy to see the following, by the definition of sine and cosine:

$$R_\theta(\begin{bmatrix}1\\0\end{bmatrix}) = \begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix};$$
$$R_\theta(\begin{bmatrix}0\\1\end{bmatrix}) = \begin{bmatrix}-\sin\theta\\ \cos\theta\end{bmatrix}.$$

As a result, we see that

$$R_\theta(\begin{bmatrix}x\\y\end{bmatrix}) = x\begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix} + y\begin{bmatrix}-\sin\theta\\ \cos\theta\end{bmatrix} = \begin{bmatrix}x\cos\theta - y\sin\theta\\ x\sin\theta + y\cos\theta\end{bmatrix}.$$

If you like, we can also say that the rotation map is the matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

And the calculation process is

$$\begin{aligned} R_\theta \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= x \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} + y \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}. \end{aligned}$$

☺

1.2.4 Linear problems

Let us now zoom into an actual problem. Here is a famous problem: I put a bunch of chickens and rabbits into a barn, and I count 6 heads and 20 legs. How many chickens and rabbits are in the barn?

I'm sure you all know how to solve this problem from high school. Suppose there are x chickens and y rabbits. Then the conditions imply that $x + y = 6$ and $2x + 4y = 20$. The we solve the equations and we are done.

But now let us say we have 8 heads and 22 legs. Well, you will have to start your calculation all over again. What a big waste of time. Is there a way to definitively, once and for all, solve every possible chicken-rabbit problem?

Let us clarify our demands here. Say we are looking at 6 heads and 20 legs for our chicken-rabbit problem. We have a linear map L which is the counting process. It sends the chicken-rabbit combinations to the head-leg combinations, so $L : \mathbb{R}^2 \to \mathbb{R}^2$. Now, we have some unknown input $\boldsymbol{x}$, and its output is $\begin{bmatrix} 6 \\ 20 \end{bmatrix}$. Our goal is to find $\boldsymbol{x}$. In short, we want to solve the unknown $\boldsymbol{x}$ following equaiton:

$$L(\boldsymbol{x}) = \begin{bmatrix} 6 \\ 20 \end{bmatrix}.$$

To solve this, we should find a "reverse process" to the linear map L. Let us call this reverse process L^{-1}. Then we have the solution:

$$\boldsymbol{x} = L^{-1}(\begin{bmatrix} 6 \\ 20 \end{bmatrix}).$$

Then our goal is to understand the map L^{-1}. Imagine that we know all about L^{-1}. Then the only calculation we need to do is a simple plug-in. For 6 heads and 20 legs, the answer is simply $L^{-1}(\begin{bmatrix} 6 \\ 20 \end{bmatrix})$. For 8 heads and 22 legs, the answer is simply $L^{-1}(\begin{bmatrix} 8 \\ 22 \end{bmatrix})$. We want a formula for L^{-1}, which will solve all chicken-rabbit problems at once.

How can we understand L^{-1}? Since L is linear, it is not hard to imagine that L^{-1} is probably linear as well. And if it is linear, we understand it by writing out its matrix. So we need to figure out $L^{-1}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $L^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, the two columns of the matrix for L^{-1}.

What is $L^{-1}\begin{bmatrix}1\\0\end{bmatrix}$? This is asking: how many chickens and rabbits will give one head and no leg? Suppose we have x chickens and y rabbits, this means we are solving

$$\begin{aligned}x+y=&1,\\2x+4y=&0.\end{aligned}$$

And the answer is $x=2$ and $y=-1$. So $L^{-1}\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}2\\-1\end{bmatrix}$.

What is $L^{-1}\begin{bmatrix}0\\1\end{bmatrix}$? This is asking: how many chickens and rabbits will give no head and one leg? Similarly, we solve

$$\begin{aligned}x+y=&0,\\2x+4y=&1.\end{aligned}$$

And the answer is $x=-\frac{1}{2}$ and $y=-\frac{1}{2}$. So $L^{-1}\begin{bmatrix}0\\1\end{bmatrix}=\begin{bmatrix}-\frac{1}{2}\\\frac{1}{2}\end{bmatrix}$.

So $L^{-1}=\begin{bmatrix}2&-\frac{1}{2}\\-1&\frac{1}{2}\end{bmatrix}$.

In particular, if we have a heads and b legs, then the amount of chickens and rabbits are

$$\begin{aligned}L^{-1}\begin{bmatrix}a\\b\end{bmatrix}&=\begin{bmatrix}2&-\frac{1}{2}\\-1&\frac{1}{2}\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix}\\&=\begin{bmatrix}2a-\frac{1}{2}b\\-a+\frac{1}{2}b\end{bmatrix}.\end{aligned}$$

You can plug in various values and check the validity of this formula. Congradulations, we have solved all chicken-rabbit problems in existence.

Recall that we have $L=\begin{bmatrix}1&1\\2&4\end{bmatrix}$. In this case, we also write the following to indicate that L and L^{-1} are inverse process of each other.

$$\begin{bmatrix}1&1\\2&4\end{bmatrix}^{-1}=\begin{bmatrix}2&-\frac{1}{2}\\-1&\frac{1}{2}\end{bmatrix}.$$

We say $\begin{bmatrix}1&1\\2&4\end{bmatrix}$ and $\begin{bmatrix}2&-\frac{1}{2}\\-1&\frac{1}{2}\end{bmatrix}$ are inverse matrix of each other. As you can imagine, many linear problems can be solved en masse if we have a way to reliably and quickly find inverse matrices.

1.3 Exercises

Exercise 1.3.1. *Let us use* $\begin{bmatrix}a\\b\\c\\d\end{bmatrix}$ *to represent the linear combination "a apples + b bananas + c cherries + d dollars".*

1. *I spent* 20 *dollars to buy* 3 *apples and* 4 *cherries. What is the vector representing this change in my household?*

2. *I sold* 3 *apples and* 4 *cherries for* 20 *dollars. What is the vector representing this change in my household?*

3. *I ate 3 apples and 4 bananas. What is the vector representing this change in my household?*
4. *(Silly problem) I bought the same amount of apples and bananas for 10 dollars. Then I ate three apples, so now the amount of bananas I own is twice the amount of apples. Then I sold the remaining apples for 3 dollars. What is the vector representing the total change in my household?*

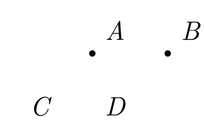
Exercise 1.3.2. *Let us use* $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$ *to represent the linear combination "a husbands + b wives + c sons + d daughters".*

1. *Describe your family (your parents are the husbands/wives) as a vector.*
2. *Describe your parents families as vectors.*
3. *Describe your ideal family structure vector for yourself in the future.*
4. *Describe the family structure vector you would wish upon your worst enemy.*

Exercise 1.3.3. *We interpret* $\begin{bmatrix} x \\ y \end{bmatrix}$ *as an arrow vector.*

1. *Move the arrows* $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ *and* $\begin{bmatrix} -2 \\ -1 \end{bmatrix}$ *around to form a triangle.*
2. *You can move the arrows* $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ *and* $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ *around, but they will never form a triangle. Why? (Hint: Check their lengths.)*
3. *Find some coefficients* a, b, c *such that*
$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$
4. *Find a triangle whose three sides are parallel to the vectors* $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ *and* $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$*. Maybe use the last subproblem as a inspiration? (Remark: the answer here is not unique, but all possible answers are similar triangles. As an extra challenge, can you see why? There is an algebraic reason and a geometric reason.)*
5. *You can move the arrows* $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ *and* $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ *around, but they will never form a triangle. Why?*

Exercise 1.3.4. *We have points* A, B, C, D *on a plane, as shown below. Once we pick an origin and draw a coordinate chart, then* A, B, C, D *will have coordinates, so they will become elements of* $\mathbb{R}^2$.



1. *Suppose* C *is the origin. Now* A, B *will have coordinates, so they will become elements of* $\mathbb{R}^2$*. Can you find the resulting point of the vector addition* $A + B$*? You should draw out the parallelogram involved in this process, and mark the resulting point* $A + B$*. (Remark: The direction of the* x, y*-axes has no effect on the result!)*

2. *Suppose C is the origin. Can you find the resulting point of the scalar multiplication $2A$?*

3. *Suppose D is the origin. Can you find the resulting point of the vector addition $A+B$?*

4. *Suppose D is the origin. Can you find the resulting point of the scalar multiplication $2A$?*

5. *Suppose B is the origin. Can you find the resulting point of the vector addition $A+B$?*

6. *Suppose B is the origin. Can you find the resulting point of the scalar multiplication $2A$?*

Exercise 1.3.5. *Let us use $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ to represent the linear combination "a apples + b bananas + c cherries". Now, when we buy these fruits, we need to pay.*

1. *Suppose the apples are 1 dollar each, bananas are 2 dollars each, and cherries are 3 dollars each. We have a map $L_1 : \mathbb{R}^3 \to \mathbb{R}$, sending each fruit combination to the resulting price. Can you write out the matrix for L_1?*

2. *Suppose the apples are 2 dollar each, bananas are 3 dollars each, and cherries are 1 dollars each. We have a map $L_2 : \mathbb{R}^3 \to \mathbb{R}$, sending each fruit combination to the resulting price. Can you write out the matrix for L_2?*

3. *(Just some observation, not a problem.) Technically, the prices for an apple, for a banana, and for a cherry are three real numbers. So conceivably, maybe we can write it as a column vector $\begin{bmatrix} p \\ q \\ r \end{bmatrix}$. But in practice, people don't do this, because linear combinations of these "price vectors" make no sense. If one supermarket has prices $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, and another supermarket has prices $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$, what is the meaning of the vector addition $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$? There is no meaning and no point. Therefore, it is NOT useful to write them as column vectors. On the other hand, as we have seen in this question, they describes a corresponding linear map, and as matrices for these linear maps, these prices form a ROW VECTOR. Conventionally, we write column vectors to represent things that can be linearly combined. And when we write row vectors, they are simply matrices for some linear maps.*

Exercise 1.3.6. *Find the matrices of the following linear planar transformations.*

1. *Let α be the line obtained by rotating x-axis around the origin counterclockwise by $\frac{\pi}{8}$. Find the matrix for the relfection about the line α.*

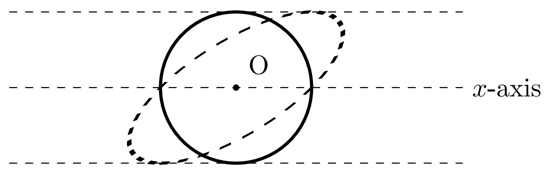
2. *Find the matrix for the rotation around the origin, that sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$.*

3. *Find the matrix for the linear map that fixes all the points on x-axis, but stretches the plane along the y-axis by a factor of 3.*

4. *Find the matrix for the linear map that fixes all the points on the line $x+y=0$, but stretches the plane along the direction of $x=y$ by a factor of 2.*

5. *Find the matrix for the linear map that sends $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$. (Hint: This map is linear. So where should it send $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$?)*

6. *Find the matrix for the linear map that sends the circle below to the dashed ellipse. Note that the point* $\begin{bmatrix}1\\0\end{bmatrix}$ *is NOT changed, while the point* $\begin{bmatrix}0\\1\end{bmatrix}$ *is sent to* $\begin{bmatrix}1\\1\end{bmatrix}$.



Exercise 1.3.7. *Let us use* $\begin{bmatrix}a\\b\\c\end{bmatrix}$ *to represent the linear combination "a burgers + b wings + c cokes". The fast food restaurant is selling the following meal combinations:* $\begin{bmatrix}1\\2\\1\end{bmatrix}$, $\begin{bmatrix}2\\3\\2\end{bmatrix}$ *and* $\begin{bmatrix}2\\0\\1\end{bmatrix}$.

1. *Write out the matrix for the linear map* $L : \mathbb{R}^3 \to \mathbb{R}^3$, *whose input is a combination of meal combos, and output is the total amount of burgers, wings and cokes contained in these meal combos.*
2. *I want to have* $\begin{bmatrix}1\\1\\1\end{bmatrix}$ *for lunch. How should I buy these meal combos to achieve this? (We allow negative coefficients.)*
3. *I want to have a single wing. How should I buy these meal combos to achieve this? (We allow negative coefficients.)*
4. *I want to have a single burger. How should I buy these meal combos to achieve this? (We allow negative coefficients.)*
5. *I want to have a single coke. How should I buy these meal combos to achieve this? (We allow negative coefficients.)*
6. *I want to have* $\begin{bmatrix}x\\y\\z\end{bmatrix}$ *for lunch for known constants* x, y, z. *How should I buy these meal combos to achieve this? (We allow negative coefficients.)*
7. *Write out the matrix for the linear map* $T : \mathbb{R}^3 \to \mathbb{R}^3$, *whose input is the desired amount of burgers, wings and cokes, and output is the linear combination of meal combos that can achieve the desired amount of foods. (T is the inverse matrix of L.)*

Exercise 1.3.8. *Are these matrix-vector multiplication defined? If it is defined, calculate the result of the matrix-vector multiplication.*

1. $\begin{bmatrix}1 & 1\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix}$.
2. $\begin{bmatrix}1 & 1\\1 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}$.
3. $\begin{bmatrix}1 & 1 & 1\\1 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix}$.

4. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

5. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

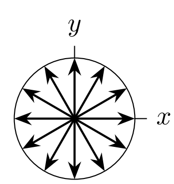
7. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. *(Can you see a pattern?)*

Exercise 1.3.9 (Matrix-Vector Multiplication)**.** 1. *Let us define a **magic matrix** to be a 3×3 matrix whose entries are $1, \dots, 9$ in some order, such that in each row, each column, and each of the two diagonals, the three entries add up to the same number. For example, a typical magic matrix is* $\begin{bmatrix} 2 & 9 & 4 \\ 7 & 5 & 3 \\ 6 & 1 & 8 \end{bmatrix}$. *For an arbitrary magic matrix M, try to find all possible values of* $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

2. *Let us define a **Sudoku matrix** to be a 9×9 matrix such that each row is made of $1, \dots, 9$, and each column is made of $1, \dots, 9$, and each of the nine 3×3 "submatrix" is also made of $1, \dots, 9$. For example, a typical Sudoku matrix is*

$$\left[\begin{array}{ccc|ccc|ccc} 5 & 3 & 4 & 6 & 7 & 8 & 9 & 1 & 2 \\ 6 & 7 & 2 & 1 & 9 & 5 & 3 & 4 & 8 \\ 1 & 9 & 8 & 3 & 4 & 2 & 5 & 6 & 7 \\ \hline 8 & 5 & 9 & 7 & 6 & 1 & 4 & 2 & 3 \\ 4 & 2 & 6 & 8 & 5 & 3 & 7 & 9 & 1 \\ 7 & 1 & 3 & 9 & 2 & 4 & 8 & 5 & 6 \\ \hline 9 & 6 & 1 & 5 & 3 & 7 & 2 & 8 & 4 \\ 2 & 8 & 7 & 4 & 1 & 9 & 6 & 3 & 5 \\ 3 & 4 & 5 & 2 & 8 & 6 & 1 & 7 & 9 \end{array}\right].$$

For an arbitrary Sudoku matrix M, try to find all possible values of $M \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$.



Exercise 1.3.10. *Look at this clock. There are twelve arrow vectors here. Assume tha that they are all unit vectors on the xy-plane.*

1. *Find the sum of all twelve vectors.*

2. *Find the sum of all vectors except the 2 o'clock vector.*

3. *Fix the end points of all vectors, but move the starting points of all vectors to the 6 o'clock location, i.e., the 12 o'clock vector is now* $\begin{bmatrix}0\\2\end{bmatrix}$*, and the 3 o'clock vector is now* $\begin{bmatrix}1\\1\end{bmatrix}$*, and so on. Now find the sum of all twelve vectors.*

Exercise 1.3.11. *Write the following vector $\boldsymbol{b}$ as the result of a matrix multiplying a vector.*

1. $\boldsymbol{b} = 2\begin{bmatrix}1\\1\end{bmatrix} + 4\begin{bmatrix}0\\1\end{bmatrix} + 5\begin{bmatrix}1\\0\end{bmatrix}$.

2. $\boldsymbol{b} = 5\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix} + 4\begin{bmatrix}5\\4\\3\\2\\1\end{bmatrix}$.

3. $\boldsymbol{b} = \begin{bmatrix}2b+a+c\\c-b\\a+b+c\\a+b\end{bmatrix}$, *where a, b, c are constants.*

4. $\boldsymbol{b} = f\left(\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}\right)$, *where $f: \mathbb{R}^5 \to \mathbb{R}^5$ is a linear map such that $f(\boldsymbol{e}_k) = k\boldsymbol{e}_{6-k}, k = 1, \ldots, 5$.*

5. ☕*Suppose we know the following rule of the weathers. If it rains one day, then the next day has* 0.8 *chance of raining. If it does not rain one day, then the next day has* 0.3 *chance of raining. Suppose today has* 0.5 *chance of raining, let* $\boldsymbol{b} = \begin{bmatrix}p\\1-p\end{bmatrix}$ *where p is the chance of raining tomorrow.*

Exercise 1.3.12. *It is also possible to do linear combination of matrices. In general, if A, B are matrices of the same size, then $A + B$ means we are adding the corresponding entries to get a new matrix. If A is a matrix and k is a scalar, then kA means we multiply k to each entry of A.*

1. *Say* $A = \begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}$ *and* $B = \begin{bmatrix}6 & 5 & 4\\3 & 2 & 1\end{bmatrix}$. *Compute $A + B$ and $A - B$.*

2. *Say* $A = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ *and* $B = \begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}$. *Compute $2A + 3B$ and $3B - 2A$.*

3. ☕*Let* $A = \begin{bmatrix}1 & 0 & 0 & 0\\2 & 2 & 0 & 0\\3 & 3 & 3 & 0\\4 & 4 & 4 & 4\end{bmatrix}$, $B = \begin{bmatrix}4 & 3 & 2 & 1\\3 & 3 & 2 & 1\\2 & 2 & 2 & 1\\1 & 1 & 1 & 1\end{bmatrix}$ *and* $C = \begin{bmatrix}1 & 2 & 3 & 4\\0 & 2 & 3 & 4\\0 & 0 & 3 & 4\\0 & 0 & 0 & 4\end{bmatrix}$. *Compute $A + B + C$.*

4. *For any matrix M, let us use $S(M)$ to denote the sum of all entries of M. Then observe the following computations*

$$S(\begin{bmatrix}4 & 3 & 2 & 1\\3 & 3 & 2 & 1\\2 & 2 & 2 & 1\\1 & 1 & 1 & 1\end{bmatrix}) = 1 + S(\begin{bmatrix}3 & 3 & 2 & 1\\3 & 3 & 2 & 1\\2 & 2 & 2 & 1\\1 & 1 & 1 & 1\end{bmatrix})$$

$$= 1 + 4 + S(\begin{bmatrix} 2 & 2 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}).$$

Continue this idea to show that $S(B) = 1^2 + 2^2 + 3^2 + 4^2$ *for the matrix* B *in the last subproblem.*

5. *Show that* $S(A) = S(B) = S(C) = 1^2 + 2^2 + 3^2 + 4^2$, *and compute* $S(A + B + C)$. *(This idea can be generalized to prove the formula* $1^2 + \cdots + n^2 = \frac{1}{6}(2n^3 + 3n^2 + n)$*.)*

Chapter 2

Linear Geometry

2.1 Geometric interpretations of a linear problem

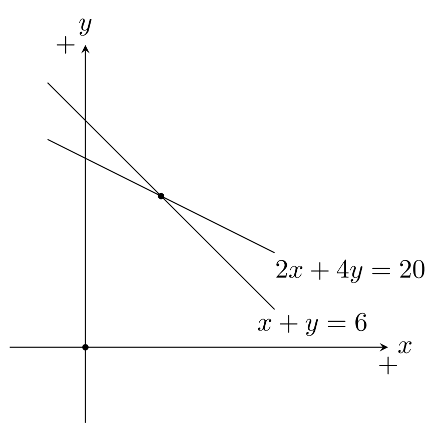
Our class is called linear algebra. Alas, this unfortunate name misled people into believing that we would mostly do algebra and calculations. WRONG! In reality, we will mostly do geometry.

Let us reconsider the chicken-rabbit problem. What geometry lies inside this problem?

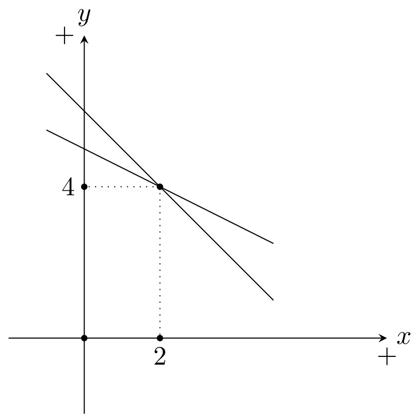
Our problem is a standard system of linear equations. We are basically trying to solve $x + y = 6$ and $2x + 4y = 20$. In matrix-vector form, we are trying to solve

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \end{bmatrix}.$$

Let us first look at this row by row. The two rows here are essentially the two equations $x + y = 6$ and $2x + 4y = 20$. On a Cartesian coordinate plane, these two equations are actually two lines.



From this perspective, each row of the equation $\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \end{bmatrix}$ is a line, and we are trying to find the intersection of the two lines. In particular, we can also get an answer by drawing these lines and measure the location of the intersection, and get $x = 2, y = 4$.



However, let us now change our perspective. Compare the two representations of the same equation:

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \end{bmatrix}.$$

$$\begin{cases} x + y = 6 \\ 2x + 4y = 20 \end{cases}.$$

As we can clearly see, the entries of the matrix are exactly in the corresponding locations as the coeficients of the two line equations. In particular, while each row of the matrix corresponds to a line equation, each column of the matrix corresponds to an unknown variable!

We can also see this more clearly via a direct calculation: each column is exactly a list of all coefficients relevant to an unknown variable.

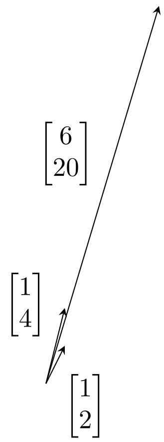
$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

Now in this view, our goal is to solve the following vector equation:

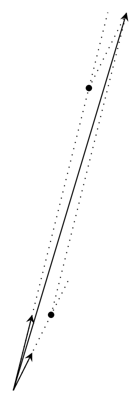
$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \end{bmatrix}.$$

We are trying to solve x, y from above. This means that we have vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, and we want to know how to combine them to get $\begin{bmatrix} 6 \\ 20 \end{bmatrix}$.

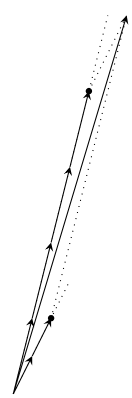
Graphically, it looks like this



How can the two arrows combine to get the big arrow? From the end point of the big arrow, we can draw parallel lines and get a parallelogram.



Can you see it? In fact, we can tell by measurement that $x = 2$ and $y = 4$.



Let us put a summary here. A linear problem will always have two geometric interpretations. If we read row by row, then we are trying to find intersections. If we read column by column, then we are trying to combine vectors. Either way, they are all the same problem.

2.2 Lines on a plane

Linear problems all have (at least two) geometric interpretations. These interpretations are wonderful as a source of guidance and intuition. But to make use of them, we need to have a sound grasp on (Cartesian) geometry itself.

Consider this question in linear algebra: say we have two linear equations on two variables, i.e., say

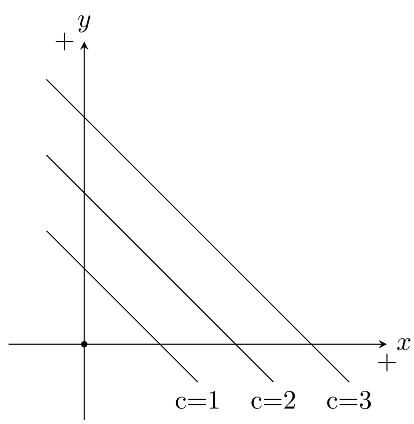
$$\begin{cases} x+y=2 \\ 2x+2y=3 \end{cases}.$$

Does it have a unique solution? Geometrically, if we take the row-view, then we are asking if two lines have a unique intersection point. In particular, we want a way to check if they are parallel. Let us try to review some relevant geometric concepts.

In planar geometry, if we try to study a line on a coordinate plane, then we can simply write out its equation. Typically, the equation of a line should be something like

$$ax+by=c.$$

Here a, b, c are known constants, a, b are not simultaneously zero, and x, y are the coordinates of points on this line. The ratio $a : b$ controls the slope (or gradient in some textbook) of this line. (We need a, b to not both be zero, so that this ratio makes sense.) And if we fix a, b and change c, then the line will move around in a parallel manner. Below are lines $x + y = c$ for various values of c.

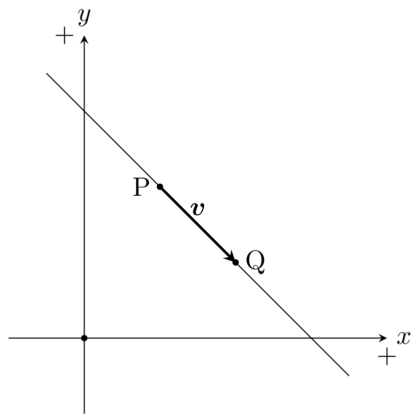


Note that $c = 0$ is exactly when the line goes through the origin, because $x = y = 0$ is always a solution to $ax + by = 0$.

In particular, given two lines, say with equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, how can we tell if they are parallel or not? Well, they are parallel if and only if their coefficients for x, y have the same ratio, i.e., $a_1 : b_1 = a_2 : b_2$.

Definition 2.2.1. *We define the normal vector to the equation $ax + by = c$ to be $\begin{bmatrix} a \\ b \end{bmatrix}$. (Then two lines are parallel if and only if the normal vectors to their equaitons are parallel.)*

Now let us take a slightly different view. Given a line we can draw an arrow on this line. This arrow vector clearly marks the direction of this line! We can easily obtain this directional vector in this manner: simply take any two points on the line, and take their difference. In the example below, we take two points P, Q, then the vector $\boldsymbol{v} = Q - P$ is a direction vector of this line.



Definition 2.2.2. *Given a line, a direction vector for this line is any non-zero arrow vector starting and ending on this line. (Then two lines are parallel if and only if their direction vectors are parallel.)*

Now we have two ways to decide parallelism: normal vector and direction vector. Are these two related?

Say our line has equation $ax+by=c$, so the normal vector is $\boldsymbol{n}=\begin{bmatrix}a\\b\end{bmatrix}$. Suppose $P=\begin{bmatrix}x_P\\y_P\end{bmatrix}$ and $Q=\begin{bmatrix}x_Q\\y_Q\end{bmatrix}$ are two points on this line, then $\boldsymbol{v}=Q-P=\begin{bmatrix}x_Q-x_P\\y_Q-y_P\end{bmatrix}$ is a direction vector. Since P,Q lies on our line, this means we have

$$ax_P+by_P=c,$$

$$ax_Q+by_Q=c.$$

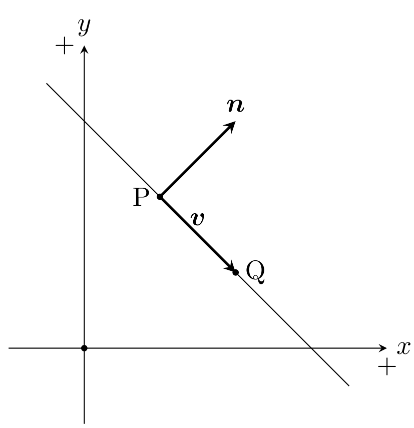
Take the difference between the equations we have

$$a(x_Q-x_P)+b(y_Q-y_P)=0.$$

What is this? This is a dot product.

$$\boldsymbol{n}\cdot\boldsymbol{v}=0.$$

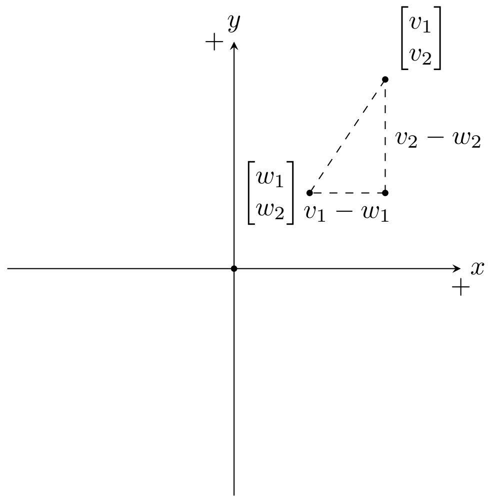
So the two vectors are orthogonal to each other!



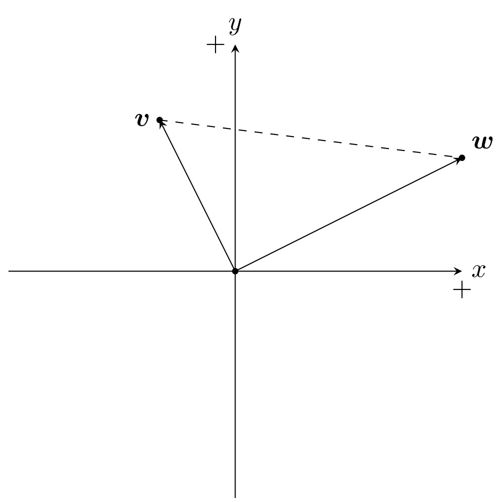
Hopefully everything here is just a review. If not, don't worry. Let us dig into these concepts.

2.3 Dot product

The importance of dot product comes from Pythagorean theorem. On the plane, given two points $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, what is their distance? According to Pythagorean theorem, the distance is $\sqrt{(v_1 - w_1)^2 + (v_2 - w_2)^2}$.



Now, suppose we have two arrow vectors $\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, when are they perpendicular (orthogonal)? Consider the following graph



We have $\boldsymbol{v} \perp \boldsymbol{w}$ if and only if the triangle above is a right triangle, i.e., it obey Pythagorean theorem. Let $||\boldsymbol{v}||$ denote the length of a vector $\boldsymbol{v}$, then the three sides of the triangle above have lengths $||\boldsymbol{v}||, ||\boldsymbol{w}||, ||\boldsymbol{v}-\boldsymbol{w}||$. So $\boldsymbol{v} \perp \boldsymbol{w}$ if and only if we have

$$||\boldsymbol{v}||^2 + ||\boldsymbol{w}||^2 = ||\boldsymbol{v} - \boldsymbol{w}||^2.$$

In terms of coordinates, this means

$$(v_1^2 + v_2^2) + (w_1^2 + w_2^2) = (v_1 - w_1)^2 + (v_2 - w_2)^2.$$

Substract $v_1^2, w_1^2, v_2^2, w_2^2$ from both sides, we have

$$0 = -2v_1w_1 - 2v_2w_2.$$

In particular, $\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \perp \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ if and only if $v_1w_1 + v_2w_2 = 0$.

Definition 2.3.1. *Given vectors $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$, say $\boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \boldsymbol{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$, we define their dot product as $\boldsymbol{v} \cdot \boldsymbol{w} = v_1w_1 + \cdots + v_nw_n$. (Or simply $\boldsymbol{v} \cdot \boldsymbol{w} = \sum v_iw_i$.)*

Proposition 2.3.2. *The length of a vector $\boldsymbol{v}$ is $||\boldsymbol{v}|| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}}$.*

Proof. Repeated Pythagorean theorem. □

Proposition 2.3.3. *Dot product is symmetric and bilinear, i.e., we have*

$$\boldsymbol{v} \cdot \boldsymbol{w} = \boldsymbol{w} \cdot \boldsymbol{v}.$$

And we have for all constants s, t and vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$

$$(s\boldsymbol{u} + t\boldsymbol{v}) \cdot \boldsymbol{w} = s(\boldsymbol{u} \cdot \boldsymbol{w}) + t(\boldsymbol{v} \cdot \boldsymbol{w}),$$

$$\boldsymbol{w} \cdot (s\boldsymbol{u} + t\boldsymbol{v}) = s(\boldsymbol{w} \cdot \boldsymbol{u}) + t(\boldsymbol{w} \cdot \boldsymbol{v}).$$

Proposition 2.3.4 (n-dimension orthogonality)**.** *In $\mathbb{R}^n$, $\boldsymbol{v} \perp \boldsymbol{w}$ if and only if $\boldsymbol{v} \cdot \boldsymbol{w} = 0$.*

Proof. By Pythagorean theorem, $\boldsymbol{v} \perp \boldsymbol{w}$ if and only if the triangle with sides $||\boldsymbol{v}||, ||\boldsymbol{w}||, ||\boldsymbol{w} - \boldsymbol{v}||$ is a right triangle. (You can draw it in three dimension to see this clearly.)

And again by Pythagorean theorem, this would happen if and only if $||\boldsymbol{v}||^2 + ||\boldsymbol{w}||^2 = ||\boldsymbol{v} - \boldsymbol{w}||^2$. This means

$$(\sum v_i^2) + (\sum w_i^2) = \sum (v_i - w_i)^2.$$

Move all left hand side terms to the right, this means

$$0 = \sum [(v_i - w_i)^2 - v_i^2 - w_i^2].$$

And this means

$$0 = \sum (-2v_iw_i).$$

And this means (by law of distribution)

$$0 = -2 \sum v_iw_i.$$

And this means

$$\sum v_iw_i = 0.$$

So this means $\boldsymbol{v} \cdot \boldsymbol{w} = 0$. All deductions in this proof go both ways, so this condition is "if and only if". □

So you see, dot product is very useful in detecting orthogonality.

Now let us go back to geometry, and use dot product as our newest tool. Consider the three dimensional space $\mathbb{R}^3$, then what is the collection of all points $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ satisfying a single equation $ax + by + cz = d$? Here a, b, c, d are some known constants, and a, b, c are not simultaneously zero.

The answer is a plane. But let us slow down and see some simple cases first.

Suppose $d = 0$. What is the solution set to $ax + by + cz = 0$? This is a dot product format! We are asking for vectors $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ such that $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$. In particular, given the known vector (and normal vector to the equation) $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, we are looking at all vectors perpendicular to it. Consequently, the solution set to $ax + by + cz = 0$ is exactly the plane perpendicular to $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and going through the origin.

Now, in general, what is $ax + by + cz = d$? Just like two-dimensional cases, $ax + by + cz = 0$ and $ax + by + cz = d$ are parallel. The change in d only translate the planes in a parallel manner. So as we can see, the solution set to $ax + by + cz = d$ is a plane perpendicular to the normal vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, and the value d controls how "high/low" is this plane.

Try to graph $x + y + z = 1$. This is a plane that goes through the points $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and you can see that it is indeed perpendicular to its normal vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

To sum up, given a plane and its equation, you can find its perpendular direction as easy as reading the coefficients! What is a direction perpendicular to the plane $x + 2y + z = 4$? It is simlpy $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. I don't even need to calculate anything, I just read.

This idea can easily be generalized.

Definition 2.3.5. *In $\mathbb{R}^n$, a hyperplane is the solution set to some equation $a_1x_1 + \cdots + a_nx_n = b$, here $x_1, \ldots, x_n$ are unknown coordinates, $a_1, ..., a_n, b$ are known constants, and $a_1, \ldots, a_n$ are not simultaneously zero. (This hyper plane is perpendicular to its normal vector $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$.)*

The dot product is useful for many things.

Proposition 2.3.6 (n-dimension orthogonality)**.** *In $\mathbb{R}^n$, $||\boldsymbol{v}||^2 = \boldsymbol{v} \cdot \boldsymbol{v}$. And given two vectors $\boldsymbol{v}, \boldsymbol{w}$, if the angle between them is θ, then $\cos\theta = \frac{\boldsymbol{v}\cdot\boldsymbol{w}}{||\boldsymbol{v}||\,||\boldsymbol{w}||}$.*

Proof. By Pythagorean theorem, $||\boldsymbol{v}||^2 = \boldsymbol{v} \cdot \boldsymbol{v}$ is obvious.

For the angle formula, note that the arrows $\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{v} - \boldsymbol{w}$ form a triangle.

The cosine theorem of a triangle gives

$$||\boldsymbol{v} - \boldsymbol{w}||^2 = ||\boldsymbol{v}||^2 + ||\boldsymbol{w}||^2 - 2||\boldsymbol{v}||\,||\boldsymbol{w}|| \cos\theta.$$

Now we have

$$||\boldsymbol{v} - \boldsymbol{w}||^2 = (\boldsymbol{v} - \boldsymbol{w}) \cdot (\boldsymbol{v} - \boldsymbol{w}) = \boldsymbol{v} \cdot \boldsymbol{v} + \boldsymbol{w} \cdot \boldsymbol{w} - 2\boldsymbol{v} \cdot \boldsymbol{w}.$$

So plug it into the cosine theorem equation,

$$\boldsymbol{v} \cdot \boldsymbol{v} + \boldsymbol{w} \cdot \boldsymbol{w} - 2\boldsymbol{v} \cdot \boldsymbol{w} = \boldsymbol{v} \cdot \boldsymbol{v} + \boldsymbol{w} \cdot \boldsymbol{w} - 2||\boldsymbol{v}||\,||\boldsymbol{w}|| \cos\theta.$$

Simplification gives

$$\cos\theta = \frac{\boldsymbol{v}\cdot\boldsymbol{w}}{||\boldsymbol{v}||\,||\boldsymbol{w}||}.$$

$\square$

Now let us go back to linear algebra. Our typical linear problem is the following: given a matrix A and a vector $\boldsymbol{b}$, we want to solve $\boldsymbol{x}$ from $A\boldsymbol{x} = \boldsymbol{b}$. Suppose A only has one row, say $A = \begin{bmatrix} a_1, \dots, a_n \end{bmatrix}$. Then according to our matrix-vector multiplication formula.

$$A\boldsymbol{x} = \begin{bmatrix} a_1, \dots, a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = a_1x_1 + \cdots + a_nx_n = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

Let us use a superscript "T" to indicate that we are writing vectors horizontally, i.e., $\begin{bmatrix} 1 \\ 2 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} 1 & 2 \end{bmatrix}$. Then we see that

$$\boldsymbol{a}\cdot\boldsymbol{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x}.$$

Here $\boldsymbol{a}^{\mathrm{T}}$ is a one row matrix. As we can see, the one-row matrix as a linear map simply means that we are taking dot product with a fixed vector. This is the meaning of all one-row matrices as linear maps. (In this sense, dot product describes the easiest kinds of linear maps.)

What if we have many rows? Let us now go back to the chicken rabbit problem. The equations are

$$\begin{cases} x + y = 6 \\ 2x + 4y = 20 \end{cases}.$$

But let us rewrite this in terms of dotproducts. We are seeking $\begin{bmatrix} x \\ y \end{bmatrix}$ such that its dot product with $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ will produce the desired value 6 and 20. (So, we now have a dot-product interpretation of the chicken-rabbit problem.)

In particular, we can see from this example that the first coordinate of $A\boldsymbol{x}$ is simply the dot product between the first row vector of A (the normal vector to the first equation/hyperplane) and the unknown $\boldsymbol{x}$.

This actually gives us some computational insights about matrix-vector multiplications! In general, given a matrix A and a vector $\boldsymbol{b}$, we want to solve $\boldsymbol{x}$ from $A\boldsymbol{x} = \boldsymbol{b}$. Now the rows of A are in fact normal vectors to various hyperplanes. Let these vectors be $\boldsymbol{h}_1, \dots, \boldsymbol{h}_m$, then we have

$$A = \begin{bmatrix} \boldsymbol{h}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{h}_m^{\mathrm{T}} \end{bmatrix}.$$

And we have a calculation rule

$$\begin{bmatrix} \boldsymbol{h}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{h}_m^{\mathrm{T}} \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} \boldsymbol{h}_1^{\mathrm{T}}\boldsymbol{x} \\ \vdots \\ \boldsymbol{h}_m^{\mathrm{T}}\boldsymbol{x} \end{bmatrix} = \begin{bmatrix} \boldsymbol{h}_1\cdot\boldsymbol{x} \\ \vdots \\ \boldsymbol{h}_m\cdot\boldsymbol{x} \end{bmatrix}$$

So this is one way to think of matrix-vector multiplications in general: we simply perform a dot product between each row of the matrix and the vector. This way we will get corresponding coordinates one by one. Here are some examples, and see if you can feel the dot product involved.

$$\begin{bmatrix}1 & 1\\ 2 & 4\end{bmatrix}\begin{bmatrix}5\\ 6\end{bmatrix} = \begin{bmatrix}1\times 5 + 1\times 6\\ 2\times 5 + 4\times 6\end{bmatrix}.$$

$$\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}2\\ 1\end{bmatrix} = \begin{bmatrix}1\times 2 + 2\times 1\\ 3\times 2 + 4\times 1\end{bmatrix}.$$

2.4 Crossing product

Let us set our eyes on a series of geometric problems. First, given two (non-parallel) vectors in $\mathbb{R}^3$, can you find a vector perpendicular to both of them?

Say we have $\boldsymbol{v} = \begin{bmatrix}v_1\\ v_2\\ v_3\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}w_1\\ w_2\\ w_3\end{bmatrix}$. Our goal is to find $\boldsymbol{x}$ such that $\boldsymbol{v}\cdot\boldsymbol{x} = \boldsymbol{w}\cdot\boldsymbol{x} = 0$. In terms of matrix-vector multiplication, it means

$$\begin{bmatrix}v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}.$$

In terms of regular high school algebra, it is this problem:

$$\begin{cases}v_1x_1 + v_2x_2 + v_3x_3 = 0\\ w_1x_1 + w_2x_2 + w_3x_3 = 0\end{cases}.$$

Now, one way to get a solution is to ELIMINATE a variable. For example, let us eliminate x_1. To get rid of x_1, we want to use our two equations, adjust the coefficient of x_1 to be the same, and then take the difference. In this case, we multiply the first equation by w_1 and multiply the second equation by v_1, and then we take the difference. This gives

$$(v_2w_1 - v_1w_2)x_2 + (v_3w_1 - v_1w_3)x_3 = 0.$$

In particular, this means $x_2 : x_3 = (v_3w_1 - v_1w_3) : (v_1w_2 - v_2w_1)$.

Similarly, if we start by eliminating the variable x_3, we would get $x_1 : x_2 = (v_2w_3 - v_3w_2) : (v_3w_1 - v_1w_3)$.

Combine both, we see that $x_1 : x_2 : x_3 = (v_2w_3 - v_3w_2) : (v_3w_1 - v_1w_3) : (v_1w_2 - v_2w_1)$.

So $\begin{bmatrix}v_2w_3 - v_3w_2\\ v_3w_1 - v_1w_3\\ v_1w_2 - v_2w_1\end{bmatrix}$ is a vector that is perpenducular to both $\boldsymbol{v}$ and $\boldsymbol{w}$.

Definition 2.4.1. *In the three-dimensional space $\mathbb{R}^3$, we define the cross product such that*

$$\begin{bmatrix}v_1\\ v_2\\ v_3\end{bmatrix} \times \begin{bmatrix}w_1\\ w_2\\ w_3\end{bmatrix} = \begin{bmatrix}v_2w_3 - v_3w_2\\ v_3w_1 - v_1w_3\\ v_1w_2 - v_2w_1\end{bmatrix}.$$

In general, if $\boldsymbol{v}, \boldsymbol{w}$ are parallel, you can check that $\boldsymbol{v}\times\boldsymbol{w} = \boldsymbol{0}$, the zero vector. If they are not parallel, then $\boldsymbol{v}\times\boldsymbol{w}$ is a vector perpendicular to both $\boldsymbol{v}$ and $\boldsymbol{w}$. (And all vectors perpendicular to both $\boldsymbol{v}, \boldsymbol{w}$ are simply multiples of $\boldsymbol{v}\times\boldsymbol{w}$.)

Now let us see another problem. Given two non-parallel planes in $\mathbb{R}^3$, how can we find its intersection? To simplify the problem, let us assume that both planes go through the origin. If their equations are $v_1x_1 + v_2x_2 + v_3x_3 = 0$ and $w_1x_1 + w_2x_2 + w_3x_3 = 0$, then we are trying to solve

$$\begin{cases}v_1x_1 + v_2x_2 + v_3x_3 = 0\\ w_1x_1 + w_2x_2 + w_3x_3 = 0\end{cases}.$$

It is the same problem as before! But now we have some geometric intuitions. Intuitively, if the two planes are not parallel, then this intersection is a line. How can we describe a line? We need a direction vector. Since the solution is $x_1 : x_2 : x_3 = (v_2w_3 - v_3w_2) : (v_3w_1 - v_1w_3) : (v_1w_2 - v_2w_1)$, therefore the direction of this line is exactly $\boldsymbol{v} \times \boldsymbol{w}$.

In short, if two planes have normal vectors $\boldsymbol{v}, \boldsymbol{w}$, then their intersection line should have direction parallel to $\boldsymbol{v} \times \boldsymbol{w}$.

Please do some calculation practices yourself to get familiar with cross product a bit.

2.5 Exercises

Exercise 2.5.1. *Recall that, whenever we have a degree one equation on two variables, then its solution set is a line on the Cartesian plane $\mathbb{R}^2$. When you draw lines, make sure to find the x, y-intercepts and mark it on the graph (i.e., the intersection of the line with x, y-axes).*

1. *Draw the lines for the equations $2x + 3y = 6$, $3x + 3y = 6$, $4x + 3y = 6$. What are their relations?*

2. *Draw the lines for the equations $2x + 3y = 6$, $2x + 2y = 6$, $2x + y = 6$. What are their relations?*

3. *Draw the lines for the equations $2x + 3y = 6$, $2x + 3y = 9$, $2x + 3y = 12$. What are their relations?*

Exercise 2.5.2. *Recall that, whenever we have a degree one equation on two variables, then its solution set is a line on the Cartesian plane $\mathbb{R}^2$. When you draw lines, make sure to find the x, y-intercepts and mark it on the graph (i.e., the intersection of the line with x, y-axes).*

For this problem, we want to investigate the following question: what would happen to the lines if we perform linear combinations of equations?

1. *Draw the lines for the equations $2x + 3y = 6$, $4x + 6y = 12$, $6x + 9y = 18$. What are their relations?*

2. *Find the intersection of the lines $2x + y = 4$ and $2x + 3y = 6$.*

3. *Let us see what happen if we add the two line equations. Draw the lines $2x + y = 4$, $2x + 3y = 6$, and $(2x + y) + (2x + 3y) = 4 + 6$. What are their relations?*

4. *Draw the lines $(2x+y)+(2x+3y) = 4+6$, $(2x+y)+2(2x+3y) = 4+12$, $(2x+y)+3(2x+3y) = 4+18$. What is the relation between them? (Challenge problem: How would you describe the tendency of change as we add more and more $2x + 3y = 6$ to our equation?)*

Exercise 2.5.3. *I want to find my soul mate. Let us imagine that souls are vectors in a 5-dimensional space, and all souls have the same length (because all human beings are equal). The compatibility between two souls is calculated by taking a dot product.*

Say my soul is the vector $\boldsymbol{x} = \frac{1}{4}\begin{bmatrix} 1 \\ 2 \\ -1 \\ -3 \\ 1 \end{bmatrix}$.

1. *Find my soul compatibility with Mr Boring, whose soul vector is* $\frac{1}{\sqrt{5}}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

2. *Find a soul vector $\boldsymbol{y}$ such that $\boldsymbol{x} \cdot \boldsymbol{y}$ is as large as possible. This is suppose to be my soul mate. What do you find? (Mind that $\boldsymbol{y}$ should also have length one.)*

3. *Find a soul vector* $\boldsymbol{y}$ *such that* $\boldsymbol{x} \cdot \boldsymbol{y}$ *is as small as possible. This is suppose to be my arch-enemy. What do you find?*

Exercise 2.5.4. *Let us explore some goofy cross products. Let* $\boldsymbol{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ *and* $\boldsymbol{w} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

1. *Find* $\boldsymbol{v} \times \boldsymbol{w}$.
2. *Find* $\boldsymbol{w} \times \boldsymbol{v}$. *Observe the calculation results, what is its relation with* $\boldsymbol{v} \times \boldsymbol{w}$*?*
3. *Find* $(\boldsymbol{v} \times \boldsymbol{w}) \times \boldsymbol{w}$.
4. *Find* $\boldsymbol{v} \times (\boldsymbol{w} \times \boldsymbol{w})$. *Is there a law of associativity for cross product?*
5. *Compare the value of* $\boldsymbol{v} \times \boldsymbol{w}$, $(\boldsymbol{v} \times \boldsymbol{w}) \times \boldsymbol{w}$, $((\boldsymbol{v} \times \boldsymbol{w}) \times \boldsymbol{w}) \times \boldsymbol{w}$. *Do you see a pattern?*
6. *Challenge: write the pattern as a mathematical theorem and prove it in general.*

Exercise 2.5.5 (Gram Schmidt Orthogonalization)**.** *Recall that a unit vector is a vector with length one. Our goal is to find three mutually orthogonal unit vectors in* $\mathbb{R}^3$. *The process here is called the Gram Schmidt Orthogonalization, where we started with three arbitrary vectors, and "shear" them so that they become mutually orthogonal unit vectors. This is one of the most important linear algebra algorithms.*

Let $\boldsymbol{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}, \boldsymbol{w} = \begin{bmatrix} -8 \\ 0 \\ -11 \end{bmatrix}$. *These three vectors are not mutually orthogonal, and not unit vectors.*

1. *Calculate* $\boldsymbol{u} \cdot \boldsymbol{u}$ *and* $\boldsymbol{u} \cdot \boldsymbol{v}$ *and* $\boldsymbol{u} \cdot \boldsymbol{w}$.
2. *Now, we aim to chanch* $\boldsymbol{v}, \boldsymbol{w}$ *along* $\boldsymbol{u}$*, so that they are orthogonal to* $\boldsymbol{u}$. *Please find real numbers* s, t *such that* $\boldsymbol{u} \perp \boldsymbol{v} + s\boldsymbol{u}, \boldsymbol{w} + t\boldsymbol{u}$. *From now on, let* $\boldsymbol{v}' = \boldsymbol{v} + s\boldsymbol{u}$ *and* $\boldsymbol{w}' = \boldsymbol{w} + t\boldsymbol{u}$.
3. *Calculate* $\boldsymbol{v}' \cdot \boldsymbol{v}'$ *and* $\boldsymbol{v}' \cdot \boldsymbol{w}'$.
4. *Now, we aim to change* $\boldsymbol{w}'$ *along* $\boldsymbol{v}'$*, so that they are orthogonal to* $\boldsymbol{v}'$. *Please find real numbers* r *such that* $\boldsymbol{v}' \perp \boldsymbol{w}' + r\boldsymbol{v}'$. *From now on, let* $\boldsymbol{w}'' = \boldsymbol{w}' + r\boldsymbol{v}'$.
5. *Are the vectors* $\boldsymbol{u}, \boldsymbol{v}', \boldsymbol{w}''$ *mutually orthogonal to each other? Verify their dot products.*
6. *Let* $\boldsymbol{u}_0 = \frac{1}{||\boldsymbol{u}||}\boldsymbol{u}$, $\boldsymbol{v}_0 = \frac{1}{||\boldsymbol{v}'||}\boldsymbol{v}'$, *and* $\boldsymbol{w}_0 = \frac{1}{||\boldsymbol{w}''||}\boldsymbol{w}''$. *Find these vectors. Congradulation! You have done your first Gram Schmidt Orthogonalization.*

Exercise 2.5.6 (The projection map)**.** *Fix a vector* $\boldsymbol{v} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.

1. *Compute* $||\boldsymbol{v}||$. *(This property of* $\boldsymbol{v}$ *is crucial! Only for unit vectors, the following will be projections.)*
2. *Let us decompose* $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ *into two vectors,* $\begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{2\sqrt{2}} \end{bmatrix}$ *and* $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{2\sqrt{2}} \end{bmatrix}$. *What are their relations to* $\boldsymbol{v}$*? (Check if they are parallel or orthogonal to* $\boldsymbol{v}$.*)*
3. *Find the length of the projection of* $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ *to the direction of* $\boldsymbol{v}$. *Compare this with* $\boldsymbol{v} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. *(Challenge: can you see why we have this pattern?)*

4. We have a "projection length" map $P : \mathbb{R}^3 \to \mathbb{R}$ such that $P(\boldsymbol{x}) = \boldsymbol{v} \cdot \boldsymbol{x}$. Show that P is linear, i.e., verify that for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$ and any $a, b \in \mathbb{R}$, then $P(a\boldsymbol{x} + b\boldsymbol{y}) = aP(\boldsymbol{x}) + bP(\boldsymbol{y})$. (This map sends each $\boldsymbol{x}$ to the length of its projection to the direction of $\boldsymbol{v}$.)

5. Find the matrix for the linear map P.

6. We have a "projection vector" map $Q : \mathbb{R}^3 \to \mathbb{R}^3$ such that $Q(\boldsymbol{x}) = (\boldsymbol{v} \cdot \boldsymbol{x})\boldsymbol{v}$. Show that Q is linear, i.e., verify that for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$ and any $a, b \in \mathbb{R}$, then $Q(a\boldsymbol{x} + b\boldsymbol{y}) = aQ(\boldsymbol{x}) + bQ(\boldsymbol{y})$. (This map sends each $\boldsymbol{x}$ to the resulting vector of its projection to the direction of $\boldsymbol{v}$.)

7. Find the matrix for the linear map Q.

8. Verify that for all $\boldsymbol{x} \in \mathbb{R}^3$, we have $Q(Q(\boldsymbol{x})) = Q(\boldsymbol{x})$.

Exercise 2.5.7. *Fix a vector* $\boldsymbol{v} = \begin{bmatrix}1\\2\\3\end{bmatrix}$. *We now investigate cross products*

1. Calculate $\boldsymbol{v} \times \boldsymbol{v}$.

2. As usual, we use $\boldsymbol{e}_i$ to denote the vector whose i-th coordinate is 1 and other coordinates are 0. Calculate $\boldsymbol{v} \times \boldsymbol{e}_1$, $\boldsymbol{v} \times \boldsymbol{e}_1$ and $\boldsymbol{v} \times \boldsymbol{e}_3$.

3. We have a map $C : \mathbb{R}^3 \to \mathbb{R}^3$ such that $C(\boldsymbol{x}) = \boldsymbol{v} \times \boldsymbol{x}$. Show that C is linear, i.e., verify that for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$ and any $a, b \in \mathbb{R}$, then $C(a\boldsymbol{x} + b\boldsymbol{y}) = aC(\boldsymbol{x}) + bC(\boldsymbol{y})$.

4. Find the matrix for the linear map C. (Pretty, yes?)

5. Let $\boldsymbol{w} = \begin{bmatrix}1\\1\\-1\end{bmatrix}$. Calculate $\boldsymbol{v} \cdot \boldsymbol{w}$, $\boldsymbol{v} \times \boldsymbol{w}$, $||\boldsymbol{v}||\,||\boldsymbol{w}||$, and $||\boldsymbol{v} \times \boldsymbol{w}||$.

Exercise 2.5.8. *Given any two arrow vectors $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^3$, then they would form a parallelogram. (The four vertices of the parallelogram will have coordinates $\boldsymbol{0}, \boldsymbol{v}, \boldsymbol{w}, \boldsymbol{v} + \boldsymbol{w}$.) What is the area of this parallelogram? By high school geometry, this should be $||\boldsymbol{v}||\,||\boldsymbol{w}||\,|\sin\theta|$, where θ is the angle between $\boldsymbol{v}$ and $\boldsymbol{w}$.*

1. For real numbers $v_1, v_2, v_3, w_1, w_2, w_3$, show that
$$(v_1^2+v_2^2+v_3^2)(w_1^2+w_2^2+w_3^2) = (v_1w_1+v_2w_2+v_3w_3)^2+(v_2w_3-v_3w_2)^2+(v_3w_1-v_1w_3)^2+(v_1w_2-v_2w_1)^2.$$

2. For vectors $\boldsymbol{v} = \begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix}$, $\boldsymbol{w} = \begin{bmatrix}w_1\\w_2\\w_3\end{bmatrix}$, show that
$$||\boldsymbol{v}||^2||\boldsymbol{w}||^2 = (\boldsymbol{v} \cdot \boldsymbol{w})^2 + ||\boldsymbol{v} \times \boldsymbol{w}||^2.$$

3. Recall that $\boldsymbol{v} \cdot \boldsymbol{w} = ||\boldsymbol{v}||\,||\boldsymbol{w}|| \cos\theta$. Show that $||\boldsymbol{v} \times \boldsymbol{w}|| = ||\boldsymbol{v}||\,||\boldsymbol{w}||\,|\sin\theta|$. (In particular, $||\boldsymbol{v} \times \boldsymbol{w}||$ is the area of the parallelogram we are interested in!)

4. Given arrow vectors $\begin{bmatrix}a\\b\end{bmatrix}$ and $\begin{bmatrix}c\\d\end{bmatrix}$ on the plane, what is the area of the parallelogram formed by these two vectors? You should do this by imagining that the plane is actually inside the space, and the vectors are $\begin{bmatrix}a\\b\\0\end{bmatrix}$ and $\begin{bmatrix}c\\d\\0\end{bmatrix}$, then do the cross product to get a formula for area.

Exercise 2.5.9. *Given any three arrow vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^3$, then they would form a parallelopiped. To see this, first move the arrow $\boldsymbol{u}$ according to the vector $\boldsymbol{v}$. This way, the arrow $\boldsymbol{u}$ would sweep through a parallelogram. Next, move this parallelogram according to the vector $\boldsymbol{w}$. Then this parallelogram would sweep through a parallelopiped. The arrows $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are three mutually adjacent edges of the parallelopiped.*

Let $\boldsymbol{u} = \begin{bmatrix}1\\1\\1\end{bmatrix}$, $\boldsymbol{v} = \begin{bmatrix}1\\1\\0\end{bmatrix}$, $\boldsymbol{w} = \begin{bmatrix}0\\2\\1\end{bmatrix}$

What is its volume? We think of the parallelogram formed by $\boldsymbol{u}, \boldsymbol{v}$ as the "base", and we should try to get the volumn by "base times height".

1. *To get the height, first we need the direction of the height, which should be perpendicular to both $\boldsymbol{u}$ and $\boldsymbol{v}$. To do this, calculate $\boldsymbol{u} \times \boldsymbol{v}$, and $\frac{\boldsymbol{u}\times\boldsymbol{v}}{||\boldsymbol{u}\times\boldsymbol{v}||}$.*
2. *To get the value of the height, we need to project $\boldsymbol{w}$ onto the direction of the height. So the height is $|\frac{\boldsymbol{u}\times\boldsymbol{v}}{||\boldsymbol{u}\times\boldsymbol{v}||} \cdot \boldsymbol{w}|$. Calculate this value.*
3. *The volume is base times height. So calculate the volume by multiplying the base area $||\boldsymbol{u} \times \boldsymbol{v}||$ with the height $|\frac{\boldsymbol{u}\times\boldsymbol{v}}{||\boldsymbol{u}\times\boldsymbol{v}||} \cdot \boldsymbol{w}|$.*
4. *(Read only) As a summary, it seems that the volume is in fact simply $|(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}|$. Note that if we started by treating $\boldsymbol{v}, \boldsymbol{w}$ as base, then we would also get a formula $|(\boldsymbol{v} \times \boldsymbol{w}) \cdot \boldsymbol{u}|$, and so on. In short, these "cross product then dot product" structures are quite symmetric. In the expression $|(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}|$, you can rearrange the order of the three vectors in whatever way you like, and the value you get is always the same: the volume of the parallelopiped.*

Exercise 2.5.10. *How can we define a line in higher dimensions, i.e., a line in $\mathbb{R}^n$? We can no longer use equations. In this problem, we would rigorously define the meaning of a line in $\mathbb{R}^n$. We shall use $n = 3$ as an example.*

Pick two points P, Q in the space $\mathbb{R}^3$, say $P = \begin{bmatrix}1\\2\\3\end{bmatrix}$, $Q = \begin{bmatrix}2\\3\\4\end{bmatrix}$. *How can we find the line through P, Q in the space?*

1. *Find the coordinates of the arrow vector from P to Q, i.e., find $Q - P$.*
2. *We call the arrow vector in the last subproblem as $\boldsymbol{v}$. Then geometrically, our line should be like this: we start at the point P, and we move in the direction $\boldsymbol{v}$ by an arbitrary amount. So we can describe it as a set $\{P + t\boldsymbol{v} : t \in \mathbb{R}\}$. This means this set contains all points of the format $P + t\boldsymbol{v}$, where t can be any real number. Please write out the coordinates of $P + t\boldsymbol{v}$ as a function of t.*
3. *Let P, Q be as before, but let* $\boldsymbol{v} = \begin{bmatrix}1\\1\\0\end{bmatrix}$ *and* $\boldsymbol{w} = \begin{bmatrix}0\\0\\1\end{bmatrix}$. *Then find the intersection of the line $\{P + t\boldsymbol{v} : t \in \mathbb{R}\}$ and the line $\{Q + s\boldsymbol{w} : s \in \mathbb{R}\}$. (Note that we use different letters for the unknown variables in the description of the two sets, because they may not have the same value. This is important to avoid mistakes!)*
4. *Find two degree one equations in three variables x, y, z, so that the common solutions to both equations are exactly the points on the line.*

Exercise 2.5.11 (Basic geometric concepts in n-dimensional space)**.** ☕ *We shall generalize the concepts in the last problem.*

Consider the space $\mathbb{R}^n$ and any non-zero $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$ with $\boldsymbol{a}, \boldsymbol{b}$ NOT parallel. (In this class, non-zero means $\boldsymbol{a}, \boldsymbol{b} \neq \boldsymbol{0}$, i.e., the coordinates cannot ALL be zero. But some coordinates are allowed to be zero.) Think of $\boldsymbol{a}, \boldsymbol{b}$ as two distinct points in an n-dimensional space. Note that the "arrow vector" starting from $\boldsymbol{a}$ and ending in $\boldsymbol{b}$ would exactly have coordinates $\boldsymbol{b} - \boldsymbol{a}$.

As we know, a line through $\boldsymbol{a}, \boldsymbol{b}$ could be defined like this: we start at $\boldsymbol{a}$, and we move in the arrow direction $\boldsymbol{b} - \boldsymbol{a}$ by an arbitrary amount, and we get a line. So it is the set $\{\boldsymbol{a} + t(\boldsymbol{b} - \boldsymbol{a}) : t \in \mathbb{R}\}$.

1. *Show that the line through $\boldsymbol{a}, \boldsymbol{b}$ is exactly the set $\{s\boldsymbol{a} + t\boldsymbol{b} : s, t \in \mathbb{R}$ and $s + t = 1\}$.*
2. ☕*Show that for any $\boldsymbol{p}$ on the line through $\boldsymbol{a}, \boldsymbol{b}$, then there is a unique pair s, t such that $\boldsymbol{p} = s\boldsymbol{a} + t\boldsymbol{b}$ and $s + t = 1$. (Hint: to prove uniqueness, suppose $s\boldsymbol{a} + t\boldsymbol{b} = s'\boldsymbol{a} + t'\boldsymbol{b}$, and show that $s = s', t = t'$. Here the fact that $\boldsymbol{a}, \boldsymbol{b}$ are NOT parallel is VERY important.)*
3. *Show that the line segment connecting $\boldsymbol{a}, \boldsymbol{b}$ is exactly the set $\{s\boldsymbol{a} + t\boldsymbol{b} : s, t \in \mathbb{R}$ and $s + t = 1$ and $0 \leq s, t \leq 1\}$. (So insides of a line segment are the "weighted averages" of the two end points.)*
4. *Give a similar definition of a plane through three points $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ in $\mathbb{R}^n$.*
5. *Give a similar definition of a triangle with vertices $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ in $\mathbb{R}^n$. (Here I want a subset of $\mathbb{R}^n$ whose elements are points inside the triangle, including the edges of the triangule.)*
6. *(Read Only) In a similar way you can define k-dimensional* ***affine subspaces*** *of $\mathbb{R}^n$. The word "affine" simply means "flat", i.e., no curvature. A* ***line*** *is a name for 1-dimensional affine subspace, a* ***plane*** *is a name for 2-dimensional affine subspace, and a* ***hyperplane*** *is a name for $n-1$-dimensional affine subspace.*

Exercise 2.5.12 (Hyperplanes and their normal vectors)**.** *We know that in $\mathbb{R}^2$, a line can be represented by a "linear" equation. For example, $x + 2y = 2$ would be some line in $\mathbb{R}^2$. What is the situation in higher dimensions?*

Consider the space $\mathbb{R}^3$ and a subset $H = \{\begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + 2y + 3z = 6\}$.

1. *Let $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ be the intersection of H with the x, y, z-axes respectively. Find the coordinates of $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$.*
2. *Using your definition in the last big problem, show that H is exactly a plane through $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$.*
3. *Show that the arrow vectors $\boldsymbol{a} - \boldsymbol{b}, \boldsymbol{b} - \boldsymbol{c}, \boldsymbol{c} - \boldsymbol{a}$ (these are all arrow vectors lying on the plane) are all perpendicular to $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. (The last vector here is obtained by extracting the coefficients $1, 2, 3$ from the equation of our plane $x + 2y + 3z = 6$.)*
4. *(Read Only) In general, any "linear" equation in $\mathbb{R}^n$ would yeild a hyperplane. Specifically, if the variables are $x_1, \ldots, x_n$, then for any coefficients $a_1, \ldots, a_n \in \mathbb{R}$ and any constant $b \in \mathbb{R}$, the solutions to the equation $a_1x_1 + \cdots + a_nx_n = b$ is a hyperplane. And the normal vector to this hyperplane is exactly $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$, which is perpendicular to this hyperplane.*

Chapter 3

Gaussian Elimination

3.1 Preliminary on Maps

We need to get some terminology straight. First we need sets. What is a set? Intuitively, a ***set*** is a collection of stuff, and these stuff inside a set are called ***elements***. This intuitive picture will serve us just fine, so let us NOT dive into the set theory stuff for the moment. We only focus on what we would need.

We typically write sets as simply a list of its elements. For example, if a set S has elements a, b, c, then we say $S = \{a, b, c\}$. When we say a is an element of S, we write $a \in S$. If a is not an element of S, we write $a \notin S$.

However, sometimes we describe a set by a property. If P is some property, then we may write $S = \{x : x \text{ satisfies property } P\}$. This means the elements of S are exactly those satisfying property P.

Finally, if T is a big set, and P is some property, we may also write $S = \{x \in T : x \text{ satisfies property } P\}$. This means the elements of S are exactly the elements in T satisfying property P.

Remark 3.1.1. *This intuitive understanding of set might run into trouble such as Russel's paradox. Learn set theory if you'd like to know more. Luckily, it will not matter much in our course.*

Example 3.1.2 (Set constructions)**.** Say $S = \{1, 2, \clubsuit\}$ and $T = \{2, 3\}$. Then we can create the ***union*** of the sets $S \cup T = \{1, 2, 3, \clubsuit\}$. We can also create the ***intersection*** of the sets $S \cap T = \{2\}$.

We can also create the ***product*** (or more specifically a ***Cartesian product***) of the sets as $S \times T = \{\begin{bmatrix} s \\ t \end{bmatrix} : s \in S \text{ and } t \in T\}$, or more specifically,

$$S \times T = \{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} \clubsuit \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} \clubsuit \\ 3 \end{bmatrix}\}.$$

Note that if $|S|$ means the number of elements in a set S, then we have $|S \times T| = |S| \times |T|$ for finite sets S, T.

Finally, we write $S \times \cdots \times S$ (n times) simply as S^n, as we did in the notation of $\mathbb{R}^n$. Note that we have $|S^n| = |S|^n$ for a finite set S. ☺

Now we need maps to connect various sets.

1. A ***map*** (or sometimes called a ***function***) is a machine that takes an element (or "input") from one set, and transform it into an element (or "output") in anther set.
2. We usually write $f : X \to Y$, if f is sending elements of X to elements of Y.
3. We call the set of inputs the ***domain*** of our map, and the set of outputs the ***codomain*** of our map.
4. If the domain and codomain are the same, we also say that this map is a ***transformation***.

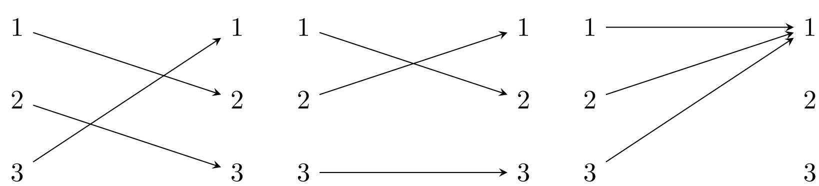
5. If f sends an input x to y, we write $f(x) = y$ or $f : x \mapsto y$. We say y is the ***image*** of x, and x is a ***pre-image*** of y.

   (IMPORTANT DETAIL: be careful of the difference of "a" and "the". When I write "the image", I am implying that such image is UNIQUE, that there is only one possible image for x. When I write "a pre-image", I am saying that maybe y has more than one pre-image. We simply don't know and make no assumptions.)

Example 3.1.3. 1. A human is a map. If we think of X as the set of all foods (inputs), and Y as the set of all, well, you know, outputs. Then $human : X \to Y$ is a map. Each particular food is transformed by a human into a particular piece of, well, you know.

2. In the world of Harry Potter, a wand is a map that sends incantations into magical spells. (Incidentally, "f" looks like a wand in shape.)

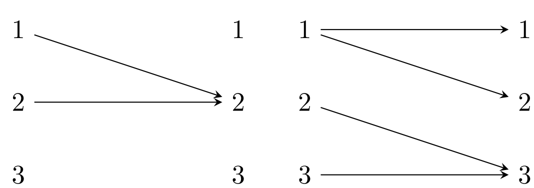
3. Let $X = \{1, 2, 3\}$. Then $f : X \to X$ with $f(1) = 2, f(2) = 3, f(3) = 1$ is a map, and $g : X \to X$ with $g(1) = 2, g(2) = 1, g(3) = 3$ is a map, and $h : X \to X$ with $h(x) = 1$ for all x is a map. In fact, let us draw some diagrams to see them.



(a) Map f (b) Map g (c) Map h

4. Of course we also know many maps from $\mathbb{R}$ to $\mathbb{R}$. Say $f(x) = x + 1, f(x) = 2x, f(x) = e^x, f(x) = \ln(x), f(x) = \sin x$, etc..

5. What is NOT a map? Well, here are two cases where $f : \{1, 2, 3\} \to \{1, 2, 3\}$ is NOT a map.



(a) 3 has no defined value (b) 1 has too many defined value

☺

Remark 3.1.4. *Many academic text in English uses the following shorthands all the time. Might as well get used to them. (They all come from Latin.)*

Some useful shorthands in any English text:

1. *The word "etc." means "and so on...". For example, "I ate some apples, bananas, cherries, etc.."*

2. *The word "i.e." means "that is", and it usually means the next words are explanations of the previous words. For example, "I went to my dream university, i.e., Tsingua University."*

3. *The word "e.g." means "for example". For example, "I like all fruits, e.g., apples, bananas, cherries, etc.."*

4. *I sometimes use "s.t." for "such that". This is not latin of course....*

5. *This is not Latin, but it is VERY important. If I write "**iff**", it does not mean "if", and it is NOT a typo. It means "if and only if". The content before "iff" and the content after "iff" should be logically equivalent.*

6. *(Honorable mention) Sometimes people write "OTW" for "otherwise", "TFAE" for "the following are equivalent", "WLOG" for "without loss of generality", "QED" for "quod erat demonstrandum" (it means the proof is finished).*

3.2 The pre-image problem

As we have seen, a typical linear problem looks like this: find unknown vector $\boldsymbol{x}$ from the equation

$$L(\boldsymbol{x}) = \boldsymbol{b}.$$

Here L is a known linear map (we know its matrix), and $\boldsymbol{b}$ is some known vector.

In short, we have an output of a map, and we try to trace back to a corresponding input. This is a very typical problem in mathematics, called the pre-image problem.

Example 3.2.1. 1. My son had red poop one day. I was super scared for a minute, but then I was reminded that he had dragon fruits for lunch. If we think of my son as a map sending foods into poops, then given the red poop, I have successfully figured out the pre-image.

2. Say $f(x) = 2x + 5$. How to solve for $f(x) = 3$? Well, $2x + 5 = 3$, and then we see that $x = -1$.

3. Say $f(x) = e^x$. How to solve for $f(x) = 1$? Well, $e^x = 1$, so we have $x = \ln 1 = 0$.

4. Say $f(x) = x^2$. How to solve for $f(x) = -1$? Well, you cannot....

5. Say $f(x) = x^2$. How to solve for $f(x) = 1$? Well, there are two solutions $x = \pm 1$. There is no UNIQUE solution!

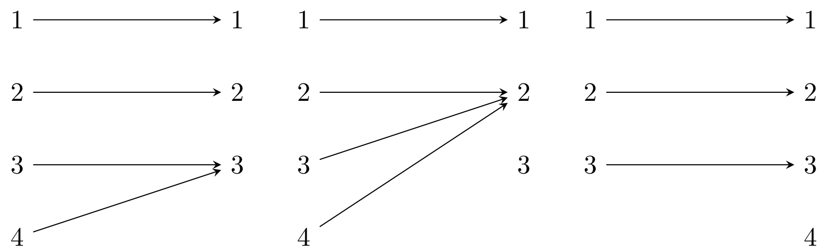
☺

Typically, for a map $f : X \to Y$, we say the set X is the domain of f, and the set Y is the codomain of f. If $f(x) = y$, then we say y is the image of x, while x is a pre-image of y. (We say "a" pre-image rather than "the" pre-image. This is because there might be more than one pre-images, and sometimes there might be no pre-images at all.)

Among the previous examples, obviously the last two scenarios are less desirable. We hope for a unique solution, but that may not happen. In this case, it is not your fault. You should instead blame the map for not being nice enough.

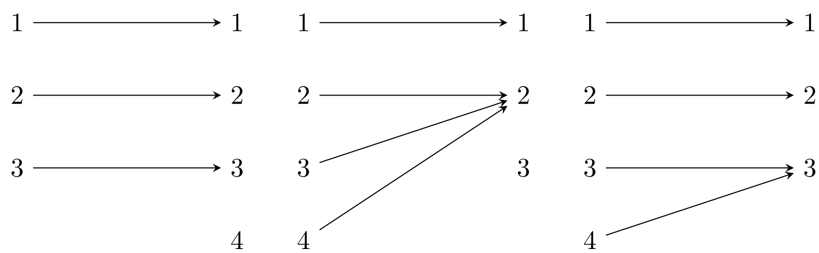
Preferably, we hope our maps would have these kinds of properties:

1. If every element in the codomain has at least one preimage, then we say the map is a ***surjective*** map. Or simply say our map is a ***surjection***. (Everything is the codomain is hit by some arrow in the diagram.)



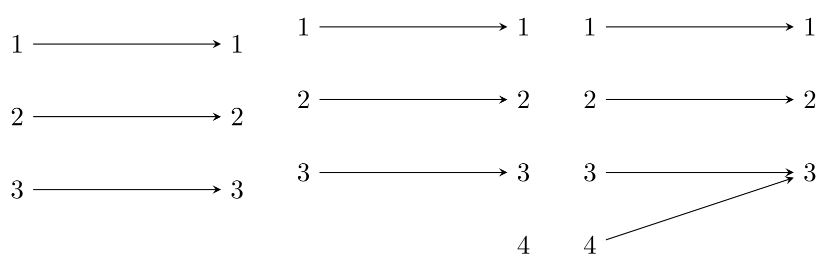
(a) Surjection (b) Not a surjection (c) Not a surjection

2. If every element in the codomain has at most one preimage, then we say the map is a ***injective*** map. Or simply say our map is an ***injection***. (Arrows never collide to the same targets in the diagram.)



(a) Injection (b) Not an injection (c) Not an injection

3. If every element in the codomain has a UNIQUE preimage (exactly one), then we say the map is a ***bijective*** map. Or simply say our map is a ***bijection***. (Arrows give a one-to-one correspondence in the diagram.)



(a) Bijection (b) Not a bijection (c) Not a bijection

Remark 3.2.2.

Bijection = each element in the codomain has exactly one pre-image.
Surjection = each element in the codomain has at least one pre-image.
Injection = each element in the codomain has at most one pre-image.
Since exactly one = "at least one" + "at most one", we immediately see that bijection = surjection + injection.

If $f : X \to Y$ is a bijection, then there is a UNIQUE map $f^{-1} : Y \to X$ such that $f(x) = y$ if and only if $f^{-1}(y) = x$. We can simply reverse the directions of all arrows used to describe f, and we would get this inverse map f^{-1}. Conversely, if a map f has an inverse map, then it must be a bijection. (Otherwise, how can you even define f^{-1}?)

To end this section, let us look at my favorite map. (I'm sure it will be your favorite as well.) It is going to be the ***identity map***.

Definition 3.2.3. *A map* id_X *is an identity map for the set* X *if its domain and codomain are both* X*, and we have* $\mathrm{id}_X(x) = x$ *always. Sometimes we write* id *for short if the underlying set is obvious.*

The identity map is the nicest and most important map ever. In particular it is also linear.

Proposition 3.2.4. *Consider the identity map* $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$*. Then* id *is linear, and its matrix is the* $n \times n$ *matrix* $\begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$*. (Here empty entries means zero.) We call this matrix the **identity matrix**.*

Proof. Recall that the identity map simply does nothing to the input. So it is linear because for all $s, t \in \mathbb{R}$ and $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$, we have

$$\mathrm{id}(s\boldsymbol{a} + t\boldsymbol{b}) = s\boldsymbol{a} + t\boldsymbol{b} = s(\mathrm{id}(\boldsymbol{a})) + t(\mathrm{id}(\boldsymbol{b})).$$

Now let us find its matrix. Since $\mathrm{id}(\boldsymbol{e}_i) = \boldsymbol{e}_i$, we see that the i-th column is simply $\boldsymbol{e}_i$. This results in the pattern as described. ☐

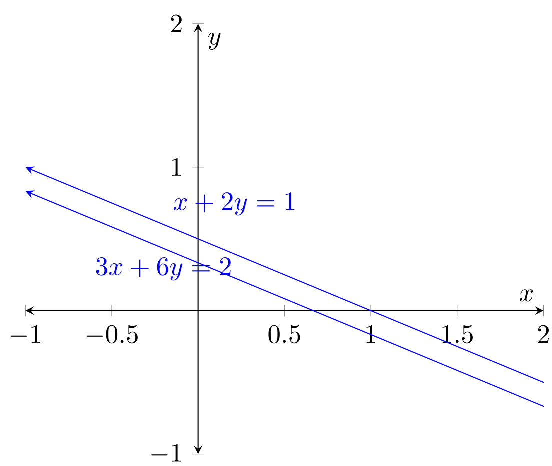
Quick test: what is $\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}$? No need to calculate at all. It is simply $\begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}$.

3.3 The "jectivity" of a linear map

We want to solve the pre-image problem $A\boldsymbol{x} = \boldsymbol{b}$, where A is a linear map/matrix, and $\boldsymbol{b}$ is a known output vector, and $\boldsymbol{x}$ is the unknown input vector that we try to solve for.

Will we have unique solutions? This depends on whether A is injective/surjective/bijective. But how can we tell? Let us see some examples.

Example 3.3.1. Consider $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. It turns out that there is no solution.



With the row view, you may graph out the two lines and try to find intersection. However, you shall see that the two lines are parallel. So there is no intersection. In particular, $\begin{bmatrix}1 & 2\\3 & 6\end{bmatrix}$ is not surjective.

In this sense, we see that for a linear map $A : \mathbb{R}^n \to \mathbb{R}^m$, if the rows of A are parallel, then we are be in trouble. Say if the seond row of A is three times the first row of A. Then for the output $A\boldsymbol{x}$, the second coordinate of $A\boldsymbol{x}$ must be three times the first coordinate of A.

Consequently, possible outputs of $A\boldsymbol{x}$ will all have extra constraints. They cannot be all of the codomain $\mathbb{R}^m$. In particular, A is not surjective. ☺

Here is another situation, where we have too many solutions.

Example 3.3.2. Consider $\begin{bmatrix}1 & 0 & 1\\0 & 1 & 2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$. It turns out that there are infinitely many solutions. And if you tries to solve it the high school way, then you will fail to find a unique answer, because you don't have enough equations.

With the row view, things happen in $\mathbb{R}^3$. You may graph out the two planes in the space, and try to find intersection. However, the two planes intersect in a line. So everything on that line is a solution.

To find this line, since the line lies on both planes, it is orthogonal to both normal vectors, i.e., it is orthogonal to both $\begin{bmatrix}1\\0\\1\end{bmatrix}$ and $\begin{bmatrix}0\\1\\2\end{bmatrix}$. So, this is a line parallel to $\begin{bmatrix}1\\2\\-1\end{bmatrix}$. (You can compute this via cross product.)

By pure luck, I observe that $\begin{bmatrix}1\\1\\1\end{bmatrix}$ satisfies both equations, i.e., it is is on this solution line. So the solution set is the unique line going through the point $\begin{bmatrix}1\\1\\1\end{bmatrix}$ in the direction of $\begin{bmatrix}1\\2\\-1\end{bmatrix}$.

If you like, you may now write your solution in parametrized form, i.e., $\begin{bmatrix}1\\1\\1\end{bmatrix} + t\begin{bmatrix}1\\2\\-1\end{bmatrix}$ for any $t \in \mathbb{R}$. Alternatively, you may also write $\begin{bmatrix}1+t\\1+2t\\1-t\end{bmatrix}$ or, more aligned with high school tradition, as

$$\begin{cases} x = 1+t,\\ y = 1+2t,\\ z = 1-t. \end{cases}$$

To summarize, in this case we have too many solutions. So the linear map $\begin{bmatrix}1 & 0 & 1\\0 & 1 & 2\end{bmatrix}$ is not injective. But can we find some intuition as to why this is happening?

It turns out that we have to look at it with the column view. With the column view, we are trying to combine columns of $\begin{bmatrix}1 & 0 & 1\\0 & 1 & 2\end{bmatrix}$ into the vector $\begin{bmatrix}2\\3\end{bmatrix}$. But in fact the first two columns would be enough. We do not really need the third vector!

This redundant columns of A seems to be the cause. We never need the third vector, because the third column vector is itself a linear combination of the first two columns.

$$\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix} + 2\begin{bmatrix}0\\1\end{bmatrix}.$$

In fact, let us say that $\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$ is some particular solution, so we have

$$x_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y_0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} +_0 z \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Now, if we increase x_0 by 1, y_0 by 2 and decrease z_0 by 1, what would happen?

On the left hand side of the equation above, we would have an extra $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, two extra $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and one less $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. These would cancel out each other! So the equation above would still hold.

In particular, $\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ would still be a solution. In fact, by the same argument, $\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ would still be a solution for any real number t.

We can say that $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ is a free direction for the solution set, i.e., if you are inside the solution set, then moving in this direction freely, you will stay inside the solution set.

This example shows that, if the columns of A are linearly related, then their relation would allow us to go from one solution to many more solutions! Therefore, $A\boldsymbol{x} = \boldsymbol{b}$ will never have a unique solution, and A cannot be injective! ☺

Here I want to draw your attention to the following phenomena, which underlies both anomalies.

Definition 3.3.3. *For a collection of vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ in $\mathbb{R}^m$, a **linear relation** is a list of coefficients $\begin{bmatrix} t_1 \\ \vdots \\ t_k \end{bmatrix} \in \mathbb{R}^k$ such that*

$$t_1\boldsymbol{v}_1 + \cdots + t_k\boldsymbol{v}_k = \mathbf{0}.$$

Example 3.3.4. Consider the following cases. What linear relations can you spot?

1. $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$.
2. $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Note that all zeroes would always give a linear relation! Can you find some linear relation that is not $\mathbf{0}$? ☺

Example 3.3.5. In fact, vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ have linear relation $\boldsymbol{w}$ if and only if

$$\begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_k \end{bmatrix} \boldsymbol{w} = \mathbf{0}.$$

Can you see why? ☺

Definition 3.3.6. *A collection of vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ in $\mathbb{R}^n$ is called **linearly independent** if the only linear relation among them is $\mathbf{0}$.*

*Otherwise, the collection is **linearly dependent**.*

(Conventionally, a single non-zero vector is linearly independent. And any collection that contains a zero vector is linearly dependent.)

Example 3.3.7. Consider the following cases. Are they linearly dependent?

1. $\begin{bmatrix}1\\0\end{bmatrix}$, $\begin{bmatrix}2\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$. Note that two vectors are parallel.

2. $\begin{bmatrix}1\\2\end{bmatrix}$, $\begin{bmatrix}3\\4\end{bmatrix}$ and $\begin{bmatrix}4\\6\end{bmatrix}$.

3. $\begin{bmatrix}1\\2\\3\end{bmatrix}$, $\begin{bmatrix}4\\5\\6\end{bmatrix}$ and $\begin{bmatrix}7\\8\\9\end{bmatrix}$. (This one is a bit hard. Stare at it for a while to see.)

4. $\begin{bmatrix}1\\2\\3\end{bmatrix}$.

5. $\mathbf{0}$.

As you can see, "parallel" is a special case of linear dependency. ☺

Example 3.3.8. Consider

$$\begin{cases} x + y = 1, \\ x + 2y = 2, \\ 2x + 3y = 3. \end{cases}$$

Note that the third equation is a linear combination of the first two (it is the sum of the first two). It is REDUNDANT!

Now consider

$$\begin{cases} x + y = 1, \\ x + 2y = 2, \\ 2x + 3y = 4. \end{cases}$$

Here we have an inconsistency. The third equation is NOT a linear combination of the previous two, but its left hand side is. In particular, the left hand side and the right hand side of the third equation are NOT the same linear combinations of the corresponding sides of the previous two equations! This creates a contradiction.

In particular, $\begin{bmatrix}1 & 1\\1 & 2\\2 & 3\end{bmatrix}$ is not surjective. Any output of this map $\begin{bmatrix}x+y\\x+2y\\2x+3y\end{bmatrix}$ must have the third coordinate equal to the sum of the first two coordinates! And if we try to solve $A\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{b}$ has no such linear relation in its coordinates, then there would be no solution.

As you can see, when rows of A are linearly dependent, then whatever linear relation they have, any output of A must have the same relation in its coordinates. In this manner, outputs of A cannot freely cover the entire codomain, as they are constrained. ☺

Example 3.3.9. Suppose A has a column relation $\boldsymbol{v}$. Then $A\boldsymbol{v} = \mathbf{0}$.

Now for any system $A\boldsymbol{x} = \boldsymbol{b}$, since A is a linear map, we immediately have

$$A(\boldsymbol{x} + t\boldsymbol{v}) = A\boldsymbol{x} + t(A\boldsymbol{v}) = \boldsymbol{b} + t\mathbf{0} = \boldsymbol{b}.$$

So $\boldsymbol{b}$ will never have just one pre-image! You can always go from one pre-image of $\boldsymbol{b}$ to infinitely many more pre-images of $\boldsymbol{b}$. So A cannot be injective. ☺

After Gaussian elimination we shall prove the following with the help of rank. For now just let me spoil the answer in advance.

1. A matrix A as a linear map is injective iff columns of A are linearly independent. In this case, a linear system $A\boldsymbol{x} = \boldsymbol{b}$ has either a UNIQUE solution, or there is no solution. (I.e., the system has at most one solution.)

2. A matrix A as a linear map is surjective iff rows of A are linearly independent. In this case, a linear system $A\boldsymbol{x} = \boldsymbol{b}$ always have at least one solution.

3. A matrix A as a linear map is bijective iff both columns and rows are linearly independent. In this case, a linear system $A\boldsymbol{x} = \boldsymbol{b}$ always have a UNIQUE solution.

3.4 Gaussian Elimination

This section is about solving a linear system.

Example 3.4.1. Let us revisit the chicken-rabbit problem. Say we want to solve $\begin{bmatrix}1 & 1\\2 & 4\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}6\\20\end{bmatrix}$. What should we do?

Well, the high school way is to solve by substitution. However, there is a much cooler way to do this.

Imagine that all chickens and rabbits are well-trained. If they hear a whistle, they will all raise a leg. Now, we started with 6 heads and 20 legs. I whistle twice, now I have 6 heads but only 8 standing legs. But now all chickens have no leg standing! All the chickens must now fall on their butt, and only rabbits are left with two leg standing each. So 8 divided by 2 is the number of rabbits, 4. We have 2 chickens and 4 rabbits.

This is the process of Gaussian elimination. ☺

For the ease of use, let us define the following notion.

Definition 3.4.2. *For a linear system $A\boldsymbol{x} = \boldsymbol{b}$, we say A is the* ***coefficient matrix*** *of the system, and we say $\begin{bmatrix}A & \boldsymbol{b}\end{bmatrix}$ is the* ***augmented matrix*** *of the system.*

Here the notation means we write A, but then put in $\boldsymbol{b}$ as the last column. Basically the augmented matrix is just the same as writing down the whole system, but we get too lazy to write the variables and the equality symbles and so on.

Now, in the process above, we started with the system $\begin{bmatrix}1 & 1 & 6\\2 & 4 & 20\end{bmatrix}$. When I whistle twice, essentially I am substracting twice the first row from the second row (we also write $r_2 \to r_2 - 2r_1$ as a short hand notation for this), and we get a new system $\begin{bmatrix}1 & 1 & 6\\0 & 2 & 8\end{bmatrix}$. (In high school terms, we are substracting multiples of one equation from the other.) However, despite being a new system, note that the solution set is the SAME as before! This perserving of solutions is the very fundation of why we did this.

Then we divide the second row by 2 (we also write $r_2 \to \frac{1}{2}r_2$), we have $\begin{bmatrix}1 & 1 & 6\\0 & 1 & 4\end{bmatrix}$. Pay attention now. The second row reads exactly $y = 4$. We have solved one variable.

Next we want to solve for x from the first equation. We substract the second row from the first row ($r_1 \to r_1 - r_2$), and we have $\begin{bmatrix}1 & 0 & 2\\0 & 1 & 4\end{bmatrix}$. Now the two lines reads $x = 2, y = 4$, which is the solution.

They key lies in the following operations, called ***elementary row operations***.

1. (Swapping) Swapping two rows. (I.e., swapping two equations.) We write $r_i \leftrightarrow r_j$.

2. (Scaling) Multiplying a row by the same scalar. (I.e., multiplying both sides of an equation by the same number.) We write $r_i \to kr_i$. (Here we require $k \neq 0$.)

3. (Shearing) Adding a multiple of one row to another row. (I.e., adding multiples of one equation to another.) We write $r_j \to r_j + kr_i$. (Here we require $i \neq j$.)

(Note that, geometrically, the first row operation only re-order the hyperplanes. The second one does not change the hyperplanes. And the third one is "rotating" hyperplanes around their common intersections.)

The magic of these three operations is that they preserve solutions.

Proposition 3.4.3. *The elementary row operations on the augmented matrix would preserve the solution set.*

Proof. Suppose X is a system of equations, and S_X is the solution set. And Y is a system of equations that can be logically decuded from X, and S_Y is the corresponding solution set.

Now, since every equation in Y can be deduced from X, therefore any solution to the system X must remain a solution to the system Y. Therefore $S_X \subseteq S_Y$.

Now we consider elementary row operations. Suppose X is a system of linear equations, and S_X is the solution set. And Y is a system of equations that can be achieved from X by elementary row operations, and S_Y is the corresponding solution set.

First of all, we already see that $S_X \subseteq S_Y$. However, note that all elementary row operations are REVERSIBLE. Therefore X can also be achieved from Y by elementary row operations. Thus $S_Y \subseteq S_X$ as well. So we must have $S_X = S_Y$. □

Here are some other proofs. Pick any one you like.

Proof. Swapping means simply switch the places of two equations. And this obviously has no effect on the solution set. Similarly, Scaling means simply multiplying both sides of a particular equation by some non-zero number. And this obviously has no effect on the solution set as well.

For shearing, here are two kinds of arguments. First is geometrical. Geometrically, changing r_j to $r_j + kr_i$ means we rotate the hyperplane for the equation r_j along the intersection of r_i and r_j. (Feel free to verify this yourself.) This will definitely preserve the intersection, which is what we actually need to find the solution set. Hence the solution set is preserved.

Alternatively, here is a more algebraic argument. For simplicity, consider the system with equations $r_i, r_j, r_j + kr_i$. Here the last equation $r_j + kr_i$ is obviously equivalent. So the system of $r_i, r_j, r_j + kr_i$ has the same solution set as the system of r_i, r_j.

However, look at $r_i, r_j, r_j + kr_i$ again. Now, we treat r_j is redundant rather than $r_j + kr_i$! This is because it can also be expressed as linear combination of others. I.e., $r_j = (r_j + kr_i) - kr_i$. Hence throwing away the redundant equation r_j, we see that $r_i, r_j, r_j + kr_i$ also has the same solution set as the system of $r_i, r_j + kr_i$.

Therefore, the system of r_i, r_j has the same solution set as the system of $r_i, r_j + kr_i$. □

Whatever argument you use, there is a really important hidden part of the proof: all elementary operations can be reversed. The inverse operation of $r_i \leftrightarrow r_j$ is itself, the inverse operation of $r_i \to kr_i$ is $r_i \to= \frac{1}{k} r_i$, and the inverse operation of $r_j \to r_j + kr_i$ is $r_j \to r_j - kr_i$.

So, whatever row operations you just did, you can always go back. This is vital for the preservation of the solution set.

Example 3.4.4. Consider the ILLEGAL row operation $r_j \to kr_j$ where $k = 0$. Since 0 has no inverse, this operation CANNOT be reversed. It actually would fail to preserve the solution set. For example, the solution set to the linear system of $x = 1$ (which is a hyperplane) is NOT the same as the solution set to the linear system $0 = 0$ (which is the whole space). Such "one-way" operations will enlarge the solution set. ☺

So, ideally, if you have a linear system, you basically just use these three operations in whatever order necessary, until we reached the solution.

Remark 3.4.5. *(OPTIONAL CONTENT)*

One can similarly define elementary column operations. What would that mean?

Well, row operations acts on equations. Column operations acts on variables. An elementary column operation would be equivalent to a change of variable.

For example, say we have a system $x + 2y = 3, x + 3y = 4$.

If we substract the first column from the second column, this would mean we are doing a change of variable $x' = x + y, y' = y$, *and now we have* $x' + y' = 3, x' + 2y' = 4$ *as desired.*

In theory one can also solve equations by keep changing variables. However, it is less efficient, because once you solved the new variables, you have to then retrace and try to resolve the old variable. So as far as linear systems are concerned, we prefer row operations.

Now, given a system of linear equations $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$, when is the end game? When would we consider it solved?

Example 3.4.6. For the chicken-rabbit problem, we have $\left[\begin{array}{cc|c} 1 & 1 & 6 \\ 2 & 4 & 20 \end{array}\right]$. We eventually got to $\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 4 \end{array}\right]$.

Now the two equations reads $x = 2, y = 4$. So we are done. We have an unique solution. ☺

So ideally, we hope to make the coefficient matrix into the identity matrix.

Intuitively, if we go from $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ to $\begin{bmatrix} I & \boldsymbol{b}' \end{bmatrix}$, where I is the identity matrix, then the system is transformed from $A\boldsymbol{x} = \boldsymbol{b}$ into $I\boldsymbol{x} = \boldsymbol{b}'$, with the solution preserved. On the other hand, the identity map always does nothing. So $\boldsymbol{I}\boldsymbol{x} = \boldsymbol{b}'$ simply means $\boldsymbol{x} = \boldsymbol{b}'$, so the answer is $\boldsymbol{b}'$.

Now, this might not be possible sometimes. Consider the following example.

Example 3.4.7. Suppose we have a system $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$. The map here is surjective but not injective, and we expect infinitely many solutions.

To solve this, we want to reduce the number of variables in the equations, i.e., we want as many zeroes as possible in the coefficient matrix. So let us first kill the w variable in the second equation by doing $r_2 \to r_2 - r_1$. This gives the system $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

As you can see now, the second equation probably cannot simplify any further. So instead, let us use this to simplify the first equation as much as possible. Doing $r_1 \to r_1 - r_2$, we get another zero entry with $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Now the two equations reads $w + x + z = 2, y + z = 2$. For each equation, if we move all but the first variable to the right side of the equation, we have $w = 2 - x - z, y = 2 - z$.

It is now reasonably clear that we can let x, z be whatever values they like, and then deduce the corresponding w and y from it. If you like, you can also write the solution set as

$$\{\begin{bmatrix} 2 - x - z \\ x \\ 2 - z \\ z \end{bmatrix} : x, z \in \mathbb{R}\}.$$

We can also write the solution set as

$$\{\begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} : x, z \in \mathbb{R}\}.$$

Geometrically, we start at a point $\begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}$ and first we can move in the direction $\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ by an arbitrary amount. So far we have a line. But then, we can move this line in the direction $\begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$ by an arbitrary amount. Moving a line in another direction would now give us a plane.

So the solution set is a two-dimensional plane, it went through the point $\begin{bmatrix}2\\0\\2\\0\end{bmatrix}$, and it is parallel to the arrow vectors $\begin{bmatrix}-1\\1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}-1\\0\\-1\\1\end{bmatrix}$. These two vectors are "free directions" in the solution set, as in "you can move freely in these directions while remaining in the solution set".

Since we can choose the value of x, z however we like, and deduce w, y from it, therefore we may call x, z as the ***free variables***, and w, z as the ***dependent variables***.

As a final observation, the free directions $\begin{bmatrix}-1\\1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}-1\\0\\-1\\1\end{bmatrix}$ are really the column relations for our matrix $\begin{bmatrix}1 & 1 & 1 & 2\\1 & 1 & 2 & 3\end{bmatrix}$. ☺

So, what we should do is to use all those row operations, and try to simplify the equation by introducing as much zeroes as possible. Then, hopefully we can choose some "free variables", and the other variables can just depend on them, i.e., "dependent variables".

Now, there is a special type of matrices where we can easily choose our free variables and dependent variables.

Definition 3.4.8. *We say a matrix is in **Row Echelon Form** if it satisfy the following property:*

1. *All zero rows are below all non-zero rows.*
2. *In each non-zero row, we call the first non-zero entry the **pivot of that row**. Then the pivots of lower rows are always to the right of previous (higher) rows.*

*We say a matrix is in **Reduced Row Echelon Form** if furthermore, all pivots are* 1*, and all entries above each pivot are zero.*

So a REF looks something like this: (Here a star simply means the entry might not be zero. And an empty entry means the entry is zero. The big dot means that the entry must be non-zero.)

$$\begin{bmatrix}
\bullet & * & \cdots & * & * & * & \cdots & * & * & * & \cdots & * & \cdots\cdots & * & * & \cdots & * \\
 & & & & \bullet & * & \cdots & * & * & * & \cdots & * & \cdots\cdots & * & * & \cdots & * \\
 & & & & & & & & \bullet & * & \cdots & * & \cdots\cdots & * & * & \cdots & * \\
 & & & & & & & & & & & & \ddots & \vdots & \vdots & & \vdots \\
 & & & & & & & & & & & & & * & * & \cdots & * \\
 & & & & & & & & & & & & & \bullet & * & \cdots & * \\
 & & & & & & & & & & & & & & & &
\end{bmatrix},$$

And a RREF looks something like this:

$$\begin{bmatrix}
1 & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & 1 & * & \cdots & * & 0 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & & & & & 1 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & & & & & & & & & \ddots & \vdots & \vdots & & \vdots \\
 & & & & & & & & & & & & & 0 & * & \cdots & * \\
 & & & & & & & & & & & & & 1 & * & \cdots & * \\
 & & & & & & & & & & & & & & & &
\end{bmatrix}.$$

Example 3.4.9. Consider the system $\begin{bmatrix}1 & 1 & 1 & 2\\ 0 & 0 & 1 & 1\end{bmatrix}\begin{bmatrix}w\\ x\\ y\\ z\end{bmatrix}=\begin{bmatrix}4\\ 2\end{bmatrix}$. This is in row Echelon form.

The the first equation says w (the pivot location) has to depend on later variables in certain way. The second equation says that y (the pivot location) has to depend on later variables in certain way. And these are the ONLY requirements!

So there are only requirements on w and y, and no requirement on x, z. Hence we can pick arbitrary x, z, and get a solution.

As you can see, REF can already decide on free variables and dependent variables. ☺

Example 3.4.10. Suppose our system is now $\begin{bmatrix}1 & 1 & 0 & 0 & 2\\ 0 & 0 & 1 & 0 & 3\\ 0 & 0 & 0 & 1 & 4\\ 0 & 0 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\\ x_5\end{bmatrix}=\begin{bmatrix}4\\ 4\\ 5\\ 0\end{bmatrix}$. This is a reduced row Echelon form. Note that the pivots are on the first, third and forth column.

Now the equations are like $x_1+x_2+2x_5=4, x_3+3x_5=4, x_4+4x_5=5, 0=0$. By moving all variables other than the left-most one to the right side of the equation, we see that we can choose the x_2, x_5 to be the free variables. As you can see here, the dependent variables correspond exactly to columns with pivots, i.e., ***pivotal columns***, whereas the free variables corresponds to columns without pivots, i.e., ***free columns***.

The point is that, each pivot, as the left-most non-zero entry of the row, will corresponds to a dependent variable specified by this very row (which represents an equation). And the requirements of RREF guarantee that this dependent variable will NOT be used in any other equations, and hence we can simply read out the solution.

Intuitively, you may think of RREF is "as close as possible" to the identity matrix. So when the identity matrix is out of reach, we go for RREF. (Also note that the identity matrix itself is a special case of RREF.) ☺

As you can see, reading the solution set from RREF is a brainless robotic process. From now on, if you turned a system's augmented matrix into RREF, we simply consider the problem solved.

Now we move on to Gaussian Elimination. The foundation of the idea is of course row operations. Starting from a system $\left[A \mid \boldsymbol{b}\right]$, we perform row operations until this augmented matrix is in reduced row echelon form. Then we are done.

However, sometimes we would use upper rows to reduce the rows below, like $r_j \to r_j + kr_i$ with $j > i$. Other times, we would use lower rows to reduce the rows above, like $r_j \to r_j + kr_i$ with $j < i$. Can we do this in a more organized way? Gaussian elimination is the attempt to always go in the following order: first, we ONLY use upper rows to reduce lower rows. We keep doing this until we reach REF. Then, we only use lower rows to reduce upper rows, and reach RREF.

Example 3.4.11. Typically Gaussian elimination works like this. We start with a system say $\left[\begin{array}{ccc|c}1 & 1 & 1 & 3\\ 2 & 4 & 4 & 10\\ 3 & 4 & 6 & 13\end{array}\right]$.

First I use ONLY the first row to reduce the rows below. The goal is to make the entry 1 in the upper left column into a pivot, so I want to kill every entry below it. We have

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 3\\ 2 & 4 & 4 & 10\\ 3 & 4 & 6 & 13\end{array}\right] \to \left[\begin{array}{ccc|c}1 & 1 & 1 & 3\\ 0 & 2 & 2 & 4\\ 0 & 1 & 3 & 4\end{array}\right].$$

Now the first row is done. Leave it alone forever. Next we want to move to the down-right entry of the previous pivot, and make it the new pivot. So we ONLY use the second row to reduce the rows below. We have

$$\left[\begin{array}{ccc|c}1 & 1 & 1 & 3\\ 0 & 2 & 2 & 4\\ 0 & 1 & 3 & 4\end{array}\right] \to \left[\begin{array}{ccc|c}1 & 1 & 1 & 3\\ 0 & 2 & 2 & 4\\ 0 & 0 & 2 & 2\end{array}\right].$$

Now we are in REF and we have finished the first portion of elimination. For the second portion, we shall only use lower rows to change upper rows. First I change all pivots to ones.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 2 & 2 & 4 \\ 0 & 0 & 2 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Now I go bottom up. We start from the last row, and kill all entries above the last pivot. So we have

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Next we look at the second last pivot, and use this row to kill all entries above this pivot. We have

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Now we are in RREF and we can simply read out the answer. ☺

As you can see, the idea is basically to keep doing row operations to get RREF. However, we do it in an organized way: top-down first, and bottom up later.

Let us see another example.

Example 3.4.12. Consider the linear system

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}.$$

Observe the row relations on the left hand side and on the right hand side, and you will see that this system has no solution. The solution set is the empty set.

What would happen if we do Gaussian elimination? We start from the augmented matrix $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 2 & 3 & 6 \end{array}\right]$. First, we declare the top left entry to be a pivot, so I want to kill every entry below it.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 2 & 2 & 3 & 6 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ & & 1 & 4 \end{array}\right].$$

Then we move on to the second row. We declare the $(2,3)$-entry to be a pivot, so I want to kill every entry below it.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ & & 1 & 4 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ & & & 3 \end{array}\right].$$

We then turn all pivots into 1. It seems that we only need to divide the third row by 3.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ & & & 3 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ & & & 1 \end{array}\right].$$

Next we go bottom up. Use the last pivot, and kill everything above it.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ & & & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ & & 1 & 0 \\ & & & 1 \end{array}\right].$$

Next we use the second to last pivot, and kill everything above it.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ & & 1 & 0 \\ & & & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{array}\right].$$

We are finally in RREF! We are done. But read out the system, we now have $x+y=0, z=0$ and $0=1$. The last equation $0=1$ indicates that there could be no solution.

In general, if a system has no solution, then its RREF will have a pivot in the last column, i.e., it would read as the equation $0=1$. ☺

So far, we had great success doing this "top-down" first, "bottom up" later approach. Unfortunately, this does not always work. Look at this example.

Example 3.4.13. We start with a system say $\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \end{array}\right]$.

First, I would use the first row to reduce the rows below, and make the upper left entry a pivot. Oops! This cannot be done, since the upper left entry is zero, and it has no ability to reduce anything. We failed.

Well, this is actually not a big deal. In this case, just pick any non-zero entry in the first column, and SWAP it with the first row. Then we can proceed as desired. In this case, we have

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 1 & 2 & 2 & 5 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right].$$

Now we want to make the $(2,2)$ entry into a pivot again. Oops! We hit zero again. What should we do? Well, we simply swap again. Keep in mind that we ONLY swap the second row with lower rows. Row one is already done, and should be left alone ever since. So we have

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Well, actually no more reduction is needed. We are already done with REF. Now we proceed with the bottom-up portion of elimination, and find RREF. ☺

As you can see, if you hit an obstruction during your elimination process, then don't worry. Just swap rows to make your pivot non-zero, and continue.

What if there is no proper rows to swap with? Consider this example.

Example 3.4.14. Consider the system $\left[\begin{array}{cccccc|c} 0 & 1 & 2 & 3 & 4 & 0 & 10 \\ 0 & 2 & 4 & 0 & 2 & 1 & 9 \\ 0 & 3 & 6 & 1 & 4 & 1 & 15 \end{array}\right]$.

First I would like to make the upper left entry a pivot. Oops, it is zero. No worry, I shall simply swap with... wait, the entire first column is zero. There is no one to swap with!

But actually this is no cause for worry. This simply means that the first pivot is NOT in the first column. So we start with the second column and proceed. So we have

$$\left[\begin{array}{cccccc|c} 0 & 1 & 2 & 3 & 4 & 0 & 10 \\ 0 & 2 & 4 & 0 & 2 & 1 & 9 \\ 0 & 3 & 6 & 1 & 4 & 1 & 15 \end{array}\right] \to \left[\begin{array}{cccccc|c} 0 & 1 & 2 & 3 & 4 & 0 & 10 \\ 0 & 0 & 0 & -6 & -6 & 1 & -11 \\ 0 & 0 & 0 & -8 & -8 & 1 & -15 \end{array}\right].$$

So $(1,2)$-entry is my first pivot. Now we move to the lower right entry, i.e., the $(2,3)$-entry, and try to make it a pivot. However, it is zero again, and there is nothing below to swap with. But this is fine. This simply means that we have no pivot in this column as well. We simply move to the next column, and make the $(2,4)$-entry a pivot instead. So we have

$$\left[\begin{array}{cccccc|c} 0 & 1 & 2 & 3 & 4 & 0 & 10 \\ 0 & 0 & 0 & -6 & -6 & 1 & -11 \\ 0 & 0 & 0 & -8 & -8 & 1 & -15 \end{array}\right] \to \left[\begin{array}{cccccc|c} 0 & 1 & 2 & 3 & 4 & 0 & 10 \\ 0 & 0 & 0 & -6 & -6 & 1 & -11 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} \end{array}\right].$$

And now we are in REF. Then we proceed with the bottom-up portion of elimination, and find RREF, which is $\left[\begin{array}{cccccc|c} 0 & 1 & 2 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right]$. ☺

Technically, some people only use the term Gaussian elimination to describe the process of going from an arbitrary matrix to REF. And people use the longer term Gauss-Jordan elimination to refer to the whole process of going all the way to RREF. It is NOT a very important distinction though.

Algorithm 3.4.15. *The* ***Gaussian Elimination*** *refers to the following process, where we start with a generic matrix A, and put it into REF.*

First, find a non-zero entry in the first column, and do a row swap to make that row the first row. (If the first column is entirely zero, then simply move on to the second column, etc..)

Next, use the first row to row-reduce all lower rows, so that all other entries in the first column are zero. We are happy with our first row now. From this point on, the first row shall NEVER change.

Next, look at the next column, find a non-zero entry in it (but not in the first row), swap that row to the second row, and row-reduce lower rows, and so on. You can see how this goes.

This process will give you a REF in the end. (Note that this is largely a "downward" process. We work out the first row, then the second row, and so on.)

Algorithm 3.4.16. *The* ***Gauss-Jordan Elimination*** *refers to the following process, where we start with a generic matrix A, and put it into RREF.*

First, we use Gaussian elimination to go from A to a REF. Next, we scale the rows so that all pivots are 1. Finally, we use each pivotal rows to clear entries above the pivots. Now we have RREF and we are done.

(In practice, this is an"upward" process. We usually work with the last pivot, clear all entries above it, then the second to last pivot, clear all entires above it, and so on. This way we avoided some redundant calculations.)

Proposition 3.4.17. *For any matrix A, we can transform it into RREF using elementary row operations. (We call the result* ***the RREF of*** *A or simply write $RREF(A)$.)*

Proof. Just use Gauss-Jordan elimination. □

Proposition 3.4.18. *We have the following situations for a linear system $A\boldsymbol{x} = \boldsymbol{b}$.*

1. *If the RREF of the augmented matrix has a pivot in the last column (the "augmented portion"), then there is no solution. (Because that corresponding equation would read $0 = 1$.)*

2. *If the RREF of the augmented matrix has no contradiction (no pivot in the "augmented portion"), and has the same number of pivots as variables, then there is a unique solution. (All variables are dependent! No free direction in the solution set.)*

3. *If the RREF of the augmented matrix has no contradiction (no pivot in the "augmented portion"), and has less pivots than variables, then there are infinitely many variables. (Not enough constraints to solve it.)*

3.5 Uniqueness of RREF

Freedom is not what it is. It is what we choose it to be.

I could be talking about life, but actually I am talking about variables. Recall that in reality, we can sometimes choose which variable is free. Say we only have a single equation $x + y = 1$, then we can write $x = 1 - y$, where y is free and x is dependent. Or we can also write $y = 1 - x$, and now x is free and y is dependent. In this sense, RREF is trying to move dependent variables to be as early as possible, and push free variables to be as late as possible.

However, if we choose free variables to be as late as possible, then there must be only one way to do it. In particular, it means we must have the following:

Proposition 3.5.1. *For any matrix A, its RREF is unique. (I.e., if we can use elementary row operations to reduce A to RREF, say X, and also use elementary row operations to reduce A to RREF, say Y, then $X = Y$.)*

To prove this, we make two key observations.

Lemma 3.5.2. *For any matrix A already in RREF, then the k-th pivotal column must be $\boldsymbol{e}_k$.*

Proof. Look at the definition of an RREF. This is trivial by definition.

$$\begin{bmatrix}
1 & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & 1 & * & \cdots & * & 0 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & & & & & 1 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & & & & & & & & & \ddots & \vdots & \vdots & & \vdots \\
 & & & & & & & & & & & & & 0 & * & \cdots & * \\
 & & & & & & & & & & & & & 1 & * & \cdots & * \\
 & & & & & & & & & & & & & & & &
\end{bmatrix}.$$

∎

Lemma 3.5.3. *For any matrix A already in RREF, then the free columns are exactly the columns that are linear combinations of columns to its left.*

Proof. Look at the definition of an RREF.

$$\begin{bmatrix}
1 & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & 1 & * & \cdots & * & 0 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & & & & & 1 & * & \cdots & * & \cdots\cdots & 0 & * & \cdots & * \\
 & & & & & & & & & & & & \ddots & \vdots & \vdots & & \vdots \\
 & & & & & & & & & & & & & 0 & * & \cdots & * \\
 & & & & & & & & & & & & & 1 & * & \cdots & * \\
 & & & & & & & & & & & & & & & &
\end{bmatrix}.$$

If we pick a free column, then it is in fact a linear combination of all the pivotal columns to its left!

And if we pick a pivot column, then it's pivot entry is the first non-zero entry on that row. Thus there is no way for earlier columns to produce a non-zero entry here! ∎

Proof of Proposition 3.5.1. We perform induction on the number of columns of A. Suppose A has only one column. Then if $A \neq \mathbf{0}$, the RREF of A must be a single pivotal column, so the RREF must be $\boldsymbol{e}_1$. It is indeed unique. Now, if $A = \mathbf{0}$, then the RREF of A can only remain $\mathbf{0}$, so again it is unique.

Now suppose the statement is true for matrices k columns. Suppose A has $k+1$ columns, say $A = \begin{bmatrix} A_k & \boldsymbol{a} \end{bmatrix}$ where A_k is the collection of its first k columns. Suppose we can row reduce A into RREF $X = \begin{bmatrix} X_k & \boldsymbol{x} \end{bmatrix}$, and we can also row reduce A into RREF $Y = \begin{bmatrix} Y_k & \boldsymbol{y} \end{bmatrix}$. Our goal is to show that $X = Y$.

First of all, since X, Y are both in RREF, we see that X_k, Y_k are both RREF of A_k. By induction hypothesis, we have $X_k = Y_k$. So now our goal is to show that $\boldsymbol{x} = \boldsymbol{y}$.

Suppose $\boldsymbol{x}$ is a free column in the RREF X. Then it is a linear combination of columns to its left. Say $X_k = Y_k = \begin{bmatrix}\boldsymbol{b}_1 & \dots & \boldsymbol{b}_k\end{bmatrix}$, and say $\boldsymbol{x} = t_1\boldsymbol{b}_1 + \cdots + t_k\boldsymbol{b}_k$. Then this implies that $X_k\boldsymbol{t} = \boldsymbol{x}$ for $\boldsymbol{t} = \begin{bmatrix}t_1\\ \vdots \\ t_k\end{bmatrix}$. In particular, $\boldsymbol{t}$ is a solution to the linear system with augmented matrix $\begin{bmatrix}X_k & \boldsymbol{x}\end{bmatrix}$.

Since the matrix $\begin{bmatrix}X_k & \boldsymbol{x}\end{bmatrix}$ can turn into A via row operations, which can turn into $\begin{bmatrix}Y_k & \boldsymbol{y}\end{bmatrix}$ via row operations, therefore the linear system with augmented matrix $\begin{bmatrix}X_k & \boldsymbol{x}\end{bmatrix}$ and the linear system with augmented matrix $\begin{bmatrix}Y_k & \boldsymbol{y}\end{bmatrix}$ must have identical solution set. So $\boldsymbol{t}$ must also be a solution to the linear system with augmented matrix $\begin{bmatrix}Y_k & \boldsymbol{y}\end{bmatrix}$. So $\boldsymbol{y} = Y_k\boldsymbol{t} = X_k\boldsymbol{t} = \boldsymbol{x}$, and thus we are done with $X = Y$.

Similarly, if $\boldsymbol{y}$ us a free column in the RREF of Y, we shall also get $\boldsymbol{x} = \boldsymbol{y}$ for the same reason.

Finally, suppose $\boldsymbol{x}$ is a pivotal column in X and $\boldsymbol{y}$ is a pivotal column in Y. Say $X_k = Y_k$ has t pivotal columns, then $\boldsymbol{x}$ must be the $t+1$-th pivotal column in X, hence it must be $\boldsymbol{e}_{t+1}$. And similarly, $\boldsymbol{y}$ must also be $\boldsymbol{e}_{t+1}$. So we are done. □

The key to the magic proof above, is the conversion between solutions to a linear system, and the linear relations among columns of a matrix. In particular, we can extract the following lemma.

Lemma 3.5.4. *Elementary row operations preserve linear relations among columns.*

Example 3.5.5. Say $\begin{bmatrix}0&1&2&3&4&0\\0&2&4&0&2&1\\0&3&6&1&4&1\end{bmatrix}$. Note how the third column is twice the second column. Now do all kinds of row operations to it, and see that this will always be the case. ☺

Proof of the Lemma. The key realization is the following: given a matrix A, its column linear relations are exactly solutions to the system $A\boldsymbol{x} = \boldsymbol{0}$!

(For example, in the $\begin{bmatrix}0&1&2&3&4&0\\0&2&4&0&2&1\\0&3&6&1&4&1\end{bmatrix}$ case, if you apply this matrix to $\begin{bmatrix}0\\-2\\1\\0\\0\\0\end{bmatrix}$, you will get $\boldsymbol{0}$. This is precisely because the third column is twice the second column. Recall that we DEFINE the matrix vector multiplication to be a linear combination of the columns of the matrix.)

Now row operations preserve solutions. Therefore, they preserve linear dependencies among columns. □

Example 3.5.6. Think about a RREF, say $X = \begin{bmatrix}0&1&2&0&1&0\\0&0&0&1&1&0\\0&0&0&0&0&1\end{bmatrix}$. You can easily see that pivotal columns are EXACTLY the columns that are NOT a linear combination of previous columns, where as free columns MUST BE a linear combination of previous pivotal columns, and the entries in the free column simply tells you the coefficients! (Furthermore, these "linear" statements must be simultaneously true for both X and A, because row operations preserves the linear relations among columns.)

Say $A = \begin{bmatrix}0&1&2&3&4&0\\0&2&4&0&2&1\\0&3&6&1&4&1\end{bmatrix}$. The first column of A is all zero, hence the first column is always going to be zero in RREF.

The second column of A is NOT a linear combination of previous columns. Hence in the RREF, it will also NOT be a linear combination of previous columns, i.e., it is a pivot column. In fact, it is the FIRST pivotal column, so the corresponding column in RREF must be $\boldsymbol{e}_1$.

The third column of A is twice the second column. So this must still be true in the RREF. So the third column of RREF must be $2\boldsymbol{e}_1$.

The fourth column of A is NOT a linear combination of previous columns. Hence it is the SECOND pivotal column, and the corresponding column in RREF must be $\boldsymbol{e}_2$.

The fifth column of A is the sum of the second and fourth column of A. Hence this relation must still be true in RREF, and the fifth column of RREF must be $\boldsymbol{e}_1 + \boldsymbol{e}_2$.

Finally, the last column of A is NOT a linear combination of previous columns. (One can see this by, say, noticing that all previous columns are orthogonal to $\begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix}$, but not the last column. I get this vector by doing a cross product of the previous two pivot columns.) Hence it is the THIRD pivotal column, and the corresponding column in RREF must be $\boldsymbol{e}_3$.

So the RREF of A has no choice but to be $\begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$.

In short, the first column of A that is NOT a linear combination of previous columns must be a pivotal column, and thus must be $\boldsymbol{e}_1$ in any RREF. The second one which is NOT a linear combination of previous columns must also be a pivotal column, and thus must be $\boldsymbol{e}_2$. And so on.

And a free column MUST be a linear combination of previous pivotal columns for some specific coefficients. RREF simply records this coefficients. So any RREF must also have the same free columns.

So RREF is unique. ☺

Definition 3.5.7. *For a matrix A, we define its* ***rank*** *to be the number of pivots in the RREF of A. (Note that this is well-defined since RREF is unique. Otherwise we cannot define this.)*

So given a linear system $A\boldsymbol{x} = \boldsymbol{b}$, what is the rank of A? What is the rank of the augmented matrix $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$?

Well, if the two rank disagree, it means we have a contradiction. (Can you see why?)

If the two ranks agree, what is this rank? On one hand, it means after simplification, how many non-zero rows we have for our system. Note that the zero rows are all REDUNDANT equations. Therefore what's left are the effective eqations.

For example, if we have $x + y = 1, 2x + 2y = 2$, then the second equation is redundant. Even though we have two equations, but effectively it is as if we only have one equation (either one will do in this case). So, rank is the same as the number of effective equations.

On the other hand, each pivot corresponds to a dependent variable. So, rank is also the number of dependent variables.

So given a coefficient matrix A of size $m \times n$ (i.e., the system $A\boldsymbol{x} = \boldsymbol{b}$ has m equations and n variables), we have the following conclusion: (Here by rank I mean the rank of A, not the augmented matrix.)

1. rank = the number of effective equations = the number of dependent variables. (The fundamental theorem of linear algebra.)

2. m - rank = the number of redundant equations.

3. n - rank = the number of free variables (in fact, the dimension of the solution set).

Here are some really interesting corollaries.

Proposition 3.5.8. *Given an $m \times n$ matrix A of rank r, then we always have $r \leq m, n$. Furthermore, $r = n$ iff the linear map defined by A is injective (whatever $\boldsymbol{b}$, we have at most one solution to $A\boldsymbol{x} = \boldsymbol{b}$), and $r = m$ iff the linear map defined by A is surjective (whatever $\boldsymbol{b}$, we have at least one solution to $A\boldsymbol{x} = \boldsymbol{b}$).*

Proof. That $r \leq m, n$ is obvious. (Just look at the pivots in RREF....)

Suppose $r = n$. Then either we have a contradiction, or we have all variables dependent variables. There is no free variable. So the solution is unique in that case.

Conversely, suppose A is injective. Then the only solution to $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$, i.e., there are NO linear dependency among the columns at all. So all columns are pivotal, and $r = n$.

Suppose $r = m$. Note that the augmented matrix also have m rows, and by adding another column, its rank cannot get smaller. So its rank is at least r and at most m. But given $r = m$, its rank must also be r. So there is no source of contradiction. We always have at least one solution.

Conversely, suppose $r < m$. This implies that on the left hand side, we have linear dependency among rows (and therefore we get redundant equations). Choose $\boldsymbol{b}$ that violates this linear dependency in its coordinates, then $A\boldsymbol{x} = \boldsymbol{b}$ will have no solution. □

Remark 3.5.9. *In fact, the rank of A is the maximal number of independent columns in A, and also the maximal number of independent rows in A. These two are always the same!*

Corollary 3.5.10. *The linear map for a matrix A is bijective iff A is a square matrix with full rank. (I.e., $m = n = r$.) In particular, a square matrix is injective iff surjective iff bijective.*

Intuitively, if the domain and codomain have the same size (same dimension), then "no squeeze" and "cover the codomain" would be the same thing.

Remark 3.5.11. *Compare the situation with sets. If X, Y are finite sets with the same amount of element, then any map $f : X \to Y$ is injective iff surjective iff bijective.*

To see this, suppose X, Y both have n elements.

Suppose f is injective. Let elements of X be $x_1, \ldots, x_n$, and let $y_i = f(x_i)$. Since f is injective, and $x_1, \ldots, x_n$ are all distinct, therefore their images $y_1, \ldots, y_n$ are all distinct. But Y only has n elements! Therefore $Y = \{y_1, \ldots, y_n\}$, and f is surjective.

Now suppose f is surjective. Let elements of Y be $y_1, \ldots, y_n$, and for each y_i pick any pre-image x_i (which we can do because f is surjective). Now, since $x_1, \ldots, x_n$ all goes to distinct images, they must themselves be distinct elements. But X only has n elements! Therefore $X = \{x_1, \ldots, x_n\}$, and f is injective.

In conclusion, f is injective iff surjective, and hence iff bijective.

The discussion above can also be thought about in an intuitively manner. Suppose X, Y have the same size. Then if f is not injective, then it will squeeze elements somewhere, hence its range will be smaller in size. So it cannot be surjective.

Conversely, if f is not surjective, then it will have smaller range than domain. So something must squeeze. So it cannot be injective.

Corollary 3.5.12. *A matrix has linearly independent columns iff its linear map is injective. It has linearly independent rows iff its linear map is surjective.*

Proof. We have column independence iff all columns are pivotal, iff $n = r$.

We have row independence iff row operations cannot produce a zero row, iff $m = r$. □

Corollary 3.5.13. *For a square matrix, its rows are linearly independent iff its columns are linearly independent (iff its corresponding linear map is bijective).*

The corollary above is a bit shocking for beginners. You might want to try a few to see what might happen. Say $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Its rows are dependent, and columns are dependent. If you change the lower right entry to 10, then rows are independent and columns are independent. For SQUARE matrices, you must always have both or nothing.

Corollary 3.5.14 (Dimensions are linearly well-established)**.** *There is no linear bijection between $\mathbb{R}^m$ and $\mathbb{R}^n$ when $m \neq n$. Furthermore, if $m > n$, then any linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ must not be surjective (smaller space cannot cover larger space), while any linear map from $\mathbb{R}^m \to \mathbb{R}^n$ must not be injective (larger space must squeeze to fit into a smaller space).*

Proof. For any linear map $f : \mathbb{R}^n \to \mathbb{R}^m$, then it corresponds to some $m \times n$ matrix. But if $m \neq n$, then it cannot be a bijection. (Other statements can be done similarly.) $\square$

This last corollary is also comparable to the situation of sets. However, the requirement of linearity is important here. If one only require continuity, then there is in fact a continuous and surjective map from $\mathbb{R}$ to $\mathbb{R}^2$. Search for "space-filling curve" if you are interested.

3.6 Exercises

Exercise 3.6.1. *Find the RREF of the following matrices.*

1. $\begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 20 & 300 \\ 1234 & 5678 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 3 & 2 & 1 \\ 2 & -3 & 0 & -2 \end{bmatrix}$.

3. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

4. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$.

5. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 11 \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \\ 7 & 8 & 9 \end{bmatrix}$.

7. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Exercise 3.6.2. *Consider the matrix* $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$. *Only use row scaling and row shearing, can you turn it into* $\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$*?*

Exercise 3.6.3. *Consider the matrix* $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

1. *For what coefficients will the columns combine to zero?*

2. *Find the RREF of* $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

3. *Switch row one and row two, and we get* $\begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$. *Then, for what coefficients will the columns combine to zero?*

4. *Find the RREF of* $\begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$.

5. *Scale row one by factor* 2*, and we get* $\begin{bmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. *Then, for what coefficients will the columns combine to zero?*

6. *Find the RREF of* $\begin{bmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

7. *Add row two to row one, and we get* $\begin{bmatrix} 5 & 7 & 9 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. *Then, for what coefficients will the columns combine to zero?*

8. *Find the RREF of* $\begin{bmatrix} 5 & 7 & 9 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Exercise 3.6.4 (Gaussian Eliminations)**.** *Solve these by Gaussian elimination, and write out the solution set explicitly. (So don't just stop at RREF for this problem. Write out the set.)*

1. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

3. $\begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix}$.

Exercise 3.6.5 (Cross product and linear system in space)**.** *Consider the system* $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 15 \end{bmatrix}$.

1. *Find the RREF of its augmented matrix.*

2. *Its solution set is a line. What is a direction vector of this line?*

3. *Compute* $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \times \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$. *Compare with the answer above. Can you explain the pattern?*

Exercise 3.6.6 (Situations of the solution set)**.** *Find a constant b that satisfies the condition.*

1. $\begin{bmatrix} 1 & 2 \\ 3 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ *has no solution.*

2. $\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ b \end{bmatrix}$ *has no solution.*

3. $\begin{bmatrix} 2 & 5 & 1 \\ 4 & b & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$ *has no solution.*

4. $\begin{bmatrix} b & 3 \\ 3 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \end{bmatrix}$ *has infinitely many solutions.*

5. $\begin{bmatrix} 2 & b \\ 4 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 16 \\ c \end{bmatrix}$ *has infinitely many solutions. (For this problem, find constants b, c that satisfy the condition.)*

6. $\begin{bmatrix} 1 & b & 0 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ *has a non-zero solution.*

7. $\begin{bmatrix} b & 2 & 3 \\ b & b & 4 \\ b & b & b \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ *has a non-zero solution. (For this problem, find three possible values for b that satisfies the condition.)*

Exercise 3.6.7 (Solving a hardcore linear system)**.** ☕*For each real value of p, find all solutions to the following system:*

$$\begin{bmatrix} p & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ p \\ p^2 \end{bmatrix}.$$

(Hint: first replace the first row by the sum of three rows.)

Exercise 3.6.8 (Find the system given a solution set???)**.** *You don't truely understand something, until you can reverse engineer it.*

For a matrix A, suppose the solution set to $A\boldsymbol{x} = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ is $\{ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mid s, t \in \mathbb{R}\}$.

1. *Can you find the values of $A \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, A \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ by simply observing the solution set? No calculation should be required, just simple conceptual understandings should be enough.*

2. *What does the subproblem above tell you about the columns of A and their relations? Find A.*

Exercise 3.6.9 (Alternative method to the last problem)**.** *For a matrix A, suppose the solution set to $A\boldsymbol{x} = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ is $\{ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mid s, t \in \mathbb{R}\}$. Let us try an alternative method.*

1. *From the solution set's description, how many free variables should you have? How many dependent variables?*

2. *Write out RREF of the AUGMENTED matrix of this system by simply reading at the solution set. (Pivotal columns should be obvious. For the free columns, think about potential column relations of $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$.)*

3. *Find A by doing row operations to RREF.*

Exercise 3.6.10 (Rank-one matrices)**.** *If a matrix has all rows parallel, must it also have all columns parallel? (As a convention, we treat* **0** *as parallel to everything.) Give a counter-example or prove why.*

Exercise 3.6.11 (Arithmetic sequences in Matrices)**.** *For a given matrix, suppose the columns are all arithmetic sequences. (For examle, something like* $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 3 \\ 5 & 4 & 2 \\ 7 & 5 & 1 \end{bmatrix}$*.)*

1. *Assume that the first row of such a matrix is* $\boldsymbol{a}^{\mathrm{T}}$ *and the second row is* $\boldsymbol{b}^{\mathrm{T}}$*. Can you deduce all lower rows?*

2. ☕*Show that such a matrix must have rank at most 2.*

Exercise 3.6.12 (Is a column a linear combination of previous columns?)**.** *During our class, when we prove the uniqueness of RREF, we encountered the following problem: how to decide whether a column is a linear combination of previous columns or not?*

For the following problem, first reduce them into a standard linear system $A\boldsymbol{x} = \boldsymbol{b}$*, and then solve it.*

1. *For the matrix* $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \end{bmatrix}$*, how can we write the third column as a linear combination of the previous two columns?*

2. *For the matrix* $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \end{bmatrix}$*, how can we write the forth column as a linear combination of the second column and the third column?*

3. *For the matrix* $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \end{bmatrix}$*, how can we write the fifth column as a linear combination of the third column and the fourth column? (Maybe by this time, you don't actually need to calculate to see the answer?)*

4. *For the matrix* $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 14 & 13 & 15 \end{bmatrix}$ *(this is NOT the same matrix as before), how can we write the third column as a linear combination of the previous two columns?*

5. *(Read only) In general, one may go from* $\begin{bmatrix} A \mid \boldsymbol{b} \end{bmatrix}$ *to RREF by doing elimination. One can also attempt to deduce the column relations of A, and simply read out the RREF. However, doing so would require solve a linear system for EACH column of A, as you can see from this problem. So it is much slower that Gaussian elimination in practice.*

Exercise 3.6.13 (Column operations and variable substitution)**.** *Consider a linear system with two equations,*

$$\begin{cases} x \; + y = 3, \\ x + 2y = 5. \end{cases}$$

Then the augmented matrix is $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 5 \end{bmatrix}$*. Now, suppose I do a substitution of variables, say I set* $y' = y + x$*. Then the equations are now*

$$\begin{cases} \qquad y' = 3, \\ -x + 2y' = 5. \end{cases}$$

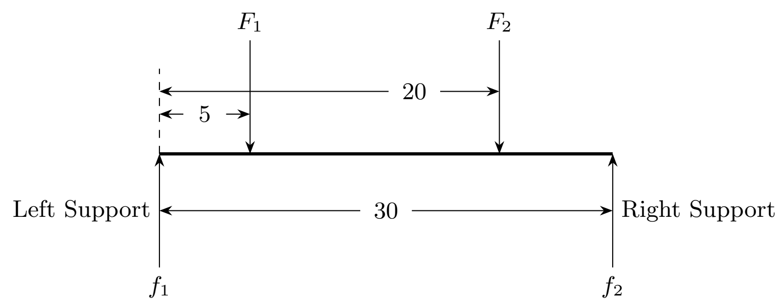
And the new augmented matrix is $\begin{bmatrix} 0 & 1 & 3 \\ -1 & 2 & 5 \end{bmatrix}$. *As you can see, this corresponds to a COLUMN operation, where we substract the second column from the first column!*

In general, variable substitutions are column operations. Find the elementary COLUMN operations on the augmented matrix that corresponds to the following change of variable.

1. *New variables are* $x' = y, y' = x$.
2. *New variables are* $x' = 2x, y' = y$.
3. *New variables are* $x' = x, y' = 2x + y$.
4. *New variables are* $x' = x + 1, y' = y$. *(Recall that we are looking at the AUGMENTED matrix.)*

Exercise 3.6.14 (Above the Bridge)**.** *This is a problem in physics, but it is a linear system. The problem assumes that you understand the Law of the lever.*

Two heavy objects are placed on top of a bridge. Their weights are F_1 *and* F_2*, and their locations are shown below. The bridge has two supporting columns, a left one and a right one. Since we put weights on the bridge, the supporting columns will provide upward support forces* f_1, f_2. *Suppose* f_1, f_2 *are* 45 *newtons and* 15 *newtons. Can you find* F_1 *and* F_2*?*



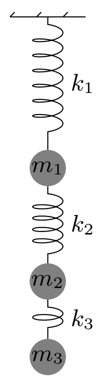
1. *Think of the bridge surface as a lever, with the left support point as pivot. The bridge is not turning, so by the Law of the lever, this gives us an equation. Write out this equation.*
2. *Think of the bridge surface as a lever, with the right support point as pivot. The bridge is not turning, so by the Law of the lever, this gives us an equation. Write out this equation.*
3. *Write out the augmented matrix for this system, and find its RREF, and find the value of* F_1, F_2.

Exercise 3.6.15 (Below the Spring)**.** *This is a problem in physics, but it is a linear system. The problem assumes that you understand Hooke's law of the spring.*

We have three springs and balls hooked together as shown belown. (We ignore the size and shape of the balls, and we ignore the mass of the springs.) The three springs have characteristics $k_1 = 1, k_2 = 2, k_3 = 3$ *(i.e., stiffness). The balls have masses* m_1, m_2, m_3. *We assume the gravitational acceleration is* $g = 10$.

Suppose the springs are elongated by lengths $60, 20, 10$ *from the top spring to the bottom spring. Find the masses of the balls.*

1. *The top spring is in equilibrium, so its pulling force would equal to the total weight of the three balls. Write out this equation.*
2. *The second spring is in equilibrium, so its pulling force would equal to the total weight of the two lower balls. Write out this equation.*
3. *The third spring is in equilibrium, so its pulling force would equal to the weight of the lowest ball. Write out this equation.*
4. *Write out the augmented matrix for this system, and find its RREF, and find the value of* m_1, m_2, m_3.


k_1
m_1
k_2
m_2
k_3
m_3

Chapter 4

The Algebra of Linear Maps

4.1 Map Composition

Given a map $f : X \to Y$ and $g : Y \to Z$, their ***composition*** is a map $g \circ f : X \to Z$, such that $g \circ f(x) = g(f(x))$.

Proposition 4.1.1. *Map composition is associative, i.e.,* $(f \circ g) \circ h = f \circ (g \circ h)$ *for maps* $f : Y \to Z, g : X \to Y, h : W \to X$.

Proof. Just compute directly by definition.

$$(f \circ g) \circ h(x) = (f \circ g)(h(x)) = f(g(h(x))) = f((g \circ h)(x)) = f \circ (g \circ h)(x).$$

□

Graphically, map composition is obviously associative. Just look at this graph:

$$\begin{tikzcd}
W \arrow[r, "h"'] \arrow[rrr, bend left, "(f\circ g)\circ h = f\circ(g\circ h)"] \arrow[rr, bend right, "g\circ h"'] & X \arrow[r, "g"'] \arrow[rr, bend right, "f\circ g"'] & Y \arrow[r, "f"'] & Z
\end{tikzcd}$$

So we have associativity checked. What about other properties? Say, do we have commutativity? The answer is NO.

Example 4.1.2. Here are some examples.

1. Say $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = e^x, g(x) = 2x$. You can easily verify that $f \circ g$ is NOT the same as $g \circ f$, since $f \circ g(x) = e^{2x}$ while $g \circ f(x) = 2e^x$. Composition is NOT commutative.

2. Say $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x + 1, g(x) = 3x + 2$. Compute $f \circ g$ and $g \circ f$, what would you see? What if we change g into $g(x) = 3x + 1$?

3. What condition on the injectivity or surjectivity of f, g would guarantee that $f \circ g$ is injective? surjective? bijective? Find proofs and counter examples if you like.

☺

Example 4.1.3. Your mother's father is not the same as your father's mother. There is no commutativity.

But your mother's (father's mother) is the same as your (mother's father)'s mother. So there is associativity. ☺

Example 4.1.4. Here are some food for thought. Try to prove the following statements yourself.

1. If f, g are both injective, then $f \circ g$ is injective.
2. If f, g are both surjective, then $f \circ g$ is surjective.
3. If f, g are both bijective, then $f \circ g$ is bijective.
4. If $f \circ g$ is injective, then g is injective. (But f may not be injective.)
5. If $f \circ g$ is surjective, then f is surjective. (But g may not be surjective.)
6. If $f \circ g$ is bijective, then g is injective and f is surjective. (But we know no more.)

For the last three statements, here is an example to think about.

$$\begin{tikzcd}
1 \arrow[r] & 1 \arrow[r] & 1 \\
2 \arrow[r] & 2 \arrow[r] & 2 \\
3 \arrow[r] & 3 \arrow[r] & 3 \\
 & 4 \arrow[ur] &
\end{tikzcd}$$

☺

Remark 4.1.5. *Here we provide proof to two statements that will be useful later.*

If $f \circ g$ is surjective, then let us show that f is surjective.

The standard proof for surjectivity is like this: you pick an arbitrary element of the codomain, and then you go find a pre-image. That's it. (So everything has a pre-image.)

For each y in the codomain of f (which is also the codomain of $f \circ g$), we can find a pre-image for the map $f \circ g$, say x. Then $f(g(x)) = y$. But then $g(x)$ is a pre-image of y for the map f. So we have found a pre-image for arbitrary y. So f is surjective.

If $f \circ g$ is injective, then let us show that g is injective.

The standard proof for injectivity is like this: you show that any two inputs with the same image must actually be the same input. (So different inputs must go to different images.)

Suppose $g(x) = g(y)$ for two inputs x, y, and our goal is to prove $x = y$. Now we apply f to both sides of $g(x) = g(y)$, and we have $f \circ g(x) = f \circ g(y)$. But since $f \circ g$ is injective, $f \circ g(x) = f \circ g(y)$ means that $x = y$. This is what we want, so we are done! This shows that g is injective.

Now, in the example above, we see that if $f \circ g$ is bijective, then we can NOT guarantee that f, g are bijective. However, in some special cases, this can be done.

Corollary 4.1.6. *If X, Y, Z are finite sets, all with the same number of elements. Then for any $g : X \to Y$ and $f : Y \to Z$, if $f \circ g$ is bijective, then both f and g are bijective.*

Proof. If $f \circ g$ is bijective, then g is injective and f is surjective. But since X, Y, Z all have the same number of elements, g is injective iff bijective, and f is surjective iff bijective. So f, g are both bijective. $\square$

Remark 4.1.7. *The corresponding case for linear maps will be true. For any linear maps $f, g : \mathbb{R}^n \to \mathbb{R}^n$ (note that the domain and codomain have the same dimension! So both must be square matrices!), if $f \circ g$ is bijective, then f, g are both bijective. We will prove this later.*

Now that we have all the terminology, let us look at my favorite map, the identity map $\mathrm{id}_X : X \to X$.

Proposition 4.1.8. *1. The identity map is always bijective, and it is its own inverse map.*

2. *For any set, its identity map is UNIQUE.*

3. $\mathrm{id} \circ g = g$ *whenever the composition is defined, and* $g \circ \mathrm{id} = g$ *whenever the composition is defined.*

4. *If a map* f *satisfies the condition that* $f \circ g = g$ *whenever the composition is defined, or that* $g \circ f = g$ *whenever the composition is defined, then* f *must be the identity map for its domain. (This is a necessary and sufficient condition.)*

Proof. Only the last one is non-trivial. Suppose the domain of f is X and the codomain is Y. Pick $g = \mathrm{id}_X$, then $f \circ g$ is defined, and $f \circ g = g$ as required. However, since g is the identity map on X, we also have $f \circ g = f$. Hence $f = g$. Done. ☐

Proposition 4.1.9. *If we have a pair of maps* $f : X \to Y$ *and* $g : Y \to X$*, such that* $f \circ g = \mathrm{id}_Y, g \circ f = \mathrm{id}_X$*, then* f, g *are inverse of each other.*

Proof. If $f(x) = y$, then $g(y) = g \circ f(x) = \mathrm{id}_X(x) = x$.
If $g(y) = x$, then $f(x) = f \circ g(y) = \mathrm{id}_Y(y) = y$.
So $f(x) = y$ iff $g(y) = x$. ☐

Remark 4.1.10. *However, note that we need BOTH* $f \circ g = \mathrm{id}_Y$ *AND* $g \circ f = \mathrm{id}_X$*. Merely one of them is not enough.*

For example, $f : \{0\} \to \mathbb{R}$ *such that* $f(0) = 0$*, and* $g : \mathbb{R} \to \{0\}$ *such that* $g(x) = 0$ *for all* $x \in \mathbb{R}$*. Then* $g \circ f = \mathrm{id}_{\{0\}}$*. Yet,* f, g *are not inverse of each other. (Because* $f \circ g \neq \mathrm{id}_{\mathbb{R}}$*).*

4.2 Matrix Multiplication

We are doing linear maps from $\mathbb{R}^m$ to $\mathbb{R}^n$. Recall that we have learned about matrix-vector multipications. if $A = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, then we have

$$\begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \sum v_i \boldsymbol{a}_i.$$

So $A\boldsymbol{v}$ is simply a linear combination of columns of A with respect to the coefficients given as coordinates of $\boldsymbol{v}$.

Now consider the first coordinate of the output. Given a linear combination of columns of A, to find the first coordinate, we actually only need the first coordinate of EACH column, and combine them accordingly.

This gives rise to a different (but equivalent) formula. If $A = \begin{bmatrix} \boldsymbol{r}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{r}_m^{\mathrm{T}} \end{bmatrix}$, then see if you can verify the following formula

$$\begin{bmatrix} \boldsymbol{r}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{r}_m^{\mathrm{T}} \end{bmatrix} \boldsymbol{v} = \begin{bmatrix} \boldsymbol{r}_1^{\mathrm{T}}\boldsymbol{v} \\ \vdots \\ \boldsymbol{r}_m^{\mathrm{T}}\boldsymbol{v} \end{bmatrix}.$$

So effectively, it is like we are doing a "dot product" of $\boldsymbol{v}$ with each row of A.

Example 4.2.1. Here is a simple example. Let us see the two ways of computing a matrix-vector multiplication.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 18 \end{bmatrix}.$$

$$\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix}\\ \begin{bmatrix}3 & 4\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix}\end{bmatrix} = \begin{bmatrix}8\\18\end{bmatrix}.$$

Personally, I just imagine that for $A\boldsymbol{v}$, then $\boldsymbol{v}$ is a brick, and the rows of A are the faces of the people that I hate. So I take the brick, and smash it onto those faces one by one. Here "smash" would mean taking a dot product. ☺

4.2.1 Composition of Linear Maps

Recall that, the matrix $A = \begin{bmatrix}1 & 1\\ 2 & 4\end{bmatrix}$ represents the "counting" process in the chicken-rabbit cage problem. Given these four enties, we now know how to count. And given a vector $\boldsymbol{v} = \begin{bmatrix}x\\y\end{bmatrix}$ that represents x chickens and y rabbits, the vector $A\boldsymbol{v}$ represents the number of heads and leg. Note that as a linear map, we have $A : \mathbb{R}^2 \to \mathbb{R}^2$.

Now suppose we are furthermore selling these heads and legs for money, and suppose the rest of the body parts has no value. (Ma La Tu Tou and Pao Jiao Feng Zhua....) Anyway, say each head is worth 5 yuan and each leg 7 yuan. How much would x chickens and y rabbits worth in total? Note that this evaluation process is also linear, and it should be a linear map from $\mathbb{R}^2$ to $\mathbb{R}$, i.e., it should be a 1 by 2 matrix.

To do this, we need the evaluation map $B = \begin{bmatrix}5 & 7\end{bmatrix}$ applied to $A\boldsymbol{v}$. The evaluation map has this matrix because it sends $\boldsymbol{e}_1$, i.e., one head, to 5, while it sends $\boldsymbol{e}_2$, i.e., one leg, to 7.

So with the chicken-rabbit input of $\boldsymbol{v} = \begin{bmatrix}x\\y\end{bmatrix}$, we should get $A\boldsymbol{v} = \begin{bmatrix}x+y\\2x+4y\end{bmatrix}$ of head-leg output, and the total worth of them would be $B(A\boldsymbol{v}) = 5(x+y) + 7(2x+4y) = 19x + 33y$.

In particular, we see that the composition $B \circ A$ is still linear, and it has a matrix of $\begin{bmatrix}19 & 33\end{bmatrix}$

Proposition 4.2.2. *The composition of linear maps is still linear.*

Proof. Say f, g are linear and has a well-defined composition. Then $f \circ g(a\boldsymbol{v} + b\boldsymbol{w}) = f(g(a\boldsymbol{v} + b\boldsymbol{w})) = f(ag(\boldsymbol{v}) + bg(\boldsymbol{w})) = af(g(\boldsymbol{v})) + bf(g(\boldsymbol{w})) = a(f \circ g)(\boldsymbol{v}) + b(f \circ g)(\boldsymbol{w})$. □

Definition 4.2.3. *Given two matrix A, B, say as linear maps we have $A : \mathbb{R}^n \to \mathbb{R}^m, B : \mathbb{R}^d \to \mathbb{R}^n$, then we can do their composition. The composition of these two linear maps is what we define to be the matrix multiplication AB.*

Proposition 4.2.4. *The matrix multiplication AB is well-defined iff the number of columns of A is the same as the number of rows of B.*

This is because the codomain of B must match with the domain of A. So when you do matrix multiplications, you should always expect to see something like below. You can almost imagine that the n here in the middle just got canceled away.

$$m\left\{\overbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}^{n}\right. \; n\left\{\overbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}^{d}\right. = m\left\{\overbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}^{d}\right. .$$

$$(m \times n \text{ matrix})(n \times d \text{ matrix}) = (m \times d \text{ matrix})$$

Now we move on to the computations.

Proposition 4.2.5. *Let us write B in terms of its columns, $B = \begin{bmatrix}\boldsymbol{b}_1 & \dots & \boldsymbol{b}_n\end{bmatrix}$. Then $AB = \begin{bmatrix}A\boldsymbol{b}_1 & \dots & A\boldsymbol{b}_n\end{bmatrix}$.*

Proof. Note that the i-th column of AB should be $(AB)\boldsymbol{e}_i$. By definition of map composition, this is $A(B(\boldsymbol{e}_i))$, and $B\boldsymbol{e}_i$ must be the i-th column of B, i.e., $\boldsymbol{b}_i$. So $(AB)\boldsymbol{e}_i = A\boldsymbol{b}_i$ is the i-th column of AB. □

In particular, feel free to verify now that $\begin{bmatrix}5 & 7\end{bmatrix}\begin{bmatrix}1 & 1\\ 2 & 4\end{bmatrix} = \begin{bmatrix}19 & 33\end{bmatrix}$, as in our chicken-rabbit example. We also see immediately that matrix-vector multiplication is a special case of matrix-matrix multiplication, if we simply think of the vector as a matrix with a single column.

Remark 4.2.6. *As a side note, you also see that given an $m \times n$ matrix A, then you can only multiply a column vector (i.e., $n \times 1$ matrix) to the right of A, and only multiply a row vector (i.e., $1 \times m$ matrix) to the left of A.*

Example 4.2.7. To rotate $\mathbb{R}^2$ by an angle of θ counter clockwise, this linear map has a matrix of $R_\theta = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$. To reflect $\mathbb{R}^2$ about the line $x = y$, this linear map has a matrix of $F = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$. (Can you see why F is this matrix?)

What if we rotate and then reflect? Then we are doing FR_θ. Doing this computation, you shall see we have $FR_\theta = \begin{bmatrix}\sin\theta & \cos\theta\\ \cos\theta & -\sin\theta\end{bmatrix}$.

What if we reflect and then rotate? Again computation gives $R_\theta F = \begin{bmatrix}-\sin\theta & \cos\theta\\ \cos\theta & \sin\theta\end{bmatrix}$.

Can you see the relation between the two? In fact we shall always have $R_\theta F = FR_{-\theta}$ for whatever rotation R_θ and reflection F. (Here we require the rotation to be around the origin, and the reflection to be fixing the origin.)

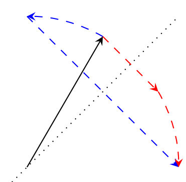


Figure 4.2.1: The blue process is FR_θ while the red process is $R_{-\theta}F$.

Note that rotations before and after a reflecion are like mirror images of each other. So intuitively, it should be clear that rotation by θ before a reflection is the same as rotation by $-\theta$ after the reflection. ☺

Above example shows that matrix multiplication (i.e., linear map composition) fails to be commutative. But we do have associativity. Warning: Associativity is the source of MANY magic in linear algebra. Keep it in mind.

Proposition 4.2.8. *We have $(AB)C = A(BC)$.*

Proof. This is because map composition is associative. End of proof just by definition.

If you want a more computational proof, here it is. Write C in terms of its columns $C = \begin{bmatrix}\boldsymbol{c}_1 & \dots & \boldsymbol{c}_n\end{bmatrix}$. Then

$$A(BC) = A(B\begin{bmatrix}\boldsymbol{c}_1 & \dots & \boldsymbol{c}_n\end{bmatrix}) = A\begin{bmatrix}B\boldsymbol{c}_1 & \dots & B\boldsymbol{c}_n\end{bmatrix} = \begin{bmatrix}AB\boldsymbol{c}_1 & \dots & AB\boldsymbol{c}_n\end{bmatrix} = (AB)\begin{bmatrix}\boldsymbol{c}_1 & \dots & \boldsymbol{c}_n\end{bmatrix} = (AB)C.$$

□

Remark 4.2.9. *In the old Chinese textbooks (not the one by myself), this was done like this. They introduce matrix multiplication without any talk of map composition, and simply throw a formula to the students as the definition of matrix multiplication.*

$$\begin{bmatrix}a_{11} & \dots & a_{1n}\\ \vdots & \ddots & \vdots\\ a_{m1} & \dots & a_{mn}\end{bmatrix}\begin{bmatrix}b_{11} & \dots & b_{1d}\\ \vdots & \ddots & \vdots\\ b_{n1} & \dots & b_{nd}\end{bmatrix} = \begin{bmatrix}a_{11}b_{11} + a_{12}b_{21} + \dots + a_{1n}b_{n1} & \dots & a_{11}b_{1d} + a_{12}b_{2d} + \dots + a_{1n}b_{nd}\\ \vdots & \ddots & \vdots\\ a_{m1}b_{11} + a_{m2}b_{21} + \dots + a_{mn}b_{n1} & \dots & a_{m1}b_{1d} + a_{m2}b_{2d} + \dots + a_{mn}b_{nd}\end{bmatrix}.$$

Next, they literally compute AB and then $(AB)C$. Then they compute BC and $A(BC)$. And they manipulate the formulat to show that the two are the same.

It is my opinion that this is a counter-intuitive and counter-productive approach to prove associativity.

Example 4.2.10. From this point on, we don't need matrix-vector multiplication any more. We only need matrix multiplication. Here are somethings to think about.

Suppose $\boldsymbol{v} = \begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}1\\2\\3\end{bmatrix}$. Then what is $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w}$? This is simply the dot product, and it yields $1+4+9=14$. Alternatively, you can also think of this as a matrix multiplication of a 1×3 matrix $\boldsymbol{v}^{\mathrm{T}}$ with a 3×1 matrix $\boldsymbol{w}$, and it gives you the 1×1 matrix which is simply the number $\begin{bmatrix}14\end{bmatrix}$.

Now consider $\boldsymbol{v}\boldsymbol{w}^{\mathrm{T}}$. This is the matrix multiplication of a 3×1 matrix with a 1×3 matrix, hence it gives a 3×3 matrix as a result! This would be $\boldsymbol{v}\boldsymbol{w}^{\mathrm{T}} = \boldsymbol{v}\begin{bmatrix}1 & 2 & 3\end{bmatrix}$.

Recall the formula

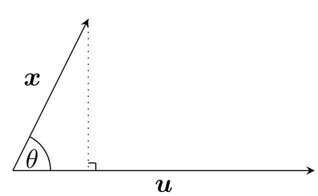
$$A\begin{bmatrix}\boldsymbol{b}_1 & \boldsymbol{b}_2 & \boldsymbol{b}_3\end{bmatrix} = \begin{bmatrix}A\boldsymbol{b}_1 & A\boldsymbol{b}_2 & A\boldsymbol{b}_3\end{bmatrix}.$$

So we see that

$$\boldsymbol{v}\begin{bmatrix}1 & 2 & 3\end{bmatrix} = \begin{bmatrix}\boldsymbol{v}1 & \boldsymbol{v}2 & \boldsymbol{v}3\end{bmatrix} = \begin{bmatrix}1 & 2 & 3\\2 & 4 & 6\\3 & 6 & 9\end{bmatrix}.$$

In general, $\begin{bmatrix}v_1\\\vdots\\v_m\end{bmatrix}\begin{bmatrix}w_1 & \dots & w_n\end{bmatrix} = \begin{bmatrix}v_1w_1 & \dots & v_1w_n\\\vdots & \ddots & \vdots\\v_mw_1 & \dots & v_mw_n\end{bmatrix}$. In particular, the (i,j)-entry of $\boldsymbol{v}\boldsymbol{w}^{\mathrm{T}}$ is simply v_iw_j. ☺

Example 4.2.11 (Projection Formula versus Projection map)**.** Given a unit vector $\boldsymbol{u}$, how to do a projection of a vector $\boldsymbol{x}$ to the direction of $\boldsymbol{u}$?



Let θ be their angle, then by high school geometry, we know the length of this projection is $||\boldsymbol{x}||\cos\theta$. After the projection process, the vector $\boldsymbol{x}$ would change into the vector $(||\boldsymbol{x}||\cos\theta)\boldsymbol{u}$.

By definition of angle, we know that $\cos\theta = \frac{\boldsymbol{x}\cdot\boldsymbol{u}}{||\boldsymbol{x}||||\boldsymbol{u}||}$. Since $\boldsymbol{u}$ is a unit vector, we simply and see that

$$(||\boldsymbol{x}||\cos\theta)\boldsymbol{u} = (\boldsymbol{x}\cdot\boldsymbol{u})\boldsymbol{u}.$$

So the projection formula is simply $(\boldsymbol{x}\cdot\boldsymbol{u})\boldsymbol{u}$, or $(\boldsymbol{u}^{\mathrm{T}}\boldsymbol{x})\boldsymbol{u}$.

Now let us review this process. "Projection to the direction of $\boldsymbol{u}$" is a map, and it sends any input vector $\boldsymbol{x}$ to the resulting vector of projection $(\boldsymbol{x}\cdot\boldsymbol{u})\boldsymbol{u}$, i.e., $P_{\boldsymbol{u}}:\mathbb{R}^n\to\mathbb{R}^n$ such that $P_{\boldsymbol{u}}(\boldsymbol{x}) = (\boldsymbol{u}^{\mathrm{T}}\boldsymbol{x})\boldsymbol{u}$. It is very easy to verify that this is linear, so what is its matrix?

Recall scalar multiplication $k\boldsymbol{v}$ and $\boldsymbol{v}k$ for some $\boldsymbol{v}\in\mathbb{R}^n$, say $n\neq 1$. The former is not matrix multiplication, because k only has one column while $\boldsymbol{v}$ has n rows. However, the latter IS a matrix multiplication, since $\boldsymbol{v}$ only has one column, and k has one row, and the way of calculation is exactly as expected!

In particular, if you started with $(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w})\boldsymbol{u}$, then it should NOT be equal to $\boldsymbol{v}^{\mathrm{T}}(\boldsymbol{w}\boldsymbol{u})$ (which is an illegal anyway). This is because $(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w})$ and $\boldsymbol{u}$ here are NOT doing a matrix multiplication, and this is merely a scalar multiplication. However, you can write $(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w})\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w}) = (\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}})\boldsymbol{w}$. Now $\boldsymbol{u}(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w})$ is a legal matrix multiplciation, so we can proceed to use the associativity of matrix multiplication.

How to find the matrix $P_{\boldsymbol{u}}$? Well, first we try to reorganize $(\boldsymbol{u}^{\mathrm{T}}\boldsymbol{x})\boldsymbol{u}$ into matrix multiplications as much as possible

$$(\boldsymbol{u}^{\mathrm{T}}\boldsymbol{x})\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{u}^{\mathrm{T}}\boldsymbol{x}).$$

Next we use associativity to get

$$\boldsymbol{u}(\boldsymbol{u}^{\mathrm{T}}\boldsymbol{x}) = (\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}})\boldsymbol{x}.$$

So we see that $P_{\boldsymbol{u}}(\boldsymbol{x}) = (\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}})\boldsymbol{x}$. In particluar, $P_{\boldsymbol{u}} = \boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$.

In general, if $\boldsymbol{w}$ is not a unit vector, what is the projection to the direction of $\boldsymbol{w}$? It should be the same as projection to the direction of $\frac{1}{||\boldsymbol{w}||}\boldsymbol{w}$, which is now a unit vector. So we have

$$P_{\boldsymbol{w}} = (\frac{1}{||\boldsymbol{w}||}\boldsymbol{w})(\frac{1}{||\boldsymbol{w}||}\boldsymbol{w})^{\mathrm{T}} = \frac{1}{||\boldsymbol{w}||^2}\boldsymbol{w}\boldsymbol{w}^{\mathrm{T}} = \frac{\boldsymbol{w}\boldsymbol{w}^{\mathrm{T}}}{\boldsymbol{w}^{\mathrm{T}}\boldsymbol{w}}.$$

What a pretty formula!

Be careful here. The denominator is NOT a matrix!!! You can NEVER put a matrix or a vector in the denominator. Only numbers are allowed to do so. Computationally, we are dividing each entry of the matrix $\boldsymbol{w}\boldsymbol{w}^{\mathrm{T}}$ by the number $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{w}$. ☺

4.2.2 Rows, Columns, and Entries of a matrix

From now on, we shall always use $\boldsymbol{e}_i$ to denote a vector whose i-th coordinate is 1, and all other coordinates are zero.

We already know the following by definition of matrices as linear maps. (If you forget, reveiw chapter one.)

Proposition 4.2.12. *The i-th column of a matrix A is $A\boldsymbol{e}_i$.*

But what about rows?

Proposition 4.2.13. *The i-th row of a matrix A is $\boldsymbol{e}_i^{\mathrm{T}}A$.*

For example, calculate $\begin{bmatrix}0 & 1 & 0\end{bmatrix}\begin{bmatrix}1 & 2 & 3 & 4\\5 & 6 & 7 & 8\\9 & 10 & 11 & 12\end{bmatrix}$. When we apply the row vector to each column, it is as if we are taking a dot product, and we simply get the second coordinate of each column, which gives the second row.

Remark 4.2.14. *Say A is a matrix and $\boldsymbol{v}$ is a (vertical) vector, then $A\boldsymbol{v}$ is again a (vertical) vector.*

In comparison, if we have a row vector $\boldsymbol{v}^{\mathrm{T}}$, then $\boldsymbol{v}^{\mathrm{T}}A$ is again a row vector.

In fact, you may check that $A\boldsymbol{v}$ is a linear combination of columns of A according to coordinates of $\boldsymbol{v}$. Similarly, $\boldsymbol{v}^{\mathrm{T}}$ is a linear combination of rows of A according to coordinates of $\boldsymbol{v}^{\mathrm{T}}$.

In the example $\begin{bmatrix}0 & 1 & 0\end{bmatrix}\begin{bmatrix}1 & 2 & 3 & 4\\5 & 6 & 7 & 8\\9 & 10 & 11 & 12\end{bmatrix}$, we are taking none of the first row, one copy of the second row, and none of the third row. So the result of this linear combination is simply the second row.

Remark 4.2.15. *If the columns of A are linearly dependent, then $A\boldsymbol{v} = \boldsymbol{0}$ for some nonzero vector $\boldsymbol{v}$. Similarly, if the rows of A are linearly dependent, then $\boldsymbol{v}^{\mathrm{T}}A = \boldsymbol{0}^{\mathrm{T}}$ for some nonzero $\boldsymbol{v}^{\mathrm{T}}$.*

It all works out pretty symmetrically. Just remember, "columns are to the right of A, while rows are to the left of A".

Corollary 4.2.16. *Suppose $A = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}\\\vdots\\\boldsymbol{a}_m^{\mathrm{T}}\end{bmatrix}$. We have matrix multiplication $AB = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}\\\vdots\\\boldsymbol{a}_m^{\mathrm{T}}\end{bmatrix}B = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}B\\\vdots\\\boldsymbol{a}_m^{\mathrm{T}}B\end{bmatrix}$.*

Proof. This is because $\boldsymbol{e}_i^{\mathrm{T}}(AB) = (\boldsymbol{e}_i^{\mathrm{T}}A)B = \boldsymbol{a}_i^{\mathrm{T}}B$. $\square$

This is the "dual" picture to our previous go-to way to do matrix multiplication, i.e., $A\begin{bmatrix}\boldsymbol{b}_1 & \dots & \boldsymbol{b}_n\end{bmatrix} = \begin{bmatrix}A\boldsymbol{b}_1 & \dots & A\boldsymbol{b}_n\end{bmatrix}$.

Let us now finally shift our attention to entries.

Proposition 4.2.17. *The (i, j) entry of A is $\boldsymbol{e}_i^{\mathrm{T}}A\boldsymbol{e}_j$.*

Proof. Note that $\boldsymbol{e}_i^{\mathrm{T}}(A\boldsymbol{e}_j)$ is the i-th row of the j-th column of A, so we are done.

Also note how associativity is lurking here. We can alternatively look at $(\boldsymbol{e}_i^{\mathrm{T}}A)\boldsymbol{e}_j$, which is the j-th column of the i-th row of A, and it is the same entry. $\square$

Proposition 4.2.18. *The (i, j) entry of AB is the dot product between the i-th row of A and the j-th column of B.*

Proof. $\boldsymbol{e}_i^{\mathrm{T}}(AB)\boldsymbol{e}_j = (\boldsymbol{e}_i^{\mathrm{T}}A)(B\boldsymbol{e}_j)$. Simple associativity. $\square$

In particular, if we write A in rows and B in columns, we have

$$AB = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}\end{bmatrix}\begin{bmatrix}\boldsymbol{b}_1 & \dots & \boldsymbol{b}_n\end{bmatrix} = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{b}_1 & \dots & \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{b}_n \\ \vdots & \ddots & \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{b}_1 & \dots & \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{b}_n\end{bmatrix}.$$

If you truely like things to be computational, you can further write matrix multiplication in terms of entries.

$$AB = \begin{bmatrix}a_{11} & \dots & a_{1r} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mr}\end{bmatrix}\begin{bmatrix}b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{r1} & \dots & b_{rn}\end{bmatrix} = \begin{bmatrix}\sum_i a_{1i}b_{i1} & \dots & \sum_i a_{1i}b_{in} \\ \vdots & \ddots & \vdots \\ \sum_i a_{mi}b_{i1} & \dots & \sum_i a_{mi}b_{in}\end{bmatrix}.$$

Example 4.2.19. Let us do a calculation for fun.

A ***lower triangular matrix*** is a matrix whose entries above the diagonal are all zero. For example, say $L = \begin{bmatrix}1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1\end{bmatrix}$. Similarly, an ***upper triangular matrix*** is a matrix whose entries below the diagonal are all zero. For example, say $U = \begin{bmatrix}1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1\end{bmatrix}$. (Note that row echelon forms are all upper triangular.)

Calculation gives $LU = \begin{bmatrix}1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4\end{bmatrix}$. Pretty, yes?

Let us try to see another pretty sight. A Pascal's lower triangular matrix is a lower triangular matrix, where the first column has all 1, the diagonal entries are all 1, and each entry is the sum of the entry above it and the entry to its upper left. Say $L = \begin{bmatrix}1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1\end{bmatrix}$.

Similarly, a Pascal's upper triangular matrix is a lower triangular matrix, where the first row has all 1, the diagonal entries are all 1, and each entry is the sum of the entry to the left and the entry to its upper left. Say $U = \begin{bmatrix}1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1\end{bmatrix}$.

Calculation gives $LU = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$. Pretty, yes? This is the symmetric Pascal's matrix, where the first row and first column has all 1, and each entry is the sum of the entry to the left and the entry above it.

These pretty structures are of curse NOT coincidences. (Nothing in math is a coincidence.) However, their nature is more combinatorial than linear algebra, so let us leave it at that. ☺

We like triangular matrices for the following reason.

Proposition 4.2.20. *The product of two upper (lower) triangular matrices is still upper (lower) trianguler. Furthermore, the diagonal entries of the product is the entry-wise product of the diagonal entries of the two matrices. In short, we should have*

$$\begin{bmatrix} a_{11} & * & * \\ & \ddots & * \\ & & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & * & * \\ & \ddots & * \\ & & b_{nn} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & * & * \\ & \ddots & * \\ & & a_{nn}b_{nn} \end{bmatrix}.$$

Proof. Note that a triangular matrix must be a square matrix. So if two triangular matrices could multiply, then they must have the same number of columns and rows.

Suppose A, B are upper triangular, say $A = \begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_n^{\mathrm{T}} \end{bmatrix}$ and $B = \begin{bmatrix} \boldsymbol{b}_1 & \dots & \boldsymbol{b}_n \end{bmatrix}$. Then the (i,j)-entry of AB is $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{b}_j$. This is a dot product. Say $\boldsymbol{a}_i^{\mathrm{T}} = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 & a_i & \dots & a_n \end{bmatrix}$ and $\boldsymbol{b}_j = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_j \\ 0 \\ \vdots \\ 0 \end{bmatrix}$.

Let us consider a lower-triangular entry, which is the case when $i > j$. Since A, B are upper triangular, the first $(i-1)$ entries of $\boldsymbol{a}_i^{\mathrm{T}}$ are zero, and the last $(n-j)$-entries of $\boldsymbol{b}_j$ are zero. So we have

$$\begin{bmatrix} \overbrace{0 \quad \dots \quad 0}^{i-1} & a_i & \dots & a_n \end{bmatrix} \left.\begin{bmatrix} \vdots \\ b_j \\ 0 \\ \vdots \\ 0 \end{bmatrix}\right\} n-j = \overbrace{a_1b_1 + \dots + \cdots}^{\text{First } i-1 \text{ terms are } 0} \underbrace{\cdots + \dots + a_nb_n}_{\text{Last } n-j \text{ terms are } 0}.$$

Since $i > j$, we have $(i-1) + (n-j) \geq n$. So all terms are zero. So all (i,j)-entry of AB with $i > j$ are zero. Hence AB is still upper triangular.

When $i = j$, then we have

$$\begin{bmatrix} \overbrace{0 \quad \dots \quad 0}^{i-1} & a_i & \dots & a_n \end{bmatrix} \left.\begin{bmatrix} \vdots \\ b_i \\ 0 \\ \vdots \\ 0 \end{bmatrix}\right\} n-i = \overbrace{a_1b_1 + \dots}^{\text{First } i-1 \text{ terms are } 0} + a_ib_i + \underbrace{\dots + a_nb_n}_{\text{Last } n-i \text{ terms are } 0} = a_ib_i.$$

So the (i,i)-entry of AB is just $a_{ii}b_{ii}$.

The case of lower triangular matrices is similar. ☐

Let us have one last example. What if we multiply a matrix with itself?

Definition 4.2.21. *The k-th power of a matrix A, written as A^k, is A multiplying itself k times.*

Example 4.2.22. Let $A = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$. This is a very important matrix, the simplest shearing matrix.

By calculation, you can see that $A^2 = \begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}, A^3 = \begin{bmatrix}1 & 3\\0 & 1\end{bmatrix}$ and so on. See a pattern? Let us prove it here with mathematical induction.

I claim $A^k = \begin{bmatrix}1 & k\\0 & 1\end{bmatrix}$. The base case when $k = 1$ is trivial, so we only need the inductive step now. Suppose the statement is true for k, let us prove it for $k+1$.

We have $A^{k+1} = A^k A = \begin{bmatrix}1 & k\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix} = \begin{bmatrix}1 & k+1\\0 & 1\end{bmatrix}$. So we are done. ☺

Finally, let us play a bit with the most important matrix, i.e., the identity matrix.

Proposition 4.2.23. *Consider the $n \times n$ identity matrix I. For all $n \times m$ matrix A, we have $IA = A$. For all $m \times n$ matrix A, we have $AI = A$.*

By convention, for any square matrix A, we usually define $A^0 = I$. (We use this convention even if A has no inverse matrix. Even if $A = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$, we still define $A^0 = I$. This convention will ensure that $A^sA^t = A^{s+t}$ for all non-negative integer s, t and any square matrix A.)

4.2.3 Geometries of Linear Maps

Now we move on to several kinds of matrices, and hopefully also provide you with a variaty of perspectives on how to understand them. I want you to keep in mind: each specific entry is NOT important. Only collectively as a linear may, would they be important. What does the matrix behave? What is the process of this linear map? Those are the more important questions.

Let us start with some geometrically interesting maps.

Example 4.2.24. We know rotations on the plane are $R_\theta = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$. What if you compose two rotations? We have $R_\theta R_\phi = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}\cos\phi & -\sin\phi\\ \sin\phi & \cos\phi\end{bmatrix} = \begin{bmatrix}\cos\theta\cos\phi - \sin\theta\sin\phi & -\sin\theta\cos\phi - \sin\phi\cos\theta\\ \sin\theta\cos\phi + \sin\phi\cos\theta & \cos\theta\cos\phi - \sin\theta\sin\phi\end{bmatrix}$. Do these formula looks familiar to you? They SHOULD! These are basic trignometry stuff.

So upon further simplification, this gives $\begin{bmatrix}\cos(\theta+\phi) & -\sin(\theta+\phi)\\ \sin(\theta+\phi) & \cos(\theta+\phi)\end{bmatrix} = R_{\theta+\phi}$. Huh. On second thought, this is no surprise at all if you just think about the geometry. $R_\theta R_\phi$ literally means do the rotation by ϕ, then do the rotation by θ. Obviously the result is a rotation by $\theta + \phi$. In fact, this whole process could be thought of as a proof of the trigonometry sum formula.

Btw, it should also be obvious that $R_\theta^k = R_{k\theta}$ and so on.

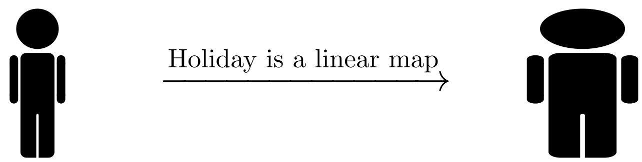
The moral of the story is this: matrix multiplication formula is fine. But sometimes, understanding the meaning of the maps can help you calculate much faster. ☺

Example 4.2.25. Flipping the plane about the line $x = y$ has a matrix of $F = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$. If you try to apply this to a vector, you see that it sends $\begin{bmatrix}x\\y\end{bmatrix}$ to $\begin{bmatrix}y\\x\end{bmatrix}$, so it literally just swap the coordinates. We have previously see that $FR_\theta = R_{-\theta}F$.

Flipping the plane about the x-axis has a matrix of $D = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$. (Can you see why?) You can again verify that $DR_\theta = R_{-\theta}D$.

In fact, for any reflection F of the plane about a line through the origin, we always have $FR_\theta = R_{-\theta}F$. Intuitively, rotating by θ means in the mirror, you are rotating by $-\theta$. Try to see this geometrically, and then see if you can verify this numerically. (In general, to reflect along the line with slope angle θ, the matrix is $\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}$.) ☺

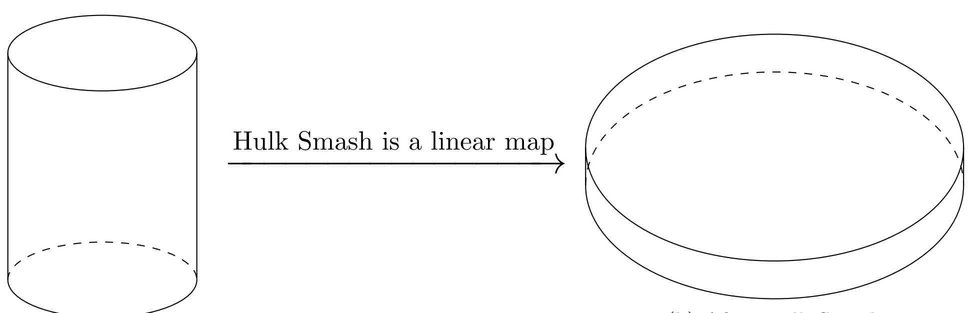
Example 4.2.26. Let us stretch the plane now. Say we stretch everything in the x-axis direction by a factor of 2. This is a linear map with matrix $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$. (Can you see why?) A picture of myself will now appear twice as fat....



(a) Before Holiday (b) After Holiday

Figure 4.2.2: Holiday $= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

Matrices like these are called ***diagonal matrices***. They stretch things along the coordinate-axes. Say if we have $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{6} \end{bmatrix}$, then this linear map is stretching the space $\mathbb{R}^3$ in the x-axis direction by a factor of 2, then in the y-axis direction by a factor of 3, and finally in the z-axis direction by a factor of $\frac{1}{6}$ (so we are squeezing in this direction).



(a) Before Hulk Smash (b) After Hulk Smash

Figure 4.2.3: Hulk Smash$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{6} \end{bmatrix}$

The identity matrix is a special case of a diagonal matrix. We are now stretching everything by a factor of 1, i.e., we leave them all unchanged. You may also think of the reflection matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ as a diagonal matrix. We are scaling things in the y-direction by a factor of -1, so things are flipped about the x-axis.

The square projection matrix (projection to the x-axis), $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, which sends $\begin{bmatrix} x \\ y \end{bmatrix}$ to $\begin{bmatrix} x \\ 0 \end{bmatrix}$, is also a special case of this. We are now squeezing everything in the y-axis direction into nothing. Hence we obtained an orthogonal projection to the x-axis.

Multiplications of diagonal matrices are always diagonal. This compares nicely with triangular matrices. (In fact, a matrix is diagonal iff it is both upper triangular and lower triangular.)

In general, for diagonal matrices we can computationally verify that $\begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix} \begin{bmatrix} b_1 & & \\ & \ddots & \\ & & b_n \end{bmatrix} = \begin{bmatrix} a_1 b_1 & & \\ & \ddots & \\ & & a_n b_n \end{bmatrix}$. Here some entries are empty, which means they are zero. Can you see this geometrically?

Algebraically, also note that for diagonal matrices, we always have commutativity. ☺

Example 4.2.27. For a parallelogram, we learned long ago that its area is "base" times "height". Now, given a parallelogram and fixed base, can you draw all parallelograms with the same height?

You will end up drawing many "sheared" versions of this parallelogram.

Consider the matrix $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$. Then for any parallelogram whose base is on the x-axis, this linear map will "shear" it into some tilted version of itself, but always with the same base and same height! In particular, a shearing always preserves their volumn. This is the shearing of the x-axis direction to the y-axis direction.

In physics, a "shearing" is a force that acts like a pair of scissors. If you observe a pair of scissors, you will realize that one blade is always on top of the other. When you use the scissors to cut things, the top blade will push things to the right, while the lower blade will push things to the left. This is exactly what $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ will tries to do to the plane.

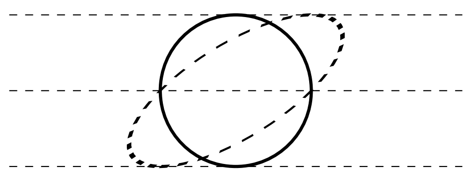


Figure 4.2.4: A shearing of a circle

In fact, a shearing will always preserve the area of whatever shape in the plane. Draw some triangles and see if you can prove that these triangles have area preserved after a shearing.

Now, consider the geometric nature of a shearing, if I do the same shearing twice, it is as if I simply sheared by twice the original amount. Hence it is geometrically very obvious that $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$.

If I did some shearing and then sheared some more, then in total I simply did a bigger shear. Hence it is also geometrically very obvious that $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a+b \\ 0 & 1 \end{bmatrix}$. Note that all shearings of the xy-plane parallel to the x-axis will commute.

However, shearings along different directions will NOT commute. For example, $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ is a shearing parallel to the y-axis. (Draw some graph and verify this yourself.)

Then we have

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

So they do not commute. ☺

4.2.4 Linear Combinations of Matrices

Matrix multiplication is a very powerful tool. It allows us to compose linear maps, understand behaviors among linear maps, understand iterations of a linear transformation, and so on so forth.

Another powerful tool is to do linear combinations of matrices.

Definition 4.2.28. *Given two maps $f, g : \mathbb{R}^n \to \mathbb{R}^m$ (for this definition, they are not required to be linear), we define $f + g : \mathbb{R}^n \to \mathbb{R}^m$ to be the map such that $(f + g)(\boldsymbol{x})$ is defined as $f(\boldsymbol{x}) + g(\boldsymbol{x})$, and for any number $k \in \mathbb{R}$, we define $kf : \mathbb{R}^n \to \mathbb{R}^m$ to be the map such that $(kf)(\boldsymbol{x}) = kf(\boldsymbol{x})$.*

This definition is very much standard. For example, if we are considering real functions from $\mathbb{R}$ to $\mathbb{R}$, say e^x and x^2, then the sum of e^x and x^2 is the function $e^x + x^2$. This is obviously the only sensible way to define such a sum.

In the case of matrices (i.e., linear maps), note that we require the domains and codomains to be the same. In particular, $A + B$ is ONLY defined when A, B have the same number of rows and the same number of columns.

Luckily for us, this is super easy to compute.

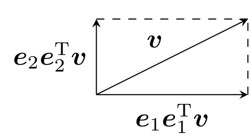
Proposition 4.2.29. *Given A, B of the same size, then the (i, j) entry of $A + B$ is simply the (i, j) entry of A plus the (i, j) entry of B. Similarly, for any $k \in \mathbb{R}$, the (i, j) entry of kA is simply k times the (i, j) entry of A.*

Proof. By definition of linear combinations of matrices, $(A + B)\boldsymbol{v} = A\boldsymbol{v} + B\boldsymbol{v}$ and $(kA)\boldsymbol{v} = k(A\boldsymbol{v})$ for all input $\boldsymbol{v}$.

Let us now figure out the j-th column of $A + B$. But we have $(A + B)\boldsymbol{e}_j = A\boldsymbol{e}_j + B\boldsymbol{e}_j$. Hence this is simply the sum of the j-th column of A and j-th column of B. Done.

Similarly, $(kA)\boldsymbol{e}_j = k(A\boldsymbol{e}_j)$. So the j-th column of kA is simply k times the j-th column of A. $\square$

Example 4.2.30. Think about the identity map $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. What does this equation mean? It tells us that the identity map is the sum of the projection to x-axis and the projection to y-axis.



In particular, if we input a vector $\boldsymbol{v}$ to the equation, we have $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\boldsymbol{v} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\boldsymbol{v} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\boldsymbol{v}$. This means each vector is the sum of its x-axis "component vector" plus its y-axis "component vector". This orthogonal decomposition of $\boldsymbol{v}$ is a very useful tool in many problems in high school physics.

Note that $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}}$ and $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \boldsymbol{e}_2\boldsymbol{e}_2^{\mathrm{T}}$. In general, for the $n \times n$ identity matrix I, we have $I = \sum \boldsymbol{e}_i\boldsymbol{e}_i^{\mathrm{T}}$, and geometrically this means any vector is the sum of its "coordinate component vectors".

In fact, take any pair of orthogonal unit vectors in $\mathbb{R}^2$, say $\boldsymbol{u} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$, $\boldsymbol{v} = \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix}$, and calculate $\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} + \boldsymbol{v}\boldsymbol{v}^{\mathrm{T}}$. What do you see? And now you know why. ☺

Example 4.2.31. Let us work out a formula for the hyperplane projection matrix in $\mathbb{R}^3$. Suppose we want to project to a hyperplane through the origin. What should I do?

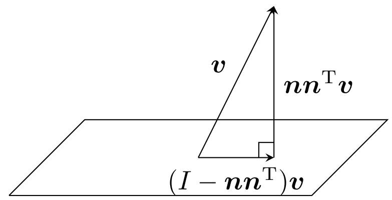


Figure 4.2.5: From $\boldsymbol{v}$ to its reflection $H\boldsymbol{v}$.

Let $\boldsymbol{n}$ be a unit normal vector. Then for any vector $\boldsymbol{v}$, we have $\boldsymbol{v} = (\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v} + (I - \boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$, where I is the identity matrix. You can verify that this is an orthogonal decomposition into a component parallel to $\boldsymbol{n}$, i.e., $(\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$, and a component orthogonal to $\boldsymbol{n}$ (and hence parallel to our plane of reflection), i.e., $(I - \boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$.

In particular, the projection matrix is $I - \boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$. ☺

Example 4.2.32. Let us work out a formula for the reflection matrix in $\mathbb{R}^3$. Suppose we want to reflect about a plane through the origin. What should I do?

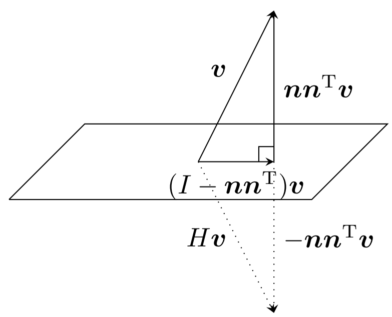


Figure 4.2.6: From $\boldsymbol{v}$ to its reflection $H\boldsymbol{v}$.

Let $\boldsymbol{n}$ be a unit normal vector. Then for any vector $\boldsymbol{v}$, we have $\boldsymbol{v} = (\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v} + (I - \boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$, where I is the identity matrix. You can verify that this is an orthogonal decomposition into a component parallel to $\boldsymbol{n}$, i.e., $(\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$, and a component orthogonal to $\boldsymbol{n}$ (and hence parallel to our plane of reflection), i.e., $(I - \boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$.

Now for the part that is parallel to plane of reflection, it should stay unchanged. For the part that is parallel to the normal vector, it shall be reflected (i.e., negated). So all in all, we want to send $\boldsymbol{v}$ to $-(\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v} + (I - \boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$. Simplifying this, we want to send $\boldsymbol{v}$ to $(I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}})\boldsymbol{v}$. Obviously this linear map is simply multiplying by the matrix $I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$.

In general, for any unit vector $\boldsymbol{n} \in \mathbb{R}^n$, for this very reason $I_n - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$ is always a reflection about a hyperplane with normal vector $\boldsymbol{n}$. This gives all possible reflections in all dimensions.

Some texts also refers to these "higher dimensional reflections" as ***Householder transformations***. ☺

So far we have been trying to focus on the geometry of linear maps. Now let us also look at some analytical perspectives.

Example 4.2.33. We learn integration, and it is basically the idea of accumulation. Suppose I track my total amount of money each month. Say I start with 1 dollar, but after one month I have 3 dollars, after two months I have 6 dollars, and after three months I have 7 dollars. I can write these data as a vector $\begin{bmatrix}1\\3\\6\\7\end{bmatrix}$.

Suppose I am NOT interested in the accumulated amount of money. Rather, I am interested in how many money can I earn each month. The " rate of increase", or in calculus terms, the "derivative" of the total amount of money. What should I do?

The key is the ***forward difference matrix***. It looks like $D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. It will send $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ to $\begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \\ x_4 - x_3 \end{bmatrix}$. So it records the initial amount, and traces the increases between terms.

In our case, $D \begin{bmatrix} 1 \\ 3 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$. I started with 1 dollar, and during the first month I earned 2 dollars. During the second month I earned 3 dollars. And during the third month I earned 1 dollar.

Obviously this matrix is very useful in statistics. But furthermore, keep in mind that essentially, D is just a discrete version of "taking derivatives". For example, consider $D \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix}$, i.e., the derivative of a exponential function is still exponential (ignoring the first coordinates).

There are some interesting properties if you apply this D to sequences. For example, D applied to an arithmetic sequence gives $D \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$. This is because the arithmetic sequence as a function has constant derivative. So after D, its coordinates are constant (ignoring the first coordinate).

Applying D to a quadratic sequence gives $D \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \\ 25 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}$, an arithmetic sequence (ignoring the first coordinate), because the derivative of a degree two polynomial is a degree one polynomial.

Finally, applying D to the Fibbonacci sequence gives the sequence again $D \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \\ 5 \\ 8 \\ 13 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 2 \\ 3 \\ 5 \end{bmatrix}$, just shifted by two terms. This is because the formula for the Fibbonacci numbers is the linear combination of two exponential functions, and hence the derivative of it is also a linear combination of these two exponential functions. ☺

Example 4.2.34. Now look at $D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. It is obviously the difference between two matrices, $D = I - J$, where I is the identity matrix, and J is the matrix with ones immediately below the diagonal, and zeroes everywhere else. What do they do?

I obviously sends a sequence to itself. The identity map never changes anything. But J is the "shift down" operator. It shifts the sequence down a term, and sends $\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$ to $\begin{bmatrix}0\\x_1\\x_2\\x_3\end{bmatrix}$. Looking at this, you can probably see why $I - J$ gives the forward difference matrix. We have the original term (preserved by I) minus the previous term (that got shifted down by J).

Similarly one can look at J^{T}, the matrix with ones immediately above the diagonal, and zeroes everywhere else. Then $I - J^{\mathrm{T}}$ is the ***backward difference matrix***. It will send $\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$ to $\begin{bmatrix}x_1-x_2\\x_2-x_3\\x_3-x_4\\x_4\end{bmatrix}$. ☺

A very interesting property of linear combinations of matrices is the law of distribution. We look back at the definition, and we have

Lemma 4.2.35 (Matrix-Vector law of distribution)**.** *For any $\boldsymbol{v} \in \mathbb{R}^n$, any $m \times n$ matrices A, B and any $x, y \in \mathbb{R}$, we have $(xA + yB)\boldsymbol{v} = x(A\boldsymbol{v}) + y(B\boldsymbol{v})$.*

Proof. This is just repeating the definition of linear combinations of maps. □

This can immediately be generalized to a full law of distribution on any matrix multiplications.

Proposition 4.2.36 (Law of distribution)**.** *For any matrices A, B, C and any $x, y \in \mathbb{R}$, we have $(xA + yB)C = x(AC) + y(BC)$ and $C(xA + yB) = xCA + yCB$ whenever the matrix multiplications/linear combinations involved are all well-defined.*

Proof. $(xA + yB)C$ is just $xA + yB$ applied to each column of C. So this immediately follows from the lemma above.

$C(xA+yB)$ is slightly trickier, but not much. Let $\boldsymbol{a}_i$ be the i-th column of A and $\boldsymbol{b}_i$ be the i-th column of B, then we know the i-th column of $xA + yB$ must be $x\boldsymbol{a}_i + y\boldsymbol{b}_i$.

So the i-th column of $C(xA+yB)$ is $C(x\boldsymbol{a}_i + y\boldsymbol{b}_i) = xC\boldsymbol{a}_i + yC\boldsymbol{b}_i$ by linearity of C. And this is exactly the i-th column of $x(CA) + y(CB)$. □

Remark 4.2.37. *By this time it should be pretty clear that "the law of distribution" is just a special case of linearity. In real numbers, when we have the law of distribution $a(b+c) = ab + ac$, we are really just saying that "multiplication by a" is linear.*

Pay special attention to the ORDER of multiplication, because for example AB might NOT be BA. So if you write $A(xB+yC) = xBA+yCA$, then it would be the WRONG formula. As a quick memorization tip, whatever is on the left before, it will still be on the left (because the map happens later in time). Similarly, whatever is on the right before, it will still be on the right.

Corollary 4.2.38. *$(A + B)^2 = A^2 + AB + BA + B^2$, and so on. If $AB = BA$, then in fact $(A + B)^2 = A^2 + 2AB + B^2$. However, if $AB \neq BA$, then we must have $(A + B)^2 \neq A^2 + 2AB + B^2$.*

Corollary 4.2.39. *$(A + I)^2 = A^2 + 2A + I$, and so on for other polynomial calculations.*

Definition 4.2.40. *Given a polynomial $p(x) = a_n x^n + \cdots + a_1 x + a_0$ and a square matrix A, we have $p(A) = a_n A^n + \cdots + a_1 A + a_0 I$.*

Corollary 4.2.41. *Given two polynomials $p(x), q(x)$ and a square matrix A, then $p(A)q(A) = q(A)p(A)$. And if $h(x) = p(x)q(x)$, then $h(A) = p(A)q(A) = q(A)p(A)$.*

Proof. Observe that polynomial is simply a linear combination of powers. For example, $p(x) = a_n x^n + \cdots + a_1 x + a_0$ means $p(x)$ is a linear combination of $x^n, \ldots, x^1, x^0$ with coefficient a_i for x^i. And $p(A)$ is simply the corresponding linear combination of powers of A.

You can summarize this proof using only one sentence: ***Since all powers of A commute, therefore all polynomials of A (i.e., linear combinations of powers of A) commute.***

Now we begin our proof. Let $p(x) = \sum_{i=0}^{m} a_i x^i$ and $q(x) = \sum_{j=0}^{n} b_j x^j$. Then we have

$$\begin{aligned}
p(A)q(A) =& (\sum_{i=0}^{m} a_i A^i)(\sum_{j=0}^{n} b_j A^j) \\
=& \sum_{i,j} a_i b_j A^i A^j \\
=& \sum_{i,j} a_i b_j A^{i+j} \\
=& \sum_{i,j} a_i b_j A^{j+i} \\
=& \sum_{i,j} a_i b_j A^j A^i \\
=& (\sum_{j=0}^{n} b_j A^j)(\sum_{i=0}^{m} a_i A^i) = q(A)p(A).
\end{aligned}$$

Now

$$h(x) = p(x)q(x) = (\sum_{i=0}^{m} a_i x^i)(\sum_{j=0}^{n} b_j x^j) = \sum_{i,j} a_i b_j x^{i+j}.$$

So from previous calculations we have

$$h(A) = \sum_{i,j} a_i b_j A^{i+j} = p(A)q(A).$$

□

Example 4.2.42. Consider the matrix $E = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. It sends $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ to $\begin{bmatrix} x_1 + x_2 \\ x_2 \\ x_3 \end{bmatrix}$. So what it does is to add the second coordinate onto the first.

Note that this matrix E differ from I at only one entry. The difference $X = E - I$ is a matrix whose $(1, 2)$ entry is 1, and all other entries are zero. It sends $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ to $\begin{bmatrix} x_2 \\ 0 \\ 0 \end{bmatrix}$. So it simply takes the second coordinate, and stuff it in the first coordinate, and clear everything else.

In particular, it is easy to see that $I + kX = \begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ will sends $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ to $\begin{bmatrix} x_1 + kx_2 \\ x_2 \\ x_3 \end{bmatrix}$. Furthermore, it is also easy to check that $X^2 = O$, the zero matrix. In particular, $X^k = O$ whenever $k \geq 2$.

As a result, we have

$$(I + X)^k = I + kX + \text{Higher degree terms} = I + kX.$$

So $E^k = \begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Hey! This is exactly how shearings would behave!

This is no surprise. Imagine what would happen if we have a parallelopiped with base on the xz-plane. The base would be fixed, and it would be tilted along the x-direction but the height is preserved (the height would be in the y-direction). This E is a 3-dimensional shear, and it preserves volumn instead of area. ☺

So to end this section, let us write out all different ways to write matrix multiplication AB.

1. $A\begin{bmatrix}\boldsymbol{b}_1 & \dots & \boldsymbol{b}_n\end{bmatrix} = \begin{bmatrix}A\boldsymbol{b}_1 & \dots & A\boldsymbol{b}_n\end{bmatrix}$.
2. $\begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}\end{bmatrix} B = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}B \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}B\end{bmatrix}$.
3. $\begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}\end{bmatrix}\begin{bmatrix}\boldsymbol{b}_1 & \dots & \boldsymbol{b}_n\end{bmatrix} = \begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{b}_1 & \dots & \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{b}_n \\ \vdots & \ddots & \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{b}_1 & \dots & \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{b}_n\end{bmatrix}$.
4. $\begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_n\end{bmatrix}\begin{bmatrix}\boldsymbol{b}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}}\end{bmatrix} = \sum \boldsymbol{a}_i\boldsymbol{b}_i^{\mathrm{T}}$.

The last one is new. Can you see why?

Proposition 4.2.43. $\begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_n\end{bmatrix}\begin{bmatrix}\boldsymbol{b}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}}\end{bmatrix} = \sum \boldsymbol{a}_i\boldsymbol{b}_i^{\mathrm{T}}$.

Proof. An entry-wise calculation proof goes like this. Say A is $m \times n$ and B is $n \times d$. The (i,j) entry of AB is simply $\sum_k a_{ik}b_{kj}$. So we have

$$AB = \begin{bmatrix}\sum_k a_{1k}b_{k1} & \dots & \sum_k a_{1k}b_{kd} \\ \vdots & \ddots & \vdots \\ \sum_k a_{mk}b_{k1} & \dots & \sum_k a_{mk}b_{kd}\end{bmatrix} = \sum_k \begin{bmatrix}a_{1k}b_{k1} & \dots & a_{1k}b_{kd} \\ \vdots & \ddots & \vdots \\ a_{mk}b_{k1} & \dots & a_{mk}b_{kd}\end{bmatrix} = \sum_k \boldsymbol{a}_k\boldsymbol{b}_k^{\mathrm{T}}.$$

So we are done.

Alternatively, to use fancy entry notation, consider the (i,j) entry $\boldsymbol{e}_i^{\mathrm{T}}AB\boldsymbol{e}_j$. We have $\boldsymbol{e}_i^{\mathrm{T}}\begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_m\end{bmatrix}\begin{bmatrix}\boldsymbol{b}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}}\end{bmatrix}\boldsymbol{e}_j =$ $\begin{bmatrix}\boldsymbol{e}_i^{\mathrm{T}}\boldsymbol{a}_1 & \dots & \boldsymbol{e}_i^{\mathrm{T}}\boldsymbol{a}_m\end{bmatrix}\begin{bmatrix}\boldsymbol{b}_1^{\mathrm{T}}\boldsymbol{e}_j \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}}\boldsymbol{e}_j\end{bmatrix}$. Now this is just a row vector with a column vector, so it calculates like a dot product and gives $\sum_k(\boldsymbol{e}_i^{\mathrm{T}}\boldsymbol{a}_k)(\boldsymbol{b}_k^{\mathrm{T}}\boldsymbol{e}_j)$. Use associativity and linearity, we have $\boldsymbol{e}_i^{\mathrm{T}}(\sum_k \boldsymbol{a}_k\boldsymbol{b}_k^{\mathrm{T}})\boldsymbol{e}_j$. So we see that AB and $\sum_k \boldsymbol{a}_k\boldsymbol{b}_k^{\mathrm{T}}$ have the same (i,j)-entry for all i,j. So they are the same matrix.

Yet alternatively, for a third proof, consider the fact that $I = \sum \boldsymbol{e}_i\boldsymbol{e}_i^{\mathrm{T}}$. Then we have

$$AB = AIB = A(\sum \boldsymbol{e}_i\boldsymbol{e}_i^{\mathrm{T}})B = \sum A\boldsymbol{e}_i\boldsymbol{e}_i^{\mathrm{T}}B = \sum(A\boldsymbol{e}_i)(\boldsymbol{e}_i^{\mathrm{T}}B) = \sum \boldsymbol{a}_i\boldsymbol{b}_i^{\mathrm{T}}.$$

Associativity is so cool.... □

4.2.5 Transpose of Matrices

Now we talk about the transpose of a matrix.

Definition 4.2.44. *Given a $m \times n$ matrix A, its **transpose** is an $n \times m$ matrix A^{T} whose (i,j) entry is the (j,i) entry of A.*

Intuitively, A^{T} is just an entry-wise "reflection" of A about the diagonal entries. Like $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$.

Proposition 4.2.45. $(A^{\mathrm{T}})^{\mathrm{T}} = A$.

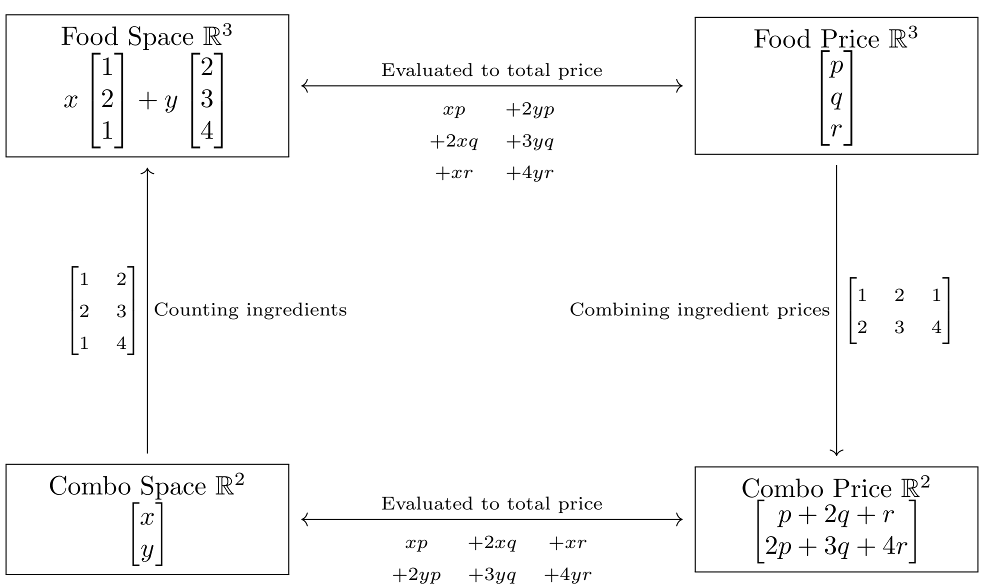
But what does this transpose mean?

It concers a deeper mathematical concept called a "dual space", which I cannot explain yet. (It is taught in next semester's class.) However, here is an example, and if you like, draw whatever intuition from it as much as you can.

Example 4.2.46. Suppose I want to burgers, chicken wings and cokes. There are two meal combos. Combo one contains 1 burger and 2 wings and 1 coke. Combo two contains 2 burger 3 wings and 4 cokes. Can you see a linear relation here?

Of course. If we have x combo one and y combo two, then we have $\begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ burgers, wings and cokes.

Now suppose the burgers, wings and cokes are p, q, r dollars each. Then the combos have prices $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$.



So, we are combining foods into meal combos. On the content side, we have A. On the evaluation side, we have A^{T}. "Counting ingredients" and "combining prices" are transpose of each other. ☺

Remark 4.2.47. *This remark is largely philosophical. Read it if you like, and forget about it if you cannot understand....*

Transpose is suppose to bounce between objects and evaluations of objects. To understand this, we first need a philosophical perspective: an object and its evaluation is in duality.

Mathematically speaking, given a set X, we can call a function $f : X \to \mathbb{R}$ an evaluation on X. Now let Y be the set of all evaluations on X.

Given $f \in Y$, obviously it evaluates $x \in X$ via $x \mapsto f(x)$. However, given $x \in X$, we can also use it to evaluate any $f \in Y$ via $f \mapsto f(x)$.

Consider this. Say that we have students doing two math tests. Then we have a function f sending students to their scores on the first test, and a function g sending students to their scores on the second test. So who is evaluating whom? An obvious answer is that functions are evaluating inputs. Each test evaluates students to the corresponding test score.

But one can also argue that the students are also evaluating the tests. For a fixed student, by doing both tests, this person is evaluating the differences between the tests, by giving different scores.

Such is the duality, where evaluations and "evaluatees" can switch places if we switch perspective. Transpose, philosophically speaking, is such a perspective switch, where we are switching the roles of evaluations and "evaluatees". What is a vector $\boldsymbol{v}$? Think of this as an object. What is a row vector $\boldsymbol{w}^{\mathrm{T}}$? Think of this as an evaluation of vectors. Then if A sends vectors forward to vectors, we have A^{T} sending evaluations backward to evaluations.

Now, philosophy and understanding aside, the biggest computational consequence of transpose is to make rows into columns and columns into rows. Recall that when we do AB, we are doing dot products of rows of A and columns of B. So if we do transpose, we are reversing the order of multiplication.

Proposition 4.2.48. *We have $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$*

Proof. This is an entry-wise proof.

Write A in rows by $A = \begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}} \end{bmatrix}$ and write B in columns by $B = \begin{bmatrix} \boldsymbol{b}_1 & \dots & \boldsymbol{b}_n \end{bmatrix}$. Then the (i, j) entry of AB is $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{b}_j$.

Now note that $A^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$ and $B^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{b}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_m^{\mathrm{T}} \end{bmatrix}$. So the (i, j) entry of $B^{\mathrm{T}}A^{\mathrm{T}}$ is $\boldsymbol{b}_i^{\mathrm{T}}\boldsymbol{a}_j$, which by commutativity of dot product equals to $\boldsymbol{a}_j^{\mathrm{T}}\boldsymbol{b}_i$. This is the (j, i) entry of AB. So $B^{\mathrm{T}}A^{\mathrm{T}} = (AB)^{\mathrm{T}}$.

Intuitively, since matrix multiplication uses the rows of the first matrix to dot the columns of the second matrix, by taking transpose (i.e., reversing the role of rows and columns), the order of multiplication is reversed.

(Also recall the idiom "rows to the left and columns to the right". If we switch the roles of rows and columns, then we are switching left and right. So the order of multiplication is reversed.) ☐

Proof. This is a cooler (albeit longer) proof. Again we uses special case that we know to be true, to establish a more general case.

First, by commutativity of dot product, we know $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = \boldsymbol{w}^{\mathrm{T}}\boldsymbol{v}$. Note that transpose does nothing to 1 by 1 matrices. So $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = \boldsymbol{w}^{\mathrm{T}}\boldsymbol{v} = (\boldsymbol{w}^{\mathrm{T}}\boldsymbol{v})^{\mathrm{T}}$, and this is a special case of our desired statement.

Second, consider $(A\boldsymbol{x})^{\mathrm{T}}$. Say $A = \begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}} \end{bmatrix}$, then

$$\begin{aligned} &(A\boldsymbol{x})^{\mathrm{T}} \\ =&\left(\begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}} \end{bmatrix} \boldsymbol{x}\right)^{\mathrm{T}} \\ =&\begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{x} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{x} \end{bmatrix}^{\mathrm{T}} \end{aligned}$$

$$
\begin{aligned}
&= \begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{x} & \dots & \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{x} \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{x}^{\mathrm{T}}\boldsymbol{a}_1 & \dots & \boldsymbol{x}^{\mathrm{T}}\boldsymbol{a}_m \end{bmatrix} \\
&= \boldsymbol{x}^{\mathrm{T}} \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_m \end{bmatrix} \\
&= \boldsymbol{x}^{\mathrm{T}} A^{\mathrm{T}}.
\end{aligned}
$$

So our statement is good when we have matrix multiplying vectors.

Finally, consider the full case. Say $B = \begin{bmatrix} \boldsymbol{b}_1 & \dots & \boldsymbol{b}_n \end{bmatrix}$. Then

$$
\begin{aligned}
&(AB)^{\mathrm{T}} \\
=&(A \begin{bmatrix} \boldsymbol{b}_1 & \dots & \boldsymbol{b}_n \end{bmatrix})^{\mathrm{T}} \\
=& \begin{bmatrix} A\boldsymbol{b}_1 & \dots & A\boldsymbol{b}_n \end{bmatrix}^{\mathrm{T}} \\
=& \begin{bmatrix} (A\boldsymbol{b}_1)^{\mathrm{T}} \\ \vdots \\ (A\boldsymbol{b}_n)^{\mathrm{T}} \end{bmatrix} \\
=& \begin{bmatrix} \boldsymbol{b}_1^{\mathrm{T}} A^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}} A^{\mathrm{T}} \end{bmatrix} \\
=& \begin{bmatrix} \boldsymbol{b}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}} \end{bmatrix} A^{\mathrm{T}} \\
=& B^{\mathrm{T}} A^{\mathrm{T}}.
\end{aligned}
$$

The idea of using special case to prove a general case is a powerful one, and the "linear perspective" rather than "entry perspective" is nice, so I want you to see this proof. □

Remark 4.2.49. *Note that 1 by 1 matrices are unchanged by taking transpose. So we have* $\boldsymbol{e}_i^{\mathrm{T}} A \boldsymbol{e}_j = (\boldsymbol{e}_i^{\mathrm{T}} A \boldsymbol{e}_j)^{\mathrm{T}} = \boldsymbol{e}_j^{\mathrm{T}} A^{\mathrm{T}} \boldsymbol{e}_i$*, so the* (i,j) *entry of* A *is the same as the* (j,i) *entry of* A^{T}*, as expected.*

Another fine property of transpose is that it is linear.

Proposition 4.2.50. *We have* $(xA + yB)^{\mathrm{T}} = xA^{\mathrm{T}} + yB^{\mathrm{T}}$

Proof. This is trivial. Just name the entries and compute. Say $A = (a_{ij})_{m\times n}, B = (b_{ij})_{m\times n}$, then both sides has (i,j) entry $xa_{ji} + yb_{ji}$.

Or if you would like to practice fancy ways to write entries, you can do this, where I repeatedly used the fact that transpose of 1 by 1 matrix is itself:

$$
\boldsymbol{e}_i^{\mathrm{T}}(xA+yB)^{\mathrm{T}}\boldsymbol{e}_j = \boldsymbol{e}_j^{\mathrm{T}}(xA+yB)\boldsymbol{e}_i = x\boldsymbol{e}_j^{\mathrm{T}}A\boldsymbol{e}_i + y\boldsymbol{e}_j^{\mathrm{T}}B\boldsymbol{e}_i = x\boldsymbol{e}_i^{\mathrm{T}}A^{\mathrm{T}}\boldsymbol{e}_j + y\boldsymbol{e}_i^{\mathrm{T}}B^{\mathrm{T}}\boldsymbol{e}_j = \boldsymbol{e}_i^{\mathrm{T}}(xA^{\mathrm{T}}+yB^{\mathrm{T}})\boldsymbol{e}_j.
$$

□

4.3 Gaussian Elimination via Matrix Multiplication

4.3.1 Elementary Matrices

Now it is time to review Gaussian eliminations. What are those row operations?

Consider the shear $E = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. We know it adds the second coordinate onto the first of the input vector. Now imagine applying E to a matrix on the left, EA. What would happen?

Well, say A in columns is $A = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$. Then $EA = E\begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix} = \begin{bmatrix} E\boldsymbol{a}_1 & \dots & E\boldsymbol{a}_n \end{bmatrix}$. So we are adding the second coordinate onto the first of EACH column vector. Effectively, we are adding the second row of A to the first row of A, i.e., $r_1 \to r_1 + r_2$. This is a row operation!

Remember how elementary operations preserve linear relations among the columns? That means exactly that these row operations are linear. So all of them are in fact matrix multiplications.

Definition 4.3.1. *The following matrices are called elementary matrices. (They corresponds to elementary row operations when we multiply them to the left of some matrix.) Here we assume $i \neq j$ are two indices.*

1. *A swap matrix is a matrix P_{ij} which has 1 on the (i, j) and (j, i) entry, 0 on the (i, i) and (j, j) entry, and otherwise identical to the identity matrix. For example, over $\mathbb{R}^7$,* $P_{25} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$.

2. *A scale matrix is a diagonal matrix D with all diagonal entries non-zero. (We don't want to multiply both sides of an equation by zero....)*

3. *A shear matrix is a matrix E_{ij}^k with $i \neq j$, whose (i, j) entry is k, and is otherwise identical to the identity matrix.*

You can check yourself that they corresponds exactly to elementar row operations. The correspondence is like this:

1. $P_{ij}A$ means applying the row operation $r_i \leftrightarrow r_j$ to A.

2. DA means applying the row operation $r_i \to d_i r_i$ to A for all i, where d_i is the i-th diagonal entry of D.

3. $E_{ij}^k A$ means applying the row operation $r_i \to r_i + k r_j$ to A.

All of these should be remembered. However, if you have trouble remember them, just think this: for any elementary matrix E, we have $E = EI$. So the look of E as a matrix is exactly how the identity matrix I would be transformed, if we apply the corresponding row operation.

Remark 4.3.2. *So what are we doing when we do Gaussian elimination? We started with augmented matrix $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$. When we perform a row operation, we are multiplying some elementary matrix E to the left of it. So we are doing $E\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} = \begin{bmatrix} EA & E\boldsymbol{b} \end{bmatrix}$, and we now have a new augmented matrix.*

Equivalently, consider the equations $A\boldsymbol{x} = \boldsymbol{b}$. We can also multiply E from the left to both sides of the equations. This gives $EA\boldsymbol{x} = E\boldsymbol{b}$.

So as you can see, doing a row operations on the augmented matrix is EXACTLY the same as simply multiplying a matrix to both sides of the equation.

Example 4.3.3. Immediately I know that $E_{ij}^x E_{ik}^y = E_{ik}^y E_{ij}^x$. Why? Because it is obvious if you think of them as row operations. Similarly, for different indices i, j, k, l, we have $E_{ij}^x P_{kl} = P_{kl} E_{ij}^x$, because the corresponding row operations don't even touch each other. And so on so forth.

(In general, parallel things commute. Disjoint things commute. Entangled things are more likely to fail to commute.)

You can also easily see that $(E_{ij}^k)^t = E_{ij}^{kt}, E_{ij}^x E_{ij}^y = E_{ij}^{x+y}$, which justify the exponent notation. We even have E_{ij}^k and E_{ij}^{-k} as inverse map of each other. (All by simply looking at the meaning as row operations.)

In this sense, many calculations about these elementary matrices can simply be done without any entry-wise calculations. Just think what would happen as row operations. ☺

Example 4.3.4. In the example above, we have spotted many commuting behavior. However, there are also many non-commuting behaviors. Consider $E_{12} = \begin{bmatrix}1&1&0\\0&1&0\\0&0&1\end{bmatrix}$ and $E_{23} = \begin{bmatrix}1&0&0\\0&1&1\\0&0&1\end{bmatrix}$.

A calculation would reveal that $E_{12}E_{23} = \begin{bmatrix}1&1&1\\0&1&1\\0&0&1\end{bmatrix}$, while $E_{23}E_{12} = \begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}$. Notice the difference in $(1,3)$ entry.

Essentially, we can think of it like this. Say $A = \begin{bmatrix}\boldsymbol{r}_1^{\mathrm{T}}\\\boldsymbol{r}_2^{\mathrm{T}}\\\boldsymbol{r}_3^{\mathrm{T}}\end{bmatrix}$. $E_{12}E_{23}$ means we first do E_{23} and add the third row to the second row, so now we have $\begin{bmatrix}\boldsymbol{r}_1^{\mathrm{T}}\\\boldsymbol{r}_2^{\mathrm{T}}+\boldsymbol{r}_3^{\mathrm{T}}\\\boldsymbol{r}_3^{\mathrm{T}}\end{bmatrix}$. Then we do E_{12} and add the second row to the first, and we have $\begin{bmatrix}\boldsymbol{r}_1^{\mathrm{T}}+\boldsymbol{r}_2^{\mathrm{T}}+\boldsymbol{r}_3^{\mathrm{T}}\\\boldsymbol{r}_2^{\mathrm{T}}+\boldsymbol{r}_3^{\mathrm{T}}\\\boldsymbol{r}_3^{\mathrm{T}}\end{bmatrix}$. Note that the original third row is eventually carried over to the first row.

In comparison, $E_{23}E_{12}$ means we do E_{12} first and add the second row to the first, and now we have $\begin{bmatrix}\boldsymbol{r}_1^{\mathrm{T}}+\boldsymbol{r}_2^{\mathrm{T}}\\\boldsymbol{r}_2^{\mathrm{T}}\\\boldsymbol{r}_3^{\mathrm{T}}\end{bmatrix}$. Then we do E_{23} and add the third row to the second, and now we have $\begin{bmatrix}\boldsymbol{r}_1^{\mathrm{T}}+\boldsymbol{r}_2^{\mathrm{T}}\\\boldsymbol{r}_2^{\mathrm{T}}+\boldsymbol{r}_3^{\mathrm{T}}\\\boldsymbol{r}_3^{\mathrm{T}}\end{bmatrix}$. As you can see, this order of operations means that the original third row never made it into the first row.

The difference in the $(1,3)$-entry between the matrices $\begin{bmatrix}1&1&1\\0&1&1\\0&0&1\end{bmatrix}$ and $\begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}$ means exactly this question: does the original third row make it into the first row or not? ☺

Now this is not the end of it. For an elementary matrix E, what if we do AE?

Consider AE_{12}^1 for 3 by 3 matrices. Note that $E_{12}^1 = \begin{bmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_1+\boldsymbol{e}_2 & \boldsymbol{e}_3\end{bmatrix}$. So if we write A in columns $A = \begin{bmatrix}\boldsymbol{a}_1 & \boldsymbol{a}_2 & \boldsymbol{a}_3\end{bmatrix}$, then we haven

$$AE_{12}^1 = A\begin{bmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_1+\boldsymbol{e}_2 & \boldsymbol{e}_3\end{bmatrix} = \begin{bmatrix}A\boldsymbol{e}_1 & A\boldsymbol{e}_1+A\boldsymbol{e}_2 & A\boldsymbol{e}_3\end{bmatrix} = \begin{bmatrix}\boldsymbol{a}_1 & \boldsymbol{a}_1+\boldsymbol{a}_2 & \boldsymbol{a}_3\end{bmatrix}.$$

So we are adding the first column to the second, i.e., $c_2 \to c_2 + c_1$! Remember the saying "row left column right"? Well congratulations, it just got a new meaning. For these elementary matrices, if applied to the left, then they are elementar row operations. If applied to the right, then they are in fact elementar column operations.

1. AP_{ij} means applying the column operation $c_i \leftrightarrow c_j$ to A.
2. AD means applying the column operation $c_i \to d_i c_i$ to A for all i, where d_i is the i-th diagonal entry of D.
3. AE_{ij}^k means applying the column operation $c_j \to c_j + kc_i$ to A.

Pay special attention to the shearings! For E_{ij}^k, as row operation it is $r_i \to r_i + kr_j$, where as column operations it is $c_j \to c_j + kc_i$, the meaning of the indices is reversed!

Remark 4.3.5. *There is more that is reversed. Consider the matrix multiplication of several elementary matrices, $E_1E_2\dots E_k$. Which operation happen first?*

As row operations, we are doing $E_1E_2\dots E_kA$, so actually E_k happens first, then E_{k-1}, and so on. But as column operations, we are doing $AE_1E_2\dots E_k$. So actually E_1 happens first, then E_2, and so on.

Here is an extra funny non-trivial thing to think about.

Example 4.3.6. Look at $\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$. Let us first do $r_2 \to r_1 + r_2$ and have $\begin{bmatrix}1 & 2 & 3\\ 5 & 7 & 9\end{bmatrix}$, and then do $c_3 \to c_3 + c_2 + c_1$ and have $\begin{bmatrix}1 & 2 & 6\\ 5 & 7 & 21\end{bmatrix}$.

What if we do $c_3 \to c_3 + c_2 + c_1$ first, and then do $r_2 \to r_1 + r_2$? Then we first get to $\begin{bmatrix}1 & 2 & 6\\ 4 & 5 & 15\end{bmatrix}$, and then we have $\begin{bmatrix}1 & 2 & 6\\ 5 & 7 & 21\end{bmatrix}$. We get to the same result.

A row op and a column op always commute. ☺

Proposition 4.3.7. *Given a matrix A, if we first do a row operation and then do a column operation, it is the same as first doing that column operation, then do that row operation.*

Proof. Say the row operation is represented by matrix E, and the column operation is represented by matrix F. Then we are saying $(EA)F = E(AF)$, which is true by associativity. □

4.3.2 Inverse Matrices

Proposition 4.3.8. *The inverse map of a bijective linear map is linear.*

Proof. Consider $f(f^{-1}(a\boldsymbol{v} + b\boldsymbol{w})) = a\boldsymbol{v} + b\boldsymbol{w} = af(f^{-1}(\boldsymbol{v})) + bf(f^{-1}(\boldsymbol{w})) = f(af^{-1}\boldsymbol{v} + bf^{-1}(\boldsymbol{w}))$. Now apply f^{-1} to both sides, and we have $f^{-1}(a\boldsymbol{v} + b\boldsymbol{w}) = af^{-1}\boldsymbol{v} + bf^{-1}(\boldsymbol{w})$ as desired. □

Definition 4.3.9. *We say a square matrix is* ***invertible*** *if it is bijective as a linear map. The matrix for its inverse map is denoted as A^{-1}, the* ***inverse*** *of A.*

In particular, a matrix is invertible if and only if it is bijective as linear maps. Since there is NO bijection between spaces of different dimension, only square matrices could have the possibility of being invertible. In fact, we already know the following when we talk about ranks.

Proposition 4.3.10. *Suppose A is a square matrix. Then TFAE*

1. *A is invertible (bijective).*
2. *A is injective.*
3. *A is surjective.*

These are enough to get us the following basic properties of matrices.

Proposition 4.3.11. *If A, B are square matrices (very important condition), then AB invertible implies that A, B are both invertible.*

Proof. If the composition AB is bijective, then B must be injective. (This is simply a property of any (not necessarily linear) maps.) But since B is square, this means that B is bijective.

Similarly, if the composition AB is bijective, then A must be surjective. (This is simply a property of any (not necessarily linear) maps.) But since A is square, this means that A is bijective. □

Corollary 4.3.12. *For square matrices A, B, if $AB = I$, then A, B are invertible and they are inverse of each other.*

Proof. I is bijective, so A, B are both invertible. Now apply A^{-1} to both sides from the left, we have $B = A^{-1}$.

Keep in mind that order of multiplications matter! If we were to apply A^{-1} to both sides from the right, we would have $ABA^{-1} = A^{-1}$, which would fail to tell us anything. □

About the order of multiplication, we have a very useful calculation law here.

Proposition 4.3.13. *If A, B are invertible (bijective), then AB is invertible. We have $(AB)^{-1} = B^{-1}A^{-1}$.*

Proof. If A, B are both bijective, then obviously their composition is also bijective.

$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$. So we are done. □

Note how the order of multiplication is reversed! This fact is true for all maps, not just linear ones. If you wear your sock and then wear your shoe, then you must take off your shoe and then take of your sock. The order is reversed. If you open a door and go out, then to undo it, you have to get in and then close the door. Again the order is reversed.

This is a very neat comparison with transpose. In fact, inverse and transpose play nicely with each other. Recall the following:

Proposition 4.3.14. *Suppose A is any matrix (maybe non-square). Then TFAE (short hand for "the followings are equivalent")*

1. *A is injective. ($\forall \boldsymbol{b}$, $A\boldsymbol{x} = \boldsymbol{b}$ has at most one solution.)*
2. *The columns of A are linearly independent.*

Proposition 4.3.15. *Suppose A is any matrix (maybe non-square). Then TFAE*

1. *A is surjective. ($\forall \boldsymbol{b}$, $A\boldsymbol{x} = \boldsymbol{b}$ has at least one solution.)*
2. *The rows of A are linearly independent.*

This immediately implies the following statement.

Corollary 4.3.16. *A is injective iff A^{T} is surjective, and A is surjective iff A^{T} is injective*

Proof. A is injective iff columns of A are linearly independent iff rows of A^{T} are linearly independent iff A^{T} is surjective.

The other one is very similar. □

Combining the two, we have the following.

Proposition 4.3.17. *A is invertible iff A^{T} is invertilbe, and $(A^{\mathrm{T}})^{-1} = (A^{-1})^{\mathrm{T}}$.*

Proof. $A^{\mathrm{T}}(A^{-1})^{\mathrm{T}} = (A^{-1}A)^{\mathrm{T}} = I^{\mathrm{T}} = I$. Yay. □

We are now at a very good place to talk about the law of cancellation. Suppose you have real numbers $x, y, z \in \mathbb{R}$, and you have an equation $xy = xz$. What would you do to simplify this? Surely you would "cancel" the x on both sides, and obtain $y = z$. This is called the law of cancellation. However, this is ONLY true if $x \neq 0$, and NOT true if $x = 0$.

Does matrix multiplications satisfy the law of cancellation? Consider these examples.

Example 4.3.18. Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. You can verify that $AA = O$. Yet we also have $AO = OA = O$. Thus we have $AA = AO$ and $AA = OA$. You can easily see that BOTH indicates that the law of cancellation FAILS for matrices.

In general, the term ***law of left cancellation*** refers to the phenomenon of $AB = AC$ implying $B = C$. For some A, this is true. For some other A, this is false.

Similarly, the term ***law of right cancellation*** refers to the phenomenon of $BA = CA$ implying $B = C$. For some A, this is true. For some other A, this is false. ☺

Proposition 4.3.19. *The law of left cancellation is true for A iff A as a linear map is injective. Dually, the law of right cancellation is true for A iff A as a linear map is surjective.*

Proof. Suppose the law of left cancellation is true for A. For any inputs $\boldsymbol{v}, \boldsymbol{w}$, if $A\boldsymbol{v} = A\boldsymbol{w}$, then applying left cancellation gives $\boldsymbol{v} = \boldsymbol{w}$. So A is injective.

Conversely, suppose that A is injective. Suppose $AB = AC$. Then for any input $\boldsymbol{v}$ for B and C, we would have $(AB)\boldsymbol{v} = (AC)\boldsymbol{v}$. Associativity then gives us $A(B\boldsymbol{v}) = A(C\boldsymbol{v})$. Now we use injectivity of A, which gives us $B\boldsymbol{v} = C\boldsymbol{v}$. So B and C give the same output for all inputs. Hence $B = C$. So the law of left cancellation is true for A.

So we have proven the statement about left cancellation. What about right cancellation? Well, observe that A has the law of left cancellation iff A^{T} has the law of right cancellation. At the same time, A is injective iff A^{T} is surjective. So we are done. ☐

As you can see, being injective, surjective or bijective will have a very tangible impact on our calculations with matrices.

Let us now see some examples of invertible matrices and transpose. Note that all matrices, even non-square matrices, have transpose. In contrast, even some square matrices has no inverse.

Example 4.3.20.

1. The inverse of the identity matrix is itself, obviously.
2. R_θ and $R_{-\theta}$ are inverse of each other. Funnily, they are also transpose of each other. Huh.
3. $I - \boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$, i.e., projections to a hyperplane with unit normal vector $\boldsymbol{u}$, is not invertible. Apply this to $\boldsymbol{u}$ and you get $\mathbf{0}$, so this is not injective.
4. Householder transformations are inverse of themselves. (Because they are reflections.) Calculate and see: $(I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}})^2 = I$. (Is the inverse also the transpose?)
5. In the case of rotations and reflections, note that these maps preserves orthogonality. They send orthogonal inputs to orthogonal outputs. Later we shall see that transpose and orthogonality are very closely related.
6. Say $D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$. Then it is invertible iff all d_i are non-zero, and its inverse in that case is $D^{-1} = \begin{bmatrix} d_1^{-1} & & \\ & \ddots & \\ & & d_n^{-1} \end{bmatrix}$. On the other hand, the transpose of D is just D itself.
7. Swaps are inverses of themselves. (This is in fact a special case of a Householder transformation.)
8. E_{ij}^k and E_{ij}^{-k} are inverse of each other. They are NOT transpose of each other. This is a very good place to look at the difference between transpose and inverse. E_{ij}^k as a row operation is $r_i \to r_i + kr_j$. Its transpose E_{ji}^k is $r_j \to r_j + kr_i$, whereas its inverse is E_{ij}^{-k}, which is $r_i \to r_i - kr_j$.

☺

In particular, all elementary matrices are invertible. As a result, we always have $A\boldsymbol{x} = \boldsymbol{b}$ if and only if $EA\boldsymbol{x} = E\boldsymbol{b}$ for any elementary matrix E. This explains right away why row operations fix solution set. And why don't we use column operations? Because it is hard to slide in E after A in this equation.

Now, if we are faced with a linear system $A\boldsymbol{x} = \boldsymbol{b}$, the most obvious reaction would be applying A^{-1} to both sides, and get $\boldsymbol{x} = A^{-1}\boldsymbol{b}$. But how to find the inverse? We already know how: Gaussian elimination.

Proposition 4.3.21. *Given an invertible $n \times n$ matrix A, then its RREF is I.*

Proof. Informally this is obvious. Given any system, say the augmented matrix is $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$, then with A bijective, we should have a unique solution. All variables solved, which can only come from $\begin{bmatrix} I & AnswerVector \end{bmatrix}$.

Formally, consider any system $A\boldsymbol{x} = \boldsymbol{b}$. We have no free variables because A is injective, and no contradiction because A is surjective. So all n pivots are in the n columns of A. This gives no choice except that A will be row reduced to I. □

Corollary 4.3.22. *Any invertible matrix is a product of elementary matrices.*

Proof. Consider row reducing A to I. Say we used elementary row operations with matrices $E_1, \ldots, E_k$. Then $E_k \ldots E_1 A = I$. So $A = E_1^{-1} \ldots E_k^{-1}$. □

Above corollary shows that, if B is any invertible matrix, then BA is essentially the same as applying some sequence of row operations on A. And similarly AB is the same as applying some sequence of column operations on A.

Corollary 4.3.23. *Given an invertible matrix A, consider the matrix $\begin{bmatrix} A & I \end{bmatrix}$. (This means we put A and I side to side to make a big rectangular matrix.) Then its RREF is $\begin{bmatrix} I & A^{-1} \end{bmatrix}$.*

Proof. Obviously $\begin{bmatrix} I & A^{-1} \end{bmatrix}$ is a RREF, so we only need to show that the two are row-equivalent, i.e., they can be transformed into each other using elementary row operations.

But if A is invertible, then A^{-1} is a sequence of row operations. Note that applying row operations to $\begin{bmatrix} A & I \end{bmatrix}$ is equivalent to applying the same row operations to A and to I simultaneously. So we have the following matrix multiplication calculation

$$A^{-1} \begin{bmatrix} A & I \end{bmatrix} = \begin{bmatrix} A^{-1}A & A^{-1}I \end{bmatrix} = \begin{bmatrix} I & A^{-1} \end{bmatrix}.$$

□

So this is how to find the inverse of a matrix.

Example 4.3.24. Consider the chicken rabbit matrix $\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$. To find its inverse, we consider the augmented $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix}$. Now to RREF, we do $r_2 \to r_2 - 2r_1$, $r_2 \to \frac{1}{2} r_2$, and $r_1 \to r_1 - r_2$. This gives $\begin{bmatrix} 1 & 0 & 2 & -\frac{1}{2} \\ 0 & 1 & -1 & \frac{1}{2} \end{bmatrix}$. So the inverse to our original matrix is $\begin{bmatrix} 2 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$. ☺

Proposition 4.3.25. *For an invertible 2×2 matrix, we have*

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

So we swap the diagonal entries, and negate the non-diagonal ones. In particular, 2×2 matrix here is invertible iff $ad - bc \neq 0$.

Proof. We have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & ad \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix}.$$

So if $ad - bc \neq 0$, we see that $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ multiply with $\frac{1}{ad-bc} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ would give us the identity matrix. So this matrix is invertible and we have found its inverse.

If $ad - bc = 0$, then the two column vectors are parallel. So A is not bijective. □

There is no easy formula for higher dimension cases. (Oh there is a formula alright, it is just too ugly to be useful.)

4.3.3 Inverses of Triangular Matrices

Here let us give pay special attention to triangular matrices.

Example 4.3.26. You can easily check the following.

$$\begin{bmatrix}1&0&0&0\\2&1&0&0\\3&0&1&0\\4&0&0&1\end{bmatrix}^{-1}=\begin{bmatrix}1&0&0&0\\-2&1&0&0\\-3&0&1&0\\-4&0&0&1\end{bmatrix}.$$

$$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&2&1&0\\0&3&0&1\end{bmatrix}^{-1}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&-2&1&0\\0&-3&0&1\end{bmatrix}.$$

So sometimes, we can just negate the non-diagonal entries??? However, this is not always the case. See the next example. ☺

Example 4.3.27. Let us calculate $\begin{bmatrix}1&0&0&0\\1&1&0&0\\1&1&1&0\\1&1&1&1\end{bmatrix}^{-1}$. Consider $\begin{bmatrix}1&0&0&0&1&0&0&0\\1&1&0&0&0&1&0&0\\1&1&1&0&0&0&1&0\\1&1&1&1&0&0&0&1\end{bmatrix}$. Now we do $r_2 \to r_2 - r_1, r_3 \to r_3 - r_1, r_4 \to r_4 - r_1$, and then we do $r_3 \to r_3 - r_2, r_4 \to r_4 - r_2$, and finally we do $r_4 \to r_4 - r_3$. We would have $\begin{bmatrix}1&0&0&0&1&0&0&0\\0&1&0&0&-1&1&0&0\\0&0&1&0&0&-1&1&0\\0&0&0&1&0&0&-1&1\end{bmatrix}$.

So $\begin{bmatrix}1&0&0&0\\1&1&0&0\\1&1&1&0\\1&1&1&1\end{bmatrix}^{-1}=\begin{bmatrix}1&0&0&0\\-1&1&0&0\\0&-1&1&0\\0&0&-1&1\end{bmatrix}$

As you can see, we only used upper rows to reduce lower rows, which is neat.

Similarly, we have $\begin{bmatrix}1&1&1&1\\0&1&1&1\\0&0&1&1\\0&0&0&1\end{bmatrix}^{-1}=\begin{bmatrix}1&-1&0&0\\0&1&-1&0\\0&0&1&-1\\0&0&0&1\end{bmatrix}$, and this process would only involve using lower rows to reduce upper rows.

Recall our pretty pattern $LU=\begin{bmatrix}1&1&1&1\\1&2&2&2\\1&2&3&3\\1&2&3&4\end{bmatrix}$. Its inverse would be $U^{-1}L^{-1}=\begin{bmatrix}2&-1&0&0\\-1&2&-1&0\\0&-1&2&-1\\0&0&-1&1\end{bmatrix}$.

(This last matrix is in fact related to heat distributions and wave distributions and so on. Its output coordinates are like $2x_i - (x_{i-1} + x_{i+1})$, so it is comparing each coordinate with the average of neighboring coordinates. Take heat distribution for example, if a place has higher temperature than its neighbors, then it will cool down.) ☺

What is the key difference?

Example 4.3.28. Consider $A=\begin{bmatrix}1&0&0&0\\2&1&0&0\\3&0&1&0\\4&0&0&1\end{bmatrix}$. What are the row operations this matrix represents?

To figure this out, we just need to know what are the row operations to go from I to $AI = A$.

$$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix} \to \begin{bmatrix}1&0&0&0\\2&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$$
$$\to \begin{bmatrix}1&0&0&0\\2&1&0&0\\3&0&1&0\\0&0&0&1\end{bmatrix}$$
$$\to \begin{bmatrix}1&0&0&0\\2&1&0&0\\3&0&1&0\\4&0&0&1\end{bmatrix}.$$

The three row operations are $E_{21}^2, E_{31}^3, E_{41}^4$, and they commute! And to find A^{-1}, it turns out we can simply negate all non-diagonal entries.

On the other hand, consider $\begin{bmatrix}1&1&1&1\\0&1&1&1\\0&0&1&1\\0&0&0&1\end{bmatrix}$. What are the row operations for this matrix?

$$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix} \to \begin{bmatrix}1&0&0&0\\1&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$$
$$\to \begin{bmatrix}1&0&0&0\\1&1&0&0\\1&1&1&0\\0&0&0&1\end{bmatrix}$$
$$\to \begin{bmatrix}1&0&0&0\\1&1&0&0\\1&1&1&0\\1&1&1&1\end{bmatrix}.$$

The three row operations are E_{21}, E_{32}, E_{43}, and they do not commute! And to find A^{-1}, there is no easy way. You CANNOT simply negate all non-diagonal entries. ☺

For triangular matrices, inverses are sometimes easy to find, sometimes hard. But the key is always about the row operations they represent. So first, let us device a standard way to decompose a triangular matrix into row operations.

Definition 4.3.29. *A triangular matrix (upper or lower) is* ***unit triangular*** *if all diagonal entries are 1.*

Example 4.3.30. Consider $A = \begin{bmatrix}1&&&\\2&1&&\\3&4&1&\\5&6&7&1\end{bmatrix}$. How can I write this into a series of elementary matrices?

$$\begin{bmatrix}1&&&\\2&1&&\\3&4&1&\\5&6&7&1\end{bmatrix} = \begin{bmatrix}1&&&\\2&1&&\\&&1&\\&&&1\end{bmatrix}\begin{bmatrix}1&&&\\&1&&\\3&&1&\\&&&1\end{bmatrix}\begin{bmatrix}1&&&\\&1&&\\&&1&\\5&&&1\end{bmatrix}\begin{bmatrix}1&&&\\&1&&\\&4&1&\\&&&1\end{bmatrix}\begin{bmatrix}1&&&\\&1&&\\&&1&\\&6&&1\end{bmatrix}\begin{bmatrix}1&&&\\&1&&\\&&1&\\&&7&1\end{bmatrix}.$$

We may call this a standard elementary decomposition of a unit lower triangular matrices. Each non-diagonal entry of A is exactly an operation, and the order of these row operations goes from lower right to upper left.

Also observe that all shearings here uses the upper row to change the lower row! ☺

Proposition 4.3.31. *An invertible matrix is unit upper triangular iff it can be written as a series of row operations made ONLY of shearings, and ONLY use lower rows to change upper rows. (No swaps or scaling involved, and no using upper rows to change lower rows.)*

Similarly, an invertible matrix is unit lower triangular iff it can be written as a series of row operations made ONLY of shearings, and ONLY use upper rows to change lower rows. (No swaps or scaling involved, and no using lower rows to change upper rows.)

Proof. Let us use unit upper triangular matrices for a change of flavor.

Given $U = \begin{bmatrix} 1 & a_{12} & \dots & a_{1n} \\ & \ddots & \ddots & \vdots \\ & & \ddots & a_{n-1,n} \\ & & & 1 \end{bmatrix}$, we aim to show that $U = E_1 \dots E_k$ for shearings $E_1, \dots, E_k$ that only uses lower rows to change upper rows. This means $U = E_1 \dots E_k I$. So the question is now this: starting from the identity matrix, can we find shearings $E_1, \dots, E_k$ that only uses lower rows to change upper rows, and reach U?

Well, this is actually quite easy. Start from the second second column and work your way to the right, and you can see that this is always possible. For example, to reach $\begin{bmatrix} 1 & 2 & 3 & 4 \\ & 1 & 2 & 3 \\ & & 1 & 2 \\ & & & 1 \end{bmatrix}$, we can simply do

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \xrightarrow{r_1 \to r_1 + 2r_2} \begin{bmatrix} 1 & 2 & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \xrightarrow[r_2 \to r_2 + 2r_3]{r_1 \to r_1 + 3r_3} \begin{bmatrix} 1 & 2 & 3 & \\ & 1 & 2 & \\ & & 1 & \\ & & & 1 \end{bmatrix} \xrightarrow[\substack{r_2 \to r_2 + 3r_4 \\ r_3 \to r_3 + 2r_4}]{r_1 \to r_1 + 4r_4} \begin{bmatrix} 1 & 2 & 3 & 4 \\ & 1 & 2 & 3 \\ & & 1 & 2 \\ & & & 1 \end{bmatrix}.$$

For lower unit triangular matrices, you can do the similar thing (or simply take transpose). □

Corollary 4.3.32. *All unit triangular matrices are invertible.*

How about non-unit triangular matrices?

Proposition 4.3.33. *A (lower or upper) trianglular matrix is invertible if and only if its diagonal entries are all non-zero.*

Proof. For upper triangular matrices, if all diagonal entries are non-zero, then we are already in REF and we see that we have full pivots. So it is invertible.

Conversely, suppose say the (i, i) entry is zero. (For example, consider $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 & 9 \\ 0 & 0 & 0 & 10 & 11 \\ 0 & 0 & 0 & 12 & 13 \\ 0 & 0 & 0 & 0 & 14 \end{bmatrix}$.)

Then the first i columns only uses $i - 1$ rows, so they could produce at most $i - 1$ pivots. (The first three columns in the example above contains only two pivots.) And the last $n - i$ columns obviously could produce at most $n - i$ pivots. (The last two columns in the example above contains only two pivots.) So we have at most $(i - 1) + (n - i) = n - 1$ pivots, which is less than full. So our matrix is NOT invertible.

For a lower triangular matrix, note that it is the transpose of an upper triangular matrix. □

So we have the following intuitions:

1. Think of a unit upper triangular matrix as a series of shearings using lower rows to reduce upper rows.
2. Think of a unit lower triangular matrix as a series of shearings using upper rows to reduce lower rows.
3. A non-unit triangular matrix is simply a unit triangular matrix plus some row scaling.
4. Can you figure out the corresponding understanding of unit triangular matrices as column operations?

In particular, here are some nice corollaries.

Corollary 4.3.34. *Products of unit upper (lower) triangular matrices are still unit upper (lower) triangular. The inverse of a unit upper (lower) triangular matrix is still unit upper (lower) triangular.*

Proof. Products of unit upper (lower) triangular matrices means we do all the corresponding row operations, and they are all shearings using the lower rows to change upper rows. So the result is still unit upper (lower) triangular.

The inverse to $r_i \to r_i + kr_j$ is $r_i \to r_i - kr_j$. In particular, if we are using lower rows to change upper rows (i.e., $i > j$), then the inverse is also using lower rows to change upper rows. So the inverse is still unit upper (lower) triangular. □

Corollary 4.3.35. *If a matrix is upper (lower) triangular and invertible, then its inverse is also upper (lower) triangular with diagonal entries inverted.*

Proof. Just also invert the row scaling as well. □

4.3.4 LU decomposition

We are returning to Gaussian elimination now with our new found toys and perspectives. What is Gaussian elimination?

Example 4.3.36. Suppose we have $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 5 \\ 4 & 6 & 8 \end{bmatrix}$. To do Gaussian elimination, first we shall attempt a top-down process where we use upper rows to reduce lower rows. In this case, we want $r_2 \to r_2 - 2r_1, r_3 \to r_3 - 4r_1$. This is the same as multiplying $L_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$ to the left of A. (Recall that the looks of L_1 is exactly what L_1 shall do to I, so you can quickly see that this is the right matrix.)

So we have $L_1A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 4 \end{bmatrix}$. Next we want to do $r_3 \to r_3 - 2r_2$, so we are multiplying $L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ and get $L_2L_1A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -2 \end{bmatrix}$. This is our REF, and obviously it is upper triangular. Let us call this U.

So we have $L_2L_1A = U$. Reorganize this, we see that $A = LU$ where $L = L_1^{-1}L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$ is lower triangular and U is upper triangular. ☺

If you recall, our idea of elimination is to first go top down, using upper rows to reduce lower rows as much as possible, until we reach REF. This means we are using lower triangular shearing matrices, and the result we get to, an REF, is an upper triangular thing.

Suppose we got lucky, and have no need to use row swaps. Then we have lower triangular shearings $L_1, \dots, L_k$, such that $L_k \dots L_1 A = U$ for some upper triangular U. Then in particular, let $L = L_1^{-1} \dots L_k^{-1}$, we see that $A = LU$ for a lower triangular L and an upper triangular U.

All in all, this L records how to row reduce A, and U is the resulting REF.

Definition 4.3.37. *We say a square matrix A has an LU decomposition if $A = LU$ for a lower triangular L and upper triangular U.*

In essence, LU decomposition is just the matrix way of writing Gaussian elimination. Given the LU decomposition, it is very easy to tell what steps should you do. You can really just read it out.

Example 4.3.38. Suppose we have $A = LU$ as $\begin{bmatrix}1&1&1\\2&3&5\\4&6&8\end{bmatrix} = \begin{bmatrix}1&0&0\\2&1&0\\4&2&1\end{bmatrix}\begin{bmatrix}1&1&1\\0&1&3\\0&0&-2\end{bmatrix}$.

When we row reduce, the new row operations E would act as $EA = (EL)U$. So you can just focus on reducing L to I, and when you have reduced L to I successfully, A would turn into the REF matrix U automatically.

Looking at L column by column (left to right). Then we should do

$$L = \begin{bmatrix}1&0&0\\2&1&0\\4&2&1\end{bmatrix} \xrightarrow[r_3 \to r_3 - 4r_1]{r_2 \to r_2 - 2r_1} \begin{bmatrix}1&0&0\\0&1&0\\0&2&1\end{bmatrix} \xrightarrow{r_3 \to r_3 - 2r_2} \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} = I.$$

Note that even without any thinking, I can literally read out the steps by reading the entries column by column (left to right).

Now we do the same thing to A, and we should reach U. Indeed, we have

$$A = \begin{bmatrix}1&1&1\\2&3&5\\4&6&8\end{bmatrix} \xrightarrow[r_3 \to r_3 - 4r_1]{r_2 \to r_2 - 2r_1} \begin{bmatrix}1&1&1\\0&1&3\\0&2&4\end{bmatrix} \xrightarrow{r_3 \to r_3 - 2r_2} \begin{bmatrix}1&1&1\\0&1&3\\0&0&-2\end{bmatrix} = U.$$

In conclution, simply read out the entries of L, and we should know how to do elimination on A. Conversely, given an elimination process from an invertible matrix A to its REF, then we can read out the unit lower triangular matrix L from the elimination process. ☺

The LU decomposition is among the MOST widely used matrix decompositions. This is not just because gaussian elimination is useful, it is also because triangular matrices are super nice. For an $n \times n$ triangular matrix A, $A\boldsymbol{x} = \boldsymbol{b}$ can be solved in about $\frac{1}{2}n^2$ calculations, while in general solving $A\boldsymbol{x} = \boldsymbol{b}$ for a generic $n \times n$ matrix A takes about $\frac{1}{3}n^3$ calculations.

Remark 4.3.39. *Suppose we are going from A to its RREF. How many calculations do we need? Here if an entry changed once we count it as one calculation, and I am ignoring lower degree terms. And swapping does not count as calculations because no value is changed, only the storage location is swapped.*

If A is triangular, say lower triangular, then each entry below the diagonal would change exactly once during elimination. So we need a total of $(n-1) + (n-2) + \cdots + 1$ calculations. Then we scale the diagonal entries to 1, which takes at most n calculations. So we need a total of $n + \cdots + 1 = \frac{1}{2}n^2 + \frac{1}{2}n$ calculations.

If A is generic, since we ignored swapping, we can assume that we can just go forward elimination and then back. To use the first row to reduce all rows below, generically we need $n(n-1)$ calculations. Next, we use the second row to reduce below. Note that the first column is empty now, so we only need $(n-1)(n-2)$ calculations. And so on so forth. This takes $n(n-1) + (n-1)(n-2) + \cdots + 2 \times 1 + 1 \times 0$ calculations. Now we are left with an upper triangular matrix, which takes $n + \cdots + 1$ calculations. So combine the two, we need $n^2 + \cdots + 1^2 = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n$ calculations.

So instead of solving $A\boldsymbol{x} = \boldsymbol{b}$, suppose we already know that $A = LU$. Then first we can solve $\boldsymbol{y}$ from $L\boldsymbol{y} = \boldsymbol{b}$, and then we can solve $\boldsymbol{x}$ from $U\boldsymbol{x} = \boldsymbol{y}$. This would take about n^2 steps instead of about $\frac{1}{3}n^3$ steps. When n is large, this makes a huge difference.

Remark 4.3.40. *However, keep in mind of this: even though solving $LU\boldsymbol{x} = \boldsymbol{b}$ is easy, finding $A = LU$ is not. In fact, finding $A = LU$ is exactly the Gaussian elimination, so it takes about $\frac{1}{3}n^3$ calculations. So what is the point after all?*

The point is that in practice, say we are doing a CT scan, then the A is fixed, while the scan result $\boldsymbol{b}$ is unknown before hand. Suppose we have many future patients, then solving $A\boldsymbol{x} = \boldsymbol{b}_1, A\boldsymbol{x} = \boldsymbol{b}_2, \ldots, A\boldsymbol{x} = \boldsymbol{b}_r$ one by one would takes $\frac{r}{3}n^3$ calculations. On the other hand, if we do $A = LU$ before hand, then we only need $\frac{1}{3}n^3 + \frac{r}{2}n^2$ calculations, which saves time. Or simply put, LU decomposition is basically doing most of Gaussian elimination beforehand, without even knowing what $\boldsymbol{b}$ is.

Now, how about finding A^{-1} and then simply apply A^{-1} to all these $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_r$? Well that works. But how would you find the inverse? You would use Gaussian elimination, which in essence is just $A = LU$. So finding $A = LU$ beforehand is basically the same as finding A^{-1} before hand, except that LU decomposition can be done even for non-invertible matrices.

Now, LU decomposition might not be unique. For example, the zero matrix multiplying any upper triangular matrix would give the same answer. However, the uniqueness do exist under some special perspective.

Definition 4.3.41. *The LDU decomposition of A is $A = LDU$ where L is UNIT lower triangular, U is UNIT upper triangular, and D is diagonal.*

Intuitively, L is the forward elimination that gives you the REF. Then we scale pivots to 1, which is D, scaling of rows. Finally U represent the backward process that gives RREF.

Theorem 4.3.42. *If A is invertible and has LU decomposition, then the LDU decomposition of A is unique.*

Proof. Suppose $L_1D_1U_1 = A = L_2D_2U_2$ are two LDU decompositions. I aim to show that $L_1 = L_2, U_1 = U_2, D_1 = D_2$. Note that since A is invertible, all matrices involved here are invertible.

The key is to manipulate the equation into $L_2^{-1}L_1D_1 = D_2U_2U_1^{-1}$. Now the left hand side is lower triangular, but the right hand side is upper triangular! How can a matrix be both lower triangular and upper triangular? Well it has to be diagonal, of course.

Furthermore, $L_2^{-1}L_1$ is in fact unit lower triangular, with ones on the diagonal. Now $L_2^{-1}L_1D_1$ means multiplying columns of $L_2^{-1}L_1$ with corresponding entries of D_1. So diagonal entries of $L_2^{-1}L_1D_1$ will just be the diagonal entries of D_1. But we already know that $L_2^{-1}L_1D_1$ is diagonal! So we must have $L_2^{-1}L_1D_1 = D_1$. Since D_1 is invertible, we can simplify this to $L_1 = L_2$.

Similarly we have $U_1 = U_2$ on the right hand side. Finally, $L_2^{-1}L_1D_1 = D_2U_2U_1^{-1}$ is now $D_1 = D_2$. Done. ☐

Fun proof, yes?

Remark 4.3.43. *Why LU decomposition? Why not UL decomposition? This is largely a matter of convention.*

Suppose $A = LU$. Then $A^{-1} = U^{-1}L^{-1}$, so you see that an LU decomposition of A corresponds to an UL decomposition for A^{-1}. So, if you want to do UL decomposition for A, you are essentially doing LU decomposition to A^{-1}.

4.3.5 Diagonally Dominant Matrices

Now, is it possible to just recognize an invertible matrix by sight? Most of the time, no. However, we have seen some special cases, like diagonal matrices, triangular matrices and so on. These we can just tell by sight. Here is another kind.

Example 4.3.44. Imagine that a matrix has big entries on the diagonal, and tiny entries off the diagonal. Must it be invertible?

In 2×2 case, we have $\begin{bmatrix} \text{big} & \text{tiny} \\ \text{tiny} & \text{big} \end{bmatrix}$. It is quite obvious that the two columns are NOT parallel, hence the columns are linearly independent. So our square matrix must be invertible.

What about the 3×3 case? We have $\begin{bmatrix} \text{big} & \text{tiny} & \text{tiny} \\ \text{tiny} & \text{big} & \text{tiny} \\ \text{tiny} & \text{tiny} & \text{big} \end{bmatrix}$. To be invertible, it means it shall have full pivots (on the diagonal) after Gaussian elimination.

Imagine the first step of such an elimination. We would need to substract $\frac{\text{tiny}}{\text{big}}$ times the first row from rows below. This shall yeild something like

$$\begin{bmatrix} \text{big} & \text{tiny} & \text{tiny} \\ 0 & \text{big} - \frac{\text{tiny}}{\text{big}}\text{tiny} & \text{tiny} - \frac{\text{tiny}}{\text{big}}\text{tiny} \\ 0 & \text{tiny} - \frac{\text{tiny}}{\text{big}}\text{tiny} & \text{big} - \frac{\text{tiny}}{\text{big}}\text{tiny} \end{bmatrix}.$$

But as you can imagine, $\frac{\text{tiny}}{\text{big}}\text{tiny}$ is probably even more tiny. So it shall have little effect on the entries. We should expect big entries to remain big, and tiny entries to remain tiny. Hence we have something like

$$\begin{bmatrix} \text{big} & \text{tiny} & \text{tiny} \\ 0 & \text{big} & \text{tiny} \\ 0 & \text{tiny} & \text{big} \end{bmatrix}.$$

The next elimination step would similarly create

$$\begin{bmatrix} \text{big} & \text{tiny} & \text{tiny} \\ 0 & \text{big} & \text{tiny} \\ 0 & 0 & \text{big} \end{bmatrix}.$$

So we see that we shall have full pivots. Such a matrix must be invertible. This idea can easily be generalized to larger matrices (but the precise definition of "big" here is unfortunately quite blurry). ☺

Definition 4.3.45. *A square matrix $A = (a_{ij})_{n\times n}$ is (row) diagonally dominant if in each row, the absolute value of the diagonal entry is larger than the sum of absolute values of all other entries, i.e., $|a_{ii}| > \sum_{j\neq i} |a_{ij}|$ for each i. (We can similarly define column diagonally dominant matrices.)*

Intuitively, a diagonally dominant matrix is basically a matrix that is "close to being diagonal".

Proposition 4.3.46. *A (row or column) diagonally dominant matrix is invertible.*

Before we prove it, let us first figure out when shall we see these matrices.

Example 4.3.47. Diagonally dominant matrix usually apear on things that are "mostly stable".

Let us predict the weather. Each hour, the weather could be sunny, cloudy or rainy. Furthermore, given any hour, the weather next hour is likely to be the same.

Suppose we know the following:

1. If an hour is sunny, then the next hour has 90% chance to still be sunny, and 10% chance to be cloudy.
2. If an hour is cloudy, then the next hour has 60% chance to be cloudy, and 20% chance to be rainy.
3. If an hour is rainy, then the next hour has 50% chance to be rainy, 30% chance to be cloudy, and 20% chance to be rainy.

A probability chart might look like this:

	Sunny Now	Cloudy Now	Rainy Now
Sunny Next	0.9	0.2	0.2
Cloudy Next	0.1	0.6	0.3
Rainy Next	0	0.2	0.5

.

Note that things are mostly stable because one hour is very likely to have the same weather as the next hour. We also have a natural matrix here. What linear map would this matrix represent?

Suppose a given hour has x, y, z chance to be sunny, cloudy or rainy respectively. We may write this info as a vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Then the next hour, we shall have $x\begin{bmatrix} 0.9 \\ 0.1 \\ 0 \end{bmatrix} + y\begin{bmatrix} 0.2 \\ 0.6 \\ 0.2 \end{bmatrix} + z\begin{bmatrix} 0.2 \\ 0.3 \\ 0.5 \end{bmatrix}$ chance of being sunny, cloudy or rainy.

To put this together, we see that the matrix $A = \begin{bmatrix} 0.9 & 0.2 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0 & 0.2 & 0.5 \end{bmatrix}$ is the transition process. It sends the probability distribution of an hour $\boldsymbol{v}$ to the probability distribution of the next hour $A\boldsymbol{v}$. In a sense, A helps us predict the future.

Note that A is (row) diagonally dominant. The diagonal entries specifically refers to the chances of a sunny hour remains sunny, a cloudy hour remains cloudy, and a rainy hour remains rainy. And these situations dominates. Essentially, this is because the weather usually likes to remain unchanged between two consecutive hours. As with most continuous process.

In situations such as heat diffusions, wave transmissions, probability evolutions and so on, you will always see such diagonally dominant matrices.

So what does it mean that these matrices can be inverted? Well, you can watch a film "Tenet" by Noland. When he creates a world where entropy is reversed, bullets go back into guns, cars drive backwards, flames freeze things, he is in a sense using the fact that these evolution matrices are invertible. ☺

How to show that something is invertible? Well, you do Gaussian elimination and see if you get full pivots, i.e., see if the RREF is the identity matrix. This can be rephrased into the following statement.

Lemma 4.3.48. *A square matrix A is invertible iff $\boldsymbol{x} = \boldsymbol{0}$ is the only solution to $A\boldsymbol{x} = \boldsymbol{0}$.*

(All outputs have unique pre-images iff the zero output has a unique pre-image.)

Proof. If A is invertible, then it is injective. So $A\boldsymbol{x} = \boldsymbol{0}$ has at most one solution, and therefore it has to be $\boldsymbol{0}$.

Now suppose $A\boldsymbol{x} = \boldsymbol{0}$ has a unique solution $\boldsymbol{x} = \boldsymbol{0}$. This means the augmented matrix $\begin{bmatrix} A & \boldsymbol{0} \end{bmatrix}$ has an RREF of $\begin{bmatrix} I & \boldsymbol{0} \end{bmatrix}$. So A has full pivots, and hence A is invertible. ☐

(Some textbook might WRONGLY tell you to look at the determinant of the matrix. However, finding determinant is almost always slower than doing elimination. So in practice you should NOT do that.)

Example 4.3.49. Now why are these things invertible?

Consider $\begin{bmatrix} 0.9 & 0.2 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0 & 0.2 & 0.5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.9x + 0.2y + 0.2z \\ 0.1x + 0.6y + 0.3z \\ 0.2y + 0.5z \end{bmatrix}$. Being diagonally dominant means each output coordinate depends the MOST on the corresponding input coordinates, and less on other input coordinates.

To check for invertibility, let us consider how to make the output zero.

The first output coordinate is $0.9x + 0.2y + 0.2z$. Note that, if x has the largest non-zero absolute value among x, y, z ($|x| \geq |y|$ and $|x| \geq |z|$), then this cannot be zero. y, z are already small, and they got multiplied with non-diagonal entries, which are also small. x is already large, and it got multiplied with the diagonal entry, which is also large.

In particular, any solution to $\begin{bmatrix} 0.9 & 0.2 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0 & 0.2 & 0.5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \boldsymbol{0}$ must NOT have x with the largest non-zero absolute value among x, y, z.

But similarly, the solution cannot have y with the largest non-zero absolute value, and cannot have z with the largest non-zero absolute value. We are left with one choice, $x = y = z = 0$. ☺

Proof of the Propositions. Suppose A is row diagonally dominant, and $\boldsymbol{x} \neq \boldsymbol{0}$. Say $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ where x_i has the largest absolute value among all coordinates. Then let us consider the i-th coordinate of the output.

If $A = (a_{ij})_{n\times n}$, then the i-th coordinate of the output is

$$|\sum_j a_{ij}x_j| \geq |a_{ii}||x_i| - \sum_{j\neq i}|a_{ij}||x_j| \geq |a_{ii}||x_i| - \sum_{j\neq i}|a_{ij}||x_i| = (|a_{ii}| - \sum_{j\neq i}|a_{ij}|)|x_i| > 0.$$

So $A\boldsymbol{x} \neq 0$. We cannot combine columns of A non-trivially to get $\mathbf{0}$. So A has linearly independent column, and A is injective and hence bijective.

Finally, if A is column diagonally dominant, then A^{T} is row diagonally dominant, so we are done. □

If you check out the Chinese textbook by Liang, Tian and myself, there is also a cool example on heat distribution on wires about diagonally dominant matrix.

4.4 Block Matrices

4.4.1 Meaning of Blocks

There are two major techniques when it comes to matrices. The first one is decompositions, and we have already seen it in the form of LU decompositions or LDU decompositions. Given a complicated map, we decompose it into the composition of a chain of simpler maps.

The second one is to utilize block matrices, which is what we shall do here. We start with a mystery.

Example 4.4.1. Consider the transformation on $\mathbb{R}^3$ that rotate the xy-plane by 45 degree counterclockwise and stretch the z-axis by a factor of 2. So by looking at images of $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$, we see that the matrix is $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

This is NOT a diagonal matrix. However, it is a ***block diagonal matrix***. If you divide it into four blocks as $\left[\begin{array}{cc|c} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \hline 0 & 0 & 2 \end{array}\right]$, you see that the non-diagonal blocks are all zero.

Now if you take inverse, we have $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$. Hey, this is just as if we are taking inverse of each diagonal block!

If we take the inverse of a diagonal matrix, we just invert each diagonal entry. It seems that, if we take the inverse of a block diagonal matrix, we can just invert each diagonal block.

In fact, the two blocks here behaves "independently". The upper left block only uses the first two coordinates to change the first two coordinates (rotating the xy-plane), while the lower right block only uses the third coordinate to change the third coordinate (stretching the z-axis). No wonder they got inverted independently. ☺

As you can see, block matrices are NOT just a formality in grouping entries. It in fact has meanings. Each individual block is in fact a linear "submap" in some sense.

Example 4.4.2. Consider a map sending foods to nutrients. Say we have foods: apples, bananas, meat. And we have nutrients: fibers, proteins, suger. Then this map is a matrix A, such that if we have $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ apples, bananas and meat, then we have $A\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ fibers, proteins and suger. Obviously A is a 3 by 3 matrix.

Now consider the block form $A = \left[\begin{array}{cc|c} a & b & c \\ \hline d & e & f \\ g & h & i \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where A_{ij} represent the corresponding blocks.

What does A_{11} do? It sends fruits to the low calory nutrients they contain. What does A_{21} do? It send fruits to the high calory nutrients they contain. What does A_{12} do? It sends meat to the low calory nutrients it contains. What does A_{22} do? It send meat to the high calory nutrients it contains.

$$\begin{tikzcd}
\text{fruits} \arrow[r, "A_{11}"] \arrow[dr, "A_{21}"] & \text{low calory} \\
\text{meat} \arrow[r, "A_{22}"] \arrow[ur, "A_{12}"] & \text{high calory}
\end{tikzcd}.$$

And what is A? A as a linear map is simply the collection of these four linear maps. ☺

Intuitively, when we have a block matrix, we are grouping input coordinates and output coordinates. The block A_{ij} records how the j-th group of inputing coordinates effect the i-th group of outputing coordinates.

Example 4.4.3. Consider $\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 1 & 2 \\ \hline 0 & 0 & 1 \end{array}\right]$. Note that the lower left block is zero. This means the first two input coodinates does NOT effect the third output coordiante.

Indeed we have $\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 1 & 2 \\ \hline 0 & 0 & 1 \end{array}\right] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y+z \\ x+y+2z \\ z \end{bmatrix}$.

$$\begin{tikzcd}
\mathbb{R}^2 \arrow[r, "A_{11}"] & \mathbb{R}^2 \\
\mathbb{R} \arrow[r, "A_{22}"] \arrow[ur, "A_{12}"] & \mathbb{R}
\end{tikzcd}.$$

This is a ***block upper triangular matrix***.

In particular, block diagonal means each groups of coordinates only effect themselves. In particular, instead of one system, it is more like many separate independent systems, one for each diagonal block. Here is a picture for $\left[\begin{array}{cc|c} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \hline 0 & 0 & 2 \end{array}\right]$.

$$\begin{tikzcd}
\mathbb{R}^2 \arrow[r, "A_{11}"] & \mathbb{R}^2 \\
\mathbb{R} \arrow[r, "A_{22}"] & \mathbb{R}
\end{tikzcd}.$$

☺

Definition 4.4.4. *A* ***block diagonal matrix*** *is a matrix whose block form is* $\begin{bmatrix} A_1 & & \\ & \ddots & \\ & & A_k \end{bmatrix}$ *for square matrices* $A_1, \dots, A_k$.

A ***block upper triangular matrix*** *is a matrix whose block form is* $\begin{bmatrix} A_1 & * & * \\ & \ddots & * \\ & & A_k \end{bmatrix}$ *for square matrices* $A_1, \dots, A_k$. *One can similarly define a* ***block lower triangular matrix***.

Remark 4.4.5. *Note that we usually require the diagonal blocks to be square. For example,* $\begin{bmatrix}1&0&0\\2&0&0\\0&3&4\end{bmatrix}$ *is NOT considered a block diagonal matrix or a block upper triangular matrix.*

We want nice formula such as $\begin{bmatrix}A_1 & * & * \\ & \ddots & * \\ & & A_k\end{bmatrix}^{-1} = \begin{bmatrix}A_1^{-1} & * & * \\ & \ddots & * \\ & & A_k^{-1}\end{bmatrix}$*, so we want these blocks to be square.*

We are now ready to do some calculation to make these ideas rigorous.

Proposition 4.4.6. $\begin{bmatrix}A\\B\end{bmatrix}\boldsymbol{x} = \begin{bmatrix}A\boldsymbol{x}\\B\boldsymbol{x}\end{bmatrix}$ *given that A, B has as much columns as coordinates of $\boldsymbol{x}$.*

Proof. Write A, B in row vectors and this is trivial. Here is a graph if A is $m_1 \times n$ and B is $m_2 \times n$.

$$\begin{tikzcd}
\mathbb{R}^n \arrow[r, "A"] \arrow[dr, "B"'] & \mathbb{R}^{m_1} \\
& \mathbb{R}^{m_2}
\end{tikzcd}.$$

∎

Proposition 4.4.7. $\begin{bmatrix}A & B\end{bmatrix}\begin{bmatrix}\boldsymbol{x}\\\boldsymbol{y}\end{bmatrix} = A\boldsymbol{x} + B\boldsymbol{y}$ *given that A has as much columns as coordinates of $\boldsymbol{x}$, and B has as much columns as coordinates of $\boldsymbol{y}$.*

Proof. Write A, B in column vectors and this is trivial. Here is a graph if A is $m \times n_1$ and B is $m \times n_2$.

$$\begin{tikzcd}
\mathbb{R}^{n_1} \arrow[r, "A"] & \mathbb{R}^m \\
\mathbb{R}^{n_2} \arrow[ur, "B"'] &
\end{tikzcd}.$$

∎

Lemma 4.4.8. *For matrices (in block form) $A = \begin{bmatrix}A_1 & A_2\end{bmatrix}$ and $B = \begin{bmatrix}B_1\\B_2\end{bmatrix}$, where A_1 has the same number of columns as the number of rows of B_1, and A_2 has the same number of columns as the number of rows of B_2. Then $AB = \begin{bmatrix}A_1 & A_2\end{bmatrix}\begin{bmatrix}B_1\\B_2\end{bmatrix} = A_1B_1 + A_2B_2$. (Just like how a horizontal vector acts on a vertical vector.)*

Proof. For any input vector $\boldsymbol{v}$, we have

$$AB\boldsymbol{v} = \begin{bmatrix}A_1 & A_2\end{bmatrix}\begin{bmatrix}B_1\\B_2\end{bmatrix}\boldsymbol{v} = \begin{bmatrix}A_1 & A_2\end{bmatrix}\begin{bmatrix}B_1\boldsymbol{v}\\B_2\boldsymbol{v}\end{bmatrix} = A_1B_1\boldsymbol{v} + A_2B_2\boldsymbol{v}$$

So in short, $AB\boldsymbol{v} = (A_1B_1 + A_2B_2)\boldsymbol{v}$.

Here is a nice graph to see it. From the domain $\mathbb{R}^n$ of AB to the codomain $\mathbb{R}^m$ of AB, there are two "routes", one is via A_1B_1, and the other is via A_2B_2. So in total we just have $A_1B_1 + A_2B_2$.

$$\begin{tikzcd}
\mathbb{R}^n \arrow[r, bend left, "B_1"] \arrow[rrr, bend left, "B_2"] & & \mathbb{R}^a \arrow[rrrr, bend right, "A_1"'] & \oplus & \mathbb{R}^b \arrow[rr, bend right, "A_2"'] & & \mathbb{R}^m
\end{tikzcd}$$

∎

Remark 4.4.9. *Compare this with the formula of* $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = \sum v_i w_i$, *and* $\begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix} \begin{bmatrix} \boldsymbol{b}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{b}_n^{\mathrm{T}} \end{bmatrix} = \sum \boldsymbol{a}_i \boldsymbol{b}_i^{\mathrm{T}}$.

Lemma 4.4.10. *For matrices (in block form)* $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$ *and* $B = \begin{bmatrix} B_1 & B_2 \end{bmatrix}$, *where* A_1, A_2 *has the same number of columns as the number of rows of* B_1, B_2. *Then* $AB = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \end{bmatrix} = \begin{bmatrix} A_1B_1 & A_1B_2 \\ A_2B_1 & A_2B_2 \end{bmatrix}$. *(Just like how a column vector acts on a row vector.)*

Proof. For any input vector $\begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix}$, we have

$$\begin{aligned}
& AB \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix} \\
&= \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix} \\
&= \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} (B_1\boldsymbol{x} + B_2\boldsymbol{y}) \\
&= \begin{bmatrix} A_1B_1\boldsymbol{x} + A_1B_2\boldsymbol{y} \\ A_2B_1\boldsymbol{x} + A_2B_2\boldsymbol{y} \end{bmatrix} \\
&= \begin{bmatrix} \begin{bmatrix} A_1B_1 & A_1B_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix} \\ \begin{bmatrix} A_2B_1 & A_2B_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix} \end{bmatrix} \\
&= \begin{bmatrix} A_1B_1 & A_1B_2 \\ A_2B_1 & A_2B_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{bmatrix}.
\end{aligned}$$

Here is a nice graph to see it. Say the domain of AB is $\mathbb{R}^{n_1+n_2}$, and codomain is $\mathbb{R}^{m_1+m_2}$, and the codomain of B and domain of A is $\mathbb{R}^d$. Then the only route from $\mathbb{R}^{n_i}$ to $\mathbb{R}^{m_j}$ is A_jB_i.

$$\begin{tikzcd}
\mathbb{R}^{n_1} \arrow[dr, "B_1"] & & \mathbb{R}^{m_1} \\
& \mathbb{R}^d \arrow[ur, "A_1"] \arrow[dr, "A_2"] & \\
\mathbb{R}^{n_2} \arrow[ur, "B_2"] & & \mathbb{R}^{m_2}
\end{tikzcd}.$$

□

You can see the trend here. You can simply pretend that these blocks are "entries", and just do it as if you are doing matrix multiplications. You'll be just fine.

(But be careful of the order of multiplication. Whatever is originally on the left, it will end up on the left. E.g. $\begin{bmatrix} A_1 & A_2 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = A_1B_1 + A_2B_2$ is correct, while $\begin{bmatrix} A_1 & A_2 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = B_1A_1 + B_2A_2$ is WRONG.)

We merely do a simple case here, since it contains all the crucial ideas for a generalization.

Proposition 4.4.11. *Sometimes we can divide a matrix into tiny blocks. Then matrix multiplications can be done block-wise, as if the blocks are just entries. For example, if* $A = \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix}$ *and* $B = \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$, *where* A_1 *is* m_1 *by* n_1, A_2 *is* m_1 *by* n_2, A_3 *is* m_2 *by* n_1, A_4 *is* m_2 *by* n_2, B_1 *is* n_1 *by* r_1, B_2 *is* n_1 *by* r_2, B_3 *is* n_2 *by* r_1, B_4 *is* n_2 *by* r_2.

Then $AB = \begin{bmatrix} A_1B_1 + A_2B_3 & A_1B_2 + A_2B_4 \\ A_3B_1 + A_4B_3 & A_3B_2 + A_4B_4 \end{bmatrix}$. *Just as you would expect from doing multiplications of 2 by 2 matrices.*

Essentially we can divide A and B into as many blocks as we want, and multiply A and B by pretending that each block is just some number, as long as all block multiplications involved are well-defined.

Proof. Think of A as two column blocks and B as two row blocks, we have

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1 \\ A_3 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \end{bmatrix} + \begin{bmatrix} A_2 \\ A_4 \end{bmatrix} \begin{bmatrix} B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1B_1 + A_2B_3 & A_1B_2 + A_2B_4 \\ A_3B_1 + A_4B_3 & A_3B_2 + A_4B_4 \end{bmatrix}.$$

A picture looks like this:

$$\begin{tikzcd}
\mathbb{R}^{n_1} \arrow[r, "B_1"] \arrow[dr, "B_3"] & \mathbb{R}^{d_1} \arrow[r, "A_1"] \arrow[dr, "A_3"] & \mathbb{R}^{m_1} \\
\mathbb{R}^{n_2} \arrow[r, "B_4"'] \arrow[ur, "B_2"] & \mathbb{R}^{d_2} \arrow[r, "A_4"'] \arrow[ur, "A_2"] & \mathbb{R}^{m_2}
\end{tikzcd}.$$

As you can see from above, to move from $\mathbb{R}^{n_1}$ to $\mathbb{R}^{m_1}$, you can do A_1B_1 or A_2B_3. So the corresponding upper left block after the multiplication is $A_1B_1 + A_2B_3$. ☐

As you can imagine, this is probably NOT going to make computation easier. However, it will make certain special case easier. For example you can have the following:

1. $\begin{bmatrix} A_1 & O \\ O & A_2 \end{bmatrix} \begin{bmatrix} B_1 & O \\ O & B_2 \end{bmatrix} = \begin{bmatrix} A_1B_1 & O \\ O & A_2B_2 \end{bmatrix}$. Here O means a block of all zero entries.
2. When invertible, $\begin{bmatrix} A & O \\ O & B \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & O \\ O & B^{-1} \end{bmatrix}$.
3. Products of block triangular matrices are block trianguler, and the diagonal blocks of the product is the product of corresponding diagonal blocks.

We end this section with an super mysterious special scenario, where there is no block in sight, yet the essense of a block matrix is still there.

Example 4.4.12. Sometimes there are "hidden blocks".

Consider the matrix $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 5 & 0 & 3 \end{bmatrix}$. You can check that its inverse is $\begin{bmatrix} 3 & 0 & -1 \\ 0 & \frac{1}{2} & 0 \\ -5 & 0 & 2 \end{bmatrix}$. Curiously, you may also check that $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ and $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ are inverse of each other. Hey! It is as if in the original matrix $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 5 & 0 & 3 \end{bmatrix}$, the four corner entries form a "secret" block and the middle entry form the other block, and we are block diagonal!

How can this be? Recall the REASON behind the behavior of block diagonal matrices. It is because groups of coordinates are independent of each other. Now consider $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 5 & 0 & 3 \end{bmatrix}$, and you can see that the first coordinate and the third coordinate will NOT effect the second coordinate, and the second coordinate will NOT effect the first and third coordinate. So it indeed have the desired behavior. It is "secretly" block diagonal. ☺

4.4.2 Block Elimination and Block Inverse

Recall that elementary row operations are essentialy multiplying matrices from the left. Now sometimes we get lazy, and we will try to do MANY operations together, and do a BLOCK row operation.

Example 4.4.13. Consider $\begin{bmatrix} O & I_a & O \\ I_a & O & O \\ O & O & I_b \end{bmatrix}$. When acting on matrices with $2a+b$ rows from the left, this matrix will swap the first a rows with next a rows. In particular, $\begin{bmatrix} O & I_a & O \\ I_a & O & O \\ O & O & I_b \end{bmatrix}\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} B \\ A \\ C \end{bmatrix}$ where A, B have a rows each, and C has b rows. ☺

Example 4.4.14. Consider $\begin{bmatrix} I_a & O \\ X & I_b \end{bmatrix}$ and $\begin{bmatrix} A \\ B \end{bmatrix}$ where A has a rows, B has b rows and X is any $b \times a$ matrix. We have $\begin{bmatrix} I_a & O \\ X & I_b \end{bmatrix}\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} A \\ B+XA \end{bmatrix}$. We are adding X times the first block row to the second block row. ☺

Example 4.4.15. Consider $\begin{bmatrix} X & O \\ O & Y \end{bmatrix}\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} XA \\ YB \end{bmatrix}$. This is block row scaling. ☺

Similarly, if we apply block swapping matrices, block shearing matrices and block diagonal matrices to the right, then we are doing block column operations.

Example 4.4.16. How to find the inverse of $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$, provided that A is invertible?

We can first row reduce downward as we are used to. To kill C, I need to substract CA^{-1} copies of the top block rom from the bottom block row. (Note that $A^{-1}C$ would be WRONG here. The order matters.)

This gives $\begin{bmatrix} I & O \\ -CA^{-1} & I \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ O & D-CA^{-1}B \end{bmatrix}$. Note that the matrix on the right hand side is invertible iff both diagonal blocks are invertible.

This yields the following theorem. ☺

Definition 4.4.17. *For a block matrix $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ where A, D are square, the* ***Schur complement*** *of A is $D - CA^{-1}B$ when A is invertible, and the Schur complement of D is $A - BD^{-1}C$ when D is invertible.*

Proposition 4.4.18. *For a block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, if A and its Schur complement are both invertible, then M is invertible with inverse $\begin{bmatrix} I & -A^{-1}B \\ O & I \end{bmatrix}\begin{bmatrix} A^{-1} & O \\ O & (D-CA^{-1}B)^{-1} \end{bmatrix}\begin{bmatrix} I & O \\ -CA^{-1} & I \end{bmatrix}$.*

The proof is basically finishing the block LDU decomposition and then invert each matrix. The LDU decomposition is

$$M = \begin{bmatrix} I & O \\ CA^{-1} & I \end{bmatrix}\begin{bmatrix} A & O \\ O & (D-CA^{-1}B) \end{bmatrix}\begin{bmatrix} I & A^{-1}B \\ O & I \end{bmatrix}.$$

Don't memorize this formula. Rather, just remember the idea of block operations, and don't be surprised when you do block operations and see a Schur complement somewhere.

4.4.3 (Optional) Woodburry formula and Sherman-Morrison formula

Block matrix multiplications are no doubt super helpful, but due to the non-commutativity, expressions could also get super long. For example, when A and its Schur complements are invertible, we have an expression for the inverse of $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$. Multiply out the previous results, we have

$$\begin{bmatrix} I & -A^{-1}B \\ O & I \end{bmatrix}\begin{bmatrix} A^{-1} & O \\ O & (D-CA^{-1}B)^{-1} \end{bmatrix}\begin{bmatrix} I & O \\ -CA^{-1} & I \end{bmatrix}$$

$$= \begin{bmatrix} A^{-1} & -A^{-1}B(D-CA^{-1}B)^{-1} \\ O & (D-CA^{-1}B)^{-1} \end{bmatrix} \begin{bmatrix} I & O \\ -CA^{-1} & I \end{bmatrix}$$
$$= \begin{bmatrix} A^{-1}+A^{-1}B(D-CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D-CA^{-1}B)^{-1} \\ -(D-CA^{-1}B)^{-1}CA^{-1} & (D-CA^{-1}B)^{-1} \end{bmatrix}.$$

But when D and its Schur complements are invertible, we have another expression for the inverse of $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$. Symetrically, this is

$$\begin{bmatrix} (A-BD^{-1}C)^{-1} & -D^{-1}C(A-BD^{-1}C)^{-1} \\ -(A-BD^{-1}C)^{-1}BD^{-1} & D^{-1}+D^{-1}C(A-BD^{-1}C)^{-1}BD^{-1} \end{bmatrix}$$

Compare the upper left block in both expressions you would obtain the famous Woodbury formula, true whenever A, D are invertible and B, C have the correct number of rows and columns:

$$(A-BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D-CA^{-1}B)^{-1}CA^{-1}.$$

The formula is powerful and useful in some fields. Although it looks horrible, note that we have Schur complements everywhere. In fact, the ugly portion $(A^{-1}B)(D-CA^{-1}B)^{-1}(CA^{-1})$, the three portions corresponds exactly to the LDU decomposition entries. Of course, there is no need to memorize this though. (I won't test it in the final.) Whenever you need this in the future, just google it again to get this formula.

Example 4.4.19. Let us do some special case of this. Suppose $A = I, B = \boldsymbol{e}_i, C = \boldsymbol{e}_j^{\mathrm{T}}, D = -I$ where $i \neq j$. Then we have a formula $(I - \boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}})^{-1} = I + \boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}}$. Is this true?

Well it is. These are just shearing matrices for a single row operation, and obviously the inverse of that shearing behaves this way. Say when $i = 2, j = 1$, we have $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$.

Let us be more general. Consider the inverse $I + \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$. Forget about the precise Woodburry formula, and rather just remember this: its inverse is $I - k\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ for some number k. Now look at the equation $(I + \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}})(I - k\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}) = I$, and you can solve and get $k = \frac{1}{1+\boldsymbol{v}^{\mathrm{T}}\boldsymbol{u}}$.

The formula $(I + \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}})^{-1} = I - \frac{\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}}{1+\boldsymbol{v}^{\mathrm{T}}\boldsymbol{u}}$ is much more useful and easier to remember.

Now substitute $\boldsymbol{u}$ by $A^{-1}\boldsymbol{u}$, and multiply A^{-1} on both sides from the right, we have $(A + \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}})^{-1} = A^{-1} - \frac{(A^{-1}\boldsymbol{u})(\boldsymbol{v}^{\mathrm{T}}A^{-1})}{1+\boldsymbol{v}^{\mathrm{T}}A^{-1}\boldsymbol{u}}$. This is sometimes called the Sherman-Morrison formula.

Matrices that looks like $\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ are also called rank-one matrices sometimes, because they can only have one pivot. Don't memorize the precise Sherman-Morrison. Rather, keep in mind of what it is trying to say: if we change a matrix A by a rank-one matrix, then its inverse would also change by some rank-one matrix. In fact, the Woodburry formula can be think of as a rank r version of this. ☺

Example 4.4.20. Here is another nice applicatin of the Woodburry formula. Apply this to $(I + AB)^{-1}$, and we have $(I + AB)^{-1} = I - A(I + BA)^{-1}B$. Note that A, B here does not need to be square. As long as AB is square, we are fine. Also note that AB, BA might be suqare matrices of different dimensions.

In particular, this implies that $I + AB$ is invertible iff $I + BA$ is invertible. ☺

Remark 4.4.21. *Here is an alternative proof of the fact above, that $(I + AB)^{-1} = I - A(I + BA)^{-1}B$. What is the fundamental relation between $I + AB$ and $I + BA$? It is the following fact, called a push-over identity: $A(I + BA) = (I + AB)A$ and $(I + BA)B = B(I + AB)$. Can you see the "pushing over"?*

Now inverting the $I + AB, I + BA$, and we have the push-over identity in the form of $(I + AB)^{-1}A = A(I + BA)^{-1}$.

To complete the proof, we have $I = (I + AB)^{-1}(I + AB) = (I + AB)^{-1} + (I + AB)^{-1}AB$. Now use the above identity, and we have the desired formula.

Remark 4.4.22. *This remark is entirely optional. By the summation of geometric series, we know that for real numbers $|x| < 1$, we have the formula*

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \ldots.$$

Apply this to AB or BA, we have

$$(I+AB)^{-1} = I - AB + ABAB - ABABAB + ...$$

$$(I+BA)^{-1} = I - BA + BABA - BABABA + ...$$

Now note that $A(I+BA)^{-1}B$ is exactly $AB - ABAB + ABABAB - ....$ This is exactly a part of the formula for $(I+AB)^{-1}$, wow! Hence we have $(I+AB)^{-1} = I - A(I+BA)^{-1}B$.

The arguments so far are not rigorous yet, and there are some convergence issues to think about. Nevertheless, it gives a very nice intuition about this fact.

4.4.4 Symmetric Matrices and LDL$^{\mathrm{T}}$ Decomposition

You surely have wondered about this. Some matrices are very pretty, in that they equal to their own transpose, say $\begin{bmatrix}1&2&3\\2&4&5\\3&5&6\end{bmatrix}$.

Definition 4.4.23. *A matrix A is said to be* ***symmetric*** *if $A = A^{\mathrm{T}}$.*

So these are matrices like $\begin{bmatrix}1&2\\2&3\end{bmatrix}$, where the entries are symmetric about the diagonal.

Proposition 4.4.24. *For any square matrix A, $A + A^{\mathrm{T}}$ is symmetric.*

For any $m \times n$ matrix A, AA^{T} and $A^{\mathrm{T}}A$ are symmetric.

Proof. Direct calculation. ☐

Now it sure looks pretty on the outside. Luckily, it is also pretty on the inside.

Proposition 4.4.25. *If $A = A^{\mathrm{T}}$ is invertible, and it has an LU decomposition, then in fact we have $A = LDL^{\mathrm{T}}$ where L is unit lower triangular, and D is diagonal.*

Proof. We know the LDU decomposition is unique. Now consider $A = LDU$ and $A = A^{\mathrm{T}} = (LDU)^{\mathrm{T}} = U^{\mathrm{T}}D^{\mathrm{T}}L^{\mathrm{T}}$, and note that these are two LDU decomposition for A. So we must have $U = L^{\mathrm{T}}$. ☐

We sometimes call this the LDL$^{\mathrm{T}}$ decomposition. It is in fact faster than the traditional elimination (only take half as much time). Why? Because symmetricity means we only need to work with entries below the diagonal.

Example 4.4.26. Suppose we have $\begin{bmatrix}1&2&3&4\\2&5&6&7\\3&6&8&9\\4&7&9&10\end{bmatrix}$. For the first step of the elimination, we use the first row to reduce below, and we have $\begin{bmatrix}1&2&3&4\\0&1&0&-1\\0&0&-1&-3\\0&-1&-3&-6\end{bmatrix}$. Huh. The lower right block is still symmetric!

In fact, this is no coincidence. Say our symmetric matrix is $\begin{bmatrix}a&\boldsymbol{v}^{\mathrm{T}}\\\boldsymbol{v}&S\end{bmatrix}$. Consider the elimination $\begin{bmatrix}1&\mathbf{0}^{\mathrm{T}}\\-\frac{1}{a}\boldsymbol{v}&I\end{bmatrix}\begin{bmatrix}a&\boldsymbol{v}^{\mathrm{T}}\\\boldsymbol{v}&S\end{bmatrix} = \begin{bmatrix}a&\boldsymbol{v}^{\mathrm{T}}\\\mathbf{0}&S-\frac{1}{a}\boldsymbol{v}\boldsymbol{v}^{\mathrm{T}}\end{bmatrix}$, we see that the lower right block must still be symmetric.

So we do not need to compute all entries during our elimination. Starting with $\begin{bmatrix}1&2&3&4\\2&5&6&7\\3&6&8&9\\4&7&9&10\end{bmatrix}$, to do the first step of the elimination, we only need to calculate the entries below the diagonal and get

$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & & \\ 0 & 0 & -1 & \\ 0 & -1 & -3 & -6 \end{bmatrix}$, and we can then fill in the blank according to the symmetricity without a thought. So for big matrices, we save about half the time. ☺

4.4.5 When do we have an LU Decomposition

Definition 4.4.27. *For a matrix A, its upper left $k \times k$ entries form its k**-th leading principal submatrix**.*

Theorem 4.4.28. *An invertible matrix A has an LU decomposition if and only if all its leading principal submatrices are invertible. (I.e., upper left square block of various sizes are all invertible.)*

Example 4.4.29. If the upper left entry is zero, then Gaussian elimination needs a swap right away. So of course there is no LU decomposition.

Consider $\begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. We now use top row to eliminate below, and we have $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & -3 \\ 0 & 1 & -6 \end{bmatrix}$. And now we need a swap, so no LU decomposition. This is precisely because the $\begin{bmatrix} 2 & 2 \end{bmatrix}$ on the second row is a multiple of $\begin{bmatrix} 1 & 1 \end{bmatrix}$ on the first row, i.e., the second leading principal submatrix is NOT invertible. (Dependent rows = NOT surjective.) ☺

Proof of Sufficiency. If $A = LU$ and A invertible, it follows that L, U are both invertible. So the diagonal entries of L, U are all non-zero.

Now consider the block form $L = \begin{bmatrix} L_{11} & O \\ L_{21} & L_{22} \end{bmatrix}, U = \begin{bmatrix} U_{11} & U_{12} \\ O & U_{22} \end{bmatrix}$, here $L_{11}, L_{22}, U_{11}, U_{22}$ are all triangular with non-zero diagonal entries, and hence all invertible.

Then we have $A = LU = \begin{bmatrix} L_{11} & O \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} \\ O & U_{22} \end{bmatrix}$. Ignore the rest, just focus on the upper left block of this multiplication and we see that the k-th leading principal submatrix is exactly $L_{11}U_{11}$. Hence it must be invertible. □

Proof of Necessity. Say A is $n \times n$. We do this by induction on n. The case $n = 1$ is trivial because A would be a non-zero number.

Let us do the inductive step. We write block form $A = \begin{bmatrix} A_{n-1} & \boldsymbol{v} \\ \boldsymbol{w}^{\mathrm{T}} & a \end{bmatrix}$. If all leading principal submatrices of A are invertible, then all leading principal submatrices of A_{n-1} are invertible. So by induction hypothesis, we have $A_{n-1} = L_{n-1}U_{n-1}$. In particular, L_{n-1}^{-1} is the elimination we would do on A_{n-1}.

Now we block eliminate A. First we have $\begin{bmatrix} I & \mathbf{0} \\ -\boldsymbol{w}^{\mathrm{T}}A_{n-1}^{-1} & 1 \end{bmatrix} A = \begin{bmatrix} A_{n-1} & \boldsymbol{v} \\ 0 & a - w^{\mathrm{T}}A_{n-1}^{-1}\boldsymbol{v} \end{bmatrix}$. Next we eliminate A_{n-1}, so we need to use L_{n-1}^{-1}. This leads to $\begin{bmatrix} L_{n-1}^{-1} & \mathbf{0} \\ \mathbf{0}^{\mathrm{T}} & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ -\boldsymbol{w}^{\mathrm{T}}A_{n-1}^{-1} & 1 \end{bmatrix} A = \begin{bmatrix} U_{n-1} & L_{n-1}^{-1}\boldsymbol{v} \\ 0 & a - w^{\mathrm{T}}A_{n-1}^{-1}\boldsymbol{v} \end{bmatrix}$. Now we end up with an upper triangular matrix. So reorganizing the terms, A has the following LU decomposition:

$$A = \begin{bmatrix} L_{n-1} & \mathbf{0} \\ \boldsymbol{w}^{\mathrm{T}}A_{n-1}^{-1}L_{n-1} & 1 \end{bmatrix} \begin{bmatrix} U_{n-1} & L_{n-1}^{-1}\boldsymbol{v} \\ 0 & a - w^{\mathrm{T}}A_{n-1}^{-1}\boldsymbol{v} \end{bmatrix}$$

□

4.4.6 Permutations and PLU Decomposition

Now, what if a matrix has no LU decomposition? Well, you swap the rows around, and then do the elimination.

Definition 4.4.30. *A matrix is a* ***permutation matrix*** *if as a row operations it will simply permute the rows. (As a linear map, it will simply permute the coordinates of the input vector.)*

Consider that the appearance of a matrix must exactly be what it will do to I as a row operation, we immediately have this corollary.

Corollary 4.4.31. *A matrix is a permutation matrix if and only if it has rows of I in any order, if and only if it has the column of I in any order, if and only if it has exactly a single non-zero entry of 1 in each row and in each column.*

Example 4.4.32. Here are all possible 3 by 3 permutation matrices.

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix},\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix},\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix},\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix},\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix},\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}.$$

The corresponds to "change nothing", $r_1 \leftrightarrow r_2$, $r_1 \leftrightarrow r_3$, $r_2 \leftrightarrow r_3$, cyclically $r_3 \to r_2 \to r_1 \to r_3$, and cyclically $r_1 \to r_2 \to r_3 \to r_1$. ☺

As you can see, a permutation matrix is determined by the location of ones in its entries. And we have the following observation.

Lemma 4.4.33. *The (i, j) entry of a permutation matrix P is 1 if and only if, as a row operation, P sends the j-th row to the i-th row.*

Proof. The (i, j) entry of a permutation matrix P is 1 if and only if its j-th column is $\boldsymbol{e}_i$, if and only if $P\boldsymbol{e}_j = \boldsymbol{e}_i$. □

Here is a trivial proposition, cosider the meaning of permutation matrices as row operations.

Proposition 4.4.34.

1. *There are $n!$ permutation matrices of size $n \times n$.*
2. *Multiplications of permutation matrices are permutation matrices.*
3. *Inverse of a permutation matrix is the transpose of that permutation matrix.*
4. *Every permutation matrix is a product of swapping matrices.*

Proof. First two are trivial.

Let us prove that the inverse of a permutation matrix is the transpose. Suppose the (i, j) entry of a permutation matrix P is 1. What does this mean? From the fact that $PI = P$, this means that the j-th row of the identity matrix is changed into the i-th row. In particular,

$$\begin{aligned}&(i,j) \text{ entry of } P^{\mathrm{T}} \text{ is } 1\\ \Leftrightarrow&(j,i) \text{ entry of } P \text{ is } 1\\ \Leftrightarrow&P \text{ as a row operation send the } i\text{-th row to the } j\text{-th row}\\ \Leftrightarrow&P^{-1} \text{ as a row operation send the } j\text{-th row to the } i\text{-th row}\\ \Leftrightarrow&(i,j) \text{ entry of } P^{-1} \text{ is } 1.\end{aligned}$$

Now let us prove the last one. Instead of proving it, let me give you an algorithm to do this. (The rigorous proof is just to use mathematical induction to show that this works always. I'll leave that to you.)

Suppose we want to permute $(1, 2, 3, 4, 5)$ to $(4, 5, 2, 1, 3)$. We swap to construct our desired result in the left most location first:

$$(1, 2, 3, 4, 5) \to (4, 2, 3, 1, 5) \to (4, 1, 3, 2, 5) \to (4, 1, 2, 3, 5).$$

Now the first location is done, we move on to the second location.

$$(4,1,2,3,5) \to (4,1,2,5,3) \to (4,1,5,2,3) \to (4,5,1,2,3).$$

Then we fix the third location.

$$(4,5,1,2,3) \to (4,5,2,1,3).$$

And we are done.

You can see how by swapping I can succeed one by one by focusing on one location at a time. This process also has a "Gaussian elimination" feel to it. $\square$

Now we can make our statement about matrices without LU decomposition.

Definition 4.4.35. *A **PLU decomposition** of a matrix A means writing $A = PLU$ where P is a permutation matrix, L is a lower triangular matrix, and U is an upper triangular matrix.*

Theorem 4.4.36. *Every invertible matrix has a PLU decomposition.*

Proof. Say A is invertible $n \times n$ matrix. We again proceed by induction on n. The case $n = 1$ is again trivial. Let us look at the inductive step now.

If A is invertible, its first column cannot all be zero. So we can find a swap P_1 such that $P_1 A$ has nonzero $(1,1)$ entry. Say $A = \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \boldsymbol{w} & A_{n-1} \end{bmatrix}$ where $a \neq 0$.

Next elimination gives $P_1 A = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & A_{n-1} - \frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}} \end{bmatrix}$.

Since $a \neq 0$ and A invertible, we see that $A_{n-1} - \frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}}$ must be invertible and $(n-1) \times (n-1)$. So it has a PLU decomposition, say $A_{n-1} - \frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}} = P_{n-1} L_{n-1} U_{n-1}$.

So to go on elimination, we need to permute the bottom $n-1$ rows of $\begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & A_{n-1} - \frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}} \end{bmatrix}$, and we now have

$$\begin{aligned} P_1 A &= \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix} \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & L_{n-1}U_{n-1} \end{bmatrix} \\ P_1 A &= \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & L_{n-1} \end{bmatrix} \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & U_{n-1} \end{bmatrix} \\ A &= P_1^{-1} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & L_{n-1} \end{bmatrix} \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & U_{n-1} \end{bmatrix}. \end{aligned}$$

Now we are almost done, except that $\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix}$ is in the wrong order, and they do not commute! What should we do? We multiply and decompose.

$$\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}\boldsymbol{w} & P_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}P_{n-1}^{-1}\boldsymbol{w} & I \end{bmatrix}$$

So we have our desired PLU decomposition

$$A = (P_1^{-1} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & P_{n-1} \end{bmatrix})(\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \frac{1}{a}P_{n-1}^{-1}\boldsymbol{w} & I \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & L_{n-1} \end{bmatrix}) \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & U_{n-1} \end{bmatrix}.$$

Here the parenthesis helps you see the P,L,U portion of the decomposition. $\square$

Don't memorize these proofs. The point is to show you how block multiplication and block elimination works, so focus on that. If you look at all the proofs, for the criteria of LU decomposition and for the existence of PLU decomposition, you see that the key is just to proceed with the elimination, and write the next step in block form.

4.5 Exercises

4.5.1 Exercises for Matrix Multiplication

Exercise 4.5.1 (Dimensions in matrix multiplications)**.** *Say* A, B, C *are* $3\times 5, 5\times 3, 3\times 1$ *matrices respectively, then which of* $BA, AB, ABAB, BAC, BABC$ *are well-defined?*

Exercise 4.5.2. *Calculate the following matrix multiplications. Or maybe it is undefined!*

1. $\begin{bmatrix}1 & 2 & 3 & 4\\ 5 & 6 & 7 & 8\end{bmatrix}\begin{bmatrix}1 & 2 & 3 & 4\\ 5 & 6 & 7 & 8\end{bmatrix}$.

2. $\begin{bmatrix}1 & 2 & 3 & 4\\ 5 & 6 & 7 & 8\end{bmatrix}\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\\ 1 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$.

3. $\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$.

4. $\begin{bmatrix}0 & 1 & 0\end{bmatrix}\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$.

5. $\begin{bmatrix}1 & -2 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$.

6. $\begin{bmatrix}1 & 2 & 3 & 4\\ 5 & 6 & 7 & 8\end{bmatrix}\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$.

7. $\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}\begin{bmatrix}2\end{bmatrix}$.

8. $\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}\begin{bmatrix}1 & 2 & 3 & 4\end{bmatrix}$.

9. $\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 3 & 9\end{bmatrix}\begin{bmatrix}3\\ 2\\ 1\end{bmatrix}$.

10. $\begin{bmatrix}1 & 2 & 1\\ & 1 & 1\\ & & 1\end{bmatrix}\begin{bmatrix}x^2\\ x\\ 1\end{bmatrix}$.

11. $\begin{bmatrix}1 & 2 & 1\\ & 1 & 1\\ & & 1\end{bmatrix}\begin{bmatrix}(x+1)^2\\ x+1\\ 1\end{bmatrix}$.

12. $\begin{bmatrix}1 & 2 & 1\\ & 1 & 1\\ & & 1\end{bmatrix}\begin{bmatrix}(x+2)^2\\ x+2\\ 1\end{bmatrix}$. *(Do you see a pattern here?)*

Exercise 4.5.3. *Calculate the following matrix multiplications.*

1. $\begin{bmatrix} 1 & 1 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & 1 \end{bmatrix}$. *(Can you generalize this to an $n \times n$ pattern?)*

2. $\begin{bmatrix} & & 1 \\ & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & \\ 1 & & \end{bmatrix}$. *(Can you generalize this to an $n \times n$ pattern?)*

3. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^2$. *(Can you generalize this to an $n \times n$ pattern?)*

4. $\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}^2$. *(Can you generalize this to an $n \times n$ pattern?)*

5. $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}^2$. *(Can you generalize this to an $n \times n$ pattern?)*

Exercise 4.5.4. *Calculate the following matrix multiplications.*

1. *Suppose $\boldsymbol{u}, v, w$ are mutually orthogonal unit vectors in $\mathbb{R}^5$. Compute*

$$\begin{bmatrix} \boldsymbol{u}^{\mathrm{T}} \\ \boldsymbol{v}^{\mathrm{T}} \\ \boldsymbol{w}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \boldsymbol{u} & \boldsymbol{v} & \boldsymbol{w} \end{bmatrix}.$$

2. *Compute* $\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}^2$. *(This is actually the square of a reflection.)*

3. $\begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix} \begin{bmatrix} \cos\phi & \sin\phi \\ \sin\phi & -\cos\phi \end{bmatrix}$. *This is a product of two reflections, and the result is a rotation. What is the angle of this rotation? (Can you also find a geometric proof?)*

Exercise 4.5.5 (Powers of a matrix)**.** *Calculate the following matrix powers.*

1. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^2$.

2. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^3$.

3. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{2020}$.

4. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{-1}$.

5. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^2$.

6. $\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}^3$.

7. $\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}^{2020}$.

8. $\begin{bmatrix}1&-1&-1&-1\\-1&1&-1&-1\\-1&-1&1&-1\\-1&-1&-1&1\end{bmatrix}^2$.

9. $\begin{bmatrix}1&-1&-1&-1\\-1&1&-1&-1\\-1&-1&1&-1\\-1&-1&-1&1\end{bmatrix}^3$.

10. $\begin{bmatrix}1&-1&-1&-1\\-1&1&-1&-1\\-1&-1&1&-1\\-1&-1&-1&1\end{bmatrix}^{2020}$.

11. $\begin{bmatrix}1&-1&-1&-1\\-1&1&-1&-1\\-1&-1&1&-1\\-1&-1&-1&1\end{bmatrix}^{-1}$.

Exercise 4.5.6 (Commutativity trouble)**.** *Let* $A = \begin{bmatrix}1&1\\0&1\end{bmatrix}, B = \begin{bmatrix}1&0\\1&1\end{bmatrix}$.

1. *Calculate AB and BA. Are they the same?*

2. *Calculate $(A+B)^2, A^2+2AB+B^2$. Are they the same?*

3. *Calculate $(AB)^2, A^2B^2$. Are they the same?*

Exercise 4.5.7 (Polynomials of a matrix)**.** *Suppose for a matrix A we have $A^2 = A$.*

1. *Simplify the polynomials A^3+2A^2-A-I and $A^2+3A+4I$ into the format $sA+tI$ for some constants $s,t \in \mathbb{R}$. (Read only: note that you can use this strategy to simplify all possible polynomials of A into the format $sA+tI$.)*

2. ☕*Show that $I+2A$ is invertible by finding its inverse. (Hint: The inverse is also a polynomial of A.)*

Exercise 4.5.8. *The **trace** of an $n \times n$ matrix A is the sum of all of its diagonal entries. We denote it as* $\mathrm{trace}(A)$*. If you like, you can write* $\mathrm{trace}(A) = \sum_{i=1}^{n} \boldsymbol{e}_i^{\mathrm{T}} A \boldsymbol{e}_i$.

1. *Show that* $\mathrm{trace}(xA+yB) = x\,\mathrm{trace}(A) + y\,\mathrm{trace}(B)$.

2. *Show that* $\mathrm{trace}(I_n) = n$.

3. *Show that* $\mathrm{trace}(A) = \mathrm{trace}(A^{\mathrm{T}})$.

4. *For a unit vector $\boldsymbol{u}$, show that* $\operatorname{trace}(\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}) = 1$ *and* $\operatorname{trace}(I_n - \boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}) = n-1$. *(For projections, trace simply tells you the dimension of the projection range.)*

5. *Show that, if* $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$, *then* $\operatorname{trace}(A^{\mathrm{T}}B) = \operatorname{trace}(BA^{\mathrm{T}}) = \sum a_i b_i$. *(This is a "dot product" for matrices.)*

6. *Show that* $\operatorname{trace}(\boldsymbol{v}\boldsymbol{w}^{\mathrm{T}}) = \operatorname{trace}(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{v})$.

7. ☕*Can you prove that* $\operatorname{trace}(AB) = \operatorname{trace}(BA)$ *in general? Here A is $m \times n$ and B is $n \times m$. (Hint: break down A into columns and B into rows, and use sub-problem 6. Alternatively, exploit the symmetry of the "dot product" in sub-problem 5, but you will need to generalize the formula in sub-problem 5 to arbitrary non-square matrices first.)*

8. ☕*Prove that for any square matrices A, B, then $AB - BA$ CANNOT be the identity matrix. (However, in infinite dimensional spaces, $AB - BA = I$ is related to the famous Heisenberg uncertainty principle in physics.) (Hint: Obviously use previous subproblems.)*

Exercise 4.5.9. *A permutation is a way to rearrange the orders of objects. For example, if we have three objects $1, 2, 3$, then a permutation is a bijective map $\sigma : \{1, 2, 3\} \to \{1, 2, 3\}$. Maybe it sends $1, 2, 3$ to $2, 1, 3$ respectively. This concept is very important in games and computer science.*

One way to study a permutation is to study a corresponding matrix. A permutation matrix is a linear map that permutes the standard basis vectors. For example, $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ *corresponds to the permutation* $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) \mapsto (\boldsymbol{e}_2, \boldsymbol{e}_1, \boldsymbol{e}_3)$. *We see that $\boldsymbol{e}_3$ is unchanged, so this is a* ***fixed point*** *of our permutation. In this example, we have one fixed point in the standard basis vectors.*

1. *Find a 5×5 permutation matrix P such that $P^k \neq I$ for all $1 \leq k \leq 5$, but $P^6 = I$.*

2. *For a permutation matrix P, prove that* $\operatorname{trace}(P)$ *is the number of fixed standard basis vectors for the corresponding permutation.*

3. ☕*Consider the compositions of two permutations P_1P_2 and P_2P_1. Show that they always have the same number of fixed standard basis vectors. (Hint: Look at the trace problem above.)*

Exercise 4.5.10. *It's a cold winter, and you sell matches for money. In the first day, you sold x_1 matches for p_1 dollar each. In the second day, you sold x_2 matches for p_2 dollar each. So on so forth. Let* $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

1. *Let $\boldsymbol{v} \in \mathbb{R}^n$ be the "total matches sold" vector, so its i-th coordinate is the total number of matches you have sold from day 1 to day i. Find a matrix A such that $A\boldsymbol{v} = \boldsymbol{x}$.*

2. *Let $\boldsymbol{y} \in \mathbb{R}^n$ be the "revenue" vector, so its i-th coordinate is your revenue for the i-th day. Find a matrix B such that $B\boldsymbol{x} = \boldsymbol{y}$.*

3. *Let $\boldsymbol{w} \in \mathbb{R}^n$ be the "total revenue" vector, so its i-th coordinate is the total revenue you've earned from day 1 to day i. Find a matrix C such that $C\boldsymbol{y} = \boldsymbol{w}$.*

4. *Calculate CBA. (This matrix will send the "total matches sold" vector to the "total revenue" vector.)*

Exercise 4.5.11 (Wacky Psychology)**.** *Suppose each person has two innate properties, ability level (AL) and self-doubt (SD). For example, a person, say John, may have an AL of 3 and SD of 1. This gives a 2 dimensional space, and every person corresponds to some vector in this space. In particular, we can think of John as the vector* $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

1. *If we give a person a praise, then we boost his/her AL by 10%, and reduce his/her self-doubt by 20%. Can you write out a matrix that describes the effect of a praise? If we were to give John a praise, what would his AL and SD be?*

2. *If we scold a person, his/her self-doubt would be increased by 20%, and their AL would decrease by twice the amount of their original self-doubt. Can you write out a matrix that describes the effect of a scolding? If we were to scold John, what would his AL and SD be?*

3. *According to our model, which one is better: To first scold a person, and then praise the person. Or to first praise the person, then scold the person?*

4. *Describe a process that would reduce John's ability level to be smaller than 0.11, while keeping his ability level positive throughout this process.*

4.5.2 Exercises for Gaussian Elimination via Matrices

Exercise 4.5.12 (Matrix multiplications and row/column operations)**.** *Calculate the following matrix multiplications*

1. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 4 & 0 \end{bmatrix}$.

3. $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 \\ 1 & 3 & 3 & 1 & 0 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$.

4. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}^k$ *for an integer k. Find the formula depending on k.*

Exercise 4.5.13 (Elementary shearings)**.** *In $\mathbb{R}^4$, let $X_{ij} = \boldsymbol{e}_i \boldsymbol{e}_j^{\mathrm{T}}$ be the corresponding 4 by 4 matrix. (Note that the elementary shearing matrix is $E_{ij}^k = I + kX_{ij}$.)*

1. *Write out X_{13}, X_{32}, X_{12}.*

2. *Calculate $X_{13}X_{32}$ and $X_{32}X_{13}$. (See if you can do this via the fact that $X_{ij} = \boldsymbol{e}_i \boldsymbol{e}_j^{\mathrm{T}}$, and use associativity. Dot products between these standard basis vectors are super easy.)*

3. *Verify that $X_{ij}^2 = O$ when $i \neq j$. (Not part of the problem, but you may realize that this implies that $(E_{ij}^1)^k = (I + X_{ij})^k = I + kX_{ij} = E_{ij}^k$.)*

4. *For any two square matrices A, B, show that $AB = BA$ if and only if $(A-I)(B-I) = (B-I)(A-I)$. (Hence to study the commutativity behavior between elementary shearings, we only need to study the commutativity of these X_{ij}.)*

Exercise 4.5.14 (Shifting operators)**.** *Let $J = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.*

1. *Going from A to JA, would J be shifting things up? down? left? right?*
2. *Going from A to AJ, would J be shifting things up? down? left? right?*
3. *Given the Pascal's symmetric matrix* $P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$, *calculate $PJ + J^{\mathrm{T}}P$. Can you explain the "coincidence"?*
4. *Calculate J, J^2, J^3, J^4. (Also note that their appearances is exactly how they would shift the identity matrix.)*
5. *Show that $I - J$ has inverse $I + J + J^2 + J^3$.*
6. *Calculate $(J + I)^2, (J + I)^3, (J + I)^4$. (Not part of the problem, but can you see a pattern for the entries? Can you prove the pattern for $(J + I)^k$?)*
7. ☕*Describe the set of all matrices that commutes with J.*

Exercise 4.5.15 (Gaussian elimination to find inverse)**.** *Find inverse matrices. You may start with $\begin{bmatrix} A & I_n \end{bmatrix}$ and work your way to $\begin{bmatrix} I_n & A^{-1} \end{bmatrix}$, or you may do whatever that works.*

1. $\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$.
2. $\begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$. *(Compare this with above. Merely a single-entry difference, yet the inverses look drastically different. However, is the difference rank one?)*
3. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. *(Horses move like this in chess. This pattern is also useful in five-in-a-row, i.e., Wǔ Zǐ Qí.)*
4. $\begin{bmatrix} & & 1 \\ & \iddots & \\ 1 & & \end{bmatrix}$.
5. $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{bmatrix}$.
6. $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$.
7. $\begin{bmatrix} 1 & -a & 0 & 0 \\ 0 & 1 & -b & 0 \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Exercise 4.5.16. *Find a unit lower triangular matrix L such that $L^{-1}A$ is in REF. Then for square matrices, find the unique LDU decomposition of A.*

1. $A = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 3 & 4 & 1 \end{bmatrix}$.

2. $A = \begin{bmatrix} 1 & 2 & 3 \\ & 1 & 4 \\ & & 1 \end{bmatrix}$.

3. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}$.

4. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix}$.

5. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$.

6. $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$.

Exercise 4.5.17 (Constraints through matrix multiplications)**.** *In each cases, find a matrix B that satisfies the given condition.*

1. $AB = 4A$ *for all 3 by 3 matrices A.*

2. $BA = 4A$ *for all 3 by 3 matrices A.*

3. *Every row of BA is the first row of A for all 3 by 3 matrices A.*

4. *Every entry of AB is the average of the entries in the corresponding row of A for all 3 by 3 matrices A.*

5. $B^2 \neq O$, $B^3 = O$. *(Just find any example of such a matrix is enough.)*

6. $B\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} B$. *Find all possible B for this problem.*

Exercise 4.5.18 (Weird "triangular" matrices)**.** *An $n \times n$ matrix is called a **northwest** matrix if all entries below the anti-diagonal from $(1, n)$ entry to $(n, 1)$ entry are zero. Similarly, we can define a **southeast** matrix. Now, if A is northwest, what about $A^{\mathrm{T}}, A^{-1}, A^2$? What if we multiply a northwest matrix with a southeast matrix?*

4.5.3 Exercises for Block Matrices

Exercise 4.5.19. *Calculate the following matrix multiplications.*

1. $\begin{bmatrix} 1 & 1 & & \\ & 1 & & \\ & & 1 & \\ & & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & & \\ & 1 & & \\ & & 1 & \\ & & 1 & 1 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 1 & -1 & \\ & 1 & & -1 \\ -1 & & 1 & -1 \\ & -1 & & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 3 & 5 & 7 \\ 1 & 1 & 2 & 3 \end{bmatrix}$.

3. $\begin{bmatrix} 1 & & & \\ & 1 & & \\ 1 & & -1 & \\ 1 & 1 & & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 6 & 8 & 10 & 12 \end{bmatrix}$.

Exercise 4.5.20. *Calculate the following block matrix multiplications if possible, or decide that they are undefined.*

1. $\begin{bmatrix} A & B \\ B & A \end{bmatrix} \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix}$. *Assume* $A\boldsymbol{v} = \boldsymbol{e}_1, A\boldsymbol{w} = \boldsymbol{e}_2, B\boldsymbol{v} = \boldsymbol{e}_3, B\boldsymbol{w} = \boldsymbol{e}_4$, *and* A, B *have four rows each.*

2. $\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix} \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix}$. *All vectors are in* $\mathbb{R}^2$.

3. $\begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix} \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix}$. *All vectors are in* $\mathbb{R}^2$.

4. $\begin{bmatrix} A & \\ A & A \end{bmatrix} \begin{bmatrix} A & A \\ & A \end{bmatrix}$ *where* A *is* 2×2.

5. $\begin{bmatrix} A & \\ A & A \end{bmatrix} \begin{bmatrix} A & A \\ & A \end{bmatrix}$ *where* A *is* 2×3.

Exercise 4.5.21 (Block Operations)**.** *Calculate the inverse of the following matrices. (Try not to use the ugly formula. Try do block operations from scratch for practice.)*

1. $\begin{bmatrix} I_n & O \\ A & I_m \end{bmatrix}$. *(This is lower triangular.)*

2. $\begin{bmatrix} O & I_m \\ I_n & A \end{bmatrix}$. *(This is block-southeast.)*

3. $\begin{bmatrix} O & A \\ B & O \end{bmatrix}$ *where* A, B *are invertible.*

4. $\begin{bmatrix} A & C \\ O & B \end{bmatrix}$ *where* A, B *are invertible. (This is block upper triangular.)*

Exercise 4.5.22. *A matrix* A *is* ***skew symmetric*** *if* $A^{\mathrm{T}} = -A$.

1. *If* A *is* 3×3 *and skew symmetric, show that* $A\boldsymbol{x} = \boldsymbol{v} \times \boldsymbol{x}$ *for some* $\boldsymbol{v}$ *depending only on* A. *(Here cross product is defined as* $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{bmatrix}$. *In* $\mathbb{R}^3$, *one can verify that the vector* $\boldsymbol{v} \times \boldsymbol{w}$ *is always perpendicular to both* $\boldsymbol{v}$ *and* $\boldsymbol{w}$. *Feel free to do the dot product and check this.)*

2. *Find a non-zero matrix* A *such that* $\begin{bmatrix} O & A \\ A & O \end{bmatrix}$ *is skew symmetric but not symmetric.*

3. *Find a* $2n \times 2n$ *skew symmetric matrix* A *such that* $A^2 = -I$ *for each* n. *(Remark: For real numbers, you may never have* $x^2 = -1$. *But for real matrices,* $A^2 = -I$ *is possible.)*

4. *Show that for any square matrix A, you can write it as the sum of a symmetric matrix and a skew symmetric matrix.*

Exercise 4.5.23. *Suppose the RREF of A is R, and the rows of A are linearly independent. Find the RREF of the following matrices.*

1. $\begin{bmatrix} A & 2A \end{bmatrix}$.
2. $\begin{bmatrix} A \\ 2A \end{bmatrix}$.
3. $\begin{bmatrix} A & A \\ A & A \end{bmatrix}$.
4. $\begin{bmatrix} A & A \\ O & A \end{bmatrix}$, *where the rank of A is equal to the number of rows of A.*

Exercise 4.5.24 (We can study affine maps using matrices as well)**.** *A map $f : \mathbb{R}^n \to \mathbb{R}^n$ is called an* ***affine map*** *if you can find an $n \times n$ matrix A and a vector $\boldsymbol{b} \in \mathbb{R}^n$ such that $f(\boldsymbol{x}) = A\boldsymbol{x} + \boldsymbol{b}$. (It does not fix the origin, so it is not linear, merely affine. For example, $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = ax + b$ is an affine map.)*

1. *Show that $\begin{bmatrix} A & \boldsymbol{b} \\ \mathbf{0}^{\mathrm{T}} & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ 1 \end{bmatrix} = \begin{bmatrix} f(\boldsymbol{x}) \\ 1 \end{bmatrix}$. Let us define M_f as $\begin{bmatrix} A & \boldsymbol{b} \\ \mathbf{0}^{\mathrm{T}} & 1 \end{bmatrix}$, and call this the matrix for the affine map.*
2. *Show that $M_f M_g = M_{f \circ g}$.*
3. *Show that f is invertible iff A is invertible, and $M_f^{-1} = M_{f^{-1}}$. Also find the block inverse M_f^{-1}.*

Exercise 4.5.25 (Hidden Block Diagonal Matrix)**.** *Define* $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \triangle \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & a_{12} & 0 \\ 0 & b_{11} & 0 & b_{12} \\ a_{21} & 0 & a_{22} & 0 \\ 0 & b_{21} & 0 & b_{22} \end{bmatrix}$

1. *Find a matrix X such that for all A, B, $X(A \triangle B)X^{-1} = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$.*
2. *Show that $(A_1 \triangle A_2)(B_1 \triangle B_2) = (A_1B_1) \triangle (A_2B_2)$. (The first sub-problem might help.)*
3. *Show that when A, B are invertible, $(A \triangle B)^{-1} = A^{-1} \triangle B^{-1}$. (The first sub-problem might help.)*

Exercise 4.5.26 (The place where SM formula can be used)**.** *Suppose A has inverse $\begin{bmatrix} 1 & 9 & 0 \\ 1 & 0 & 8 \\ 0 & 1 & 1 \end{bmatrix}$. If I increase the $(1, 3)$ entry of A by 1, what is the new inverse? (Hint: It is easier to use Sherman-Morrison formula.)*

Part II

Abstract Structures

Chapter 5

Abstract Vector Space

5.1 Motivation

We have mastered $\mathbb{R}^n$. Now we move on to abstract vector spaces. As you shall soon discover, abstract vector spaces are "pretty much" the same as $\mathbb{R}^n$, and upon picking a basis, they behave exactly like $\mathbb{R}^n$. In some sense, there is nothing to learn. However, they represent a shift in perspective which is more geometric, more spatial, and more fundamental.

In short, an (abstract) vector space refers to any set in which we can do linear combinations. We call elements of vector spaces as vectors. Also, maps between vector spaces that respect the linear structures are linear maps.

There are several reasons why we cannot do everything in $\mathbb{R}^n$.

Example 5.1.1 (Infinite dimensional vector spaces)**.** At the start of this class, we have an example where we add or substract sound waves. This example is pretty much about the same kind of phenomena: sometimes we can do linear combinations of certain objects, yet these objects cannot be written in coordinates.

Let V represent the space of all infinitely differentiable real functions. (I.e., functions $f : \mathbb{R} \to \mathbb{R}$ such that $f^{(k)}(x)$ exists everywhere for all k.) Functions such as $\mathrm{e}^x, \sin x, \cos x$ and polynomials are all elements of V.

Now, if several functions are infinitely differentiable, then their linear combinations are infinitely differentiable. (More specifically, via the formula $(af + bg)'(x) = af'(x) + bg'(x)$.) In particular, we see that elements of V can linearly combine into some other elements of V. V is a place where you can do linear algebra.

We can also study linear maps. Let $D : V \to V$ be the map sending each f to f'. Then you can also immediately see that this is a linear map. (Again via the formula $(af + bg)'(x) = af'(x) + bg'(x)$.) One might also study the map $M : V \to V$ that sends $f(x)$ to $xf(x)$.

Here is a funny formula, which is secretly related to Heisenberg's uncertainty principle in physics. We have $DM - MD = I$, where $I : V \to V$ is the identity map sending f to itself. To see this, note that $[(DM - MD)f](x) = (DMf)(x) - (MDf)(x) = (xf(x))' - xf'(x) = f(x)$. This is very interesting though, because you can prove that over $\mathbb{R}^n$, we could never have linear maps $A, B : \mathbb{R}^n \to \mathbb{R}^n$ such that $AB-BA = I$.

So our space V is unlike $\mathbb{R}^n$ for any n. It is in fact an infinite dimensional space, and there is no "standard basis" in the conventional sense. Elements of V can be linearly combined, yet they do not have "coordinates". Linear maps can be computed, yet they do not have "matrices". ☺

Luckily for us, infinite dimensional spaces are NOT the main concern of our class. Almost everything in our class should be finite dimensional. But even in that case, we still need abstract vector spaces.

Example 5.1.2 (Subspaces)**.** A subspace is, loosely speaking, a vector space contained within another larger vector space.

Consider the plane $\mathbb{R}^2$. Let $V = \{\begin{bmatrix} x \\ x \end{bmatrix} \in \mathbb{R}^2 : x \in \mathbb{R}\}$, the subset of all vectors whose coordinates are the same. This is a straight line. Let $W = \{\begin{bmatrix} x \\ -x \end{bmatrix} \in \mathbb{R}^2 : x \in \mathbb{R}\}$, the subset of all vectors whose coordinates add up to zero. This is also a straight line.

You can quickly verify that any linear combinations of vectors in V should stay in V. Such things are called "subspaces" of $\mathbb{R}^2$. Similarly, W is also a subspace, as linear combinations of vectors in W should stay in W.

Sometimes, we do not care about the whole space $\mathbb{R}^2$. Rather, I just want to study a map sending elements of V to elements of W. Imagine that V is an elastic band, and I would like to "relocate" this elastic band into the position of W, and then stretch it by a scale of three.

In effect, we have a map $f : V \to W$ that sends $\begin{bmatrix} x \\ x \end{bmatrix}$ to $\begin{bmatrix} 3x \\ -3x \end{bmatrix}$. You can quickly verify that

$$f(a\begin{bmatrix} x \\ x \end{bmatrix} + b\begin{bmatrix} y \\ y \end{bmatrix}) = \begin{bmatrix} 3ax + 3by \\ -3ax - 3by \end{bmatrix} = af(\begin{bmatrix} x \\ x \end{bmatrix}) + bf(\begin{bmatrix} y \\ y \end{bmatrix}).$$

So our map f is linear.

However, note that V, W are both lines (one-dimensional). So, should f be some 1×1 matrix? But that would be absurd. It sends a vector with two coordinates to a vector with two coordinates. Maybe it is a 2×2 matrix? But then its domain and codomain are both one-dimensional. Hmm.

Now, you may say "screw this, I'm just going to use a 2×2 matrix to represent f." Fine. Note that the matrix $\begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix}$ would indeed send every $\begin{bmatrix} x \\ x \end{bmatrix}$ to $\begin{bmatrix} 3x \\ -3x \end{bmatrix}$. Aha! Is this the matrix for f?

However, also note that the matrix $\begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$ would indeed send every $\begin{bmatrix} x \\ x \end{bmatrix}$ to $\begin{bmatrix} 3x \\ -3x \end{bmatrix}$. And so does the matrix $\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, which is not even invertible, and so does infinitely many other matrices. And all these matrices have very different properties and behaviors. Yet they somehow ALL represent the linear map f? Impossible!

The trouble is again related to the standard basis. When we write elements of V, W in coordinates, we are expressing them as linear combinations of the standard basis, i.e., $\begin{bmatrix} a \\ b \end{bmatrix}$ really stands for $a\boldsymbol{e}_1 + b\boldsymbol{e}_2$. However, the standard basis vectors are OUTSIDE of V or W.

As you can imagine, it might NOT be a good idea to express elements inside of V as linear combinations of vectors outside of V. By using outside vectors to express inside vectors, V, W are one-dimensional spaces, yet their elements have more than one coordinates. You have redundant information flying around. And everything is messed up accordingly in the ensuing calculations.

In short, you CANNOT rely on the regular "coordinate" argument anymore, and you should abandon the standard basis. In other words, you would have to think of V, W as abstract vector spaces.

Maybe we can just name vectors in V, W according to their distance to the origin. Then $\begin{bmatrix} a \end{bmatrix} \in V$ now represents a vector in V with distance a to the origin (upper right being positive), and $\begin{bmatrix} b \end{bmatrix} \in W$ now represents a vector in W with distance b to the origin (lower right being positive). In this way, we can write a matrix $L = \begin{bmatrix} 3 \end{bmatrix}$, and it just multiply this distance by 3. ☺

Example 5.1.3 (Change of basis)**.** Let us prove that the three medians of a triangle are concurrent, i.e., sharing the same intersection point. I'm sure you have a bazzilion ways to do it in highschool. However, let us try to do it with brute force computation!

Now, generically, our triangle ABC lies in $\mathbb{R}^2$, but the three vertices could be anywhere! So the coordinate should be like $\begin{bmatrix} a_x \\ a_y \end{bmatrix}$, $\begin{bmatrix} b_x \\ b_y \end{bmatrix}$ and $\begin{bmatrix} c_x \\ c_y \end{bmatrix}$, where all six numbers are unknown. Now, we might try to just go ahead and compute with this. We may find the three midpoints, and then find the line equations of the three median, and then intersect them to see what happens. Or, we might as well kill ourselves, since this would surely be one ugly computation process. Let us give up for now.

So what can we do? Well, a better way is to "forget" about your old coordinate system. Just erase your x-axis and y-axis and such. And we shall simply re-define our coordinate system. How? Well, a coordinate system requires three things: You need to pick an origin, then you need to pick the directions for your x-axis and y-axis, i.e., you need to pick the vectors $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$.

Let us now define the origin to be A. Nice! Now A has coordinates $\begin{bmatrix}0\\0\end{bmatrix}$. Then let us define $\begin{bmatrix}1\\0\end{bmatrix}$ to be the vector AB and $\begin{bmatrix}0\\1\end{bmatrix}$ as the vector AC. Then now $B=\begin{bmatrix}1\\0\end{bmatrix}$ and $C=\begin{bmatrix}0\\1\end{bmatrix}$.

Huh, now the calculations are super easy. The three mid points of the three sides of the triangle are $D=\begin{bmatrix}1/2\\1/2\end{bmatrix}, E=\begin{bmatrix}0\\1/2\end{bmatrix}, F=\begin{bmatrix}1/2\\0\end{bmatrix}$. The three medians are $AD: x=y, BE: x+2y=1$, and $CF: 2x+y=1$. You can check that all three lines went through the point $\begin{bmatrix}1/3\\1/3\end{bmatrix}$. End of proof.

In other words, if the original coordinates (tied to the standard basis) suck, then we would like to change coordinate (by using a different set of vectors as "basis"). As a result of that, $\mathbb{R}^n$ is no longer the same $\mathbb{R}^n$, and every vector before and after the basis-change would have different coordinates. Nevertheless, the underlying ABSTRACT space and abstract vectors are the same space. The space did not change at all, and the vectors did not change at all. We simply changed the names (coordinates).

I hope this example illustrates the difference between ABSTRACT entities (independent of coordinates), and the NOMINAL entities (dependent of coordinates). ☺

So what is abstraction? My favorite explanation is one by Aristotle. He thinks that abstraction is about "forgetting intentionally". For example, say I am looking at a football, and I'm trying to reach the abstract concept of "sphere". What should I do? According to Aristotle, I should intentially "forget" its color, "forget" its smell, "forget" its texture and its location, and so on until the only thing that I still remember is its shape, the sphere.

(You see, I have a terrible memory, and I forget things all the time. That is probably why I am good at mathematics.)

But how can forgetting things help us? In the example above, we intentionally FORGET the origin and the coordinate axes, so that we can re-pick our own. Given $\mathbb{R}^n$, if you forget where the origin is, and you also forget where the standard basis vectors are, then you would reach some abstract entity.

In our case, according to the level of abstraction, we have the following concepts:

1. $\mathbb{R}^n$ is, well, $\mathbb{R}^n$. We know it already. It comes with a GIVEN origin, and a standard basis $\boldsymbol{e}_1, ..., \boldsymbol{e}_n$. Now, the location of the origin and the direction of these standard basis elemtents might be inconvenient, but there's nothing much you can do about it, because at the level of $\mathbb{R}^n$, coordinates must be fixed, and the standard basis vectors must be fixed.

2. Now, we can choose to "forget" where the basis is. Then we arrive at an abstract vector space V, i.e., it is like $\mathbb{R}^n$, but we forget where the standard basis vectors are. Then as a result, you CANNOT write things in coordinates anymore, since you no longer remember which vectors are the standard basis vectors. Given an element of V, you may write $\boldsymbol{v} \in V$, but you do not have a coordinate expression. Nevertheless, given $\boldsymbol{v}, \boldsymbol{w}$, you can do linear combinations of them just fine. You would still have $2\boldsymbol{v} + \boldsymbol{v} = 3\boldsymbol{v}$, and you don't need coordinates to do that.

3. (Optional) Now starting from abstract vector spaces, you can further "forget" where the origin is. Oops. Then we arrive at an ***affine space***. It is just like $\mathbb{R}^n$, but not only you forget to remember where the basis vectors are, you also forget where the origin is. In particular, you CANNOT add vectors or scale vectors now. Why? For example, scaling $\boldsymbol{v}$ means the arrow from the origin to $\boldsymbol{v}$ is tripled in length, but we don't know where the origin is! So what can we do? Well, for example, we

can find "mid-points". Given two vector $\boldsymbol{v}, \boldsymbol{w}$, you can find their midpoint by doing $\frac{1}{2}\boldsymbol{v} + \frac{1}{2}\boldsymbol{w}$, and geometrically you can see that this process does NOT require the origin. It is essentially still the same set as $\mathbb{R}^n$ with the same algebraic structure, but you just forget where the origin is. (Our class does NOT study affine spaces. You can check out the subject of affine geometry if you like.)

So, why abstract? Why forget? The short answer is that we forget in order to remember, or re-choose. The more you forget, the more freedom you get to enjoy.

Hopefully now you have a vague sense of what abstract vector spaces are. On the plane, we can fix a vector $\boldsymbol{v}$. Under some basis, it will have a "name" as maybe $\begin{bmatrix}1\\2\end{bmatrix}$. Under a different basis, it will have a different "name" as maybe $\begin{bmatrix}3\\4\end{bmatrix}$. But despite the different names, the underlying vector is the same $\boldsymbol{v}$. We want to focus on the essential vector rather than to focus on the superficial name.

5.2 How to study a vector space

In short, a vector space is a set V where the concept of "linear combination" is defined, and linear combinations of elements are still elements of V. In particular, we need to check the following two conditions, then we can say V is a vector space over a field $\mathbb{F}$:

1. If $\boldsymbol{v}, \boldsymbol{w} \in V$, then we need to verify that $\boldsymbol{v} + \boldsymbol{w} \in V$.
2. If $\boldsymbol{v} \in V$, and we pick any scalar $k \in \mathbb{F}$, then we need to verify that $k\boldsymbol{v} \in V$. (Here $\mathbb{F}$ is usually $\mathbb{R}$ or $\mathbb{C}$, depending on how we define V.)

Example 5.2.1. These are note vector spaces.

1. (Addition not closed) Take the dual cone $z^2 = x^2 + y^2$ in the space $\mathbb{R}^3$. Here $\pm\boldsymbol{e}_1 + \boldsymbol{e}_3$ are both in it, but their sum $2\boldsymbol{e}_3$ is NOT in this cone. So addition is not defined on the cone. (And can only be defined on the larger set $\mathbb{R}^3$.) (However, scalar multiplication is fine though.)
2. (Scalar multiplication not closed) Take the first quadrant of $\mathbb{R}^2$. This is NOT a vector space, because if $\boldsymbol{v} \neq \boldsymbol{0}$ is in it, then $(-1)\boldsymbol{v}$ is NOT in it. So scalar multiplication is NOT defined on this set, but rather, only on the larger set $\mathbb{R}^2$.

☺

(The rules above are OK 90% of the time. However, to have a more rigorous proof of something being a vector space, then you will need to read the optional sections about axioms of vector spaces.)

Example 5.2.2. Here is a nice contrast. In particular, a vector space must contain the zero vector.

1. Take $W = \{\begin{bmatrix}x\\y\\z\end{bmatrix} : x + y + z = 0\}$. This is a vector space, which you can verify yourself.
2. Take $W = \{\begin{bmatrix}x\\y\\z\end{bmatrix} : x + y + z = 1\}$. This is NOT a vector space, since $\boldsymbol{e}_1, \boldsymbol{e}_2 \in W$ but $\boldsymbol{e}_1 + \boldsymbol{e}_2 \notin W$. And in particular, $\boldsymbol{0} \notin W$. (This is a translation of the vector space above. These are called affine spaces, not vector spaces.)

☺

Elements of an abstract vector space V will not have innate coordinates. So if we want to do computations, there is usually only one way to study a vector space: by finding a basis for it, and thus turn it into $\mathbb{R}^n$.

Definition 5.2.3. *We say a collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in V$ is a* ***basis*** *of V (plural form is "bases") if any vector of V is a UNIQUE linear combination of them.*

Definition 5.2.4. *We say a vector space V has dimension k if it has a basis made of k vectors.*

Remark 5.2.5. *In our class we require a basis to only contain finitely many elements. We say a space is finite dimensional if there is a basis. For infinite dimensional spaces, they might not have a basis at all. We may construct something called a Hammel basis, but there will be some curious situations that is very different from finite dimensional spaces.*

Example 5.2.6. Our favorite example of a vector space is of course $\mathbb{R}^n$, and our favorite basis for $\mathbb{R}^n$ is the standard basis, i.e., $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$. Every vector is a unique linear combination of them.

In fact, if $\boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, then what are the coordinates $v_1, \ldots, v_n$? They are precisely the coefficents when we write $\boldsymbol{v}$ as a linear combination of the standard basis vectors. ☺

Example 5.2.7. Basic calculus shows that the solutions to the differential equation $f''(x) = -f(x)$ must be $a \sin x + b \cos x$ for some $a, b \in \mathbb{R}$. So the solution space is the space of functions $\{a \sin x + b \cos x : a, b \in \mathbb{R}\}$, and you can see that this is a vector space. A basis is made of $\sin x, \cos x$, so this space has dimension two. ☺

Example 5.2.8. Consider the space of 2×3 matrices. A basis is made of the following matrices

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We can also say the basis vectors here are $\boldsymbol{e}_i \boldsymbol{e}_j^{\mathrm{T}}$ for $1 \leq i \leq 2$ and $1 \leq j \leq 3$. This is actually the preferred way to write the basis above.

Therefore, this is a six-dimensional space. ☺

Example 5.2.9. Given three objects, say apple, banana and cherry, a person's preferences about them can be written as a "preference circle". For example, the following graph indicates that a person likes apple more than banana, and banana more than cherry.

$$\begin{tikzcd}
Apple \arrow[rr, bend left, "1"] & & Banana. \arrow[dl, bend left, "1"] \\
& Cherry \arrow[ul, bend left, "-1"] &
\end{tikzcd}$$

Here 1 means the arrow goes from the more preferred option to the less preferred option, and -1 means the opposite. A person can have six possible preferences, see if you can write them all out. (Note that a person's preference must be contradiction-free, i.e., transitive. If I prefer A to B, and B to C, then I prefer A to C.)

Now, given many people with their preference cycles, we can add them "coordinate-wise" like this:

$$\begin{tikzcd}
Apple \arrow[rr, bend left, "a_1"] & & Banana \arrow[dl, bend left, "b_1"] \\
& Cherry \arrow[ul, bend left, "c_1"] &
\end{tikzcd}
+
\begin{tikzcd}
Apple \arrow[rr, bend left, "a_2"] & & Banana \arrow[dl, bend left, "b_2"] \\
& Cherry \arrow[ul, bend left, "c_2"] &
\end{tikzcd}
=
\begin{tikzcd}
Apple \arrow[rr, bend left, "a_1+a_2"] & & Banana. \arrow[dl, bend left, "b_1+b_2"] \\
& Cherry \arrow[ul, bend left, "c_1+c_2"] &
\end{tikzcd}$$

In effect, we are polling the results of peoples preference, to get the preference of the whole population. Say if we add up the preference cycles of many people, and it turns out that the arrow from apple to banana is positive, this means more people prefer apple to banana than the oppositive. As you can see, this gives a vector space, the "population preference space". Can you find a basis made of three vectors?

A basis is in fact made of these three:

$$\begin{tikzcd} Apple \arrow[rr, bend left, "1"] & & Banana \arrow[dl, bend left, "1"] \\ & Cherry \arrow[ul, bend left, "-1"] & \end{tikzcd} \quad \begin{tikzcd} Apple \arrow[rr, bend left, "-1"] & & Banana \arrow[dl, bend left, "1"] \\ & Cherry \arrow[ul, bend left, "1"] & \end{tikzcd} \quad \begin{tikzcd} Apple \arrow[rr, bend left, "1"] & & Banana. \arrow[dl, bend left, "-1"] \\ & Cherry \arrow[ul, bend left, "1"] & \end{tikzcd}$$

To see uniqueness, say we have an arbirary preference population cycle such as this:

$$\begin{tikzcd} Apple \arrow[rr, bend left, "a"] & & Banana. \arrow[dl, bend left, "b"] \\ & Cherry \arrow[ul, bend left, "c"] & \end{tikzcd}$$

We can decompose it uniquely as a linear combination of the previous basis with coefficients $\frac{a+b}{2}, \frac{b+c}{2}, \frac{c+a}{2}$.

Also, here is a fun thing to try: try to combine many preference cycles such that all three arrows are positive. So in your population, more people like apple than banana, and more people like banana than cherry, and more people like cherry than apple. This is called a voting paradox, and it can actually happen. (This shows why american elections are flawed, when you are forced to choose between two candidates rather than from all candidates. Politicians exploit this all the time.) ☺

The point of finding a basis is to find coordinates.

Definition 5.2.10. *Given a vector space V and a basis $\mathcal{B} = (\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$, then for each vector $\boldsymbol{v} \in V$, if $\boldsymbol{v} = a_1\boldsymbol{v}_1 + \cdots + a_n\boldsymbol{v}_n$, then we say the **coordinates** of $\boldsymbol{v}$ under $\mathcal{B}$ is $\boldsymbol{v}_{\mathcal{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$.*

*We call the map $\boldsymbol{v} \to \boldsymbol{v}_{\mathcal{B}}$ as the **coordinate map** under the basis $\mathcal{B}$.*

Remark 5.2.11. *(IMPORTANT) Note that the ORDER of the basis vectors will matter! If we change the order of the basis vectors, we are changing the order of the coordinates for all vectors!*

Example 5.2.12. In the solution space V to the differential equation $f''(x) = -f(x)$, if we pick $\sin x$ and $\cos x$ as basis $\mathcal{B}$, then the solution $g(x) = 2\sin x + 2\cos x$ will have coordinates $g_{\mathcal{B}} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

However, you may check that $\sin(x + \frac{\pi}{4}), \cos(x + \frac{\pi}{4})$ is also a basis $\mathcal{C}$ for V. (Because $\sin(x + \frac{\pi}{4}) = \frac{\sqrt{2}}{2}\sin(x) + \frac{\sqrt{2}}{2}\cos(x)$ and $\cos(x + \frac{\pi}{4}) = \frac{\sqrt{2}}{2}\cos(x) - \frac{\sqrt{2}}{2}\sin(x)$.) Under this basis, the coordinates for $g(x)$ will now be $g_{\mathcal{C}} = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}$.

As you can see, coordinates are ephemeral, i.e., they change all the time. Think of the vector g as a person, and coordinates of g as "names" of g. A person can change its name all the time, but it is still the same person. When we want to understand a solution to the differential equation $f''(x) = -f(x)$, it is usually better to deal with the abstract vector g. The coordinates are only temporary tools to help us do some calculations, but they are not innate to g. Coordinates depends on our choice of basis $\mathcal{B}$, and $\mathcal{B}$ has nothing to do with g. ☺

We now see some examples of vector spaces for some very important distinctions.

Example 5.2.13 (Subspaces of $\mathbb{R}^n$)**.** Consider $\mathbb{R}^3$ and the plane W defined by $x+y+z=0$. If $\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ lies on this plane (i.e., $x_1+y_1+z_1 = x_2+y_2+z_2 = 0$), then obviously their sum still lies on this plane, and their scalar multiples would also lie on this plane. So this is a subspace.

Any line through the origin in $\mathbb{R}^n$ is a vector space. Any plane through the origin is a vector space. Any solution set to $A\boldsymbol{x} = \mathbf{0}$ for any matrix A is a vector space, because $A\boldsymbol{x} = \mathbf{0}$ and $A\boldsymbol{y} = \mathbf{0}$ would imply that $A(s\boldsymbol{x} + t\boldsymbol{y}) = \mathbf{0}$.

In fact, these are ***subspaces*** of $\mathbb{R}^n$, because they are subsets who are also vector spaces using the vector addition and scalar multiplication inherited from $\mathbb{R}^n$.

In contrast, take the plane $x+y+z=1$ in $\mathbb{R}^3$. Then $\boldsymbol{e}_1, \boldsymbol{e}_2$ is in it, but $2\boldsymbol{e}_1, \boldsymbol{e}_1 + \boldsymbol{e}_2$ are not in it. This is NOT a vector space. In particular, any subspace must contain the origin of the ambient space.

In short, only planes (and lines and hyperplanes and so on) that contains the origin can be subspaces. Those that do not contain the origin are not. (They are called affine spaces, because they are translations of vector spaces.) ☺

Example 5.2.14 (Coordinates and Nature)**.** Consider $\mathbb{R}^3$ and the plane W defined by $x+y+z=0$. This is a subspace, and as a plane it should have dimension two.

Now pick an element of W, say $\boldsymbol{w} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. Wait, if $\boldsymbol{w} \in W$, and W is two dimensional, then why would $\boldsymbol{w}$ have three coordinates, not two?

The correct view is to treat $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ not as coordinates, but rather as the nature of $\boldsymbol{w}$. To find the coordinates of $\boldsymbol{w}$ in W, we first need to pick a basis for W. Say the basis $\mathcal{B}$ made of $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. Then the coordinates of $\boldsymbol{w}$ are $\boldsymbol{w}_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, because $\boldsymbol{w}$ is the sum of the first basis vector and twice the second basis vector.

When we are dealing with vectors in $\mathbb{R}^n$, since we almost always us the standard basis, the nature of any vector $\boldsymbol{v} \in \mathbb{R}^n$ would coincide with the coordinates of $\boldsymbol{v}$ under the standard basis. However, if we are dealing with a vector $\boldsymbol{w}$ in some subspace W of $\mathbb{R}^n$, then the nature of $\boldsymbol{w}$ is a list of n real numbers, but the coordinates of $\boldsymbol{w}$ will depend on our choice of basis for W, and the coordinates will be a list of $\dim W$ real numbers. ☺

Here is a definition used above.

Definition 5.2.15. *A subset W of a vector space V is a* ***subspace*** *if it is also a vector space itself, using the same vector addition and scalar multiplication as V.*

Here let us see some more exotic examples of subspaces. In particular, a set may be both a vector space and NOT a vector space, depending on how you define your vector addition and scalar multiplication.

Example 5.2.16. (Optional)

Consider the plane $x+y+z=1$ in the space $\mathbb{R}^3$. We already know that this is NOT a vector space under usual vector addition and scalar multiplication. But what if we change the definition of scalar multiplication and vector addition?

Suppose we define $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \boxplus \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a+x-1 \\ b+y \\ c+z \end{bmatrix}$, and $k \boxtimes \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} ka-k+1 \\ kb \\ kc \end{bmatrix}$. Then you can go ahead and verify that this gives a vector space structure for the plane $x+y+z=1$.

This is not an aribtrary structure. This structure is called "declare $\begin{bmatrix}1\\0\\0\end{bmatrix}$ to be the origin", and thus transforming an affine space into a vector space. Can you see why? Check that $\boldsymbol{e}_1$ is indeed the "zero vector" for $\boxplus$.

Let V be the plane $x+y+z=1$ with $\boxplus, \boxtimes$ as its operations, then it is a vector space. Let W be the plane $x+y+z=0$, so it is also a vector space. Consider the map $f: W \to V$ that goes like $f(\boldsymbol{x}) = \boldsymbol{x}+\boldsymbol{e}_1$. Then you can in fact verify that $f(a\boldsymbol{x}+b\boldsymbol{y}) = a \boxtimes f(\boldsymbol{x}) \boxplus b \boxtimes f(\boldsymbol{y})$. So this f is a linear map. In fact, this is a bijection with inverse $g(\boldsymbol{x}) = \boldsymbol{x}-\boldsymbol{e}_1$.

In short, the "vector space" V is obtained by shifting W while preserving its linear structure faithfully. ☺

Remark 5.2.17. *Here the requirement is used to rule out the following weird phenomena: Sometimes V is a vector space, and under the vector addition and scalar multiplication of V, W is NOT a subspace. (Say $V=\mathbb{R}^3$ and W is the plane given by $x+y+z=1$.)*

However, we may define some weird and exotic vector addition and scalar multiplication, which shall make W into a vector space. (We shall see this construction later.) Then W by itself is now a vector space. Nevertheless, it is still NOT a subspace of V, because it does not have the same vector addition and scalar multiplication as V. In fact W will never be a subspace of V.

Proposition 5.2.18. *A subset W of V is a subspace of V if and only if it is closed under vector addition and scalar multiplication of V. (In particular, for any $\boldsymbol{v},\boldsymbol{w}\in V$ and $k\in\mathbb{R}$, we should have $\boldsymbol{v}+\boldsymbol{w}, k\boldsymbol{v}\in V$.)*

All subspaces of $\mathbb{R}^n$ are quite easy to spot. They are all k-dimensional flat things that go through the origin.

Corollary 5.2.19. *A subspace W must contain the zero vector of V. And the* ***inclusion map*** *$\iota: W\to V$ defined as $\boldsymbol{w}\mapsto\boldsymbol{w}$ is linear. (As a side note, the inclusion map is always injective, but maybe not surjective, unlike the identity map.)*

Note that, if W uses a different definition of vector addition and scalar multiplication, then the inclusion map is not going to be linear any more.

Definition 5.2.20 (Useful Subspaces for Matrices)**.** *Given an $m\times n$ matrix A, the following four subspaces are sometimes called the* ***fundamental subspaces for*** *A.*

1. $\mathrm{Ran}(A)$*, the* ***range*** *or* ***column space*** *which contains all vectors that are images of the linear map A. (I.e., all possible linear combinations of columns of A). This is a subspace of $\mathbb{R}^m$.*
2. $\mathrm{Ker}(A)$*, the* ***kernel*** *or* ***zero set*** *or* ***null space****, which contains all solutions $\boldsymbol{x}$ to $A\boldsymbol{x}=\boldsymbol{0}$. This is a subspace of $\mathbb{R}^n$.*
3. $\mathrm{Ran}(A^{\mathrm{T}})$ *is the row space of A. (And also the column space of A^{T}.)*
4. $\mathrm{Ker}(A^{\mathrm{T}})$ *is the left kernel of A. (And also the kernel of A^{T}.) This is called the left kernel of A because $\boldsymbol{x}\in\mathrm{Ker}(A^{\mathrm{T}})$ if and only if $\boldsymbol{x}^{\mathrm{T}}A=\boldsymbol{0}^{\mathrm{T}}$.*

Proposition 5.2.21. $\mathrm{Ran}(A)$ *and* $\mathrm{Ker}(A)$ *are indeed subspaces.*

Proof. Take any $\boldsymbol{v},\boldsymbol{w}\in\mathrm{Ran}(A)$ and any $a,b\in\mathbb{R}$. Then by definition of range, $\boldsymbol{v}=A\boldsymbol{x}$ for some $\boldsymbol{x}$ and $\boldsymbol{w}=A\boldsymbol{y}$ for some $\boldsymbol{y}$. Then $a\boldsymbol{v}+b\boldsymbol{w}=A(a\boldsymbol{x}+b\boldsymbol{y})\in\mathrm{Ran}(A)$. So this is indeed a subspace.

Take any $\boldsymbol{v},\boldsymbol{w}\in\mathrm{Ker}(A)$ and any $a,b\in\mathbb{R}$. Then by definition of range, $A\boldsymbol{v}=A\boldsymbol{w}=\boldsymbol{0}$. Then $A(a\boldsymbol{v}+b\boldsymbol{w})=aA\boldsymbol{v}+bA\boldsymbol{w}=a\boldsymbol{0}+b\boldsymbol{0}=\boldsymbol{0}$. □

Proposition 5.2.22. *For an $m\times n$ matrix A, the corresponding linear map is surjective if and only if* $\mathrm{Ran}(A)=\mathbb{R}^m$*. It is injective if and only if* $\mathrm{Ker}(A)=\{\boldsymbol{0}\}$*.*

Proof. The first statement is literally the definition of surjectivity.

For the second statement, if A is injective, then $A\boldsymbol{x} = \mathbf{0}$ will only have one solution. Therefore it has to be $\boldsymbol{x} = \mathbf{0}$. So $\operatorname{Ker}(A) = \{\mathbf{0}\}$. Conversely, suppose $\operatorname{Ker}(A) = \{\mathbf{0}\}$. Suppose $A\boldsymbol{v} = A\boldsymbol{w}$, then $A(\boldsymbol{v} - \boldsymbol{w}) = \mathbf{0}$, and hence $\boldsymbol{v} - \boldsymbol{w} \in \operatorname{Ker}(A)$ and it has to be the zero vector. So $\boldsymbol{v} = \boldsymbol{w}$. Hence A is injective. ☐

Now we move away from subspaces, and onto more abstract spaces and linear maps.

Example 5.2.23 (Spaces of Matrices)**.**

1. All $m \times n$ matrices is a vector space $M_{m\times n}$, because we can add matrices and scale matrices. This space is mn dimensional, and a basis is made of matrices of the form $\boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}}$, i.e., it has only a single entry 1 and all other entries are zeros.
2. In fact, given any vector spaces V, W, the set of linear maps from V to W is a vector space $\mathcal{L}(V, W)$. We can add linear maps or scale a linear map in the obvious manner. (I.e., $(f + g)(\boldsymbol{v}) = f(\boldsymbol{v}) + g(\boldsymbol{v}), (kf)(\boldsymbol{v}) = k(f(\boldsymbol{v}))$.)
3. The ***trace*** of a square matrix is the sum of its diagonal entries. This is a linear map from $M_{n\times n} \to \mathbb{R}$.
4. The transpose is a linear map from $M_{m\times n}$ to $M_{n\times m}$.
5. The space of all $n \times n$ upper triangular matrices is a vector space and in fact a subspace of $M_{n\times n}$. Can you find its dimension and find a basis?
6. Transpose can be restricted to a map from the space of upper triangular matrices to the space of lower triangular matrices. (This will be a linear bijection.)
7. The space of unit upper triangular matrices is NOT a vector space. If U_1, U_2 are unit upper triangular, then $U_1 + U_2$ is NOT. (It also does not contain the zero matrix, hence not a subspace.)
8. The space of all $n \times n$ invertible matrices is NOT a vector space. If A is invertible, then $A - A$ is not. (It also does not contain the zero matrix, hence not a subspace.)
9. The space of all $n \times n$ symmetric matrices is a vector space. Can you find its dimension and find a basis? The transpose map restricted to this domain and codomain is the identity map.
10. Matrix inversion is NOT a linear map, because in general, $(A + B)^{-1} \neq A^{-1} + B^{-1}$.
11. Let us say a 3×3 matrix is a ***magic matrix*** if entries in each row, each column and each of the two diagonals add up to the same number. For example, we have the famous $\begin{bmatrix} 2 & 9 & 4 \\ 7 & 5 & 3 \\ 6 & 1 & 8 \end{bmatrix}$. Then all such matrices form a subspace of $M_{3\times 3}$. Can you see why?

☺

Example 5.2.24 (Spaces of Matrices)**.**

1. Consider the derivative map $D : V \to V$, where V is the space of infinitely differentiable functions. Then $D^2 + I$ is also a linear map from V to V, such that $(D^2 + I)f = D^2 f + If = f'' + f$. In particular, $\operatorname{Ker}(D^2 + I)$ is the solution set to the differential equation $f'' = -f$.
2. Consider the transpose map $T : M \to M$, where M is the space of $n \times n$ matrices. Then $I - T : M \to M$ is also a linear map, where $(I - T)(A) = A - A^{\mathrm{T}}$. So $\operatorname{Ker}(I - T)$ is the space of symmetric matrices, and $\operatorname{Ran}(I - T)$ is in fact the space of skew-symmetric matrices, i.e., matrices A such that $A^{\mathrm{T}} = -A$.

☺

So far, we have been concerning ourselves with real vector spaces. However, we sometimes might want to change the realm of coefficients, say maybe we want to allow all complex numbers to be coefficients. This will NOT affect vector addition, but when we do scalar multiplications, we can scale by more choices.

Example 5.2.25. 1. The set of complex numbers $\mathbb{C}$ is a real vector space (of dimension 2 and basis 1 and i), since we can do real linear combinations of complex numbers.

2. However, the set of complex numbers $\mathbb{C}$ is also a complex vector space. If we allow all complex numbers to be scalars, then all elements of $\mathbb{C}$ are scalar multiples of 1. So 1 is itself a basis for $\mathbb{C}$, and $\mathbb{C}$ has complex dimension one. We can use $\mathbb{C}^n$ to denote the standard n-dimensional complex vector spaces, and everything works pretty much the same as $\mathbb{R}^n$, except that now we have complex numbers in place of real numbers.

3. The set of real numbers $\mathbb{R}$ is an infinite dimensional vector space over the set of rational numbers $\mathbb{Q}$. This example is just for fun.

☺

A large part of calculus concerns linear maps on function spaces. These are infinite dimensional spaces

Example 5.2.26. 1. The space of all continuous functions from any domain D to $\mathbb{R}$, $\mathcal{C}(D)$, is a vector space. Here we add or scale functions in the obvious manner, i.e., $(f+g)(x) = f(x)+g(x)$ and $(kf)(x) = kf(x)$ for all $x \in D$. Here D can be $\mathbb{R}$, or any interval, or the set of all people with f sending each person to their age/height/salary, etc..

2. The space of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ is a vector space. Here we add or scale functions in the obvious manner.

3. The space of all smooth functions (i.e., infinitely differentiable) from $\mathbb{R}$ to $\mathbb{R}$, $\mathcal{C}^\infty(\mathbb{R})$, is a vector space. Here we add or scale functions in the obvious manner. (Usually $\mathcal{C}^k(\mathbb{R})$ means the space of all continuously k-times differentiable real functions.)

4. The differentiation map $\frac{d}{dx}$ from the space of continuously differentiable real function $\mathcal{C}^1(\mathbb{R})$ (functions whose derivatives are continuous) to the space of continuous functions $\mathcal{C}(\mathbb{R})$ is a linear map.

5. The integration map from a to b for any $a, b \in \mathbb{R}$ is a linear map $\int_a^b$ from the space $\mathcal{C}([a,b])$ to $\mathbb{R}$. (Just an optional side note. If vectors in $\mathbb{R}^n$ are "column vectors", then linear maps $\mathbb{R}^n \to \mathbb{R}$ are "row vectors". In this perspective, elements (functions) in $\mathcal{C}([a,b])$ are "columns", then the integration $\int_a^b$ is a "row".)

6. The limit operation $\lim_{n\to\infty}$ is a linear map from the space of convergent sequences to $\mathbb{R}$.

☺

5.3 Basis and Dimensions

5.3.1 Linear Combination Map and Coordinate Map

In advanced mathematics, teacher would try to show you that everything is a map. In linear algebra, I endeavor to show you that everything is a linear map. In particular, a basis is a linear map.

What does a basis do? If we fix a basis for a vector space V, then everything in the space will now have coordinates. Maybe some vector $\boldsymbol{v} \in V$ now have coordinates $\begin{bmatrix}1\\2\\3\end{bmatrix}$. Maybe some other vector $\boldsymbol{w} \in V$ now have coordinates $\begin{bmatrix}4\\5\\6\end{bmatrix}$. In particular, a basis for V is a map from V to $\mathbb{R}^n$, as it turns elements of V into elements of $\mathbb{R}^n$.

Given a vector space V and a basis $\mathcal{B}$, we can write each vectors in V into coordinates. This gives rise to a natural map:

Definition 5.3.1. *For any ordered basis $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ in a vector space V, we define the* ***coordinate map*** *to be the map $(-)_{\mathcal{B}}$ from V to $\mathbb{R}^n$ such that each input vector is sent to its coordinates under $\mathcal{B}$. I.e., we send $\boldsymbol{v}$ to $\boldsymbol{v}_{\mathcal{B}}$.*

Note that the order of the basis vectors is IMPORTANT, because if we change the order of basis vectors, the resulting coordinates will also change order.

Let us also define the inverse process.

Definition 5.3.2. *For any ordered collection of vectors $\mathcal{C} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ in a vector space V, we define the* ***linear combination map*** *to be the map from $\mathbb{R}^k$ to V such that input coordinates are used as coefficients to do linear combinations of these vectors.*

We sometimes lazily use $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ to denote this map, so we have $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k) \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix} = \sum a_i \boldsymbol{v}_i$, which agrees with our matrix multiplication instincts. We also simply write $\sum a_i \boldsymbol{v}_i = \mathcal{C} \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix}$ to denote this linear combination.

Note the similarity between this and matrix-vector multiplications. However, the map $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ is technically not a matrix, merely an abstract linear map. And $\boldsymbol{v}_i$ are abstract vectors that are usually NOT a column of numbers. Nonetheless, the i-th "column" $\boldsymbol{v}_i$ is exactly the image of $\boldsymbol{e}_i \in \mathbb{R}^k$ under the map $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$, which is why the formula looks familiar.

Example 5.3.3. Consider $V = \mathbb{R}^2$, and we have a collection $\mathcal{C} = (\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix})$. Then this $\mathcal{C}$ would induce a linear combination map $\mathcal{C} : \mathbb{R}^2 \to V$, such that

$$\mathcal{C} \begin{bmatrix} a \\ b \end{bmatrix} = (\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}) \begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

In particular, the matrix for $\mathcal{C}$ is simply $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$. The so-called linear combination map is what we've been doing all along. The i-th vector in the collection $\mathcal{C}$ is exactly the image of $\boldsymbol{e}_i$ under the linear combination map $\mathcal{C}$. ☺

From now on, if we say we pick a basis $\mathcal{B}$, then this is an ordered collection of vectors $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ that form a basis. Then we can also use the notation $\mathcal{B}$ for the linear combination map $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n) : \mathbb{R}^n \to V$. We no longer distinguish between a collection and its corresponding linear combination map. The two are the same thing, just like matrices and linear maps.

Now, for any vector $\boldsymbol{v} \in V$ with basis $\mathcal{B}$ for V, clearly $\boldsymbol{v}$ is the linear combinations of vectors in $\mathcal{B}$ with coefficents as $\boldsymbol{v}_{\mathcal{B}}$. In particular, we always have $\boldsymbol{v} = \mathcal{B}(\boldsymbol{v}_{\mathcal{B}})$ for all $\boldsymbol{v} \in V$. The composed process of $\boldsymbol{v} \mapsto \boldsymbol{v}_{\mathcal{B}} \mapsto \mathcal{B}(\boldsymbol{v}_{\mathcal{B}})$ is the same as the identity process. Therefore, we see that the coordinate map is always the inverse of the linear combination map.

Proposition 5.3.4. *For a vector space V and its basis $\mathcal{B}$, the coordinate map with respect to the basis $\mathcal{B}$ is the inverse of the linear combination map, i.e., the coordinate map is $\mathcal{B}^{-1}$. (If you like, we can say $\boldsymbol{v}_{\mathcal{B}}$ is a shorthand for $\mathcal{B}^{-1}\boldsymbol{v}$.)*

Proof. See discussion above. □

Corollary 5.3.5. *For a vector space V and its basis $\mathcal{B}$, the map $\mathcal{B} : \mathbb{R}^n \to V$ is a linear bijection. In particular, the coordinate map is also linear.*

Proof. $\mathcal{B}$ is linear because it is doing linear combinations. And $\mathcal{B}$ is bijective because it has an inverse. And finally, the coordinate map is the inverse of a linear map, so it is also linear. □

In short, what is a basis? A basis is a linear bijection that connectes V with $\mathbb{R}^n$. The converse is also true.

Proposition 5.3.6. *$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ form a basis for V iff the linear combination map $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is bijective.*

Proof. $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ being surjective means all vectors in V are linear combinations of these vectors.

$(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ being injective means distinct linear combinations of these vectors give distinct results. In particular, if a vector in V is a linear combination of these vectors, then the coefficients are unique.

Now the statement looks just like a trivial repetition of the definition.... □

Extracting from this proof, we have some really interesting new concepts.

Definition 5.3.7. *For any ordered collection of vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ in a vector space V, we say they are linearly independent if distinct linear combinations of them gives distinct results. (I.e., linear combination map $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is injective.)*

Definition 5.3.8. *For any ordered collection of vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ in a vector space V, we say they are spanning if all vectors of V are linear combination of them. (I.e., linear combination map $(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is surjective.) We also say that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ span the space V.*

Corollary 5.3.9. *$\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ form a basis iff it is linearly independent and spanning.*

These concepts are defined exactly the same as the cases in $\mathbb{R}^n$, and they behave the same way.

Example 5.3.10. Intuitively, linear independence means no redundancy.

Say your goal is to use vectors to span $\mathbb{R}^2$. (I.e., we want to use some vectors to express all other vectors.) You pick $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_1 + \boldsymbol{e}_2$. Oops, the last vector is redundant. So these three vectors are NOT linearly independent.

Spanning means we have enough vectors to span everything.

Together, if you have enough vectors to span everything, and none is redundant, then you have a basis. ☺

Example 5.3.11. In the context of $\mathbb{R}^n$, given a bunch of vectors, do they form a basis?

Say I give you $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\9\end{bmatrix}$. To see if they form a basis for $\mathbb{R}^3$, we just need to check if its linear combination map is bijective. The linear combination map of this collection is $\begin{bmatrix}1&4&7\\2&5&8\\3&6&9\end{bmatrix}$ and it is not bijective. So our vectors here do not form a basis for $\mathbb{R}^3$. ☺

5.3.2 Existence of Basis and Uniqueness of Dimension

The idea is to grab "good" vectors one by one, until they form a basis. However, we need to make sure that the vectors we grab are linearly independent. Here are some useful lemmas for future use.

Lemma 5.3.12 (Criteria for independence)**.** *TFAE:*

1. *Vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ are linearly independent.*
2. *$\sum a_i \boldsymbol{v}_i = \boldsymbol{0}$ implies that all $a_i = 0$.*
3. *None of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ is zero and none is a linear combination of the rest.*

Proof. The proofs here are not the most efficient ones. However, it might be beneficial for you to see more flavors of proofs.

We prove the equivalence cyclically. If they are linearly independent (i.e., all vectors has unique coordinates), then $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ is injective. So if $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\begin{bmatrix}a_1\\ \vdots\\ a_k\end{bmatrix} = \mathbf{0} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\mathbf{0}$, we must have $\begin{bmatrix}a_1\\ \vdots\\ a_k\end{bmatrix} = \mathbf{0}$, i.e., all a_i are zero.

Now suppose $\sum a_i\boldsymbol{v}_i = \mathbf{0}$ implies that all $a_i = 0$. Suppose for contradiction that $\boldsymbol{v}_i = \sum_{j\neq i} a_j\boldsymbol{v}_j$ (this include the case of $\boldsymbol{v}_i = \mathbf{0}$), then we have $\boldsymbol{v}_i - \sum_{j\neq i} a_j\boldsymbol{v}_j = \mathbf{0}$. And this is a linear combination of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ and the coefficients are not all zero. In particular, $a_i = 1 \neq 0$. Contradiction, Hence none of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is zero and none is a linear combination of the rest.

Now suppose none of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is zero and none is a linear combination of the rest. Let us prove the injectivity of $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$. If $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\begin{bmatrix}a_1\\ \vdots\\ a_k\end{bmatrix} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\begin{bmatrix}b_1\\ \vdots\\ b_k\end{bmatrix}$. Suppose for contradiction that some $a_i \neq b_i$, say WLOG (without loss of generality) that $a_1 \neq b_1$. Then $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\begin{bmatrix}a_1\\ \vdots\\ a_k\end{bmatrix} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\begin{bmatrix}b_1\\ \vdots\\ b_k\end{bmatrix}$ implies that $\mathbf{0} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)\begin{bmatrix}a_1 - b_1\\ \vdots\\ a_k - b_k\end{bmatrix} = \sum(a_i - b_i)\boldsymbol{v}_i$, and therefore $\boldsymbol{v}_1 = \sum_{i\neq 1}\frac{a_i-b_i}{a_1-b_1}\boldsymbol{v}_i$. So $\boldsymbol{v}_1 = \mathbf{0}$ or is a linear combination of the rest, which cannot happen. So we have $a_1 = b_1$, and similarly $a_i = b_i$ for all i. Hence $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ is injective, the vectors are linearly independent. □

Here the best tool is the second criterion. The third is hard to use, but it paints a nice conceptual picture, and explains the reason for the word "independence".

Let us see how the criterion is used in practice.

Example 5.3.13. I claim that the functions $\sin x, \cos x, \sin(2x), \cos(2x)$ are linearly independent. (Note that they are non-linearly dependent, e.g., $\sin^2 + \cos^2 = 1$ and so on. But linearly, they are independent.)

To see this, suppose there is a way to combine them to zero (the zero function), say

$$(\sin x, \cos x, \sin(2x), \cos(2x))\begin{bmatrix}a\\ b\\ c\\ d\end{bmatrix} = 0.$$

Then the same relation must be true at all possible values of x. Take $x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$, then we have

$$\begin{bmatrix}\sin(0) & \cos(0) & \sin(2\times 0) & \cos(2\times 0)\\ \sin(\frac{\pi}{4}) & \cos(\frac{\pi}{4}) & \sin(2\times\frac{\pi}{4}) & \cos(2\times\frac{\pi}{4})\\ \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) & \sin(2\times\frac{\pi}{2}) & \cos(2\times\frac{\pi}{2})\\ \sin(\frac{3\pi}{4}) & \cos(\frac{3\pi}{4}) & \sin(2\times\frac{3\pi}{4}) & \cos(2\times\frac{3\pi}{4})\end{bmatrix}\begin{bmatrix}a\\ b\\ c\\ d\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\\ 0\end{bmatrix}.$$

However, the matrix here is $\begin{bmatrix}0 & 1 & 0 & 1\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 & 0\\ 1 & 0 & 0 & -1\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & -1 & 0\end{bmatrix}$, which is invertible. So $\begin{bmatrix}a\\ b\\ c\\ d\end{bmatrix} = \mathbf{0}$. ☺

Lemma 5.3.14 (Independence extension lemma)**.** *(This is probably the MOST USEFUL theoretical lemma. People in real life use this to find basis for vector spaces they face in real problems.)*

If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ are linearly independent, and $\boldsymbol{v}_{k+1}$ is NOT a linear combination of them, then $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{k+1}$ are linearly independent.

Proof. Suppose $\sum_{i=1}^{k+1} a_i \boldsymbol{v}_i = \mathbf{0}$. Suppose $a_{k+1} \neq 0$, then we have $\boldsymbol{v}_{k+1} = \sum_{i=1}^{k}(-\frac{a_i}{a_{k+1}})\boldsymbol{v}_i$, impossible. So $a_{k+1} = 0$. But then we have $\sum_{i=1}^{k} a_i \boldsymbol{v}_i = \mathbf{0}$, so by linear independence of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$, we see all $a_i = 0$. ☐

So with this tool in mind, let us prove the existence of a basis and a dimension.

Definition 5.3.15. *A vector space is* ***finite dimensional*** *if it cannot have infinitely many linearly independent vectors. Otherwise it is* ***infinite dimensional****.*

Theorem 5.3.16. *A finite dimensional vector space V has a basis.*

Proof. If $V = \{0\}$, then by convension we say $\varnothing$ is its basis. This is just a convention.

If $V \neq \{0\}$, then pick any nonzero vector $\boldsymbol{v}_1$. If this single vector is spanning, then it is a basis all by itself, so we are done.

If it is not spanning, then there are vectors in V that is NOT a multiple of $\boldsymbol{v}_1$. Say let us pick any one, $\boldsymbol{v}_2$. Then by the independence extension lemma, $\boldsymbol{v}_1, \boldsymbol{v}_2$ is linearly independent. If it is now spanning, then we have found a basis, done.

If it is not spanning, then there are vectors in V that is NOT a linear combination of $\boldsymbol{v}_1, \boldsymbol{v}_2$. Say let us pick any one, $\boldsymbol{v}_3$. Then by the independence extension lemma, $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ is linearly independent. If it is now spanning, then we have found a basis, done.

If not, we just keep going. We cannot keep going forever, because by definition we cannot have infinitely many linearly independent vectors. This process must terminate somewhere, and when it terminates, we have a basis. ☐

Example 5.3.17. Let us see this in action. Consider the vector space $\mathcal{P}_2$, the space of polynomials of degree at most 2. How to find a basis?

First we pick any non-zero element, say the constant polynomial 1. Does this spans $\mathcal{P}_2$? Not yet. Obviously it only spans all constant polynomials.

So we pick anything that is NOT a constant polynomial, say x. Now, does $1, x$ span $\mathcal{P}_2$? Not yet. Their linear combination can reach any degree 0 or degree 1 polynomial, but they can never combine to give you a degree two polynomial.

So we pick anything that is NOT a degree 0 or degree 1 polynomial, say x^2. Now they span everything.

So we have a basis $1, x, x^2$ for our space $\mathcal{P}_2$.

Note that this gives a bijection, the linear combination map $(1, x, x^2) : \mathbb{R}^3 \to \mathcal{P}_2$ that sends $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ to $a + bx + cx^2$. Note that the "columns" of the map $(1, x, x^2)$ are not really columns, but merely abstract vectors.

Conversely, for any polynomial $a + bx + cx^2 \in \mathcal{P}_2$, we can try to find the coordinate of this polynomial under the basis $1, x, x^2$. This is obviously just $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Hence we have a ***coordinate map*** $C : \mathcal{P}_2 \to \mathbb{R}^3$ that sends $a + bx + cx^2$ to its coordinates $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ under the basis $1, x, x^2$. Obviously C and $(1, x, x^2)$ are inverse map of each other. We always have such a coordinate map, because by the definition of basis, $(1, x, x^2)$ is always a linear bijection. ☺

So it seems that any finite dimensional vector space V has a bijective linear map to some $\mathbb{R}^n$. We want to call this n the dimension of the space. But this needs one more theorem.

Theorem 5.3.18 (Dimension is well-defined)**.** *In a finite dimensional vector space, any two basis contain the same number of elements. And this number is called the* ***dimension*** *of this space.*

Proof. Suppose V has a basis with m vectors, then there is a bijective linear map between $\mathbb{R}^m$ and V. Now suppose V has another basis with n vectors. Then there is a bijective linear map between $\mathbb{R}^n$ and V. So there is a bijective linear map between $\mathbb{R}^n$ and $\mathbb{R}^m$, i.e., an invertible $m \times n$ matrix! But this could only happen when $m = n$. $\square$

So we have established the following idea: An abstract vector space is either infinite dimensional, or it has a basis of n vectors where n is by definition the dimension of this space. Note that a space usually have many many basis, but all of them has the same number of vectors.

Example 5.3.19. Another basis for $\mathcal{P}_2$ is $1, x+1, (x+1)^2$. For any polynomial $p(x)$, we can express it as a unique linear combination of these three. Note that $1 = 1, x = (x+1) - 1$ and $x^2 = (x+1)^2 - 2(x+1) + 1$, so everything that is spanned by $1, x, x^2$ is spanned by $1, x+1, (x+1)^2$. Furthermore, it is obvious that for $1, x+1, (x+1)^2$, each one is NOT a linear combination of previous ones. So by independence extension lemma they are linearly independent.

To find coordinates for $p(x) \in \mathcal{P}_2$, suppose $p(x) = a + b(x+1) + c(x+1)^2$. Then $p(-1) = a$ and $p'(-1) = b$ and $p''(-1) = 2c$. So the coordinates are $\begin{bmatrix} p(-1) \\ p'(-1) \\ \frac{p''(-1)}{2} \end{bmatrix}$ or $\begin{bmatrix} \frac{p(-1)}{0!} \\ \frac{p'(-1)}{1!} \\ \frac{p''(-1)}{2!} \end{bmatrix}$ if you like to generalize this to higher degree polynomials. (The similarity to Tayler expansion in calculus is NOT a coincidence, but out of the scope of this class. We don't explore that here in our linear algebra class.)

Take P_2, the space of polynomials of degree at most 2. Take a vector $\boldsymbol{w} = 3x^2 + 4x + 1$ and basis $1, x+1, (x+1)^2$. Then the linear combination map is $L : \mathbb{R}^3 \to P_2$ that sends $\boldsymbol{e}_1, \boldsymbol{e}_2$ and $\boldsymbol{e}_3$ to $1, x+1, (x+1)^2$. we have $L^{-1}(\boldsymbol{w}) = \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix}$ by our previous formula. You can check that we indeed have $3x^2 + 4x + 1 = -2(x+1) + 3(x+1)^2$.

The basis $1, x+1, (x+1)^2$ is preferable than $1, x, x^2$ when x is near -1 most of the time. ☺

So, it turns out that a vector space is either infinite dimensional, or essentially $\mathbb{R}^n$ because it has a linear bijection to $\mathbb{R}^n$. In linear algebra, we call a bijective linear map an ***isomorphism***. This is a very important concept in all of mathematics, it means two things are structurally the same, and they only differ in names.

Example 5.3.20. Most practically, isomorphisms allows us to "transfer" calculations. Say we have a bijective linear map $L : V \to W$. Then to do calculation in V, it is enough to do the corresponding calculation in W.

Suppose I want to add $\boldsymbol{v}_1 \in V$ and $\boldsymbol{v}_2 \in V$, but I don't know how to do this. I only know how to add things in W. Well, no worry. First I send then into W by applying L. Now I add $L(\boldsymbol{v}_1)$ and $L(\boldsymbol{v}_2)$, which I know how to do, because I'm now in W. Then I apply the inverse of L. So I see that $\boldsymbol{v}_1 + \boldsymbol{v}_2 = L^{-1}(L(\boldsymbol{v}_1) + L(\boldsymbol{v}_2))$.

Most practically, if we find a basis for V, this means we have a bijection $V \leftrightarrow \mathbb{R}^n$. Then any calculation we want to do in V, we can simply do it in $\mathbb{R}^n$ instead. Just write vectors of V in terms of coordinates, and now we are in $\mathbb{R}^n$, where everything is familiar. After we finish calculation in $\mathbb{R}^n$, just do the linear combination according to our chosen basis in V to go back to V. ☺

Remark 5.3.21 (Optional Remarks on Isomorphisms)**.** *This is the idea of* ***isomorphism****. For any two things with structures, we say they are isomorphic if, from some perspective (i.e., through some bijective function), I can almost pretend that they are the same thing (i.e., the bijective function happens to matches the structures on both sides exactly).*

For another example of an isomorphism, in Chinese we know that yi plus yi is er. Then I build a function, called a dictionary, that maps yi to one, er to two, and so forth. Then I see that one plus one is two, this is still correct. So the dictionary is a function that matches the mathematical structure of things. Then USING this isomorphism, if I prove some mathematical theorem in Chinese, then I don't really need to prove it in English again. I know the mathematical theorem will be true in English as well, I know this even without doing any translation myself. The very EXISTENCE of a dictionary already guarantees that the same mathematical theorem should be true in English as well.

The idea is this: as far as linear structures are concerned, the existence of a bijective linear map means the two things have IDENTICAL linear strucutre. So if I FIX a bijective map between two vector spaces, then I can PRETEND that they are identical. I'll be fine as long as I always go back and forth according to the same bijection.

And under this pretending, I can see that the two things are practically indistinguishable. Say there is a bijective linear map between V, W. Then if V has dimension n, then W must also have dimension n. If all triangles in V have concurrent medians, then all triangles in W have concurrent medians. If everyone living in V loves Chinese food in a linear way, then everyone living in W loves Chinese food in the same linear way. In short, whenever I prove some big theorem in V about its linear structure, then the same theorem must be true for W as well. Let me repead: as long as I FIX this bijective linear map, then they are indistinguishable, they are identical, there are no difference between them.

However, if one day, I decide to go back and forth using ANOTHER linear bijection between them, then all of a sudden, all previous "pretending" no longer works. This is just like if I pick another basis in V, then the expressions are now all different, and angles and length of vectors are different, and so on. It is like waking up from a dream....

So this is how it works. Say you prove something about V (say V has dimension n). To transfer this property to W, you first build a bijective linear map between them. Now W have the same property from the perspective of this linear map. Then you show that this property of W is INNATE, that it is really just about W itself, independent of any choice of basis and independent of how W is connected to other spaces. Then W has this property INDEPENDENT from the bijection of our choice. (You see why being independent of choice of basis is important.)

Coordinates of a vector, entries of a matrix, angles and dot products, these things are all DEPENDENT of basis. They are all illusions. The true linear property are those independent of basis. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, then the $(1,1)$ entry of A will change under a different basis, but the surjectivity of A shall never change. It will always remain surjective.

All in all, the correct way to say this is the following: Not all finite dimensional spaces are $\mathbb{R}^n$, but each of them looks like some $\mathbb{R}^n$ for some n.

Here are some easy result that follows from the power of isomorphism:

Theorem 5.3.22. *For any vector space V, suppose it has a basis of n elements. Then we have the following facts:*

1. *Every basis must have exactly n vectors.*
2. *A collection of more than n vectors must be linearly DEPENDENT.*
3. *A collection of less than n vectors cannot be spanning.*
4. *A collection is a basis if and only if it is linearly independent and has n vectors.*
5. *A collection is a basis if and only if it is spanning and has n vectors.*
6. *Every linearly independend collection of vectors can be extended to a basis.*
7. *Every spanning system contains a basis.*

Proof. First statement is already done.

For the second and third statement, pretend your space is $\mathbb{R}^n$, and we have a collection $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$. Then the linear combination map goes from $\mathbb{R}^k$ to $\mathbb{R}^n$, hence it is an $n \times k$ matrix. So if $k > n$, it cannot be injective (and the collection cannot be linearly independent). If $k < n$, then it cannot be surjective (and the collection cannot be spanning).

For the fourth and fifth statement, this is because an $n \times k$ matrix is a bijection if and only if it is square and injective, and if and only if it is square and surjective.

For the last two statements, simply combine the results above. Say we have a linearly independent collection. Then as long as I have less than n vectors, i shall repeatedly use independence extension lemma, until I reach n vectors, and at that moment I must have a basis.

If we have a spanning collection, then as long as I have more than n vectors, i shall have redundant vectors. I throw them away, until I reach n vectors, and at that moment I must have a basis. □

Remark 5.3.23. *So say we are in an n dimensional space. To find a basis, we first pick any $\boldsymbol{v}_1$. Then we pick any $\boldsymbol{v}_2$ linearly independent from $\boldsymbol{v}_1$. Then we pick any $\boldsymbol{v}_3$ linearly independent from $\boldsymbol{v}_1, \boldsymbol{v}_2$, and so on. As soon as we hit $\boldsymbol{v}_n$, we are done. This must be a basis. There is no need to check if it is spanning. (Interesting question to think about: When I pick my $\boldsymbol{v}_3$, why $\boldsymbol{v}_2$ cannot be a linear combination of $\boldsymbol{v}_1$ and $\boldsymbol{v}_3$?)*

Alternatively, say we have lots of vectors that would certainly span our n dimensional space. Then we can keep throwing away redundant vecotrs (vectors that are linear combination of others). When we have n vectors left, we must arrive at a basis. There is no need to check linear independency.

Example 5.3.24. Here is an interesting application. Say we want to solve differential equation $f''' - 6f'' + 11f' - 6f = 0$. How can we find all possible solutions?

A useful fact is the following: consider a differential equation of the form

$$f^{(k)} + a_{k-1}f^{(k-1)} + \cdots + a_1 f' + a_0 f = 0.$$

Then all such differential equations will have solution space of dimension k (the highest order of derivetives involved).

Assuming this fact, then the solution space to $f''' - 6f'' + 11f' - 6f = 0$ is three dimensional. Then, by trial and error, we find out that $\mathrm{e}^x, \mathrm{e}^{2x}, \mathrm{e}^{3x}$ all satisfy this differential equation, and they are linearly independent. Hooray! This must be a basis. So the solution set is made of all linear combinations of them. ☺

Here we have a special corollary for subspaces.

Definition 5.3.25. *For vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$, their* ***span*** *is the subspace of all linear combinations of them.*

Corollary 5.3.26. *If vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ are linearly independent, then their span has dimension k.*

Remark 5.3.27. *With the idea above, we see that the independency extension lemma is the formal rigorous statement of the following informal intuition: for an m dimensional subspace, if it moves in a new direction (new vector not contained in the original subspace), then we get an $m+1$ dimensional subspace. Point moves to make a line, and a line moves to make a plane, these are all special cases of the lemma.*

And linear dependence happen exactly like this: we add a new vector to our collection, but it fails to contribute a new dimension!

5.3.3 (Optional) Infinite dimensional spaces

Just some examples to show you what might happen.

Example 5.3.28.

1. Consider the space of all sequences, $\mathbb{R}^{\mathbb{N}}$. Say $(1, 1, 1, ...)$ is a sequence, and $(3, 1, 4, 1, 5, 9, 2, 6, ...)$ is a sequence, and $(\frac{2}{1}, \frac{3}{2}, \frac{4}{3}, ...)$ is also a sequence, and so on. We can add sequences by adding them term-wise, and we can multiply a real number to a sequence by mutliply the number term-wise. This makes $\mathbb{R}^{\mathbb{N}}$ a vector space. This vector space is INFINITE DIMENSIONAL. You can try to come up with a basis, and you can try $(1, 0, 0, ...), (0, 1, 0, ...)$ and so forth. Surely, it seems like any sequence is like some infinite linear combination of them, no? But are they really basis? Not really. Infinite dimensional spaces works differently.

2. Why are they not really a basis? First let us note that the set S_c of all converging sequences is a subspace in $\mathbb{R}^{\mathbb{N}}$. For example, sequence $(\frac{2}{1}, \frac{3}{2}, \frac{4}{3}, ...)$ is converging with limit 1. This is easy to see,

because if $(a_n), (b_n)$ are converging to values a and b, then $(a_n + b_n)$ would converge to $a + b$, and $k(a_n)$ would converge to ka. So this is indeed a subspace. This is also NOT the whole space, because there are non-convergin sequences out there. Now you see that the vectors $(1, 0, 0, ...), (0, 1, 0, ...)$ are all contained in S_c. So they span AT MOST S_c. Their span, by definition of a subspace, will always stay in S_c. Since there are divergent sequences like $(1, 2, 3, ...)$ in $\mathbb{R}^{\mathbb{N}} - S_c$, there is simply no way this sequence would be in the span of vectors contained entirely in S_c. In fact, $(1, 0, 0, ...), (0, 1, 0, ...)$ does NOT even span S_c, because they all have limit zero, so all of their linear combinations could only have limit zero.

3. Suffice to say, there is no such thing as an "infinite" linear combination. Whenever we do a linear combination, we are always only allowed to use finitely many vectors, even if we have an infinite collection of vectors at our hands. Otherwise we may run into paradoxical phenomina as explained above.

4. Consider the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$, denoted sometimes as $C(\mathbb{R})$. For two functions $f : x \mapsto f(x)$ and $g : x \mapsto g(x)$, we can define their sum as $f + g : x \mapsto f(x) + g(x)$. We can also multiply a function with a real number as $kf : x \mapsto k(f(x))$. This makes $C(\mathbb{R})$ a real vector space. This is NOT finite dimensional.

5. Consider the set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ that is periodic with period 2π. Then the Fourier theorem says the following: any such function f must be the sum of some series $f(x) = c + \sum_n a_n \sin(nx) + \sum_n b_i \cos(nx)$. The sums go through all possible positive integers. This statement has a very "basis"-feel to it. Namely, f seems like a linear combination of the constant function and the sines and cosines. Furthermore, you might in fact do "dot products" for these functions, and define $f \cdot g = \int_0^{2\pi} f(x)g(x)dx$. Then you will see that the constant function, sines and cosines are mutually orthogonal! This is not just a "basis", but in fact an orthogonal "basis". This is the start of Fourier analysis. It is NOT a true basis in the traditional sense. Not every function is a linear combination of them. However, every function is a LIMIT of linear combinations of them. (We do not always have such nice things for infinite dimensional basis, btw.)

☺

5.4 Entries of a Linear Map and Changing Basis

5.4.1 Entries of a Linear Map

Matrices are linear maps between $\mathbb{R}^n$ and $\mathbb{R}^m$. But what about abstract linear maps on abstract vector spaces?

If we have a linear map $L : V \to W$, then there is no matrix right away. However, recall that picking a basis would essentially turn the vector space into $\mathbb{R}^n$. So, if we pick a basis for V and pick a basis for W, then V is now effectively $\mathbb{R}^n$ and W is now effectively $\mathbb{R}^m$. Then L is now some map from $\mathbb{R}^n$ to $\mathbb{R}^m$. Hence it is now a matrix.

Keep in mind that any matrix expression of L will depend on our choice of basis for the domain and codomain. In particular, there is no THE matrix for L, since we may pick different basis, and then L may appear to be different matrices. Just as a vector may have different coordiantes under different basis, a linear map may have different matrices under different basis.

Definition 5.4.1. *Given a basis $\mathcal{B}$ for a space V and a basis $\mathcal{C}$ for a space W, the matrix for a linear map $L : V \to W$ is $L_{\mathcal{C}\leftarrow\mathcal{B}} = \mathcal{C}^{-1}L\mathcal{B}$.*

So if the inputs are coordinates under $\mathcal{B}$, we first combine coordinates into the actual vector via the linear combination map $\mathcal{B}$, then we map through L, and finally we express the abstract image in terms of coordinates via the coordinate map $\mathcal{C}^{-1}$. As you can see, this is basically just L, expressed as a matrix using the basis of the domain and the basis of the codomain.

To summarize the idea above, think about the following diagram:

$$\begin{tikzcd} V \arrow[r, "L"] & W \\ \mathbb{R}^n \arrow[u, "\mathcal{B}"] \arrow[r, "L_{\mathcal{C}\leftarrow\mathcal{B}}"] & \mathbb{R}^m \arrow[u, "\mathcal{C}"'] \end{tikzcd}$$

Consider this very straight forward proposition.

Proposition 5.4.2. *Given a linear map $L : V \to W$, basis $\mathcal{B}$ for V and basis $\mathcal{C}$ for W, then for any $\boldsymbol{v} \in V$, we have $L_{\mathcal{C}\leftarrow\mathcal{B}}\boldsymbol{v}_{\mathcal{B}} = (L\boldsymbol{v})_{\mathcal{C}}$.*

Proof. This should be intuitively trivial. For a rigorous proof, we have

$$L_{\mathcal{C}\leftarrow\mathcal{B}}\boldsymbol{v}_{\mathcal{B}} = \mathcal{C}^{-1}L\mathcal{B}\mathcal{B}^{-1}\boldsymbol{v} = \mathcal{C}^{-1}L\boldsymbol{v} = (L\boldsymbol{v})_{\mathcal{C}}.$$

□

We also have the following nice result.

Proposition 5.4.3. *Given a linear map $L : V \to W$, and a linear map $T : W \to U$, say we have basis $\mathcal{B}$ for V, basis $\mathcal{C}$ for W, and basis $\mathcal{D}$ for U. Then we have $(TL)_{\mathcal{D}\leftarrow\mathcal{B}} = T_{\mathcal{D}\leftarrow\mathcal{C}}L_{\mathcal{C}\leftarrow\mathcal{B}}$.*

Proof. This should be intuitively trivial. For a rigorous proof, we have

$$T_{\mathcal{D}\leftarrow\mathcal{C}}L_{\mathcal{C}\leftarrow\mathcal{B}} = \mathcal{D}^{-1}T\mathcal{C}\mathcal{C}^{-1}L\mathcal{B} = \mathcal{D}^{-1}TL\mathcal{B} = (TL)_{\mathcal{D}\leftarrow\mathcal{B}}.$$

□

Now, to find the matrix $L_{\mathcal{C}\leftarrow\mathcal{B}}$ of a linear map L, we do not really use the formula $\mathcal{C}^{-1}L\mathcal{B}$. Rather, we go back to the basics in this class: we compute the matrix $L_{\mathcal{C}\leftarrow\mathcal{B}}$ column by column, by inspecting $L_{\mathcal{C}\leftarrow\mathcal{B}}\boldsymbol{e}_i$ for each i.

Example 5.4.4. Suppose we have a linear map $M : \mathcal{P}_2 \to \mathcal{P}_3$, that multiply each polynomial by x. How can we understand this linear map?

One way is the following: we pick a basis in $\mathcal{P}_2$ and in $\mathcal{P}_3$. Then the two vector spaces are now looking like $\mathbb{R}^3$ and $\mathbb{R}^4$. Then we see that our linear map corresponds to a 4 by 3 matrix.

To find this matrix, we again ONLY inspect the image of $\boldsymbol{e}_i$ to get its columns. We always do this to find whatever matrix.

Suppose we pick basis $1, x, x^2 \in \mathcal{P}_2$ and $1, x, x^2, x^3 \in \mathcal{P}_3$, then $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$ in $\mathcal{P}_2$ are in fact $1, x, x^2$. They would go to x, x^2, x^3, which are $\boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4$ in the codomain coordinates. So the corresponding matrix for M is $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Suppose we pick basis $2, x+1, x^2 \in P_2$ and $x+1, x-1, x^2-x, x^3+x^2$. Compute the image of $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3$. They are in fact $2, x+1, x^2$ in the domain. So they would go to $2x, x^2+x, x^3$ in the codomain. Note that $2x = (x+1)+(x-1)$, and $x^2+x = (x^2-x)+(2x) = (x^2-x)+(x+1)+(x-1)$. Finally,

$$x^3 = (x^3+x^2) - x^2 = (x^3+x^2) - (x^2-x) - x = (x^3+x^2) - (x^2-x) - \frac{1}{2}(x+1) - \frac{1}{2}(x-1).$$

So these corresponds to vectors in the codomain with coordinates $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ -1 \\ 1 \end{bmatrix}$. Then the matrix for our linear map is now $\begin{bmatrix} 1 & 1 & -\frac{1}{2} \\ 1 & 1 & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$.

Either way, since we have the matrix, we can now compute whatever we need about M. For example, if the input is $2+(x+1)+(x^2)$, then the output coordinates should be $\begin{bmatrix}1 & 1 & -\frac{1}{2}\\ 1 & 1 & -\frac{1}{2}\\ 0 & 1 & -1\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}\frac{3}{2}\\ \frac{3}{2}\\ 0\\ 1\end{bmatrix}$, which represents the vector $\frac{3}{2}(x+1)+\frac{3}{2}(x-1)+(x^3+x^2)$. ☺

Example 5.4.5. Suppose we have a linear map $D : \mathcal{P}_3 \to \mathcal{P}_3$, then under the obvious easy basis $1, x, x^2, x^3$ for both the domain and the codomain, what is the matrix of D?

Well, $D(1) = 0, D(x) = 1, D(x^2) = 2x, D(x^3) = 3x^2$. So the matrix for D is $\begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 0 & 3\\ 0 & 0 & 0 & 0\end{bmatrix}$.

As you can imagine, if we simply pick the space $\mathcal{P}$ of all polynomials, then under the "obvious basis" $1, x, x^2, \ldots$, then the matrix for the derivative map $D : \mathcal{P} \to \mathcal{P}$ is an infinite matrix like $\begin{bmatrix}0 & 1 & & \\ & 0 & 2 & \\ & & 0 & \ddots\\ & & & \ddots\end{bmatrix}$. And at the same time, the map "multiplication by x" $M : \mathcal{P} \to \mathcal{P}$ is also an infinite matrix like $\begin{bmatrix}0 & & & \\ 1 & 0 & & \\ & 1 & 0 & \\ & & \ddots & \ddots\end{bmatrix}$.

You can directly compute and see that $DM - MD = I$. Here DM is D shifted to the left, and MD is D shifted down. ☺

In real life, we humans would simply pick basis and operates at the abstract level. Then computers would go find the matrices and coordinates and calculate for us.

5.4.2 Change of Basis

The point of an abstract vector space is to change basis, so let us see how it can be done computationally.

Example 5.4.6. Again take our example of $\mathcal{P}_2$ with $\boldsymbol{w} = 3x^2 + 4x + 1$. It is $\begin{bmatrix}0\\-2\\3\end{bmatrix}$ under the basis $\mathcal{B} = (1, x+1, (x+1)^2)$.

Now similarly, for the basis $\mathcal{C} = (1, x-1, (x-1)^2)$, we can compute and have $\boldsymbol{w} = \mathcal{C}\begin{bmatrix}8\\10\\3\end{bmatrix}$. How can one convert between these two basis?

In general, suppose $\boldsymbol{v}$ in basis $\mathcal{B}$ has coordinates $\begin{bmatrix}a\\b\\c\end{bmatrix}$, then what is its coordinates in basis $\mathcal{C}$? This means given $\boldsymbol{v} = \sum a_i \boldsymbol{v}_i$, how can I express $\boldsymbol{v}$ in another basis $\boldsymbol{w}_i$?

To achieve this transition, we need to know what the basis vectors $\boldsymbol{v}_i$ looks like in basis vectors $\boldsymbol{w}_i$, and substitute. Note that In our case, $1 = 1, x+1 = 2+(x-1), (x+1)^2 = 4+4(x-1)+(x-1)^2$. So if $\boldsymbol{v} = a + b(x+1) + c(x+1)^2$, then it will be $\boldsymbol{v} = a + 2b + b(x-1) + 4c + 4c(x-1) + c(x-1)^2$ and it has new coordinates $\begin{bmatrix}a+2b+4c\\ b+4c\\ c\end{bmatrix}$.

What is the relation between the two? We see that the new coordinates are in fact $\begin{bmatrix} a+2b+4c \\ b+4c \\ c \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, a matrix times the old coordinates! Why is that? ☺

Such a matrix is called a change of coordinates matrix. But why is this a matrix?

Given a vector space V and two bases $\mathcal{B}, \mathcal{C}$, consider the linear combination maps $\mathcal{B}, \mathcal{C}$. These two maps are bijections. So we have the following graph:

$$\begin{tikzcd} & V & \\ \mathbb{R}^n \arrow[ur, "\mathcal{B}"] & & \mathbb{R}^n \arrow[ul, "\mathcal{C}"'] \end{tikzcd}$$

$$\begin{tikzcd} & \boldsymbol{v} & \\ \boldsymbol{v}_{\mathcal{B}} \arrow[ur, "\mathcal{B}"] & & \boldsymbol{v}_{\mathcal{C}} \arrow[ul, "\mathcal{C}"'] \end{tikzcd}$$

So we can clearly see that the change of coordinate process is $\boldsymbol{v}_{\mathcal{C}} = \mathcal{C}^{-1}\mathcal{B}\boldsymbol{v}_{\mathcal{B}}$. Since the linear combination maps $\mathcal{B}, \mathcal{C}$ are linear bijections, we see that the change of coordinate process is $\mathcal{C}^{-1}\mathcal{B}$, which is a linear map. Furthermore, the inputs and outputs are both elements of $\mathbb{R}^n$, so it is an $n \times n$ matrix.

Let us take another approach.

Corollary 5.4.7. *Given a vector space V and bases $\mathcal{B}, \mathcal{C}$, then $I_{\mathcal{C}\leftarrow\mathcal{B}}\boldsymbol{v}_{\mathcal{B}} = \boldsymbol{v}_{\mathcal{C}}$. (So the change of coordinate matrix is actually the matrix for the identity map.)*

Proof. $I_{\mathcal{C}\leftarrow\mathcal{B}}\boldsymbol{v}_{\mathcal{B}} = (I\boldsymbol{v})_{\mathcal{C}} = \boldsymbol{v}_{\mathcal{C}}$.

For an alternative proof, recall that the change of coordinate matrix should be $\mathcal{C}^{-1}\mathcal{B}$. But we have $I_{\mathcal{C}\leftarrow\mathcal{B}} = \mathcal{C}^{-1}I\mathcal{B} = \mathcal{C}^{-1}\mathcal{B}$. □

This should not surprise us at all. After all, the change of basis process do NOT really change the under lying vector. It IS the identity map. But because the input vector was expressed in the old basis, and the output vector is the same vector but under the new basis, therefore the change is precisely $I_{\mathcal{B}\to\mathcal{C}}$.

Definition 5.4.8. *Given two basis $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ and $\mathcal{C} = (\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$ of a vector space V, then we define the **change of coordinate matrix** as a matrix $I_{\mathcal{C}\leftarrow\mathcal{B}}$, i.e., the matrix for the identity map from basis $\mathcal{B}$ to $\mathcal{C}$.*

Now, the fact that $I_{\mathcal{C}\leftarrow\mathcal{B}}$ is a matrix is MORE IMPORTANT that the fact that it is $\mathcal{C}^{-1}\mathcal{B}$ or any other expression. Because we have a very straight forward way to find a matrix: its i-th column is simple the image of $\boldsymbol{e}_i$!

So if $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n), \mathcal{C} = (\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$, then the i-th column of $I_{\mathcal{C}\leftarrow\mathcal{B}}$ is $\mathcal{C}^{-1}\mathcal{B}\boldsymbol{e}_i = \mathcal{C}^{-1}\boldsymbol{v}_i$, i.e., the coordinates of $\boldsymbol{v}_i$ in the basis $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$.

Remark 5.4.9 (Algorithm to find change of coordinate matrix)**.** *Express each old basis vector in terms new basis vectors, and that is the corresponding column for your change of coordinate matrix. Review the example at the start of this subsection to get a better feel of this.*

Alternatively, you can also express new basis vectors in terms of old basis vectors (sometimes the problem tells you how to do this explicitly). Then you will obtain the basis transition matrix, and the inverse matrix is the change of coordinate matrix.

Example 5.4.10. Again take our example of $\mathcal{P}_2$ with basis $\mathcal{B} = (1, x+1, (x+1)^2)$ and basis $\mathcal{C} = (1, x-1, (x-1)^2)$. How can one convert coordinates in $\mathcal{B}$ into coordinates in $\mathcal{C}$?

The columns of the change of coordinate matrix should be the vectors of the old basis expressed in coodinates in the new basis. Observe that

$$\begin{array}{rcl} 1 & = & 1 \\ x+1 & = & 2+(x-1) \\ (x+1)^2 & = & 4+4(x-1)+(x-1)^2 \end{array}.$$

So the change of coordinate matrix is $I_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$, where we read the three columns from the coordinates above. ☺

Now, sometimes there is another way to do this. By taking a closer look you might see that $(1, x+1, (x+1)^2) = (1, x-1, (x-1)^2) \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$. (Just interpret the multiplication here as a "block multiplication" type of thing.) Such a matrix is called a basis transition matrix.

Note how the very same matrix appears both as a basis transition matrix AND a change of coordinate matrix, but in OPPOSITE directions. The matrix $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ is the change of coordinates matrix from the OLD coordinates to the NEW coordinates (multiplied from the left to a coordinate vector), while it is also the basis transition matrix from the NEW basis to the OLD basis (multiplied from the right to a linear combination map of the basis).

Definition 5.4.11. *Given a collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ in V and an $m \times n$ matrix $A = \begin{bmatrix} \boldsymbol{a}_1 & \ldots & \boldsymbol{a}_n \end{bmatrix}$ (so the entries of A are all real numbers rather than abstract vectors), we define $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m)A = (\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$ where $\boldsymbol{w}_i = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m)\boldsymbol{a}_i$. So we simply do linear combination of these $\boldsymbol{v}_i$ according to each column of A, and get n vectors out of it.*

Definition 5.4.12. *Given two basis $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ and $\mathcal{C} = (\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$ of a vector space V, then we define the* ***basis transition matrix*** *as a matrix $T_{\mathcal{B}\to\mathcal{C}}$ such that $\mathcal{B}T_{\mathcal{B}\to\mathcal{C}} = \mathcal{C}$.*

Proposition 5.4.13. *If $\mathcal{B} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ and $\mathcal{C} = (\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$ are two bases of the same vector space V, then the $T_{\mathcal{C}\to\mathcal{B}} = I_{\mathcal{C}\leftarrow\mathcal{B}} = \mathcal{C}^{-1}\mathcal{B}$, and it is an invertible $n \times n$ matrix when $\dim V = n$.*

Proof. Look at the following diagram.

$$\begin{tikzcd} & V & \\ \mathbb{R}^n \arrow[ur, "\mathcal{B}"] \arrow[rr, "I_{\mathcal{C}\leftarrow\mathcal{B}}"] & & \mathbb{R}^n \arrow[ul, "\mathcal{C}"'] \end{tikzcd}$$

Note that a vector $\boldsymbol{v}$ lives in the V in the top, and its coordinates in $\mathcal{B}$ lives in the bottom left, and its coordinates in $\mathcal{C}$ lives in the bottom right. it is obvious that do perform a change of coordinates $I_{\mathcal{C}\leftarrow\mathcal{B}}$, it is the same as applying the linear map $\mathcal{C}^{-1}\mathcal{B}$. note that this map goes from $\mathbb{R}^n$ to $\mathbb{R}^n$, so it corresponds to an $n \times n$ matrix.

Now consider the basis transition map from $\mathcal{C}$ to $\mathcal{B}$, $T_{\mathcal{C}\to\mathcal{B}}$. It is clear that we want $\mathcal{B} = \mathcal{C}T_{\mathcal{C}\to\mathcal{B}}$. So we immediately see that $T_{\mathcal{C}\to\mathcal{B}} = \mathcal{C}^{-1}\mathcal{B}$. □

Corollary 5.4.14. *The change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ and the basis transition matrix from $\mathcal{B}$ to $\mathcal{C}$ are inverse of each other.*

Remark 5.4.15. *You should also note that the strange situation of $T_{\mathcal{C}\to\mathcal{B}} = I_{\mathcal{C}\leftarrow\mathcal{B}}$ is a result of associativity.*

Given $\boldsymbol{v} \in V$, we have $\mathcal{B}\boldsymbol{v}_{\mathcal{B}} = \boldsymbol{v} = \mathcal{C}\boldsymbol{v}_{\mathcal{C}}$ by definition. But the left hand side is also $(\mathcal{C}T_{\mathcal{C}\to\mathcal{B}})\boldsymbol{v}_{\mathcal{B}}$, whereas the right hand side is also $\mathcal{C}(I_{\mathcal{C}\leftarrow\mathcal{B}}\boldsymbol{v}_{\mathcal{B}})$.

Example 5.4.16. Suppose I want to buy x burgers and y cokes. Now combo A sells 2 burger and 1 coke, while combo B sells 3 burger and 4 cokes. How many combos do I need?

Well, note that $(comboA, comboB) = (burger, cokes)\begin{bmatrix}2 & 3\\1 & 4\end{bmatrix}$. So $\begin{bmatrix}2 & 3\\1 & 4\end{bmatrix}$ is the basis transition matrix from single orders to combos. Therefore, the change of coordinate map from single orders to combos would be $\begin{bmatrix}2 & 3\\1 & 4\end{bmatrix}^{-1} = \frac{1}{5}\begin{bmatrix}4 & -3\\-1 & 2\end{bmatrix}$. So I need $\frac{4}{5}x - \frac{3}{5}y$ combo A and $-\frac{1}{5}x + \frac{2}{5}y$ combo B.

Note that a matrix inversion process is involved. Typically, a problem will look like this: we start with an old basis and old coordinates. The problem tells you how the old basis would combine into the new basis. Then the problem asks how the old coordinates would change into the new coordinates.

In effect, they give you $T_{\mathcal{B}\to\mathcal{C}}$ for free, and ask you for $I_{\mathcal{C}\leftarrow\mathcal{B}}$. These two are just inverse matrix of each other. ☺

Drawing diagrams is a very practical way to help us figure things out. Consider this:

Proposition 5.4.17. *Given three basis $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, let C_{ij}, T_{ij} be the change of coordinates map and the transition map from $\mathcal{B}_i$ to $\mathcal{B}_j$. Then $C_{13} = C_{23}C_{12}$ (note that this will multiply things from their left, so C_{12} happens first) and $T_{13} = T_{12}T_{23}$ (note that this will multiply things from their right, so T_{12} happens first).*

Proof. Conceptually this is obvious. But we can also stare at the following diagram until we see this.

$$\begin{tikzcd}
& V & \\
\mathbb{R}^n \arrow[ur, "\mathcal{B}_1"] \arrow[r, "C_{12}"] \arrow[rr, bend right, "C_{13}"'] & \mathbb{R}^n \arrow[u, "\mathcal{B}_2"] \arrow[r, "C_{23}"] & \mathbb{R}^n \arrow[ul, "\mathcal{B}_3"']
\end{tikzcd}$$

Also keep in mind that $C_{ij} = T_{ji}$.

Another alternative is to compute it directly, say $C_{23}C_{12} = \mathcal{B}_3^{-1}\mathcal{B}_2\mathcal{B}_2^{-1}\mathcal{B}_1 = \mathcal{B}_3^{-1}\mathcal{B}_1 = C_{13}$. □

Example 5.4.18 (Hen and rabbit again?)**.** The matrix $\begin{bmatrix}1 & 1\\2 & 4\end{bmatrix}$ is invertible. Is this a change of basis? YES!

The content of the cage (the underlying abstract entity) never changes. However, I should think of the content of the cage as a linear combination of ... what? This is precisely what the hen-rabbit problem is about.

If we describe the content of the cage using the "basis" of $(\text{head}, \text{leg})$, I get coordinates $\begin{bmatrix}6\\20\end{bmatrix}$. If we describe the content of the cage using the "basis" of $(\text{hen}, \text{rabbit})$, what coordinate should I get?

We have a basis transition $(\text{hen}, \text{rabbit}) = (\text{head}, \text{leg})\begin{bmatrix}1 & 1\\2 & 4\end{bmatrix}$. So the answer is $\begin{bmatrix}1 & 1\\2 & 4\end{bmatrix}^{-1}\begin{bmatrix}6\\20\end{bmatrix}$. ☺

Let us take a moment and think about a change of basis in $\mathbb{R}^n$ itself. We of course always start with the standard basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$. Suppose we want to change to a new basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^n$. How would I do that?

Proposition 5.4.19. *The change of coordinate matrix to go from the standard basis in $\mathbb{R}^n$ to a new basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ is $\begin{bmatrix}\boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n\end{bmatrix}^{-1}$.*

Proof. Note that for vectors in $\mathbb{R}^n$, $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ has exactly the same meaning as $\begin{bmatrix}\boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n\end{bmatrix}$. So we have $(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)\begin{bmatrix}\boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n\end{bmatrix} = I\begin{bmatrix}\boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n\end{bmatrix} = \begin{bmatrix}\boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n\end{bmatrix} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$. So $\begin{bmatrix}\boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n\end{bmatrix}$ is the basis transition matrix from the standard basis to the new basis. □

Corollary 5.4.20. *An invertible matrix A can be thought of as a change of coordinate matrix in $\mathbb{R}^n$, to go from the standard basis to a new basis, i.e., the columns of A^{-1}.*

Here come the mind blasting moment. ALL invertible matrices might simply be seen as some change of basis process!

Proposition 5.4.21. *Say we started with an n-dimensional vector space V and a basis $\mathcal{B}$. Then for any $n \times n$ invertible matrix A, it is $C_{\mathcal{B}\to\mathcal{C}}$ for some new basis $\mathcal{C}$.*

Proof. Simply set $\mathcal{C} = \mathcal{B}A^{-1}$, and we are done. □

Remark 5.4.22. *Whenever you see an invertible matrix, then under some perspective, it will represent a change of basis.*

So an invertilbe matrix is a change of basis, but it is also a bijective linear map? How can the two be the same? Say you have a pen in your hand. You can rotate the pen (apply the linear map), or you can tilt your head (change of basis), and the pen would result in the same posture as far as your eyes can see. The two process are the same invertible matrix.

5.4.3 Change of basis for a Linear Map

Here is a thought: given a matrix for our linear map under some bases, can I convert it using the change of basis formula, to get the matrix in another basis?

Proposition 5.4.23. *Given bases $\mathcal{B}_1, \mathcal{B}_2$ for a space V and a basis $\mathcal{C}$ for a space W, then for a linear map $L : V \to W$ we have $L_{\mathcal{C}\leftarrow\mathcal{B}_2} = L_{\mathcal{C}\leftarrow\mathcal{B}_1} I_{\mathcal{B}_1\leftarrow\mathcal{B}_2}$.*

Similarly, given a basis $\mathcal{B}$ for a space V and bases $\mathcal{C}_1, \mathcal{C}_2$ for a space W, then for a linear map $L : V \to W$ we have $L_{\mathcal{C}_2\leftarrow\mathcal{B}} = I_{\mathcal{C}_2\leftarrow\mathcal{C}_1} L_{\mathcal{C}_1\leftarrow\mathcal{B}}$.

Proof. The proofs are trivial. For example, we have $L_{\mathcal{C}\leftarrow\mathcal{B}_1} I_{\mathcal{B}_1\leftarrow\mathcal{B}_2} = (LI)_{\mathcal{C}\leftarrow\mathcal{B}_2} = L_{\mathcal{C}\leftarrow\mathcal{B}_2}$.

Alternatively, look at the diagram below.

$$
\begin{tikzcd}
\mathbb{R}^n \arrow[r, "L_{\mathcal{C}_1\leftarrow\mathcal{B}_1}"] \arrow[d, "\mathcal{B}_1"] & \mathbb{R}^m \arrow[d, "\mathcal{C}_1"'] \\
V \arrow[r, "L"] & W \\
\mathbb{R}^n \arrow[u, "\mathcal{B}_2"] \arrow[r, "L_{\mathcal{C}_2\leftarrow\mathcal{B}_2}"] \arrow[uu, bend left, "I_{\mathcal{B}_1\leftarrow\mathcal{B}_2}"] & \mathbb{R}^m \arrow[u, "\mathcal{C}_2"'] \arrow[uu, bend right, "I_{\mathcal{C}_1\leftarrow\mathcal{C}_2}"']
\end{tikzcd}
$$

You can immediately see that for $L_{\mathcal{C}_1\leftarrow\mathcal{B}_2}$, for example, we need to go from bottom left to upper right. So $L_{\mathcal{C}_1\leftarrow\mathcal{B}_2} = L_{\mathcal{C}_1\leftarrow\mathcal{B}_1} I_{\mathcal{B}_1\leftarrow\mathcal{B}_2}$. The other one goes from upper left to bottom right, and can be proven similarly. □

Remark 5.4.24. *In particular, change of basis in the domain corresponds to multiplying an invertible matrix on the RIGHT, while change of basis in the codomain corresponds to multiplying an invertible matrix on the LEFT.*

Example 5.4.25. Note that

$$(1, x, x^2) \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = (2, x+1, x^2),$$

$$(1, x, x^2, x^3) \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = (x+1, x-1, x^2-x, x^3+x^2).$$

As a result, using the corresponding change of coordinate matrix, the matrix of our previous linear map under the ugly basis can be computed as

$$\begin{bmatrix}1 & -1 & 0 & 0\\ 1 & 1 & -1 & 0\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix}^{-1}\begin{bmatrix}0 & 0 & 0\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}2 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 1 & -\frac{1}{2}\\ 1 & 1 & -\frac{1}{2}\\ 0 & 1 & -1\\ 0 & 0 & 1\end{bmatrix}.$$

Hooray, this works! ☺

In fact, the process of "diagram chase" would allow us to create many "change of basis" formula for matrices of linear maps. Let us look at this diagram:

$$\begin{tikzcd}
\mathbb{R}^n \arrow[r, "L_{\mathcal{C}_1\leftarrow\mathcal{B}_1}"] \arrow[d, "\mathcal{B}_1"'] & \mathbb{R}^d \arrow[r, "T_{\mathcal{D}_1\leftarrow\mathcal{C}_1}"] \arrow[d, "\mathcal{C}_1"] & \mathbb{R}^m \arrow[d, "\mathcal{D}_1"] \\
U \arrow[r, "L"] & V \arrow[r, "T"] & W \\
\mathbb{R}^n \arrow[r, "L_{\mathcal{C}_2\leftarrow\mathcal{B}_2}"] \arrow[u, "\mathcal{B}_2"] & \mathbb{R}^d \arrow[r, "T_{\mathcal{D}_2\leftarrow\mathcal{C}_2}"] \arrow[u, "\mathcal{C}_2"'] & \mathbb{R}^m \arrow[u, "\mathcal{D}_2"']
\end{tikzcd}$$

By snaking around the diagram, you can get results like

$$(T\circ L)_{\mathcal{D}_1\leftarrow\mathcal{B}_2} = I_{\mathcal{D}_1\leftarrow\mathcal{D}_2}T_{\mathcal{D}_2\leftarrow\mathcal{C}_2}I_{\mathcal{C}_2\leftarrow\mathcal{C}_1}L_{\mathcal{C}_1\leftarrow\mathcal{B}_1}I_{\mathcal{B}_1\leftarrow\mathcal{B}_2}.$$

I'm not even going to write this as a proposition or to prove it. Just stare at these diagrams and you should be able to see this.

5.4.4 Row and Column operations and the Rank Normal Form

Now let us not forget the point of doing change of basis. The point is to gain more interpretations and to understand linear algebra better.

Suppose we have a matrix $A:\mathbb{R}^n\to\mathbb{R}^m$. Now maybe A looks ugly. If I perform some change of basis, hopefully A will look pretty, and we can solve problems involving A much better, yes? So how to do this? And how simple can we simplify A?

Now, change of basis in $\mathbb{R}^n$ are simply multiplication by invertible matrices. So we can reformulate our problem into this:

Question 5.4.26. *Given an $m\times n$ matrix A, can you find invertible $m\times m$ matrix R and invertible $n\times n$ matrix C such that BAC is nice?*

Now, an invertible matrix on the left is simply a series of row operations, while an invertible matrix on the right is simply a series of column operations! In fact, combining our intuitions on row/column operations and change of basis, we have the following:

1. A row operation is a change of basis in the codomain.

2. A column operation is a change of basis in the domain.

3. RREF means doing a change of basis to the codomain, so that your system looks as simple as possible.

So how would we simplify a matrix A via change of basis? Say $A=\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & 2 & 2 & 2\\ 1 & 1 & 1 & 2\\ 1 & 1 & 1 & 1\end{bmatrix}$. First we row reduce to get RREF $\begin{bmatrix}1&0&0&0\\0&1&1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}$. Next we column reduce, using pivots to kill the rest of each row. We now have $\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&0&0\end{bmatrix}$. Finally, let us throw zero columns to the right, and we have $\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&0\end{bmatrix}$. Huh.

Proposition 5.4.27. *Given any $m \times n$ matrix A with rank r, we can find invertible $m \times m$ matrix R and invertible $n \times n$ matrix C such that $RAC = \begin{bmatrix}I_{r\times r}&O\\O&O\end{bmatrix}$. We say this is the **rank normal form** or **rank revealing form** for our matrix A.*

Proof. What we just did. Starting from A, we first change basis in codomain, i.e., row reduce A, until we reach RREF. Then change basis in domain, i.e., column reduce by using pivots to kill the rest of each row. Now only pivots are left. And then they are column-swapped to stay in the upper left block. □

Note that we write $\begin{bmatrix}I_{r\times r}&O\\O&O\end{bmatrix}$ for simplicity, but technically it could be $\begin{bmatrix}I&O\end{bmatrix}$ or $\begin{bmatrix}I\\O\end{bmatrix}$ or I, depending on whether we have zero columns or zero rows at all.

Corollary 5.4.28. *Given any linear map, by finding the right basis for the domain and for the codomain, the map will have matrix $\begin{bmatrix}I&O\\O&O\end{bmatrix}$ or $\begin{bmatrix}I&O\end{bmatrix}$ or $\begin{bmatrix}I\\O\end{bmatrix}$ or I. We say this is the **rank normal form** for your linear map L.*

Think about the meaning of these statements. We realized that ANY linear map, by choosing the correct basis, will look like the matrix $\begin{bmatrix}I_{r\times r}&O\\O&O\end{bmatrix}$ or $\begin{bmatrix}I&O\end{bmatrix}$ or $\begin{bmatrix}I\\O\end{bmatrix}$ or I.

So it turns out that, as far as finite dimensions goes, if we always pick good bases, then the abstract world is extremely beautiful and simplistic. We have ONLY FOUR kinds of linear maps:

1. Each finite dimensional vector space looks like $\mathbb{R}^n$ for some n.
2. A linear injection must look like $\begin{bmatrix}I\\O\end{bmatrix}$ under the right basis for the domain and the codomain. Geometrically, this is an inclusion map. For example $\begin{bmatrix}1&0\\0&1\\0&0\end{bmatrix}$ would simply send $\begin{bmatrix}x\\y\end{bmatrix}$ to $\begin{bmatrix}x\\y\\0\end{bmatrix}$. It represent the process of including the xy-plane into $\mathbb{R}^3$.
3. A linear surjection must look like $\begin{bmatrix}I&O\end{bmatrix}$ under the right basis for the domain and the codomain.Geometrically, this is a projection map. For example $\begin{bmatrix}1&0&0\\0&1&0\end{bmatrix}$ would simply send $\begin{bmatrix}x\\y\\z\end{bmatrix}$ to $\begin{bmatrix}x\\y\end{bmatrix}$. It represent the process of projecting $\mathbb{R}^3$ onto the xy-plane.
4. A linear bijection must look like the identity matrix under the right basis for the domain and the right basis for the codomain.
5. A linear map that is neither injective nor surjective looks like $\begin{bmatrix}I&O\\O&O\end{bmatrix}$ under the right basis for the domain and the codomain.

Furthermore, here are some interesting ramifications.

Theorem 5.4.29 (Full rank decomposition)**.** *For any $m \times n$ matrix A with rank r, we can find an $m \times r$ matrix B with rank r (so it is injective), and an $r \times n$ matrix C with rank r (so it is surjective), such that $A = BC$.*

Proof. Pick the right basis, then $A = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$.

However, we have

$$\begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix} = \begin{bmatrix} I_{r\times r} \\ O \end{bmatrix} \begin{bmatrix} I_{r\times r} & O \end{bmatrix}.$$

So set $B = \begin{bmatrix} I_{r\times r} \\ O \end{bmatrix}$ and $C = \begin{bmatrix} I_{r\times r} & O \end{bmatrix}$, and we are done. □

Remark 5.4.30. *This is part of a larger phenomenon. For any map f between sets, then we can always find an injective map g and a surjective map h such that $f = g \circ h$.*

What does this mean geometrically? It means that ALL linear maps A behave like this:

1. First, we do a surjective linear map C, i.e., we collapse a few dimensions of the domain, until only r dimensions left.
2. Then, we do an injective linear map B, i.e., we use an inclusion map to put the leftover from the last step into a larger space, the codomain.

Computationally, if all basis we choose are nice, then ALL linear maps $A : \mathbb{R}^n \to \mathbb{R}^m$ of rank r behave like this:

1. Given an input $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$, we drop a few coordinates and get $\begin{bmatrix} x_1 \\ \vdots \\ x_r \end{bmatrix}$.
2. Then, we add a few zero coordinates, and get $\begin{bmatrix} x_1 \\ \vdots \\ x_r \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^m$.

So, all linear maps are simply "collapse a few dimensions, and then put into a larger space." From this intuition, the following corollary is completely trivial.

Corollary 5.4.31. *The rank of a matrix A is the same as* $\dim(\mathrm{Ran}(A))$.

Proof. We don't even need the full rank decomposition. Say A has rank r. Simply pick the right basis, then $A = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$. Now look, its range (column space) is spanned by $\boldsymbol{e}_1, \dots, \boldsymbol{e}_r$ and therefore has dimension r.

Obviously changing basis could only change the names of vectors, and could not change the dimension of $\dim(\mathrm{Ran}(A))$. □

Our previous proofs show that, given a matrix A with rank r, we can find a rank normal form also with rank r. But is it possible to find some rank normal form of A with a different rank? This is NOT possible. In particular, the rank normal form is unique, as shown below.

Lemma 5.4.32. *For any $m \times n$ matrix A, pick any invertible $m \times m$ matrix R and invertible $n \times n$ matrix C, then A and RAC must have the same rank.*

Proof. No change of basis could possibly change $\dim(\mathrm{Ran}(A))$ in any way. $\square$

Corollary 5.4.33. *The rank normal form of a matrix is unique. And two matrices A, B have the same rank if and only if we can find invertible $m \times m$ matrix R and invertible $n \times n$ matrix C, such that $B = RAC$.*

In particular, the rank of a matrix COMPLETELY characterizes the linear map. If two matrices are both $m \times n$ and have the same rank, then they are the SAME linear map, differ only by a change of basis in the domain and a change of basis in the codomain.

We end this section by a useful decompositon in practice.

Proposition 5.4.34 (Rank-one decomposition)**.** *A matrix of rank r is the sum of r rank-one matrices.*

Proof. Given a matrix A with rank r, we first do the full rank decomposition $A = BC$. Note that B has r columns, and C has r rows. Say $B = \begin{bmatrix} \boldsymbol{b}_1 & \dots & \boldsymbol{b}_r \end{bmatrix}$ and $C = \begin{bmatrix} \boldsymbol{c}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{c}_r \end{bmatrix}$. Then we have

$$A = BC = \begin{bmatrix} \boldsymbol{b}_1 & \dots & \boldsymbol{b}_r \end{bmatrix} \begin{bmatrix} \boldsymbol{c}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{c}_r \end{bmatrix} = \sum_{i=1}^{r} \boldsymbol{b}_i \boldsymbol{c}_i^{\mathrm{T}}.$$

Each $\boldsymbol{b}_i \boldsymbol{c}_i^{\mathrm{T}}$ is clearly a rank-one matrix. So we are done. $\square$

As a side note, let us briefly talk about the rank of an abstract linear map. This is mostly trivial, but we establish to be technically more rigorous.

Definition 5.4.35. *Given a linear map L, we define its rank* $\mathrm{rank}(L)$ *as* $\dim \mathrm{Ran}(L)$*.*

Recall that $\mathrm{Ran}(L)$ is the collection of all images of L, and they form a natural subspace of the codomain. Its dimension is what we call the rank of L.

Proposition 5.4.36. *The rank of A as a matrix (i.e., number of pivots) is the same as the rank of A as a linear map.*

Proof. Notice that A and $\mathrm{RREF}(A)$ can be thought of as the same linear map, differ only by a change of basis in the codomain. So their range have the same dimension. Now the dimension of $\mathrm{RREF}(A)$ is obviously the number of pivots. $\square$

Example 5.4.37. Let A be a matrix such that each column is an arithmetic sequence. Then each column is of the form $a \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + d \begin{bmatrix} 0 \\ \vdots \\ n-1 \end{bmatrix}$. So the range of A is contained in the span of $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ \vdots \\ n-1 \end{bmatrix}$. So it is at most 2 dimensional.

So A has rank at most two. ☺

Let us review the full rank decomposition process. If $L : V \to W$, how can I decompose this into a surjective linear map followed by an injective linear map?

Well, first of all, we can restricte the codomain to $\mathrm{Ran}(L)$. Then the map $L' : V \to \mathrm{Ran}(L)$ would be surjective, where L' is essentially the same as L, and they only differ in the choice of codomain.

Next, we have an inclusion map $\iota : \mathrm{Ran}(L) \to W$, since $\mathrm{Ran}(L)$ must be a subspace of W.

Finally, see that $L = \iota \circ L'$ as desired.

5.5 (Optional) Alternative proof of Woodbury formula

Let us give an application of rank normal form by proving the Woodbury formula. The idea is very straight forward: we change basis to make the maps look nice.

Theorem 5.5.1. *For any $m \times n$ matrix A and $n \times m$ matrix B, we have $I_m + AB$ invertible if and only if $I_n + BA$ invertible, and we have a formula $(I_m + AB)^{-1} = I_m - A(I_n + BA)^{-1}B$.*

Proof. Note that as a linear map, A goes from $\mathbb{R}^n$ to $\mathbb{R}^m$ and B goes from $\mathbb{R}^m$ to $\mathbb{R}^n$.

Let us change basis in $\mathbb{R}^m$ and $\mathbb{R}^n$ so that A looks nice. This means we have $A = P \begin{bmatrix} I_r & O \\ O & O \end{bmatrix} Q$ for some invertible matrices P, Q. Now, under these change of bases, $B = Q^{-1} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} P^{-1}$ for some B_1, B_2, B_3, B_4.

So $I_m + AB = I_m + P \begin{bmatrix} B_1 & B_2 \\ O & O \end{bmatrix} P^{-1} = P \begin{bmatrix} B_1 + I_r & B_2 \\ O & I_{m-r} \end{bmatrix} P^{-1}$. So this is invertible if and only if $B_1 + I_r$ is invertible.

Similarly, $I_n + BA = Q^{-1} \begin{bmatrix} B_1 + I_r & O \\ B_3 & I_{n-r} \end{bmatrix} Q$. So this is invertible if and only if $B_1 + I_r$ is invertible. So we have $I_m + AB$ invertible if and only if $I_n + BA$ invertible.

Now for the formula, note that $(I_n + BA)^{-1} = Q^{-1} \begin{bmatrix} (B_1 + I_r)^{-1} & O \\ -B_3(B_1 + I_r)^{-1} & I_{n-r} \end{bmatrix} Q$. So $A(I_n + BA)^{-1}B = P \begin{bmatrix} (B_1 + I_r)^{-1}B_1 & (B_1 + I_r)^{-1}B_2 \\ O & O \end{bmatrix} P^{-1}$.

In contrast, $(I_m + AB)^{-1} = P \begin{bmatrix} (B_1 + I_r)^{-1} & -(B_1 + I_r)^{-1}B_2 \\ O & I_{m-r} \end{bmatrix} P^{-1}$. So we see that $A(I_n + BA)^{-1}B + (I_m + AB)^{-1} = P \begin{bmatrix} (B_1 + I_r)^{-1}(B_1 + I_r) & O \\ O & I_{m-r} \end{bmatrix} P^{-1} = I_m$. So we are done. ☐

5.6 Rank-Nullity Theorem and Subspace Algebra

For a linear map L, recall that $\mathrm{Ker}(L)$ is the solution set to $L\boldsymbol{x} = \mathbf{0}$, while $\mathrm{Ran}(L)$ is the subspace of all its images. Both of these are subspaces.

Now under the right basis, L looks like its rank normal form. So it is very easy to see the following:

Theorem 5.6.1 (Rank-Nullity Theorem)**.** $\dim(\mathit{domain}(L)) = \mathrm{rank}(L) + \dim \mathrm{Ker}(L) = \dim \mathrm{Ran}(L) + \dim \mathrm{Ker}(L)$

Note that if the domain is $\mathbb{R}^n$, then for any system $L\boldsymbol{x} = \boldsymbol{b}$, we see that $\dim(\mathit{domain}(L))$ is the total number of variables. The rank of L is the number of dependent variables. Finally, $\dim \mathrm{Ker}(L)$ means all possible free directions to move inside the solution set of $L\boldsymbol{x} = \mathbf{0}$, i.e., the number of free variables.

Hence, This is the abstract statement of our previously established fact that "the number of dependent variable + the number of free variable = the number of all variables". Nevertheless, here is another formal proof.

Proof. Pick nice basis, and poof! We can now assume WLOG (without loss of generality) that L is the matrix $\begin{bmatrix} I_r & O \\ O & O \end{bmatrix}$.

Its range is spanned by $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_r$, and its kernel is spanned by $\boldsymbol{e}_{r+1}, \ldots, \boldsymbol{e}_n$. So we are done by counting. ☐

Remark 5.6.2. *Intuitively, if L starts with n dimensions, and kills $n - r$ dimensions, its range is left with r dimension.*

Here are some extreme cases.

Corollary 5.6.3. *L is injective if and only if* $\mathrm{Ker}(L) = \{\mathbf{0}\}$*, if and only if* $\mathrm{Ran}(L)$ *has the same dimension as the domain. L is surjective if and only if* $\mathrm{Ran}(L)$ *is the same as the codomain, if and only if* $\dim \mathrm{Ker}(A)$ *is the dimension of the domain minus the dimension of the codomain.*

Proof. Obviously L is surjective if and only if $\mathrm{Ran}(L)$ is the same as the codomain. Using rank-nullity, the statement on $\dim \mathrm{Ker}(A)$ is immediate.

For the injective case, note that $L\boldsymbol{v} = L\boldsymbol{w}$ iff $L(\boldsymbol{v} - \boldsymbol{w}) = \mathbf{0}$ iff $\boldsymbol{v} - \boldsymbol{w} \in \mathrm{Ker}(L)$. If the kernel is trivial, we have $\boldsymbol{v} = \boldsymbol{w}$, so L is injective. Conversly if L is injective and $L\boldsymbol{v} = \mathbf{0} = L\mathbf{0}$, we have $\boldsymbol{v} = \mathbf{0}$.

So L is injective if and only if $\mathrm{Ker}(L)$ is trivial, if and only if $\dim \mathrm{Ker}(L) = 0$, if and only if $\dim \mathrm{Ran}(L)$ is the same as the dimension of the domain. $\square$

The idea here actually immediately gives us a picture of how solution sets are like.

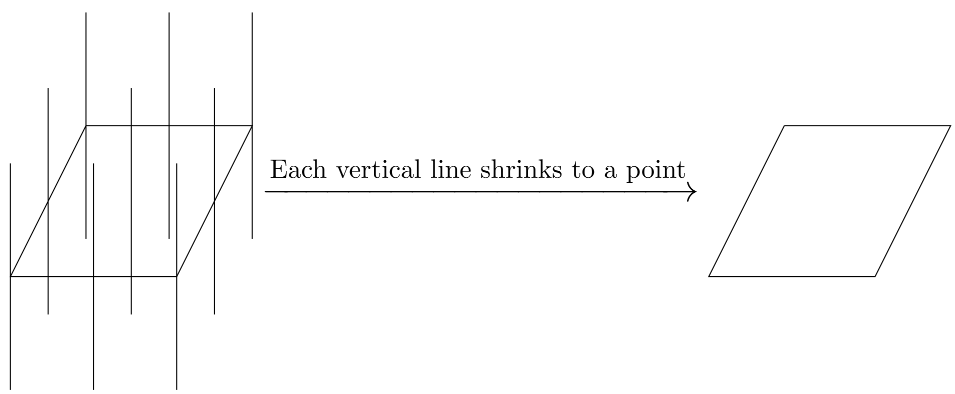
Corollary 5.6.4 (Description to all solutions to a linear system)**.** *For any linear map $L : V \to W$ and any $\boldsymbol{w} \in W$, the solution set for the equation $L\boldsymbol{x} = \boldsymbol{w}$ is a translation of* $\mathrm{Ker}(L)$*. More rigorously, if $\boldsymbol{x}_0$ is a solution, then the solution set is* $\boldsymbol{x}_0 + \mathrm{Ker}(L) = \{\boldsymbol{x}_0 + \boldsymbol{v} \mid \boldsymbol{v} \in \mathrm{Ker}(L)\}$.

Proof. If $L\boldsymbol{y} = \boldsymbol{w}$ too, then $L\boldsymbol{x}_0 = L\boldsymbol{y}$, so $\boldsymbol{y} - \boldsymbol{x}_0 \in \mathrm{Ker}(L)$. So $\boldsymbol{y} \in \boldsymbol{x}_0 + \mathrm{Ker}(L)$. $\square$

Example 5.6.5. Consider $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. It sends $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ to $\begin{bmatrix} x \\ y \end{bmatrix}$, so it corresponds to the projection of $\mathbb{R}^3$ to the xy-plane. Its kernel is simply the z-axis.

So for any $\boldsymbol{b} \in \mathbb{R}^2$, what is the solution set to $A\boldsymbol{x} = \boldsymbol{b}$? It must be a parallel translation of the z-axis. It is simply some vertical line in $\mathbb{R}^3$. In particular, if we move $\boldsymbol{b}$ around, the solution set would also move around in a parallel manner.

In particular, the geometric behavior of A is this: in the domain $\mathbb{R}^3$, look at all the lines parallel to the z-axis. A would shrink each line to a point. Hence the range of A is just the xy-plane.

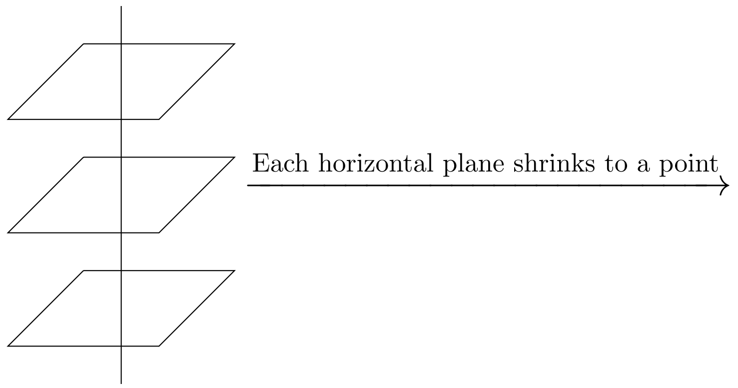


You can also see that the kernel has dimension one, the domain has dimension three. By shrinking everything parallel to the kernel, we end up with a range with dimension two.

Similarly, consider $A = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$. It sends $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ to z, so it corresponds to the projection of $\mathbb{R}^3$ to the z-axis. Its kernel is simply the xy-plane.

So for any $b \in \mathbb{R}$, what is the solution set to $A\boldsymbol{x} = b$? It must be a parallel translation of the xy-plane. These are just various horizontal planes in $\mathbb{R}^3$. In particular, if we move b around, the solution set would also move around in a parallel manner.

In particular, the geometric behavior of A is this: in the domain $\mathbb{R}^3$, look at all the planes parallel to the xy-plane. A would shrink each plane to a point. Hence the range of A is just the z-axis.



You can also see that the kernel has dimension two, the domain has dimension three. By shrinking everything parallel to the kernel, we end up with a range with dimension one. ☺

Remark 5.6.6. *Now we have a clearer intuition on an arbitrary linear map.*

Say L has kernel dimension d. Then in fact, it is collapsing all d-dimensional affine subspaces parallel to $\mathrm{Ker}(L)$ *to points. Each d-dimensional object is now turned into a 0-dimensional object. So our n dimensional domain is turned in to the $n-d$ dimensional range.*

With our most powerful tool established, we are now free to study subspaces. The first important part is to study containment relation.

Proposition 5.6.7. *Let V be a finite dimensional space. If $W \subseteq V$, then* $\dim W \leq \dim V$*. If $W \subseteq V$ and* $\dim W = \dim V$*, then in fact $W = V$.*

Proof. Say $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ are linearly independent vectors inside W. Note that, the definition of linear independence does not really need W. It simply tries to combine these vectors and see if we can get $\mathbf{0}$. Therefore, they are also linearly independent inside V. In particular, if V is finite dimensional, then W must also be finite dimensional.

If we pick a basis $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ for W, then they are linearly independent and they are in V. So we can extend it to a basis of V, and thus $\dim W \leq \dim V$.

In particular, if V is also k-dimensional, then $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ is a linearly independent collection of k vectors in V, and thus also a basis for V. So $V = W$.

For an alternative proof, consider the inclusion map $inc : W \to V$, which is linear. So say A is a matrix for it under some basis. Then the rank of A is at most the number of rows of A. But the former is just $\dim(W)$ while the latter is $\dim(V)$. So $\dim(W) \leq \dim(V)$. Furthermore, if the two are equal, then A is a square matrix. So it is injective (since it is inclusion) and square, hence bijective. But if the inclusion map is bijective, then it is in fact the identity map. So $V = W$. □

We already uses this from time to time without pointing it out. Now let us specifically consider subspaces of $\mathbb{R}^n$.

Corollary 5.6.8. *Any subspace W of $\mathbb{R}^n$ is* $\mathrm{Ran}(A)$ *for some A and* $\mathrm{Ker}(B)$ *for some B.*

Proof. Pick a basis $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k \in W \subseteq \mathbb{R}^n$ for W. Now $W = \mathrm{Ran}\begin{bmatrix}\boldsymbol{w}_1 & \ldots & \boldsymbol{w}_k\end{bmatrix}$. Done.

Now extend $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ to a basis $\mathcal{C} = \boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ for V (via the independence extension lemma, essentially). Note that for any vector $\boldsymbol{x} \in \mathbb{R}^n$, then $\boldsymbol{x} \in W$ if and only if the last $n-k$ coordinates of $\boldsymbol{x}_{\mathcal{C}}$ are zero. Let B be a map from $\mathbb{R}^n$ to $\mathbb{R}^{n-k}$, such that B sends each $\boldsymbol{x}$ to the last $n-k$ coordinates of $\boldsymbol{x}_{\mathcal{C}}$. Then immediately $\mathrm{Ker}(B) = W$, and B is an $(n-k) \times n$ matrix. End of proof.

(To see a more constructive version of the later portion of the proof, let $C = \begin{bmatrix}\boldsymbol{w}_1 & \ldots & \boldsymbol{w}_n\end{bmatrix}^{-1}$ be the coordimate map for this new basis, and $P = \begin{bmatrix}O & I_{n-k}\end{bmatrix}$ be the projection map sending $\begin{bmatrix}x_1 \\ \vdots \\ x_n\end{bmatrix}$ to $\begin{bmatrix}x_{k+1} \\ \vdots \\ x_n\end{bmatrix}$.

Then $\boldsymbol{v} \in W$ iff it only needs vectors $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ among the basis $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$, iff its last $n-k$ coordinates under the new basis are all zero, iff $PC\boldsymbol{v} = \mathbf{0}$. So $W = \operatorname{Ker}(PC)$.) $\square$

Now, given some subspaces, we are interested in how to make more subspaces.

Definition 5.6.9. *Given two subspaces U, W in V, their **intersection** $U \cap W$ is, well, um, their intersection. Their **sum** (or their **span**) $U + W$ is the set $\{\boldsymbol{u} + \boldsymbol{w} \mid \boldsymbol{u} \in U, \boldsymbol{w} \in W\}$.*

Proposition 5.6.10. *Given two subspaces, their intersection and their sum are subspaces.*

Proof. Suppose $\boldsymbol{u}, \boldsymbol{w} \in U \cap W$, and take any $a, b \in \mathbb{R}$.

Since $\boldsymbol{u}, \boldsymbol{w} \in U$ and U is a subspace, therefore $a\boldsymbol{u} + b\boldsymbol{w} \in U$. Similarly, since $\boldsymbol{u}, \boldsymbol{w} \in W$ and W is a subspace, therefore $a\boldsymbol{u} + b\boldsymbol{w} \in W$. Hence $a\boldsymbol{u} + b\boldsymbol{w} \in U \cap W$.

Suppose $\boldsymbol{v}_1, \boldsymbol{v}_2 \in U + W$, and take any $a, b \in \mathbb{R}$.

For $i = 1, 2$, since $\boldsymbol{v}_i \in U + W$, we must have $\boldsymbol{v}_i = \boldsymbol{u}_i + \boldsymbol{w}_i$ for some $\boldsymbol{u}_i \in U, \boldsymbol{w}_i \in W$. Then $a\boldsymbol{v}_1 + b\boldsymbol{v}_2 = a(\boldsymbol{u}_1 + \boldsymbol{w}_1) + b(\boldsymbol{u}_2 + \boldsymbol{w}_2) = (a\boldsymbol{u}_1 + b\boldsymbol{u}_2) + (a\boldsymbol{w}_1 + b\boldsymbol{w}_2)$. Since the first summand is in U and the second summand is in W, the result is in $U + W$. $\square$

In the specific case of subspaces of $\mathbb{R}^n$, we have the following nice block matrx interpretations of sum and intersections.

Proposition 5.6.11. $\operatorname{Ran}(A) + \operatorname{Ran}(B) = \operatorname{Ran}\begin{bmatrix} A & B \end{bmatrix}$, *and* $\operatorname{Ker}(A) \cap \operatorname{Ker}(B) = \operatorname{Ker}\begin{bmatrix} A \\ B \end{bmatrix}$.

Proof. $\operatorname{Ran}(A) + \operatorname{Ran}(B)$ means the all possible linear combinations of (all possible linear combinations of columns of A) and (all possible linear combinations of columns of B). $\operatorname{Ran}\begin{bmatrix} A & B \end{bmatrix}$ means the all possible linear combinations of columns of A and columns of B. Wait. Why do we even need to prove this? This is totally trivial!

$\boldsymbol{x} \in \operatorname{Ker}(A) \cap \operatorname{Ker}(B)$ means $A\boldsymbol{x} = B\boldsymbol{x} = \mathbf{0}$. On the other hand, $\boldsymbol{x} \in \operatorname{Ker}\begin{bmatrix} A \\ B \end{bmatrix}$ means $\begin{bmatrix} A\boldsymbol{x} \\ B\boldsymbol{x} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$. Wait. Why do we even need to prove this? Again, this is totally trivial! $\square$

Generically, the following also provide some nice intuitions. In particular, they are analogous to the situations of subsets.

Proposition 5.6.12. *$U \cap W$ is the largest subspace contained in both, and $U + W$ is the smallest subspace containing both.*

Proof. Pretty straightforward by definition.

If V is a subspace containing both U and W, surely it contains all $\boldsymbol{u} + \boldsymbol{w}$ where $\boldsymbol{u} \in U$ and $\boldsymbol{w} \in W$. So $V \supseteq U + W$.

If V is a subspace inside both U and W, surely it is inside $U \cap W$. So $V \subseteq V \cap W$. $\square$

Now, it might feel like sums and intersections of subspaces bear some similarity to unions (smallest subset containing both subsets) and intersections (largest subset inside both subsets) of subsets. However, there are some crucial distinctions.

Example 5.6.13. Subspace algebra can be very tricky, and very dangerous. Recall that, normally, for ordinary subsets, then we have distrubutativity of $\cup$ and $\cap$ over each other, namely, $A \cup (B \cap C) = (A \cup B) \cup (A \cup C)$ and $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$. Draw some Vayne diagram to see this.

However, suppose we look at $\mathbb{R}^2$, let U be the line $x = 0$, let V be the line $y = 0$ and let W be the line $x = y$. Note that any two of them can intersect only at the origin, and any two of them would span the whole space.

$U + (V \cap W)$ may NOT equal to $(U + V) \cap (U + W)$. The first one is U, while the second one is the whole $\mathbb{R}^2$.

$U \cap (V + W)$ may NOT equal to $U \cap V + U \cap W$. The first one is U, while the second one is the origin. ☺

However, sometimes the analogy is true. Recall that for two subsets, we have $|A \cup B| = |A| + |B| - |A \cap B|$. (This is called the ***inclusion-exclusion principle*** sometimes.) In particular, if A, B are both big but $A \cup B$ is not big enough, then $A \cup B$ would be "too crowded", and thus A, B must have big intersections. The same goes for subspaces

Theorem 5.6.14 (Inclusion-Exclusion Principle for Subspaces)**.** $\dim(V+W) = \dim V + \dim W - \dim(V \cap W)$.

Proof. The first proof is done by basis-extension and counting. Suppose $\dim(V \cap W) = k, \dim V = a, \dim W = b$.

Pick a basis $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k$ for $V \cap W$. Extend this to a basis of V as $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{a-k}$. Similarly, extend the linearly independent collection $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k$ to a basis of W as $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_{b-k}$. I claim that $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{a-k}, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_{b-k}$ form a basis for $V + W$. (This would then imply that $\dim(V + W) = k + (a-k) + (b-k) = a + b - k$, establishing the desired result.)

Spanning is obvious, so we only prove linear independence here. Suppose $\sum a_i \boldsymbol{u}_i + \sum b_i \boldsymbol{v}_i + \sum c_i \boldsymbol{w}_i = \mathbf{0}$. Then $\sum a_i \boldsymbol{u}_i + \sum b_i \boldsymbol{v}_i = -\sum c_i \boldsymbol{w}_i$. Note that the left hand side is in V, while the right hand side is in W, so both sides are in $V \cap W$!

So $\sum c_i \boldsymbol{w}_i \in V \cap W \subseteq W$. However, every vector in W must be a unique linear combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_{b-k}$, and if the vector is actually in $V \cap W$, then the coordinates for $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{b-k}$ must all be zero. Hence, since $\sum c_i \boldsymbol{w}_i \in V \cap W \subseteq W$, we see that all c_i are zero.

Then $\sum a_i \boldsymbol{u}_i + \sum b_i \boldsymbol{v}_i = \mathbf{0}$. But since $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{a-k}$ is a basis for V, this implies that all a_i and all b_i are zero.

So $\sum a_i \boldsymbol{u}_i + \sum b_i \boldsymbol{v}_i + \sum c_i \boldsymbol{w}_i = \mathbf{0}$ implies that all coefficients are zero. Hence $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{a-k}, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_{b-k}$ is a linearly independent collection of vectors. So it is indeed a basis.

Now let us do an alternative (hardcore but optional) proof. This proof is arguably more abstract, but it is what advanced mathemagicians (ha!) typically would do: we solve a problem by building TONS of structures around the problem, and see what would happen.

Consider the following chain

$$V \cap W \xrightarrow{D} V \times W \xrightarrow{S} V + W$$

Let us explain the space $V \times W$ and the maps D and S.

Here the space $V \times W$ refers to the vector space $\{\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix} : \boldsymbol{v} \in V, \boldsymbol{w} \in W\}$ where addition and scalar multiplication are done in the obvious manner. Clearly if $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_a$ is a basis for V and $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_b$ is a basis for W, then a basis for $V \times W$ is $\begin{bmatrix} \boldsymbol{v}_i \\ \mathbf{0} \end{bmatrix}$ and $\begin{bmatrix} \mathbf{0} \\ \boldsymbol{w}_j \end{bmatrix}$ for all i, j. So $\dim(V \times W) = \dim V + \dim W$.

The map D sends $\boldsymbol{x}$ to $\begin{bmatrix} \boldsymbol{x} \\ -\boldsymbol{x} \end{bmatrix}$. And the map S sends $\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix}$ to $\boldsymbol{v} + \boldsymbol{w}$. These are all linear maps, as you can verify. Furthermore, it is obvious by definition that D is injective and S is surjective.

On extra remark, we also have $\mathrm{Ran}(D) = \mathrm{Ker}(S)$. (This is called ***exact***. The spaces $V \cap W, V \times W, V + W$ form a classical short exact sequence, which is an important phenomenon in all algebra studies.) To see this, first of all obviously $S \circ D$ is the zero map (it sends everything to $\mathbf{0}$). Hence $\mathrm{Ran}(D) \subseteq \mathrm{Ker}(S)$. Conversely, if $\boldsymbol{v} \in \mathrm{Ker}(S)$, then $S(\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix}) = \mathbf{0}$, then $\boldsymbol{v} + \boldsymbol{w} = \mathbf{0}$, hence $\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix} = \begin{bmatrix} \boldsymbol{v} \\ -\boldsymbol{v} \end{bmatrix} = D(\boldsymbol{v}) \in \mathrm{Ran}(D)$. Hence $\mathrm{Ker}(S) \subseteq \mathrm{Ran}(D)$. So we have the desired "exactness".

We have finished building. Now to prove the statement, just use rank-nullity twice, once on D and once on S. The rank-nullity on D gives

$$\dim(V \cap W) = \dim \mathrm{Ran}(D) + \dim \mathrm{Ker}(D) = \dim \mathrm{Ran}(D) = \dim \mathrm{Ker}(S).$$

Here $\dim \mathrm{Ker}(D) = 0$ because D is injective.

The rank-nullity on S gives

$$\dim(V) + \dim(W) = \dim(V \times W) = \dim \mathrm{Ker}(S) + \dim \mathrm{Ran}(S) = \dim(V \cap W) + \dim(V + W).$$

So we are done.

Intuitively, short exact sequences are used to decompose algebraic structures. We are pointing out that $V \times W$ is in some sense composed of $V \cap W$ and $V + W$. $\square$

Remark 5.6.15. *(Entirely optional remark)*

The second proof above is actually the same as the popular proof for the inclusion exclusion principle for subsets. For subsets, a proof is typically like this: we "pretent" that there is no intersection and count elements of X and elements of Y. Now when we put these elements into $X \cup Y$, we would have overlaps. The overlaps are $X \cap Y$. Hence $|X \cup Y| = |X| + |Y| - |X \cap Y|$.

For our proof here, the space $V \times W$ is the linear algebra way of "pretending" that the two spaces have zero intersection. In particular, $\dim(V \times W) = \dim V + \dim W$.

For subsets, when we put X, Y into $X \cup Y$, we are performing inclusion maps. Here the map $S : V \times W \to V + W$ is actually the natural linear extension of both inclusion maps $V \to V + W$ and $W \to V + W$.

Finally, for subsets, we would try to look at overlaps of X, Y in $X \cup Y$. Correspondingly, overlaps for the map $S : V \times W \to V + W$ can be studied as the kernel of S. (Which is the collection of inputs with common image $\mathbf{0}$, i.e., they "overlap" at $\mathbf{0}$.)

Here is an immediate application.

Corollary 5.6.16. *If $\dim(W) = n - 1$ is a subspaces of n-dimensional space V, then for any subspace U of V, either $U \subseteq W$ or $\dim(U \cap W) = \dim(U) - 1$.*

This essentially reads that any hyperplane would always cut your dimension by one. Say you are in $\mathbb{R}^n$ and have k hyperplanes in generic position. ("Generic position" means their equations have no redundancy or contradiction.) What is the dimension of their intersection? It would be $n - k$, as each plane cut the dimension by 1.

In particular, n variables with k effective equations means your solution set is an $n - k$ dimensional (affine) space.

Here is another application of the inclusion-exclusion formula.

Corollary 5.6.17. *If $V \cap W = \{\mathbf{0}\}$, then $\dim(V + W) = \dim V + \dim W$.*

A special case of above is the concept of complement subspaces.

Definition 5.6.18. *We say two subspaces U, W of V are complements if $U + W = V$ and $U \cap W = \{\mathbf{0}\}$.*

This is just the analogy of complements for subsets, where we require $S \cup T$ to be the whole set and $S \cap T = \varnothing$.

Corollary 5.6.19. *If U, W are complements of each other in V, then $\dim V = \dim U + \dim W$.*

Another useful phenomenon is that they provides unique decomposition of vectors, similar to $\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}1\\2\\0\end{bmatrix} + \begin{bmatrix}0\\0\\3\end{bmatrix}$. (This is the unique decomposition where we think of the xy-plane and the z-axis as complementary subspaces.)

Corollary 5.6.20. *If U, W are complementary subspaces of V, then for any $\boldsymbol{v} \in V$, we can find unique $\boldsymbol{u} \in U$ and unique $\boldsymbol{w} \in W$ such that $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{w}$.*

Proof. Since $U + W = V$, such decomposition must exist.

Now suppose $\boldsymbol{u}_1 + \boldsymbol{w}_1 = \boldsymbol{v} = \boldsymbol{u}_2 + \boldsymbol{w}_2$. Then rearrange this gives $\boldsymbol{u}_1 - \boldsymbol{u}_2 = \boldsymbol{w}_2 - \boldsymbol{w}_1$. Now the left hand side is in U and the right hand side is in W, so both are in $U \cap W = \{0\}$. Hence both sides are zero, $\boldsymbol{u}_1 = \boldsymbol{u}_2, \boldsymbol{w}_1 = \boldsymbol{w}_2$. $\square$

Now again, the analogy holds at the amount of dimensions, but fail in terms of actual subspaces. In particular complements of subspaces are NOT unique. For example, in $\mathbb{R}^2$, the y-axis and the line $x = y$ are both complements to the x-axis. (This is the same example that fails the law of distribution among subspaces. This is NOT a coincidence, but we don't need to dwell on it too much at the moment.)

5.7 How to find Kernel

In this section we introduce a method to find the kernel of a matrix. Given a matrix A, we aim to find a basis for $\mathrm{Ker}(A)$.

First of all, you should already know how to do this. $\mathrm{Ker}(A)$ is exactly the solution set to $A\boldsymbol{x} = \mathbf{0}$. So this is basically Gaussian elimination.

Example 5.7.1. Suppose A, we first reduce it to RREF, say $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$. Then we see the second and fourth columns are free. So the solution set must be $\begin{bmatrix} * \\ x \\ * \\ z \end{bmatrix}$, where the star portion depends on x, z, and the variables x, z are free. Using these equations we see that the star portion must be $\begin{bmatrix} -2x-3z \\ x \\ -4z \\ z \end{bmatrix}$. So the solution set is $\mathrm{Ker}(A) = \{\begin{bmatrix} -2x-3z \\ x \\ -4z \\ z \end{bmatrix} : x, z \in \mathbb{R}\}$.

But what is a basis for this space? Note that $\begin{bmatrix} -2x-3z \\ x \\ -4z \\ z \end{bmatrix} = x\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z\begin{bmatrix} -3 \\ 0 \\ -4 \\ 1 \end{bmatrix}$. So $\mathrm{Ker}(A)$ is spanned by $\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 0 \\ -4 \\ 1 \end{bmatrix}$. We also see that these two vectors are linearly independent. Therefore they form a basis for $\mathrm{Ker}(A)$.

So we are done. As you can see, the Gaussian elimination process is enough to find a basis for the kernel, where each basis vector corresponds to a free variable. ☺

Now look at the example above. The basis vectors are columns of $\begin{bmatrix} 2 & 3 \\ -1 & 0 \\ 0 & 4 \\ 0 & -1 \end{bmatrix}$, while the RREF of the matrix is $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$. You can easily see that the entries $2, 3, 4$ are in common, and in somewhat corresponding locations. Is there a way to utilize this correspondence directly?

There is indeed a way. First we force the RREF into a "square" matrix such that all pivots are on the diagonal, i.e., we change $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$ into $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Then we substract by identity, and we get

$\begin{bmatrix}0 & 2 & 0 & 3\\0 & -1 & 0 & 0\\0 & 0 & 0 & 4\\0 & 0 & 0 & -1\end{bmatrix}$. Now the non-zero columns are exactly the basis vectors for the kernel.

The rest of the section aims to prove that this is a valid method.

Definition 5.7.2. *Given a matrix A, we get its RREF X, delete all zero rows, then add in zero-rows so that all pivots of X are on the diagonal. We call the resulting matrix A' the square-RREF of A.*

Proposition 5.7.3. *If A' is the square-RREF of A, then* $\mathrm{Ker}(A) = \mathrm{Ker}(A')$.

Proof. Note that elementary row operations, deleting zero rows, or adding zero rows do not change the solution set. □

Lemma 5.7.4. *If A' is the square-RREF of A, then it is an upper triangular matrix. And for each index i, either the i-th column of A' is $\boldsymbol{e}_i$, or the i-th row of A' is zero.*

Proof. Before the proof, you may stare at $\begin{bmatrix}1 & 2 & 0 & 3\\0 & 0 & 0 & 0\\0 & 0 & 1 & 4\\0 & 0 & 0 & 0\end{bmatrix}$ for some intuition about this lemma.

Now we begin the proof. By construction of A', each row of A' is either entirely zero, or has the first non-zero entry on the diagonal. Therefore it is upper triangular.

First, note that elementary row operations, deleting zero rows, or adding zero rows do not change whether a column is pivotal or free. Therefore, A has i-th column pivotal iff $\mathrm{RREF}(A)$ has i-th column pivotal, iff A' has i-th column pivotal.

Now, since A' is upper triangular, if its (i,i)-entry is non-zero, then its i-th column cannot be a linear combination of previous columns. So it is pivotal. So the i-th column of A is a pivotal column. By definition of RREF, if the i-th column of A is pivotal, then the pivot entry in the i-th column of $\mathrm{RREF}(A)$ is 1, and all non-pivot entries of the i-th column of $\mathrm{RREF}(A)$ must be zero. Since we obtained A' from $\mathrm{RREF}(A)$ by adding zero rows, therefore the i-th column of A' is $\boldsymbol{e}_i$.

On the other hand, pivots in the i-th column of $\mathrm{RREF}(A)$ is moved down to the (i,i)-entry of A' by construction. So if the (i,i)-entry of A' is zero, then $\mathrm{RREF}(A)$ has no pivot in the i-th column. So when we create A' from $\mathrm{RREF}(A)$, the entire i-th row of A' is a newly inserted zero row. □

Lemma 5.7.5. *If A' is the square-RREF of A, we take its diagonal entries and make a diagonal matrix D. Then $A'D = D$ and $DA' = A'$.*

Proof. Note that by the last lemma, for each index i, either $A'\boldsymbol{e}_i = \boldsymbol{e}_i$ (which implies that the (i,i)-entry of D is 1), or $\boldsymbol{e}_i^{\mathrm{T}}A' = \mathbf{0}^{\mathrm{T}}$ (which implies that the (i,i)-entry of D is 0).

Suppose for an index i, $A'\boldsymbol{e}_i = \boldsymbol{e}_i$. Then the (i,i)-entry of D is 1. Since D is diagonal, this means $D\boldsymbol{e}_i = \boldsymbol{e}_i$ and $\boldsymbol{e}_i^{\mathrm{T}}D = \boldsymbol{e}_i^{\mathrm{T}}$. So $A'D\boldsymbol{e}_i = A'\boldsymbol{e}_i = \boldsymbol{e}_i = D\boldsymbol{e}_i$ and $\boldsymbol{e}_i^{\mathrm{T}}DA' = \boldsymbol{e}_i^{\mathrm{T}}A'$. In particular, D and $A'D$ have the same i-th column, and A' and DA' have the same i-th row.

Now suppose for an index i, $\boldsymbol{e}_i^{\mathrm{T}}A' = \mathbf{0}^{\mathrm{T}}$. Then the (i,i)-entry of D is 0. Since D is diagonal, this means $D\boldsymbol{e}_i = \mathbf{0}$ and $\boldsymbol{e}_i^{\mathrm{T}}D = \mathbf{0}^{\mathrm{T}}$. So $A'D\boldsymbol{e}_i = A'\mathbf{0} = \mathbf{0} = D\boldsymbol{e}_i$ and $\boldsymbol{e}_i^{\mathrm{T}}DA' = \mathbf{0}^{\mathrm{T}}A' = \mathbf{0}^{\mathrm{T}} = \boldsymbol{e}_i^{\mathrm{T}}A'$. In particular, D and $A'D$ have the same i-th column, and A' and DA' have the same i-th row.

Either way, all columns of D and $A'D$ are identical, so $D = A'D$. And all rows of A' and DA' are identical, so $A' = DA'$. □

Lemma 5.7.6. *If A' is the square-RREF of A, then $(A')^2 = A'$.*

Proof. $(A')^2 = A'A' = A'(DA') = (A'D)A' = DA' = A'$. □

Lemma 5.7.7. *If A' is the square-RREF of A, then* $\mathrm{Ran}(A' - I) \subseteq \mathrm{Ker}(A)$.

Proof. Since $(A')^2 = A'$, we see that $A'(A' - I) = 0$. So $\mathrm{Ran}(A' - I) \subseteq \mathrm{Ker}(A') = \mathrm{Ker}(A)$. □

Theorem 5.7.8. *If A' is the square-RREF of A, then* $\operatorname{Ran}(A' - I) = \operatorname{Ker}(A)$. *Furthermore, non-zero columns of* $\operatorname{Ran}(A' - I)$ *are linearly independent and hence form a basis for* $\operatorname{Ker}(A)$.

Proof. We already have $\operatorname{Ran}(A' - I) \subseteq \operatorname{Ker}(A)$. To see $\operatorname{Ran}(A' - I) = \operatorname{Ker}(A)$, it is enough to show that $\dim \operatorname{Ran}(A' - I) \geq \dim \operatorname{Ker}(A)$.

Suppose the domain of A is n-dimensional, and A has rank r. Then A' is $n \times n$, and it has r diagonal entires with value 1 and $n - r$ diagonal entries of value 0. So $A' - I$ has $n - r$ diagonal entries of value -1. So by the lemma below, $\operatorname{rank}(A' - I) \geq n - r = \dim \operatorname{Ker}(A)$.

Now for each i, $A' - I$ has zero i-th column iff A' has i-th column $\boldsymbol{e}_i$, iff the (i, i)-entry of A' is 1, iff the i-th column of A is pivotal. So there are a total of r such columns. So $A' - I$ has exactly $n - r$ non-zero columns, and their span has dimension $\dim \operatorname{Ran}(A' - I) \geq n - r$. Hence all these columns are linearly independent. $\square$

5.8 (Optional) Rank Inequalities

Here are somethings that are very useful sometimes, if you want to do rank estimates for some generic linear maps. Nevertheless, they are not an essential part of this class, and is not used for other contents in this lecture notes. We present them here, along with an application: to decide when a square matrix (not necessarily invertible) can have LU decompositions.

Definition 5.8.1. *Given a linear map $L : V \to W$ and a subspace $U \subseteq W$, the pullback of U via L is the subspace is $L^{-1}(U) = \{\boldsymbol{v} \mid L(\boldsymbol{v}) \in U\}$.*

Dually, given a subspace $U \subseteq V$, the pushforward of U via L is the subspace $L(U) = \{L(\boldsymbol{v}) \mid \boldsymbol{v} \in U\}$.

These can be thought of as generalizations of range and kernel. The range is the pushforward of the domain. The kernel is the pullback of $\{\boldsymbol{0}\}$. In particular, here is a very interesting example.

Proposition 5.8.2. *For any matrices A, B, we have* $\operatorname{Ran}(AB) = A(\operatorname{Ran}(B))$ *and* $\operatorname{Ker}(AB) = B^{-1}(\operatorname{Ker}(A))$.

Proof. These are very clear just from the definition. For example, we have $\operatorname{Ran}(AB) = \{AB\boldsymbol{v} \mid \boldsymbol{v} \in \mathbb{R}^n\} = A\{B\boldsymbol{v} \mid \boldsymbol{v} \in \mathbb{R}^n\} = A \operatorname{Ran}(B)$. $\square$

Let us establish some dimension results that is enough for all our future computations.

Lemma 5.8.3 (Range and Kernal of restrictions of maps)**.** *Given a linear map $L : V \to W$ and a subspace U of W, let L' be the restriction of L with domain $L^{-1}(U)$ and codomain U. Then* $\operatorname{Ran}(L') = \operatorname{Ran}(L) \cap U$ *and* $\operatorname{Ker}(L') = \operatorname{Ker}(L)$.

Given a linear map $L : V \to W$ and a subspace U of V, let L' be the restriction of L with domain U and codomain $L(U)$. Then $\operatorname{Ran}(L') = L(U)$ *and* $\operatorname{Ker}(L') = \operatorname{Ker}(L) \cap U$.

Proof. The proofs are mostly trivial, but lengthy (because every equality between sets $X = Y$ need to be proven twice as $X \subseteq Y$ and $Y \subseteq X$, hence there are 8 things to prove...). Feel free to skip these proofs.

For both statements, beware of the key relation between L and L'. If $\boldsymbol{x}$ is in the domain of L', then we must have $L'\boldsymbol{x} = L\boldsymbol{x}$ by definition. If $\boldsymbol{x}$ is not in the domain of L', then $L'\boldsymbol{x}$ is not defined while $L\boldsymbol{x}$ could be defined.

Let us prove the first paragraph.

Therefore, pick any $\boldsymbol{x} \in \operatorname{Ker}(L')$. Then $L'\boldsymbol{x} = \boldsymbol{0}$. Hence $L\boldsymbol{x} = \boldsymbol{0}$ and $\boldsymbol{x} \in \operatorname{Ker}(L)$. Conversely, if $\boldsymbol{x} \in \operatorname{Ker}(L)$, then $\boldsymbol{x} \in L^{-1}(\{\boldsymbol{0}\}) \subseteq L^{-1}(U)$, hence $L'\boldsymbol{x}$ is defined. So $L'\boldsymbol{x} = L\boldsymbol{x} = \boldsymbol{0}$ and thus $\boldsymbol{x} \in \operatorname{Ker}(L')$.

Pick any $\boldsymbol{b} \in \operatorname{Ran}(L')$, then $\boldsymbol{b} = L'\boldsymbol{x}$ for some $\boldsymbol{x}$ in the domain of L'. Hence $\boldsymbol{b} = L\boldsymbol{x} \in \operatorname{Ran}(L)$. And since $\boldsymbol{x} \in L^{-1}(U)$, we see that $\boldsymbol{b} = L\boldsymbol{x} \in U$. So $\boldsymbol{b} \in \operatorname{Ran}(L) \cap U$.

Conversely, say $\boldsymbol{b} \in \operatorname{Ran}(L) \cap U$. Since $\boldsymbol{b} \in \operatorname{Ran}(L)$, by definition this means $\boldsymbol{b} = L\boldsymbol{x}$ for some $\boldsymbol{x}$. But then since $\boldsymbol{b} \in U$, this means that $\boldsymbol{x} \in L^{-1}(U)$. Hence L' is defined on $\boldsymbol{x}$, and $L'\boldsymbol{x} = L\boldsymbol{x} = \boldsymbol{b}$. So $\boldsymbol{b} \in \operatorname{Ran}(L')$.

Now let us prove the second paragraph.

$L(U)$ is literally defined as the set of images of $L\boldsymbol{u}$ for $\boldsymbol{u} \in U$, hence L' is surjective by construction. So we have $\operatorname{Ran}(L') = L(U)$.

If $\boldsymbol{x} \in \operatorname{Ker}(L')$, then $L'\boldsymbol{x} = \mathbf{0}$, hence $\boldsymbol{x}$ is defined (i.e., $\boldsymbol{x} \in U$), and hence $L\boldsymbol{x} = L'\boldsymbol{x} = \mathbf{0}$. So we also have $\boldsymbol{x} \in \operatorname{Ker}(L)$. So $\boldsymbol{x} \in \operatorname{Ker}(L) \cap U$.

Conversely, say $\boldsymbol{x} \in U \cap \operatorname{Ker}(L)$. Then $L\boldsymbol{x} = \mathbf{0}$, and L' is defined on $\boldsymbol{x}$. Therefore $L'\boldsymbol{x} = L\boldsymbol{x} = \mathbf{0}$. So $\boldsymbol{x} \in \operatorname{Ker}(L')$. ☐

Proposition 5.8.4 (Dimensions of pullbacks and pushforwards of subspaces)**.**

$$\dim L^{-1}(U) = \dim(U \cap \operatorname{Ran}(L)) + \dim \operatorname{Ker}(L).$$

$$\dim L(U) = \dim U - \dim(U \cap \operatorname{Ker}(L)).$$

Proof. Just apply rank-nullity on L' in the previous lemma. ☐

You don't need to remember the formula above. Personally I simply keep in mind that all such things can be done via rank-nullity, and I search for these formulas online (or re-deduce them myself) when I need them.

Corollary 5.8.5. *We have many rank inequalities among matrices:*

1. $\operatorname{rank}(A + B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$.
2. $\operatorname{rank}(A) + \operatorname{rank}(B) - n \leq \operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B))$. *(Here n is the dimension of the domain of A which is also the codomain of B.)*
3. *(Optional)* $\max(\operatorname{rank}(A), \operatorname{rank}(B)) \leq \operatorname{rank}(\begin{bmatrix} A & B \end{bmatrix}) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$.
4. *(Optional)* $\max(\operatorname{rank}(A), \operatorname{rank}(B)) \leq \operatorname{rank}(\begin{bmatrix} A \\ B \end{bmatrix}) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$.
5. *(Optional)* $\operatorname{rank}(\begin{bmatrix} A & O \\ O & B \end{bmatrix}) = \operatorname{rank}(A) + \operatorname{rank}(B)$.
6. *(Optional)* $\operatorname{rank}(\begin{bmatrix} A & O \\ C & B \end{bmatrix}) \geq \operatorname{rank}(A) + \operatorname{rank}(B)$.

Proof. The goal is NOT to prove these things. Rather, the goal is to practice using rank-nullity on subspaces.

1. Obvious because all columns of $A + B$ are linear combinations of columns of A and columns of B, so $\operatorname{Ran}(A + B) \subseteq \operatorname{Ran}(A) + \operatorname{Ran}(B)$. Take dimensions and $\dim \operatorname{Ran}(A + B) \leq \dim(\operatorname{Ran}(A) + \operatorname{Ran}(B)) \leq \dim \operatorname{Ran}(A) + \dim \operatorname{Ran}(B)$. The last inequality is inclusion-exclusion principle.
2. Obviously any image via AB is in particular some image of A, so $\operatorname{Ran}(AB) \subseteq \operatorname{Ran}(A)$, which gives $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$.

   Now $\operatorname{Ran}(AB) = \{AB\boldsymbol{v} \mid \boldsymbol{v} \in \mathbb{R}^n\} = A\{B\boldsymbol{v} \mid \boldsymbol{v} \in \mathbb{R}^n\} = A\operatorname{Ran}(B)$. By the pushforward formula (equivalently, by rank-nullity on the map from $\operatorname{Ran}(B)$ to $A\operatorname{Ran}(B)$), we have $\dim A\operatorname{Ran}(B) = \dim \operatorname{Ran}(B) - \dim(\operatorname{Ran}(B) \cap \operatorname{Ker}(A))$. This immediatley established $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$.

   By the same equation, we can also have $\dim A\operatorname{Ran}(B) \geq \dim \operatorname{Ran}(B) - \dim \operatorname{Ker}(A) = \dim \operatorname{Ran}(B) + \dim \operatorname{Ran}(A) - n$. So we are done.
3. For $\begin{bmatrix} A & B \end{bmatrix}$ both sides are obvious, because $\operatorname{Ran}\begin{bmatrix} A & B \end{bmatrix} = \operatorname{Ran}(A) + \operatorname{Ran}(B)$.
4. Let n be the dimension of the common domain. Now $\operatorname{Ker}(\begin{bmatrix} A \\ B \end{bmatrix}) = \operatorname{Ker}(A) \cap \operatorname{Ker}(B)$. Take dimensions we have $\min(\dim \operatorname{Ker}(A), \dim \operatorname{Ker}(B)) \geq \dim \operatorname{Ker}(\begin{bmatrix} A \\ B \end{bmatrix}) = \dim \operatorname{Ker}(A) + \dim \operatorname{Ker}(B) - \dim(\operatorname{Ker}(A) + \operatorname{Ker}(B)) \geq \dim \operatorname{Ker}(A) + \dim \operatorname{Ker}(B) - n$. Now use the fact that rank is n minus the dimension of the kernel, and these converts to the desried equation.

5. It looks quite obvious that $\operatorname{Ran}(\begin{bmatrix} A \\ O \end{bmatrix})$ and $\operatorname{Ran}(\begin{bmatrix} O \\ B \end{bmatrix})$ have trivial intersection, while the former has the same dimension as $\operatorname{Ran}(A)$, and the latter has the same dimension as $\operatorname{Ran}(B)$.

6. Suppose A has RREF A' and rank a, and B has RREF B' and rank b. Then by row operations, $\operatorname{rank}(\begin{bmatrix} A & O \\ C & B \end{bmatrix}) = \operatorname{rank}(\begin{bmatrix} A' & O \\ * & B' \end{bmatrix})$. Now, deleting columns would obviously only reduce the dimension of the column space, so it would only decrease rank. By dropping all non-pivotal columns, we have $\operatorname{rank}(\begin{bmatrix} A' & O \\ * & B' \end{bmatrix}) \geq \operatorname{rank}(\begin{bmatrix} I_a & O \\ O & O \\ * & I_b \\ * & O \end{bmatrix})$. By further block row operations, we have $\operatorname{rank}(\begin{bmatrix} I_a & O \\ O & O \\ * & I_b \\ * & O \end{bmatrix}) = \operatorname{rank}(\begin{bmatrix} I_a & O \\ O & O \\ O & I_b \\ O & O \end{bmatrix}) = a + b$. So we are done.

□

Remark 5.8.6. *Note that $\begin{bmatrix} A & B \end{bmatrix}$ and $\begin{bmatrix} A \\ B \end{bmatrix}$, although similar in terms of rank inequalities, might NOT have the same rank. For example, take $\begin{bmatrix} \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \end{bmatrix}$.*

The meaning behind the proof is sometimes more important than the actual results. For example, consider the statement $\operatorname{rank}(A) + \operatorname{rank}(B) - n \leq \operatorname{rank}(AB)$. In the proof, we essentially established that $\dim \operatorname{Ran}(AB) \geq \dim \operatorname{Ran}(B) - \dim \operatorname{Ker}(A)$. So if A killed k dimensions, then performing the linear map A after B, A would kill at most k dimensions in $\operatorname{Ran}(B)$. Hence the resulting space $\operatorname{Ran}(AB)$ has a dimension at most $\dim \operatorname{Ran}(B) - \dim \operatorname{Ker}(A)$.

The inequality $\dim \operatorname{Ran}(AB) \geq \dim \operatorname{Ran}(B) - \dim \operatorname{Ker}(A)$ is sometimes more important than the actual rank inequality. For example, one can easily see the following fun statement, by using this inequality repeatedly. (Useful next semester for the establishement of Jordan canonical form.)

Proposition 5.8.7. *If A is $n \times n$ and $A^k = O$, then $\dim \operatorname{Ker}(A) \geq \frac{n}{k}$.*

Proof. Intuitively, if $\dim \operatorname{Ker}(A) < \frac{n}{k}$, then for $\operatorname{Ran}(A^k) = A^k \operatorname{Ran}(I_n)$, each application of A to the subspace would kill less than $\frac{n}{k}$ dimensions, and a total of k steps would kill less than n dimensions. So $\dim \operatorname{Ran}(A^k) > 0$, there must be surviving dimensions.

Rigorous proof goes like this: If $A^k = 0$, then we have

$$\begin{aligned} 0 &= \dim \operatorname{Ran}(A^k) \\ &\geq \dim \operatorname{Ran}(A^{k-1}) - \dim \operatorname{Ker}(A) \\ &\geq \dim \operatorname{Ran}(A^{k-2}) - 2\dim \operatorname{Ker}(A) \\ &\geq \cdots \\ &\geq \dim \operatorname{Ran}(A^0) - k \dim \operatorname{Ker}(A) = n - k \operatorname{Ker}(A). \end{aligned}$$

Hence $\operatorname{Ker}(A) \geq \frac{n}{k}$. □

A quick corollary to the rank inequalities.

Corollary 5.8.8. *If U is an upper triangular matrix with k non-zero diagonal entries, then $\operatorname{rank}(U) \geq k$.*

Proof. This is basically $\operatorname{rank}(\begin{bmatrix} A & B \\ & C \end{bmatrix}) \geq \operatorname{rank}(A) + \operatorname{rank}(C)$ plus mathematical induction. □

Here is one last inequality between ranks that is sometimes useful.

Proposition 5.8.9 (The Inclusion-Exclusion Principle for maps)**.** *For any $m \times d$ matrix A, $d \times k$ matrix B and $k \times n$ matrix C, we have* $\operatorname{rank}(AB) + \operatorname{rank}(BC) \leq \operatorname{rank}(ABC) + \operatorname{rank}(B)$.

Intuitively, ABC is the union of AB and BC, while B is the intersection of AB and BC. Hence this is a version of "Inclusion-Exclusion Principle" for maps. This intuition also inspires a very beautiful proof. For subspaces, we have $\dim(V) + \dim(W) = \dim(V + W) + \dim(V \cap W)$, and the proof relies on the construction of an intermediate space $V \times W$ with dimension $\dim(V) + \dim(W)$. This idea can be used for all variant versions of "inclusion-exclusion principle". In this case, how to build an intermediate map for AB and BC? We want a way to combine the two maps independently, and one way to do this is via the diagram below, which represents the two submaps of a big map.

$$\begin{tikzcd}
\operatorname{Dom}(B) \arrow[r, "AB"] & \operatorname{Cod}(A) \\
\operatorname{Dom}(C) \arrow[r, "BC"] & \operatorname{Cod}(B)
\end{tikzcd}.$$

So we pick the domain as the space $\operatorname{Dom}(B) \times \operatorname{Dom}(C)$, and the codomain as $\operatorname{Cod}(A) \times \operatorname{Cod}(B)$, then we can treat AB and BC as the diagonal blocks of a big map, i.e., the "independent union" of both maps. However, how to handle the intersection map B? From the diagram above, there is an obvious place to put it, and it gives us a new diagram

$$\begin{tikzcd}
\operatorname{Dom}(B) \arrow[r, "AB"] \arrow[dr, "B"] & \operatorname{Cod}(A) \\
\operatorname{Dom}(C) \arrow[r, "BC"] & \operatorname{Cod}(B)
\end{tikzcd}.$$

So we now have a map $M : \operatorname{Dom}(B) \times \operatorname{Dom}(C) \to \operatorname{Cod}(A) \times \operatorname{Cod}(B)$ with block structure $M = \begin{bmatrix} AB & \\ B & BC \end{bmatrix}$.

Proof of IEP for maps. Consider $\begin{bmatrix} AB & \\ B & BC \end{bmatrix}$. (The combination of AB and BC is represented as the diagonal blocks, while the intersection of AB and BC is represented by the lower left block.) We already know that $\operatorname{rank}(AB) + \operatorname{rank}(BC) \leq \operatorname{rank}(\begin{bmatrix} AB & \\ B & BC \end{bmatrix})$.

On the other hand, by row and column block operations we have $\begin{bmatrix} AB & \\ B & BC \end{bmatrix} \to \begin{bmatrix} & -ABC \\ B & BC \end{bmatrix} \to \begin{bmatrix} & -ABC \\ B & \end{bmatrix}$, so $\operatorname{rank}(\begin{bmatrix} AB & \\ B & BC \end{bmatrix}) = \operatorname{rank}(\begin{bmatrix} & -ABC \\ B & \end{bmatrix}) = \operatorname{rank}(B) + \operatorname{rank}(-ABC) = \operatorname{rank}(B) + \operatorname{rank}(ABC)$. So we are done. □

If you think the proof above is too "magical", then here is another proof using subspaces.

Alternative Proof. All these maps have many different domains and codomains. For example, it makes little sense to add $\dim \operatorname{Ran}(AB)$ and $\dim \operatorname{Ran}(BC)$, because $\operatorname{Ran}(AB)$ and $\operatorname{Ran}(BC)$ are not in the same underlying vector space. To make meaningful progress, we need to fix a specific underlying vector space. Say we pick the domain of A.

We rearrange the inequality as $\operatorname{rank}(BC) - \operatorname{rank}(ABC) \leq \operatorname{rank}(B) - \operatorname{rank}(AB)$. (We immediatey see that the Inclusion-Exclusion Principle for maps is a generalization of the fact that $\operatorname{rank}(BC) \leq \operatorname{rank}(B)$.)

Now $\operatorname{rank}(AB) = \dim \operatorname{Ran}(AB) = \dim A(\operatorname{Ran}(B)) = \dim \operatorname{Ran}(B) - \dim(\operatorname{Ker}(A) \cap \operatorname{Ran}(B))$. So the right hand side is simply $\dim(\operatorname{Ker}(A) \cap \operatorname{Ran}(B))$. Similarly, the left hand side is simply $\dim(\operatorname{Ker}(A) \cap \operatorname{Ran}(BC))$.

So we now want to prove that $\dim(\mathrm{Ker}(A) \cap \mathrm{Ran}(BC)) \leq \dim(\mathrm{Ker}(A) \cap \mathrm{Ran}(B))$. Since we know $\mathrm{Ran}(BC) \subseteq \mathrm{Ran}(B)$, we have $\mathrm{Ker}(A) \cap \mathrm{Ran}(BC) \subseteq \mathrm{Ker}(A) \cap \mathrm{Ran}(B)$, and hence the dimension inequality is also true. ☐

The alternative proof here is more revealing. Ultimately, the Inclusion-Exclusion Principle for maps is about comparing the subspaces $\mathrm{Ker}(A) \cap \mathrm{Ran}(BC)$ with $\mathrm{Ker}(A) \cap \mathrm{Ran}(B)$.

Also note that taking $B = I$, from the inequality $\mathrm{rank}(AB) + \mathrm{rank}(BC) \leq \mathrm{rank}(ABC) + \mathrm{rank}(B)$, we can also deduce that $\mathrm{rank}(A) + \mathrm{rank}(C) \leq \mathrm{rank}(AC) + n$, which we know before.

5.9 (Optional) When can we LU

We know that, for an invertible matrix A, an LU decomposition is possible if and only if all principal submatrices are invertible. But what if A is not invertible? An LU decomposition might still be possible. For example, the zero matrix is the zero matrix multiplied by the zero matrix. Here is a more exotic example.

Example 5.9.1. Consider $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. If you attempt to do Gaussian elimination, surely you need to row swap right away! So can there still be LU decompositions?

There IS. We have $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. The non-zero portion of $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ can "shift-up" to satisfy the upper triangular requirement, and the "shift-down" operator is lower triangular. ☺

For a generic (maybe not invertible) square matrix A, when can we perform an LU decomposition?

Theorem 5.9.2. *A square matrix A has an LU decomposition if and only if the following is true: for all k, if $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ where A_{11} is $k \times k$, then we have* $\mathrm{rank}(A_{11}) + k \geq \mathrm{rank}(\begin{bmatrix} A_{11} & A_{12} \end{bmatrix}) + \mathrm{rank}(\begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix})$.

Proof. Let us only prove the forward direction. Suppose $A = LU$ for lower triangular L and upper triangular U, and we perform block decompositions $L = \begin{bmatrix} L_{11} & \\ L_{21} & L_{22} \end{bmatrix}$ and $U = \begin{bmatrix} U_{11} & U_{12} \\ & U_{22} \end{bmatrix}$, where L_{11}, U_{11} are $k \times k$.

Now $\mathrm{rank}(\begin{bmatrix} A_{11} & A_{12} \end{bmatrix}) = \mathrm{rank}(\begin{bmatrix} L_{11}U_{11} & L_{11}U_{12} \end{bmatrix}) = \mathrm{rank}(L_{11} \begin{bmatrix} U_{11} & U_{12} \end{bmatrix}) \leq \mathrm{rank}(L_{11})$, and similarly $\mathrm{rank}(\begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix}) \leq \mathrm{rank}(U_{11})$. So $\mathrm{rank}(\begin{bmatrix} A_{11} & A_{12} \end{bmatrix}) + \mathrm{rank}(\begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix}) \leq \mathrm{rank}(L_{11}) + \mathrm{rank}(U_11) \leq \mathrm{rank}(L_{11}U_{11}) + k = \mathrm{rank}(A_{11}) + k$. So we are done.

For the other direction, the proof is basically by mathematical induction on n. When $n = 1$, again the case is trivial.

Suppose $n > 1$. We write $A = \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ \boldsymbol{w} & A_{n-1} \end{bmatrix}$, and suppose that our conditions on rank of submatrices are all true.

Case 1:

Suppose $a \neq 0$. Then $A = \begin{bmatrix} 1 & \\ \frac{1}{a}\boldsymbol{w} & I_{n-1} \end{bmatrix} \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ & A_{n-1} - \frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}} \end{bmatrix}$. We now aim to show that $A_{n-1} - \frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}}$ satisfy the induction hypothesis and thus has an LU decomposition.

For any positive integer $k < n-1$, let $A_{n-1} = \begin{bmatrix} A_k & B_k \\ C_k & D_k \end{bmatrix}$, and let $\boldsymbol{v}^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{v}_k^{\mathrm{T}} & (\boldsymbol{v}')^{\mathrm{T}} \end{bmatrix}$, and let $\boldsymbol{w} = \begin{bmatrix} \boldsymbol{w}_k \\ \boldsymbol{w}' \end{bmatrix}$. Here A_k is $k \times k$, and $\boldsymbol{v}_k, \boldsymbol{w}_k \in \mathbb{R}^k$. Then our condition on A states that $\mathrm{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} \\ \boldsymbol{w}_k & A_k \end{bmatrix}) + k + 1$

$\geq \mathrm{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} \\ \boldsymbol{w}_k & A_k \\ \boldsymbol{w}' & C_k \end{bmatrix}) + \mathrm{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} & (\boldsymbol{v}')^{\mathrm{T}} \\ \boldsymbol{w}_k & A_k & B_k \end{bmatrix})$. So we have the deduction

$$
\begin{aligned}
&\operatorname{rank}(A_k - \frac{1}{a}\boldsymbol{w}_k\boldsymbol{v}_k^{\mathrm{T}}) + k \\
=&\operatorname{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} \\ & A_k - \frac{1}{a}\boldsymbol{w}_k\boldsymbol{v}_k^{\mathrm{T}} \end{bmatrix}) + k - 1 \\
=&\operatorname{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} \\ \boldsymbol{w}_k & A_k \end{bmatrix}) + k - 1 \\
\geq&\operatorname{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} \\ \boldsymbol{w}_k & A_k \\ \boldsymbol{w}' & C_k \end{bmatrix}) + \operatorname{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} & (\boldsymbol{v}')^{\mathrm{T}} \\ \boldsymbol{w}_k & A_k & B_k \end{bmatrix}) - 2 \\
=&\operatorname{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} \\ & A_k - \frac{1}{a}\boldsymbol{w}_k\boldsymbol{v}_k^{\mathrm{T}} \\ & C_k - \frac{1}{a}\boldsymbol{w}'\boldsymbol{v}_k^{\mathrm{T}} \end{bmatrix}) + \operatorname{rank}(\begin{bmatrix} a & \boldsymbol{v}_k^{\mathrm{T}} & (\boldsymbol{v}')^{\mathrm{T}} \\ & A_k - \frac{1}{a}\boldsymbol{w}_k\boldsymbol{v}_k^{\mathrm{T}} & B_k - \frac{1}{a}\boldsymbol{w}_k(\boldsymbol{v}')^{\mathrm{T}} \end{bmatrix}) - 2 \\
=&\operatorname{rank}(\begin{bmatrix} A_k - \frac{1}{a}\boldsymbol{w}_k\boldsymbol{v}_k^{\mathrm{T}} \\ C_k - \frac{1}{a}\boldsymbol{w}'\boldsymbol{v}_k^{\mathrm{T}} \end{bmatrix}) + \operatorname{rank}(\begin{bmatrix} A_k - \frac{1}{a}\boldsymbol{w}_k\boldsymbol{v}_k^{\mathrm{T}} & B_k - \frac{1}{a}\boldsymbol{w}_k(\boldsymbol{v}')^{\mathrm{T}} \end{bmatrix}).
\end{aligned}
$$

So our conditions on rank of submatrices are also true for $A_{n-1}-\frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}}$. By induction hypothesis, $A_{n-1}-\frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}}$ has LU decompositions, say $A_{n-1}-\frac{1}{a}\boldsymbol{w}\boldsymbol{v}^{\mathrm{T}} = L_{n-1}U_{n-1}$. Then $A = \begin{bmatrix} 1 & \\ \frac{1}{a}\boldsymbol{w} & I_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \\ & L_{n-1} \end{bmatrix} \begin{bmatrix} a & \boldsymbol{v}^{\mathrm{T}} \\ & U_{n-1} \end{bmatrix}$. So we are done.

Case 2:

Now suppose $a = 0$. Then our condition on A states that $1 = \operatorname{rank}(a) + 1 \geq \operatorname{rank}(\boldsymbol{v}^{\mathrm{T}}) + \operatorname{rank}(\boldsymbol{w})$. So at least one of $\boldsymbol{v}, \boldsymbol{w}$ is the zero vector. So A must have a zero first column or a zero first row.

Suppose that $\boldsymbol{v} \neq \mathbf{0}$, then $\boldsymbol{w} = \mathbf{0}$. Say the i-th coordinate of $\boldsymbol{v}$ is the first non-zero coordinate of $\boldsymbol{v}$. Then by row operations, we can find a lower triangular matrix L such that $LA = \begin{bmatrix} 0 & \boldsymbol{v}^{\mathrm{T}} \\ \mathbf{0} & A' \end{bmatrix}$ where the i-th column of A' is entirely zero. Again we aim to show that A' satisfy the induction hypothesis and thus has an LU decomposition.

For any positive integer $k < n-1$, let $A' = \begin{bmatrix} A_k & B_k \\ C_k & D_k \end{bmatrix}$, and let $\boldsymbol{v} = \begin{bmatrix} \boldsymbol{v}_k & \boldsymbol{v}' \end{bmatrix}$. Here A_k is $k \times k$, and $\boldsymbol{v}_k \in \mathbb{R}^k$. Let $t = 1$ if $k \geq i$, and $t = 0$ if $k < i$. So we have the deduction

$$
\begin{aligned}
&\operatorname{rank}(A_k) + k \\
=&\operatorname{rank}(\begin{bmatrix} 0 & \boldsymbol{v}_k^{\mathrm{T}} \\ & A_k \end{bmatrix}) + k - t \\
\geq&\operatorname{rank}(\begin{bmatrix} 0 & \boldsymbol{v}_k^{\mathrm{T}} \\ & A_k \\ & C_k \end{bmatrix}) + \operatorname{rank}(\begin{bmatrix} 0 & \boldsymbol{v}_k^{\mathrm{T}} & (\boldsymbol{v}')^{\mathrm{T}} \\ & A_k & B_k \end{bmatrix}) - t - 1 \\
=&\operatorname{rank}(\begin{bmatrix} A_k \\ C_k \end{bmatrix}) + \operatorname{rank}(\begin{bmatrix} A_k & B_k \end{bmatrix}).
\end{aligned}
$$

So we are done.

Case 3:

Suppose $a = 0$ and $\boldsymbol{w} \neq \mathbf{0}$, then $\boldsymbol{v} = \mathbf{0}$. This case is the same as above, where we simply take transpose of everything.

Case 4:

Finally, suppose $a = 0$ and $\boldsymbol{v} = \boldsymbol{w} = \mathbf{0}$. Say the i-th row is the first row of A that is not entirely zero. Then let A' be obtained from A by swapping the i-th row and first row of A. Clearly $A = LA'$ where $L = I + \boldsymbol{e}_i\boldsymbol{e}_1^{\mathrm{T}} - \boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}}$, which is lower triangular. So A has LU decomposition if A' has LU decomposition.

We aim to show that A' is in case 2 above, so we are done. Obviously A' has zero first column and non-zero first row, so all we need is to show that the submatrices of A' satisfies the rank condition.

For any positive integer $k \leq i$, the first k rows of A' has rank exactly 1, and the first k columns of A' has rank at most $k-1$ since the first column of A' is zero. So if $A' = \begin{bmatrix} A'_{11} & A'_{12} \\ A'_{21} & A'_{22} \end{bmatrix}$ where A'_{11} is $k \times k$, we have $\text{rank}(A'_{11}) + k \geq k \geq \text{rank}(\begin{bmatrix} A'_{11} & A'_{12} \end{bmatrix}) + \text{rank}(\begin{bmatrix} A'_{11} \\ A'_{21} \end{bmatrix})$.

For any positive integer $k > i$, then the first k columns of A' must have the same rank as the first k columns of A (since they differ by a row swap), and the first k rows of A' must also have the same rank as the first k rows of A, because they are made of the same rows. Finally, the upper left $k \times k$ block of A' would also have the same rank as the upper left $k \times k$ block of A, again because they are made of the same rows. So if $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $A' = \begin{bmatrix} A'_{11} & A'_{12} \\ A'_{21} & A'_{22} \end{bmatrix}$ where A_{11}, A'_{11} are $k \times k$, then we have $\text{rank}(A_{11}) = \text{rank}(A'_{11})$, $\text{rank}(\begin{bmatrix} A_{11} & A_{12} \end{bmatrix}) = \text{rank}(\begin{bmatrix} A'_{11} & A'_{12} \end{bmatrix})$, and $\begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} = \begin{bmatrix} A'_{11} \\ A'_{21} \end{bmatrix}$. So by the rank condition on A we have $\text{rank}(A_{11}) + k \geq \text{rank}(\begin{bmatrix} A_{11} & A_{12} \end{bmatrix}) + \text{rank}(\begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix})$, which implies $\text{rank}(A'_{11}) + k \geq \text{rank}(\begin{bmatrix} A'_{11} & A'_{12} \end{bmatrix}) + \text{rank}(\begin{bmatrix} A'_{11} \\ A'_{21} \end{bmatrix})$. So A' is indeed in case 2, and it has an LU decomposition. $\square$

5.10 (Optional) Axioms of Abstract Vector Spaces

Informally, we want to define abstract vector space as a set, in which you can do linear combinations. Now let us ponder the exact meaning of this, and hopefully we shall arrive at a collection of axioms that give us a good definition. You can safely skip most of this subsection and jump right to the end to see the definition. Meanwhile, if you are ever curious why we use these definitions, then here is a relatively detailed discourse of them.

Keep in mind of this key idea: All the laws serves one purpose: we want to simplify ALL expressions into linear combinations. Look at the following expressions. What calculation law would you want to have, in order to simplify them into linear combinations of the form $a\boldsymbol{v} + b\boldsymbol{w}$?

1. $(\boldsymbol{v} + \boldsymbol{w}) + \boldsymbol{v}$.
2. $2(\boldsymbol{v} + \boldsymbol{w})$.
3. $2(3\boldsymbol{v})$.
4. $2\boldsymbol{v} + 3\boldsymbol{v}$.
5. $\boldsymbol{v} + (-1)\boldsymbol{v}$.
6. ....

Example 5.10.1 (Associativity and Commutativity)**.**

Say I have $\boldsymbol{w} + \boldsymbol{v} + \boldsymbol{w}$. The obvious thing to do is to simplify it into $\boldsymbol{v} + 2\boldsymbol{w}$. But what is actually involved? Specifically, we need to do the following:

$$(\boldsymbol{w} + \boldsymbol{v}) + \boldsymbol{w} = (\boldsymbol{v} + \boldsymbol{w}) + \boldsymbol{w} = \boldsymbol{v} + (\boldsymbol{w} + \boldsymbol{w}) = \boldsymbol{v} + 2\boldsymbol{w}.$$

Here we mainly talk about the first two steps. The first is called commutativity, and the second is called associativity. It is trivially obvious that we want these properties to be part of the definition of a vector space.

But if you are curious, let us discuss them further. For new students of mathematics, a common misconception is that maybe commutativity is stronger than associativity. That is WRONG. The two are independent.

Pause and think about this. Given $(a+b)+c$, if we only have associativity, can you write out all possible equivalent re-formulations for this formula? How many are there? What if we only have commutativity?

Eitherway, without associativity or commutativity of addition, $(\boldsymbol{w}+\boldsymbol{v})+\boldsymbol{w}$ will NEVER be able to be simplified into $\boldsymbol{v}+2\boldsymbol{w}$. Then the very concept of linear combination will no longer work the way we want. ☺

Remark 5.10.2 ((Optional) Operation Order and Operand Order)**.**

Associativity states that $(a+b)+c = a+(b+c)$, so we may change the order of the operations. But Commutativity states that $(a+b)+c = (b+a)+c$. Here the order of operations are NOT changed, because we are always adding a and b first, then we add the result with c. What is allowed by commutativity is to change the order of OPERANDS (or summands), i.e., the objects of the operation (or the sum).

In short, if we want to change the calculation order of the "+" symbols, we need associativity. If we need to change the order of a, b, c, then we need commutativity. One deals with operations while fixing the operands, the other deals with operands while fixing the operations, so their actions are completely independent of each other.

Case in point, for our example $(\boldsymbol{w}+\boldsymbol{v})+\boldsymbol{w}$, without commutativity, association will only ever change the order in which we compute the addition symbols, but the order of summands will always remain $\boldsymbol{w}, \boldsymbol{v}, \boldsymbol{w}$. So the two w cannot be moved together to be simplified. On the other hand, without associativity, commutativity will only ever change the order of $\boldsymbol{w}, \boldsymbol{v}, \boldsymbol{w}$, but the order of computation is unchanged. So we will ALWAYS compute the sum of $\boldsymbol{v}$ and $\boldsymbol{w}$ first, and never able to compute $\boldsymbol{w}+\boldsymbol{w}$ first as we would have liked.

Of course, these discussion is only relevant if you aspire to learn more algebra after this class. For the purpose of this class, we want vector addition to be both. So no order matters any way.

Example 5.10.3 (Zero Vector)**.**

Say I have $0\boldsymbol{v}+0\boldsymbol{w}$. Surely I would like to simplify that to $\boldsymbol{0}$, the zero vector, yes?

Afterall, the zero vector is special. All linear maps must send the zero vector to the zero vector, and all vectors are parallel to the zero vector, and so on. You just have to have a zero vector.

Now, what does a zero vector do? The defining trait of the zero vector is the fact that $\boldsymbol{0}+\boldsymbol{v}=\boldsymbol{v}$ for all other $\boldsymbol{v}$. Note that this immediately implies that the zero vector is unique. If $\boldsymbol{0}, \boldsymbol{0}'$ are two zero vectors, then $\boldsymbol{0} = \boldsymbol{0}+\boldsymbol{0}' = \boldsymbol{0}'$. ☺

Example 5.10.4 (Negation of vectors)**.** Our study is called "linear algebra", and "linear" is the adjective form of "line". A line extends both ways to infinity. In particular, given any direction, there must be an opposite direction.

In short, for any vector $\boldsymbol{v}$, there must be a vector $\boldsymbol{w}$ such that $\boldsymbol{v}+\boldsymbol{w}=\boldsymbol{0}$. We call this the negation of $\boldsymbol{v}$, and denote it as $-\boldsymbol{v}$. The concept of negations immediately allow us to SUBSTRACT vectors. We define $\boldsymbol{x}-\boldsymbol{y}$ to be $\boldsymbol{x}+(-\boldsymbol{y})$.

Now one can immediately see that given any vector $\boldsymbol{v}$, its negation is unique. If $\boldsymbol{w}, \boldsymbol{w}'$ are both its negations, then $\boldsymbol{w} = \boldsymbol{w}+\boldsymbol{v}+\boldsymbol{w}' = \boldsymbol{w}'$.

The existence of negations of vectors also indicates that we have the law of cancellations when it comes to vector additions, i.e., if $\boldsymbol{v}+\boldsymbol{w} = \boldsymbol{u}+\boldsymbol{w}$, then $\boldsymbol{v}=\boldsymbol{u}$. ☺

All the axioms so far only concerns with vector addition. Now let us move to scalar multiplications

Example 5.10.5 (Scalar Multiplications are linear on scalars)**.** All calculational laws serves to simplify expressions into linear combinations. Say we see $2\boldsymbol{v}+3\boldsymbol{v}$? Surely we want to simplify that into $5\boldsymbol{v}$. This requires a law of distribution on scalars, i.e., $(a+b)\boldsymbol{v} = a\boldsymbol{v}+b\boldsymbol{v}$.

Finally, what about $2(3\boldsymbol{v})$? Surely we want to simplify that into $6\boldsymbol{v}$. We want a weird law of "associativity for scalar multiplications" that goes like $(ab)\boldsymbol{v} = a(b\boldsymbol{v})$. However, looking at this from another angle, and it would not be weird anymore: let $m_{\boldsymbol{v}}$ be the map that sends a real number x to the vector $x\boldsymbol{v}$, then $m_{\boldsymbol{v}}(ax+by) = (ax+by)\boldsymbol{v} = (ax)\boldsymbol{v}+(by)\boldsymbol{v} = a(x\boldsymbol{v})+b(y\boldsymbol{v}) = am_{\boldsymbol{v}}(x)+bm_{\boldsymbol{v}}(y)$, i.e., $m_{\boldsymbol{v}}$ is linear.

So don't think of these as law of distributions or some weird law of associativity. The best way to think about this is that, for fixed $\boldsymbol{v}$, the output $k\boldsymbol{v}$ is linear on the input k. ☺

Example 5.10.6 (Scalar Multiplications are linear on vectors)**.** On the other hand, what if we see $2(\boldsymbol{v}+\boldsymbol{w})$? Surely we want this to be $2\boldsymbol{v}+2\boldsymbol{w}$, right? So we need a law of distribution on vectors, i.e., $k(\boldsymbol{v}+\boldsymbol{w}) = k\boldsymbol{v}+k\boldsymbol{w}$.

In particular, let m_k be the map that sends vectors $\boldsymbol{v}$ to $k\boldsymbol{v}$. Then $m_k(a\boldsymbol{v} + b\boldsymbol{w}) = k(a\boldsymbol{v} + b\boldsymbol{w}) = k(a\boldsymbol{v}) + k(b\boldsymbol{w}) = (ak)\boldsymbol{v} + (bk)\boldsymbol{w} = a(k\boldsymbol{v}) + b(k\boldsymbol{w}) = am_k(\boldsymbol{v}) + bm_k(\boldsymbol{w})$. So m_k is linear.

Again, I think the best way to think about this is that, for fixed k, the output $k\boldsymbol{v}$ is linear on the input $\boldsymbol{v}$. ☺

We see that scalar multiplication is linear on each of the two inputs. This sort of thing is called ***bilinear***.

Now there are some interesting consequences for the bilinearity of scalar multiplications.

Example 5.10.7 (One is respected)**.** Consider $1(k\boldsymbol{v}) = (1 \times k)\boldsymbol{v} = k\boldsymbol{v}$. It is reasonably obvious that when we multiply a vector by 1, we do not want to change that vector. However, what if you have $1\boldsymbol{v}$ and no k to be a mediator? Then we do not know what might happen here.

Note that this has to be a separate axiom. Nothing else implies this. If one were to define, say, $1\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$, then check carefully and see that all previous axioms are NOT violated. ☺

Example 5.10.8 (Zero is respected)**.** We have $0\boldsymbol{v} + \boldsymbol{v} = (0 + 1)\boldsymbol{v} = \boldsymbol{v}$. Substract $\boldsymbol{v}$ from both sides, we see that $0\boldsymbol{v} = \boldsymbol{0}$ for all vectors.

In fact, since $k \mapsto k\boldsymbol{v}$ is linear, 0 has to goes to the zero vector. ☺

Example 5.10.9 (Minus one is respected)**.** Note that $\boldsymbol{v}+(-1)\boldsymbol{v} = (1-1)\boldsymbol{v} = \boldsymbol{0}$. So we see that $(-1)\boldsymbol{v} = -\boldsymbol{v}$ is the negation indeed. ☺

Let us now define an abstract vector space (over $\mathbb{R}$).

Definition 5.10.10. *An abstract vector space is a set V together with two operations, a vector addition $+ : V \times V \to V$ and a scalar multiplication $m : \mathbb{R} \times V \to V$, such that the following is true:*

Vector addition gives an abelian group:

1. *(Associative and Commutative) Vector addition is associative and commutative.*
2. *(Zero Vector) There is a zero vector $\boldsymbol{0} \in V$ such that $\boldsymbol{0} + \boldsymbol{v} = \boldsymbol{v}$ for all $\boldsymbol{v} \in V$.*
3. *(Negation Vector) For any vector $\boldsymbol{v} \in V$, there is a unique negation $-\boldsymbol{v} \in V$ such that $\boldsymbol{v} + (-\boldsymbol{v}) = \boldsymbol{0}$. (This also gives the law of cancellation for vector additions.)*

Scalar multiplication is bilinear:

1. *(Linear in scalars) $(ab)\boldsymbol{v} = a(b\boldsymbol{v})$ and $(a + b)\boldsymbol{v} = a\boldsymbol{v} + b\boldsymbol{v}$ for all $a, b \in \mathbb{R}, \boldsymbol{v} \in V$.*
2. *(Linear in vectors) $a(\boldsymbol{v} + \boldsymbol{w}) = a\boldsymbol{v} + a\boldsymbol{w}$ for all $a \in \mathbb{R}, \boldsymbol{v}, \boldsymbol{w} \in V$.*
3. *(Respect identity) $1\boldsymbol{v} = \boldsymbol{v}$.*

Remark 5.10.11. *Uniqueness of the zero vector and the negation vector are consequences of their definition and associativity. The negation axiom also allows us to define vector substractions via $\boldsymbol{v} - \boldsymbol{w} = \boldsymbol{v} + (-\boldsymbol{w})$.*

The (linear in scalars) axiom shows that $1\boldsymbol{v} + 0\boldsymbol{v} = (1 + 0)\boldsymbol{v} = 1\boldsymbol{v}$. Now use the (Negation vector) axiom to cancel the $1\boldsymbol{v}$ on both sides, we see that $0\boldsymbol{v} = \boldsymbol{0}$.

Now use the (respect identity) axiom, we have $(-1)\boldsymbol{v} + \boldsymbol{v} = (-1)\boldsymbol{v} + 1\boldsymbol{v} = (-1 + 1)\boldsymbol{v} = 0\boldsymbol{v} = \boldsymbol{0}$. Hence $(-1)\boldsymbol{v}$ is the negation of $\boldsymbol{v}$.

Theorem 5.10.12 (Optional)**.** *In a vector space V, if we have vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ and scalars $a_1, \dots, a_t$ such that they are all combined into a single vector using vector addition and scalar multiplication, then we can simplify the expression into $b_1\boldsymbol{v}_1 + \cdots + b_k\boldsymbol{v}_k$ for some scalars $b_1, \dots, b_k$.*

Proof. Say the starting formula is $f(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t)$ which involves s operations (i.e., scalar addition and vector multiplication). We do this by mathematical induction on the number of operations s. If the number of operation involved is zero, then we have a single vector $\boldsymbol{v}_1$ without any operation, and we have $\boldsymbol{v}_1 = 1\boldsymbol{v}_1$ by the last axiom of vector space, so we are done with $b_1 = 1$.

Suppose the statement is true when the number of operations is less than s. Now suppose we have s operations in total.

If the last operation is a scalar multiplication, WLOG say scalar multiplicatino by a_t, then the expression must be $f(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) = a_t g(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t)$. Now g involves $s-1$ operations, and hence by induction hypothesis $g(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) = b'_1\boldsymbol{v}_1 + \cdots + b'_k\boldsymbol{v}_k$ for some scalars $b'_1, \ldots, b'_k$. Therefore

$$\begin{aligned}
f(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) =& a_t g(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) \\
=& a_t(b'_1\boldsymbol{v}_1 + \cdots + b'_k\boldsymbol{v}_k) \\
=& a_t(b'_1\boldsymbol{v}_1) + \cdots + a_t(b'_k\boldsymbol{v}_k) \\
=& (a_t b'_1)\boldsymbol{v}_1 + \cdots + (a_t b'_k)\boldsymbol{v}_k.
\end{aligned}$$

Note that we must involve the fact that scalar multiplication is linear on the vector, and the "weird association law" that $a(b\boldsymbol{v}) = (ab)\boldsymbol{v}$.

If the last operation is a vector addition, then we must have $f(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) = g(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) + h(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t)$. Now g, h must both have less than s operations, therefore by induction hypothesis $g(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) = c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k$ for some scalars $c_1, \ldots, c_k$, and $b(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) = c'_1\boldsymbol{v}_1 + \cdots + c'_k\boldsymbol{v}_k$ for some scalars $c'_1, \ldots, c'_k$. So we have

$$\begin{aligned}
f(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) =& g(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) + h(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, a_1, \ldots, a_t) \\
=& (c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k) + (c'_1\boldsymbol{v}_1 + \cdots + c'_k\boldsymbol{v}_k) \\
=& (c_1 + c'_1)\boldsymbol{v}_1 + \cdots + (c_k + c'_k)\boldsymbol{v}_k.
\end{aligned}$$

Note that we must use the law of association, the law of commutativity, and the fact that scalar multiplication is linear on the scalar.

So we are done. □

Remark 5.10.13. *In the proof above, we have used ALL axioms except for the zero vector axiom and the negation vector axiom.*

The zero vector axiom is in fact the case of the theorem above when $k = 0$. If there are no vectors, the linear combination of no vector must be $\mathbf{0}$.

The negation axiom is in fact redundant. Given $\mathbf{v}$, its negation must be $(-1)\mathbf{v}$, which you can verify from other axioms.

You don't have to memorize these axioms. Rather, just remember the purpose of them. ALL of them serves the same purpose: whatever you do with vector additions and scalar multiplications, in the end it is just a linear combination. To have a nice concept of "linear combination" is the purpose of all of them.

Now this immediately prompts ups to define linear maps, which are maps that perserve vector space structures.

Definition 5.10.14. *Given two vector space V, W, a map $f : V \to W$ is linear if $f(\boldsymbol{v} + \boldsymbol{w}) = f(\boldsymbol{v}) + f(\boldsymbol{w})$ and $f(k\boldsymbol{v}) = kf(\boldsymbol{v})$. I.e., f preserves linear combinations.*

Now, the following example illustrates that it is VERY important to figure out what is this underlying vector space, and VERY important to see if linear combinations make sense. And sometimes (when linear combinations make no sense) certain things are best NOT treated as a vector space.

Example 5.10.15 (Optional)**.** The spectral colors (colors on a rainbow) depends only on their wavelength. Think about the wavelength as elements of $\mathbb{R}$, then we see that all spectral colors are subsets of the vector space $\mathbb{R}$. In this sense, color is one-dimensional.

However, what about those RGB nonsense? In a computer, a color is usually encoded with a TRIPLE of real numbers, indicating the amount of red, green and blue. Does that not make color three dimensional?

It turns out that there are NON-spectral colors, and this is peculier to human. In the eyes of a human there are three types of cone cells, and they are activated by roughly the color red, green and blue respectively. Depending on how the three types of cone cells are activated, our brain will decide its color.

For example, we have the color violet, and the color purple, and they appear similar in our eyes. Violet feels a little more blue-ish, and it is in fact a spectral color. It has objectively NO relation with red or blue, and it is simply a different wave length. Purple is essentially a mixture of red and blue. So why do they appear similar in our eyes?

It turns out that when we see violet, weirdly our "red" cone sell is activated by a tiny bit. So, even though violet has nothing to do with red, our brain is convinced that we see a hint of redness in it. For this reason, human thinks violet and purple are similar. There are many animals with different numbers and kinds of cone cells, and to them, maybe violet and purple are NOT similar at all. "Foolish humans...." They would think.

So, as far as us humans are concerned, since we have three kinds of cone cells, our color perception depends on how much each type of cone cells is activated. So we need three real numbers to express our color perception. So even though objective spectral colors form a one-dimensional structure, our human color perception form a three-dimensional structure.

Color is objective, but the perception of color (whether two colors are "close") is subjective.

Furthermore, before we declare that we are working in a vector space, we need to think about the meaning of vector addition and scalar multiplication. On the spectral color line, what does it mean to add wavelength? Probably no meaning. So it might be best that we DON'T think of it as a vector space. We need the continuous structure of $\mathbb{R}$ more than its linear structure.

On the space $\mathbb{R}^3$ of human color perceptions, we add RGB coefficients all the time in computers. It is a bit similar to mixing colored lights. So it is OK to treat this as a vector space. ☺

5.11 (Optional) Axioms and mathematical structures

Defining mathematical structures via a bunch of axioms is very important to mathematicians. For non-mathematicians, they are less useful. But for mathematicians, this is the best way to differentiate different structures, and identify similar structures. Here are some mathematical structures for those interested.

(These are absolutely not required. You don't need to know any of these.)

Definition 5.11.1. *A **group** is a set G with an operation $G \times G \to G$ (which may be written multiplicatively or additively, however you like), such that the following axioms hold. (Here the result of an operation of g and h is written simply as gh.)*

1. *(Associativity) $(xy)z = x(yz)$ for all $x, y, z \in G$.*
2. *(Identity) There is an element $e \in G$ such that $eg = ge = g$ for all $g \in G$. (We call e the identity element.)*
3. *(Inverse) For each $g \in G$, we can find $h \in G$ such that $gh = hg = e$.*

Here associativity is the heart of all mathematical wonders. The latter two axioms guarantee that we have the law of cancellation. Finally, the uniqueness of the identity element and the uniqueness of the inverse element can be deduced from the axioms.

Example 5.11.2.

The set of integers $\mathbb{Z}$ with the operation of addition is a group.

The set of integers $\mathbb{Z}$ with the operation of multiplication is NOT a group, because many elements have no multiplicative inverse. We see here that merely specifying the set is NOT enough. From now on, we shall write things like $(\mathbb{Z}, +)$ and $(\mathbb{Z}, \times)$ to specify which operations are in use. So $(\mathbb{Z}, +)$ is a group and $(\mathbb{Z}, \times)$ is not a group.

The set of natural numbers $(\mathbb{N}, +)$ is NOT a group, because many elements have no additive inverse.

The rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, complex numbers $\mathbb{C}$ are all groups (under addition).

Throwing away zero, then $(\mathbb{Q} - \{0\}, \times)$ is in fact a group. The same is true for $(\mathbb{R} - \{0\}, \times)$ and $(\mathbb{C} - \{0\}, \times)$.

Any vector space V with vector addition form a group.

All invertible diagonal $n \times n$ matrices with matrix multiplication form a group. ☺

Note that for all the groups above, we have an extra law, the law of commutativity.

Definition 5.11.3. *A group is an* ***abelian group*** *if we have the law of commutativity. (Abel is the name of a famous mathematician. "Abelian" is the adjective of "Abel".)*

What about non-abelian groups? They are also super important!

Example 5.11.4.

All invertible $n \times n$ matrices with matrix multiplication form a group. If $n \neq 1$, this group is NOT abelian.

All invertible upper triangular $n \times n$ matrices with matrix multiplication form a group. If $n \neq 1$, this group is NOT abelian. You can also just look at unit upper triangular matrices, and it is still a group, and if $n \neq 1, 2$, this group is NOT abelian.

All permutations on n objects from a group. When $n \neq 1, 2$, this is NOT abelian.

In fact, for any set X, all bijective functions $f : X \to X$ form a group. (The operation is function composition.) Usually this is not abelian

We can also add extra conditions here. All continuous bijective function $f : \mathbb{R} \to \mathbb{R}$ form a group (with function composition as the group operation).

All symmetries of a triangle form a non-abelian group. There are six elements here, three reflections and three rotations (here we treat the identity symmetry as a "zero rotation'). ☺

The last example is very revealing. Ultimately, one might see the study of groups as the study of symmetries. Symmetries of geometric shapes, symmetries of the universe, symmetries among various symmetries themselves, you name it!

Now, groups are super important, and the axioms are very few. However, this makes them HARD to study. With only a few axioms, there are many possible ways a group might look like. The study of finite non-abelian groups is still pretty much a big mystery to us. Nevertheless, they already help us solve many difficult problems.

Two prominant kinds are the various "straight edge and compass" problems (e.g., can you trisect an arbitrary angle using a compass and an (unmarked) ruler?), and the statement "polynomials of degree 5 and above have no radical solutions." (I.e., no nice formula exists. Say if the polynomial has degree 2, we have $\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$. But for polynomials of degree 5 and above, no such formula exists.) These are done by studying potential symmetries among solutions to an algebraic equation.

Remark 5.11.5. *The mathematician most famous for inventing most of group theory is a French super genius called Galois. He also solve the polynomial problem mentioned above.*

He died when he was 20 years old, duelling for a woman he loved. Yet in his short life, his contribution to math is more than most mathematicians' whole life's work. Wow, he did ALL THAT before turning 20 years old? This surely makes us feel bad, yes? Comparatively, what was I doing when I was 20? Well... I guess I was also chasing after a woman, who is now my wife and the mother of my two children. Hey, I guess I'd rather have my life than Galois' after all.

His life story is quite awesome. Check it out.

In short, groups are useful, but they are usually way too hard to study. To make them easier to study, we need to add more axioms.

Definition 5.11.6. *A **field** is a set $\mathbb{F}$ with two operations $+: \mathbb{F} \times \mathbb{F} \to \mathbb{F}$ and $\times: \mathbb{F} \times \mathbb{F} \to \mathbb{F}$, such that the following axioms hold.*

1. *$(\mathbb{F}, +)$ is an abelian group. We usually write the additive identity element as 0.*
2. *$(\mathbb{F} - \{0\}, \times)$ is an abelian group. We usually write the multiplicative identity element as 1.*
3. *We have the law of distribution.*

So technically, the statement of being an "abelian group" needs four axioms. So we have a total of nine axioms for a field. But with these many axioms, they are MUCH easier to study.

Example 5.11.7.

$\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all very useful fields.

Let us define $\mathbb{F}_p$ for a prime number p as the set $\{0, \dots, p-1\}$, where addition and multiplication are done mod p. (E.g., $(p-2)+3=1$ and so on. We simply treat $p=0$ always.) Then $\mathbb{F}_p$ is a field.

If p is not a prime, then the above construction will NOT be a field. For example, if we consider mod 6, then 2 will have no multiplicative inverse.

There are other fields, but the ones above are the most important ones. ☺

Now look at the definition of a vector space again.

Definition 5.11.8. *A vector space over a field $\mathbb{F}$ is a set V together with two operations, a vector addition $+: V \times V \to V$ and a scalar multiplication $m: \mathbb{F} \times V \to V$, such that the following axioms hold. (We write the result of scalar multiplication of $k \in \mathbb{F}$ with $\boldsymbol{v} \in V$ as $k\boldsymbol{v}$.)*

1. *$(V, +)$ is an abelian group.*
2. *($\mathbb{F}$-linear structure) $(a+b)\boldsymbol{v} = a\boldsymbol{v} + b\boldsymbol{v}$ and $a(b\boldsymbol{v}) = (ab)\boldsymbol{v}$.*
3. *($\mathbb{F}$-linear structure respect V-group structure) $k(\boldsymbol{v}+\boldsymbol{w}) = k\boldsymbol{v} + k\boldsymbol{w}$, and $1\boldsymbol{v} = \boldsymbol{v}$.*

Remark 5.11.9. *Technically, to define a vector space, the group requirement means four axioms, and that $\mathbb{F}$ is a field means nine more axioms, and then we need four more for the vector space. So these 17 axioms can only mean one thing: vector spaces are super easy to learn, as we have so many axioms (tools) to help us!*

Specifically, pay attention to the second axiom here on $\mathbb{F}$-linear structures. Intuitively this says that each $\boldsymbol{v}$ is contained in a "line" made of various $k\boldsymbol{v}$, and structure-wise, this "line" looks exactly like $\mathbb{F}$. Addition on this "line" is the same as addition on $\mathbb{F}$, and multiplication on this "line" is the same as multiplication on $\mathbb{F}$. (So if $\mathbb{F} = \mathbb{R}$, this means every vector is contained in something that looks like $\mathbb{R}$, i.e., a line.)

So what is a vector space over $\mathbb{F}$? Conceptually, it is a space where elements are trapped inside various "lines". And the field $\mathbb{F}$ describes the shape and property of these "lines".

Our class focus mainly on the case of $\mathbb{R}$, so that our intuitions about lines will be spot on.

If we study vector spaces over $\mathbb{C}$ (which is geometrically a plane), then each "line" actually looks like a plane. Then we are no longer doing "linear algebra", but technically "planar algebra"....

If we study vector spaces over $\mathbb{F}_2$, then we would venture into computer science, since the all scalars are now 0 and 1. Discrete world still have geometry though. My personal favorite finite geometric object is the Fano plane, and there is a board game called "SET" which is essential investigating vector spaces over $\mathbb{F}_3$.

5.12 (Optional) Dimensionality from first principles

In the proof above, we took a huge detour. Long ago we established Gaussian elimination, and from the RREF of matrices we see that $\mathbb{R}^m$ and $\mathbb{R}^n$ cannot be isomorphic. (I.e., they have different linear structure.) This gives us that dimension for abstract vector spaces are well-defined.

However, can we do this without the detour? Can we prove this fact by simply using AXIOMS of a vector space, i.e., from first principles? Yes. Warning though, the proof is a bit abstract and hardcore. (But the idea is simple: you eat, then you poop, then you eat, then you poop, and repeat, until you eat all that is required.)

Remark 5.12.1. *The benefit of doing this in first principles is that it can be generalized to infinite dimensions. With some minor changes to the proofs below, one can show that any vector space has a "Hammal" basis, a subset H such that any vector of V is a linear combination of some (finitely many) vectors in H, and everything in H are linearly independent.*

The most powerful lemma is of course again the independence extension lemma. But let us rewrite it in a different form.

Lemma 5.12.2 (Extending Linear Independency)**.** *For any vector space V, if L is a linearly independent collection of vectors, and a vector $\boldsymbol{w}$ is not in the span of L, then $L \cup \{\boldsymbol{w}\}$ is still linearly independent. (Here L is treated as a subset of V.)*

In contrast, if you have a spanning collection, but it is NOT linearly independent, then it means you have some redundent vectors here. You may simply throw them away. However, WHAT to throw a way can be tricky.

Example 5.12.3. Say we have a collection of $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_1 + \boldsymbol{e}_2, \boldsymbol{e}_3$ in $\mathbb{R}^3$, where e_i are the standard basis. This is a spanning collection but linearly dependent. You may throw away any single one of the first three, and you will then get a basis. But you CANNOT throw away $\boldsymbol{e}_3$, since that will destroy the spanning property. ☺

So WHAT to throw away can be tricky. But here comes a nice perspective: we can look at this problem from the opposite angle. What NOT to throw away?

Lemma 5.12.4 (Extending Linear Independency with a Finite Cap)**.** *For any vector space V, if L is a linearly independent collection of vectors, and S is a FINITE spanning system, and $L \subseteq S$, then there is a basis B containing L and contained in S.*

Proof. If L is already spanning, then we are done. Suppose this is not the case.

Let W be the span of L, then $W \neq V$. In particular, S cannot be contained in W. So we can find $\boldsymbol{v} \in S \setminus W$, and $L \cup \{\boldsymbol{v}\}$ would be a linearly independent set between L and S, according to the last lemma.

If $L \cup \{\boldsymbol{v}\}$ is spanning, then this is a basis and we are done. If not, we repeat above process to throw in another $\boldsymbol{w} \in S \setminus span(L \cup \{\boldsymbol{v}\})$, and now $L \cup \{\boldsymbol{v}, \boldsymbol{w}\}$ would be a linearly independent set between L and S. If this is now spanning, then we have a basis. If not, keep doing this.

Since S is finite, either we find a basis somewhere along the process above, or we keep getting larger and larger linearly independent subset of S, until we fill up all of S. But since all sets obtained this way are linearly independent, we would see that S is itself linearly independent. Then S is a basis. ☐

Remark 5.12.5. *We already know that, for some spanning collections, there might be some redundancy that we can throw away. The above lemma states that we actually have some freedom in choosing what NOT to throw away. You can choose to KEEP a linearly independent subset in the spanning collection, and ONLY throw away vectors among the rest in your spanning collection.*

Btw, setting $L = \varnothing$ implies that any finite spanning collection contains a basis.

As you can see, a linearly independent collection is "may or may not be enough", while a dependent spanning collection is "may or may not be redundant". This leads to the following lemma.

Lemma 5.12.6 (Any Finite Spanning System has no less elements than any Linear Independent Set)**.** *For any vector space V, if L is a linearly independent collection of vectors, and S is a FINITE spanning system, then $|L| \leq |S|$.*

Proof. Suppose $L \subseteq S$, then obviously we are done. So we proceed by induction from here. We shall do induction on how many elements of L is NOT in S. The base step is when $L \subseteq S$, and we have already said that this is trivial.

We assume that we have proven that, if L only contains at most t elements not in S, then we are done. Let us proceed to try to prove it for the case when L only contains $t+1$ elements not in S.

Pick any $\boldsymbol{v} \in L \setminus S$. Then consider $S \cup \{\boldsymbol{v}\}$. Obviously $S \cup \{\boldsymbol{v}\}$ is still spanning, and it contains a linearly independent subset $L \cap (S \cup \{\boldsymbol{v}\})$, so there is a basis B between $L \cap (S \cup \{\boldsymbol{v}\})$ and $S \cup \{\boldsymbol{v}\}$.

Now how many elements does L have that is not in B? You will see that now B contains all of $L \cap S$ and also contains $\boldsymbol{v}$, so L now have at most t elements not in B, rather than $t+1$. And B is spanning. So by induction hypothesis, $|L| \leq |B|$. Now we only need to prove that $|B| \leq |S|$.

Indeed, since S is spanning, $S \cup \{\boldsymbol{v}\}$ is linearly DEPENDENT since $\boldsymbol{v}$ can be written as a linear combination of the rest. So, it is NOT a basis, so B is a proper subset of $S \cup \{\boldsymbol{v}\}$. So $|B| < |S| + 1$, which means that $|B| \leq |S|$.

To sum up, we have inductively shown that, as long as L is finite, $|L| \leq |S|$.

What if L is infinite to begin with? Then pick any subset $L' \subseteq L$ with $|S| + 1$ elements. Now L' is finite and linearly independent, so we must have $|L'| \leq |S|$. But that would be a contradiction with our choice of L! So such a case is impossible. So if some spanning collection S is finite, then any linearly independent collection L must also be finite. □

Remark 5.12.7. *So an intuitive proof goes like the following: You pick an element $\boldsymbol{v}$ in L but not in S. Consider $S \cup \{\boldsymbol{v}\}$. (S eat something new.) Now $S \cup \{\boldsymbol{v}\}$ must have redundancies, and you can throw away a redundant vector while KEEPING the linearly independent subset $L \cap (S \cup \{\boldsymbol{v}\})$. (S pooped away the redundancy.) This way we obtain a new set S' that is of the same size as S, but now S' contains one more elements of L than S does. Repeat this, and eventually we can put all of L into some spanning set of the same size as S. So we are done. (S finished eating all of L while pooping away $|L|$ irrelevant redundancies.)*

Or, to put into simple words, how to achieve anything in life? You need to go one step at a time, and keep throwing away redundant things.

There are other ways to prove this lemma. But our proof here is easily generalizable, that it also applied to mathematical areas other than linear algebra.

NOTE that, what if a vector space has NO finite spanning sets at all, that ALL spanning sets are infinite? Then in that case, this lemma proves nothing at all. However, with some twists and transfinite induction (a version of mathematical induction that goes beyond infinity), the idea will carry over.

5.13 Exercises

Exercise 5.13.1. *Which of the following are linearly independent?*

1. $\sin x, \sin(x+1), \sin(x+2)$.
2. $\sin x, \sin(2x), \cos(x), \cos(2x)$.
3. $\mathrm{e}^x, \mathrm{e}^{x+1}, \mathrm{e}^{x+2}$.
4. $\mathrm{e}^x, \mathrm{e}^{2x}, \mathrm{e}^{3x}$.
5. $x^2, (x+1)^2, (x+2)^2$.
6. $1, x, x^2, x^3$
7. $(\sin x)^2, \cos(2x), 1$.

8. $1+\mathrm{i}, 1-\mathrm{i}$. *(The field of scalars is $\mathbb{R}$.)*

9. $1+\mathrm{i}, 1-\mathrm{i}$. *(The field of scalars is $\mathbb{C}$.)*

10. $1, \pi$. *(The field of scalars is $\mathbb{R}$.)*

11. $1, \pi$. *(The field of scalars is $\mathbb{Q}$.)*

12. $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}, \begin{bmatrix}-1 & 1\\ 1 & -1\end{bmatrix}$.

13. Burger, Wing, Coke, and a combo (made of one burger and two wings).

14. Burger, Wing, Coke, and Nothing. (Here "nothing" is the zero vector.)

15. Seven Pokémons. (Each Pokémon has six stats, so the Pokémon space is 6 *dimensional.)*

Exercise 5.13.2 (Finding Kernel Algorithm)**.** *First, a quick summary of a section in the lecture notes.*

Let me show you a magical way to find a basis for the kernel of a matrix. Given a matrix A, how can we find $\mathrm{Ker}(A)$*? Here is an algorithm that always works.*

First, we perform Gaussian elimination to get $\mathrm{RREF}(A)$*, say* $\mathrm{RREF}(A) = \begin{bmatrix}1 & 2 & 0 & 3\\ 0 & 0 & 1 & 4\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$.

Next, we throw away zero rows and get $\begin{bmatrix}1 & 2 & 0 & 3\\ 0 & 0 & 1 & 4\end{bmatrix}$*. Note that at this stage, the number of columns is always at least the number of rows (can you see why?).*

Next we add zero-rows so that all the pivots are on the diagonal, so we have a square matrix $A' = \begin{bmatrix}1 & 2 & 0 & 3\\ 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 4\\ 0 & 0 & 0 & 0\end{bmatrix}$.

Finally, I claim that non-zero columns of $A' - I = \begin{bmatrix}0 & 2 & 0 & 3\\ 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 4\\ 0 & 0 & 0 & -1\end{bmatrix}$ *is a basis for* $\mathrm{Ker}(A)$*. This would always work.*

The proof is in the lecture notes.

For this problem, find a basis for the following spaces.

1. $\mathrm{Ker}(\begin{bmatrix}1 & 1\\ 2 & 4\end{bmatrix})$.

2. $\mathrm{Ker}(\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix})$.

3. $\mathrm{Ker}(\begin{bmatrix}1 & 2 & 3 & 4\\ 5 & 6 & 7 & 8\end{bmatrix})$.

4. $\mathrm{Ker}(\begin{bmatrix}1 & 1 & 1 & 1 & 1\\ 3 & 2 & 1 & 1 & -3\\ 0 & 1 & 2 & 2 & 6\\ 5 & 4 & 3 & 3 & 1\end{bmatrix})$.

5. $\mathrm{Ker}(\begin{bmatrix}1 & -2 & 1 & 1 & -1\\ 2 & 1 & -1 & -1 & -1\\ 1 & 7 & -5 & 5 & 5\\ 3 & -1 & -2 & 1 & -1\end{bmatrix})$.

6. $\mathrm{Ker}(\begin{bmatrix}2 & 1 & -1 & -1 & -1\\ 1 & -1 & 1 & 1 & -2\\ 3 & 3 & -3 & -3 & 4\\ 4 & 5 & -5 & -5 & 7\end{bmatrix})$.

7. $\mathrm{Ker}(\begin{bmatrix}I_{n\times n} & I_{n\times n}\end{bmatrix})$.

8. $\mathrm{Ker}(\begin{bmatrix}I_{n\times n} & I_{n\times n} & I_{n\times n}\end{bmatrix})$.

9. $\mathrm{Ker}(\begin{bmatrix}I_{n\times n} & I_{n\times n}\\ O & O\end{bmatrix})$.

Exercise 5.13.3. *These are essentially calculus problems, but I want you to feel the linear algebra in them.*

Find a basis for the following vector spaces, and indicate their dimension. (Maybe you see a pattern on the dimensions.)

1. *Solutions to* $f' = f$.
2. *Solutions to* $f'' = f$.
3. *Solutions to* $f' = 0$.
4. *Solutions to* $f'' = 0$.
5. *Solutions to* $f''' = 0$.
6. *Solutions to* $f' = 4f$.
7. *Solutions to* $f'' = 4f$.
8. *Solutions to* $f'' = f'$.

Exercise 5.13.4. *Find the matrices for the following linear maps under the given basis.*

1. V *is the span of functions* $1, \frac{1}{x}, x\sin(\frac{1}{x})$*, and the linear map is* $\lim_{x\to\infty} : V \to \mathbb{R}$.
2. V *is the space of* 2×2 *skew symmetric matrices, i.e., matrices such that* $A^{\mathrm{T}} = -A$*, and* T *is the transpose map from* V *to* V*. Pick any basis for* V.
3. V *is the span of functions* $1, x+1, x^2$*, and* $D : V \to V$ *is the derivative map.*
4. V *is the space of* 2×3 *matrices with basis* $\mathbf{e}_i\boldsymbol{e}_j^{\mathrm{T}}$*. The linear map sends* X *to* $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}X$.

Exercise 5.13.5. *For the following vector spaces* V *and their basis* $\mathcal{B}, \mathcal{C}$*, please find the change of coordinate matrix from* $\mathcal{B}$ *to* $\mathcal{C}$.

1. V *is the span of* $\mathcal{B} = (1, x, x^2, x^3)$. $\mathcal{C} = (1, x+2, (x+2)^2, (x+2)^3)$.
2. V *is the span of* $\mathcal{B} = (\sin(x), \sin(x+1))$. $\mathcal{C} = (\sin(x+2), \sin(x+3))$.
3. V *is the space of* 2×3 *matrices. Let* $A = \begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$ *and* $B = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$ *and* $C = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}$.

$$\mathcal{B} = (A, \begin{bmatrix}I_{2\times 2} & \mathbf{0}\end{bmatrix}, \begin{bmatrix}\mathbf{0} & I_{2\times 2}\end{bmatrix}, BA, AC, AC^2).$$

$$\mathcal{C} = (\boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}}, \boldsymbol{e}_1\boldsymbol{e}_2^{\mathrm{T}}, \boldsymbol{e}_1\boldsymbol{e}_3^{\mathrm{T}}, \boldsymbol{e}_2\boldsymbol{e}_1^{\mathrm{T}}, \boldsymbol{e}_2\boldsymbol{e}_2^{\mathrm{T}}, \boldsymbol{e}_2\boldsymbol{e}_3^{\mathrm{T}}).$$

4. V *is the space of* 2×2 *symmetric matrices.*

$$\mathcal{B} = (\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}).$$

$$\mathcal{C} = (\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}).$$

5. $V = \mathbb{R}^3$. *Vectors in* $\mathcal{B}$ *are columns of an invertible matrix* B. *Vectors in* $\mathcal{C}$ *are columns of an invertible matrix* C.

Exercise 5.13.6. *In chemistry, many crystal structures are NOT squares, but hexagons. So maybe we don't want to use the basis* $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ *for* $\mathbb{R}^2$. *What basis should you pick? What are the change of coordinate matrices back and forth?*

Exercise 5.13.7. *Find the basis for* $\mathrm{Ran}(A), \mathrm{Ran}(A^{\mathrm{T}}), \mathrm{Ker}(A), \mathrm{Ker}(A^{\mathrm{T}})$, *where we have*

$$A = \begin{bmatrix}1 & & \\6 & 1 & \\9 & 8 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 3 & 4\\ & 1 & 2 & 3\\ & & 1 & 2\end{bmatrix}.$$

Exercise 5.13.8. *Suppose the solution set to* $A\boldsymbol{x} = \begin{bmatrix}1\\3\end{bmatrix}$ *is* $\{\begin{bmatrix}1\\0\end{bmatrix} + k\begin{bmatrix}0\\1\end{bmatrix} : k \in \mathbb{R}\}$. *Find* A.

Exercise 5.13.9 (Vector spaces and basis). *(This problem requires you to know the axioms of a vector space in the optional section.)*

For the following, if it is not a vector space, just point out which axiom of vector spaces is failed by what vectors. If it is a vector space, you don't need to verify axioms. Just find its dimension instead, and find a basis if it is finite dimensional. (All spaces are over $\mathbb{R}$ *unless specifically mentioned.)*

1. V *is the set of all discontinuous functions* $f : \mathbb{R} \to \mathbb{R}$ *with its usual addition and scalar multiplication.*

2. *The solution set of* $x^2 + y^2 \leq z^2$ *in* $\mathbb{R}^3$. *(This is the pair of cones but we also include the insides of the two cones.)*

3. ☕*The set of subsets of* $X = \{1, 2, 3\}$, *where we define* $S + T$ *for subsets* S, T *as the* ***symmetric difference****, i.e.,* $S + T = \{x \in X \mid x \in S \text{ but not } T, \text{ or } x \in T \text{ but not } S\}$.

   This is a vector space defined over $\mathbb{F}_2$, *where the vectors are subsets, and scalar-vector multiplication is simply* $0S = \varnothing$ *and* $1S = S$ *for all subsets* S.

   (Here $\mathbb{F}_2 = \{0, 1\}$ *such that* $0 = 0+0 = 1+1 = 0 \times 1 = 1 \times 0 = 0 \times 0$ *and* $1 = 0+1 = 1+0 = 1 \times 1$. *You can also think "even" and "odd" in place of* 0 *and* 1. *Then all these calculations would make sense.)*

4. $J = \begin{bmatrix}0 & 1 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix}$ *and* V *is the space of all matrices that commutes with* J. *(You can use the results of previous HW directly.)*

5. $A = \begin{bmatrix}1 & 0\\0 & 2\end{bmatrix}$ *and* V *is the space of all matrices that commutes with* A.

6. ☕V *is the space of all* 3×3 *magic matrices. (I.e., matrices like the zero matrix, or like* $M = \begin{bmatrix}2 & 9 & 4\\7 & 5 & 3\\6 & 1 & 8\end{bmatrix}$, *where each row, each column, and each of the two diagonal adds up to the same number.)*

   (Hint: Flip all columns of M. *Flip all rows of* M. *Take transpose of* M. *Take reflection of* M *along the anti-diagonal. Take differences of these things. Take linear combinations of these things. Try until you figure this out.)*

Exercise 5.13.10 (Solution space to a differential equation)**.** *Consider the space V of all real functions f that solves the differential equation $f''' - 6f'' + 11f' - 6f = 0$. You may use the fact that V is spanned by $\mathrm{e}^x, \mathrm{e}^{2x}, \mathrm{e}^{3x}$.*

1. *Show that the matrix $A = \begin{bmatrix} \mathrm{e}^0 & \mathrm{e}^0 & \mathrm{e}^0 \\ \mathrm{e}^1 & \mathrm{e}^2 & \mathrm{e}^3 \\ \mathrm{e}^2 & \mathrm{e}^4 & \mathrm{e}^6 \end{bmatrix}$ is invertible.*

2. *Show that $\mathrm{e}^x, \mathrm{e}^{2x}, \mathrm{e}^{3x}$ are linearly independent, so that they form a basis for V. (Hint: if they are dependent, then what does that tell you about the matrix A in the last subproblem?)*

3. ☕*Given any $f \in V$, find the coordinates of f under this basis in terms of $f(0), f(1), f(2)$. (Hint: use the matrix A in the first sub-problem. You may simply write A^{-1} without calculating the inverse matrix.)*

Exercise 5.13.11 (Quotient Space)**.** *Recall that a line in $\mathbb{R}^3$ is a subset $L = \{\boldsymbol{p} + t\boldsymbol{v} : t \in \mathbb{R}\}$ for fixed $\boldsymbol{p}, \boldsymbol{v} \in \mathbb{R}^3$. This is a line through $\boldsymbol{p}$ and in the direction of $\boldsymbol{v}$.*

Consider the space $V = \mathbb{R}^3$, and fix a line W through the origin. Let V/W be the space of all lines in V parallel to W. For any two lines $L_1, L_2 \in V/W$, we define $L_1 + L_2 := \{\boldsymbol{p} + \boldsymbol{q} : \boldsymbol{p} \in L_1, \boldsymbol{q} \in L_2\}$. And for any scalar $k \in \mathbb{R}$ and any line $L \in V/W$, we define $kL := \{k\boldsymbol{p} : \boldsymbol{p} \in L\}$ when $k \neq 0$, and define $0L := W$.

1. *Show that for any two lines $L_1, L_2 \in V/W$, we still have $L_1 + L_2 \in V/W$. And for any line $L \in V/W$ and any $k \in \mathbb{R}$, we still have $kL \in V/W$.*

2. *Show that for any two lines $L_1, L_2 \in V/W$ and any $k \in \mathbb{R}$, we have $k(L_1 + L_2) = kL_1 + kL_2$. (You don't need to verify the other axioms. But in fact we also have other axioms as well, so V/W is a vector space.)*

3. *Which line in V/W is the "zero vector"?*

4. *Find the dimension of V/W and find a basis for V/W. (Hint: Pick any $\boldsymbol{w}$ that spans W, and extend it to a basis.)*

Exercise 5.13.12 (Addition and Scalar multiplication on polynomials mod $p(x)$)**.** *(☕This problem is a bit more advanced and abstract.) Let V be the space of all polynomials. Fix a polynomial $p(x) = x^2 + 3x + 2$.*

1. *For any finitely many polynomials, can you always find another polynomial that is NOT in their span? (This means V is infinite dimensional.)*

2. *Let W be the collections of all polynomials that contain $p(x)$ as a factor. (By convention, the zero polynomial contains all other polynomial as factors.) Is W a subspace?*

3. ☕*For any $r(x) \in V$, we write $[r(x)]$ as the subset $\{r(x) + p(x)q(x) : q(x) \in V\}$, i.e., all polynomials whose remainder after divided by $p(x)$ is the same as $r(x)$. We define $[r_1(x)] + [r_2(x)]$ as the subset $\{f_1(x) + f_2(x) : f_1(x) \in [r_1(x)]$ and $f_2(x) \in [r_2(x)]\}$. For $k \neq 0$, we define $k[r(x)]$ as the subset $\{kf(x) : f(x) \in [r(x)]\}$ for scalar $k \in \mathbb{R}$, and we define $0[r(x)] = [0]$. Show that $[r_1(x)] + [r_2(x)] = [r_1(x) + r_2(x)]$ and $k[r(x)] = [kr(x)]$.*

4. ☕*We use V/W to denote the set of all subsets $[r(x)]$ with addition and scalar multiplication as specified above. This is a finite dimensional vector space (you don't need to verify this, but feel free to do so). Find its dimension and find a basis.*

Exercise 5.13.13 (Space of matrices)**.** *Consider $M_{2\times 2}$ be the space of 2×2 matrices. We fix a basis $X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, X_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, X_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, X_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. We also fix a matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.*

1. *Consider the linear map $L_A : M_{2\times 2} \to M_{2\times 2}$ that sends X to AX. Write out the matrix for this linear map under the given basis. (Preferably as a block matrix. The block form would make it easier to imagine generalizations to higher dimensions.)*

2. *Consider the linear map $R_A : M_{2\times 2} \to M_{2\times 2}$ that sends X to XA. Write out the matrix for this linear map under the given basis. (Preferably as a block matrix. The block form would make it easier to imagine generalizations to higher dimensions.)*
3. *Calculate the 4×4 matrix multiplication $L_A R_A$ and $R_A L_A$. (You may use block matrix multiplication to facilitate the process.)*
4. *Can you see why $L_A R_A = R_A L_A$ without any calculation? (Hint: Think about the meaning of these linear maps. L_A means multiplying A to X from the left, while R_A means multiplying A to X from the right. Which law of matrix multiplication is this?)*
5. *Find a basis of $M_{2\times 2}$ made of invertible matrices, and find the change of coordinate matrix to this new basis.*

Exercise 5.13.14 (Change of basis and linear maps)**.** *Let V be a space with three bases $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ and W be a space with two bases $\mathcal{C}_1, \mathcal{C}_2$, and a bijective linear map $L : V \to W$. We use $I_{j\leftarrow i}$ to mean a change of coordinate map from $\mathcal{B}_i$ to $\mathcal{B}_j$ or from $\mathcal{C}_i$ to $\mathcal{C}_j$, and $L_{j\leftarrow i}$ to mean the matrix for L from $\mathcal{B}_i$ to $\mathcal{C}_j$. Which of the following formula about matrices $L_{j\leftarrow i}, I_{j\leftarrow i}$ are true? (No proof needed for correct ones. But for wrong ones, just briefly point out what is wrong with it.)*

1. $L_{1\leftarrow 1} = I_{1\leftarrow 2} L_{2\leftarrow 1}$.
2. $(L_{1\leftarrow 1})^{-1} = (L_{2\leftarrow 1})^{-1} I_{1\leftarrow 2}$.
3. $(L_{1\leftarrow 1})^{-1} = (L^{-1})_{2\leftarrow 1} I_{1\leftarrow 2}$.
4. $L_{2\leftarrow 2} = I_{2\leftarrow 1} L_{1\leftarrow 1} (I_{2\leftarrow 1})^{-1}$.
5. $L_{2\leftarrow 2} = L_{2\leftarrow 1} I_{1\leftarrow 1} L_{1\leftarrow 2}$.
6. $L_{1\leftarrow 3} = I_{1\leftarrow 2} L_{2\leftarrow 1} I_{1\leftarrow 3} I_{3\leftarrow 2} (L_{1\leftarrow 2})^{-1} I_{1\leftarrow 2} L_{2\leftarrow 1} I_{1\leftarrow 2} I_{2\leftarrow 3}$.

Exercise 5.13.15 (Complex numbers)**.** *Consider the complex plane $\mathbb{C}$ as a vector space over $\mathbb{R}$. We know it is a two dimensional real vector space with a basis $1, \mathrm{i}$.*

1. *For the complex number $w = 2 + 3\mathrm{i}$, consider the map $M_w : \mathbb{C} \to \mathbb{C}$ that maps input z to output wz. Find the matrix for M_w under the basis $1, \mathrm{i}$.*
2. *For any two complex numbers w, z, do the matrix multiplication to verify that $M_w M_z = M_{wz}$.*
3. *Can you see without computation that $M_w M_z = M_{wz}$? Which law of complex number multiplication is this? (Hint: Think about the meaning of these linear maps.)*
4. *Consider $\mathbb{F} = \{\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \mid a, b \in \mathbb{R}\}$, and find a bijective map $\phi : \mathbb{C} \to \mathbb{F}$ that satisfies $\phi(w + z) = \phi(w) + \phi(z), \phi(wz) = \phi(w)\phi(z)$. (Hint: surely the previous sub-problems are building up to something, yes?)*
5. *(Reading only) ϕ above is called a field isomorphism. It shows that $\mathbb{F}$ is also a field, and $\mathbb{F}$ and $\mathbb{C}$ are essentially the same field, just with different names for each element. Working on the field $\mathbb{C}$ is identical to working on the field $\mathbb{F}$. In a sense, you may think of "complex numbers" as a specific kind of 2×2 real matrices in disguise.*

Exercise 5.13.16 (Equation space and its subspace)**.** *let V be the space of linear equations on three variables, with elements such as $ax + by + cz = d$ for arbitrary real numbers a, b, c, d. So the vectors here are actually equations. We can do linear combination of these equations, i.e., $2(2x + 3y - z = 3) + 3(x - y = 2) = (7x + 3y - 2z = 12)$.*

(Note that since $x = 1$ and $x = 2$ are both elements of V, and V is a vector space, therefore V must include its difference, i.e., $0 = 1$. This equation $0 = 1$ is also an element of V, and it is indeed of the form $ax + by + cz = d$ where $a = b = c = 0$ and $d = 1$.)

1. *Find a basis of V, and find the dimension of V.*

2. *Suppose we have a basis made of equations $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$. What is the solution set when all equations $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ are satisfied?*

3. *Fix a point $\boldsymbol{p}$ in $\mathbb{R}^3$. Let W be the subset of V, made of all equations that contains this point $\boldsymbol{p}$ in its solution set. Is W a subspace? Prove or give counter examples. (Hint: to show that something is a subspace, you just need to show that $\boldsymbol{v}+\boldsymbol{w}$ and $k\boldsymbol{v}$ are still in W when $\boldsymbol{v}, \boldsymbol{w}$ is in W.)*

4. *Consider $x=1, y=2, z=3, x+y+z=7$. Are they linearly independent? Prove or provide a linear relation. (Note: you might want to try this proof for fun. First prove that the first three are linearly independent. For the last equation, take the subspace of all equations whose solution set contains the point $\begin{bmatrix}1\\2\\3\end{bmatrix}$. Then this subspace contains $x=1, y=2, z=3$, but fails to contain the last equation $x+y+z=7$. This gives you a way to use independence extension lemma.)*

5. *Consider $x=1, x=2, x=3$. Are they linearly independent? Prove or provide a linear relation.*

6. *(Not part of the HW) If you are curious, you may consider ways to generalize the statements here. What if we fix a line L, and let W be made of all equations that contains this line L in its solution set? What if we include more variables and go to higher dimensions?*

Exercise 5.13.17 (Effective number of chemical reactions)**.** *We have three chemical reactions when burning charcoals, $C+O_2=CO_2$, $2C+O_2=2CO$, and $2CO+O_2=2CO_2$. Note that these are NOT really equations. Rather, they reflect the process that the left hand side is transformed into the right hand side during the chemical reaction involved. So we don't solve them. Rather, we do their linear combinations according to how much of each reaction occurred. Then we shall see what's converted into what in total.*

For example, $(2C+O_2=2CO)+(2CO+O_2=2CO_2)=2(C+O_2=CO_2)$, and this means first we burn C to get CO, and then all the CO are burned and converted into CO_2. The total effect of combining $2C+O_2=2CO$ and $2CO+O_2=2CO_2$ is to do $C+O_2=CO_2$ twice.

1. *Let W be the space of linear combinations of the molecules C, CO, CO_2, O_2. The chemical reaction $C+O_2=CO_2$ is a map sending each $\boldsymbol{w} \in W$ to $\boldsymbol{w}-C-O_2+CO_2$. Is this map linear?*

2. *Consider the maps corresponding to the chemical reactions $C+O_2=CO_2$, $2C+O_2=2CO$, and $2CO+O_2=2CO_2$. Are these reactions linearly independent? What is the dimension of their span?*

3. *Let $M=\begin{bmatrix}-1&-2&0\\0&2&-2\\1&0&2\\-1&-1&-1\end{bmatrix}$, i.e., the three columns corresponds to the changes induced by the three chemical reactions on W with basis C, CO, CO_2, O_2. Find invertible matrices R, C such that RMC is in rank normal form, and find the rank r of M. (So we see that the three equations $C+O_2=CO_2$, $2C+O_2=2CO$, and $2CO+O_2=2CO_2$ only have r effective equations.)*

Exercise 5.13.18 (Algorithm to find full-rank decomposition)**.** *We know that, given any $m \times n$ matrix A of rank r, then $A=BC$ where B is $m \times r$ and injective, and C is $r \times n$ and surjective.*

Suppose $A=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$.

1. *Delete all non-pivotal columns of A (columns of A that are linear combinations of previous columns), and we obtain a matrix B. Find B. Why must the rank of A equal to the rank of B?*

2. *Find the reduced row echelon form of A and delete all zero rows, and we obtain a matrix R. Can you find the rank of R?*

3. *Compute BR, what do you have? (This gives an algorithm to perform full rank decompositon, if you ever need to.)*

4. *(Not part of the homework) Can you see why you have this phenomena? This process can be generalized to an arbitrary A.*

Exercise 5.13.19 (Meaning of rank-one decomposition)**.** *This problem concerns a recommendation algorithm used in some video platform websites. We are greatly simplifying the process, of course.*

Suppose we have seven people, A, B, ..., G, and they watched six videos on bilibili, a, b, ..., f. Suppose for each video, they can click "like" or "dislike", and this data is collected into the following matrix.

$$M = \begin{array}{c|c} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \hline \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 & 1 \end{bmatrix} \end{array}.$$

Here 1 means the corresponding person likes the corresponding video, -1 means dislike, and 0 means no response.

Let us suppose that a, b, c are all educational videos, while c, d, e, f are all funny videos. So we have two categories for a video: educational or funny. Note that each person seems to react reasonably consistent about videos in each category. The exception is c, which is both educational and funny, and whether a person likes or dislikes seems to be influenced by other factors.

1. *What is the rank of this 7×6 matrix?*

2. *Build a matrix X such that its two columns corresponds to the two categories of videos, and its seven rows corresponds to the seven people, and the entries records whether the corresponding person likes, dislikes or has no response to the corresponding category in general.*

3. *Build a matrix Y such that its two rows corresponds to the two categories of videos, and its six columns corresponds to the six videos. And an entry is 1 if the video is in this category, zero if this video is NOT in this category.*

4. *Check to see that $M - XY$ has rank 1. (This represents portions of M that cannot be explained by categories of the videos.)*

5. *Write M as the sum of three rank one matrices, where the first one reflects how educational videos contributes to M, the second one reflects how funny videos contributes to M, and the third one reflects how other factors contributes to M.*

Exercise 5.13.20 (More Inclusion-Exclusion Principals)**.** 1. *Prove that for subsets X, Y, Z, we have $|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|$.*

2. *Find an example of subspaces X, Y, Z such that $\dim(X + Y + Z) \neq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)$.*

3. *Prove that for subspaces X, Y, Z, we have $\dim(X + Y + Z) \leq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)$. (Hint: Use the regular IEP for subspaces repeatedly. Prove that $(X \cap Z) + (Y \cap Z) \subseteq (X + Y) \cap Z$.)*

Chapter 6

Inner Product Space

Abstraction is fun and nice and useful, but they have missing pieces. In particular, we do not have lengths or angles.

Example 6.0.1. Consider the space of polynomials of degree at most one, $\mathcal{P}_1$. Then under the basis $(1, x)$, the elements 2 and $3x$ in $\mathcal{P}_2$ have coordinates $\begin{bmatrix}2\\0\end{bmatrix}, \begin{bmatrix}0\\3\end{bmatrix}$. In particular, they seem to have length 2 and 3, and it seems like they are perpendicular.

However, under the basis $(2+3x, 3x)$, then the elements 2 and $3x$ in $\mathcal{P}_2$ now have coordinates $\begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$. Now they have length $\sqrt{2}$ and 1, and they are no longer perpendicular.

As you can see, "length" and "angles" are NOT defined for an abstract vector spaces.

Similarly, you also lose transpose. Consider $M : \mathcal{P}_1 \to \mathcal{P}_1$ which is $a + bx \mapsto -b + ax$. Under the basis $(1, x)$, its matrix is $\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}$, whose inverse equal to its transpose. However, under the basis $(1 + x, x)$, then its matrix is $\begin{bmatrix}-1 & -1\\2 & 1\end{bmatrix}$, whose inverse is $\begin{bmatrix}1 & 1\\-2 & -1\end{bmatrix}$, which is not the transpose. ☺

In practice, we would often want to use length and angles. However, we might still want to reserve an option to change basis. This gives rise to the definition of an inner product space.

In the sense of abstractions, an inner product space is pretty much $\mathbb{R}^n$ where you forget about your basis, but you remembered the lengths and angles. If you then also forget about the lengths and angles, then you have reached an abstract vector space.

Conversely, you can also think of an inner product space as an abstract vector space with an extra structure, the inner product (e.g. dot product), which allows you to do lengths and angles. And you can think of $\mathbb{R}^n$ as an inner product space with an extra structure, i.e., a fixed basis.

In practice, we shall see that (finite dimensional) inner product spaces are pretty much just $\mathbb{R}^n$, but you are allowed to change basis as long as the length and angle structures are preserved in the process, i.e., orthonormal change of basis.

6.1 Fundamental Theorem of Linear Algebra

Here let us plug in the missing pieces in the case of $\mathbb{R}^n$, utilizing the dot product structure. Keep in mind that these statements will NOT have easy analogues for abstract linear maps. (The analogues exist, but they require much deeper knowledge to describe.)

We now go back to the world of $\mathbb{R}^n$. Let us refrain from all change of basis at the moment, and always use the standard basis. By doing so, we have lengths, angles, dot products, and transpose, and all that, i.e., the missing pieces are now all back.

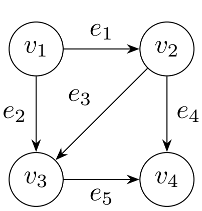
Given an $m \times n$ matrix A, there are four ***fundamental subspaces*** related to it.

1. The range of A, $\mathrm{Ran}(A)$;
2. The kernel of A, $\mathrm{Ker}(A)$;
3. The ***left range*** or ***row space*** of A, which is in fact $\mathrm{Ran}(A^{\mathrm{T}})$;
4. The ***left null space*** or ***left kernel*** of A, which is in fact $\mathrm{Ker}(A^{\mathrm{T}})$.

The names are somewhat revealing. For example, what is the subspace spanned by all rows of A? Well, these rows are just columns of A^{T}, so the space is simply $\mathrm{Ran}(A^{\mathrm{T}})$. You can also verify this: $\boldsymbol{b} \in \mathrm{Ran}(A^{\mathrm{T}})$ if and only if $\boldsymbol{b}^{\mathrm{T}} = \boldsymbol{x}^{\mathrm{T}} A$ for some $\boldsymbol{x}$. This explains the name of "left range". Similarly, for the left kernel, you can verify that $\boldsymbol{x} \in \mathrm{Ker}(A^{\mathrm{T}})$ if and only if $\boldsymbol{x}^{\mathrm{T}} A = \mathbf{0}^{\mathrm{T}}$.

These things have practical meanings in real life applications.

Example 6.1.1. Consider an electric circuit in Figure 6.1.1.



$$M = \begin{array}{c c} \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} & \begin{array}{c} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{array} \\ \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} & \end{array}$$

Figure 6.1.1: Graph of electric circuit and its incidence matrix M

Such a structure is mathematically called a directed graph, where we have some vertices, and some arrows (electrical wires) that goes from some vertex to another vertex. Here we think of the arrows as actual electrical wires. Given electrical potentials on the vertices, the difference between electrical potentials would induce voltages on wires. Then given the resistance of each wire, we can find out the electrical current on each wire. This is a classical situation in the study of electrical circuit.

(Why do we have arrows on the wires? Well, when we measure electrical currents, if the electricity flow in the arrow direction, we can declare this to be a "positive current", and if it flows in the opposite direction, we can declare this to be a "negative current".)

Whenever we have a graph, we can draw its incidence matrix M, where each column corresponds to a vertex, and each row corresponds to an arrow. The (i, j) entry of M is -1 if the i-th arrow starts at the j-th vertex, 1 if the i-th arrow ends at the j-th vertex, and 0 otherwise.

Now, what linear map does M represents? First of all, we see that M would send a generic input $\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$ to $\begin{bmatrix} u_2 - u_1 \\ u_3 - u_1 \\ u_3 - u_2 \\ u_4 - u_2 \\ u_4 - u_3 \end{bmatrix}$. In particular, if the inputs are electrical potentials at each vertex, then the output are their differences (i.e., voltages) on each wire!

Electrical potentials on vertices $\xrightarrow{M}$ Voltages on wires

In particular, if we assign electrical potentials on vertices arbitrarily, then $\mathrm{Ran}(M)$ is the space of all possible voltages on wires in this circuit.

On the other hand, $\mathrm{Ker}(M)$ is the set of electrical potentials that gives zero voltages everywhere. Huh. Without any computation, just by physics intuition, you can see that $\mathrm{Ker}(M) = \{ \begin{bmatrix} u \\ u \\ u \\ u \end{bmatrix} : u \in \mathbb{R}\}$, i.e., when all electrical potentials are identical. This should be the case as long as my graph for electrical circuit is a connected graph.

Now, what about the range and kernel of M^T?

(You might notice this formula in physics: voltage times current is power. We have many wires here. Let $\boldsymbol{v}$ be the vector recording the voltages on these wires, and let $\boldsymbol{c}$ be the vector recording the currents on these wires, then $\boldsymbol{v} \cdot \boldsymbol{c}$ is the total power of the circuit, a scalar! Hence if M acts on the voltage side, then M^{T} acts on the current side.)

Let $\boldsymbol{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix}$ be a vector recording the electrical currents on each wire. Then note that $M^{\mathrm{T}}\boldsymbol{c} = \begin{bmatrix} -c_1 - c_2 \\ c_1 - c_3 - c_4 \\ c_2 + c_3 - c_5 \\ c_4 + c_5 \end{bmatrix}$. What are these output coordinates? Well, look at the second one. The vertex v_2 has wires e_1 flowing into it, and wires e_3, e_4 flowing out of it, and the second coordinate here is exactly recording that! In particular, the coordinates record the net inflow of electrical currents at each vertex. (Keep in mind that each c_i could be positive or negative, depending on the direction of the current.) Feel free to verify this yourself.

Electrical currents on wires $\xrightarrow{M^{\mathrm{T}}}$ Net current inflow on vertices

If you think about this, WITHOUT outside influence, when the electrical flows are stable, we would expect each vertex to have zero net inflow, i.e., whatever flows into a vertex, it must then flow out. This is ***Kirchhoff's current law***, which says that stable currents must be in $\mathrm{Ker}(M^{\mathrm{T}})$.

If you compute $\mathrm{Ker}(M^{\mathrm{T}})$ (say via Gaussian elimination), this space is in fact spanned by $\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$. You can see that the both corresponds to closed loops in the graph (where -1 means we are going against an arrow). So stable currents simply means currents travels in loops. For example, the vector $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$ is also in the kernel of M^{T}, and it also corresponds to a loop. (Note that it is the sum of the two loop vectors listed previously. Geometrically, this also makes sense. The two triangle loops would add up to the square loop, where the diagonal edge got cancelled, because the two triangle loops travel in opposite directions on this edge.)

There is also ***Kirchhoff's voltage law***, which states that possible voltages (i.e. elements of $\mathrm{Ran}(M)$)

must add up to zero on closed loops. Indeed, if $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} \in \operatorname{Ran}(M)$, then Gaussian elimination gives

$$\left[\begin{array}{cccc|c} -1 & 1 & 0 & 0 & u_1 \\ -1 & 0 & 1 & 0 & u_2 \\ 0 & -1 & 1 & 0 & u_3 \\ 0 & -1 & 0 & 1 & u_4 \\ 0 & 0 & -1 & 1 & u_5 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -u_1 \\ 0 & -1 & 1 & 0 & u_2 - u_1 \\ 0 & 0 & -1 & 1 & u_1 - u_2 + u_4 \\ 0 & 0 & 0 & 0 & u_1 - u_2 + u_3 \\ 0 & 0 & 0 & 0 & u_2 - u_1 + u_5 - u_4 \end{array}\right].$$

So $\boldsymbol{u} \in \operatorname{Ran}(M)$ if and only if $u_1 - u_2 + u_3 = 0$ and $u_2 - u_1 + u_5 - u_4 = 0$, i.e., it is perpendicular to the two loop vectors $\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$. Note that these two loop vectors would actually span all loops, i.e., they span the space $\operatorname{Ker}(M^{\mathrm{T}})$.

Wait, we have obtained a funny result here. Elements of $\operatorname{Ran}(M)$ are exactly those that are perpendicular to $\operatorname{Ker}(M^{\mathrm{T}})$.

The same relation is also present for $\operatorname{Ran}(M^{\mathrm{T}})$ and $\operatorname{Ker}(M)$. Given $\operatorname{Ran}(M^T)$, the elements of $\operatorname{Ran}(M^{\mathrm{T}})$ are possible net inflows at vertices. However, if currents flow out of a vertex, then they must flow into some other vertex. So if $\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} \in \operatorname{Ran}(M^{\mathrm{T}})$, then we should have $w_1 + w_2 + w_3 + w_4 = 0$, i.e., they should be perpendicular to $\operatorname{Ker}(M)$ (which is spanned by $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$). ☺

Our goal for this section, the fundamental theorem, is basically a theorem about relations between these fundamental subspaces. First of all, by rank-nullity, we already know that $\dim \operatorname{Ran}(A) + \dim \operatorname{Ker}(A) = n$ and $\dim \operatorname{Ran}(A^{\mathrm{T}}) + \dim \operatorname{Ker}(A^{\mathrm{T}}) = m$. What else do we know?

Proposition 6.1.2. *A and A^{T} have the same rank.*

Proof. Suppose the rank of A is r. Suppose the rank normal form of A is $RAC = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$ for invertible matrices R, C. Taking transpose, we have $C^{\mathrm{T}}A^{\mathrm{T}}R^{\mathrm{T}} = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$. And since $C^{\mathrm{T}}, R^{\mathrm{T}}$ are still invertible, we immediately see that the rank normal form of A^{T} is $\begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$, and hence its rank is r as well. □

(Note that this is essentially the old statement that "the number of effective equations = the number of dependent variables".)

(The porposition above also says the following fact: for a matrix A, if its columns span a space with dimension r, then its rows span a space with dimension r.)

Remark 6.1.3. *Be careful, you cannot say "pick basis and assume that $A = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$. This is becasue if you change basis arbitrarily, then A^{T} might not corresponds to the transpose anymore.*

Indeed, to go to the range normal form, A and A^{T} must actually use DIFFERENT change of basis. One needs R and C, while the other uses C^{T} and R^{T}.

So, given an $m \times n$ matrix A with rank r, then $r = \dim \mathrm{Ran}(A) = \dim \mathrm{Ran}(A^{\mathrm{T}})$, $\dim \mathrm{Ker}(A) = n - r$ and $\dim \mathrm{Ker}(A^{\mathrm{T}}) = m - r$. So we have all the dimensions.

But hey, note that $\mathrm{Ran}(A)$ and $\mathrm{Ker}(A)$ are NOT in the same space. The former is in the codomain, while the latter the domain. So in fact $\mathrm{Ran}(A)$ and $\mathrm{Ker}(A^{\mathrm{T}})$ are both in the codomain of A, while $\mathrm{Ran}(A^{\mathrm{T}})$ and $\mathrm{Ker}(A)$ are both in the domain of A. Do the subspaces in the same space have some special relation?

Yes they do. We need to introduce the concept of an orthogonal complement.

Definition 6.1.4. *We say subsapces V, W of $\mathbb{R}^n$ are orthogonal complements if they are complement as subspaces, and all vectors in V are orthogonal to all vectors in W.*

Proposition 6.1.5. $\mathrm{Ran}(A)$ *and* $\mathrm{Ker}(A^{\mathrm{T}})$ *are orthogonal complements of each other, and* $\mathrm{Ran}(A^{\mathrm{T}})$ *and* $\mathrm{Ker}(A)$ *are orthogonal complements of each other.*

Proof. It is enough to prove the first statement, because applying the first statement to A^{T} gives the second one.

Let us first show that they are orthogonal. Intuitively this is obvious. $\mathrm{Ker}(A^{\mathrm{T}})$ is made of vectors perpendicular to all rows of A^{T}, i.e., all columns of A. But $\mathrm{Ran}(A)$ is made of vectors that are linear combinations of these columns. So of course they are orthogonal subspaces.

To be more formal, pick any $\boldsymbol{v} \in \mathrm{Ran}(A)$, say $\boldsymbol{v} = A\boldsymbol{u}$ for some $\boldsymbol{u}$, and pick any $\boldsymbol{w} \in \mathrm{Ker}(A^{\mathrm{T}})$. Then $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = (A\boldsymbol{u})^{\mathrm{T}}\boldsymbol{w} = \boldsymbol{u}^{\mathrm{T}}A^{\mathrm{T}}\boldsymbol{w} = \boldsymbol{u}^{\mathrm{T}}\mathbf{0} = 0$. So they are orthogonal subspaces.

Now let us show that they are complements. They are orthogonal, so obviously they have trivial intersection. Or to be more rigorous, suppose $\boldsymbol{v} \in \mathrm{Ran}(A) \cap \mathrm{Ker}(A^{\mathrm{T}})$, then $\boldsymbol{v}$ must be orthogonal to itself. Then $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v} = 0$, and thus $\boldsymbol{v}$ has zero length. So it must be $\mathbf{0}$.

But the dimensions of the two subspaces add up to the dimension of the ambient space, so by inclusion-exclusion principle, their sum IS the ambient space. So they are complements.

More rigorously, if A is $m \times n$ rank r, then $\dim \mathrm{Ran}(A) = r$ while $\dim \mathrm{Ker}(A^{\mathrm{T}}) = m - \dim \mathrm{Ran}(A^{\mathrm{T}}) = m - r$. With trivial intersection, this means $\dim(\mathrm{Ran}(A) + \mathrm{Ker}(A^{\mathrm{T}})) = m = \dim \mathbb{R}^m$. Since the subspace has the same dimension with the whole space, we see that the two are the same. $\square$

This immediately opens up a lot of interesting properties for orthogonal complements. Unlike regular complements, the orthogonal complement is unique.

Proposition 6.1.6. *For any subspace V of $\mathbb{R}^n$, then $V^{\perp} = \{\boldsymbol{w} \mid \boldsymbol{w}^{\mathrm{T}}\boldsymbol{v} = 0 \forall \boldsymbol{v} \in V\}$ is the only orthogonal complement of V. In particular, the orthogonal complement always exists and is unique.*

Proof. Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ be a basis for V, and let $A = \begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_k \end{bmatrix}$ (note that A must be $n \times k$ with rank k). Now $\mathrm{Ran}(A) = V$.

But also, $\boldsymbol{v} \in \mathrm{Ker}(A^{\mathrm{T}})$ iff $A^{\mathrm{T}}\boldsymbol{v} = \mathbf{0}$ iff $\boldsymbol{v}$ is orthogonal to all $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ iff $\boldsymbol{v} \in V^{\perp}$. So $\mathrm{Ker}(A^{\mathrm{T}}) = V^{\perp}$. We see that $V, V^{\perp}$ are indeed orthogonal complements.

Now we go for uniqueness. Suppose W is also an orthogonal complement of V. Then on one hand, all vectors of W are orthogonal to all vectors of V, so $W \subseteq V^{\perp}$. On the other hand, since $W, V^{\perp}$ are both complement subspaces to V, they both have dimension $n - \dim(V)$. So they are the same. $\square$

Corollary 6.1.7. $(V^{\perp})^{\perp} = V$.

Proof. $V, (V^{\perp})^{\perp}$ are both orthogonal complements of $V^{\perp}$, so by uniqueness we have equality. $\square$

For example, the xy-plane and the z-axis in $\mathbb{R}^3$ are the unique orthogonal complements of each other. The orthogonal complement is a MUCH better analogy to the complements of subsets. In particular, we have the de Morgan law:

Proposition 6.1.8. $(V + W)^{\perp} = V^{\perp} \cap W^{\perp}, (V \cap W)^{\perp} = V^{\perp} + W^{\perp}$

Proof. Pick basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_a$ for V and basis $\boldsymbol{w}_1, \dots, \boldsymbol{w}_b$ for W. Let $A = \begin{bmatrix}\boldsymbol{v}_1 & \dots & \boldsymbol{v}_a\end{bmatrix}$ and $B = \begin{bmatrix}\boldsymbol{w}_1 & \dots & \boldsymbol{w}_b\end{bmatrix}$. So $\operatorname{Ran}(A) = V, \operatorname{Ran}(B) = W, \operatorname{Ker}(A^{\mathrm{T}}) = V^{\perp}, \operatorname{Ker}(B^{\mathrm{T}}) = W^{\perp}$.

Now we already know sum and intersections are related to block matrices. We have $V+W = \operatorname{Ran}\begin{bmatrix}A & B\end{bmatrix}$ and $V^{\perp} \cap W^{\perp} = \operatorname{Ker}\begin{bmatrix}A^{\mathrm{T}}\\ B^{\mathrm{T}}\end{bmatrix}$. Then we immediately see that they are orthogonal complements because $\begin{bmatrix}A & B\end{bmatrix}^{\mathrm{T}} = \begin{bmatrix}A^{\mathrm{T}}\\ B^{\mathrm{T}}\end{bmatrix}$.

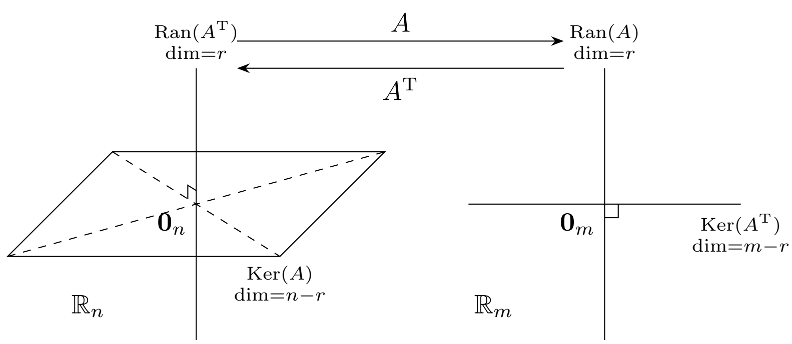
To see the second formula, just take orthogonal complement of the first formula on both sides. ☐

Hope you have fun. You can definitely see how the abstract results on rank-nullity, inclusion-exclusion principle, and subspace algebra mixes really well with concrete things in $\mathbb{R}^n$ such as block matrices and transpose.

So now we have the complete fundamental theorem of linear algebra (FTLA for short).

Theorem 6.1.9. *Given any $m \times n$ matrix A, then $\operatorname{Ran}(A)^{\perp} = \operatorname{Ker}(A^{\mathrm{T}})$ and $\operatorname{Ker}(A)^{\perp} = \operatorname{Ran}(A^{\mathrm{T}})$, and $\dim \operatorname{Ran}(A) = \dim \operatorname{Ran}(A^{\mathrm{T}})$.*

This is simply a collection of all previous results. We can see graphically here for the matrix $A = \begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$ as the following



Here are some examples of how FTLA relates to real problems.

Example 6.1.10. Again consider Figure 6.1.1. Here recall that $\operatorname{Ker}(M^{\mathrm{T}})$ is the space of currents that gives zero net inflow at each vertex, i.e., currency that satisfy Kirchhoff's current law, while $\operatorname{Ran}(M)$ is the space of voltages that sums to zero on closed loops, i.e., voltages that satisfy Kirchhoff's voltage law. In a stable case, both laws should be satisfied. The total power of our electrical circuit is the dot product of the voltage vector and the current vector, which now must be zero, because the two subspaces are orthogonal complements. Huh.

Why is that? This is because we have no battery in our system. With no battery, the electrical circuit is essentially "dead", so there can be no power output.

Suppose on each edge their is a batter providing b_i extra voltage to flow in the positive direction. Then we have a battery vector $\boldsymbol{b}$. Assume that $\boldsymbol{x}$ is the electrical potential vector and $\boldsymbol{y}$ is the current vector, then we have $\boldsymbol{b} - M\boldsymbol{x} = R\boldsymbol{y}$, where R is the diagonal matrix whose diagonal entries are the resistence on each wire. Furtheremore we know that $M^{\mathrm{T}}\boldsymbol{y} = \mathbf{0}$ given Kirchhoff's current law. So we can solve $\begin{bmatrix}R & M\\ M^{\mathrm{T}} & O\end{bmatrix}\begin{bmatrix}\boldsymbol{y}\\ \boldsymbol{x}\end{bmatrix} = \begin{bmatrix}\boldsymbol{b}\\ \mathbf{0}\end{bmatrix}$ to find stable solutions to our system. ☺

(The rest of this subsection is entirely optional.)

Let us insert a corollary before we proceed with the last example.

Corollary 6.1.11. $\operatorname{Ker}(A^{\mathrm{T}}A) = \operatorname{Ker}(A)$.

Proof. Suppose $\boldsymbol{v} \in \operatorname{Ker}(A)$. Then $A^{\mathrm{T}}A\boldsymbol{v} = A^{\mathrm{T}}\mathbf{0} = \mathbf{0}$, so $\boldsymbol{v} \in \operatorname{Ker}(A^{\mathrm{T}}A)$.

Conversely, $\boldsymbol{v} \in \operatorname{Ker}(A^{\mathrm{T}}A)$ means $A^{\mathrm{T}}A\boldsymbol{v} = \mathbf{0}$, which means $A\boldsymbol{v} \in \operatorname{Ker}(A^T)$. So in particular, $A\boldsymbol{v} \in \operatorname{Ker}(A^{\mathrm{T}}) \cap \operatorname{Ran}(A) = \{\mathbf{0}\}$. So $A\boldsymbol{v} = \mathbf{0}$, $\boldsymbol{v} \in \operatorname{Ker}(A)$.

Intuitively, the image coming from A would perfectly dodge the kernel of A^{T}. So $A^{\mathrm{T}}A$ kills exactly thos killed by A in the first step, and the second step A^{T} would fail to kill anything new. ☐

Example 6.1.12. (Optional)

Now, block row operation gives $\begin{bmatrix} R & M_G \\ O & -M_G^{\mathrm{T}}R^{-1}M_G \end{bmatrix}$, and then block column operation gives $\begin{bmatrix} R & O \\ O & -M_G^{\mathrm{T}}R^{-1}M_G \end{bmatrix}$. Note that since resistance are usually all non-zero, R has full rank, i.e., $\operatorname{rank}(R) = 5$. Since diagonal entries of R^{-1} are all positive, we have $R^{-1} = D^2$ for some diagonal D. Then $\operatorname{Ker}(M_G^{\mathrm{T}}R^{-1}M_G) = \operatorname{Ker}((DM_G)^{\mathrm{T}}(DM_G)) = \operatorname{Ker}(DM_G) = \operatorname{Ker}(M_G)$ because D is bijective. This is spanned by $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$ and hence 1-dimensional, so $M_G^{\mathrm{T}}R^{-1}M_G$ has rank $4 - 1 = 3$. So $\begin{bmatrix} R & M_G \\ O & -M_G^{\mathrm{T}}R^{-1}M_G \end{bmatrix}$ is a 9×9 matrix of rank 8.

Note that $\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$ is in the kernel, which has dimension 1, so it in fact spans the kernel. This means our system, given battery inputs, will have unique solution up to shifting all electrical potential by the same constant. ☺

Example 6.1.13. (Optional)

Let us consider a surprise example from light out puzzle. This also serves as an example of how things work over the field $\mathbb{F}_2$, where we only have coeficient $0, 1$ and we think $1 + 1 = 0$.

Say we have a single row of five tiles. Each tile is either lit up or light out. Whenever we press a tile, then this tile and its adjacent tiles change status. At any given time, if we use 1 to represent a lit-up tile and 0 to represent a tile whose light is off, then the status of tiles is a vector in $(\mathbb{F}_2)^5$. (So we have $2^5 = 32$ possible initial configurations.) For example, $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ means the first, second and forth tile are llit up.

We can press some of the tiles, and this input translates to a vector $\boldsymbol{p} \in (\mathbb{F}_2)^5$. For example, $\boldsymbol{p} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ means we are pressing the first, second and forth tile. Then since each tile when pressed would change the status of itself and all adjacent tiles, we have a matrix $M = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$, and the total change of status would be $M\boldsymbol{p}$. If the lights started with status $\boldsymbol{v}$, then after pressing $\boldsymbol{p}$, we would end up with status $M\boldsymbol{p} + \boldsymbol{v}$. We would turn off the light iff $M\boldsymbol{p} = -\boldsymbol{v}$. Note that over $\mathbb{F}_2$, we have $1 = -1$ for scalars, so we are looking to solve $\boldsymbol{p}$ from $M\boldsymbol{p} = \boldsymbol{v}$.

Now, M is not invertible. You may check that $M\begin{bmatrix}1\\1\\0\\1\\1\end{bmatrix}=\mathbf{0}$. In fact, gaussian elimination gives $rref(M)=$ $\begin{bmatrix}1&0&0&0&1\\0&1&0&0&1\\0&0&1&0&0\\0&0&0&1&1\\0&0&0&0&0\end{bmatrix}$. So $\mathrm{Ker}(M)$ is in fact spanned by $\begin{bmatrix}1\\1\\0\\1\\1\end{bmatrix}$.

Now by rank nullity theorem, this means we have $\dim \mathrm{Ran}(M) = 5 - \dim \mathrm{Ker}(M) = 4$. So not all initial status can be solved. In fact, since our matrix is symmetric, we have $\mathrm{Ran}(M) = \mathrm{Ran}(M^{\mathrm{T}}) = \mathrm{Ker}(M)^{\perp}$ by FTLA. So a status $\begin{bmatrix}a\\b\\c\\d\\e\end{bmatrix}$ can be solved iff it is "orthogonal" to $\begin{bmatrix}1\\1\\0\\1\\1\end{bmatrix}$, i.e., $a+b+d+e=0$. This means among the first, second, forth, fifth lights, only even number of them are lit up. Note that since $\mathrm{Ran}(M)$ is 4 dimensional, only $2^4 = 16$ initial status out of $|(\mathbb{F}_2)^5| = 2^5 = 32$ possible initial status are solvable.

Furthermore, given any solution $\boldsymbol{p}$ to an initial status, then $\boldsymbol{p}+\begin{bmatrix}1\\1\\0\\1\\1\end{bmatrix}$ is also an solution. We have exactly two solutions to any initial status.

Check out the english wikipedia page of "Light Out (Game)", where they listed the two vectors the span the kernel of M when there are 5×5 array of tiles. (So the space is $(\mathbb{F}_2)^{25}$ instead of $(\mathbb{F}_2)^5$. For cosmetic reasons, they write the vectors N_1, N_2 like matrices, when they are in fact a column vector in $(\mathbb{F}_2)^{25}$.) So any initial status can have 4 solutions, and only 2^{23} initial cases are solvable out of a total of 2^{25}. ☺

6.2 Inner Product Space

Now abstract vector spaces are sometimes annoying. They have no angle, no dot product, no transpose. We want these things sometimes. If abstraction is forgetting, then sometimes it seems that we have forgotten too much.

So we need some intermediate structure. This is inner product space, i.e., an abstract vector space where you can do "dot product", yet there is no standard basis.

Definition 6.2.1. *An inner product space is an abstract vector space V over $\mathbb{R}$ equipped with an* ***inner product****, which is an operation $\langle -,-\rangle$ that sends two vectors $\boldsymbol{v},\boldsymbol{w}$ of V into a scalar $\langle\boldsymbol{v},\boldsymbol{w}\rangle$ in $\mathbb{R}$, such that*

1. *(Symmetric) $\langle\boldsymbol{v},\boldsymbol{w}\rangle = \langle\boldsymbol{w},\boldsymbol{v}\rangle$ for all $\boldsymbol{v},\boldsymbol{w}\in V$.*
2. *(Bilinear) $\langle a\boldsymbol{u}+b\boldsymbol{v},\boldsymbol{w}\rangle = a\langle\boldsymbol{u},\boldsymbol{w}\rangle + b\langle\boldsymbol{v},\boldsymbol{w}\rangle$ for all $a,b\in\mathbb{R},\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}\in V$. And same thing for the other side.*
3. *(Positive Definite) $\langle\boldsymbol{v},\boldsymbol{v}\rangle\geq 0$ with equality iff $\boldsymbol{v}=\mathbf{0}$.*

The inner product here essentially serves as an abstract analogue to the "dot product". If you think dot product, then all these properties are obviously needed. The last one, in particular, allows us to define the ***length*** of an abstract vector $\boldsymbol{v}$ in an inner product space as $\|\boldsymbol{v}\| = \sqrt{\langle\boldsymbol{v},\boldsymbol{v}\rangle}$.

So, why is this enough? Remember, a major goal of this is to have angles, because orthogonality is very useful in many situations (and also required for building FTLA in abstract spaces). Traditionally in $\mathbb{R}^n$, if two vectors have angle θ, then we can work out θ from the fact that $\cos\theta = \frac{\boldsymbol{v}\cdot\boldsymbol{w}}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|}$. (Surely you've seen this formula, at least for $\mathbb{R}^2$?) So we would to define angles via the formula $\cos\theta = \frac{\boldsymbol{v}\cdot\boldsymbol{w}}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|}$. But for this to work at all, we need the following famous inequality.

Theorem 6.2.2 (Cauchy-Schwarz Inequality)**.** *For any $\boldsymbol{v}, \boldsymbol{w}$ in an inner product space, we have $\langle \boldsymbol{v}, \boldsymbol{w}\rangle^2 \leq \langle \boldsymbol{v}, \boldsymbol{v}\rangle\langle \boldsymbol{w}, \boldsymbol{w}\rangle$, with equality iff $\boldsymbol{v}, \boldsymbol{w}$ are colinear.*

There are a million proofs of this inequality, so you are very welcome to track them down online. The english wikipedia has two proofs at least. Here I present my favorite, not because it is simple, but because it is intuitive.

Definition 6.2.3. *We write $\boldsymbol{v} \mathbin{/\!/} \boldsymbol{w}$ if one is a multiple of the other. We write $\boldsymbol{v} \perp \boldsymbol{w}$ if $\langle \boldsymbol{v}, \boldsymbol{w}\rangle = 0$ in our inner product space.*

Lemma 6.2.4. *Given any $\boldsymbol{v}, \boldsymbol{w}$, let $P_{\boldsymbol{v}}(\boldsymbol{w})$ be $\frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}$. (We call $P_{\boldsymbol{v}}$ the **projection to $\boldsymbol{v}$**.) Then $\boldsymbol{v} \mathbin{/\!/} P_{\boldsymbol{v}}(\boldsymbol{w})$ and $\boldsymbol{v} \perp \boldsymbol{w} - P_{\boldsymbol{v}}(\boldsymbol{w})$.*

Proof. $\boldsymbol{v} \mathbin{/\!/} P_{\boldsymbol{v}}(\boldsymbol{w})$ is obvious. By definition $P_{\boldsymbol{v}}(\boldsymbol{w}) = \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}$ is a multiple of $\boldsymbol{v}$.

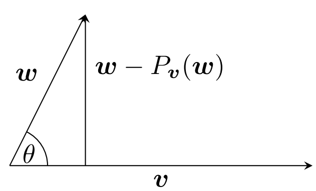
Now let us move on to perpendicularity.

$$\langle \boldsymbol{v}, \boldsymbol{w} - P_{\boldsymbol{v}}(\boldsymbol{w})\rangle = \langle \boldsymbol{v}, \boldsymbol{w} - \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}\rangle = \langle \boldsymbol{v}, \boldsymbol{w}\rangle - \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\langle \boldsymbol{v}, \boldsymbol{v}\rangle = \langle \boldsymbol{v}, \boldsymbol{w}\rangle - \langle \boldsymbol{v}, \boldsymbol{w}\rangle = 0.$$

□

Where did we get the formula $P_{\boldsymbol{v}}(\boldsymbol{w}) = \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}$? Well, we literally borrowed this from the case of $\mathbb{R}^n$, and we simply swapped the dot products with inner products. And (unsurprisingly) it works in exactly the same way.

Proof of Cauchy-Schwarz. Intuitively, the angle between $\boldsymbol{v}, \boldsymbol{w}$ can be found by looking at the right triangle made by $\boldsymbol{w}, P_{\boldsymbol{v}}(\boldsymbol{w}), \boldsymbol{w} - P_{\boldsymbol{v}}(\boldsymbol{w})$. We need all three edges to be "well-defined vectors", in which case I must have my well-defined angle. Then Cauchy-Schwarz would be true.



In this setting, "well-defined vectors" just mean they don't violate the positive definiteness. It turns out that we only need to check the last one. (This also the edges most related to the measurement of the angle.) We have

$$\begin{aligned}
0 \leq& \langle \boldsymbol{w} - P_{\boldsymbol{v}}(\boldsymbol{w}), \boldsymbol{w} - P_{\boldsymbol{v}}(\boldsymbol{w})\rangle \\
=& \langle \boldsymbol{w}, \boldsymbol{w}\rangle - 2\langle \boldsymbol{w}, \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}\rangle + \langle \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}, \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\boldsymbol{v}\rangle \\
=& \langle \boldsymbol{w}, \boldsymbol{w}\rangle - 2\frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}\langle \boldsymbol{w}, \boldsymbol{v}\rangle + (\frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle})^2\langle \boldsymbol{v}, \boldsymbol{v}\rangle \\
=& \langle \boldsymbol{w}, \boldsymbol{w}\rangle - \frac{\langle \boldsymbol{v}, \boldsymbol{w}\rangle^2}{\langle \boldsymbol{v}, \boldsymbol{v}\rangle}.
\end{aligned}$$

Rearrange terms and we are done. It is also easy to see that we have equality iff $\boldsymbol{w} = P_{\boldsymbol{v}}(\boldsymbol{w})$ iff $\boldsymbol{w} \mathbin{/\!/} \boldsymbol{v}$. □

This establishes angles in an inner product space. We say two vectors $\boldsymbol{v}, \boldsymbol{w}$ has ***angle*** $\arccos(\frac{\langle \boldsymbol{v},\boldsymbol{w}\rangle}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|})$. Here the range of arccos function is $[0, \pi]$. It is easy to check that two non-zero vectors have angle zero iff they are in the same direction, angle π iff they are in the opposite direction, and angel $\frac{\pi}{2}$ iff they are orthogonal. All is as expected.

If you play some more, you can get more things that you would expect. Here are some optional things you might want to try or skip.

Proposition 6.2.5 (Triangle inequality). $\|\boldsymbol{v} + \boldsymbol{w}\| \leq \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$

Proof. Square both sides, then compare both sides. You will see that this is just Cauchy-Schwarz in disguise.

$$\|\boldsymbol{v}+\boldsymbol{w}\|^2 - (\|\boldsymbol{v}\| + \|\boldsymbol{w}\|)^2 = \langle \boldsymbol{v}+\boldsymbol{w}, \boldsymbol{v}+\boldsymbol{w}\rangle - \langle \boldsymbol{v},\boldsymbol{v}\rangle - \langle \boldsymbol{w},\boldsymbol{w}\rangle - 2\|\boldsymbol{v}\|\|\boldsymbol{w}\| = 2\langle \boldsymbol{v},\boldsymbol{w}\rangle - 2\|\boldsymbol{v}\|\|\boldsymbol{w}\| \leq 0.$$

The last inequaltiy is due to Cauchy-Schwarz. ☐

Proposition 6.2.6 (Pythagorean Theorem (Gou Gu Ding Li)). $\|\boldsymbol{v} + \boldsymbol{w}\|^2 = \|\boldsymbol{v}\|^2 + \|\boldsymbol{w}\|^2$ *if and only if* $\boldsymbol{v} \perp \boldsymbol{w}$.

Proof. In general, we have

$$\|\boldsymbol{v}+\boldsymbol{w}\|^2 = \langle \boldsymbol{v}+\boldsymbol{w}, \boldsymbol{v}+\boldsymbol{w}\rangle = \langle \boldsymbol{v},\boldsymbol{v}\rangle + \langle \boldsymbol{w},\boldsymbol{w}\rangle + 2\langle \boldsymbol{v},\boldsymbol{w}\rangle.$$

Now orthogonality gives $\langle \boldsymbol{v},\boldsymbol{w}\rangle = 0$, so we are done. ☐

Here is a very important corollary of Pythagorean Theorem.

Proposition 6.2.7. *If $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ are mutually orthogonal (i.e. pairwise orthogonal) non-zero vectors in an inner product space V, then they are linearly independent.*

Proof. Suppose $\sum a_i\boldsymbol{v}_i = \boldsymbol{0}$. Then $0 = \|\sum a_i\boldsymbol{v}_i\|^2 = \sum\|a_i\boldsymbol{v}_i\|^2$ because the vectors involved are mutually orthogonal. So $0 = \sum a_i^2\|\boldsymbol{v}_i\|^2$. Now since all $\|\boldsymbol{v}_i\|^2$ are positive, we must have all $a_i = 0$.

Here is an alternative proof NOT using Pythagorean theorem. If $\sum a_i\boldsymbol{v}_i = \boldsymbol{0}$, we simply take inner product on both sides with $\boldsymbol{v}_j$. Then we would get $a_j\|\boldsymbol{v}_j\|^2 = 0$, hence $a_j = 0$. Since this is true for all j, we are done again. ☐

Remark 6.2.8. *Recall that for a collection of vectors, pairwise independent does NOT imply collective independence.*

However, pairwise orthogonal would indeed imply collective independence. This is amazingly useful.

THe next is not an inequality, but an equality.

Proposition 6.2.9 (Polarization identity). $\langle \boldsymbol{v},\boldsymbol{w}\rangle = \frac{1}{2}(\|\boldsymbol{v}+\boldsymbol{w}\|^2 - \|\boldsymbol{v}\|^2 - \|\boldsymbol{w}\|^2)$

Proof. Straightforward to verify. ☐

This identity has an important meaning: it means that any length structure would in fact induce an angle structure. This has some interesting ramifications for many study of geometry, and also the study of infinite dimensional spaces, but sadly we do not explore them in this class.

Nevertheless, applying this polariztion identity in the context of dot product, we have some very interesting results.

Lemma 6.2.10. *For $m \times n$ matrices A, B, if $\boldsymbol{v}^{\mathrm{T}}A\boldsymbol{w} = \boldsymbol{v}^{\mathrm{T}}B\boldsymbol{w}$ for all $\boldsymbol{v} \in \mathbb{R}^m, \boldsymbol{w} \in \mathbb{R}^n$, then $A = B$.*

Proof. By assumption we have $\boldsymbol{e}_i^{\mathrm{T}}A\boldsymbol{e}_j = \boldsymbol{e}_i^{\mathrm{T}}B\boldsymbol{e}_j$. So A, B have identical entries. ☐

Lemma 6.2.11. *For $n \times n$ symmetric matrices A, B, if $\boldsymbol{v}^{\mathrm{T}}A\boldsymbol{v} = \boldsymbol{v}^{\mathrm{T}}B\boldsymbol{v}$ for all $\boldsymbol{v} \in \mathbb{R}^n$, then $A = B$.*

Proof. Look at this adaptation of the famous polarization identity on inner products.

$$\boldsymbol{v}^{\mathrm{T}} A \boldsymbol{w} = \frac{1}{2}[(\boldsymbol{v}+\boldsymbol{w})^{\mathrm{T}} A (\boldsymbol{v}+\boldsymbol{w}) - \boldsymbol{v}^{\mathrm{T}} A \boldsymbol{v} - \boldsymbol{w}^{\mathrm{T}} A \boldsymbol{w}].$$

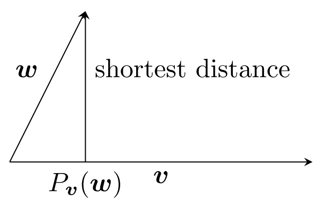
Hence if $\boldsymbol{v}^{\mathrm{T}} A \boldsymbol{v} = \boldsymbol{v}^{\mathrm{T}} B \boldsymbol{v}$ for all $\boldsymbol{v} \in \mathbb{R}^n$, then $\boldsymbol{v}^{\mathrm{T}} A \boldsymbol{w} = \boldsymbol{v}^{\mathrm{T}} B \boldsymbol{w}$ for all $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$. So $A = B$. ☐

As you have seen here, we have a new way to think about a symmetric matrix A. In the past, we have been treating matrices as linear maps. However, given a symmetric matrix A, you can also think of it as a ***bilinear map*** that sends a pair of vectors $\boldsymbol{v}, \boldsymbol{w}$ to the scalar $\boldsymbol{v}^{\mathrm{T}} A \boldsymbol{w}$. The polarization identity is true for all symmetric bilinear maps, and hence we have the results above.

The last result is not very useful in this class, but very useful in real life applications.

Proposition 6.2.12 (Projection minimizes distance)**.** *$\|\boldsymbol{w} - x\boldsymbol{v}\| \geq \|\boldsymbol{w} - P_{\boldsymbol{v}}(\boldsymbol{w})\|$ for all $x \in \mathbb{R}$, with equality if and only if $x\boldsymbol{v} = P_{\boldsymbol{v}}(\boldsymbol{w})$. In short, on the line spanned by $\boldsymbol{v}$, $P_{\boldsymbol{v}}(\boldsymbol{w})$ is the closest to $\boldsymbol{w}$.*

Proof. What the proposition is saying is very intuitive. On the line spanned by $\boldsymbol{v}$, the closest point to the point $\boldsymbol{w}$ is the projection of $\boldsymbol{w}$ to the line.



Consider all possible distance from $\boldsymbol{w}$ to the line spanned by $\boldsymbol{v}$. The square of this distance is, for some unknown x, $\|\boldsymbol{w} - x\boldsymbol{v}\|^2 = \langle \boldsymbol{w}, \boldsymbol{w} \rangle - 2x\langle \boldsymbol{w}, \boldsymbol{v} \rangle + x^2 \langle \boldsymbol{v}, \boldsymbol{v} \rangle$. Take derivative and find minimum, and we see that the minimum is reached exactly when $x = \frac{\langle \boldsymbol{w}, \boldsymbol{v} \rangle}{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$. So we are done.

In fact, plug this in and find minimum, and it would be $\langle \boldsymbol{w}, \boldsymbol{w} \rangle - \frac{\langle \boldsymbol{v}, \boldsymbol{w} \rangle^2}{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$. Now this minimum is still a square length, so it is non-negative. Then this would give a proof of Cauchy-Schwartz. (Most textbook uses this proof.) ☐

Let us see some example of abstract inner product space.

Example 6.2.13.

1. $\mathbb{R}^n$ with dot product. Obvious. We also call this inner product space the ***Euclidean space***.

2. $\mathbb{R}^n$ where we define $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} D \boldsymbol{w}$, and D is a fixed diagonal matrix with positive diagonal entries.

3. In general, if we define $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} A \boldsymbol{w}$ on $\mathbb{R}^n$ for some fixed matrix A, then symmetricity of $\langle -, - \rangle$ is just the same as the symmetricity of A. Bilinearity is immediate. We say a symmetric matrix is ***positive definite*** if $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} A \boldsymbol{w}$ is positive definite, and therefore an inner product. Some example is $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$. You can check that $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x^2 + 4xy + 5y^2 = (x+2y)^2 + y^2 \geq 0$, and with equality iff $x = y = 0$.

4. Consider $\mathbb{R}^4$ with $D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$. This matrix is NOT positive definite, because $\boldsymbol{e}_4^{\mathrm{T}} D \boldsymbol{e}_4 = -1 < 0$. So $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} D \boldsymbol{w}$ is NOT an inner product, merely a ***symmetric bilinear form***. This structure is still important though. For example, we see that $\| \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} \| = \sqrt{x^2 + y^2 + z^2 - t^2}$, and this is the length

structure used for special relativity. In general, spaces with bilinear forms of all kinds are useful and important. We only study inner product space because it is easier and intuitive.

☺

Let us see some more abstract example.

Example 6.2.14. 1. Let V be the space of continuous real functions defined on the interval $[0, 2\pi]$. Define $\langle f, g\rangle = \int_0^{2\pi} f(x)g(x)\,\mathrm{d}x$. Check and see that now V is an inner product space. Also check and see that $\sin(x), \cos(x), \sin(2x), \cos(2x), \dots$ are all orthogonal to each other. So we are giving V a structure where different frequencies are orthogonal to each other. This is the starting point of Fourier analysis.

2. Let V be the space of real random variables with finite variance. So an element $X \in V$ is basically a random number. Its expected value $\mathbb{E}(X)$ is the "average value", and its variance is $\mathrm{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2)$. This measures how much X vary from being constant. And we only consider random variables X where $\mathrm{Var}(X) < \infty$.

   For two random variables X, Y, we can (using polarization identity) define covariance as $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))]$. This being positive means X, Y are usually large together or small together, i.e., they are most likely to change in the same direction. This being negative means they are most likely to change in opposite directions. Hence the name "covariance". Now, V with covariance is NOT an inner product space. You can check that Cov is bilinear and symmetric, and $\mathrm{Cov}(X, X) \geq 0$, but $\mathrm{Cov}(X, X) = 0$ does NOT imply $X = 0$. Rather, X could simply be any constant. So close. We say that Cov is NOT positive definite, but ***positive semi-definite***. It is a ***semi-inner product***, and V with Cov is merely a ***semi-inner product space***.

3. Nonetheless, people define correlation between two random variables as $\rho = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$. What is this? This is the cosine of the angle! Note that $\rho = 1$ if and only if $X = kY + b$ for some $k > 0$, and $\rho = -1$ if and only if $X = kY + b$ for some $k < 0$. So basically, in a semi-inner product space, things behave pretty much the same as an inner product space, as long as you ignore vectors whose length is zero (constant variables, in this case).

4. Let W be the space of random variables whose value is a random number in the interval $[0, 1]$, with expected value zero. NOW W and covariance is a genuin inner product space. In statistics, when we get a bunch of data, often people would like to "center" the data, i.e., substract each data by the average. Why do this? Because a data is in V which is only a semi-inner product space, but after we center the data, it now lives in W which is a genuin inner product space.

5. This is just an observation here. Suppose two random variables are INDEPENDENT, then of course their covariance is zero, i.e., they are orthogonal in the (semi-)inner product structure. Be careful though, the converse is not true. Zero covariance variables might still be dependent, when the dependence is non-linear. Nevertheless, when you see data that are orthogonal, it does not hurt to pause and think if they are independent or not.

☺

Here is a more interesting example, in lieu of previous discussions on infinite dimenisional spaces in the last chapter.

Example 6.2.15. This is again an optional example. It aims to further explain why $\boldsymbol{e}_i$ in the sequence space fail to be a basis.

Let us consider space S of all real sequences. Consider $\boldsymbol{e}_i \in V$ where $\boldsymbol{e}_i$ is the sequence $(0, \dots, 0, 1, 0 \dots)$ where the 1 is at the i-th location. We already know that these vectors cannot span $\mathbb{R}^{\mathbb{N}}$. What can they actually span?

Algebraically, only finite linear combinations are allowed. And you can quickly see that finite linear combinations of these $\boldsymbol{e}_i$ must only have finitely many non-zero terms, i.e., they are actually finite sequences.

Let us call this S_{fin}. This is also an infinite dimensional vector space, and it is strictly smaller than the space S. In particular, sequences such as $(1, \frac{1}{2}, \frac{1}{3}, \dots)$ are NOT spanned by these $\boldsymbol{e}_i$, as they have infinitely many terms.

However, let us consider adding some geometry, by giving a length/angle structure to sequences. For two sequences $(a_0, a_1, \dots)$ and $(b_0, b_1, \dots)$, how would you do a "dot product"? Well, one obvious way to do this is to define $\langle (a_0, a_1, \dots), (b_0, b_1, \dots) \rangle = \sum_{n\in\mathbb{N}} a_n b_n$, simply the infinite analogue of a dot product.

However, consider the sequence $(1, 1, \dots)$. Then what is the length of this sequence? It is $\sqrt{\langle (1, 1, \dots), (1, 1, \dots) \rangle} = \sqrt{\infty} = \infty$. Hey! That is bad.

As a result, consider the space ℓ^2 of all sequences $(a_0, a_1, \dots)$, such that $\sum a_n^2 < \infty$. You can verify yourself that this is indeed a subspace of S, and we can make it an inner product space by using the dot product above.

Now consider $(1, \frac{1}{2}, \frac{1}{3}, \dots)$ again. It is not a linear combination of those $\boldsymbol{e}_i$. However, we can "approximate" this sequence using elements of S_{fin}, by looking as the sequences $(1, 0, \dots), (1, \frac{1}{2}, 0, \dots), \dots$. In particular, $(1, \frac{1}{2}, \frac{1}{3}, \dots)$ is a limit of linear combinations of these $\boldsymbol{e}_i$ under our new length structure! Indeed, we can compute and see that

$$\lim_{n\to\infty} \|(1, \frac{1}{2}, \frac{1}{3}, \dots) - \sum_{k=1}^{n} \frac{1}{k}\boldsymbol{e}_k\| = \lim_{n\to\infty} \sum_{k=n+1}^{\infty} \frac{1}{k^2} = 0.$$

So even though we cannot obtain all vectors using merely linear combinations, we can obtain all vectors in ℓ^2 using linear combinations and geometry (limits)!

You may also see that sequences such as $(3, 1, 4, 1, 5, 9, 2, 6, \dots)$ are still NOT obtainable using these $\boldsymbol{e}_i$. The above limit will fail to converge to zero for this sequence still. ℓ^2 is the "geometric span" of these $\boldsymbol{e}_i$, under our given length/angle structure.

You need to be careful though. Geometry depends on your definition of inner products. For example, let us now define a new inner product on sequence spaces $\langle (a_0, a_1, \dots), (b_0, b_1, \dots) \rangle = \sum_{n\in\mathbb{N}} \frac{1}{2^n} a_n b_n$. Now you can verify that sequences such as $(3, 1, 4, 1, 5, 9, 2, 6, \dots)$ are now also limits of linear combinations of these $\boldsymbol{e}_i$! The "geometric span" of these $\boldsymbol{e}_i$ will be much larger than before. On the other hand, if you define a new inner product as $\langle (a_0, a_1, \dots), (b_0, b_1, \dots) \rangle = \sum_{n\in\mathbb{N}} n a_n b_n$, then even the sequence $(1, \frac{1}{2}, \frac{1}{3}, \dots)$ will fail to be a limit of linear combinations of these $\boldsymbol{e}_i$.

The point is this: for infinite linear combinations, convergence depends on geometry, which usually depends on your definition of inner products. ☺

6.3 (Optional) Adjoint: Abstract "transpose"

To conclude this section, let us recall that we have another goal with the introduction of an inner product structure. I.e., transpose. How to relate transpose to inner products?

Proposition 6.3.1. *Given a linear map $L : V \to W$ between finite dimensional inner product spaces, there is a UNIQUE linear map $L^* : W \to V$ such that $\langle L\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, L^*\boldsymbol{w} \rangle$. (We call this the **adjoint** of A.)*

Before we prove this, think about what would happen in a Euclidean space ($\mathbb{R}^n$ with dot product). There we see that $\boldsymbol{v}^{\mathrm{T}} A^{\mathrm{T}} \boldsymbol{w} = (A\boldsymbol{v})^{\mathrm{T}} \boldsymbol{w}$. So L^* is just basically the transpose of L, but generalized into an abstract setting.

Let us first establish this "transpose" on vectors.

Lemma 6.3.2 (Vector Transpose)**.** *Given any linear map α from a finite dimensional inner product space V to $\mathbb{R}$, there is a UNIQUE vector $\boldsymbol{w}$ such that $\alpha(\boldsymbol{v}) = \langle \boldsymbol{w}, \boldsymbol{v} \rangle$ for all $\boldsymbol{v}$. We write $\alpha = \langle \boldsymbol{w}, - \rangle$ (or $\alpha = \langle \boldsymbol{w}|$, as physicists prefer).*

(Physisists also like to call $\langle -, - \rangle$ as a "braket". So they call this evaluation thing $\langle \boldsymbol{w}|$ a "bra", and a regular vector $\boldsymbol{v}$ a "ket". They also sometimes write $|\boldsymbol{v}\rangle$ for regular vectors.)

In short, transpose on a vector means "this guy is no longer a vector. Instead, it is now waiting to EAT another vector." This duality between linear evaluations and vectors are also sometimes called the (finite dimensional) Riesz representation theorem, i.e., any linear evaluation can be "represented" by a vector.

Proof of Lemma. Let V^* be the space of linear maps from V to $\mathbb{R}$. (Check yourself that this is indeed a vector space.) Define the map $\langle -| : V \to V^*$ by sending $\boldsymbol{w}$ to $\langle \boldsymbol{w}|$, the corresponding linear evaluations. Our goal is to show that this is bijective.

To see injectivity, suppose $\langle \boldsymbol{v}|$ is the zero map. Then $\langle \boldsymbol{v}, \boldsymbol{w}\rangle = 0$ for all $\boldsymbol{w} \in V$. In particular, $\langle \boldsymbol{v}, \boldsymbol{v}\rangle = 0$, so we have $\boldsymbol{v} = \mathbf{0}$. (Intuitively, only the zero vector can be perpendicular to all vectors.) This shows that $\langle -|$ is injective.

Now, note that if $n = \dim V$, then V^* is the space of linear maps from a n-dimensional space to a 1-dimensional space. By picking basis, this is the space of $1 \times n$ matrices. So $\dim V^* = n$. But $\operatorname{Ran}(\langle -|)$ is also n dimensional by rank-nullity, so $\langle -|$ is surjective. ☐

Proof of proposition. For any $\boldsymbol{w} \in W$, consider $\langle L(-), \boldsymbol{w}\rangle : V \to \mathbb{R}$. This is a linear evaluation! So in fact, it is $\langle \boldsymbol{x}|$ for a unique $\boldsymbol{x} \in V$. We define $L^*(\boldsymbol{w}) = \boldsymbol{x}$ in this mannor, so we now have a well-defined map $L^* : W \to V$. Also we immediately have $\langle L\boldsymbol{v}, \boldsymbol{w}\rangle = \langle \boldsymbol{v}, L^*\boldsymbol{w}\rangle$ by construction.

We only need to show that L^* is linear. Note that $\langle L^*(a\boldsymbol{u} + b\boldsymbol{w}), \boldsymbol{v}\rangle = \langle a\boldsymbol{u} + b\boldsymbol{w}, L\boldsymbol{v}\rangle = a\langle \boldsymbol{u}, L\boldsymbol{v}\rangle + b\langle \boldsymbol{w}, L\boldsymbol{v}\rangle = a\langle L^*\boldsymbol{u}, \boldsymbol{v}\rangle + b\langle L^*\boldsymbol{w}, \boldsymbol{v}\rangle = \langle aL^*\boldsymbol{u} + bL^*\boldsymbol{w}, \boldsymbol{v}\rangle$, so we see that as linear evaluations we have $\langle L^*(a\boldsymbol{u} + b\boldsymbol{w})| = \langle aL^*\boldsymbol{u} + bL^*\boldsymbol{w}|$. By injectivity of $\langle -|$, we have $L^*(a\boldsymbol{u} + b\boldsymbol{w}) = aL^*\boldsymbol{u} + bL^*\boldsymbol{w}$ as desired.

Finally, let us show that such L^* is unique. Suppose T is another linear map suc that $\langle L\boldsymbol{v}, \boldsymbol{w}\rangle = \langle \boldsymbol{v}, T\boldsymbol{w}\rangle$, then $\langle T\boldsymbol{w}, \boldsymbol{v}\rangle = \langle L^*\boldsymbol{w}, \boldsymbol{v}\rangle$ for all $\boldsymbol{v}, \boldsymbol{w}$. So $T\boldsymbol{w} = L^*\boldsymbol{w}$ again by injectivity of $\langle -|$. ☐

So we have a well-defined concept of "transpose" for maps between inner product spaces. Here are some very nice corollaries.

Corollary 6.3.3. $(A^*)^* = A$.

Proof. Note that $\langle \boldsymbol{v}, (A^*)^*\boldsymbol{w}\rangle = \langle A^*\boldsymbol{v}, \boldsymbol{w}\rangle = \langle \boldsymbol{v}, A\boldsymbol{w}\rangle$. So $A, (A^*)^*$ are both adjoint of A^*. By the uniqueness of adjoint, we have $(A^*)^* = A$. ☐

Corollary 6.3.4. *If $\langle \boldsymbol{v}, A\boldsymbol{w}\rangle = \langle \boldsymbol{v}, B\boldsymbol{w}\rangle$ for all $\boldsymbol{v}, \boldsymbol{w}$, then we have $A = B$.*

Proof. Again they are both adjoint of A^*. ☐

6.4 Gram Matrices and Cholesky decomposition

We make a bold statement here. We claim that, as a matter of fact, all finite dimensional inner product spaces are Euclidean. (After picking the right basis, your inner product is actually the same as the dot product.)

But first let us see a matrix phenomenon.

Example 6.4.1. Consider $\begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. After calculation, you shall see that this is $a_1b_1 + 2a_1b_2 + 3a_2b_1 + 4a_2b_2$. Hey, the (i,j)-entry of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is exactly the coefficient for a_ib_j. This is NOT a coincidence.

Think about this. When we multiply $\begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, a_1 will multiply entries in the first row, while a_2 will multiply entries in the second row. And when we multiply $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, b_1 will multiply entries in the first column, and b_2 will multiply entries in the second column. All in all, who will multiply the (i,j)-entry of the matrix? Well, it is in the i-th row and j-th column, so it is multiplied by a_i and b_j. This is why the (i,j)-entry of the matrix ends up as the coefficient for a_ib_j. ☺

Proposition 6.4.2. $\begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} c_{11} & \dots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{n1} & \dots & c_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i,j} c_{ij} a_i b_j.$

Proof.

$$\begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} C \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = (\sum_i a_i \boldsymbol{e}_i^{\mathrm{T}}) C (\sum_j b_j \boldsymbol{e}_j) = \sum_{i,j} a_i b_j (\boldsymbol{e}_i^{\mathrm{T}} C \boldsymbol{e}_j).$$

Here $\boldsymbol{e}_i^{\mathrm{T}} C \boldsymbol{e}_j$ is the (i,j) entry of C. □

Now let us again look at an inner product space. Given a (finite dimensional) inner product space V, it is first and foremost an abstract vector space. So after picking some basis, it is just $\mathbb{R}^n$. How would the inner product interact with this?

Say $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ is a basis. Then given any vector in V, say $\boldsymbol{a} = \sum a_i \boldsymbol{v}_i$ and $\boldsymbol{b} = \sum b_j \boldsymbol{a}_j$, their inner product is $\langle \boldsymbol{a}, \boldsymbol{b} \rangle = \langle \sum_i a_i \boldsymbol{v}_i, \sum_j b_j \boldsymbol{v}_i \rangle = \sum_{i,j} a_i b_j \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$. It is clear that, in fact, by knowing the values of $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$ for all i, j, we will know the whole inner product!

Let us go further. If you look closely, you might see that

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle = \sum_{i,j} a_i b_j \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix} \begin{bmatrix} \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle & \dots & \langle \boldsymbol{v}_1, \boldsymbol{v}_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \boldsymbol{v}_n, \boldsymbol{v}_1 \rangle & \dots & \langle \boldsymbol{v}_n, \boldsymbol{v}_n \rangle \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}.$$

So if we pick basis and turn all vectors into coordinates, then the inner product is represented by some symmetric matrix $G = \begin{bmatrix} \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle & \dots & \langle \boldsymbol{v}_1, \boldsymbol{v}_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \boldsymbol{v}_n, \boldsymbol{v}_1 \rangle & \dots & \langle \boldsymbol{v}_n, \boldsymbol{v}_n \rangle \end{bmatrix}$, called the ***Gram matrix*** for this basis. And given vectors with coordinates $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$, the inner product is just $\boldsymbol{v}^{\mathrm{T}} G \boldsymbol{w}$.

Definition 6.4.3. *For any finite dimensional inner product space and any basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$, the corresponding **Gram matrix** is the symmetric matrix whose (i,j) entry is $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$.*

We have caught a glimps of the important fact: any finite dimensional inner product space, after picking a basis, becomes simply $\mathbb{R}^n$ with exotic inner product $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} G \boldsymbol{w}$ for some nice symmetric matrix G. Or we can simply write $(\mathbb{R}^n, G)$ for short, indicating that the abstract vector space is $\mathbb{R}^n$, but instead of the dot product, we use the exotic inner product $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} G \boldsymbol{w}$.

Remark 6.4.4. *In case you did not notice this, the Euclidean space ($\mathbb{R}^n$ with dot product) is essentially the case when G is the identity matrix.*

Definition 6.4.5. *A symmetric matrix S is positive definite if $\boldsymbol{x}^{\mathrm{T}} S \boldsymbol{x} \geq 0$ with equality if and only if $\boldsymbol{x} = \boldsymbol{0}$.*

It is obvious that a symmetric matrix G is positive definite if and only if $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} G \boldsymbol{w}$ is indeed an inner product.

Proposition 6.4.6. *For any finite dimensional inner product space and any basis, the corresponding Gram matrix is positive-definite.*

Proof. Trivial by definition of positive definite matrices. □

In conclusion, we see that any finite dimensional inner product space, upon picking a basis, is the same as $(\mathbb{R}^n, G)$ for some positive-definite matrix G. The study of inner product spaces is now the study of positive-definite matrices.

Example 6.4.7. If A is invertible, then $AA^{\mathrm{T}}, A^{\mathrm{T}}A$ are both positive definite. They are obviously symmetric. Furthermore, if $\boldsymbol{x}^{\mathrm{T}}A^{\mathrm{T}}A\boldsymbol{x} = \|A\boldsymbol{x}\| \geq 0$ with equality if and only if $A\boldsymbol{x} = \mathbf{0}$ if and only if $\boldsymbol{x} = \mathbf{0}$ (A is invertible).

For a non-zero real number x, we know x^2 is positive. The fact that $AA^{\mathrm{T}}, A^{\mathrm{T}}A$ are both positive-definite are natural generalizations of this. (In contrast, A^2 is not guaranteed to be anything. Pick rotation matrix by $\frac{\pi}{2}$ on $\mathbb{R}^2$, and you shall see that $\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}^2 = -I$ is thoroughly negative.) ☺

Now, suppose we have a symmetric matrix S. How would I know if it is positive definite or not? Let us see an example first.

Example 6.4.8. Consider $A = \begin{bmatrix}1 & 2 & 3\\2 & 6 & 8\\3 & 8 & 14\end{bmatrix}$. I claim that A is positive definite, i.e., $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} \geq 0$ for all $\boldsymbol{x}$, with equality if and only if $\boldsymbol{x} = \mathbf{0}$.

Suppose $\boldsymbol{x} = \begin{bmatrix}x\\y\\z\end{bmatrix}$. Then first we want to show that $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} \geq 0$. We have $\begin{bmatrix}x & y & z\end{bmatrix} A \begin{bmatrix}x\\y\\z\end{bmatrix} = x^2 + 6y^2 + 14z^2 + 4xy + 6xz + 16yz$. To show that this is always non-negative, we may try to complete the squares. First let us group up everything that is related to x into a square, and we have $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = (x+2y+3z)^2 + 2y^2 + 5z^2 + 4yz$. Next, for the rest, let us group everything related to y and have $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = (x+2y+3z)^2 + 2(y+z)^2 + 3z^2$. Then the only thing left is z^2, which is by itself a square. So we see that $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} \geq 0$, with equality only if $x + 2y + 3z = y + z = z = 0$. And this condition can be solved as $x = y = z = 0$. So indeed, A is positive definite.

But let us rethink this process one more time. What are we doing when we complete the square? Let us think about this. Originally, we have variables x, y, z. Now I want to make a linear change of variables, where $x' = x + 2y + 3z$, and $y' = y + z$ and $z' = z$. Then I realize that $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = (x')^2 + 2(y')^2 + 3(z')^2 = \begin{bmatrix}x' & y' & z'\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\0 & 2 & 0\\0 & 0 & 3\end{bmatrix}\begin{bmatrix}x'\\y'\\z'\end{bmatrix}$. So we indeed have a sum of squares.

Observe that $\begin{bmatrix}x'\\y'\\z'\end{bmatrix} = \begin{bmatrix}1 & 2 & 3\\0 & 1 & 1\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$. In another words, we have $U = \begin{bmatrix}1 & 2 & 3\\0 & 1 & 1\\0 & 0 & 1\end{bmatrix}$ and $D = \begin{bmatrix}1 & 0 & 0\\0 & 2 & 0\\0 & 0 & 3\end{bmatrix}$, such that $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = (U\boldsymbol{x})^{\mathrm{T}}D(U\boldsymbol{x})$. So in fact $A = U^{\mathrm{T}}DU$. The process of completing squares on the polynomial $x^2 + 6y^2 + 14z^2 + 4xy + 6xz + 16yz$ corresponds to the process of writing A as $U^{\mathrm{T}}DU$. This is the LDU decomposition of A!

Imagine this. Suppose we want to complete the square. Then we would perform some change of variables where x, y, z becomes x', y', z' so that $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x}$ becomes a linear combination of squares of x', y', z' (with positive coefficients). This change of variable thing can be achieved as $\begin{bmatrix}x'\\y'\\z'\end{bmatrix} = E\begin{bmatrix}x\\y\\z\end{bmatrix}$ for some INVERTIBLE E. If completion of square can be achieved, then we have $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = (E\boldsymbol{x})^{\mathrm{T}}D(E\boldsymbol{x})$ for some diagonal matrix D with positive diagonal entries. In particular, we have $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = \boldsymbol{x}^{\mathrm{T}}(E^{\mathrm{T}}DE)\boldsymbol{x}$. So in fact $A = E^{\mathrm{T}}DE$.

But what is this E? Any invertible matrix can be thought of as a series of row operations or column operations. So $(E^{-1})^{\mathrm{T}}AE^{-1} = D$, i.e., we are reducing A to a diagonal matrices via "symmetric" row and column operations! So starting with A, if we add the first column to the second column, then we immediately add the first row to the second row. So on so forth, until we get a diagonal matrix. If you always do your reduction top to bottom and left to right, then you end up with $A = LDL^{\mathrm{T}}$.

Think about the following. We start with $A = \begin{bmatrix}1 & 2 & 3\\2 & 6 & 8\\3 & 8 & 14\end{bmatrix}$. The corresponding polynomial for $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x}$ is $x^2 + 6y^2 + 14z^2 + 4xy + 6xz + 16yz$. Now we substract the first row from the second row and the first column from the second column (note that the order does not matter, because $(BA)C = B(AC)$). Then we end up with $A' = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 5 \\ 3 & 5 & 14 \end{bmatrix}$. This corresponds to the transformation $x^2 + 6y^2 + 14z^2 + 4xy + 6xz + 16yz = (x+y)^2 + 3y^2 + 14z^2 + 2(x+y)y + 6(x+y)z + 10yz$. So it works! These "symmetric row+column operations" are exactly corresponding to the change of variables in the polynomial $\boldsymbol{x}^{\mathrm{T}} A\boldsymbol{x}$! ☺

First of all, there is a change of basis formula here.

Proposition 6.4.9. *On an abstract inner product space V, suppose the Gram matrix for the basis $\mathcal{B}$ is $G_{\mathcal{B}}$, and the Gram matrix under the basis $\mathcal{C}$ is $G_{\mathcal{C}}$, then we have $I_{\mathcal{B}\leftarrow\mathcal{C}}^{\mathrm{T}} G_{\mathcal{B}} I_{\mathcal{B}\leftarrow\mathcal{C}} = G_{\mathcal{C}}$.*

Proof. For any $\boldsymbol{v}, \boldsymbol{w} \in V$, we have

$$\langle \boldsymbol{v}, \boldsymbol{w}\rangle = \boldsymbol{v}_{\mathcal{B}}^{\mathrm{T}} G_{\mathcal{B}} \boldsymbol{w}_{\mathcal{B}} = \boldsymbol{v}_{\mathcal{C}}^{\mathrm{T}} I_{\mathcal{B}\leftarrow\mathcal{C}}^{\mathrm{T}} G_{\mathcal{B}} I_{\mathcal{B}\leftarrow\mathcal{C}} \boldsymbol{w}_{\mathcal{C}}.$$

We also have

$$\langle \boldsymbol{v}, \boldsymbol{w}\rangle = \boldsymbol{v}_{\mathcal{C}}^{\mathrm{T}} G_{\mathcal{C}} \boldsymbol{w}_{\mathcal{C}}.$$

Since $\boldsymbol{v}_{\mathcal{C}}, \boldsymbol{w}_{\mathcal{C}}$ are arbitrary, we have the result $I_{\mathcal{B}\leftarrow\mathcal{C}}^{\mathrm{T}} G_{\mathcal{B}} I_{\mathcal{B}\leftarrow\mathcal{C}} = G_{\mathcal{C}}$. □

Now let us example the LDU property of a positive definite matrix. To start, we at least want a positive definite matrix S to be invertible.

Proposition 6.4.10. *Positive definite matrices are invertible.*

Proof. If S is not invertible, say $\boldsymbol{x}$ is a nonzero vector in the kernel, then $\boldsymbol{x}^{\mathrm{T}} S\boldsymbol{x} = \boldsymbol{0}$, so S is not positive definite. □

But we also have another nice property of positive definiteness.

Proposition 6.4.11. *If A is positive definite, then all leading principal submatrices are positive definite.*

Proof. Let A_k be the k-th LPS. Then $\boldsymbol{x}^{\mathrm{T}} A_k \boldsymbol{x} = \begin{bmatrix} \boldsymbol{x}^{\mathrm{T}} & \boldsymbol{0}^{\mathrm{T}} \end{bmatrix} A \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{0} \end{bmatrix} \geq 0$, with equality if and only if $\begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{0} \end{bmatrix} = \boldsymbol{0}$, if and only if $\boldsymbol{x} = \boldsymbol{0}$. □

Remark 6.4.12. *In fact all "principal submatrices" are positive definite. We have not defined this term yet.... But for example, all diagonal entries are positive.*

Corollary 6.4.13. *If A is positive definite, then it has LU decomposition (in fact LDL^{T} decomposition, since A is symmetric).*

Corollary 6.4.14. *A is positive definite if and only if it is invertible with an LDL^{T} decomposition $A = LDL^{\mathrm{T}}$, and all diagonal entries of D are positive.*

Proof. ***⇒:***

We only need to show that D has positive diagonal entries. Note that the i-th diagonal entry is $\boldsymbol{e}_i^{\mathrm{T}} D\boldsymbol{e}_i = ((L^{\mathrm{T}})^{-1}\boldsymbol{e}_i)^{\mathrm{T}} A((L^{\mathrm{T}})^{-1}\boldsymbol{e}_i) \geq 0$, and equality cannot hold because $(L^{\mathrm{T}})^{-1}\boldsymbol{e}_i \neq \boldsymbol{0}$ (L is unit triangular and thus invertible).

⇐:

Note that obviously D is positive definite, because $\boldsymbol{x}^{\mathrm{T}} D\boldsymbol{x}$ is a positive linear combination of squares of all coordiates. Now $\boldsymbol{x}^{\mathrm{T}} A\boldsymbol{x} = (L\boldsymbol{x})^{\mathrm{T}} D(L\boldsymbol{x}) \geq 0$, and with equality if and only if $L\boldsymbol{x} = \boldsymbol{0}$. But L is unit triangular and thus invertible. So $L\boldsymbol{x} = \boldsymbol{0}$ if and only if $\boldsymbol{x} = \boldsymbol{0}$. □

So how to see if a matrix is positive definite? Well first it needs to be symmetric, which is easy to check. Then we do Gaussian elimination (which corresponds to an attempt to "complete the squares") to get the LDL$^{\mathrm{T}}$ decomposition. If we fail to get an LU decompositon (have to swap at some point), then A is NOT positive definite. If we get the LDL$^{\mathrm{T}}$ decomposition, then look at the diagonal entries of D. If they are all positive, then $\boldsymbol{x}^{\mathrm{T}} A\boldsymbol{x}$ as a polynomial is a positive sum of squares, so A is positive definite.

This has an interesting consequence:

Theorem 6.4.15 (Cholesky decomposition)**.** *If A is positive definite, then $A = LL^{\mathrm{T}}$ for an invertible lower triangular L. If we require the diagonal entries of L to be positive, then the decomposition is unique.*

Proof. We have $A = LDL^{\mathrm{T}}$. Since diagonal entries of D are positve, taking their square roots, we have diagonal D' whose square is D. Now $A = (LD')(LD')^{\mathrm{T}}$.

For uniqueness, note that $A = LDL^{\mathrm{T}}$ is unique. Suppose $A = BB^T$ for some invertible lower triangular matrix B with positive diagonal entries, then we can write $B = B'D'$ where B' is unit lower triangular, and D' is diagonal with positive diagonal entries. Then $A = B'D'D'(B')^{\mathrm{T}}$. By uniqueness of the LDU decomposition, we have $B' = L$ and $(D')^2 = D$. Since D' is diagonal with positive diagonal entries, each entry must be the unique square root of the corresponding entry of D. So D' is uniquely determined by D. So $B = LD'$ is also uniquely determined,. ☐

Remark 6.4.16. *We previously have seen that if A is invertible, then AA^{T} is positive definite. Now we see that if G is positive definite, then it is in fact AA^{T} for some invertible A. This is the generalization of the fact that all positive numbers are squares.*

Corollary 6.4.17. *Any finite dimensional inner product space is isomorphic to the Euclidean space.*

Proof. We know any finite dimensional space is isomorphic to $(\mathbb{R}^n, G)$ for some positive definite G. Then $G = LL^{\mathrm{T}}$ for an invertible upper triangular L.

Now we do the change of basis to get a Gram matrix $L^{-1}G(L^{-1})^{\mathrm{T}} = I$, and we are now isomorphic to $(\mathbb{R}^n, I)$. ☐

Well, as it turned out, finite dimensional inner product spaces are all the same as Euclidean spaces!

Remark 6.4.18. *It might now seem like a waste of time to introduce inner product spaces at all. They are all Euclidean!*

However, the point lies in two things. First of all, for infinite dimensional spaces, you have no choice but to do it abstractly. This involves super important spaces like function spaces and random variable spaces. And in many cases, your choice of inner product will endow the infinite dimenisonal space with different geometry.

Secondly, just like abstract vector spaces allow us to focus on invariant things under a change of basis, inner product space allows us to focus on invariant things under a change of orthonormal basis. We shall now study orthonormal basis in the next section.

6.5 Orthonormal Basis

Now, given an inner product space, we know it is isomorphic to a Euclidean space. This means by picking the right basis, the inner product will BE the dot product. Say $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ is this basis. Then if the identity matrix is our Gram matrix, we must conclude that $\|\boldsymbol{q}_i\| = 1$ for all i, and $\boldsymbol{q}_i \perp \boldsymbol{q}_j$ whenever $i \neq j$.

Definition 6.5.1. *An **orthogonal** basis (OGB for short) in an inner product space is $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ where all vectors are orthogonal to each other.*

*An **orthonormal** basis (ONB for short) is an orthogonal basis where, in addition, all vectors are unit vectors. (I.e., vectors with length one.)*

Let us take Euclidean space as an example. The standard basis is obviously a good orthonormal space. But sometimes we do need other orthonormal basis.

Example 6.5.2. Say I am pushing a box down a slope. Then we have pushing force, friction, normal force and gravity acting on the box. It is best NOT to choose the horizontal direction and vertical direction as orthonormal basis. Rather, it is better to choose the direction parallel to the slope and orthogonal to the slope as orthonormal basis. This way, three out of four forces lies on the coordinate axis, so computations are easier. ☺

A really nice property of ONB is that their coordinates are super easy to find.

Proposition 6.5.3. *Given an orthonormal basis $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ of V, then for any $\boldsymbol{v} \in V$, we have $\boldsymbol{v} = \sum \langle \boldsymbol{v}, \boldsymbol{q}_i \rangle \boldsymbol{q}_i$. In particular, the coordinate map for the basis is $(\boldsymbol{q}_1, \dots, \boldsymbol{q}_n)^{-1} = \begin{pmatrix} \langle \boldsymbol{q}_1 | \\ \vdots \\ \langle \boldsymbol{q}_n | \end{pmatrix}$ that sends each $\boldsymbol{v} \in V$ to $\begin{bmatrix} \langle \boldsymbol{q}_1, \boldsymbol{v} \rangle \\ \vdots \\ \langle \boldsymbol{q}_n, \boldsymbol{v} \rangle \end{bmatrix}$.*

Proof. Suppose $\boldsymbol{v} = \sum a_i \boldsymbol{q}_i$. Now for each i, we take inner product with $\boldsymbol{q}_i$ on both sides, we have $\langle \boldsymbol{v}, \boldsymbol{q}_i \rangle = a_i$ as desired. $\square$

Corollary 6.5.4. *If $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ is an OGB, then for any $\boldsymbol{v} \in V$, the coordinates of $\boldsymbol{v}$ for $\boldsymbol{v}_i$ is $\frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle}$.*

Proof. Let $\boldsymbol{q}_i = \frac{1}{\|\boldsymbol{v}_i\|} \boldsymbol{v}_i$, then $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ form an orthonormal basis. So $\boldsymbol{v}_i = \sum \langle \boldsymbol{v}, \boldsymbol{q}_i \rangle \boldsymbol{q}_i = \sum \frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i$. $\square$

Again, keep in minds that $\frac{\langle \boldsymbol{v}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i$ is simply the projection of $\boldsymbol{v}$ to the direction of $\boldsymbol{v}_i$. So essentially, we are saying that any vector $\boldsymbol{v}$ equals to the sum of its projections into orthogonal directions.

Remark 6.5.5. *Recall that physicists sometimes like to write $\langle \boldsymbol{v} |$ for the linear map that sends each input $\boldsymbol{w}$ to the output $\langle \boldsymbol{v}, \boldsymbol{w} \rangle$.*

Given ONB $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ of V, then for any $\boldsymbol{v} \in V$, we have

$$\boldsymbol{v} = \sum \langle \boldsymbol{v}, \boldsymbol{q}_i \rangle \boldsymbol{q}_i = \sum | \boldsymbol{q}_i \rangle \langle \boldsymbol{q}_i | (\boldsymbol{v}) = (\sum | \boldsymbol{q}_i \rangle \langle \boldsymbol{q}_i |)(\boldsymbol{v}).$$

Since this is true for all $\boldsymbol{v}$, we can conclude that $\sum | \boldsymbol{q}_i \rangle \langle \boldsymbol{q}_i | = I$ the identity map. This is actually not something new. Note that under dot product, $\langle \boldsymbol{q}_i |$ is simply $\boldsymbol{q}_i^{\mathrm{T}}$. Recall that ages ago, we have an example

$$\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \end{bmatrix} + \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \begin{bmatrix} -\frac{4}{5} & \frac{3}{5} \end{bmatrix} = I.$$

This is exactly an example of the formula $\sum | \boldsymbol{q}_i \rangle \langle \boldsymbol{q}_i | = I$. This represents the process of the decomposition of a vector $\boldsymbol{v}$ into orthogonal components.

Example 6.5.6. Consider the OGB $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$. (To make this an ONB, we would need to divide them all by 2.)

Given a vector, say $\boldsymbol{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. To find coordinates with respect to the new basis, we simply take the inner product (which in this case is simply the dot product). For example, $\frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ gives the coefficient for the basis vector $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. It appears that we simply get the average of $1, 2, 3, 4$.

In general, we can see that

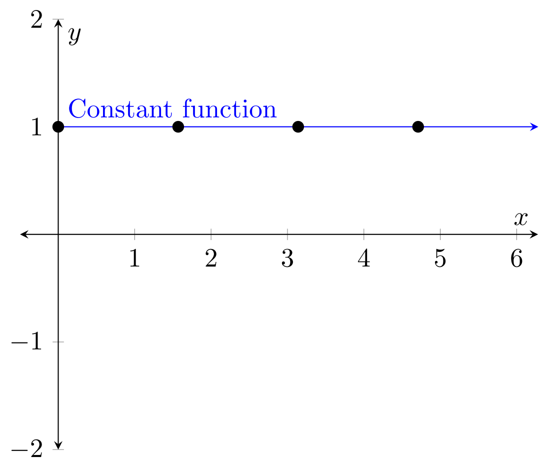
$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \frac{a+b+c+d}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{a-b+c-d}{4}\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \frac{a+b-c-d}{4}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} + \frac{a-b-c+d}{4}\begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

☺

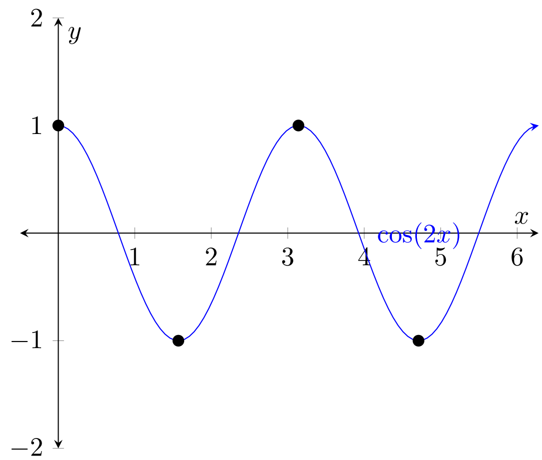
Example 6.5.7. Now is the time to introduce a famous orthogonal basis, the Haar wavelet basis. The basis $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$ is an example of Haar wavelet basis in dimension four. In general, this is an OGB where each vector only uses ± 1 as its coordinates.

Why the name "wavelet"? Well, we have the following correspondence.

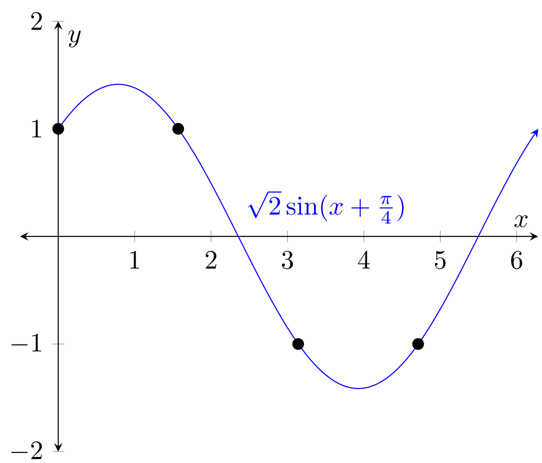
1. The vector $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ corresponds to the constant wave, i.e., the function $f(x) = 1$ for all x.



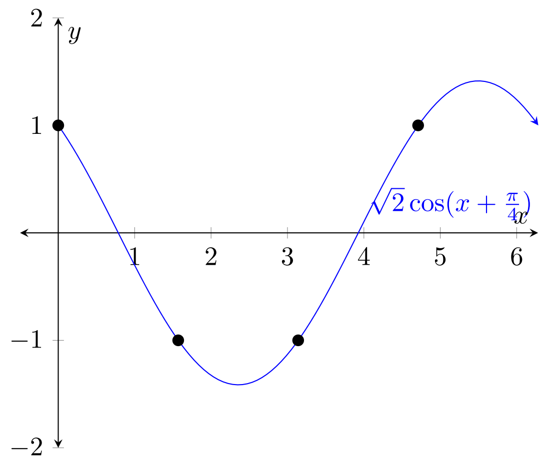
2. The vector $\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$ corresponds to the wave $\cos(2x)$, and we simply sample its value at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.



3. The vector $\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$ corresponds to the wave $\sqrt{2}\sin(x + \frac{\pi}{4})$, and again we simply sample its value at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.



4. The vector $\begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$ corresponds to the wave $\sqrt{2}\cos(x + \frac{\pi}{4})$, and again we simply sample its value at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.



As you can see, they are essentially discrete version of waves, where we have some extra constants to make sure that coordinates are ± 1 for these vectors.

Haar wavelet basis is widely used for jpeg image processing. Here we merely take $\mathbb{R}^{16}$ as an example, but keep in mind that in practice it is usually a much larger space.

Suppose we have 16 pixels in greyscale, so we can represent each pixel by a number, the "grey-ness". Then we could use the vector $\begin{bmatrix} a_1 \\ \vdots \\ a_{16} \end{bmatrix}$ to represent the following picture.

a_1	a_2	a_5	a_6
a_3	a_4	a_7	a_8
a_9	a_{10}	a_{13}	a_{14}
a_{11}	a_{12}	a_{15}	a_{16}

Now suppose the picture takes up too much space! So we want to store less numbers, and maybe the picture is a little blurry, but we can still recognize the general shapes. Say we "blend" two adjacent pixels by takining their average. Let $b_i = \frac{a_{2i-1}+a_{2i}}{2}$, then we are replacing the picture with $\begin{bmatrix} a_1 \\ \vdots \\ a_{16} \end{bmatrix}$ with $\begin{bmatrix} b_1 \\ b_1 \\ b_2 \\ b_2 \\ \vdots \\ b_8 \\ b_8 \end{bmatrix}$. The picture now looks like

b_1	b_1	b_3	b_3
b_2	b_2	b_4	b_4
b_5	b_5	b_7	b_7
b_6	b_6	b_8	b_8

You can imagine that our picture is a little blurry now, but when there are many many pixels, then averaging each pair would have neglegible effect on your picture in general. But what if we still want to compress the size of the data? Maybe I can again take the average of adjacent b_i's. Let $c_i = \frac{b_{2i-1}+b_{2i}}{2}$, then we are replacing the picture with $\begin{bmatrix} a_1 \\ \vdots \\ a_{16} \end{bmatrix}$ with $\begin{bmatrix} c_1 \\ c_1 \\ c_1 \\ c_1 \\ c_2 \\ \vdots \\ c_4 \end{bmatrix}$. The picture now looks like

c_1	c_1	c_2	c_2
c_1	c_1	c_2	c_2
c_3	c_3	c_4	c_4
c_3	c_3	c_4	c_4

Now the picutre is a lot more blurry, but we saved a lot of data storage. We only need to store four numbers. If we have many many pixels, you can continue this process for ever to get more and more blurry pictures with less and less storage need.

With this compression process in mind, let us look at the Haar wavelet basis. In the case of $\mathbb{R}^4$, they are columns $\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3, \boldsymbol{c}_4$ of the matrix $H_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$.

Now given a picture with 4 pixels, say

a_1	a_2
a_3	a_4

Then we have decomposition

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \frac{a_1+a_2+a_3+a_4}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{a_1+a_2-a_3-a_4}{4}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} + \frac{a_1-a_2+a_3-a_4}{4}\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \frac{a_1-a_2-a_3+a_4}{4}\begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

Look at the first term $\frac{a_1+a_2+a_3+a_4}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. This is exactly the picture when I averaged ALL pixels! Now we only need to store one number, and the picture is too blurry to see anything.

Now look at the first two terms.

$$\frac{a_1+a_2+a_3+a_4}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{a_1+a_2-a_3-a_4}{4}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{a_1+a_2}{2} \\ \frac{a_1+a_2}{2} \\ \frac{a_3+a_4}{2} \\ \frac{a_3+a_4}{2} \end{bmatrix}.$$

This is exactly the picture where I averaged the top two pixels and the bottom two pixels!

Now look at the first term and the third term. We have

$$\frac{a_1+a_2+a_3+a_4}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{a_1-a_2+a_3-a_4}{4}\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{a_1+a_3}{2} \\ \frac{a_2+a_4}{2} \\ \frac{a_1+a_3}{2} \\ \frac{a_2+a_4}{2} \end{bmatrix}.$$

This is exactly the picture where I averaged the left two pixels and the right two pixels!

As you can see, given a picture, we can first write it under the Haar wavelet basis. Then "blur adjacent pixels" can be done simply by dropping coordinates. The more coordinate you drop, the more blurry your picture can be.

So how can one construct this Haar wavelet basis in general? One can iterate the process of $H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}$. Feel free to check that H_{2n} would still be symmetric and has orthogonal columns. In particular, in the 16 pixel case, we have the Haar wavelet basis:

$$\begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\
1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\
1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\
1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\
1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\
1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\
1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\
1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\
1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 \\
1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\
1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 \\
1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 \\
1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1
\end{bmatrix}.$$

Note that the Haar wavelet basis is NOT ONB, merely OGB. To make it orthonormal, one needs to use columns of $\frac{1}{\sqrt{n}}H_n$ instead. ☺

Here we see an interesting matrix H_n whose columns are orthogonal and unit vectors. In general, the following types of matrices are very important.

6.6 Orthogonal Matrices

From now on, let us focus on the Euclidean space!

Let us do an alternative characterization of these matrices. If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ form an ONB, then the inverse of $(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$ is the coordinate map, which is $\begin{bmatrix} \langle \boldsymbol{v}_1| \\ \vdots \\ \langle \boldsymbol{v}_n| \end{bmatrix}$ as we have seen. Under dot product, the "bra" simply means transpose. So we have an interesting discovery. In $\mathbb{R}^n$ with the dot product, if the columns of $U = \begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n \end{bmatrix}$ form an ONB, then $U^{-1} = \begin{bmatrix} \boldsymbol{v}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{v}_n^{\mathrm{T}} \end{bmatrix} = U^{\mathrm{T}}$.

Definition 6.6.1. *We say an $n \times n$ invertible matrix A is an **orthogonal matrix** if $A^{-1} = A^{\mathrm{T}}$.*

Now, what does it mean to have $A^{\mathrm{T}}A = I$? Suppose $A = \begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n \end{bmatrix}$, then the (i, j) entry of $A^{\mathrm{T}}A$ would be $\boldsymbol{v}_i^{\mathrm{T}}\boldsymbol{v}_j$. So $A^{\mathrm{T}}A = I$ implies that columns of A are mutually orthogonal normal vectors. Similarly, $AA^{\mathrm{T}} = I$ implies that rows of A are mutually orthogonal normal vectors.

Proposition 6.6.2. *For an $n \times n$ invertible matrix A, TFAE:*

1. *A is orthogonal.*
2. *Rows of A form an orthonormal basis to the Euclidean space $\mathbb{R}^n$.*
3. *Columns of A form an orthonormal basis to the Euclidean space $\mathbb{R}^n$.*
4. *$\langle A\boldsymbol{v}, A\boldsymbol{w}\rangle = \langle \boldsymbol{v}, \boldsymbol{w}\rangle$ for all $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$. (Note that the inner product here is just dot product.)*
5. *$\|A\boldsymbol{v}\| = \|\boldsymbol{v}\|$ for all $\boldsymbol{v} \in \mathbb{R}^n$.*

Proof. We are given that A is square. So $A^{-1} = A^{\mathrm{T}}$ iff $AA^{\mathrm{T}} = I$ iff $A^{\mathrm{T}}A = I$. By writing A in columns, we see that $A^{\mathrm{T}}A = I$ iff columns of A form ONB. Similarly, by writing A in rows, we see that $AA^{\mathrm{T}} = I$ iff rows of A form ONB. So it is already obvious tha the first three statements are equivalent.

Now, if A is orthogonal, then $\langle A\boldsymbol{v}, A\boldsymbol{w}\rangle = \boldsymbol{v}^{\mathrm{T}}A^{\mathrm{T}}A\boldsymbol{w} = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = \langle \boldsymbol{v}, \boldsymbol{w}\rangle$. So the first one implies the forth one. Conversely, if $\langle A\boldsymbol{v}, A\boldsymbol{w}\rangle = \langle \boldsymbol{v}, \boldsymbol{w}\rangle$, then $\boldsymbol{v}^{\mathrm{T}}(A^{\mathrm{T}}A)\boldsymbol{w} = \boldsymbol{v}^{\mathrm{T}}I\boldsymbol{w}$ for all $\boldsymbol{v}, \boldsymbol{w}$. This means $A^{\mathrm{T}}A = I$.

Finally, the last statement is also equivalent because due to polarization identity, length determines angle. □

In general, orthogonal matrices would map the standard basis to some orthonormal basis. In particular, above proposition shows that orthogonal matrices as a linear map is exactly a "rigid motion" that preserves length and angles. (This means we have some rotation+reflection going on.) In particular, we have the following:

Lemma 6.6.3. *If A, B are orthogonal, then AB, A^{-1} are orthogonal.*

Proof. Obviously $A^{-1}(A^{-1})^{\mathrm{T}} = A^{-1}(A^{\mathrm{T}})^{-1} = (A^{\mathrm{T}}A)^{-1} = I$. And we have $(AB)(AB)^{\mathrm{T}} = A(BB^{\mathrm{T}})A^{\mathrm{T}} = AA^{\mathrm{T}} = I$.

There are alternative proofs. For example, $\|AB\boldsymbol{v}\| = \|B\boldsymbol{v}\| = \|\boldsymbol{v}\|$ for all $\boldsymbol{v}$ because A, B are orthogonal. But then this means AB is orthogonal. □

Another interpretation of an orthogonal matrix is the following: they corresponds to change of basis between orthonormal basis.

Proposition 6.6.4. *In an inner product space V, the change of coordinate matrix from an orthonormal basis to another orthonormal basis is an orthogonal matrix. Conversely, given any orthonormal basis $\mathcal{B}$, then $\mathcal{B}U$ is an orthonormal basis if U is orthogonal.*

Proof. WLOG suppose that V is the Euclidean space. Then any orthonormal basis forms an orthogonal matrix. Say the two orthonormal basis are columns of A, B. Note that both A, B are orthogonal and therefore invertible. Then because $A(A^{-1}B) = B$, we see that the basis transition matrix from A to B is $A^{-1}B$, and therefore the change of coordinate matrix is just $B^{-1}A$, which is still orthogonal.

For the "converse" part, say an orthonormal basis are columns of A, then for orthogonal U, AU is still orthogonal. So the columns of AU still form an orthonormal basis. □

Here is a preview of a future important theorem, that we do not proof at the moment.

Theorem 6.6.5 (Singular Value Decomposition)**.** *For any matrix A, we can find orthogonal matrices U, V such that $UAV = \begin{bmatrix} \Sigma & O \\ O & O \end{bmatrix}$ where Σ is diagonal.*

This is a super upgrade from rank normal form, which says there must be some bases to make our map pretty. Now we see that, in fact, there must be some orthogonal bases to make our map pretty. We can make our map pretty in a way that is also compatible with the inner product structure of the domain and codomain.

Remark 6.6.6. *In practice, people only perform change of basis using orthogonal matrices, because such change of basis process would preserve the length of error term.*

For example, suppose we have a vector $\boldsymbol{v}$. Under some basis, we measured it and get its coordinate $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. However, measurements are never precise. So technically, we have $\boldsymbol{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \boldsymbol{e}$ for some vector $\boldsymbol{e}$ recording the tiny error.

Suppose now I want to use a new basis. Let A be the change of coordinate matrix, and say $\begin{bmatrix}1\\2\\3\end{bmatrix}$ is changed into $\begin{bmatrix}4\\5\\6\end{bmatrix}$. However, due to the error term, the ACTUAL coordinates of $\boldsymbol{v}$ in the new basis should be $A(\begin{bmatrix}1\\2\\3\end{bmatrix}+\boldsymbol{e}) = \begin{bmatrix}4\\5\\6\end{bmatrix}+A\boldsymbol{e}$.

If A is an orthogonal matrix, then $\|A\boldsymbol{e}\| = \|\boldsymbol{e}\|$. In particular, if the initial error is tiny, then after the change of coordinates, the resulting error is still tiny.

This is not true for non-orthogonal matrices. For example, if $A = \begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}$ and the error term is $\boldsymbol{e} = \begin{bmatrix}0\\0.1\end{bmatrix}$, then $A\boldsymbol{e} = \begin{bmatrix}0.2\\0.1\end{bmatrix}$. The size of error is now $\|A\boldsymbol{e}\| = \frac{\sqrt{5}}{10}$, whereas the original error has size $\frac{1}{10}$. The size of error is more than doubled.

For this reason, people NEVER perform non-orthonormal change of basis. We do not want the error term to be magnified.

Let us conclude this section by some more examples of orthogonal matrices.

Example 6.6.7. Rotation matrices $R_\theta = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$ are orthogonal matrices, because it preserves length.

What about 3D rotations? Any rotation in $\mathbb{R}^3$ must be rotating around a line. Suppose we pick an orthonormal basis $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ such that $\boldsymbol{q}_1$ is the axis of rotation, and we rotate around this by rotating the plane $\operatorname{span}(\boldsymbol{q}_2, \boldsymbol{q}_3)$ via R_θ. Then under this orthonormal basis, our rotation looks like $\begin{bmatrix}1 & \mathbf{0}^{\mathrm{T}}\\ \mathbf{0} & R_\theta\end{bmatrix}$.

So, what does this looks like under the original basis, i.e., the standard basis? Since change of coordinate maps are simply arbitrary orthogonal matrices, we are looking at $U\begin{bmatrix}1 & \mathbf{0}^{\mathrm{T}}\\ \mathbf{0} & R_\theta\end{bmatrix}U^{-1}$ for some arbitrary orthogonal matrix U, where $U = \begin{bmatrix}\boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3\end{bmatrix}$. Here we first use $U^{-1} = U^{\mathrm{T}} = \begin{bmatrix}\boldsymbol{q}_1^{\mathrm{T}}\\ \boldsymbol{q}_2^{\mathrm{T}}\\ \boldsymbol{q}_3^{\mathrm{T}}\end{bmatrix}$ to change coordinates from the standard basis to our new orthonormal basis, then use the matrix under our new basis, and finally use U to change the coordinates back to the standard basis.

All in all, we see that 3D rotations are ALL $U\begin{bmatrix}1 & \mathbf{0}^{\mathrm{T}}\\ \mathbf{0} & R_\theta\end{bmatrix}U^{-1}$ for some orthogonal matrix U.

We are going to see structures like BAB^{-1} a lot. For linear transformations where the domain is the same as the codomain, and the domain and codomain are required to change basis simultaneously, then we would always go from A to BAB^{-1} for some invertible B, i.e., the change of coordinate matrix. ☺

So we have figured out all 2D and 3D rotation matrices. What about higher dimensions?

Definition 6.6.8. *A* ***Givens rotation*** *is a matrix G_{ij}^{θ} whose $(i,i),(i,j),(j,i),(j,j)$ entries form R_θ, and the rest looks exactly like the identity matrix.*

Obviously the Givens rotations are rotations. They rotate the $x_i x_j$-plane while fixing all other coordinate axis. As a result, their compositions are all rotations. It turns out that this is it. We don't have enough time to do this in class though, so we shall leave it at that.

Example 6.6.9. What about reflections, which always preserve the length as well? For example, we know that $\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ is an orthogonal matrix and in fact a reflection about the line $x = y$ on $\mathbb{R}^2$.

Suppose we want to reflect about a hyperplane with unit normal vector $\boldsymbol{q}_1$. Then we pick any orthonormal basis $\boldsymbol{q}_2, \ldots, \boldsymbol{q}_n$ for the hyper plane, and then $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ would form an orthonormal basis for $\mathbb{R}^n$. Under this basis, the reflection would send $\boldsymbol{q}_1$ to $-\boldsymbol{q}_1$, and preserve the rest. So the matrix under this basis is $\begin{bmatrix} -1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$.

As a result, a generic reflection must be $U \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} U^{-1}$ for any orthogonal matrix U. Note that the matrix in the middle is in fact $I - 2\boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}}$. So a reflection must be $U(I-2\boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}})U^{-1} = U(I-2\boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}})U^{\mathrm{T}} = UU^{\mathrm{T}} - 2U\boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}}U^{\mathrm{T}} = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$, where $\boldsymbol{u}$ is the first column of U, i.e., $\boldsymbol{q}_1$, the unit normla vector to our hyperplane.

In fact we have already seen this before. The reflection to a hyper plane is exactly $I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ where $\boldsymbol{u}$ is the unit normal vector. ☺

Definition 6.6.10. *A **Householder transformation** is $I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$ for some unit vector $\boldsymbol{n}$. Or abstractly on an inner product space, it is $I - 2|\boldsymbol{n}\rangle\langle\boldsymbol{n}|$ for some unit vector $\boldsymbol{n}$.*

It is easy to verify that this is indeed an orthogonal matrix. In fact its inverse and transpose are both itself.

Here is an interesting result that provide some nice intuitions. The proof is optional and not required.

Theorem 6.6.11 (Geometric meaning of an orthogonal matrix)**.** *An orthogonal matrix is either a product of Givens rotations (and thus a rotation it self), or is the product of a series of Givens rotations and a Householder transformation (and thus a reflection then rotation).*

6.7 (Optional) Geometrix meaning of an orthogonal matrix

To see this proof, first we shall need a few lemmas. In the following, by rotation we shall always means a product of a series of Givens rotations.

Lemma 6.7.1. *If R is a rotation, then R^{-1}, $\begin{bmatrix} R & O \\ O & I \end{bmatrix}$, $\begin{bmatrix} I & O \\ O & R \end{bmatrix}$ are still rotations.*

Proof. Just verify for all Givens rotations. Then use the fact that inverse matrices of a product is the product with the inverses with reversed order. And for a product of block diagonal matrices, we can can simply compute the product on each diagonal block. □

Lemma 6.7.2. *Given any two vectors $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$ of the same length and $n \geq 2$, then there is a rotation R such that $R\boldsymbol{v} = \boldsymbol{w}$.*

Proof. We proceed by induction. When $n = 2$, this is obvious.

For generic n, suppose $\boldsymbol{v} = \begin{bmatrix} v_1 \\ \boldsymbol{v}' \end{bmatrix}$. Then by induction hypothesis, I can find R_1 a rotation on $\mathbb{R}^{n-1}$ such that $R_1\boldsymbol{v}' = \begin{bmatrix} \|\boldsymbol{v}'\| \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. Then we see that $\begin{bmatrix} 1 & \boldsymbol{0}^{\mathrm{T}} \\ \boldsymbol{0} & R_1 \end{bmatrix} \boldsymbol{v} = \begin{bmatrix} v_1 \\ \|\boldsymbol{v}'\| \\ 0 \\ \vdots \\ 0 \end{bmatrix}$.

Now I find a rotation R_2 such that $R_2 \begin{bmatrix} v_1 \\ \|\boldsymbol{v}'\| \end{bmatrix} = \begin{bmatrix} \|\boldsymbol{v}\| \\ 0 \end{bmatrix}$. Then $\begin{bmatrix} R_2 & O \\ O & I \end{bmatrix} \begin{bmatrix} v_1 \\ \|\boldsymbol{v}'\| \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \|\boldsymbol{v}\|\boldsymbol{e}_1$.

So composing the two, we have found a rotation R such that $R\boldsymbol{v} = \|\boldsymbol{v}\|\boldsymbol{e}_1$. Similarly, we can find a rotation R' such that $R'\boldsymbol{w} = \|\boldsymbol{w}\|\boldsymbol{e}_1$. Since the two vectors have the same length, we see that $R\boldsymbol{v} = R'\boldsymbol{w}$, so $(R')^{-1}R$ is the desired rotation. □

Lemma 6.7.3. *If H is a Householder transformation, then* $\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & H \end{bmatrix}$ *is a householder transformation.*

Proof. Suppose $H = I_n - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ for some unit vector H. We now look at $\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & H \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & I_n - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} \end{bmatrix} = I_{n+1} - \begin{bmatrix} 0 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} \end{bmatrix} = I_{n+1} - 2 \begin{bmatrix} 0 \\ \boldsymbol{u} \end{bmatrix} \begin{bmatrix} 0 \\ \boldsymbol{u} \end{bmatrix}^{\mathrm{T}}$. And it is easy to verify that $\begin{bmatrix} 0 \\ \boldsymbol{u} \end{bmatrix}$ is still a unit vector. □

Now we can proceed to prove the theorem. It is of course a proof by induction. Let me write the theorem again for clarity.

Theorem 6.7.4 (Geometric meaning of an orthogonal matrix)**.** *An orthogonal matrix is either a product of Givens rotations (and thus a rotation it self), or is the product of a series of Givens rotations and a Householder transformation (and thus a reflection then rotation).*

Proof. When $n = 1$ or $n = 2$ this is trivial. We proceed by induction.

For generic n, consider any orthogonal matrix Q, and let its first column be $\boldsymbol{q}$. This is a unit vector. So we can find a rotation R such that $R\boldsymbol{q} = \boldsymbol{e}_1$. So $RQ = \begin{bmatrix} 1 & \star \\ \mathbf{0} & \star \end{bmatrix}$.

Since RQ is still an orthogonal matrix, its columns must still be orthogonal to each other. In particular, the upper right block must actually be zero. So we have $RQ = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & Q' \end{bmatrix}$ for some matrix Q'. Further more, again because RQ is orthogonal, $\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & Q' \end{bmatrix}^{-1} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & Q' \end{bmatrix}^{\mathrm{T}}$, and therefore we see that $\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & (Q')^{-1} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & (Q')^{\mathrm{T}} \end{bmatrix}$. So we see that Q' is orthognal and with smaller dimension.

So by induction, Q' is either a rotation (in which case Q is a rotation), or a rotation times a Householder transformation (in which case Q is a rotation times a Householder transformation). □

Here is a nice generalization to the affine case. We can now understand ALL distance preserving transformations on $\mathbb{R}^n$, even the non-linear ones. We define ***distance*** between two vectors $\boldsymbol{v}, \boldsymbol{w}$ to simply be $\|\boldsymbol{v} - \boldsymbol{w}\|$.

Lemma 6.7.5 (Distance ratio and convex linear combinations)**.** *In $\mathbb{R}^n$, suppose $\|\boldsymbol{v} - \boldsymbol{w}\| = r$. Then for any $0 \leq t \leq 1$, the point $t\boldsymbol{v} + (1-t)\boldsymbol{w}$ has distance $(1-t)r$ to the point $\boldsymbol{v}$, and has distance tr to the point $\boldsymbol{w}$.*

Conversely, if $\boldsymbol{x}$ has distance $(1-t)r$ to the point $\boldsymbol{v}$ and has distance tr to the point $\boldsymbol{w}$ for some $0 \leq t \leq 1$, then $\boldsymbol{x} = t\boldsymbol{v} + (1-t)\boldsymbol{w}$.

Proof. Note that $\boldsymbol{v} - (t\boldsymbol{v} + (1-t)\boldsymbol{w}) = (1-t)(\boldsymbol{v} - \boldsymbol{w})$, and $\boldsymbol{w} - (t\boldsymbol{v} + (1-t)\boldsymbol{w}) = t(\boldsymbol{w} - \boldsymbol{v})$. Hence the distance ratio is exactly $(1-t)$ to t. Given total distance r between $\boldsymbol{v}, \boldsymbol{w}$, the rest is obvious.

For the second portion, suppose $\boldsymbol{x}$ has distance $(1-t)r$ to the point $\boldsymbol{v}$ and has distance tr to the point $\boldsymbol{w}$ for some $0 \leq t \leq 1$. Intuitively, this means that $\boldsymbol{x}$ is in the ball around $\boldsymbol{v}$ with radius tr, and also in the ball around $\boldsymbol{w}$ with radius $(1-t)r$. But since $\boldsymbol{v}, \boldsymbol{w}$ have a distance of $tr + (1-t)r$, the two balls have unique intersection, and then we are done.

Let us do this more rigorously. The triangle inequality $\|\boldsymbol{v}-\boldsymbol{w}\| \leq \|\boldsymbol{v}-\boldsymbol{x}\| + \|\boldsymbol{w}-\boldsymbol{x}\|$ achieved equality. This means the corresponding Cauchy-Schwarz inequality (by squaring both sides and simplify) between $\boldsymbol{v}-\boldsymbol{x}$ and $\boldsymbol{w}-\boldsymbol{x}$ achieves equality. Hence these two are parallel. In particular, $\boldsymbol{x}$ lies on the line through the two points $\boldsymbol{v}, \boldsymbol{w}$. Since we have distance requirement with $tr, (1-t)r \leq r$, $\boldsymbol{x}$ must lie on the line segment between $\boldsymbol{v}, \boldsymbol{w}$.

If either $\boldsymbol{v}-\boldsymbol{x}$ or $\boldsymbol{w}-\boldsymbol{x}$ is the zero vector, then the statement is trivial. Suppose they are both non-zero. Let $-k(\boldsymbol{v}-\boldsymbol{x}) = \boldsymbol{w}-\boldsymbol{x}$ for some $k \geq 0$. Rearrange and we have $\boldsymbol{x} = \frac{k}{k+1}\boldsymbol{v} + \frac{1}{k+1}\boldsymbol{w}$. By the first portion of our lemma, we see that the distance between $\boldsymbol{x}$ and $\boldsymbol{v}$ should be $\frac{1}{k+1}r$. Hence $1-t = \frac{1}{k+1}$ and $t = \frac{k}{k+1}$. So we see that $\boldsymbol{x} = t\boldsymbol{v} + (1-t)\boldsymbol{w}$. ☐

Theorem 6.7.6 (Classification of isometries on $\mathbb{R}^n$)**.** *If $f : \mathbb{R}^n \to \mathbb{R}^n$ is any map that preserves distance, i.e., $\|\boldsymbol{v}-\boldsymbol{w}\| = \|f(\boldsymbol{v}) - f(\boldsymbol{w})\|$, then we must have $f(\boldsymbol{v}) = A\boldsymbol{v} + \boldsymbol{b}$ for some constant vector $\boldsymbol{b}$ and orthogonal matrix A.*

Proof. For any points $\boldsymbol{v}, \boldsymbol{w}$ and any $0 \leq t \leq 1$, let us first show that $f(t\boldsymbol{v} + (1-t)\boldsymbol{w}) = tf(\boldsymbol{v}) + (1-t)f(\boldsymbol{w})$.

Set $r = \|\boldsymbol{v}-\boldsymbol{w}\|$. Then $t\boldsymbol{v} + (1-t)\boldsymbol{w}$ has distance $(1-t)r$ to $\boldsymbol{v}$, and distance tr to $\boldsymbol{w}$.

Now we apply the map f, then $f(\boldsymbol{v}), f(\boldsymbol{w})$ shall also have distance r. The point $f(t\boldsymbol{v} + (1-t)\boldsymbol{w})$ must still have distance $(1-t)r$ to $f(\boldsymbol{v})$, and distance tr to $f(\boldsymbol{w})$. But this in turn implies that $f(t\boldsymbol{v} + (1-t)\boldsymbol{w}) = tf(\boldsymbol{v}) + (1-t)f(\boldsymbol{w})$.

The lemma below shows that this implies that $f(\boldsymbol{v}) = A\boldsymbol{v} + \boldsymbol{b}$ for some constant vector $\boldsymbol{b}$ and some matrix A. Note that we must have $f(\boldsymbol{0}) = \boldsymbol{b}$. In particular, we have

$$\|A\boldsymbol{x}\| = \|f(\boldsymbol{x}) - \boldsymbol{b}\| = \|f(\boldsymbol{x}) - f(\boldsymbol{0})\| = \|\boldsymbol{x} - \boldsymbol{0}\| = \|\boldsymbol{x}\|.$$

So A corresponds to a linear map that preserves distance. Hence it is an orthogonal matrix. ☐

Lemma 6.7.7 (Classification of all affine maps)**.** *If $f : \mathbb{R}^n \to \mathbb{R}^m$ satisfies $f(t\boldsymbol{v} + (1-t)\boldsymbol{w}) = tf(\boldsymbol{v}) + (1-t)f(\boldsymbol{w})$ for all $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$ and all $0 \leq t \leq 1$, then $f(\boldsymbol{x}) = A\boldsymbol{x} + \boldsymbol{b}$.*

Proof. (Intuitively, an n-dimensional affine space is basically $\mathbb{R}^n$, but you are NOT allowed to do linear combinations in general. You can only perform "convex linear combinations", which means $t\boldsymbol{v} + (1-t)\boldsymbol{w}$ with $0 \leq t \leq 1$. An affine map is therefore a map that preserves convex linear combinations.)

Set $\boldsymbol{b} = f(\boldsymbol{0})$, and set $g(\boldsymbol{x}) = f(\boldsymbol{x}) - \boldsymbol{b}$. Let us show that g is linear, then we are done. To start, we have $g(\boldsymbol{0}) = \boldsymbol{0}$. So far so good. Now let us check scalar multiplications.

For $0 \leq k \leq 1$, we have $k\boldsymbol{v} = k\boldsymbol{v} + (1-k)\boldsymbol{0}$. Hence $f(k\boldsymbol{v}) = kf(\boldsymbol{v}) + (1-k)f(\boldsymbol{0})$. Rearrange terms and use the fact that $\boldsymbol{b} = f(\boldsymbol{0})$, we have $f(k\boldsymbol{v}) - \boldsymbol{b} = k(f(\boldsymbol{v}) - \boldsymbol{b})$. Hence $g(k\boldsymbol{v}) = kg(\boldsymbol{v})$.

For $k > 1$, we have $\boldsymbol{v} = \frac{1}{k}(k\boldsymbol{v}) + (1-\frac{1}{k})\boldsymbol{0}$. Using similar idea to above, we would obtain $g(\boldsymbol{v}) = \frac{1}{k}g(k\boldsymbol{v})$.

For $k < 0$, we have $\boldsymbol{0} = (\frac{1}{1-k})(k\boldsymbol{v}) + (1 - \frac{1}{1-k})\boldsymbol{v}$. Using similar idea to above, we would obtain $g(\boldsymbol{0}) = (\frac{1}{1-k})g(k\boldsymbol{v}) + (1-\frac{1}{1-k})g(\boldsymbol{v})$. Rearrange and use the fact that $g(\boldsymbol{0}) = \boldsymbol{0}$, we would get $g(k\boldsymbol{v}) = kg(\boldsymbol{v})$ again.

All in all, we have $g(k\boldsymbol{v}) = kg(\boldsymbol{v})$ for all $k \in \mathbb{R}$.

Now let us check vector addition. Consider $g(\boldsymbol{v}+\boldsymbol{w})$. Note that $\boldsymbol{v}+\boldsymbol{w} = \frac{1}{2}(2\boldsymbol{v}) + \frac{1}{2}\boldsymbol{w}$. Hence $f(\boldsymbol{v}+\boldsymbol{w}) = \frac{1}{2}f(2\boldsymbol{v}) + \frac{1}{2}f(2\boldsymbol{w})$. Rearrange this, and we get $g(\boldsymbol{v}+\boldsymbol{w}) = \frac{1}{2}g(2\boldsymbol{v}) + \frac{1}{2}g(2\boldsymbol{w}) = g(\boldsymbol{v}) + g(\boldsymbol{w})$. So we are done. ☐

In short, any distance-preserving transformation on $\mathbb{R}^n$ must be a composition of translations, reflections and rotations.

6.8 (Optional) Rotations and Skew-Symmetric Matrices

These two kinds of matrices are closely related to each other. Let us see how. This is among the best ways to understand and represent rotations.

Definition 6.8.1. *We say a matrix A is **skew-symmetric** if $A^{\mathrm{T}} = -A$.*

Example 6.8.2. If $A^{\mathrm{T}} = -A$, then look at the diagonal entries on the left hand side and the right hand side. We must conclude that all diagonal entries for A are zero.

So if A is 2×2 and skew-symmetric, then it can only be something like $A = \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}$. Note that this is invertible as long as $a \neq 0$.

If A is 2×2 and skew-symmetric, then it can only be something like $A = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}$. Note that this NEVER invertible. If $a = b = c = 0$, then this is the zero matrix. If not, then $A \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}$, so the kernel is non-zero. ☺

Definition 6.8.3. *Given any matrix A, we define the matrix e^A to be the series $I + A + \frac{A^2}{2!} + \dots$.*

This series always converge. However, proving it is beyond the scope of the class. Let us just take it for granted for now.

Example 6.8.4. Recall that $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$, and $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

Now consider a matrix $A = \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}$. Note that this is a skew-symmetric matrix. We now have

$$\begin{aligned}
\mathrm{e}^A &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix} + \frac{1}{2!}\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}^2 + \frac{1}{3!}\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}^3 + \dots \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix} + \begin{bmatrix} -\frac{a^2}{2!} & 0 \\ 0 & -\frac{a^2}{2!} \end{bmatrix} + \begin{bmatrix} 0 & \frac{a^3}{3!} \\ -\frac{a^3}{3!} & 0 \end{bmatrix}^3 + \dots \\
&= \begin{bmatrix} 1 - \frac{a^2}{2!} + \dots & -a + \frac{a^3}{3!} - \dots \\ a - \frac{a^3}{3!} + \dots & 1 - \frac{a^2}{2!} + \dots \end{bmatrix} \\
&= \begin{bmatrix} \cos(a) & -\sin(a) \\ \sin(a) & \cos(a) \end{bmatrix}.
\end{aligned}$$

In particular, the exponential matrix for any 2 by 2 skew symmetric matrix is a rotation matrix. ☺

Proposition 6.8.5. *If $AB = BA$, then $\mathrm{e}^{A+B} = \mathrm{e}^A \mathrm{e}^B$.*

Proof. At an abstract level, the exponential map and its properties are derived from the basic properties of addition and multiplication. So if we have commutativity, then matrix addition and matrix multiplication now satisfies all the properties of regular addition and regular multiplication of real numbers. So the exponential map would exhibit exactly the same behavior as the one for regular real numbers.

To be more rigorous, here is the computational proof.

$$\begin{aligned}
\mathrm{e}^A \mathrm{e}^B &= (\sum_{i=0}^{\infty} \frac{1}{i!} A^i)(\sum_{j=0}^{\infty} \frac{1}{j!} B^j) \\
&= \sum_{i,j=0}^{\infty} \frac{1}{i!j!} A^i B^j \\
&= \sum_{i,j=0}^{\infty} \frac{C_{i+j}^i}{(i+j)!} A^i B^j \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{i=0}^{k} C_k^i A^i B^{k-i}
\end{aligned}$$

$$=\sum_{k=0}^{\infty}\frac{1}{k!}(A+B)^k$$
$$=e^{A+B}.$$

Can you spot the step that uses $AB = BA$? (In particular, if $AB \neq BA$, this computation would be false at that step.) $\square$

Just be careful that in general, $e^{A+B} \neq e^A e^B$ when $AB \neq BA$.

Theorem 6.8.6. *If A is skew-symmetric, then e^A is orthogonal.*

Proof. Note that $A^T = -A$, so in particular we have $AA^T = -A^2 = A^T A$. So $e^A e^{A^T} = e^{A+A^T} = e^O = I$, the identity map.

All I need to do now is to show that $(e^A)^T = e^{A^T}$. But this is obvious.

$$(e^A)^T = (I + A + \frac{A^2}{2!} + \dots)^T = I + A^T + \frac{(A^T)^2}{2!} + \dots = e^{A^T}.$$

So e^A is orthogonal. $\square$

Note that e^A is in fact always a rotation. And in fact, this goes both ways. If Q is a rotation (no reflection involved), then $Q = e^A$ for some skew-symmetric A. However, these are trickier to prove and we don't have the tools to do it yet.

Example 6.8.7. Consider any generic 3 by 3 skew symmetric matrix $A = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}$. Note that $A\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}$.

Now consider the rotation e^A. Applying this to $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, we see that

$$e^A\begin{bmatrix} a \\ b \\ c \end{bmatrix} = (I + A + \dots)\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

All the A portion of the series would NOT contribute at all, because they multiply $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ to zero. In particular, e^A is a rotation around the line containing $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$. (In fact, it is a rotation with radian degree $\|\begin{bmatrix} a \\ b \\ c \end{bmatrix}\|$. So if $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ has length 2π, then $e^A = I$. However, this is much easier to do after we have eigenvalues, so we don't prove it here.)

As you can see, in practice, the orthogonal matrices are usually ugly and hard to compute. Those cosines and sines would give ugly numbers everywhere. However, their "logarithm", the skew symmetric matrices are much prettier and intuitive, and the entries would have immediate geometric meanings behind them.

Also note that here we in fact have $A\boldsymbol{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \times \boldsymbol{v}$. So we see that cross product is the logarithm of rotations. This is why in physics and multivariable calculus, things involving rotations are usually related to the cross product. ☺

6.9 Gram-Schmidt Orthogonalization and QR decomposition

6.9.1 First Perspective: Algorithm to find an orthonormal basis

So far, we are using the existence of an ONB for free. We know they exist, because any inner product space is isomorphic (as inner product spaces) to the Euclidean space. But how to specifically find one?

In practice, people usually start with a random (non-orthonormal) basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, and tries to transform them into an ONB. This is the famous Gram-Schmidt Orthogonalization process.

Example 6.9.1. How can I make a bunch of vectors orthogonal to each other?

Suppose I have a stack of papers, but they are NOT stacked up-right. Rather, they are stacked in a tiled way, forming a parallelepiped. The three edges are $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$, where the first two are edges of a paper, and the third vector is the direction of how they are stacked.

In particular, although $\boldsymbol{v}_1, \boldsymbol{v}_2$ are orthogonal to each other, $\boldsymbol{v}_3$ is currently NOT orthogonal to $\boldsymbol{v}_1, \boldsymbol{v}_2$, and my parallelepiped is NOT a rectangular box.

Now this stack is annoying to carry. So I take the stack, and hit the desk with it on its side. Then I hit the desk with it on the other side. Now I will have a rectangular box stack, and it is now easy to carry.

When I hit the stack on its $\boldsymbol{v}_2$ side, I am doing a SHEARING (preserving the base and height) to make sure that $\boldsymbol{v}_3$ is now orthogonal to $\boldsymbol{v}_1$. And then when I hit the stack on the other side, now $\boldsymbol{v}_3$ is also sheared to be orthogonal to $\boldsymbol{v}_2$. Then I am done. The staking direction is finally orthogonal to the other two vectors.

This is the Gram-Schmidt orthogonalization process. ☺

Let us look at another example for more computational detail.

Example 6.9.2. Suppose we only have two vectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ spanning a two dimensional inner product space. To make them orthogonal, we need to shear $\boldsymbol{v}_2$ along the direction of $\boldsymbol{v}_1$ (i.e., adding multiples of $\boldsymbol{v}_1$ to $\boldsymbol{v}_2$), until it is orthogonal to $\boldsymbol{v}_1$. How much shearing do we need?

Suppose the shearing we need is $(\boldsymbol{v}_1, \boldsymbol{v}_2)\begin{bmatrix}1 & k\\ 0 & 1\end{bmatrix} = (\boldsymbol{v}_1, \boldsymbol{v}_2 + k\boldsymbol{v}_1)$. Then we must have $\langle \boldsymbol{v}_1, \boldsymbol{v}_2 + k\boldsymbol{v}_1\rangle = 0$. Solving for k, we see that we need $k = -\frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_2\rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1\rangle}$.

So the Gram Schmidt orthogonalization process would transform the basis $\boldsymbol{v}_1, \boldsymbol{v}_2$ to the OGB $\boldsymbol{v}_1, \boldsymbol{v}_2 - \frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_2\rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1\rangle}\boldsymbol{v}_1$. To make it ONB, just then divide each vector by its lengh, and we are done. ☺

Proposition 6.9.3. *Given any linearly independent $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ in an abstract vector space, let*

$$\begin{aligned}
\boldsymbol{w}_1 &= \boldsymbol{v}_1\\
\boldsymbol{w}_2 &= \boldsymbol{v}_2 - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_2\rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1\rangle}\boldsymbol{w}_1\\
\boldsymbol{w}_3 &= \boldsymbol{v}_3 - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_3\rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1\rangle}\boldsymbol{w}_1 - \frac{\langle \boldsymbol{w}_2, \boldsymbol{v}_3\rangle}{\langle \boldsymbol{w}_2, \boldsymbol{w}_2\rangle}\boldsymbol{w}_2\\
&\vdots\\
\boldsymbol{w}_k &= \boldsymbol{v}_k - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_k\rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1\rangle}\boldsymbol{w}_1 - \frac{\langle \boldsymbol{w}_2, \boldsymbol{v}_k\rangle}{\langle \boldsymbol{w}_2, \boldsymbol{w}_2\rangle}\boldsymbol{w}_2 - \cdots - \frac{\langle \boldsymbol{w}_{n-1}, \boldsymbol{v}_k\rangle}{\langle \boldsymbol{w}_{n-1}, \boldsymbol{w}_{n-1}\rangle}\boldsymbol{w}_{n-1}.
\end{aligned}$$

Then $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ are all non-zero and mutually orthogonal. Let $\boldsymbol{q}_i = \frac{\boldsymbol{w}_i}{\|\boldsymbol{w}_i\|}$, then $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$ is an orthonormal collection of vectors.

*If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ is a basis, then $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ is OGB and $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$ is ONB. We call this ONB the**Gram-Schmidt orthogonalization** of our original basis.*

Proof. Let us prove orthogonality by induction. Suppose $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_i$ are pairwise orthogonal. (The initial case $i = 1$ is trivial.)

Then consider $\boldsymbol{w}_{i+1}$. For any $j \leq i$, we have

$$\langle \boldsymbol{w}_{i+1}, \boldsymbol{w}_j \rangle = \langle \boldsymbol{v}_{i+1} - \sum_{k=1}^{i} \frac{\langle \boldsymbol{w}_k, \boldsymbol{v}_{i+1} \rangle}{\langle \boldsymbol{w}_k, \boldsymbol{w}_k \rangle} \boldsymbol{w}_k, \boldsymbol{w}_j \rangle = \langle \boldsymbol{v}_{i+1}, \boldsymbol{w}_j \rangle - \sum_{k=1}^{i} \frac{\langle \boldsymbol{w}_k, \boldsymbol{v}_{i+1} \rangle}{\langle \boldsymbol{w}_k, \boldsymbol{w}_k \rangle} \langle \boldsymbol{w}_k, \boldsymbol{w}_j \rangle = \langle \boldsymbol{v}_{i+1}, \boldsymbol{w}_j \rangle - \frac{\langle \boldsymbol{w}_j, \boldsymbol{v}_{i+1} \rangle}{\langle \boldsymbol{w}_j, \boldsymbol{w}_j \rangle} \langle \boldsymbol{w}_j, \boldsymbol{w}_j \rangle = 0.$$

So $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{i+1}$ are also mutually orthogonal. So by induction, we see that $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ are mutually orthogonal.

Finally, since we went from the basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ of the space $\operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ to the collection $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ using only shearings (invertible operations), the result must still be a basis of the space $\operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$. So $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k$ are all non-zero. The rest of the proposition is obvious. ☐

Remark 6.9.4. *Note that we started with vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$. To proceed, we first make sure that all later vectors are orthogonal to the first vector. So we have*

$$\begin{aligned}
\boldsymbol{w}_1 = & \boldsymbol{v}_1 \\
& \boldsymbol{v}_2 - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1 \rangle} \boldsymbol{w}_1 \\
& \boldsymbol{v}_3 - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_3 \rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1 \rangle} \boldsymbol{w}_1 \\
& \vdots \\
& \boldsymbol{v}_n - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_n \rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1 \rangle} \boldsymbol{w}_1.
\end{aligned}$$

Now all later vectors are orthogonal to the first vector. Then we do the shearing to make sure that all later vectors are orthogonal to the second vector. So we have

$$\begin{aligned}
\boldsymbol{w}_1 = & \boldsymbol{v}_1 \\
\boldsymbol{w}_2 = & \boldsymbol{v}_2 - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1 \rangle} \boldsymbol{w}_1 \\
& \boldsymbol{v}_3 - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_3 \rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1 \rangle} \boldsymbol{w}_1 - \frac{\langle \boldsymbol{w}_2, \boldsymbol{v}_3 \rangle}{\langle \boldsymbol{w}_2, \boldsymbol{w}_2 \rangle} \boldsymbol{w}_2 \\
& \vdots \\
& \boldsymbol{v}_n - \frac{\langle \boldsymbol{w}_1, \boldsymbol{v}_n \rangle}{\langle \boldsymbol{w}_1, \boldsymbol{w}_1 \rangle} \boldsymbol{w}_1 - \frac{\langle \boldsymbol{w}_2, \boldsymbol{v}_n \rangle}{\langle \boldsymbol{w}_2, \boldsymbol{w}_2 \rangle} \boldsymbol{w}_2.
\end{aligned}$$

Now all later vectors are orthogonal to the second vector. So on so forth. This is the process of Gram Schmidt orthogonalization.

6.9.2 Second Perspective: QR decomposition

Now, take a closer look. When we shear, we are ALWAYS substracting multiples of earlier vectors from later vectors! For example, in $\mathbb{R}^2$, to go from $\begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{bmatrix}$ to $\begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 - \frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle} \boldsymbol{v}_1 \end{bmatrix}$, we only need to use the first column to reduce the second column. This corresponds to a column operation that is upper triangular! In particular, $\begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{bmatrix} \begin{bmatrix} 1 & -\frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 - \frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle} \boldsymbol{v}_1 \end{bmatrix}$.

Think of Gram-Schmidt as an series of operations on columns of the invertible matrix $A = \begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n \end{bmatrix}$, then we are multiplying a series of upper triangular matrices to the right of it, and we get OGB $\begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n \end{bmatrix} U = \begin{bmatrix} \boldsymbol{w}_1 & \ldots & \boldsymbol{w}_n \end{bmatrix}$. Now we divide each column by its length, and we get $\begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_n \end{bmatrix} UD = \begin{bmatrix} \boldsymbol{w}_1 & \ldots & \boldsymbol{w}_n \end{bmatrix} D =$

$\begin{bmatrix} \boldsymbol{q}_1 & \dots & \boldsymbol{q}_n \end{bmatrix}$ where D is the diagonal matrix represending the division of each column by its length. And the end result here should be an ONB. In particular, the matrix $Q = \begin{bmatrix} \boldsymbol{q}_1 & \dots & \boldsymbol{q}_n \end{bmatrix}$ is an orthogonal matrix. So $AUD = Q$. We can rearrange this as $A = QR$ where $R = D^{-1}U^{-1}$ is upper triangular.

So now we get the matrix-decomposition perspective of Gram-Schmidt. (Just like Gaussian elimination and LU decomposition are essentially the same, Gram-Schmidt and QR below are essentially the same.)

Theorem 6.9.5 (QR decomposition)**.** *Let A be any $n \times k$ matrix with linearly independent columns (injection). Then we can find an $n \times k$ matrix Q with orthonormal columns and an upper triangular $k \times k$ matrix R such that $A = QR$. If we require all diagonal entries of R to be positive, then this decomposition is unique.*

In particular, if A is invertible, then $A = QR$ for some orthogonal matrix Q and upper triangular matrix R.

Proof. If A has linearly independent columns, then the Gram-Schmidt process means we do AU for some upper triangular U (which represents the shearings and the final scaling of each column), and now the columns of AU are orthonormal. So $Q = AU$ is a matrix with orthonormal columns. So $A = QU^{-1}$ is the desired QR decomposition.

We delay the proof of uniqueness for now. Later perspectives will make the proof of uniqueness trivial. $\square$

Remark 6.9.6. *The QR decomposition IS the Gram-Schmidt orthogonalization. How to find the QR decomposition? One way is to first work out Q by Gram Schmidt orthogonalization, then R follows from the shearing process.*

Example 6.9.7. Suppose we have $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \boldsymbol{v}_3 = \begin{bmatrix} 4 \\ 2 \\ 2 \\ 0 \end{bmatrix}, \boldsymbol{v}_4 = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$. So we started with $A = \begin{bmatrix} 1 & 2 & 4 & 4 \\ 1 & 2 & 2 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$.

We start our shearing. First we use the first column to take a dot product with EVERY column. The dot product with itself is 4, while the dot product with other columns are $4, 8, 4$. So we divide the later dot products by the dot product with itself, and get ratios $\frac{4}{4}, \frac{8}{4}, \frac{4}{4}$. These are the coefficients for the shearing!

To shear everything in A to be orthogonal to $\boldsymbol{v}_1$, we end up with

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & -1 \\ 1 & -1 & 0 & -1 \\ 1 & -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Note that the upper triangular matrix has entries taken exactly from the dot product ratios.

Here the upper triangular matrix is recording my column operations using the first column to reduce all the columns to the right.

Next we shear the last two columns to be orthogonal to the second column. Similarly, we use the second column and take dot products with the second, third and fourth column, and look at the ratio of the dot products. We end up with

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & -1 \\ 1 & -1 & 0 & -1 \\ 1 & -1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 1 & -1 & -2 \\ 1 & -1 & 1 & -0 \\ 1 & -1 & -1 & -0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Finally, we shear the last column to be orthogonal to the third. We end up with

$$\begin{bmatrix}1 & 1 & 1 & 2\\ 1 & 1 & -1 & -2\\ 1 & -1 & 1 & -0\\ 1 & -1 & -1 & -0\end{bmatrix} = \begin{bmatrix}1 & 1 & 1 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix}.$$

So we have

$$A = \begin{bmatrix}1 & 1 & 1 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1\end{bmatrix}\begin{bmatrix}1 & 1 & 2 & 1\\ 0 & 1 & 1 & 1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix}.$$

Now normalize these columns, we get the Haar wavelet basis.
All in all, we get the following QR decomposition:

$$\begin{bmatrix}1 & 1 & 2 & 2\\ 1 & 1 & 1 & 0\\ 1 & 0 & 1 & 0\\ 1 & 0 & 0 & 0\end{bmatrix} = (\frac{1}{2}\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1\end{bmatrix})\begin{bmatrix}2 & 2 & 4 & 2\\ 0 & 2 & 2 & 2\\ 0 & 0 & 2 & 2\\ 0 & 0 & 0 & 2\end{bmatrix}.$$

☺

6.9.3 Third Perspective: Cholesky decomposition

Let us spend a little time here to discuss what it means to have orthonormal columns.

Example 6.9.8. Consider $Q = \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{bmatrix}$. This is a matrix with orthonormal columns. Calculation shows that $Q^{\mathrm{T}}Q = I$, yet Q is not an orthogonal matrix. Why? Because Q is not a square matrix, and $QQ^{\mathrm{T}} = \begin{bmatrix}1 & & \\ & 1 & \\ & & 0\end{bmatrix} \neq I$.

For square matrices, $Q^{\mathrm{T}}Q = I$ implies that Q is orthogonal. For non-square matrix, $Q^{\mathrm{T}}Q = I$ only means that Q has orthonormal columns, but may have redundant rows, as shown in the lemma below. ☺

Lemma 6.9.9. *A matrix A has orthonormal columns iff $A^{\mathrm{T}}A = I$. A matrix A has orthonormal rows iff $AA^{\mathrm{T}} = I$.*

Proof. Write $A = \begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_n\end{bmatrix}$, then the (i, j)-entry of $A^{\mathrm{T}}A$ is exactly $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{a}_j$. So $A^{\mathrm{T}}A = I$ means $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{a}_j = 0$ when $i \neq j$, and $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{a}_j = 1$ when $i = j$. So we are done.

The other statement about orthonormal rows is simply the same thing applied to A^{T}. □

The previous proof of QR decomposition and description of Gram-Schmidt orthogonalization is very "algorithmic" in nature. They basically goes along the say line of reason and logic of Gaussian elimination. Instead of using upper rows to kill lower rows, to achieve something triangular, we now use left columns to shear right columns into orthogonal position. (And thus R is upper triangular.)

However, you are no longer a high school student anymore. The hall mark of a mature learner is to learn the SAME thing through DIFFERENT persepctives. Everything we do in this class, you can understand them as an algorithm (Gram Schmidt), or as a matrix decomposition (QR), or as a change of basis process between bases, or as a geometric relation of subspaces, or as an induction process on block matrices, or sometimes even as a mathematical version of physics statements or facts from other sciences.

So now we turn to more perspectives and descriptions and proofs of the Gram-Schmidt process and QR decomposition.

Let me deliver a fatal blow to your sanity: We have actually proven QR decomposition a long long time ago!

Alternative proof of QR decomposition. Given any matrix A with independent columns, then $A^{\mathrm{T}}A$ is positive definite. So by Cholesky decomposition, there is a unique lower triangular matrix L with positive diagonal entries such that $A^{\mathrm{T}}A = LL^{\mathrm{T}}$.

This immediately implies that $L^{-1}A^{\mathrm{T}}A(L^{-1})^{\mathrm{T}} = I$. So we see that $A(L^{-1})^{\mathrm{T}}$ has orthonormal columns, say $Q = A(L^{-1})^{\mathrm{T}}$. Then $A = QL^{\mathrm{T}}$. We are done. $\square$

This proof makes so much sense, because previously, we go from a Gram matrix G to the dot product via some change of basis. If columns of A are the basis, then $A^{\mathrm{T}}A$ is precisely the gram matrix, and the change of basis implemented by L^{T} gives the orthnormal basis Q whose Gram matrix is now I. The following three process are the same:

1. The process of going from some arbitrary invertible matrix to an orthogonal matrix. ($A = QR$)
2. The process of going from some arbitrary basis to an orthonormal basis. (Column view of $A = QR$.)
3. The process of going from some arbitrary Gram matrix to the identity matrix. ($A^{\mathrm{T}}A = R^{\mathrm{T}}Q^{\mathrm{T}}QR = R^{\mathrm{T}}R$, the Cholesky decomposition.)

Let us also prove uniqueness of the QR decomposition here.

Proof of the uniqueness of QR decomposition. Given any matrix A with independent columns, let $A = Q_1R_1 = Q_2R_2$ are two QR decompositions. Then we have $A^{\mathrm{T}}A = R_1^{\mathrm{T}}Q_1^{\mathrm{T}}Q_1R_1 = R_1^{\mathrm{T}}R_1$, and similarly $A^{\mathrm{T}}A = R_2^{\mathrm{T}}R_2$. But then these are two Cholesky decompositions of the same positive definite matrix. So by uniqueness of the Cholesky decomposition, $R_1 = R_2$. And hence we also have $Q_1 = AR_1^{-1} = AR_2^{-1} = Q_2$. $\square$

6.9.4 Fourth Perspective: Geometry of the subspace chain

Suppose we started with A, and we perform Gram-Schmidt to get Q. However, we are too lazy to record the shearing process, so we forget to have R. What to do?

Well, recall that the Gram-Schmidt process $A = QR$ is essentially a change of basis process. The i-th column of R should exactly be the coordinates of the i-th column of A under the basis Q.

Example 6.9.10. If you go from A to Q, but forget to record your column operations. How to find R now?

Well, one can also retroactively solve R from $A = QR$, which implies that $R = Q^{-1}A = Q^{\mathrm{T}}A$.

Writing A, Q in columns $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ and $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$, we see that the (i, j) entry of R is simply $\boldsymbol{q}_i^{\mathrm{T}}\boldsymbol{v}_j$. For example, when $n = 3$, the QR decomposition looks like:

$$\begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{bmatrix} = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \end{bmatrix} \begin{bmatrix} \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_3 \\ \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_3 \\ \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_3 \end{bmatrix}.$$

This is also obvious since $\boldsymbol{q}_i^{\mathrm{T}}\boldsymbol{v}_j$ is the i-th coordinate of $\boldsymbol{v}_j$ under the orthonormal basis. The i-th column of R should exactly be the coordinates of the i-th column of A under the basis Q.

Using previous example, you can compute and verify that indeed,

$$(\frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}) \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 4 & 2 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

(Here the matrix $\frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$ is symmetric and orthogonal, so it is its own inverse.)

Here is a mystery though. Can you see why must the matrix $\begin{bmatrix} \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_3 \\ \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_3 \\ \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_3 \end{bmatrix}$ always be upper triangular? It is not obvious right away. ☺

Let us unravel this mystery.

Proposition 6.9.11. *Suppose Gram Schmidt brings the basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ to the orthonormal basis $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$. Then* $\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_i) = \operatorname{span}(\boldsymbol{q}_1, \dots, \boldsymbol{q}_i)$.

Proof. This comes from the Gram Schmidt shearing process. We created $\boldsymbol{q}_i$ only using $\boldsymbol{v}_1, \dots, \boldsymbol{v}_i$, so obviously $\operatorname{span}(\boldsymbol{q}_1, \dots, \boldsymbol{q}_i) \subseteq \operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_i)$. But since both collections are linearly independent, they have the same dimension. So we have it. □

This immediatly shows that, conversely, $\boldsymbol{v}_j$ is a linear combination of $\boldsymbol{q}_1, \dots, \boldsymbol{q}_j$, and do not use $\boldsymbol{q}_{j+1}, \dots, \boldsymbol{q}_n$ at all. So its coordinates $\boldsymbol{q}_i^{\mathrm{T}}\boldsymbol{v}_j$ when $i > 0$ are all zero. So $\begin{bmatrix} \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{v}_3 \\ \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{v}_3 \\ \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_1 & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_2 & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{v}_3 \end{bmatrix}$ is upper triangular.

Remark 6.9.12. *If you think carefully, buried beneath this is the fact that the inverse of an upper triangular matrix is upper triangular. If $\boldsymbol{q}_i$ only uses $\boldsymbol{v}_1, \dots, \boldsymbol{v}_i$ for each i, then $\boldsymbol{v}_i$ only uses $\boldsymbol{q}_1, \dots, \boldsymbol{q}_i$ for each i.*

The observation here also gives us a geometric way to achieve Gram-Schmidt.

Example 6.9.13. Again consider any basis $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ in $\mathbb{R}^3$. We can actually find $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ geometrically, without any calculation!

Here is how. First of all, obviouosly $\boldsymbol{q}_1 = \frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}$. This is easy.

Now $\boldsymbol{q}_2$ must be a linear combination of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. So it lies in the plane $\operatorname{span}(\boldsymbol{v}_1, \boldsymbol{v}_2)$. Inside this plane, we should also pick $\boldsymbol{q}_2$ that is perpendicular to the line $\operatorname{span}(\boldsymbol{v}_1)$.

Hey! Given a plane and a line inside it, there are ONLY TWO unit vectors in the plane normal to the line. One is in the same "half-plane" as $\boldsymbol{v}_2$, the other is in the opposite "half-plane". So $\boldsymbol{q}_2$ must be the one in the same "half-plane" as $\boldsymbol{v}_2$. This choice is unique.

Now consider $\boldsymbol{q}_3$. It has to lie in the space $\mathbb{R}^3 = \operatorname{span}(\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3)$, and it must be orthogonal to the plane $\operatorname{span}(\boldsymbol{q}_1, \boldsymbol{q}_2) = \operatorname{span}(\boldsymbol{v}_1, \boldsymbol{v}_2)$. Now, in the space, there are ONLY TWO unit vectors orthogonal to the plane, one in each "half-space". We let $\boldsymbol{q}_3$ be the one in the same "half-space" as $\boldsymbol{v}_3$.

Then $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ must be the result of Gram-Schmidt. We simply have no other choice in each step. ☺

So now we proceed with the subspace version of Gram-Schmidt. As you shall see, the Gram-Schmidt does NOT depends on the specific vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ at all. It ONLY depends on the CHAIN of SUBSPACES they generate.

Proposition 6.9.14. *Given a chain of subspaces $\{\boldsymbol{0}\} \subset V_1 \subset V_2 \subset \cdots \subset V_n$ where $\dim V_k = k$, then there is an orthonormal basis $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ such that $\operatorname{span}(\boldsymbol{q}_1, \dots, \boldsymbol{q}_k) = V_k$. This basis is unique up to sign. (I.e., all such orthonormal basis must be $\pm\boldsymbol{q}_1, \dots, \pm\boldsymbol{q}_n$.)*

If it helps, just imagine that V_i is $\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_i)$ if you like.

Proof. To start, V_1 is one dimensional. So pick any unit vector in it and we are done. Note that in an one-dimensional space, there are only TWO possible choice of unit vectors, and they are negation of each other. So we get $\pm\boldsymbol{q}_1$.

Next, we want to find $\boldsymbol{q}_2$. To do this, we need to find a vector in V_2 that is perpendicular to V_1. However, since V_2 is a plane and V_1 is a line, there are ONLY two unit vectors perpendicular to V_1. So we get $\pm\boldsymbol{q}_2$.

This goes on until the end. At each step, when we pick $\boldsymbol{q}_k$, we are trying to find a vector in V_k perpendicular to V_{k-1}. But since V_{k-1} is only one dimension less than V_k, it is a hyperplane in V_k, and has only two unit normal vectors. So we get $\pm\boldsymbol{q}_k$. □

One technical lemma is needed here. We prove it below.

Lemma 6.9.15. *Let V be an inner product space with dimension n, and let W be a subspace with dimension $n-1$. Then there are only two unit vectors in V orthogonal to W.*

Proof. All vectors perpendicular to W are in $W^{\perp}$, which must be one dimensional. Hence $W^{\perp}$ contains only two unit vectors. ☐

In short, according to this view, we started with $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$. And each $\boldsymbol{q}_k$ is basically the unit normal vector of the hyperplane $\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_{k-1})$ in $\operatorname{span}(\boldsymbol{v}_1, \dots, \boldsymbol{v}_k)$, and there are only two such vectors. But since we require $A = QR$ to have R with positive diagonal entries, we are requiring that $\langle \boldsymbol{q}_k, \boldsymbol{v}_k \rangle > 0$. So we see that only one choice of unit normal vector could make this positive, and the other one makes this negative (the two choices of unit normal vectors are negations of each other). So we choose the positive one.

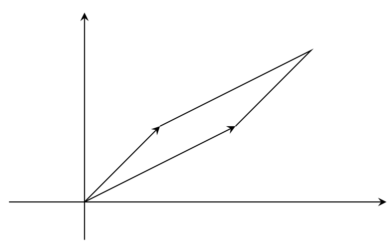
6.9.5 Fifth Perspective: Parallelotope

What is a matrix? So far, we have several interpretations.

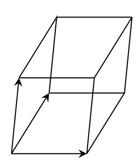
1. We can think of an invertibla matrix A as a basis, by treating its columns as basis vectors.
2. We can also think of any matrix A as a linear map, sending $\boldsymbol{v}$ to $A\boldsymbol{v}$.
3. We can also think of any matrix A as a bilinear map, sending a pair $\boldsymbol{v}, \boldsymbol{w}$ to $\boldsymbol{v}^{\mathrm{T}} A \boldsymbol{w}$. (If A is furthermore symmetric and positive definite, then this bilinear map is an inner product.)
4. We can think of an invertibla matrix A as a change of coordinate matrix or basis transition matrix. So in this sense, A is doing nothing, merely changing the names of objects.

Now let us use a new interpretation. A could represent a parallelotope.

Example 6.9.16. Consider $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$. We may think of the two column vectors as two edges of a parallelogram.



What if $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$? Then A represents a parallelopiped in the space $\mathbb{R}^3$.



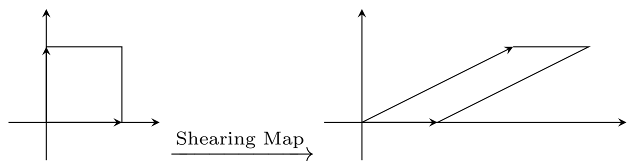
What about $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$? Well, we have two edges, and they live in $\mathbb{R}^3$. Therefore, this is a parallelogram in space!

The term ***parallelotope*** is the generalization of parallelograms and parallelopipeds. In general, an $m \times n$ matrix A can represent an n-dimensional parallelotopes in $\mathbb{R}^m$. ☺

Example 6.9.17. Consider the matrix equation

$$\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & \\ & 1\end{bmatrix} = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}.$$

Let us interpret the first $\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ as a linear map, and the other two matrices in the equation as parallelograms. Then we are saying that the shearing map $\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$ is sending the unit square $\begin{bmatrix}1 & \\ & 1\end{bmatrix}$ to the parallelogram $\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$. Indeed we have



☺

Now, so far we have been focused on the relation between A and Q in the decomposition $A = QR$, and think of R as the transformation process (basis transition matrix). What if we think of Q as the transformation process, and consider A and R as parallelotopes? These two views make a very interesting comparison.

Example 6.9.18. Suppose $A = \begin{bmatrix}1 & 1\\ 1 & 2\end{bmatrix}$, $Q = \begin{bmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$ and $R = \begin{bmatrix}\sqrt{2} & \frac{3}{2}\sqrt{2}\\ 0 & \frac{1}{2}\sqrt{2}\end{bmatrix}$. Here you can check that $A = QR$, and R is obviously upper triangular with positive diagonal entries, and Q is orthogonal because it is a rotation by 45 degree counterclockwise.

Now let us treat A as a parallelogram in $\mathbb{R}^2$ whose edges are columns of A (and one vertex is at the origin). Draw this yourself. We see that $Q^{-1}A = R$. This means, if we apply Q^{-1} to the columns of A (edges of your parallelogram), you get columns of R (another parallelogram). In short, if you rotate the parallelogram by A via Q^{-1}, i.e., clockwise by 45 degree, you get a parallelogram whose first edge is on the positive x-axis (because R is upper triangular and the first diagonal entry of R is positive), and its also on the upper half plane (because the second diagonal entry of R is also positive).

In general, you may think of $A = QR$ as such a process. Say $n = 3$. Imagine that A is a parallelepiped. We are rotation/reflecting the A parallelepiped to get the R parallelepiped. First we rotate the parallelepiped so that its first edge is in the positive x-axis. Now we keep this first edge fixed, but we rotate it around the x-axis so that its second edge is now on the positive xy-plane. Now the "base" of our parallelepiped is fixed, so we may choose to reflect it or not, so that the third edge is in the positive half space. All these rotations and reflections are Q^{-1}, and they transform A into a parallelepiped whose three edges are like $\begin{bmatrix}\textit{positive} & * & *\\ 0 & \textit{positive} & *\\ 0 & 0 & \textit{positive}\end{bmatrix}$, i.e., R.

Since Q is a rigid motion, the resulting parallelepiped by R has the same shape as A, but the location is now unique.

This process generalizes to higher dimensions very easily. (The higher dimension versions of a parallelogram or parallelepiped is called a parallelotope.) ☺

The idea that Q is a rigid motion transforming the parallelotope A to a parallelotope R turns out to be VERY IMPORTANT. It turns out that it gives rise to a BETTER way to do Gram-Schmidt than Gram-Schmidt. (Provide greater numerical stability, i.e., the calculations would NOT magnify error term.) The idea is to use a series of Householder transformations, i.e., reflections, to go from A to R. Furthermore, since we are working on parallelotopes, why restrict ourselves to n-dim parallelotopes? One can in fact look at n-dimensional parallelotopes in $\mathbb{R}^m$ and start doing rigid motions.

The following is a more general version of QR decomposition. However, pay special attention to the proof, because that is what most computers would actually do to FIND the QR decomposition.

Theorem 6.9.19. *Given any $m \times n$ matrix A with $m \geq n$, we have QR decomposition $A = QR$ where Q is $m \times m$ and orthogonal, and $R = \begin{bmatrix} R_1 \\ O \end{bmatrix}$ is an $m \times n$ block matrix where R_1 is $n \times n$ and upper triangular with NON-NEGATIVE diagonal entries. If A has full rank (i.e., injective), then R_1 has positive diagonal entries.*

Proof. Suppose $n = 1$. Then A is a single vector $\boldsymbol{v}$, and Q is any orthogonal matrix with first column $\frac{1}{\|\boldsymbol{v}\|}\boldsymbol{v}$, and $R = \|\boldsymbol{v}\|$. All is trivial.

By induction, suppose $n > 1$. Let $\boldsymbol{a}$ be the first column of A, and find a Householder transformation H such that $H\boldsymbol{a} = \|\boldsymbol{a}\|\boldsymbol{e}_1$. This is possible due to the lemma below this proof.

Then $HA = \begin{bmatrix} \|\boldsymbol{a}_1\| & b^{\mathrm{T}} \\ \mathbf{0} & A_{n-1} \end{bmatrix}$. (Geometrically, we perfomed a relfection so that the first edge of the parallelotope is now on the positive x_1-axis.)

Here A_{n-1} had $n-1$ columns, so by induction $A_{n-1} = Q_{n-1}R_{n-1}$ as desired, where $R_{n-1} = \begin{bmatrix} R_1 \\ O \end{bmatrix}$ is an $(m-1) \times (n-1)$ block matrix where R_1 is $(n-1) \times (n-1)$ and upper triangular with non-negative diagonal entries.

Now consider $\begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & Q_{n-1}^{-1} \end{bmatrix} HA = \begin{bmatrix} \|\boldsymbol{a}_1\| & b^{\mathrm{T}} \\ \mathbf{0} & R_1 \\ \mathbf{0} & O \end{bmatrix}$. Now the upper portion $\begin{bmatrix} \|\boldsymbol{a}_1\| & b^{\mathrm{T}} \\ \mathbf{0} & R_1 \end{bmatrix}$ is $n \times n$ and upper triangular, with non-negative diagonal entries. So we are done.

Finally, if A has full rank and $A = QR$, then since Q is invertible, $R = \begin{bmatrix} R_1 \\ O \end{bmatrix}$ must also have full rank. So R_1 can only have positive diagonal entries. ☐

Lemma 6.9.20. *If $\|\boldsymbol{v}\| = \|\boldsymbol{w}\|$ in $\mathbb{R}^n$, then there is a Householder matrix $H = I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$ for a unit vector $\boldsymbol{n}$ such that $H\boldsymbol{v} = \boldsymbol{w}$.*

Proof. If $\boldsymbol{v} = \boldsymbol{w} = \mathbf{0}$, pick $H = I$ and we are done. From now on, assume that $\boldsymbol{v}, \boldsymbol{w} \neq \mathbf{0}$.

Let us first get some intuition. Draw the arrow $\boldsymbol{v}$ and $\boldsymbol{w}$, and imagine this reflection process. It is obvious that $\boldsymbol{v} - \boldsymbol{w}$ must be perpendicular to the hyperplane of reflection.

Now we start our proof. Let $\boldsymbol{n} = \frac{\boldsymbol{v}-\boldsymbol{w}}{\|\boldsymbol{v}-\boldsymbol{w}\|}$, and set $H = I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$. Then direct computation would yeild $H\boldsymbol{v} = \boldsymbol{w}$. Yay. ☐

Remark 6.9.21. *In the end, if you use the formula $H = I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$ and the fact that $\boldsymbol{n} = \frac{\boldsymbol{v}-\boldsymbol{w}}{\|\boldsymbol{v}-\boldsymbol{w}\|}$, we have $H = I - 2\frac{(\boldsymbol{v}-\boldsymbol{w})(\boldsymbol{v}-\boldsymbol{w})^{\mathrm{T}}}{(\boldsymbol{v}-\boldsymbol{w})^{\mathrm{T}}(\boldsymbol{v}-\boldsymbol{w})}$. This is not surprising at all, since we already know that $\frac{(\boldsymbol{v}-\boldsymbol{w})(\boldsymbol{v}-\boldsymbol{w})^{\mathrm{T}}}{(\boldsymbol{v}-\boldsymbol{w})^{\mathrm{T}}(\boldsymbol{v}-\boldsymbol{w})}$ is the projection to the direction $\boldsymbol{v} - \boldsymbol{w}$.*

Note that the proof above is NOT just a proof, but in fact a computationally viable and excellent way to compute the QR decomposition. We started with a big parallelotope. First we reflect it so that the first edge is on the positive x_1-axis. Then induction means we again perform a reflection, so that the first two edges are on the positive half of the x_1x_2-plane. So on so forth, until we are done and get R.

(I would really love to make an animiation showing an example of such process.... But I do not know how....)

For many cases, when trying to find the QR decomposition, a computer would usually just reflect away. Orthogonal matrices usually tends to "keep small errors small". Whereas shearings R might magnify errors. So applying Q^{-1} to A to find R is a MUCH better idea, compared with applying R^{-1} and find Q. By doing a series of reflections, we can find the QR decomposition of A without enlarge any initial errors.

Remark 6.9.22. *In practice, the traditional Gram-Schmidt would require about $2mn^2$ addition / substraction / multiplication / divisions, and n square root calculations. On the other hand, using a series of Householders takes about $2mn^2 - \frac{2}{3}n^3$ addition / substraction / multiplication / divisions, and n square root calculations. It is both faster and more stable.*

The stuff below is super optional! I would strongly advise you to skip it. I add it here for completeness only.

Let us see how many calculations are involved.

Suppose we are doing regular Gram schmidt. We start with an $m \times n$ matrix $\begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_n \end{bmatrix}$. First let us use the first column to change later columns. First we need to calculate $\langle \boldsymbol{v}_1, \boldsymbol{v}_i \rangle$ for each i, and this takes m multiplications and $m-1$ additions for each i, hence a total of mn multiplications and $(m-1)n$ additions. Then we compute $\frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}$ for all $i \neq 1$, which takes $n-1$ multiplications (we treat division as a multiplication). Next we calculate $\frac{\langle \boldsymbol{v}_1, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}\boldsymbol{v}_1$ for each i and substract this from the i-th column. This takes m multiplications and m additions for each column, giving a total of $m(n-1)$ multiplications and $m(n-1)$ additions. We used a total of $mn + n - 1 + m(n-1) = 2mn + n - m - 1$ multiplications/divisions, and $(m-1)n + m(n-1) = 2mn - n - m$ additions/substractions. Let us just say $4mn$ calculations for short.

Next we use the second columns to reduce later columns. So we are essentially dealing with a matrix with m rows and $n-1$ columns, so it takes about $4m(n-1)$ calculations. By this pattern, we see that Gram-Schmidt takes a total of $4m(n + \dots + 1)$ calculations, which is about $2mn^2$. Now we have OGB, but to get ONB, we need to calculate the length, which requires $(2m+1)n$ operations and n square root calculations. Note that $(2m+1)n << 2mn^2$, so we can ignore that for simplicity.

Now let us look at the Householder approach. Given first column $\boldsymbol{v}_1$, first we calculate $\boldsymbol{v}_1^{\mathrm{T}}\boldsymbol{v}_1$, which takes $2m-1$ calculations. Next we take square root to get $\|\boldsymbol{v}_1\|$, and we compute $\boldsymbol{n} = \boldsymbol{v}_1 - \|\boldsymbol{v}_1\|\boldsymbol{e}_1$, which is a single substraction. (Note that $\boldsymbol{n}$ is not a unit vector, but we shall deal with it later.) So far we have $2m$ calculations and one square root operations.

Next, we want to apply $H = I - 2\frac{\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}}$ to each column. We can calculate $\frac{2}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}}$ with $2m$ calculations, and multiply this to $\boldsymbol{n}$ takes another m calculations.

So the i-th column $\boldsymbol{v}_i$ should turn into $\boldsymbol{v}_i - 2\boldsymbol{n}\frac{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{v}_i}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}}$, and the quantity $2\boldsymbol{n}\frac{1}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}}$ is already pre-calculated. The dot product $\boldsymbol{n}^{\mathrm{T}}\boldsymbol{v}_i$ takes $2m-1$ calculations, and multiply this to $2\boldsymbol{n}\frac{1}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}}$ takes m calculations, and substract this from $\boldsymbol{v}_i$ takes another m. This takes a total of $4m-1$ calculations for each $i \neq 1$. Hence this is a total of $(4m-1)(n-1)$ calculations.

The whole process so far takes $5m + (4m-1)(n-1)$ calculations, which is about $4mn$, and one square root. Now we have completed our induction step, and we no longer need to touch the first row or the first column. So we do this to the $(m-1) \times (n-1)$ lower right matrix left, and this takes about $4(m-1)(n-1)$ calculations. By induction, we need $4mn + 4(m-1)(n-1) + \dots + 4(m-n)(n-n)$. Compare this with the previous Gram-Schmidt calculation of $4mn + 4m(n-1) + \dots + 4m(n-n)$, and you see where the saving happened.

A more specific calculation shows that $4mn + 4(m-1)(n-1) + \dots + 4(m-n)(n-n) = 4(m-n)(n + \dots + 1) + 4(n^2 + \dots + 1) = 2(m-n)n^2 + \frac{4}{3}n^3 = 2mn^2 - \frac{2}{3}n^3$.

Now, note that for $A = Q\begin{bmatrix} R_1 \\ O \end{bmatrix}$, the lower block of zeros means many rows on the right of Q are NOT used at all. If we write $Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}$ accordingly, we see that $A = Q_1R_1$. So we have the following corollary:

Corollary 6.9.23. *Given any $m \times n$ matrix A with $m \geq n$, we have QR decomposition $A = QR$ where Q is $m \times n$ matrix with orthonormal columns, and R is $n \times n$ and upper triangular with positive diagonal entries.*

What if you have an $m \times n$ matrix A with $m < n$? Then A^{T} has a QR decomposition $A^{\mathrm{T}} = QR$, and we see that $A = R^{\mathrm{T}}Q^{\mathrm{T}}$ where R^{T} is lower triangular, and Q^{T} is still orthogonal. So we have this so-called RQ decomposition instead.

Furthermore, recall that for Gram-Schmidt, we are working on columns of A from left to right. What if you do this from right to left? Then you have $A = QR$ where R is lower triangular. In conclusion, you can have QR (if $m \geq n$) or RQ (if $m \leq n$), and R maybe upper or lower triangular. In our class, for simplicity, we usually just do QR where R is upper triangular.

6.10 Projections and Applications

6.10.1 Algebraic Projections

The study of orthogonality cannot be complete without the study of orthogonal projections, which is our goal here. However, we take a slight detour and investigate a special property of matrices. This property will end up surprisingly connected to projections.

Definition 6.10.1. *A linear map $P : V \to V$ (i.e., a square matrix) is **idempotent** if $P^2 = P$.*

Note that $P^2 = P$ immediately implies that $P^k = P$ for all positive integer k. ("idem" means "itself", and "potent" means "power". So "idempotent" literally means that its powers are all itself.) However, P might not be invertible. In fact, most of the time it is not.

Example 6.10.2. If P is invertible, we see that $P^2 = P$ implies $P = I$. So the only invertible idempotent matrix is I.

Apart from I, all other idempotent matrices are NOT invertible. Another easy example is the zero matrix O. We clearly have $O^2 = O$. ☺

As will be evident in the next examples, trace is an important aspect of the study of idempotent matrices.

Definition 6.10.3. *We define the **trace** of a square matrix A to be* $\mathrm{trace}(A) = \sum a_{ii}$*, the sum of diagonal entries of A.*

Proposition 6.10.4. $\mathrm{trace}(AB) = \mathrm{trace}(BA)$ *for any $m \times n$ matrix A and $n \times m$ matrix B.*

Proof. See homework. □

Remark 6.10.5. $\mathrm{trace}(AB) = \mathrm{trace}(BA)$ *means we can "cyclically permute" matrix multiplications while preserving the trace. For example* $\mathrm{trace}(ABC) = \mathrm{trace}(A(BC)) = \mathrm{trace}((BC)A) = \mathrm{trace}(BCA)$. *(Note that non-cyclic permutations might change the trace. We do not have* $\mathrm{trace}(ABC) = \mathrm{trace}(ACB)$ *in general. Take* $A = C^{\mathrm{T}} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ *for example.)*

Let us see some examples of idempotent matrices. They are indeed projections!

Example 6.10.6. Consider a diagonal matrix D whose diagonal entries are all 0 or 1. You can also immediately see that it has $D^2 = D$. In fact, these are the only idempotent diagonal matrices. (Can you see why? This is essentially due to the fact that $x^2 = x$ has only two solutions, $x = 0$ and $x = 1$.)

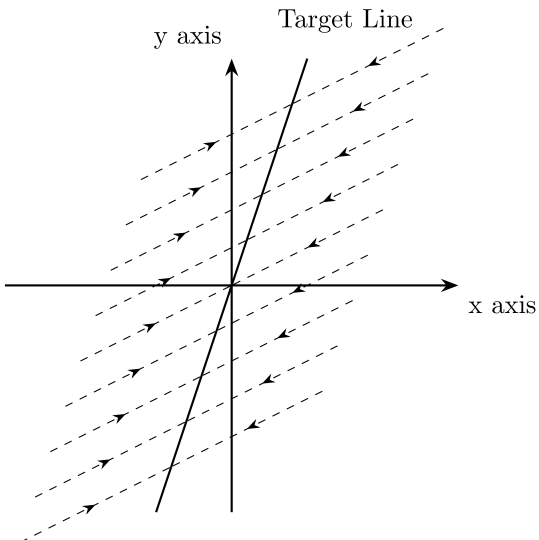
Consider such a matrix, say $D = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}$. What is this? It will send a generic vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ to $\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$. So this is a projection to the xy-plane. In fact, consider any diagonal matrix D whose diagonal entries are 0 or 1, then you see that it is a projection to some coordiate-subspaces (i.e., subspaces spanned by some coordinate axis).

It is obvious that in this case, $\mathrm{trace}(D) = \mathrm{rank}(D)$ is the dimension of the target space of your projection. ☺

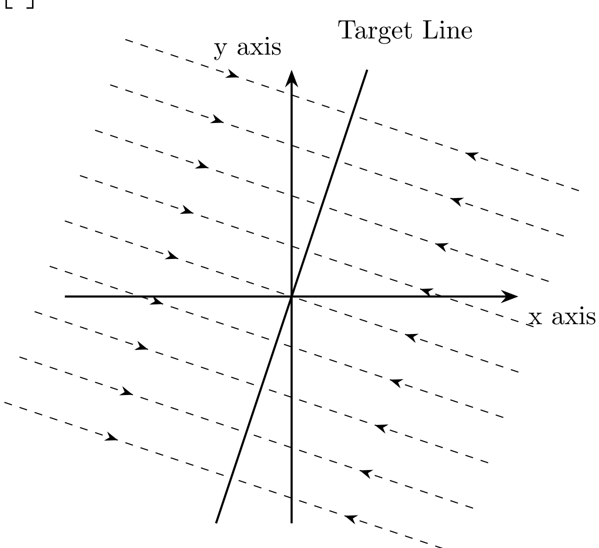
Example 6.10.7. Here let us see a non-diagonal example. Consider $P = \begin{bmatrix} -0.2 & 0.4 \\ -0.6 & 1.2 \end{bmatrix}$. You can easily verify that $P^2 = P$. What does this matrix do?

In this case, one can easily see that $\mathrm{Ran}(P)$ is spanned by $\begin{bmatrix}1\\3\end{bmatrix}$. It will map ALL vectors to the direction of $\begin{bmatrix}1\\3\end{bmatrix}$. Furthermore, you can check that $P\begin{bmatrix}1\\3\end{bmatrix} = \begin{bmatrix}1\\3\end{bmatrix}$. In fact, $P^2 = P$ literally means that P fixes its own range. This has a very "projection" feel to it.

This is what we call an ***oblique projection***. It will project everything to the line $y = 3x$ on the plane $\mathbb{R}^2$, but NOT orthogonally. Rather, every point goes to this line along the direction of $\begin{bmatrix}2\\1\end{bmatrix}$. You can check in particular that $\mathrm{Ker}(P)$ is spanned by $\begin{bmatrix}2\\1\end{bmatrix}$.



Compare this with the matrix $Q = \begin{bmatrix}0.1 & 0.3\\0.3 & 0.9\end{bmatrix}$. You will again find that $Q^2 = Q$, and $\mathrm{Ran}(Q)$ is spanned by $\begin{bmatrix}1\\3\end{bmatrix}$ as well. However, now the kernel is orthogonal to this range, and we have an ***orthogonal projection***.



Note that we have $\mathrm{trace}(P) = \mathrm{trace}(Q) = 1$, which is the dimension of the target space of our projections. This is exactly the case of diagonal idempotent matrices!

Finally, for the matrix P above, consider $I-P = \begin{bmatrix} 1.2 & -0.4 \\ 0.6 & -0.2 \end{bmatrix}$. You can easily verify that $(I-P)^2 = I-P$ as well. What does this matrix do?

In this case, one can easily see that $\operatorname{Ran}(I-P)$ is spanned by $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, so $\operatorname{Ran}(I-P) = \operatorname{Ker}(P)$. Similarly, you can verify that $\operatorname{Ker}(I-P) = \operatorname{Ran}(P)$. The matrix $I-P$ projects to exactly what P kills, and it kills exactly what P projects! it is the "complement projection" of P.

Again, you can verify that $\operatorname{trace}(I-P) = 1$, which is again the dimension of the target space of this projections. ☺

We are now ready to understand idempotent matrices. On one hand, obviously all (oblique or orthogonal) projections must be idempotent, because projecting twice is the same as projecting once. On the other hand, the following proposition states that, as linear maps, idempotent matrices are (oblique or orthogonal) projections. So the two are the same.

Proposition 6.10.8. *If $P : V \to V$ is idempotent, we have the following conclusions:*

1. $\operatorname{Ran}(P)$ *is the collection of vectors $\boldsymbol{v}$ such that $P\boldsymbol{v} = \boldsymbol{v}$.*
2. *$I-P$ is also idempotent.*
3. $\operatorname{Ran}(I-P) = \operatorname{Ker}(P)$ *and* $\operatorname{Ran}(P) = \operatorname{Ker}(I-P)$.
4. $\operatorname{Ran}(P)$ *and* $\operatorname{Ran}(I-P) = \operatorname{Ker}(P)$ *are complement subspaces. (So P and $I-P$ are projections onto complement subspaces.)*
5. *For any $\boldsymbol{v} \in V$, then $\boldsymbol{v} = P\boldsymbol{v} + (I-P)\boldsymbol{v}$ is the unique decomposition such that the first summand is in* $\operatorname{Ran}(P)$ *and the second summand is in* $\operatorname{Ran}(I-P)$.

Proof. Suppose $\boldsymbol{v} \in \operatorname{Ran}(P)$, say $\boldsymbol{v} = P\boldsymbol{w}$. Then $P\boldsymbol{v} = P^2\boldsymbol{w} = P\boldsymbol{w} = \boldsymbol{v}$. Conversely, if $\boldsymbol{v} = P\boldsymbol{v}$, then obviously $\boldsymbol{v} \in \operatorname{Ran}(P)$. So $\operatorname{Ran}(P)$ is exactly the collection of fixed points of P.

Now $(I-P)^2 = I - 2P + P^2 = I - 2P + P = I - P$, so $I-P$ is also idempotent.

Note that $\boldsymbol{v} \in \operatorname{Ran}(P)$ iff $P\boldsymbol{v} = \boldsymbol{v}$ iff $(I-P)\boldsymbol{v} = \boldsymbol{0}$, so we see that $\operatorname{Ran}(I-P) = \operatorname{Ker}(P)$. Apply this fact to the idempotent $I-P$, we get $\operatorname{Ker}(I-P) = \operatorname{Ran}(I-(I-P)) = \operatorname{Ran}(P)$.

Now let us show that $\operatorname{Ran}(P)$ and $\operatorname{Ker}(P)$ are complements. If $\boldsymbol{v} \in \operatorname{Ran}(P) \cap \operatorname{Ker}(P)$, then $\boldsymbol{v} = P\boldsymbol{v} = \boldsymbol{0}$. So the intersection is zero. On the other hand, the decomposition $\boldsymbol{v} = P\boldsymbol{v} + (I-P)\boldsymbol{v}$ holds for all $\boldsymbol{v} \in V$, so $\operatorname{Ran}(P) + \operatorname{Ran}(I-P) = V$. So $\operatorname{Ran}(P)$ and $\operatorname{Ker}(P) = \operatorname{Ran}(I-P)$ are complement subspaces.

Any decomposition according to complement subspaces must be unique. Hence the decomposition $\boldsymbol{v} = P\boldsymbol{v} + (I-P)\boldsymbol{v}$ is unique. □

The important takeaway here is that whenever you have an idempotent P, you should always think about the pair P and $I-P$. They give you two complement projections. Note that so far, we have NOT used any inner product structure in our definition of idempotent matrices and the derivation of properties. These are true for all abstract vector spaces (finite or infinite dimensional), and without an inner product structure, we cannot differentiate between oblique and orthogonal projections. We simply call them all projections, and none is preferred over others.

We are missing one last piece of the puzzle. We have seen in previous examples that $\operatorname{trace}(P) = \dim \operatorname{Ran}(P)$, it gives the dimension of the subspace we are projecting onto. This is NOT a coincidence.

Proposition 6.10.9. *If P is an idempotent square matrix, then* $\operatorname{trace}(P) = \operatorname{rank}(P)$.

Proof. Here we shall use the fact that $\operatorname{trace}(AB) = \operatorname{trace}(BA)$, whose proof is in the homework.

Suppose P is acting on $\mathbb{R}^n$, so it is a square matrix.

If $\dim \operatorname{Ran}(P) = 1$, then P is rank one, so $P = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ for some vectors $\boldsymbol{u}, \boldsymbol{v}$. Furthermore, $\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}} = P = P^2 = \boldsymbol{u}(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{u})\boldsymbol{v}^{\mathrm{T}} = (\boldsymbol{v}^{\mathrm{T}}\boldsymbol{u})P$, so we have $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{u} = 1$. It is easy to see now $\operatorname{trace}(P) = \sum v_i u_i = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{u} = 1$.

Now suppose $\dim \operatorname{Ran}(P) = r$. Then $P = UV$ where U is $n \times r$ and injective, and V is $r \times n$ and surjective. Then $UV = P = P^2 = UVUV$.

But since U is injective, we have the law of left cancellation for U. So $UV = UVUV$ implies that $V = VUV$. Also, since V is surjective, we have the law of right cancellation for V. So $V = VUV$ implies that $I_{r\times r} = VU$.

So $\operatorname{trace}(P) = \operatorname{trace}(UV) = \operatorname{trace}(VU) = \operatorname{trace}(I_{r\times r}) = r$. Note that this is essentially the same proof as above. □

Here it might seem a bit surprising that the trace is the rank. Nominally trace seems to be computed from entries, which are dependent on our choice of basis. But $\dim \operatorname{Ran}(P)$ is the rank, which is independent of choice of basis (and also independent of any inner product structure). So could it be that trace is in fact independent of basis?

Definition 6.10.10. *We say two square matrices A, B are **similar** if we can find invertible C such that $A = CBC^{-1}$.*

Note that the domain and codomain of a square matrix are the same. If we perform a change of basis simultaneously, we would result in similar matrices.

For example, consider $L : V \to V$, and V has basis $\mathcal{B}, \mathcal{C}$. Then $L_{\mathcal{C}\leftarrow\mathcal{C}} = I_{\mathcal{C}\leftarrow\mathcal{B}} L_{\mathcal{B}\leftarrow\mathcal{B}} I_{\mathcal{B}\leftarrow\mathcal{C}}$. Note that on the right side of the equation, the left-most and right-most matrices are inverse of each other. So $L_{\mathcal{C}\leftarrow\mathcal{C}}$ and $L_{\mathcal{B}\leftarrow\mathcal{B}}$ are similar matrices.

Now, trace for linear transformations is indeed independent of any choice of basis (and also independent from inner product structure).

Proposition 6.10.11. *If A, B are similar matrices, then they have the same trace.*

Proof. We have $\operatorname{trace}(CAC^{-1}) = \operatorname{trace}(C^{-1}CA) = \operatorname{trace}(A)$. □

Interpreting C as such a change of basis, then A, CAC^{-1} are referring to the same linear transformation, differ only by a change of basis. (Recall that a linear transformation is a linear map whose domain and codomain are the same.) In this sense, trace is really about the underlying linear map, and not about the nominal matrix.

Let us conclude this section now. A projection is P with $P^2 = P$, and the subspace it is projecting to is $\operatorname{Ran}(P)$, and the dimension is $\operatorname{trace}(P)$.

6.10.2 Orthogonal Projection

Previously we study projections without any reference to the inner product structure. Now, with an inner product structure, we see that most projections are probably oblique projections. However, we are more interested in orthogonal projections. These are projections that kills things orthogonal to $\operatorname{Ran}(P)$.

Definition 6.10.12. *A projection $P : V \to V$ is an **orthogonal projection** if $\operatorname{Ran}(P) \perp \operatorname{Ran}(I - P)$. (I.e., $\operatorname{Ran}(P) \perp \operatorname{Ker}(P)$.)*

The nice thing about orthogonal projections is that they give "best approximation in W" of any vector $\boldsymbol{v} \in V$. This makes them extremely useful.

Proposition 6.10.13. *If $P : V \to V$ is an orthogonal projection to $W \subseteq V$, then $P\boldsymbol{v}$ is the unique vector in W closest to $\boldsymbol{v}$. I.e., we have $\|\boldsymbol{v} - \boldsymbol{w}\| \geq \|\boldsymbol{v} - P\boldsymbol{v}\|$ for all $\boldsymbol{w} \in W$, with equality iff $\boldsymbol{w} = P\boldsymbol{v}$.*

Proof. Consider the decomposition $\boldsymbol{v} = P\boldsymbol{v} + (I - P)\boldsymbol{v}$. From the geometric meaning, we see that $P\boldsymbol{v}$ should be inside W, while $(I - P)\boldsymbol{v} \in \operatorname{Ran}(I - P) = \operatorname{Ker}(P) = \operatorname{Ran}(P)^\perp = W^\perp$.

Now our goal is to compare $\boldsymbol{v} - \boldsymbol{w}$ and $\boldsymbol{v} - P\boldsymbol{v}$. However, note that the vectors $\boldsymbol{v} - \boldsymbol{w}, \boldsymbol{v} - P\boldsymbol{v}, \boldsymbol{w} - P\boldsymbol{v}$ form a triangle. Furthremore, $\boldsymbol{w} - P\boldsymbol{v}$ is a linear combination of vectors in W, hence it is still in W. On the other hand, $\boldsymbol{v} - P\boldsymbol{v} = (I - P)\boldsymbol{v}$ is perpendicular to everything in W. Hence the triangle made of $\boldsymbol{v} - \boldsymbol{w}, \boldsymbol{v} - P\boldsymbol{v}, \boldsymbol{w} - P\boldsymbol{v}$ is a right triangle.

So by Pythagorean theorem, $\|\boldsymbol{v} - \boldsymbol{w}\|^2 = \|P\boldsymbol{v} - \boldsymbol{w}\|^2 + \|(I - P)\boldsymbol{v}\|^2 \geq \|(I - P)\boldsymbol{v}\|^2 = \|(\boldsymbol{v} - P\boldsymbol{v}\|^2$, and equality holds iff $\|P\boldsymbol{v} - \boldsymbol{w}\|^2 = 0$ iff $\boldsymbol{w} = P\boldsymbol{v}$.

(This proof is highly geometric. See if you can draw the picture. This is hilariously a geometric proof to an analytic property of an algebraic operator.) □

We have previously seen that for a subspace, it might have many oblique projections. But this is not the case any more. Another nicest thing about an orthogonal projection is that it is unique for the subspace.

Proposition 6.10.14. *Given a subspace $W \subseteq V$, then there is a unique orthogonal projection $P : V \to V$ with $\mathrm{Ran}(P) = W$.*

Proof. Uniqueness is guaranteed by the last proposition, since the image of $\boldsymbol{v}$ after orthogonal projection to W must be the unique vector in W closest to $\boldsymbol{v}$. Let us show existence.

One idea is to simply define $P\boldsymbol{v}$ as the unique vector in W closest to $\boldsymbol{v}$. Then we can manually and painstakingly verify that P is linear, and $P^2 = P$, and $W = \mathrm{Ran}(P) \perp \mathrm{Ker}(P)$.

Here is another approach. Suppose WLOG that $V = \mathbb{R}^n$. Find any ONB for W, say $\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$. Then extend this to a full ONB of the whole space $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$.

For any $\boldsymbol{v} \in V$, we have an orthogonal decomposition $\boldsymbol{v} = \boldsymbol{w}_1\boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{v} + \cdots + \boldsymbol{w}_n\boldsymbol{w}_n^{\mathrm{T}}\boldsymbol{v}$. However, in this decomposition, the portion $\boldsymbol{w}_1\boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{v} + \cdots + \boldsymbol{w}_k\boldsymbol{w}_k^{\mathrm{T}}\boldsymbol{v}$ is inside W, while the rest $\boldsymbol{w}_{k+1}\boldsymbol{w}_{k+1}^{\mathrm{T}}\boldsymbol{v} + \cdots + \boldsymbol{w}_n\boldsymbol{w}_n^{\mathrm{T}}\boldsymbol{v}$ is perpendicular to W.

In particular, let $P = \sum_{i=1}^{k} \boldsymbol{w}_i\boldsymbol{w}_i^{\mathrm{T}}$, then this is the desired projection.

Let us reformulate the approach above. (Consider this the third proof if you like.) Pick any ONB $Q = (\boldsymbol{w}_1, \dots, \boldsymbol{w}_k)$ for W, and set $P = QQ^{\mathrm{T}}$. We aim to show that P is the desired orthogonal projection to W.

To start, $\mathrm{Ker}(P) = \mathrm{Ker}(QQ^{\mathrm{T}}) = \mathrm{Ker}(Q^{\mathrm{T}}) = \mathrm{Ran}(Q)^{\perp} = W^{\perp}$.

Since P is symmetric, we also have $\mathrm{Ran}(P) = \mathrm{Ran}(P^{\mathrm{T}}) = \mathrm{Ker}(P)^{\perp} = (W^{\perp})^{\perp} = W$. So far so good.

Finally, note that the (i,j) entry of $Q^{\mathrm{T}}Q$ is $\boldsymbol{e}_i^{\mathrm{T}}Q^{\mathrm{T}}Q\boldsymbol{e}_j = \boldsymbol{w}_i^{\mathrm{T}}\boldsymbol{w}_j$. Since columns of Q form ONB, this is 1 if $i = j$ and 0 if $i \neq j$. Hence $Q^{\mathrm{T}}Q = I_{k\times k}$. So we have $P^2 = QQ^{\mathrm{T}}QQ^{\mathrm{T}} = QI_{k\times k}Q^{\mathrm{T}} = QQ^{\mathrm{T}} = P$. So P is the desired orthogonal projection. $\square$

Here is an important remark. Compare the following two formula:

1. For a unit vector $\boldsymbol{u}$, the orthogonal projection to the line spanned by $\boldsymbol{u}$ is the matrix $\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$.

2. For an ONB Q for a subspace W, the orthogonal projection to $\mathrm{Ran}(Q)$ is the matrix QQ^{T}.

It is quite obvious that the projection formula QQ^{T} is a generalization of our old formula $\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$.

Furthermore, here is something else to be careful. Note that columns of Q do NOT form a basis for the whole space, merely for a subspace. So Q is not square. If $\dim W = k$, then Q is $n \times k$ and it is injective. So $P = QQ^{\mathrm{T}}$ is an $n \times n$ matrix. We alsk know that $\mathrm{Ran}(P) = W$, hence this matrix has rank exactly k. So P is NOT invertible when $k < n$.

However, $Q^{\mathrm{T}}Q$ would be a $k \times k$ matrix. And in fact, since columns of Q form a basis for W, we have $\mathrm{Ker}(Q) = \{\mathbf{0}\}$, and hence $\mathrm{Ker}(Q^{\mathrm{T}}Q) = \mathrm{Ker}(Q) = \{\mathbf{0}\}$. So $Q^{\mathrm{T}}Q$ is always invertible. In fact, as we have seen in the calculations in the proposition above, we always have $Q^{\mathrm{T}}Q = I_{k\times k}$.

Let me again stress these two formula. If columns of Q represent an ONB for a subspace W of dimension k, then

1. QQ^{T} is the orthogonal projection to W.

2. $Q^{\mathrm{T}}Q = I_{k\times k}$.

Now, if $P^2 = P$, then it is a (possibly oblique) projection. Can we tell which projections are orthogonal? We can indeed. This turns out to be surprisingly easy.

Proposition 6.10.15. *A square matrix P is an orthogonal projection iff $P^2 = P$ and $P^{\mathrm{T}} = P$. (I.e., orthogonal projection = symmetric projection.)*

Note that we implicitly used the inner product structure because we used transpose. And here we are refering to the dot product as the inner product. For abstract inner product spaces, pick orthonormal basis and then do this.

Proof. Suppose $P^2 = P$ and $P^{\mathrm{T}} = P$. Then $\mathrm{Ran}(P) = \mathrm{Ran}(P^{\mathrm{T}}) = \mathrm{Ker}(P)^{\perp}$. Done.

Suppose P is an orthogonal projection, then it is QQ^{T} as above. Then we can easily verify that $P^2 = P$ and $P = P^{\mathrm{T}}$. □

So orthogonal projections are uniquely defined, easy to identify, and very useful. To make it even better, we have another formula.

Proposition 6.10.16. *Given any subspace $W \subseteq \mathbb{R}^n$, let $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ be a basis, and let $A = \begin{bmatrix}\boldsymbol{v}_1 & \dots & \boldsymbol{v}_k\end{bmatrix}$. Then the orthogonal projection to W is $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$.*

In general, if A is injective, then $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ is the orthogonal projection to $\mathrm{Ran}(A)$.

Proof. Since A is injective, $A^{\mathrm{T}}A$ is also injective. (See previous homework.) But $A^{\mathrm{T}}A$ is also square, so it is bijective.

Columns of A is a basis for W. To find ONB for W, we perform Gram-Schmidt on A, which means we do QR decomposition $A = QR$ where Q has orthonormal columns and R is upper triangular with positive diagonal entries. So columns of Q form ONB for W, and R is invertible.

Now

$$\begin{aligned}A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} =& QR(R^{\mathrm{T}}Q^{\mathrm{T}}QR)^{-1}R^{\mathrm{T}}Q^{\mathrm{T}} \\ =& QR(R^{\mathrm{T}}R)^{-1}R^{\mathrm{T}}Q^{\mathrm{T}} \\ =& QRR^{-1}(R^{\mathrm{T}})^{-1}R^{\mathrm{T}}Q^{\mathrm{T}} \\ =& QQ^{\mathrm{T}}.\end{aligned}$$

So this is indeed the desired orthogonal projection. □

Now the proof above may seem like magic. Let us now try to demystify it by doing the following comparision of formula.

1. Given a (non-unit) vector $\boldsymbol{v}$, the orthogonal projection to the line spanned by $\boldsymbol{v}$ is $\frac{\boldsymbol{v}\boldsymbol{v}^{\mathrm{T}}}{\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}}$.
2. Given a matrix A with independent columns, the orthogonal projection to $\mathrm{Ran}(A)$ is $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$.

Look at the denominator of $\frac{\boldsymbol{v}\boldsymbol{v}^{\mathrm{T}}}{\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}}$ and the inversed portion of $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$, you shall see that they are pretty much the same formula, with the latter generalizing the former.

Remark 6.10.17. *In the expression $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$, why don't we expand the parenthesis?*

If you try, you see that $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}} = AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}} = I$. Huh, this CANNOT be! What went wrong?

The failure of this logic lies in the fact that the identity $(AB) = B^{-1}A^{-1}$ has an ASSUMPTION! Only for square matrices this is true.

In our case, A is $n \times k$, so you CANNOT use this formula.

In fact, if $n = k$, then A being injective means it is bijective, $\mathrm{Ran}(A) = \mathbb{R}^n$ *the whole space. Then projection to the whole space obviously must be I. This is why expanding the parenthesis gives the identity matrix as our projection.*

You can prove that in general, for $n \times k$ matrix A, AA^{T} is $n \times n$ with rank the same as A, and $A^{\mathrm{T}}A$ is $k \times k$ with rank the same as A. In our case, A has rank k, hence $A^{\mathrm{T}}A$ must be invertible, and AA^{T} is not invertible when $k < n$.

6.10.3 Applications of orthogonal projections

As we have discussed before, orthogonal projections are highly useful because they gives approximations. Consider the following example.

Example 6.10.18. Suppose we want to solve the linear system:

$$\begin{bmatrix}1 & 0\\ 0 & 1\\ 1 & 1\end{bmatrix}\boldsymbol{x} = \begin{bmatrix}2\\ 1\\ 3.001\end{bmatrix}.$$

Oops, we have no solution. But do we declare failure? No we do not. Rather, we declare this to be FAKE NEWS and we firmly believe that there must be a solution. So what could happen? Well, it DOES look suspicious. We ALMOST have a solution of $\boldsymbol{x} = \begin{bmatrix}2\\1\end{bmatrix}$, if only that third coordinates of $\boldsymbol{b}$ changes a tiny little bit. In fact, everything else are integers, why the hell is this coordinate 3.001? Probably due to some error.

Sometimes we collect data, and try to solve $A\boldsymbol{x} = \boldsymbol{b}$. We know there must be a solution, because this is a real life problem where the solution must exist. However, possibly due to unaccounted noise or minor errors, it appears our system has no solution. What to do now?

In this case, one should try to find the BEST $\boldsymbol{x}$, such that $A\boldsymbol{x}$ is as close as $\boldsymbol{b}$ as possible!

So what could $A\boldsymbol{x}$ be? Well, it is obviously always in $\mathrm{Ran}(A)$. We want the vector in $\mathrm{Ran}(A)$ that is closest to $\boldsymbol{b}$, i.e., we want to project $\boldsymbol{b}$ orthogonally to the subspace $\mathrm{Ran}(A)$.

Luckily we already know how to do this! If A is injective (it usually is in practice), we just need $\boldsymbol{b}' = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$ and solve $A\boldsymbol{x} = \boldsymbol{b}'$ instead.

Let us simplify a bit. We have $\boldsymbol{b}' = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$ and $A\boldsymbol{x} = \boldsymbol{b}'$. This means $A\boldsymbol{x} = \boldsymbol{b}' = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$. Since we assumed that A is injective, by left cancellation we have $\boldsymbol{x} = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$. Yay! We now have a formula to find this $\boldsymbol{x}$, so that $A\boldsymbol{x}$ is as close to $\boldsymbol{b}$ as possible. This is the "closest" to a solution of $A\boldsymbol{x} = \boldsymbol{b}$.

Now the formula is actually not that good. To actually calculate $\boldsymbol{x}$ using this formula, you would need to calculate the inverse of some matrix, which takes a long time to do. Therefore, we can further simplify this expression to $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$ instead, and solve this using Gaussian elimination. Finally, no matrix inversion! Finding $\boldsymbol{x}$ using $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$ is in fact faster than finding $\boldsymbol{x}$ using $\boldsymbol{x} = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$. (Also, finding the inverse of a matrix would usually results in magnifying the error term, if any initial error is involved.)

To sum up, how to find the best approximated solution? Given a system $A\boldsymbol{x} = \boldsymbol{b}$ which might not have a solution, we first apply A^{T} to both sides, and then solve $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$ instead. ☺

Definition 6.10.19. *Given A, a least square solution to a system $A\boldsymbol{x} = \boldsymbol{b}$ is the solution to the system $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$.*

Why is it called the least square solution? Because that is what orthogonal projection would do.

Proposition 6.10.20. *For any linear system $A\boldsymbol{x} = \boldsymbol{b}$ which may or may not have a solution, then $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$ always have a solution. Furthremore, if $\boldsymbol{x}_0$ is a solution, then $A\boldsymbol{x}_0$ is as close to $\boldsymbol{b}$ as possible.*

Proof. First of all, let us show that there is a solution. Note that the right hand side is

$$A^{\mathrm{T}}\boldsymbol{b} \in \mathrm{Ran}(A^{\mathrm{T}}) = \mathrm{Ker}(A)^{\perp} = \mathrm{Ker}(A^{\mathrm{T}}A)^{\perp} = \mathrm{Ran}(A^{\mathrm{T}}A).$$

Hence we can always find some $\boldsymbol{x}$ such that $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$. Done.

Now suppose $\boldsymbol{x}_0$ satisfies $A^{\mathrm{T}}A\boldsymbol{x}_0 = A^{\mathrm{T}}\boldsymbol{b}$. Let us show that $A\boldsymbol{x}_0$ is the unique vector in $\mathrm{Ran}(A)$ closest to $\boldsymbol{b}$.

First we perform full rank decomposition $A = BC$, where B is injective and C is surjective. Next we perform QR decomposition $B = QR$. So Q has orthonormal columns and R is invertible. By the lemma below, we have $\mathrm{Ran}(A) = \mathrm{Ran}(Q)$, hence the projection to $\mathrm{Ran}(A)$ is just QQ^{T}.

Now $A = QRC$, where RC is surjective. Let $S = RC$ for simplicity. Plug in $A = QS$ to $A^{\mathrm{T}}A\boldsymbol{x}_0 = A^{\mathrm{T}}\boldsymbol{b}$, we have

$$S^{\mathrm{T}}Q^{\mathrm{T}}QS\boldsymbol{x}_0 = S^{\mathrm{T}}Q^{\mathrm{T}}\boldsymbol{b}.$$

This simplifies to

$$S^{\mathrm{T}}S\boldsymbol{x}_0 = S^{\mathrm{T}}Q^{\mathrm{T}}\boldsymbol{b}.$$

Since S is surjective, S^{T} is injective and has left cancellation. So $S\boldsymbol{x}_0 = Q^{\mathrm{T}}\boldsymbol{b}$. Now apply Q on both sides, we see that $A\boldsymbol{x}_0 = QQ^{\mathrm{T}}\boldsymbol{b}$. So our porposition is correct. □

Lemma 6.10.21. *For any $m \times n$ matrix A, we do the full rank decomposition $A = BC$ and then QR decomposition $B = QR$. Then* $\operatorname{Ran}(A) = \operatorname{Ran}(Q)$.

Proof. Homework. Use definition of surjectivity of C. $\square$

The conclusion is this. If you want to solve $A\boldsymbol{x} = \boldsymbol{b}$ but there is no solution, then simply solve $A^{\mathrm{T}}A\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$ instead. This gives you as close to a solution as possible.

Let me give you a practical example here.

Example 6.10.22. We uses CT scans to find tumor in a person. To simplify the problem, suppose this person only consists of four pixels, each pixel is a square with unit side length. Each pixel could be bones, flesh, blood or tumors. See the picture.

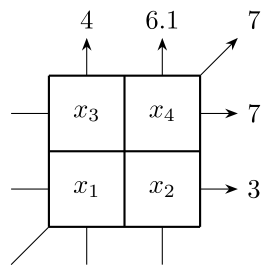


Figure 6.10.1: CT scan decay

Suppose we uses X-rays to go through these pixels, and the strength of the X-ray would decay by $1, 2, 3, 4$ units per unit length through bones, flesh, blood and tumors respectively. For example, if the lower left pixel is a bone and the lower right cell is flesh, then a ray going horizontally through the lower cells would decay by a total of $1 + 2 = 3$ units. And if the upper left cell is blood and the lower right cell is flesh, then a ray going diagonally from upper left to lower right would decay by a total of $\sqrt{2}(1 + 3) = 4\sqrt{2}$ units.

Now suppose the rate of decay in each pixel per unit length is x_1, x_2, x_3, x_4, and we uses five X-rays as shown. The total decay of each ray after measurement, is also shown. Can you figure out what is the content of each cell?

Our linear system is

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \sqrt{2} & 0 & 0 & \sqrt{2} \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 7 \\ 4 \\ 6.1 \end{bmatrix}.$$

If you attempt to solve this directly, you shall see that there is no solution. This is in fact always the case in practice. Dust in the air, trumbling patients and so on, there are always some noises that will give rise to inconsistencies.

However, we can try to find the least square solution. We consider the new system

$$\begin{bmatrix} 1 & 0 & \sqrt{2} & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & \sqrt{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \sqrt{2} & 0 & 0 & \sqrt{2} \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \sqrt{2} & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & \sqrt{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 7 \\ 4 \\ 6.1 \end{bmatrix}.$$

This simplifies to

$$\begin{bmatrix} 4 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 2 & 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 7 + 7\sqrt{2} \\ 9.1 \\ 11 \\ 13.1 + 7\sqrt{2} \end{bmatrix}.$$

Now the matrix on the left is invertible. So we solve for the least square solution, and find out that $x_1 \approx 0.950, x_2 \approx 2.075, x_3 \approx 3.025, x_4 \approx 4.000$. So we see that the four pixels x_1, x_2, x_3, x_4 corresponds to bones, flesh, blood and tumors. The tumor is located in pixel four, the upper right pixel. Problem solved.

Note that the matrix $B = \begin{bmatrix} 4 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 2 & 1 & 1 & 4 \end{bmatrix}$ depends only on the way we positioned our machine. So we are going to solve $B\boldsymbol{x} = A^{\mathrm{T}}\boldsymbol{b}$ for many different $\boldsymbol{b}$ again and again. So in practice, it is better to first perform the LU decomposition of B beforehand, and then solve for the solution according to each patient's $\boldsymbol{b}$. ☺

Example 6.10.23. Suppose we want to determine how our education effect your income. We collect data from n people, $(x_1, y_1), ..., (x_n, y_n)$, where x_i is the number of years of education received by the i-th person, and y_i is the eventual income of this person.

Suppose we believe in a linear model, that education and income should vaguely lies around some line $y = kx + b$. In this case, we aim to find the line that BEST FIT our data. Then I can claim that on average, an extra year of education shall increase your income by k.

So our model is $Y = kX + b + E$ where k, b are unknown constants and E represent factors other than education. X, Y here are random variables that represent the years of educations and income of a random person. How to find the best k, b to fit our data?

Let $\boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $\boldsymbol{u} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$, then ideally, IF the data lies on a line perfectly, we should have $\boldsymbol{y} = k\boldsymbol{x} + b\boldsymbol{u}$. Here $\boldsymbol{y}, \boldsymbol{x}, \boldsymbol{u}$ are all known, and we aim to solve for the unknown k and b.

Rearrange this, we are trying to solve $\begin{bmatrix} \boldsymbol{x} & \boldsymbol{u} \end{bmatrix} \begin{bmatrix} k \\ b \end{bmatrix} = \boldsymbol{y}$, which is a linear system. Now, of course there are other factors that influences one's income, so our data CANNOT lies on a line perfectly. So this system has no solution. What should we do? We go for the LEAST SQUARE solution instead.

So we are looking at the new system

$$\begin{bmatrix} \boldsymbol{x}^{\mathrm{T}} \\ \boldsymbol{u}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} & \boldsymbol{u} \end{bmatrix} \begin{bmatrix} k \\ b \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}^{\mathrm{T}} \\ \boldsymbol{u}^{\mathrm{T}} \end{bmatrix} \boldsymbol{y}.$$

This simplifies to

$$\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix} \begin{bmatrix} k \\ b \end{bmatrix} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}.$$

The solution is $k = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{\sum (x_i - \overline{x})^2}$ and $b = \overline{y} - k\overline{x}$, where $\overline{x} = \frac{1}{n}\sum x_i, \overline{y} = \frac{1}{n}\sum y_i$.

Now note that $k = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}$, and covariance is an "inner product" for random variables. So this is the projection formula of the random variable Y to the random variable X, wow! (Compare with the formula $\frac{\langle \boldsymbol{v}, \boldsymbol{w} \rangle}{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$ of projecting $\boldsymbol{w}$ to $\boldsymbol{v}$.) This surprisingly makes a lot of sense. If we are trying to predict Y via X, the best we can approximate is to get the projection of Y to X.

You can also check that in this case, $\mathbb{E}(E) = 0$ and $\mathrm{Var}(E)$ is minimized under this condition. So the "error term" is as small as possible.

This is a very simple case of ***linear regression***. ☺

Example 6.10.24. Does death penalty discourage murder? It seems logical. But theory must be tested by data. So let us see a way to test this.

Let $X = 1$ if a country or a state has death penalty, and $X = 0$ if a country or a state has no death penalty. Let Y be the murder rate in a state or a country. Then if you pick many many countries, you get many many data $(x_1, ..., x_n)$ and $(y_1, ..., y_n)$. Here all x_i are 0 or 1, while all y_i are some real numbers.

Let us fit these data into a line $Y = kX + b + E$ with the best possible k, b and $\mathbb{E}(E) = 0$. Then what does this mean? Well, if a country has no death penalty, then $X = 0$, and then we see that the average murder rate is b. If a country has death penalty, then $X = 1$, and then we see that the average murder rate is $b + k$. In particular, k is EXACTLY the effect of death penalty on murder rate.

After collecting data, we can compute $k = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}$, and $b = \bar{y} - \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}\bar{x}$. It turned out that, on almost all the studies conducted, k is positive!!! Oops. It turned out that death penalty is actually associated to a higher murder rate. That is very counter-intuitive. What happened?

Attempted explanation 1: Correlation is not causality. Maybe it is the other way around: Countries with higher murder rate is more likely to punish them severely, so maybe high Y value caused $X = 1$, instead of $X = 1$ causing high Y value. Unfortunately, this causality effect can be addressed with better statistic approach and better ways to collect data. (E.g., looking at countries that changed their death penalty laws, and compare the murder rate before and after.) To my knowledge, it turned out that even if we strictly investigate the causality of X on Y, k is STILL positive. So this explanation failed. Death penalty in fact lead to a higher murder rate.

Attempted Explanation 2: One argument is that death penalty encourage murder because death penalty is murder in itself. You are killing criminals, sure, but you planned and premeditated and then killed them. The message you are sending with a death penalty is that it is OK to kill when you have a justified reason. So people might think it is OK to kill when they feel justified for some reason.

Attempted Explanation 3: There is a minute but non-zero chance that all previous studies are all collectively unlucky, and we simply get the unlucky data. Maybe death penalty do deter murder. Who knows?

Also keep in mind that this is NOT an endorsement to the total revokation of death penalty. There are other philosophical concerns to think about. (E.g., Immanuel Kant.) ☺

6.11 Exercises

Exercise 6.11.1. *Suppose A is 3×4 and B is 4×5 and $AB = 0$.*

1. *Show that* $\mathrm{Ran}(B) \subseteq \mathrm{Ker}(A)$.
2. *Show that* $\mathrm{rank}(A) + \mathrm{rank}(B) \leq 4$.
3. *Show that if $A^2 = 0$, then* $\mathrm{rank}(A) \leq \frac{n}{2}$.
4. ☕*Suppose A is 3×4, B is 4×5, C is 5×6, and $ABC = 0$. What can we infer about* $\mathrm{rank}(A) + \mathrm{rank}(B) + \mathrm{rank}(C)$*?*

Exercise 6.11.2. *Recall that for any matrix A, we have* $\mathrm{Ker}(A^{\mathrm{T}}A) = \mathrm{Ker}(A)$.

1. *Do we have* $\mathrm{Ker}(AA^{\mathrm{T}}A) = \mathrm{Ker}(A)$*? Prove or show a counter example.*
2. *Do we have* $\mathrm{Ker}(A^{\mathrm{T}}AA^{\mathrm{T}}A) = \mathrm{Ker}(A)$*? Prove or show a counter example.*
3. *Do we have* $\mathrm{rank}(AA^{\mathrm{T}}) = \mathrm{rank}(A)$*? Prove or show a counter example.*
4. ☕*Suppose an $n \times n$ matrix A satisfies $AA^{\mathrm{T}}AAA^{\mathrm{T}} = 0$. What is the largest possible value of* $\mathrm{rank}(A)$*? (Hint: the previous problem on rank inequality for $A^2 = 0$ may help.)*

Exercise 6.11.3. *Given arbitrary subspaces of the domain and the codomain, could they be the four fundamental subspaces of some matrix?*

Let V be any subspace of $\mathbb{R}^n$, and W be any subspace of $\mathbb{R}^m$, and $\dim V = \dim W = d$. *We want to find a matrix whose four fundamental subspaces are $V, V^{\perp}, W, W^{\perp}$.*

1. *Let B be an $n \times d$ matrix whose column vectors form a basis for V. Show that* $\mathrm{Ran}(B) = V$ *and* $\mathrm{Ker}(B^{\mathrm{T}}) = V^{\perp}$, *and* $\mathrm{Ker}(B) = 0$.
2. *Let C be an $m \times d$ matrix whose column vectors form a basis for W. Show that* $\mathrm{Ran}(C) = W$ *and* $\mathrm{Ker}(C^{\mathrm{T}}) = W^{\perp}$, *and* $\mathrm{Ker}(C) = 0$.

3. *Show that $CB^{\mathrm{T}}\boldsymbol{x} = \mathbf{0}$ if and only if $B^{\mathrm{T}}\boldsymbol{x} = \mathbf{0}$. Similarly, $BC^{\mathrm{T}}\boldsymbol{x} = 0$ if and only if $C^{\mathrm{T}}\boldsymbol{x} = \mathbf{0}$.*

4. *Show that CB^{T} is the desired matrix, and in fact CB^{T} gives a full rank decomposition.*

Exercise 6.11.4 (Fredholm's Alternative)**.** *This is an interesting result with some future application in partial differential equations.*

We have two problems. Problem One is to solve the equations $A\boldsymbol{x} = \boldsymbol{b}$. Problem Two is to solve the eqautions $\begin{bmatrix} A^{\mathrm{T}} \\ \boldsymbol{b}^{\mathrm{T}} \end{bmatrix} \boldsymbol{y} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$.

Fredholm's Alternative is a theorem that states that: exactly one of these two problems has a solution. Not both, not neither.

1. *Suppose $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 5 \\ 5 \\ 9 \end{bmatrix}$. Verify Fredholm's Alternative here: $A\boldsymbol{x} = \boldsymbol{b}$ has no solution, while $\begin{bmatrix} A^{\mathrm{T}} \\ \boldsymbol{b}^{\mathrm{T}} \end{bmatrix} \boldsymbol{y} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$ has a solution.*

2. *In the subproblem above, if $\boldsymbol{y}$ is a solution, then show that the equations in $A\boldsymbol{x} = \boldsymbol{b}$ can be linearly combined into the equation $0 = 1$, by using coordinates of $\boldsymbol{y}$ as a coefficient. (This serves as an intuition for Fredholm's Alternative.)*

3. *If $\boldsymbol{b} \in \mathrm{Ran}(A)$, show that Problem One has a solution, while Problem Two has no solution.*

4. ☕*If $\boldsymbol{b} \notin \mathrm{Ran}(A)$, show that Problem One has no solution, while Problem Two has a solution. (Hint: consider the last non-zero row of $\mathrm{RREF}(\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix})$.)*

Exercise 6.11.5. *Given a subspace V of $\mathbb{R}^n$, how to find its orthogonal complement? First, we find a basis $A = \begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_k \end{bmatrix}$ for V. Then note that $V^{\perp} = \mathrm{Ran}(A)^{\perp} = \mathrm{Ker}(A^{\mathrm{T}})$. Therefore, we can use the kernel finding algorithm/ Gaussian elimination to find a basis for $\mathrm{Ker}(A^{\mathrm{T}})$.*

1. *Find the orthogonal complement of the span of $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$*

2. *Find a basis for the kernel of $\begin{bmatrix} 1 & 0 & 2 & 0 & 3 \\ & 1 & 4 & 0 & 5 \\ & & & 1 & 6 \end{bmatrix}$.*

3. *In $\mathbb{R}^3$, find a basis for the orthogonal complement of $\mathrm{span}(\begin{bmatrix} a \\ b \\ c \end{bmatrix})$ where $a \neq 0$. (This is a basis of the subspace with normal vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$.)*

4. *The chess matrix is*

$$A = \begin{bmatrix} r & n & b & q & k & b & n & r \\ p & p & p & p & p & p & p & p \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p & p & p & p & p & p & p & p \\ r & n & b & q & k & b & n & r \end{bmatrix}.$$

Here r, n, b, q, k, p are distinct non-zero real numbers. Find a basis for all four fundamental subspaces of A.

5. ☕*Suppose* $A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$ *and* $B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}$ *where F, G are both $n \times n$, and the four fundamental subspaces of A and B are identical. Prove that $F = G$.*

Exercise 6.11.6. *For square matrices, the range and the kernel are in the same space.*

1. *If A is symmetric, prove that* $\mathrm{Ran}(A) \perp \mathrm{Ker}(A)$.
2. *If A is skew-symmetric (i.e., $A^{\mathrm{T}} = -A$), prove that* $\mathrm{Ran}(A) \perp \mathrm{Ker}(A)$.
3. ☕*If A satisfies $A^{\mathrm{T}} = A^2$, prove that* $\mathrm{Ran}(A) \perp \mathrm{Ker}(A)$. *(As an extra challenge, can you find such a matrix A?)*
4. ☕*If A satisfies $AA^{\mathrm{T}} = A^{\mathrm{T}}A$, prove that* $\mathrm{Ran}(A) \perp \mathrm{Ker}(A)$. *(This is the definitive criteria for this phenomena. It includes all previous cases.)*

Exercise 6.11.7 (Polarization identity)**.** *This is just for fun.*

Suppose we are only allowed to do addition, substraction, and the operation $x \mapsto \frac{1}{x}$. Given two numbers x, y, can you calculate the multiplication xy?

1. *Can you produce the map $x \mapsto \frac{1}{x} - \frac{1}{x+1}$?*
2. *Can you produce the map $x \mapsto x^2$?*
3. *Can you produce the map $(x, y) \mapsto xy$?*

Exercise 6.11.8 (Orthogonal basis made of coordinates ± 1)**.** *A **Hadamard matrix** is a matrix H whose entries are all ± 1, and the columns are mutually orthogonal (but they don't have to be unit vectors). (When n is a power of 2, then an example of Hadamard matrix would be the matrix made by the Haar wavelet basis.)*

1. *For any $n \times n$ Hadamard matrix, compute $H^{\mathrm{T}}H$.*
2. *If H is a Hadamard matrix, show that $\begin{bmatrix} H & H \\ H & -H \end{bmatrix}$ is also a Hadamard matrix.*
3. *Can you find a 3×3 Hadamard matrix? Find it or show why not.*
4. *Let H be any Hadamard matrix. Then if we permute the rows and columns, or if we negate some rows and columns (i.e., multiply the row or culumn by -1), prove that the result is still a Hadamard matrix. (If two Hadamard matrices can be obtained from each other like this, then we say they are equivalent.)*
5. ☕*Prove that all 4×4 Hadamard matrices are equivalent.*
6. *(Read only) It is conjectured that a Hadamard matrix should exist for all $(4k) \times (4k)$ matrices. However, this remains unproven to this date. For examlpe, is there a 668×668 Hadamard matrix? We don't know the answer yet, as far as I can tell. The Haar wavelet basis gives a Hadamard matrix whenever n is a power of 2. But, for examlpe, try to find a Hadamard matrix when $n = 12$, if you want a challenge. And on the question of equivalence, in general, Hadamard matrices of the same size could be inequivalent. For example, there are five inequivalent 16×16 Hadamard matrices.*

Exercise 6.11.9 (A useful "non-inner product" space)**.** *Consider the space $\mathbb{R}^4$ where we define* $\langle \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ t_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ t_2 \end{bmatrix} \rangle = x_1x_2 + y_1y_2 + z_1z_2 - t_1t_2$. *This is NOT an inner product, but nevertheless, this is the structure used by special relativity. This is sometimes called a **Minkovski space-time**. We say $\boldsymbol{v} \perp \boldsymbol{w}$ if $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = 0$ under the definition above.*

1. *Find a matrix D such that $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}} D \boldsymbol{w}$. Is D symmetric? Positive Definite?*
2. ☕*For an arbitrary subspace $W \subseteq \mathbb{R}^4$, we define its "orthogonal complement" to be $W^{\perp} = \{\boldsymbol{v} \in \mathbb{R}^4 \,|\, \boldsymbol{v} \perp W\}$. Show that we always have $\dim W + \dim W^{\perp} = 4$. (Hint: use rank-nullity.)*
3. *Find a subspace strictly contained in its own "orthogonal complement". ("strictely contained" means they are not equal.)*
4. *Find a basis made of vectors of "length" zero for the Minkovski spacetime.*

Exercise 6.11.10. *Consider the following functions.*

1. $f(x, y, z) = 2x^2+4y^2+3z^2+4xy+4yz$. *Find a symmetric matrix A such that* $f(x, y, z) = \begin{bmatrix} x & y & z \end{bmatrix} A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

   Is A positive definite? If so, find the Chelosky decomposition of A, and then complete the squares for $f(x, y, z)$.
2. $f(x, y) = 4x^2 + 4xy + 2y^2 + 2y + 1$. *Find a symmetric matrix A such that* $f(x, y) = \begin{bmatrix} x & y & 1 \end{bmatrix} A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$.

   Is A positive definite? If so, find the Chelosky decomposition of A, and then complete the squares for $f(x, y)$.

Exercise 6.11.11. *Suppose $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ form an orthonormal basis for an inner product space V. Either prove that the followings are orthonormal basis, or perform Gram-Schmidt to them.*

1. $n = 3$ *and we have* $\boldsymbol{b}_1 = \frac{1}{3}(2\boldsymbol{a}_1 + 2\boldsymbol{a}_2 - \boldsymbol{a}_3), \boldsymbol{b}_2 = \frac{1}{3}(2\boldsymbol{a}_1 - \boldsymbol{a}_2 + 2\boldsymbol{a}_3), \boldsymbol{b}_3 = \frac{1}{3}(-\boldsymbol{a}_1 + 2\boldsymbol{a}_2 + 2\boldsymbol{a}_3)$.
2. $n = 5$, *and* $\boldsymbol{b}_1 = \boldsymbol{a}_1 + \boldsymbol{a}_5, \boldsymbol{b}_2 = \boldsymbol{a}_1 - \boldsymbol{a}_2 + \boldsymbol{a}_4, \boldsymbol{b}_3 = 2\boldsymbol{a}_1 + \boldsymbol{a}_2 + \boldsymbol{a}_3$.

Exercise 6.11.12. *Find bases to the following subspaces V and their orthogonal complement $V^{\perp}$ of $\mathbb{R}^4$. And write $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$ where $\boldsymbol{b}_1 \in V, \boldsymbol{b}_2 \in V^{\perp}$.*

1. *V is the hyper plane by the equation $x_1 + x_2 + x_3 - x_4 = 0$. And the vector is* $\boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.
2. *V is the span of $\boldsymbol{e}_1, \boldsymbol{e}_1 + \boldsymbol{e}_2$. And the vector is* $\boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \end{bmatrix}$.
3. *V is the span of* $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. *And the vector is* $\boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Exercise 6.11.13. *For the following matrices, find their QR decomposition, and find the matrix of orthogonal projection to their range.*

1. $\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix}$.
2. $\begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$.

3. $\begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

4. $\begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

Exercise 6.11.14. *Let* $Q = k \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$.

1. *Find k such that Q is an orthogonal matrix.*

2. *Find Q^{-1}. (Here k is unknown and Q might not be an orthogonal matrix.)*

3. *Find the orthogonal projection of* $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ *onto the span of the first column of Q.*

4. *Find the orthogonal projection of* $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ *onto the span of the first two columns of Q.*

5. *Let* $W = \frac{1}{2} \begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & 1 & -\sqrt{2} & 0 \\ 1 & -1 & 0 & \sqrt{2} \\ 1 & -1 & 0 & -\sqrt{2} \end{bmatrix}$. *And let's say k is chosen such that Q is an orthogonal matrix. Then columns of Q and columns of W are two bases of $\mathbb{R}^4$. Are they orthonormal? Find $I_{W \leftarrow Q}$ and $I_{Q \leftarrow W}$.*

Exercise 6.11.15. *Can we do QR to matrices where the columns are dependent? Let us see a special case. Say* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ *where a, c are not both zero. Suppose we have a QR decomposition $A = QR$*

1. *Prove that*
$$Q = \frac{1}{\sqrt{a^2 + c^2}} \begin{bmatrix} a & -c \\ c & a \end{bmatrix}, R = \frac{1}{\sqrt{a^2 + c^2}} \begin{bmatrix} a^2 + c^2 & ab + cd \\ 0 & ad - bc \end{bmatrix}.$$

2. *Find the QR decomposition of* $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ *and* $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ *using the formula above. For the latter one, note that R is not invertible.*

3. *If A is 3×3, this formula is much harder to do, and too complicated to be useful. But at least the formula for the first column of Q should be easy, right?*

Exercise 6.11.16. *Find the following projection matrices*

1. *Find the orthogonal projection to the plane $x - y - 2z = 0$ in $\mathbb{R}^3$.*

2. *Find the orthogonal projection to the orthogonal complement of the plane $x - y - 2z = 0$ in $\mathbb{R}^3$.*

3. *Find the orthogonal projection to the range of* $\begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$.

4. *Find the orthogonal projection to the kernel of* $\begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$. *(Hint: maybe first find the orthogonal projection to* $\mathrm{Ran}(A^{\mathrm{T}})$*.)*

5. *Find the orthogonal projection to the range of* $\frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$.

Exercise 6.11.17. *Some interesting facts about orthogonal projections.*

1. *If P is an orthogonal projection, show that the length squared of its i-th column is its (i,i) entry.*

2. *If A, B are orthogonal projections, show that $AB = BA$ if and only if AB is an orthogonal projection. Do you know to what space? (In terms of* $\mathrm{Ran}(A)$ *and* $\mathrm{Ran}(B)$*.)*

3. ☕*If A, B are orthogonal projections, show that $A+B$ is an orthogonal projection iff* $\mathrm{Ran}(A) \perp \mathrm{Ran}(B)$. *Do you know to what space? (In terms of* $\mathrm{Ran}(A)$ *and* $\mathrm{Ran}(B)$*.)*

Exercise 6.11.18. *Consider a matrix* $A = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix}$. *Let P_i be the orthogonal projection to* $\mathrm{span}(\boldsymbol{a}_1, \dots, \boldsymbol{a}_i)$.

1. *Prove that* $P_n P_{n-1} \dots P_2 P_1 = P_1$.

2. ☕*Prove that all P_i commutes.*

Exercise 6.11.19. *Let us solve the following problems by least square approximation.*

1. *Which multiple of* $\begin{bmatrix} 4 \\ 5 \\ 2 \\ 2 \end{bmatrix}$ *is closest to* $\begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$*?*

2. *Find the least square solution to* $\begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

3. *Find the least square solution to* $\begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} -4 \\ -3 \\ 3 \\ 0 \end{bmatrix}$.

Exercise 6.11.20 (Linear Regression)**.** *Suppose there are four points* $\begin{bmatrix} x_i \\ y_i \end{bmatrix}$ *on the xy-plane* $\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \end{bmatrix}, \begin{bmatrix} 4 \\ 20 \end{bmatrix}$. *Consider the following models. (Feel free to use a computer for these problems. I did not optimize the numbers in this problem.)*

1. *We wish to find the best line parallel to the x-axis to fit the data. I.e., our model is $y = b$, and we want to find the b to minimize $\sum_{i=1}^{4} |y_i - b|^2$. Find matrix A and vector $\boldsymbol{y}$ such that $A^{\mathrm{T}} A b = A^{\mathrm{T}} \boldsymbol{y}$ gives the best b as the solution.*

2. *We wish to find the best line through the origin to fit the data. I.e., our model is $y = kx$, and we want to find the k to minimize $\sum_{i=1}^{4} |y_i - kx_i|^2$. Find matrix A and vector $\boldsymbol{y}$ such that $A^{\mathrm{T}} A k = A^{\mathrm{T}} \boldsymbol{y}$ gives the best k as the solution.*

3. *We wish to find the best parabola to fit the data. I.e., our model is $y = ax^2 + bx + c$, and we want to find the a, b, c to minimize $\sum_{i=1}^{4} |y_i - ax_i^2 - bx_i - c|^2$. Find matrix A and vector $\boldsymbol{y}$ such that $A^{\mathrm{T}} A \begin{bmatrix} a \\ b \\ c \end{bmatrix} = A^{\mathrm{T}} \boldsymbol{y}$ gives the best a, b, c as the solution.*

Exercise 6.11.21 (Kalman Filter)**.** *Kalman filter is commonly used in GPS or any navigation systems. Here we do some easy special cases of it.*

Suppose we want to know a value x. But each time we measure it, there will be some noise, so we never get the accurate result. Suppose we measured n times and the results are $b_1, \ldots, b_n$. Then our best guess for x would be their average

$$\hat{x}_n = \frac{1}{n} \sum_{i=1}^{n} b_i.$$

Now we made another measurement and get b_{n+1}. Now I should have a better estimate, since I have more data now. The better estimate is

$$\hat{x}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} b_i.$$

However, a recalculation would be too slow. We would simply like to UPDATE the value of $\hat{x}_n$ to $\hat{x}_{n+1}$. Observe that

$$\hat{x}_{n+1} = \hat{x}_n + \frac{1}{n+1}(b_{n+1} - \hat{x}_n).$$

This is much faster than re-calculate the sum of $b_1, \ldots, b_{n+1}$ and divide by $n+1$. Here, observe that the relation between the update value $\hat{x}_{n+1} - \hat{x}_n$ and the deviation value $b_{n+1} - \hat{x}_n$ is linear. This is the fundation of Kalman filter.

1. *Verify that*

$$\hat{x}_{n+1} = \hat{x}_n + \frac{1}{n+1}(b_{n+1} - \hat{x}_n).$$

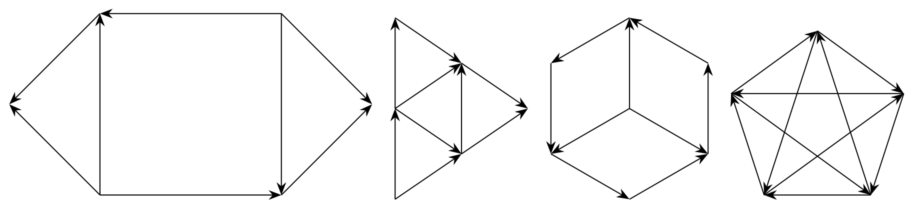
2. *Suppose $\boldsymbol{x}_0$ is the least square solution to a system $A_0 \boldsymbol{x} = \boldsymbol{b}_0$. But now we have more equations. Let $\boldsymbol{x}_1$ be the least square solution to the system $\begin{bmatrix} A_0 \\ A_1 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} \boldsymbol{b}_0 \\ \boldsymbol{b}_1 \end{bmatrix}$. Assume that $A_0^{\mathrm{T}} A_0 + A_1^{\mathrm{T}} A_1$ is invertible. Find a matrix K such that $\boldsymbol{x}_1 - \boldsymbol{x}_0 = K(\boldsymbol{b}_1 - A_1 \boldsymbol{x}_0)$.*

3. *(Read only) In practice, say A_1 is rank one, and we already know $(A_0^{\mathrm{T}} A_0)^{-1}$. Then we can simply use Sherman-Morrison to update $(A_0^{\mathrm{T}} A_0)^{-1}$ to $(A_0^{\mathrm{T}} A_0 + A_1^{\mathrm{T}} A_1)^{-1}$, so this inverse can also be found fast.*

Exercise 6.11.22 (Kirchhoff's voltage law)**.** *Consider the following electrical circuits. Label the vertices and edges yourself, write out the incidence matrix M for the circuit, and find a matrix A such that $\mathrm{Ker}(A) = \mathrm{Ran}(M)$. We require A to have as few rows as possible. (But we do not require proof. Just give the A is enough.)*

You don't need to do the last one (d). That one is just there for a challenge.

(Hint: Direct computations might take forever. The easy way is to use Kirchhoff's voltage law, which states that $\mathrm{Ran}(M)$ is made of voltages that adds up to zero on closed loops. This provides equations that vectors in $\mathrm{Ran}(M)$ should satisfy. These give many rows of A. Then, just make sure you don't use redundant equations.)



(a) (b) (c) (d)

Figure 6.11.1: Kirchhoff's Voltage Exercise

Part III

Coordinate Invariants

Chapter 7

Determinants

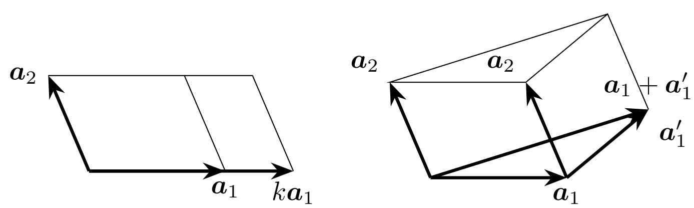
7.1 Introduction

7.1.1 Oriented Area and Oriented Volume

Given a parallelogram in $\mathbb{R}^2$, suppose two of its adjacent edges are $\boldsymbol{v} = \begin{bmatrix} a \\ c \end{bmatrix}, \boldsymbol{w} = \begin{bmatrix} b \\ d \end{bmatrix}$. Then instead we shall focus on the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Our goal here is to find the area of the parallelogram, i.e., $Area(\boldsymbol{v}, \boldsymbol{w})$ or $Area(A)$. How can we do this? First let us list some obvious properties of area.

Example 7.1.1. Consider the parallelograms $\begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix}$ and $\begin{bmatrix} k\boldsymbol{v} & \boldsymbol{w} \end{bmatrix}$ in the graph below. Obviously if I multiply an edge by k, then the area is multiplied by k. So $Area(k\boldsymbol{v}, \boldsymbol{w}) = Area(\boldsymbol{v}, k\boldsymbol{w}) = kArea(\boldsymbol{v}, \boldsymbol{w})$.



(a) Scaling (b) Vector Addition

Figure 7.1.1: Vectors and Area

Similarly, consider the case of $Area(\boldsymbol{a}_1 + \boldsymbol{a}_1', \boldsymbol{a}_2)$ above. Treating the direction of $\boldsymbol{a}_2$ as "base" and the perpendicular direction as "height", you can see that $Area(\boldsymbol{a}_1 + \boldsymbol{a}_1', \boldsymbol{a}_2) = Area(\boldsymbol{a}_1, \boldsymbol{a}_2) + Area(\boldsymbol{a}_1', \boldsymbol{a}_2)$. It seems to suggect that area is bilinear!

However, that is not the case. The CORRECT formula is $Area(k\boldsymbol{v}, \boldsymbol{w}) = Area(\boldsymbol{v}, k\boldsymbol{w}) = |k|Area(\boldsymbol{v}, \boldsymbol{w})$, because area is always positive. And we in fact have $Area(\boldsymbol{a}_1 + \boldsymbol{a}_1', \boldsymbol{a}_2) = Area(\boldsymbol{a}_1, \boldsymbol{a}_2) \pm Area(\boldsymbol{a}_1', \boldsymbol{a}_2)$, where the sign depends on whether $\boldsymbol{a}_1, \boldsymbol{a}_1'$ are on the same sides of $\boldsymbol{a}_2$ or not. (Can you draws these out?)

So area is NOT bilinear. ☺

This is an annoying situation. Area of a parallelogram is almost bilinear, but not truly. This is largely due to the need for area to be positive, as is very evident in the formula $Area(k\boldsymbol{v}, \boldsymbol{w}) = Area(\boldsymbol{v}, k\boldsymbol{w}) = |k|Area(\boldsymbol{v}, \boldsymbol{w})$.

So, if we allow negative area, then problem solved. The "signed" area, or ***oriented area*** will be bilinear.

What is oriented area? Loosely speaking, for each shape, we can "orient" the shape to be clockwise (CW) or counter-clockwise (CCW). If we give it a CCW orientation, then it will have positive area. And if we give it a CW orientation, then it will have negative area.

Example 7.1.2. You might have already encountered the concept of oriented area. For example, in calculus, a definite integral is the area below the curve. But sometimes the curve goes below the x-axis, resuting in a negative area. See the graph below.

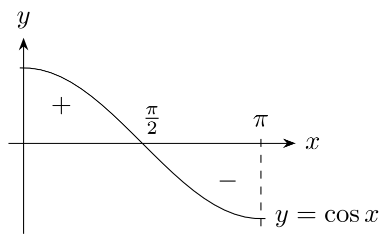


Figure 7.1.2: Oriented Area

In some sense, you can use the "clockwise" and "counterclockwise" to tell orientation. For example, if we go along the positive x-axis, you will see that we are going counterclockwise around the positive region, and clockwise around the negative region. A very informal intuition is to assume that the boundary of your region is oriented clockwise (CW) or counterclockwise (CCW), where a CCW oriented region has positive area, and a CW oriented region has a negative area. ☺

So now let us see how this oriented area could be calculated. Given two $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^2$, we use $\det(\boldsymbol{v}, \boldsymbol{w})$ or $\det \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix}$ to denote the oriented area of the corresponding parallelogram, where we go along the edge vector $\boldsymbol{v}$ first, and then go along $\boldsymbol{w}$ from the end point of $\boldsymbol{v}$ to the end point of $\boldsymbol{v} + \boldsymbol{w}$. This will give us a clear idea about whether the orientation is CW or CCW.

In particular, $\begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix}$ and $\begin{bmatrix} \boldsymbol{w} & \boldsymbol{v} \end{bmatrix}$ both represent the same parallelogram, but with opposite orientation.

Now here are some properties of oriented area.

1. (Normalized) First of all, the unit square going CCW should have area 1. I.e., we want $\det(\boldsymbol{e}_1, \boldsymbol{e}_2) = \det(I) = 1$.
2. (Bilinear) We do this because we want the oriented area to be bilinear. I.e., we want $\det(a\boldsymbol{u} + b\boldsymbol{v}, \boldsymbol{w}) = a \det(\boldsymbol{u}, \boldsymbol{w}) + b \det(\boldsymbol{v}, \boldsymbol{w})$.
3. (Anti-symmetry) If we swap the two edges, it is the same parallelogram but with reversed orientation. I.e., we want $\det(\boldsymbol{v}, \boldsymbol{w}) = -\det(\boldsymbol{w}, \boldsymbol{v})$.

Note that these properties immediately implies two interesting results:

1. (Flat parallelogram has no area) Swapping the two vectors gives $\det(\boldsymbol{v}, \boldsymbol{v}) = -\det(\boldsymbol{v}, \boldsymbol{v})$, which implies that $\det(\boldsymbol{v}, \boldsymbol{v}) = 0$. Indeed, if the two edges of the parallelogram coincide, then the area must be zero. In fact, conversely, if we have bilinearity and the fact that $\det(\boldsymbol{v}, \boldsymbol{v}) = 0$, then we can also deduce that $\det(\boldsymbol{v}, \boldsymbol{w}) + \det(\boldsymbol{w}, \boldsymbol{v}) = \det(\boldsymbol{v} + \boldsymbol{w}, \boldsymbol{v} + \boldsymbol{w}) - \det(\boldsymbol{v}, \boldsymbol{v}) - \det(\boldsymbol{w}, \boldsymbol{w}) = 0$, so we have antisymmetry. Note that we essentially used the polarization identity here.
2. (Shearing preserves area) We have $\det(\boldsymbol{v} + \boldsymbol{w}, \boldsymbol{w}) = \det(\boldsymbol{v}, \boldsymbol{w}) + \det(\boldsymbol{w}, \boldsymbol{w}) = \det(\boldsymbol{v}, \boldsymbol{w})$.

Now, to calculate $\det(A)$, we can try to swap/scale/shear columns of A, and $\det(A)$ should change accordingly. This gives us an idea to calculate this. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, assuming we do not see zero in the denominators below (I.e., $d \neq 0$ and $ad - bc \neq 0$), then we have

$$\det(A) = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \det \begin{bmatrix} a - \frac{bc}{d} & b \\ 0 & d \end{bmatrix} = \det \begin{bmatrix} a - \frac{bc}{d} & 0 \\ 0 & d \end{bmatrix} = (a - \frac{bc}{d})d \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = ad - bc.$$

This process corresponds to shearing a parallelogram until the two edges are on the coordinate axis. See the graph below, where $A = \begin{bmatrix} \boldsymbol{a}_1, \boldsymbol{a}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{b}_1, \boldsymbol{b}_2 \end{bmatrix}, \begin{bmatrix} \boldsymbol{c}_1, \boldsymbol{c}_2 \end{bmatrix}$ are the matrices in the first three steps of the above calculation.

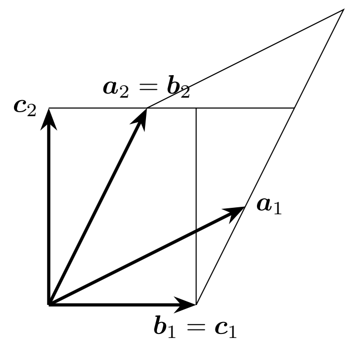


Figure 7.1.3: Column Shearing and Area

Definition 7.1.3. *Given a* 2×2 *matrix* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, *we define* $\det(A) = ad - bc$.

You can now easily verify that this satisfy all the desired relations. I.e., this is INDEED a well-defined oriented area.

Now we have tackled the two-dimensional case of parallelograms, what about parallelepipeds? Can we define an oriented volume?

Example 7.1.4. My heart is on my left side. You can rotate me, shear me, stretch me, walk me around, throw me over the moon, and my heart will always on my left side. If a person has his/her heart on the right side, then this person must not be me, right?

But one day, I lookd into a mirror. The person in the mirror will have his/her heart on the right side. As we can see, the orientation is reversed. ☺

Now, there are arguments that one can think of 3D orientations as “inward” and “outward”. Also, you can imagine the orientation refers to “positive mass” and “negative mass”, which is sometimes used to solve some problems in physics. I myself think of the positive volume as something in the REAL world, and negative volume as something in a MIRROR world. Feel free to do these things if it helps. However, as we move on into higher and higher dimensions, we will lose sight of our ability to do this. At this point, it is better to revert back to the basic of WHAT IS AN ORIENTED VOLUME.

Again, we want oriented volume to to satisfy analogous properties as oriented area. Given three vectors in $\mathbb{R}^3$, we can imagine that they form a parallelepiped. Then we want the following properties:

1. (Normalized) We want $\det(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) = \det(I) = 1$.

2. (Multilinear) We want $\det(a\boldsymbol{u} + b\boldsymbol{v}, \boldsymbol{x}, \boldsymbol{y}) = a \det(\boldsymbol{u}, \boldsymbol{x}, \boldsymbol{y}) + b \det(\boldsymbol{v}, \boldsymbol{x}, \boldsymbol{y})$, and the same for the second column and for the third column.

3. (Alternating) If we swap the two edges, it is the same parallelepiped but with reversed orientation. I.e., we want $\det(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) = -\det(\boldsymbol{u}, \boldsymbol{w}, \boldsymbol{v})$, and the same thing if we swap any pair of edges in general.

The first one is trivial. The second one can be seen geometrically. Treating the parallelogram $(\boldsymbol{x}, \boldsymbol{y})$ as the base, and the orthogonal direction as height, you shall see that $\det(\boldsymbol{u} + \boldsymbol{v}, \boldsymbol{x}, \boldsymbol{y}) = \det(\boldsymbol{u}, \boldsymbol{x}, \boldsymbol{y}) + \det(\boldsymbol{v}, \boldsymbol{x}, \boldsymbol{y})$

and $\det(k\boldsymbol{u}, \boldsymbol{x}, \boldsymbol{y}) = k\det(\boldsymbol{u}, \boldsymbol{x}, \boldsymbol{y})$. For the third one, it is easier to understand with an example. Consider the parallelepiped $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$. Reflect this about the plane $x = y$ in $\mathbb{R}^3$, and you will get the parallelepiped $(\boldsymbol{e}_2, \boldsymbol{e}_1, \boldsymbol{e}_3)$. In general, swapping a pair of edges means doing some reflection.

Again, note that these properties immediately implies two interesting results:

1. (Flat parallelepiped has no volume) If two vectors among $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are the same, then there is no volume and $\det(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) = 0$.
2. (Shearing preserves volumn) We have $\det(\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{v}, \boldsymbol{w}) = \det(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w})$.

Using these ideas, we can establish the formula. Note that the formula is a lot uglier and not nice at all.

Definition 7.1.5. *Given a* 3×3 *matrix* $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, *we define* $\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$.

There are many smart ways to memorize this formula, but honestly there is no need. Such methods are usually for 3×3 matrices only, so using them has the danger of doing higher dimensional determinants wrong.

We can go on similarly, and in general, given vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \in \mathbb{R}^n$, then the ***determinant*** of them $\det(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ is the oriented n-dimensional volumn of the corresponding n-dimensional parallelotope. We may also think of the input vectors as columns, forming an $n \times n$ square matrix A, and talk about $\det(A)$.

But the four dimension determinant formula will have 24 terms in the determinant. In general, the determinant of $n \times n$ matrices will have $n!$ terms. Instead, this is how we define determinants.

Definition 7.1.6. *We define an* $n \times n$ *determinant to be a function* $\det : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \to \mathbb{R}$ *with* n *input vectors from* $\mathbb{R}^n$*, such that*

1. *(Identity)* $\det(I) = 1$.
2. *(Multiplinear)* $\det(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ *is linear on each input* $\boldsymbol{v}_i$.
3. *(Alternating) If we swap the* i*-th input* $\boldsymbol{v}_i$ *and the* j*-th input* $\boldsymbol{v}_j$ *for any* $i \neq j$*, then the determinant is negated.*

Theorem 7.1.7. *For each* n*, determinant exists and is unique.*

We do the proof of this theorem later. For now, let us focus on some more intuitions.

Example 7.1.8. Suppose A is not invertible. Then columns of A are linearly dependent, i.e., they would fail to span $\mathbb{R}^n$ as expected, and would only span some smaller dimensional thing. As a result, the n-dimensional oriented volume of A would be zero. So $\det(A) = 0$. ☺

Example 7.1.9. Suppose we have a diagonal matrix D. Then as a parallelotope, all edges are orthogonal to each other! So the area is very easy: it is simply the product of all side lengths, i.e., all the diagonal entries of D.

We can generalize this a bit more. Consider an upper triangular matirx, say $U = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$. Taking the first edge as "base" which has length a, then the height is exactly c. So the area is exactly ac. Here b does not matter at all, because it effects neither the base nor the height. So $\det(U) = ac$.

What about $U = \begin{bmatrix} a & b & d \\ 0 & c & e \\ 0 & 0 & f \end{bmatrix}$? Taking the first two edges as "base", we see that the base area is ac, while the height is f. So the volume is $\det(U) = acf$.

You can probably see immediately that for any triangular matrix T, then $\det(T)$ is the product of diagonal entries.

Similarly, consider a block diagonal matrix $M = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$ where A, B are square. Then the A portion and the B portion are orthogonal to each other. It is not hard to see that we have $\det(M) = \det(A)\det(B)$. In fact, it is also not hard to see that $\det \begin{bmatrix} A & C \\ O & B \end{bmatrix} = \det(A)\det(B)$ as well. Taking A portion as "base", then the corresponding height is exactly $\det(B)$, and C is irrelavant. ☺

7.1.2 Volume Scaling Factor

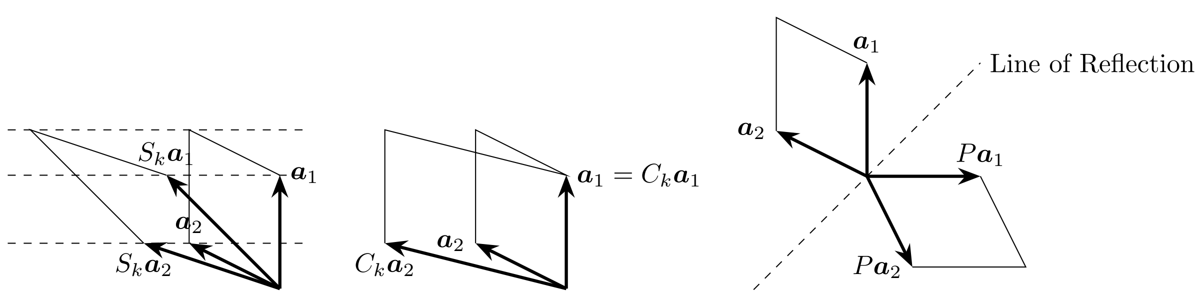
So far, by thinking of a square matrix A as an n-dimensional parallelotope in $\mathbb{R}^n$, we realized that if we do column operations, AE, then $\det(A)$ and $\det(AE)$ are related. Swapping columns would negate the determinant, shearing columns will preserve the determinant, and finally, scaling columns will scale the determinants. This subsection is devoted to row operations, $\det(EA)$.

But again, let us try to give it a meaning. Given a square matrix A, you may think of it as a linear transformation $A : \mathbb{R}^n \to \mathbb{R}^n$. Given a parallelotope $B = \begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_n \end{bmatrix}$, then $AB = \begin{bmatrix} A\boldsymbol{v}_1 & \dots & A\boldsymbol{v}_n \end{bmatrix}$ is the image of the original parallelotope after the transformation A.

So here is the big question: in general, how would A changes volumes of parallelotope? What is $\frac{\det(AB)}{\det(B)}$? Let me spoil the answer first: It is always a constant, independent of the input parallelotope B. A would simply change all parallelotope volume by the same factor. This is the volume scaling factor of A, and let us write it as $\delta(A)$. For examlpe, rotations will always have a volume scaling factor of 1, and reflections will always have a volume scaling factor of -1.

Let us see some examples here.

Example 7.1.10. Suppose we have a parallelogram, say $\begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 \end{bmatrix} = \begin{bmatrix} 0 & -0.8 \\ 1 & 0.4 \end{bmatrix}$. We can try to apply shearings $S_k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, scalings $\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$ or swapping $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ to it. As you can see from the graph below, when we apply S_{-1}, C_2, P to it, the area is changed by a fixed factor. Shearing again preserves the oriented area, swapping preserves the absolute area but changed the orientation, and finally scaling just scale the area.



(a) Shearing (b) Scaling (c) Swap

Figure 7.1.4: Row Operations and Area

Note that these row operations give an interesting contrast to column operations. For example, if we compare $\det(A)$ with $\det(AE)$ where E is a shearing, then we are shearing one edge of the parallelogram along the direction of another. However, if we are comparing $\det(A)$ with $\det(EA)$, then we are shearing the whole coordinate charts. In our case, the entire y-axis is sheared in the direction of the x-axis by our linear transformation E, and the parallelogram simply changes with it.

In these sense, the E of AE are applying changes to the edges of the parallelogram A, where as the E of EA are applying changes to the whole $\mathbb{R}^2$, and A simply changes along. If we use four matches to enclose a parallelogram A on a paper, then AE is trying to move/stetch/rotate the matches, while EA is trying to move/stetch/rotate the whole paper.

Alternatively, if you think of E of EA as a change of coordinate process, then the underlying parallelogram is in fact unchanged, and we are simply looking at A through a different perspective, e.g., squint our eyes, tilt our heads, apply a distorting mirror, these sort of things. And A will end up appearing to have a different area from before. The change from A to EA is nominal but not essential. In contrast, AE means we are essentially changing the parallelogram we are studying.

If these discussions only serves to confuses you, then you are also welcome to just understand AE and EA in terms of column/row operations.

But either way, the pattern is preserved. Shearing fixes the oriented area, scaling scales the area, and swapping flips the orientation. These are true for both EA and AE. And we have $\delta(S_k) = 1, \delta(C_k) = k, \delta(P) = -1$. ☺

So in the case of elementary matrices, they indeed have a corresponding volume scaling factor, independent of the input parallelotope. What about matrices in general?

Now recall that, given any invertible matrix A, it is the product of many elementary matrices, $A = E_1 \dots E_k$. As a result, we see that

$$\det(AB) = \det(E_1 \dots E_k B) = \delta(E_1)\det(E_2 \dots E_k B) = \cdots = \delta(E_1)\dots\delta(E_k)\det(B).$$

In particular, we see that $\delta(A) = \prod \delta(E_i)$. As you can see, this depends ONLY on A, and NOT on B at all.

Example 7.1.11. Suppose $A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$. Let us calculate $\delta(A)$.

Note that $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. The right matrix preserves the area, while the left matrix scale the area by 2. So $\delta(A) = 2$. ☺

What if A is not invertible? Then $\mathrm{Ran}(A)$ is less than n-dimensional. So any input B, AB will have zero n-dimensional volume. So $\delta(A) = 0$. Either way, we see that each linear map has a unique fixed volume scaling factor.

Definition 7.1.12. *For each $n \times n$ matrix A, we define its volumn scaling factor $\delta(A)$ to be the number such that, for any input parallelotope B, $\det(AB) = \delta(A)\det(B)$. (I.e., the linear map A scales the volumn by $\delta(A)$ always.)*

Now what is this $\delta(A)$? Note that, since AB as a linear map means we do B and then do A, obviously $\delta(AB) = \delta(A)\delta(B)$.

Corollary 7.1.13. *$\delta(AB) = \delta(A)\delta(B)$.*

Proof. For any parallelotope C, we have $\det(ABC) = \delta(A)\det(BC) = \delta(A)\delta(B)\det(C)$. Since this is true for all C, by definition $\delta(AB)$ must be the constant $\delta(A)\delta(B)$. □

For any elementary matrix E, $\delta(AE) = \delta(A)\delta(E)$ by definition. This immediatly implies the following:

1. Taking E as a swapping, we see that swapping columns of A will negate $\delta(A)$.
2. Taking E as a scaling, we see that scaling columns of A will scale $\delta(A)$.
3. Taking E as a shearing, we see that shearing columns of A will preserve $\delta(A)$.
4. Obviously $\delta(I) = 1$, as the identity map would not change anything.

These looks oddly suspicious! They are exactly the defining properties of determinant!

Proposition 7.1.14. *$\delta(A) = \det(A)$.*

Proof. We have $\det(AB) = \delta(A)\det(B)$ by definition of the volume scaling factor. Taking $B = I$, we see that $\delta(A) = \det(A)$. $\square$

So as a function on square matrices, the determinant is the same as the volume scaling factor. (Intuitively this is also obvious. To find the volume scaling factor, simply check out what would happen to a unit cube. The map A send the unit cube I to the parallelotope A, so $\delta(A) = \det(A)$.)

Finally, we also have an interesting property.

Proposition 7.1.15. $\det(A) = \det(A^{\mathrm{T}})$.

Proof. Suppose A is not invertible. Then both A, A^{T} are not invertible, so $\det(A) = 0 = \det(A^{\mathrm{T}})$.

If A is invertible, then we can reduce A to I through a series of column operations. But then we can do the corresponding row operations to A^{T}, and it will be reduced to $I^{\mathrm{T}} = I$.

Since these operations determines the determinant, we see that $\det(A) = \det(A^{\mathrm{T}})$.

To be more detailed, if $A = E_1 \dots E_k$ for elementary matrices $E_1, \dots, E_k$, then $A^{\mathrm{T}} = E_k^{\mathrm{T}} \dots E_1^{\mathrm{T}}$. Since $\det(E_i) = \det(E_i^{\mathrm{T}})$ for all elementary matrices, we have $\det(A) = \prod \det(E_i) = \prod \det(E_i^{\mathrm{T}}) = \det(A^{\mathrm{T}})$. $\square$

So now we have an alternative understanding of the determinant. We can think of $\det(A)$ as the n-volume of the n-parallelotope A, or as the volume scaling factor of the linear map A. But the latter perspective gives us the following two very interesting additional properties of the determinant.

1. $\det(AB) = \det(A)\det(B)$. This is obvious and natural from the volume scaling factor perspective.
2. $\det(A) = \det(A^{\mathrm{T}})$. This is because row operations and column operations do the same thing to $\det(A)$.

It is also very interesting if we combine this with some of our previous knowledges.

Example 7.1.16. Consider the LDU decomposition $A = LDU$. Here L and U are unit triangular, so their determinants are both one. So $\det(A) = \det(D)$. If you think about this, U, L means we are shearing the edges and also shearing the space, until we end up with a parallelotope D whose edges are on the coordinate axes. All these shearings preserves volume, so $\det(A) = \det(D)$ is simply the product of all edge lengths of D, i.e., all diagonal entries of D. (In some textbooks, i.e., Gilbert Strang, diagonal entries of D in the LDU decomposition is called the pivot values for A. This is because $L^{-1}A$ gives the row echelon form DU where pivots will be the corresponding diagonal entries of D.)

Now consider the QR decomposition $A = QR$. Q here is a rotation/reflection, so its volume scaling factor must be $\det(Q) = \pm 1$. R is an upper triangular matrix with positive diagonal entries, so $\det(R) > 0$. So in this sense, we have $\det(A) = \det(Q)\det(R)$, where $\det(R)$ is the absolute volumn of the parallelotope A, and $\det(Q)$ is the orientation of A. ☺

Remark 7.1.17. *Note that, A not only scales the volume of any input parallelotope by $\det(A)$. In fact ANY shape in $\mathbb{R}^n$, after transformation by A, will have its n-dimensional volume scaled by $\det(A)$. (To see this, chop up the shape into infinitesimal tiny parallelotopes, and take limit. This is a very standard argument in calculus.)*

This fact is crutial in multivariable calculus. In regular calculus, we integrate $\int f(x)dx$, and here dx intuitively refers to a infinitesimal tiny arrow in the direction of positive x-axis. We add up all the tiny vectors $f(x)dx$, and the result is $\int f(x)dx$.

In multivariable calculus, we integrate $\int f(x,y)dxdy$, and here $dxdy$ intuitively refers to an oriented parallelogram made by tiny vectors dx and dy. So we add up all the tiny oriented area $f(x,y)dxdy$, and the result is $\int f(x,y)dxdy$

Suppose we are doing $\int f(2x+y,y)dxdy$, and we want to do a change fo variable into $dzdy$ where $z = 2x + y$. Then $dz = 2dx + dy$ is a tiny arrow in the $\begin{bmatrix}2\\1\end{bmatrix}$ direction, and the resulting $dzdy$ will have a different oriented area from $dxdy$. Note that the map $A = \begin{bmatrix}2 & 0\\1 & 1\end{bmatrix}$ is exactly the linear transformation that changes the parallelogram $dxdy$ to $dzdy$. This means $dzdy = \det(A)dxdy = 2dxdy$. So we have $\int f(2x+y,y)dxdy = \int f(z,y)(\frac{1}{2}dzdy)$. If you miss the factor $\frac{1}{2}$, then your calculation is wrong.

7.2 Permutation Issue

7.2.1 Parity of a Permutation

Now all discussions so far depends on the existance and uniqueness of the concept of higher dimensional oriented volume. In lower dimensions, we can work it out and simply produce a formula for $\det(A)$. But for higher dimensions, we need to do this more rigorously. We want to rigorously define "oriented volume" in any dimension.

Example 7.2.1. Say we have an oriented 4-parallelotope $\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_4\end{bmatrix}$. Now $\begin{bmatrix}\boldsymbol{v}_4 & \boldsymbol{v}_1 & \boldsymbol{v}_3 & \boldsymbol{v}_2\end{bmatrix}$ describes the same parallelotope, but does it have the same orientation as before, or the opposite orientation?

Well, we need to check the number of swaps. One might do

$$\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_4\end{bmatrix} \to \begin{bmatrix}\boldsymbol{v}_4 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_1\end{bmatrix} \to \begin{bmatrix}\boldsymbol{v}_4 & \boldsymbol{v}_1 & \boldsymbol{v}_3 & \boldsymbol{v}_2\end{bmatrix}.$$

So we see that two swaps are needed. So they should have the same orientation.

But hold on a second! What if I do permutations differently? What if by some other way of swapping things, I go from $\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_4\end{bmatrix}$ to $\begin{bmatrix}\boldsymbol{v}_4 & \boldsymbol{v}_1 & \boldsymbol{v}_3 & \boldsymbol{v}_2\end{bmatrix}$ wiht odd number of swaps, hence they would have different orientation? Then we would have a contradiction at hour hand. The whole concept of oriented volume would collapse and become total nonesense.

Luckily, that may never happen. For example, let us do a different series of swaps. One might have

$$\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_4\end{bmatrix} \to \begin{bmatrix}\boldsymbol{v}_3 & \boldsymbol{v}_2 & \boldsymbol{v}_1 & \boldsymbol{v}_4\end{bmatrix} \to \begin{bmatrix}\boldsymbol{v}_3 & \boldsymbol{v}_2 & \boldsymbol{v}_4 & \boldsymbol{v}_1\end{bmatrix} \to \begin{bmatrix}\boldsymbol{v}_4 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_1\end{bmatrix} \to \begin{bmatrix}\boldsymbol{v}_4 & \boldsymbol{v}_1 & \boldsymbol{v}_3 & \boldsymbol{v}_2\end{bmatrix}.$$

Then we have four swaps, still even. ☺

So, how is this "orientation" thing defined? It seems to be someting induced by reflections. Consider a swapping matrix P, which is supposed to change orientations. Given a permutation matrix P, if it is done via an even number of swaps, then it should preserve orientation. If it is done via an odd number of swaps, then it should negate the orientation. This motivates us to do the following definition:

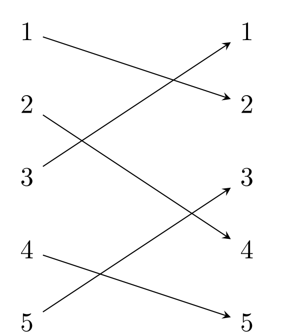
Definition 7.2.2. *A permutation is called an* ***even permutation*** *if it is the composition of even number of swaps. A permutation is called an* ***odd permutation*** *if it is the composition of odd number of swaps.*

Our goal is to rigorously show that, each permutation must be either even, or odd, but not both. This is the very fundation of orientation. Without this fact, then orientation could not ever be consistently defined. We now do this via some super fun diagrams.

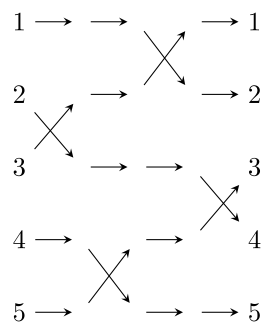
Lemma 7.2.3. *Any permutation is the composition of swaps.*

Proof. There are many many ways to do this. However, here we give a proof based on diagrams, and we merely provide an example here. You should be able to generalize this yourself.

Suppose a permutation sends $1, 2, 3, 4, 5$ to $2, 4, 1, 5, 3$. We can use the diagram below to denote this:



Did you see the crossings? Now, we repreduce these crossings via swaps, in the following way:

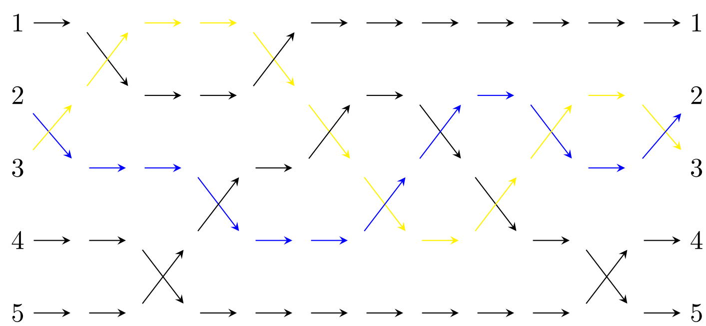


.

As you can see, we have decomposed a permutation into a series of swaps. It is easy to generalize this to any permutation. ☐

Lemma 7.2.4. *The identity permutation can only be the composition of even number of swaps, but not odd number of swaps.*

Proof. Suppose we decompose the identity permutation into a series of swaps. For illustration purpose, you can look at the following diagram:



.

Note that the composition is identity means each path goes ends up at the same number as where they started. I claim that for any such diagram, we must have an even number of "crossings" in total.

For example, consider the blue path from 2 to 2 and the yellow path from 3 to 3. Since the yellow path started below the blue path and ends up below the blue path, whenever it "crosses above", it must eventually "crosses back down". So the number of crossings between the blue path and the yellow path is even. (If you like, this is like a discrete version of intermediate value theorem in calculus.)

But the same is true for any pair of paths here. So the total number of crossings must be even. So the identity can only be the composition of even numbers of swaps. ☐

Theorem 7.2.5. *A permutation must be either even or odd, but not both.*

Proof. Suppose $P = P_1 \dots P_s = Q_1 \dots Q_t$ is the decomposition of the permutation P into swaps $P_1, \dots, P_s$ or swaps $Q_1, \dots, Q_t$. Then $I = PP^{-1} = P_1 \dots P_s Q_t \dots Q_1$. (Here note that the inverse of a swap is itself.) So the identity permutation is the composition of $s + t$ numbers of swaps. But then $s + t$ must be even, so s, t must be both even or both odd. ☐

As a result, orientation is well-defined. For the purpose of later use, let us do one last definition.

Definition 7.2.6. *For a permutation P, we define its sign* $\text{sign}(P)$ *to be* 1 *if P is even, and* -1 *if P is odd. So if P is the composition of k swaps, then* $\text{sign}(P) = (-1)^k$.

Note that many traditional Chinese textbook often refers to a concept called "inversion number", which refers to the number of inersections in our diagrams.

You should NOT focus on the number of intersections (i.e., inversion numbers). The numbers themselves are utterly useless. Only the parity (i.e., even or odd) matters.

7.2.2 (Optional) Cycle Decomposition

The methods in the last subsection is in fact NOT how people study permutations most of the time. A more useful concept is cycle decomposition. This is hard to establish for beginners, but in practis it is both easy to compute and easy to use.

First let us see an example.

Example 7.2.7. Suppose our permutation sends $1,2,3,4,5$ to $2,3,1,5,4$ respectively. Then this means we have a loop $1 \to 2 \to 3 \to 1$ and a loop $4 \leftrightarrow 5$, or "cycles". Cycle decomposition means we think of our permutation as disjoint loops (cycles) like these. It is a fact that the cycle decomposition of a permutation must exist and be unique. ☺

Theorem 7.2.8 (Cycle Decomposition)**.** *Given a permutation σ on the indices $1,2,...,n$, then we can break down the set $\{1,...,n\}$ into a union of disjoint subsets, such that P acts as a single "loop" on each subset.*

Proof. We proceed by mathematical induction. If $n=1$, this is trivial.

For generic $n>1$, consider the subset $S=\{1,\sigma(1),\sigma(\sigma(1)),...\}$. Let k be the smallest positive integer such that $\sigma^k(1)=1$. (Here σ^k means we iterate σ by k-times). Let us first see that $1,\sigma(1),...,\sigma^{k-1}(1)$ are all distinct. If for some $a<b<k$ we have $\sigma^a(1)=\sigma^b(1)$, then since permutations are bijections, we have $\sigma^{b-a}(1)=1$ and $b-a<k$, contradiction. So indeed $S=\{1,\sigma(1),\dots,\sigma^{k-1}(1)\}$ and it has k elements. Clearly σ acts as a "loop" on S. Furthermore, if $\sigma(i)\in S$, then $i\in\sigma^{-1}(S)=S$.

But then for the rest of the indices, $\{1,...,n\}-S$ has only $n-k$ elements. And if $i\notin S$, we already know that $\sigma(i)\notin S$. So σ also permutes indices in $\{1,...,n\}-S$. So by induction hypothesis, σ acts on $\{1,...,n\}-S$ as a union of disjoint "loops". So we are done. ☐

Finding a cycle decomposition is extremely fast. You simply follow the idea of the proof above, and figure it out one cycle at a time. For example, if σ sends $1,2,3,4,5$ to $4,5,1,3,2$, then we simply compute $1\mapsto\sigma(1)=4\mapsto\sigma(4)=3\mapsto\sigma(3)=1$ and we closed up the first loop. Then we move on to the first un-used index, and compute $2\mapsto\sigma(2)=5\mapsto\sigma(5)=2$ and we closed up the second loop. Done. So we can perform a cycle decomposition almost as fast as simply reading the entire permutation.

Lemma 7.2.9. *If a permutation is a single big cycle on n indices, then it is the product of $n-1$ swaps.*

Proof. We proceed by mathematical induction. If $n=1$, this is trivial.

Say the permutation σ is $1\mapsto 2\mapsto ...\mapsto n\mapsto 1$. Let $\sigma'=1\mapsto 2\mapsto ...\mapsto n-1\mapsto 1$ and $\sigma'(n)=n$, and let $\tau=n-1\mapsto n\mapsto n-1$ be a swap that fixes $1,...,n-2$. Then we can verify that σ is the composition $\sigma'\circ\tau$. So σ needs one more swap than σ', which is a product of $n-2$ swaps by induction hypothesis. So σ is the product of $n-1$ swaps. ☐

Corollary 7.2.10. *If a permutation has a cycle decomposition into $c_1,...,c_k$ cycles, then its parity is $c_1+\cdots+c_k-k$.*

Proof. Each cycle needs c_i-1 swaps. ☐

This is the best way to figure out the parity of a permutation. For super super large permutations, the diagrams in the last subsection will be an unreadable mess. Cycle decomposition is much more superior.

7.3 Uniqueness and Existence of Determinants

Definition 7.3.1. *We define a function* $\det : M_{n\times n} \to \mathbb{R}$ *to be a* ***determinant function*** *if it is multilinear (linear in each column), alternating (swapping two columns will negate value), and normalized* ($\det(I) = 1$)*.*

Note that we are going to write $\det(\boldsymbol{v}_1, \dots, \boldsymbol{v}_n)$ *sometimes, and this simply means* $\det(A)$ *where* $A = \begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_n \end{bmatrix}$

You can interpret the determinant function as n-dimensional oriented volume, or as volume scaling factor, it does not matter. Our goal is to show that this exist and is unique. So we can simply say "the determinant" instead of "a determinant function".

But before we show that it exists, let us see some examples of what is NOT a determinant.

Example 7.3.2. A multilinear function is NOT linear (unless there is only one input). For example, in $\mathbb{R}^2$, $\det(3A) \neq 3\det(A)$. Rather, it is $9\det(A)$. We CANNOT just pull out the common factor 3 from the MATRIX. Rather, we have to pull out the common factor 3 from EACH COLUMN, and thus we pulled out two 3's instead.

So linear functions like trace are NOT determinants.

Nevertheless, there are some behavioral analogies. For example, a linear combination of multilinear functions are still multilinear. For example, If f, g are multilinear, then $(f+g)(\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{w}) = f(\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{w}) + g(\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{w}) = f(\boldsymbol{u}, \boldsymbol{w}) + f(\boldsymbol{v}, \boldsymbol{w}) + g(\boldsymbol{u}, \boldsymbol{w}) + g(\boldsymbol{v}, \boldsymbol{w}) = (f+g)(\boldsymbol{u}, \boldsymbol{w}) + (f+g)(\boldsymbol{v}, \boldsymbol{w})$. ☺

Example 7.3.3. Consider the function $f : M_{n\times n} \to \mathbb{R}$ with f such that $f(A) = a_{11}a_{22}\dots a_{nn}$, i.e., it sends any matrix to the product of diagonal entries. This is multilinear. Let us just prove that it is linear in the first column. We have

$$\begin{aligned}
&f(\begin{bmatrix} b_{11}+c_{11} & a_{12} & \dots & a_{1n} \\ b_{21}+c_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1}+c_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}) \\
=&(b_{11}+c_{11})a_{22}\dots a_{nn} \\
=&b_{11}a_{22}\dots a_{nn} + c_{11}a_{22}\dots a_{nn} \\
=&f(\begin{bmatrix} b_{11} & a_{12} & \dots & a_{1n} \\ b_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}) + f(\begin{bmatrix} c_{11} & a_{12} & \dots & a_{1n} \\ c_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}).
\end{aligned}$$

You can esily verify the rest. However, this is NOT alternating. Indeed, $f(I) = 1$, but for any non-identity permutation matrix P, since it will fail to preserve all index, some diagonal entry will be zero. So $f(P) = 0$, whereas we should have $\det(P) = 1$ for even permutations and -1 for odd permutations.

Note that this is a start though. Note that if we set $f_P(A) = f(AP)$ for a fixed permutation P, then f_P is also multilinear because P merely permute the columns.

Suppose we set $g(A) = \sum_P f_P(A)$. For example, over $\mathbb{R}^3$ we have

$$g(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) = f(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) + f(\boldsymbol{u}, \boldsymbol{w}, \boldsymbol{v}) + f(\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{u}) + f(\boldsymbol{v}, \boldsymbol{u}, \boldsymbol{w}) + f(\boldsymbol{w}, \boldsymbol{u}, \boldsymbol{v}) + f(\boldsymbol{w}, \boldsymbol{v}, \boldsymbol{u}).$$

This is a linear combination of multilinear functions f_P for all P, and hence multilinear. However, it is NOT alternating. Rather, it is symmetric. As you can see, permuting columns in g would NOT change the value. What if we want something alternating? We would have to define g differently. Suppose we define

$$g(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) = f(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}) - f(\boldsymbol{u}, \boldsymbol{w}, \boldsymbol{v}) + f(\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{u}) - f(\boldsymbol{v}, \boldsymbol{u}, \boldsymbol{w}) + f(\boldsymbol{w}, \boldsymbol{u}, \boldsymbol{v}) - f(\boldsymbol{w}, \boldsymbol{v}, \boldsymbol{u}).$$

Then this shall be alternating as desired. So the desired function is $g(A) = \sum_P \pm f_P(A)$, where we add this for even permutations, and substract this for odd permutations. ☺

Theorem 7.3.4. *The determinant function exists.*

Proof. Let us start with any non-zero multilinear function $f : M_{n\times n} \to \mathbb{R}$. Say pick f such that $f(A) = a_{11}a_{22}\dots a_{nn}$, i.e., it sends any matrix to the product of diagonal entries.

Now, obviously this f is NOT a determinant function, because it is not alternating. What should we do then? WE FORCE IT!

We define a new function f_P as $f_P(A) = f(AP)$ for each permutation P. Let $\text{sign}(P)$ be 1 if P is an even permutation, and -1 if P is an odd permutation. Set $g(A) = \sum_P \text{sign}(P)f_P(A)$.

What have I done? Well, since each f_P is still multilinear, their linear combination is also multilinear. Furthermore, if S is a swap, then $g(AS) = \sum_P \text{sign}(P)f_P(AS) = \sum_P \text{sign}(P)f_{PS}(A) = -\sum_P \text{sign}(PS)f_{PS}(A) = -\sum_{P'} \text{sign}(P')f_{P'}(A) = -g(A)$. Here we set $P' = PS$ and used a change of index for the sum. So g is multilinear and alternating.

Now what about normalized? Well, check out $g(I)$. Note that $f(I) = 1$ and $f_P(I) = 0$ as long as P is not the identity permutation. So $g(I) = f(I) = 1$. So we are done. □

Remark 7.3.5. *A philosophy remark here. The initial multilinear function f does not actually matter much. Pick any f, and define $g = \sum \text{sign}(P)f_P$. Then g must be multilinear and alternating. If $g(I) = 1$, then it is the determinant and we are done. If $g(I) \neq 1, 0$, then define $h = \frac{1}{g(I)}g$, and it will be the determinant. The only potential trouble is if $g(I) = 0$. If that happens, then you need to pick a different f.*

Corollary 7.3.6 (Leibniz formula of determinants, i.e., the "big formula")**.** *A formula for a determinant function is $\det(A) = \sum_\sigma \text{sign}(\sigma)a_{\sigma(1),1}\dots a_{\sigma(n),n}$. Here σ ranges over all permutations on the index set $\{1, 2, ..., n\}$, and $a_{i,j}$ is the (i, j) entry of A.*

Proof. Just pick $f(A) = \prod a_{ii}$ and construct g as before. □

So we have established existence. Note how unwieldly this formula is. For $n \times n$ matrices, we have $n!$ terms. Oof! Don't memorize this, and don't ever use this formula, unless you have to. Also, don't let your computer compute this formula, if you don't want your computer to burn itself out. This takes forever to compute.

The only point of this formula is to show that determinants DO exist. Now a more practical approach of determinant lies in the fact that it is unique, which we now prove.

Theorem 7.3.7. *The determinant function is unique.*

Proof. Suppose f, g are two determinant functions. Suppose $f(A) = g(A)$ for some A. Then since AE is simply doing corresponding column operations on A, and since f, g are both alternating and multilinear, therefore we have $f(AE) = g(AE)$.

But we have $f(I) = 1 = g(I)$. By applying a series of column operations, I can get to any invertible matrix from I. So we have $f(A) = g(A)$ for all invertible A.

If A is not invertible, then AE will have first column zero for some E. (One column is the linear combination of others. Shear this column into zero using other columns, then swap it to the left.)

By multilinearity, we have $f(AE) = 0 = g(AE)$. So $f(A) = g(A)$ as well. □

Here is an alternative way to get the big formula. It is less conceptual and more computational. But the idea is simple: the big formula is simply this: write all inputs as linear combinations of standard basis vectors, and use multilinearity to expand everything!

Example 7.3.8. Consider $\det(A)$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We have

$$\begin{aligned}
\det(A) &= \det(\begin{bmatrix} a \\ c \end{bmatrix}, \begin{bmatrix} b \\ d \end{bmatrix}) \\
&= \det(a\boldsymbol{e}_1 + c\boldsymbol{e}_2, b\boldsymbol{e}_1 + d\boldsymbol{e}_2) \\
&= ab\det(\boldsymbol{e}_1, \boldsymbol{e}_1) + ad\det(\boldsymbol{e}_1, \boldsymbol{e}_2) + bc\det(\boldsymbol{e}_2, \boldsymbol{e}_1) + cd\det(\boldsymbol{e}_2, \boldsymbol{e}_2)
\end{aligned}$$

$$=ad - bc.$$

Again, as you can see, we get the big formula by simply expanding everything using multilinearity. ☺

Let us now do an alternative proof of the Leibniz formula.

Alternative proof of Leibniz formula. Let $\boldsymbol{a}_k$ be the k-th column of A, and let $a_{i,j}$ be the (i,j) entry of A. Then $\boldsymbol{a}_k = \sum_{i_k=1}^n a_{i_k,k}\boldsymbol{e}_{i_k}$ for each k.

Now

$$\begin{aligned}\det(A) &= \det(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n)\\ &= \det(\sum_{i_1=1}^n a_{i_1,1}\boldsymbol{e}_{i_1}, \ldots, \sum_{i_n=1}^n a_{i_n,n}\boldsymbol{e}_{i_n})\\ &= \sum_{(i_1,\ldots,i_n)} a_{i_1,1}\ldots a_{i_n,n}\det(\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}).\end{aligned}$$

Now, if a tuple $(i_1, \ldots, i_n)$ have repeated indices, then $\det(\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}) = 0$. If there is no repeated indices, then $(i_1, \ldots, i_n)$ is a permutation of $(1, \ldots, n)$, and thus it corresponds to some permutaiton P, and $\det(\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}) = \det(P) = \operatorname{sign}(P)$.

Hence we have

$$\begin{aligned}\det(A) &= \sum_{(i_1,\ldots,i_n)} a_{i_1,1}\ldots a_{i_n,n}\det(\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n})\\ &= \sum_P \operatorname{sign}(P) a_{P(1),1}\ldots a_{P(n),n}.\end{aligned}$$

Here P ranges over all possible permutations, and $P(k)$ is the number that P would permute k to. □

So here is an important conclusion: whatever the big formula can do, you can also do by simply using multilinearity. The big formula is nothing more than the ultimate expression after using multilinearity on everything. Multilinearity is enough. We shall almost NEVER use the big formula!

Example 7.3.9. We have the 3×3 determinant formula

$$\det\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{32}a_{21} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33}.$$

Can you see which term corresponds to which permutation? ☺

Example 7.3.10. We have the 2×2 determinant formula $\det\begin{bmatrix}a & b\\ c & d\end{bmatrix} = ad - bc$. So the area of the parallelogram is the difference between the area of two squares. Can you prove this statement using high school planar geometry? (Hint: geometrically speaking, multilinearity means, for example, we should do $\begin{bmatrix}a\\ c\end{bmatrix} = \begin{bmatrix}a\\ 0\end{bmatrix} + \begin{bmatrix}0\\ c\end{bmatrix}$. As a result, the parallelogram would break down into (the sum or difference of) two parallelograms. Now shear these parallelograms ultil they are rectangles.) ☺

7.4 Base, Height, Cofactor Expansion

Example 7.4.1. Given a 3×3 matrix $A = \begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$, we have

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

Boy this is ugly. However, let us try to establish some pattern here. For example, let us take all the terms involving a_{11}. Then we see $a_{11}(a_{22}a_{33} - a_{23}a_{32})$. This is exactly a_{11} times the determinant of the lower right block! We shall in this section generalize this to all matrices and all entries.

Basically, for a_{ij}, the corresponding coefficient for a_{ij} is $(-1)^{i+j}\det(A_{ij})$, where A_{ij} is the submatrix of A by removing the i-th row and j-th column. ☺

Our goal is to achieve the following: we want to arrange the big formula for $\det(A)$ as

$$\det(A) = a_{ij}(\text{Blahblah}) + (\text{Other stuff that does not involve } a_{ij}).$$

In short, we want to figure out what terms in the big formula would involve the entry a_{ij}. For this purpose, let us also use a concept from calculus.

Definition 7.4.2. *Given a function of many variables, say $f(x, y, z)$, the partial derivative $\frac{\partial}{\partial x}f$ means holding the other variables constant, and take derivative with respect to x. I.e., $\frac{\partial}{\partial x}f(x, y, z) = \lim_{\mathrm{d}x\to 0}\frac{f(x+\mathrm{d}x,y,z)-f(x,y,z)}{\mathrm{d}x}$.*

Example 7.4.3. Say $f(x, y) = x^2y$. Then $\frac{\partial}{\partial x}f = 2xy$ and $\frac{\partial}{\partial y}f = x^2$.

For determinant, we have n^2 variables $a_{11}, a_{12}, \dots, a_{nn}$. Above we showed that for 3×3 matrix A, $\frac{\partial}{\partial a_{11}}\det(A) = a_{22}a_{33} - a_{23}a_{32}$. Basically, suppose we have

$$\det(A) = a_{ij}(\text{Blahblah}) + (\text{Other stuff that does not involve } a_{ij}).$$

Then $\frac{\partial}{\partial a_{ij}}\det(A)$ is exactly the "Blahblah" portion.

In terms of calculus, we shall later show that $\frac{\partial}{\partial a_{ij}}\det(A) = (-1)^{i+j}\det(A_{ij})$. Calculus aside, the intuition is the answer to the following question: How would a particular entry a_{ij} influence $\det(A)$? The answer is via all the entriess NOT in the same column and NOT in the same row.

This makes sense, considering how the big furmula works. Each term of the big formula uses exactly one entry in each row and each column. So all the entries in the same row would NEVER be multiplied to produce a term, and the same for all entries in the same column. Their contributions to the determinant are "disjoint". ☺

Definition 7.4.4. *For a matrix A, we use A_{ij} to denote the submatrix of A by removing the i-th row and j-th column. Then we call $c_{ij} = (-1)^{i+j}\det(A_{ij})$ the (i, j) **cofactor** of A.*

*In particular, the matrix C whose (i, j) entry is c_{ij} is the **cofactor matrix** of A.*

Remark 7.4.5. *The name "cofactor" is like this: in the big formula $\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$, what terms have factor a_{11}? Then these terms make up $a_{11}(a_{22}a_{33} - a_{23}a_{32})$ or $a_{11}c_{11}$. Hence c_{11} is the "cofactor" to the factor a_{11}.*

Lemma 7.4.6. $\frac{\partial}{\partial a_{11}}(\det(A)) = \det(A_{11}) = c_{11}$.

Proof. Suppose B is the matrix obtained by increasing the $(1, 1)$ entry of A by a tiny bit $\mathrm{d}x$. We want to see how $\det(A)$ changes into $\det(B)$.

Note that A, B differ only in the first column. If A has first column $\boldsymbol{a}$, then B has first column $\boldsymbol{a} + \mathrm{d}x\boldsymbol{e}_1$. So

$$\det(B) = \det(\boldsymbol{a} + \mathrm{d}x\boldsymbol{e}_1, rest) = \det(\boldsymbol{a}, rest) + \det(\mathrm{d}x\boldsymbol{e}_1, rest) = \det(A) + \det(\mathrm{d}x\boldsymbol{e}_1, rest).$$

In particular, we see that $\det(B) - \det(A) = \det(\mathrm{d}x\boldsymbol{e}_1, rest) = \det\begin{bmatrix}\mathrm{d}x & \star\\ \mathbf{0} & A_{11}\end{bmatrix}$. This is block upper triangular, so this is $\mathrm{d}x\det(A_{11}) = c_{11}\,\mathrm{d}x$.

So $\frac{\partial}{\partial a_{11}}(\det(A)) = \lim_{\mathrm{d}x\to 0}\frac{\det(B)-\det(A)}{\mathrm{d}x} = c_{11}$. □

Proposition 7.4.7 (How entries effect determinant)**.** $\frac{\partial}{\partial a_{ij}}(\det(A)) = (-1)^{i+j}\det(A_{ij}) = c_{ij}$.

Proof. Let P_i be the permutation matrix that (as row operation) send indices 1 to 2, 2 to 3, ..., $i-1$ to i, and then i to 1. All other indices are fixed. Let P_j be defined similarly but as a column operation. Let $B = P_i A P_j$.

Then by interpreting these permutation matrices as row permutations and column permutations, one can directly verify that the $(1,1)$ entry of B, b_{11}, is exactly a_{ij}, and the submatrix B_{11} is exactly A_{ij}.

Now $\det(A_{ij}) = \det(B_{11}) = \frac{\partial}{\partial b_{11}}(\det(B)) = \frac{\partial}{\partial a_{ij}}(\det(P_i A P_j)) = \det(P_i)\frac{\partial}{\partial a_{ij}}(\det(A))\det(P_j)$. Here the last equality is because $\det(P_i), \det(P_j)$ are constants, so we can simply take them out of the derivative.

Now P_i has $i-1$ crosses in its diagram, and P_j has $j-1$ crosses in its diagram, so we see that $\det(P_i)\det(P_j) = (-1)^{i+j-2} = (-1)^{i+j}$. Here we use $i+j$ instead of $i+j-2$ because it looks prettier. Hence, $\det(A_{ij}) = (-1)^{i+j}\frac{\partial}{\partial a_{ij}}(\det(A))$. ☐

You should loosely interpret above proposition as this: the formula for determinant is $\det(A) = a_{ij}c_{ij} +$ other stuff, where "other stuff" do not use the entry a_{ij} at all. Putting these together, we would have a new formla for determinant.

Example 7.4.8.

$$\begin{aligned}\det(A) =& a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}\\ =& a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{21}(-a_{12}a_{33} + a_{13}a_{32}) + a_{31}(a_{12}a_{23} - a_{13}a_{22})\\ =& a_{11}c_{11} + a_{21}c_{21} + a_{31}c_{31}.\end{aligned}$$

Basically, each term in the big formula for $\det(A)$ must uses one entry in the first column. And if a term uses a_{i1}, then it is a term contained inside $a_{i1}c_{i1}$. ☺

Theorem 7.4.9 (Laplace expansion)**.** *Fix an index k. Then $\det(A) = a_{1k}c_{1k} + \cdots + a_{nk}c_{nk}$ (expansion via the k-th column), and similarly $\det(A) = a_{k1}c_{k1} + \cdots + a_{kn}c_{kn}$ (expansion via the k-th row).*

Proof. Look at the Lebniz formula. Each term must be has exactly ONE of $a_{1k}, \dots, a_{nk}$ as a factor. (Since each term only contains a single entry from the k-th column). In particular, by taking common factors, we can write $\det(A) = a_{1k}(\text{stuff}) + a_{2k}(\text{stuff}) + \cdots + a_{nk}(\text{stuff})$, and all the "stuff" here only uses entries NOT on the k-th column (since the one on the k-th column is already picked).

Now by taking partial derivatives, we see that $\frac{\partial}{\partial a_{ik}}(\det(A))$ is exactly the "stuff" after a_{ik}, and we have $\frac{\partial}{\partial a_{ik}}(\det(A)) = c_{ik}$. So we are done.

The expansion via rows is the same as the expansion via columns. ☐

Example 7.4.10. In general, the Laplace expansion would reduce the calculation of a single $n \times n$ determinant to the calculation of n determinants of $(n-1)\times(n-1)$ determinants. For examlpe, we have

$$\det\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix} = 1 \times \det\begin{bmatrix}5 & 6\\ 8 & 9\end{bmatrix} - 4 \times \det\begin{bmatrix}2 & 3\\ 8 & 9\end{bmatrix} + 7 \times \det\begin{bmatrix}2 & 3\\ 5 & 6\end{bmatrix}.$$

Note the sign here are alternating. Be careful not to mess it up.

Now this in general did not save too many time. It is merely doing the big formula in a more organized way. However, sometimes it will save you some time.

For examlpe, consider $\det(A) = \det\begin{bmatrix}0 & 1 & 2 & 3\\ 0 & 4 & 0 & 0\\ 5 & 6 & 7 & 8\\ 0 & 9 & 10 & 11\end{bmatrix}$. By doing Laplace expansion in the first column, we have $\det\begin{bmatrix}0 & 1 & 2 & 3\\ 0 & 4 & 0 & 0\\ 5 & 6 & 7 & 8\\ 0 & 9 & 10 & 11\end{bmatrix} = 5\det\begin{bmatrix}1 & 2 & 3\\ 4 & 0 & 0\\ 9 & 10 & 11\end{bmatrix}$. This is nice because the first column has many zeros. Keep picking rows and columns with many zeros, and we can expand along the second row and get $-20\det\begin{bmatrix}2 & 3\\ 10 & 11\end{bmatrix} = 160$.

But let us do this again by using the big formula directly. Then you shall immediately see that this again. Recall that, in the big formula, each term only uses, and must use exactly ONE entry from each column and each row. Now loot at $\det \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 4 & 0 & 0 \\ 5 & 6 & 7 & 8 \\ 0 & 9 & 10 & 11 \end{bmatrix}$. If a term in the big formula wants to be non-zero, then it MUST use the 5. There is no other non-zero alternatives. Similarly, the term MUST also use 4 in the second row. There is no other alternative.

Now since a nonzero term must use the 5 in the first column and the 4 in the second row, the rest is like $\det \begin{bmatrix} - & - & 2 & 3 \\ - & 4 & - & - \\ 5 & - & - & - \\ - & - & 10 & 11 \end{bmatrix}$. The dashed out portion does not matter, because NO NONZERO TERM will use them ever. They might as well all be zero. Now, since a non-zero term must already picked 5 and 4 here, there are only two possibilities left: pick 2 and 11, or pick 3 and 10.

So the two non-zero terms are $5\times4\times2\times11$ and $5\times4\times3\times10$. Be careful of the sign, which comes from the position of these entries. So we have $\det(A) = 5\times4\times2\times11 \det \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + 5\times4\times3\times10 \det \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. The first permutation matrix can be done in 1 swap, while the second can be done in 2 swaps. So we have $\det(A) = -5 \times 4 \times 2 \times 11 + 5 \times 4 \times 3 \times 10 = 160$.

As you can see, there is a clear equivalence between Laplace expansion and the big formula. Laplace expansion is simply grouping terms in the big formula by common factors. Intuitively, Laplace expansion is nothing more than doing the big formula in a more organized fashion.

Here is something else to think about, if you like. At least how many zeros do you need for a 4×4 matrix, to guarantee that all terms in the big formula are zero? (Answer is 4, and they must all lie on the same column or same row.)

Let us do this YET AGAIN. When we say "the non-zero terms must pick 5 in the first column", and thereby simplifying the determinant $\det \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 4 & 0 & 0 \\ 5 & 6 & 7 & 8 \\ 0 & 9 & 10 & 11 \end{bmatrix}$ into $\det \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \\ 0 & 9 & 10 & 11 \end{bmatrix}$. Hey, this is just a column operation! We can then do a row operation using the second row to shear others, and get $\det \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \\ 0 & 0 & 10 & 11 \end{bmatrix}$. And we get the results again.

See? Ultimately, the big formula is just multilinearity. So any fancy argument using the big formula must ultimately be the same as some column/row operations.

Let us do this YET YET AGAIN. Consider some row/column swaps as

$$\det \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 4 & 0 & 0 \\ 5 & 6 & 7 & 8 \\ 0 & 9 & 10 & 11 \end{bmatrix} = (-1)^3 \det \begin{bmatrix} 5 & 6 & 7 & 8 \\ 0 & 1 & 2 & 3 \\ 0 & 9 & 10 & 11 \\ 0 & 4 & 0 & 0 \end{bmatrix} = (-1)^4 \det \begin{bmatrix} 5 & 8 & 7 & 6 \\ 0 & 3 & 2 & 1 \\ 0 & 11 & 10 & 9 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

Now note that this is block upper triangular, with blocks $\begin{bmatrix} 5 \end{bmatrix}$, $\begin{bmatrix} 3 & 2 \\ 11 & 10 \end{bmatrix}$, $\begin{bmatrix} 4 \end{bmatrix}$. So the answer is $(-1)^4 \times 5 \times 4 \times \det \begin{bmatrix} 3 & 2 \\ 11 & 10 \end{bmatrix} = 160$. As you can see, why were we able to utilize the zero entries so effectively? It is precisely because these zeros allow us to make things block upper triangular, which reduced the calculation of a big determinant to be simplified into a smaller determinant.

If the zeros are smartly arranged, so that you cannot get block triangular things no matter how you permute, then none of the above methods would have worked. ☺

Recall that the Leibniz formula (big formula) is basically just writing all columns in the standard basis, and expand everything using multilinearity. And afterwards, by grouping terms in the big formula via common factors along a column or a row, we have the Laplace expansion.

So multilinearity = Leibniz formula = Laplace expansion. They are simply describing the same structure in different perspectives. The first one is the perspective of linear algebra. The Leibniz formula is from the perspective of permutation theory. And the Laplace expansion is in fact a geometric perspective.

Proposition 7.4.11 (Laplace expansion is base times height)**.** *Consider a square matrix* $A = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$ *and its cofactor matrix* $C = \begin{bmatrix} \boldsymbol{c}_1 & \dots & \boldsymbol{c}_n \end{bmatrix}$*. Then* $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{c}_j$ *is* 0 *if* $i \neq j$ *and* $\det(A)$ *if* $i = j$*. The same is true for rows.*

In particular, $\boldsymbol{c}_i$ *is perpendicular to* $\boldsymbol{a}_1, \dots, \boldsymbol{a}_{i-1}, \boldsymbol{a}_{i+1}, \dots, \boldsymbol{a}_n$*. And the length* $\|\boldsymbol{c}_i\|$ *is exactly the* $(n-1)$*-dimensional absolute volume of the* $(n-1)$*-dimensional parallelotope* $(\boldsymbol{a}_1, \dots, \boldsymbol{a}_{i-1}, \boldsymbol{a}_{i+1}, \dots, \boldsymbol{a}_n)$ *in* $\mathbb{R}^n$*.*

Proof. We already know that $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{c}_i = \det(A)$. Now, suppose we have $i \neq j$, and we aim to show that $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{c}_j = 0$.

Let us prove the case when $j = 1$ and $i \neq 1$. Imagine the Laplace expansion for the following two matrices.

$$A = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix}.$$
$$B = \begin{bmatrix} \boldsymbol{a}_i & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix}.$$

To both matrices, if we perform the Laplace expansion along the first column, the cofactors are actually identical! Therefore we have $\det(A) = \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{c}_1$ and $\det(B) = \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{c}_1$.

On the other hand, clearly B has repeated column. So $\det(B) = 0$. Hence we see that $\boldsymbol{c}_1 \perp \boldsymbol{a}_i$.

So we are done. We see that $\boldsymbol{c}_j$ is perpendicular to $\boldsymbol{a}_1, \dots, \boldsymbol{a}_{j-1}, \boldsymbol{a}_{j+1}, \dots, \boldsymbol{a}_n$, i.e., perpendicular to the "base". Then Volumn $= |\det(A)| = |\boldsymbol{a}_j^{\mathrm{T}}\boldsymbol{c}_j| = \|\text{Projection of } \boldsymbol{a}_j \text{ to } \boldsymbol{c}_j\|\|\boldsymbol{c}_j\| = \text{height} \times \|\boldsymbol{c}_j\|$. So we see that $\|\boldsymbol{c}_j\| =$ "Base Area". □

Corollary 7.4.12 (Big Formula for Inverse Matrix)**.** $C^{\mathrm{T}}A = A^{\mathrm{T}}C = \det(A)I$*. In particular, if* A *is invertible, then its inverse is* $A^{-1} = \frac{1}{\det(A)}C^{\mathrm{T}}$*. Note that this gives a formula for the inverse.*

Proof. The (i,j) entry of $C^T A$ is $\boldsymbol{c}_i^{\mathrm{T}}\boldsymbol{a}_j$. So we are done. □

Example 7.4.13. Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then the cofactor matrix is $C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$. And the inverse is $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Intuitively, the cofactors are the "entries of inverse transpose, but without denominator". ☺

The formula here may sounds nice, but it is actually too slow to compute. Each entry of C is a determinant, and you would have to calculate forever. Our old approach, i.e., Gaussian elimination on $\begin{bmatrix} A & I \end{bmatrix}$, would find A^{-1} much faster.

Corollary 7.4.14 (Cramer's rule)**.** *Suppose* A *is invertible. Then the unique solution to* $A\boldsymbol{x} = \boldsymbol{b}$ *is a vector whose* i*-th coordinate is* $x_i = \frac{\det(A_i)}{\det(A)}$*, where* A_i *is a matrix obtained by replacing the* i*-th row of* A *by* $\boldsymbol{b}$*. Note that this gives a formula for the solution of the linear system. (Too ugly to be useful though. Gaussian elimination is much faster.)*

Proof. Let us simply prove the formula for x_1. The rest are the same.

Consider $A = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix}$ and $A_1 = \begin{bmatrix} \boldsymbol{b} & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix}$. Then $\det(A_1) = \det\begin{bmatrix} x_1\boldsymbol{a}_1 + \dots + x_n\boldsymbol{a}_n & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix} = \det\begin{bmatrix} x_1\boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix}$ by shearing columns. Then we have $\det(A_1) = x_1\det\begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_n \end{bmatrix} = x_1\det(A)$. So we are done.

There is also a more geometric proof (with essentially the same idea). Note that as parallelotopes, A, A_1 share the same "base" made of $\boldsymbol{a}_2, \dots, \boldsymbol{a}_n$. So the ratio of determinant is simlpy the ratio of height. So the key is to compare how $\boldsymbol{a}_1$ contribute to the height, and how $\boldsymbol{b}$ contribute to the height.

Now height is the portion perpendicular to the base, i.e., perpendicular to $\boldsymbol{a}_2, \dots, \boldsymbol{a}_n$. If $\boldsymbol{b} = x_1\boldsymbol{a}_1 + \dots + x_n\boldsymbol{a}_n$, then $x_2\boldsymbol{a}_2 + \dots + x_n\boldsymbol{a}_n$ does NOT contribute to the height at all, because this is parallel to the base. Only $x_1\boldsymbol{a}_1$ component of $\boldsymbol{b}$ would contribute. So the height of A_1 is x_1 times the height of A. So we are done. $\square$

Remark 7.4.15. *The big formula for determinant, inverse matrix, and solutions for linear system are ALL useless for computational purpose. There are too many terms, and too many additions and multiplications to do. It is usually MUCH faster to use Gaussian elimination for all of them.*

However, the importance lies in perspectives. How would an entry contribute to the determinant? How would zero entries help simplify the determinant? How does parallelotopes has to do with solving linear systems? THESE are the main takeaways.

7.5 (Optional) Generalized Pythagorean theorem and Cauchy-Binet formula

Note that in the discussions above, we have essentially solved a geometric problem.

Corollary 7.5.1 ($(n-1)$-dimensional parallelotope in $\mathbb{R}^n$)**.** *Suppose we have an $(n-1)$-dimensional parallelotope in $\mathbb{R}^n$, whose edges form an $n \times (n-1)$ matrix A. Let A_i be the square matrix obtained by removing the i-th row of A. Then the $(n-1)$-dimensional (absolute) volumn is $\sqrt{\sum \det(A_i)^2}$.*

Proof. Consider adding an arbitrary column $\boldsymbol{x}$ to the left of A to get an $n \times n$ square matrix B. Then the $(n-1)$-parallelotope is the "base" of B using all but the first columns. Let $\boldsymbol{c}_1$ be the first column of the cofactor matrix of B, then $\|\boldsymbol{c}_1\|$ is our desired results.

Now $\boldsymbol{c}_1$ has coordinates $\pm\det(A_i)$. So we are done. $\square$

Note that A_i is obtained by deleting the i-th coordinate of all edges of the parallelotope. So they are projection image of A. For example, if $n = 3$ and A represent some parallelogram, then A_1 is the projection to yz-plane.

Corollary 7.5.2 (Generalized Pythagorean Theorem)**.** *Given a right tetrahedron (a "corner of the walls", i.e., three adjacent edges of it are mutually orthogonal to each other), the squared area of the oblique face is the sum of the squared area of the right-triangle faces.*

Any flat shape in $\mathbb{R}^3$, say it has area S, and its "shadows", i.e., projections to the three coordinate planes, has are S_{xy}, S_{yz}, S_{zx}, then $S^2 = S_{xy}^2 + S_{yz}^2 + S_{zx}^2$. (However, this is only true for flat shapes, i.e., entirely contained in an affine 2-dimensional space. Surfaces with a curvature will fail to have this.)

(You can easily see generalizations of this in higher dimensions. But the proof relies on Cauchy-Binet formula, which is poven below.)

However, in comparison, here is an interesting result.

Proposition 7.5.3 (k-dimensional parallelotope in $\mathbb{R}^n$)**.** *Consider a k-dimensional parallelotope in $\mathbb{R}^n$ (inner product is dot product), with edges forming a matrix $A = \begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_k\end{bmatrix}$. Then the k-dimensional (absolute) volumn is $\sqrt{\det(A^{\mathrm{T}}A)}$.*

Proof. Let $W = \mathrm{Ran}(A)$ and pick an orthonormal basis for $W^\perp$, $Q = \begin{bmatrix}\boldsymbol{q}_{k+1} & \dots & \boldsymbol{q}_n\end{bmatrix}$. Now all these vectors are unit vectors, mutually orthogonal, and all orthogonal to the parallelotope represented by A. As a result, $\begin{bmatrix}A & Q\end{bmatrix}$ is an n-dimensional parallelotope whose n-dimensional absolute volumn is the same as the k-dimensional absolute volume of A.

So this is $|\det\begin{bmatrix}A & Q\end{bmatrix}| = \sqrt{\det\begin{bmatrix}A & Q\end{bmatrix}^{\mathrm{T}}\det\begin{bmatrix}A & Q\end{bmatrix}} = \sqrt{\det(\begin{bmatrix}A & Q\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}A & Q\end{bmatrix})} = \sqrt{\det\begin{bmatrix}A^{\mathrm{T}}A & A^{\mathrm{T}}Q \\ Q^{\mathrm{T}}A & Q^{\mathrm{T}}Q\end{bmatrix}}$.

Now $Q^{\mathrm{T}}Q = I$ because Q has orthonormal columns. Furthermore, $A^{\mathrm{T}}Q = (Q^{\mathrm{T}}A)^{\mathrm{T}} = O$, because all columns of Q are orthogonal to the parallelotope A. So our expression simplifies to $\sqrt{\det \begin{bmatrix} A^{\mathrm{T}}A & O \\ O & I \end{bmatrix}} = \sqrt{\det(A^{\mathrm{T}}A)}$. □

Putting the two together, this hinted at a much stronger formula.

Theorem 7.5.4 (Cauchy-Binet Formula)**.** *Say A is an $n \times k$ matrix with $n \geq k$. Then $\det(A^{\mathrm{T}}A) = \sum \det(A')^2$, where A' is any $k \times k$ square submatrix of A, and we are summing all such possible suqare submatrices.*

Taking polarization identity, we in fact has $\det(A^{\mathrm{T}}B) = \sum \det(A')\det(B')$ where the sum is over all CORRESPONDING $k \times k$ square submatrices A', B' of A, B.

Proof. It turned out that it is easier to prove $\det(A^{\mathrm{T}}B) = \sum \det(A')\det(B')$. Think of both sides as taking the input $\begin{bmatrix} A & B \end{bmatrix}$. Then both sides are multilinear in columns. So, by expanding everything in terms of standard basis vectors, we only need to prove the case when all columns are some standard basis vector.

In this case, if A has repeated columns, it is easy to verify that both sides are zero. So suppose A has distinct columns. (Since columns of A are standard basis vectors, this means A is a permutation matrix.) Now note that swapping columns of A gives a minus sign on both sides, so it does not change the validity of the formula. On the other hand, simultaneously changing rows of A and B, then both sides are preserved as well.

So WLOG by swapping columns of A, and simultaneously swapping rows of A and B, we can assume that $A = \begin{bmatrix} I_k \\ O \end{bmatrix}$. Say $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$. Then both sides are $\det(B_1)$, so the statement is true. □

This concludes the generalizations of Pythagorean theorem into all dimensions. Given a k-dimensional flat shape in $\mathbb{R}^n$, then (k-dimensional volumn of this shape)^2 is the sum of all (k-dimensional volumns of its shadows on a k-dimer over all possible k-dimensional coordinate subspaces. Here, coordinate subspaces refers to subspaces spanned by the cooridnate axes.

7.6 (Optional) Determinant Tricks

7.6.1 Block eliminations

So, how to actually compute determinants? The answer is as always: Gaussian elimination. I mean, this is the source of all computations. How to solve $A\boldsymbol{x} = \boldsymbol{b}$? Gaussian elimination. How to find LU decomposition? Gaussian eliminations. How to find QR decomposition? Do Gaussian elimination on $A^{\mathrm{T}}A$.

Given a matrix A, we do Gaussian elimination, which is equivalent to $PA = LDU$. Then $\det(A) = \operatorname{sign}(P)\det(D)$, because L, U are unit triangular. $\det(D)$ is just the product of diagonal entries, because it is a rectangular parallelotope. Note that you don't have to restrict yourself to row opertions though. If needed, column operations are just fine.

Just keep in mind: shearings preserve determinant, scalings scale determinants, and swappings NEGATE the determinant. The last one is easy to forget sometimes. You can do rows or columns as you see fit.

Example 7.6.1. Consider the matrix $\begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & 2 & 0 \end{bmatrix}$. We cab swap the first row and the third row, then swap the first column and the secod colume. So we have

$$\det \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & 2 & 0 \end{bmatrix} = \det \begin{bmatrix} 0 & 2 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} = -\det \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix}.$$

This is now lower triangular, so the determinant is -2. ☺

In general, if you see no obvious clue about what to do, then Gaussian elimination is (as always) your best bet. Do NOT apply the big formula, because it would then take forever (unless we are facing some super special matrices). And in practice, no one ever use the big formula.

Now, recall that the funcation of column/row operations to simplify determinant is the fact that $\det(AB) = \det(A)\det(B)$. As a result, we can in fact use block operations to simply determinants. We have the following:

1. Block shearings preserve determinants as $\det \begin{bmatrix} I & A \\ O & I \end{bmatrix} = 1$.
2. Block scalings scale determinants as $\det \begin{bmatrix} A & \\ & I \end{bmatrix} = \det(A)$.
3. Block swaps might or might NOT change the sign. We have $\det \begin{bmatrix} & I_m \\ I_n & \end{bmatrix} = (-1)^{mn}$.

The last one might needs some justification. We want to move the m last columns to the left. The first of the m last columns, swap one by one to the left, needs a total of n swaps. Then the next one needs a total of n swaps as well. So on so forth for all m of them. So we performed a total of mn swaps. Alternatively, draw the diagram for $\begin{bmatrix} & I_m \\ I_n & \end{bmatrix}$ and see that the m parallel lines going up right would intersect ALL the n parallel lines going down right so we have a total of mn intersections.

Here is a nice application.

Proposition 7.6.2. *For any $m \times n$ matrix A and an $n \times m$ matrix B, $\det(I_m + AB) = \det(I_n + BA)$. Note that the suqare matrix on the left and on the right might have DIFFERENT sizes.*

Note that there are some deep relations between AB and BA in terms of eigenvalues, so one way is to go there. Here let me give you two other proofs.

Proof. This proof is straight forward. Change basis to simplify, then compute.

Say $A = P \begin{bmatrix} I_r & O \\ O & O \end{bmatrix} Q$ by rank normal form. This means we have changed basis in the domain and codomain of $A : \mathbb{R}^n \to \mathbb{R}^m$. So the corresponding transformation to B will be $B = Q^{-1}B'P^{-1}$ for some matrix B'. Say $B' = \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$.

Now

$$\begin{aligned}
\det(I_m + AB) &= \det(I_m + P \begin{bmatrix} B_1 & B_2 \\ O & O \end{bmatrix} P^{-1}) \\
&= \det(P(I_m + \begin{bmatrix} B_1 & B_2 \\ O & O \end{bmatrix})P^{-1}) \\
&= \det(I_m + \begin{bmatrix} B_1 & B_2 \\ O & O \end{bmatrix}) \\
&= \det \begin{bmatrix} B_1 + I_r & B_2 \\ O & I_{m-r} \end{bmatrix} \\
&= \det(B_1 + I_r).
\end{aligned}$$

Similarly, $\det(I_n + BA) = \det(I_m + Q^{-1} \begin{bmatrix} B_1 & O \\ B_3 & O \end{bmatrix} Q) = \det(B_1 + I_r)$. So we are done. ☐

Proof. This is a traditional proof using the magic of block matrices.

When you have an $m \times n$ matrix and an $n \times m$ matrix, how would you put them into a single block matrix? You don't have much option but to do $\begin{bmatrix} I_n & B \\ A & I_m \end{bmatrix}$.

Now by block row-shearing, we have $\det\begin{bmatrix} I_n & B \\ A & I_m \end{bmatrix} = \det\begin{bmatrix} I_n & B \\ O & I_m - AB \end{bmatrix} = \det(I_m - AB)$, and by column-shearing we have $\det\begin{bmatrix} I_n & B \\ A & I_m \end{bmatrix} = \det\begin{bmatrix} I_n - BA & B \\ O & I_m \end{bmatrix} = \det(I_n - BA)$. So we have $\det(I_m - AB) = \det(I_n - BA)$. Replace A by $-A$ and we are done. $\square$

Corollary 7.6.3. *$I_n + AB$ is invertible iff $I_m + BA$ is invertible. In particular, $I + \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ is invertible iff $1 + \boldsymbol{v}^{\mathrm{T}}\boldsymbol{u} \neq 0$.*

As cool as this corollary is, it suffers from the fact that it is redundant. The core of the argument is the LDU block decomposition. However, the LDU block decomposition of $\begin{bmatrix} I_n & B \\ A & I_m \end{bmatrix}$ already tells you that $I_n + AB$ is invertible iff $I_m + BA$ is invertible, in fact it gives you the FORMULA for finding the inverses, i.e., the Sherman-Morrison formula.

Nevertheless, both $\det(I_m + AB) = \det(I_n + BA)$ and the Sherman-Morrison formula is used mainly for ONE thing: low rank perturbations. If you realize that the $n \times n$ determinant you are working on is "almost" some nice matrix, where the difference has rank k which is small, then formula $\det(I + AB) = \det(I + BA)$ might reduce the computation of an $n \times n$ determinant to a $k \times k$ determinant. Let us see some examples.

Example 7.6.4. Consider $A = \begin{bmatrix} 1 + a_1 b_1 & a_1 b_2 & \dots & a_1 b_n \\ a_2 b_1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1} b_n \\ a_n b_1 & \dots & a_n b_{n-1} & 1 + a_n b_n \end{bmatrix}$. Find the determinant.

(Note that there is a tedious method of doing fancy row/column operations, and then set up inductions, and then find the generic formula. Try yourself if you want some mental exercise/torture.)

When we do mathematics, or science in general, it is VITAL to NOT ignore your first instinct and intuitions. Capture the fleeting glimps of genius perspective in your mind, and identify the related tools from that perspective, and the solution will be natural.

Think about this. If only all the ones on the diagonals are GONE, then our matrix would be $\begin{bmatrix} a_1 b_1 & \dots & a_1 b_n \\ \vdots & \ddots & \vdots \\ a_n b_1 & \dots & a_n b_n \end{bmatrix}$. This determinant is easy: it is zero (when $n > 1$). Because it has rank 1: all its columns are parallel! In fact, let $\boldsymbol{a}, \boldsymbol{b}$ be the obvious vector recording those a_i, b_i, then this matrix is $\boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}$.

In particular, $A = I + \boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}$. It is close to identity, where the difference has rank one. Time to use the formula $\det(I + AB) = \det(I + BA)$! We have $\det(A) = \det(I + \boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}) = 1 + \boldsymbol{b}^{\mathrm{T}}\boldsymbol{a} = 1 + \sum a_i b_i$. Done. ☺

Example 7.6.5. Consider $A = \begin{bmatrix} 1 + a_1 + b_1 & a_1 + b_2 & \dots & a_1 + b_n \\ a_2 + b_1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1} + b_n \\ a_n + b_1 & \dots & a_n + b_{n-1} & 1 + a_n + b_n \end{bmatrix}$. Find the determinant.

(Note that there is a tedious method of doing fancy row/column operations, and then set up inductions, and then find the generic formula. This is even worse than last one.)

Think about this. If only all the ones on the diagonals are GONE, then our matrix would be $\begin{bmatrix} a_1 + b_1 & \dots & a_1 + b_n \\ \vdots & \ddots & \vdots \\ a_n + b_1 & \dots & a_n + b_n \end{bmatrix}$. This determinant is easy: it is zero (when $n > 2$). Because it has rank 2: all its columns are spanned by $\boldsymbol{a}, \boldsymbol{u}$, where $\boldsymbol{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ and $\boldsymbol{u} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$.

In fact, let $\boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$, then this matrix is $\begin{bmatrix} \boldsymbol{a} & \boldsymbol{u} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^{\mathrm{T}} \\ \boldsymbol{b}^{\mathrm{T}} \end{bmatrix}$.

In particular, $A = I + \begin{bmatrix} \boldsymbol{a} & \boldsymbol{u} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^{\mathrm{T}} \\ \boldsymbol{b}^{\mathrm{T}} \end{bmatrix}$. It is close to identity, where the difference has rank two. Time to use the formula $\det(I + AB) = \det(I + BA)$!

We have $\det(A) = \det(I + \begin{bmatrix} \boldsymbol{a} & \boldsymbol{u} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}^{\mathrm{T}} \\ \boldsymbol{b}^{\mathrm{T}} \end{bmatrix}) = \det(I_2 + \begin{bmatrix} \boldsymbol{u}^{\mathrm{T}} & \boldsymbol{b}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{u} \end{bmatrix}) = \det \begin{bmatrix} 1 + \boldsymbol{u}^{\mathrm{T}}\boldsymbol{a} & \boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} \\ \boldsymbol{b}^{\mathrm{T}}\boldsymbol{a} & 1 + \boldsymbol{b}^{\mathrm{T}}\boldsymbol{u} \end{bmatrix} = (1 + \sum a_i)(1 + \sum b_i) - n \sum a_i b_i$. Done.

If you would like some statistical interpretation, note that we can further expand via $(1 + \sum a_i)(1 + \sum b_i) - n \sum a_i b_i = 1 + n\mathbb{E}(a) + n\mathbb{E}(b) - n^2(\mathbb{E}(ab) - \mathbb{E}(a)\mathbb{E}(b)) = 1 + n\mathbb{E}(a) + n\mathbb{E}(b) - n^2\mathrm{Cov}(a, b)$. So the degree one terms record the expected value, while the degree two term record the covariance. ☺

Example 7.6.6. This is a tough example. Many beginners will bang their head against this using row and column operations, hoping to find some magical operations or magical Laplace expansion, or hoping to set up induction, and all to no avail.

Consider $A = \begin{bmatrix} a_1 & b & \dots & b \\ b & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & b \\ b & \dots & b & a_n \end{bmatrix}$. You may try to do some operations and inductions, but the inductive formula would end up as $\det(A_n) = (a_n - b)\det(A_{n-1}) + b\prod_{i=1}^{n-1}(a_i - b)$. The generic solution is NOT easy to work out from this (albeit possible).

Instead, think about this. Wouldn't it be nice if we have a matrix where all entries are b? Then the answer is probably just zero, due to all the repeating columns. Similarly, if there are no b and all we had are the diagonal entries, then this is also nice, since determinant of a diagonal matrix is easy.

Capture this intuition and capitalize on this intuition. Our observation is essentially that our matrix A is rank one away from diagonal. Let $D = \begin{bmatrix} a_1 - b & & \\ & \ddots & \\ & & a_n - b \end{bmatrix}$ and $\boldsymbol{u} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$. Then $A = D + b\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$. Since A and D differ by a small rank, this means it is time for the formula $\det(I_m - AB) = \det(I_n - BA)$.

Suppose $a_i \neq b$ for all i, i.e., D is invertible. Then $\det(A) = \det(D + b\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}) = \det(D)\det(I + (bD^{-1}\boldsymbol{u})\boldsymbol{u}^{\mathrm{T}}) = \det(D)(1 + \boldsymbol{u}^{\mathrm{T}}(bD^{-1}\boldsymbol{u})) = [\prod(a_i - b)](1 + b\sum \frac{1}{a_i - b})$. Done. In particular, we reduced our calculation of $n \times n$ determinants to 1×1 determinants.

What if some $a_i = b$ happens? Note that in the formula $[\prod(a_i - b)](1 + b\sum \frac{1}{a_i - b})$, the denominators are "fake" , since once multiplied by $\prod(a_i - b)$, each summands of $1 + b\sum \frac{1}{a_i - b}$ would be clear of denominators. So it is in fact the same formula. Say $a_k = b$. Then the only summand without factor $(a_i - k)$ would be $b\prod_{i \neq k}(a_i - b)$, and this is the answer.

Of course, if some $a_k = b$, in fact it is easy to use row operations to make it upper triangular where the diagonals are simply the factors of $b\prod_{i \neq k}(a_i - b)$. I'll leave that method to you. ☺

Let me iterate again: the esssense of structural relation between $I + AB$ and $I + BA$ is a rank perturbation. Suppose we want to study an $n \times n$ matrix X, and we don't know how. However, it looks close to some nice matrix Y, say $X - Y$ has rank k. Then $X - Y = UV$ for some $n \times k$ matrix U and $k \times n$ matrix V. Then $X = Y + UV = Y(I_n + (Y^{-1}U)V)$. And we can utilize the relation between $I_n + (Y^{-1}U)V$ and $I_k + VY^{-1}U$. And now we have reduced the study of X, some $n \times n$ matrix, to the study of $I_k + VY^{-1}U$, some $k \times k$ matrix.

Example 7.6.7. Let us prove again that $A^{-1} = \frac{1}{\det(A)}C^{\mathrm{T}}$, without talking about the geometry of parallelopipes at all.

Consider $\frac{\partial}{\partial a_{ij}}\det(A)$. In essense, we are interested in $\det(A + \mathrm{d}x\boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}})$, i.e., what would happen to the determinant if we increase the (i, j) entry a bit. Note that this is a rank one difference from A.

So we have $\det(A+\mathrm{d}x\boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}}) = \det(A)\det(I+\mathrm{d}xA^{-1}\boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}}) = \det(A)(1+\mathrm{d}x\boldsymbol{e}_j^{\mathrm{T}}A^{-1}\boldsymbol{e}_i)$. In particular, $\frac{\partial}{\partial a_{ij}}\det(A) = \lim_{\mathrm{d}x\to 0}\frac{\det(A+\mathrm{d}x\boldsymbol{e}_i\boldsymbol{e}_j^{\mathrm{T}})-\det(A)}{\mathrm{d}x} = \lim_{\mathrm{d}x\to 0}\frac{\det(A)\,\mathrm{d}x\boldsymbol{e}_j^{\mathrm{T}}A^{-1}\boldsymbol{e}_i)}{\mathrm{d}x} = \det(A)\boldsymbol{e}_j^{\mathrm{T}}A^{-1}\boldsymbol{e}_i$.

But we also know that this is c_{ij}. So we see that $c_{ij} = \det(A)\boldsymbol{e}_j^{\mathrm{T}}A^{-1}\boldsymbol{e}_i$, i.e., the (i,j) entry of $\det(A)(A^{-1})^{\mathrm{T}}$. So $C = \det(A)(A^{-1})^{\mathrm{T}}$, from which we see that $A^{-1} = \frac{1}{\det(A)}C^{\mathrm{T}}$. ☺

Here is one final super interesting application.

Theorem 7.6.8 (Derivative of the determinant near identity is trace)**.** *Let* $f_A(t) = \det(I+tA)$. *Then* $f'_A(0) = \mathrm{trace}(A)$.

Proof. We go from I gradually towards $I+tA$. How to do this gradually? We do this one rank at a time. (One column at a time.)

Say $A = \begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_n\end{bmatrix}$. Then $A = \sum \boldsymbol{a}_i\boldsymbol{e}_i^{\mathrm{T}}$, this way we write A as a sum of n matrices of rank 1. (Can you see why?)

Now when t is infinitesimally small, we see that $\prod(I+t\boldsymbol{a}_i\boldsymbol{e}_i^{\mathrm{T}}) = I+t\sum\boldsymbol{a}_i\boldsymbol{e}_i^{\mathrm{T}}+t^2(\text{stuff}) = I+tA+t^2(\text{stuff})$, where t^2 would be too small to be relevant. So $\det(I+tA) = \det(\prod(I+t\boldsymbol{a}_i\boldsymbol{e}_i^{\mathrm{T}}))+t^2(\text{stuff})$ when t is infinitesimally small. (Note that the determinant is just a big polynomial in the entries, so the extra terms using t^2 entries would ALL have factor t^2.)

Note that $\det(\prod(I+t\boldsymbol{a}_i\boldsymbol{e}_i^{\mathrm{T}})) = \prod(\det(I+t\boldsymbol{a}_i\boldsymbol{e}_i^{\mathrm{T}})) = \prod(1+t\boldsymbol{e}_i^{\mathrm{T}}\boldsymbol{a}_i) = \prod(1+ta_{ii}) = 1+t\,\mathrm{trace}(A)+t^2(\text{stuff})$.

So all in all, we have $\det(I+tA) = 1+t\,\mathrm{trace}(A)+t^2(\text{stuff})$. Taking derivative at $t=0$, our result is obvious. ☐

Note that we did NOT use ANY property of trace other than its definition as the sum of diagonals. So in fact this gives a super cool proof that $\mathrm{trace}(AB) = \mathrm{trace}(BA)$.

Proposition 7.6.9. $\mathrm{trace}(AB) = \mathrm{trace}(BA)$. *In particular, this implies that* $\mathrm{trace}(XAX^{-1}) = \mathrm{trace}(A)$, *i.e., trace is invariange under change of basis.*

Proof. Consider $\det(I+tAB) = \det(I+tBA)$. Take derivative and done. This is my personal favorite proof. (And in fact this explains what trace and determinants are, from the perspective of Lie group and Lie algebra, though they are outside the scope of this class.) ☐

7.6.2 Shear and Expand and Induction

This is the most traditional strategy to tackle any hard and tricky determinant problem. You do row and column operations to generate as much zero entries as possible. Hopefully these zeroes will concentrate, and when they concentrate in some row or in some column, you expand along that row or that column.

Sometime, this expansion allows you to reduce some $n\times n$ determinant into some $(n-1)\times(n-1)$ determinant, and then you can use induction.

Example 7.6.10. Consider an $n\times n$ matrix $A_n = \begin{bmatrix}1 & -1 & & \\ 1 & \ddots & \ddots & \\ & \ddots & \ddots & -1 \\ & & 1 & 1\end{bmatrix}$. What is $\det(A_n)$?

Let us do Laplace expansion alongthe first column. Then we see that

$$\det(A_n) = 1\det(A_{n-1}) + 1(-\det\begin{bmatrix}-1 & O \\ & A_{n-2}\end{bmatrix}) = \det(A_{n-1})+\det(A_{n-2}).$$

In particular, if we think of the $\det(A_n)$ as a sequence for increasing n, then each term is the sum of previous two terms. Sounds familiar? This is the same induction scheme for the famous Fibonacci sequence.

Furthermore, $\det(A_1) = 1$ and $\det(A_2) = 2$. Hence this sequence goes like $1,2,3,5,8,13,\dots$. It is indeed the Fibonacci sequence.

If you have hardcore skills in working out sequences using mathematical induction, the you can find a formula. We have $\det(A_n) = \frac{\phi^{n+1}-(1-\phi)^{n+1}}{\sqrt{5}}$ where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. ☺

Example 7.6.11. Consider $\det A$ where $A = \begin{bmatrix} a_1 & b & \dots & b \\ c & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & b \\ c & \dots & c & a_n \end{bmatrix}$, where $b \neq c$. Note that the case when $b = c$ is solved previously using the formula $\det(I + AB) = \det(I + BA)$.

This determinant is notoriously tricky for beginners. Here I did NOT present the fastest solution. However, this is a faithful implement of the strategy lined out above. The purpose of doing this problem is NOT the result or speed, but rather on the strategy itself. I will intentionally try to AVOID magical row/column operations that simplify this at various places.

You see, the obvious direction to go is to, say, take the second to last column and substract this from the last column. This gives $\begin{bmatrix} a_1 & b & \dots & b & 0 \\ c & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & b & 0 \\ c & \dots & c & a_{n-1} & b - a_{n-1} \\ c & \dots & c & c & a_n - c \end{bmatrix}$.

Now we have a concentration of zeros on the last column, so we expand. This gives us:

$$\det A = (a_n - c)\det \begin{bmatrix} a_1 & b & \dots & b \\ c & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & b \\ c & \dots & c & a_{n-1} \end{bmatrix} + (-1)(b - a_{n-1})\det \begin{bmatrix} a_1 & b & \dots & b & b \\ c & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & b & b \\ c & \dots & c & a_{n-2} & b \\ c & \dots & c & c & c \end{bmatrix}.$$

Now the first term here is clearly a good set up for induction. What about the second term here?

Let $A_k = \begin{bmatrix} a_1 & b & \dots & b \\ c & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & b \\ c & \dots & c & a_k \end{bmatrix}$ and $B_k = \begin{bmatrix} a_1 & b & \dots & b & b \\ c & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & b & b \\ c & \dots & c & a_k & b \\ c & \dots & c & c & c \end{bmatrix}$. We have $\det(A_n) = (a_n - c)\det(A_{n-1}) + (a_{n-1} - b)\det(B_{n-2})$. Let us first figure out the B_k portion of this formula.

Again use the second to last column of B_k and substract this from the last column. Then we see that

$$\det(B_k) = \det \begin{bmatrix} a_1 & b & \dots & b & 0 \\ c & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & b & 0 \\ c & \dots & c & a_k & b - a_k \\ c & \dots & c & c & 0 \end{bmatrix} = (a_k - b)\det(B_{k-1}).$$

This set up the induction nicely. So we have $\det(B_k) = (a_k - b)\det(B_{k-1}) = (a_k - b)(a_{k-1} - b)\det(B_{k-2}) = \dots = [\prod_{i=1}^{k}(a_i - b)]\det(B_0)$. Note that $B_0 = \begin{bmatrix} c \end{bmatrix}$. So $\det(B_k) = c\prod_{i=1}^{k}(a_i - b)$.

Go back to A_n, we have $\det(A_n) = (a_n - c)\det(A_{n-1}) + (a_{n-1} - b)\det(B_{n-2}) = (a_n - c)\det(A_{n-1}) + c\prod_{i=1}^{n-1}(a_i - b)$. Now ideally, one must be able to deduce the generic formula from this inductive formula. In parctice, while doable, this is a bit annoying and lengthy. So instead, we cheat by using symmetry.

Note that $\det(A_n) = \det(A_n^{\mathrm{T}})$, yet the difference between A_n and A_n^{T} is simply swapping b, c. As a result of this symmetry, if we have $\det(A_n) = (a_n - c)\det(A_{n-1}) + c\prod_{i=1}^{n-1}(a_i - b)$, we must symmetrically have $\det(A_n) = (a_n - b)\det(A_{n-1}) + b\prod_{i=1}^{n-1}(a_i - c)$. Now use $(a_n - b)$ times the first equation minus $(a_n - c)$ times the second equation, we obtain the equation $(c - b)\det(A_n) = c\prod_{i=1}^{n}(a_i - b) - b\prod_{i=1}^{n}(a_i - c)$.

In particular, if $b \neq c$, then we are done. We have $\det(A_n) = \frac{c}{c-b}\prod_{i=1}^n (a_i - b) + \frac{b}{b-c}\prod_{i=1}^n (a_i - c)$. ☺

Note that this example here cannot be done via low rank perturbations, since the b portion and c portion are both triangular with almost full rank.

Here is another very tough one.

Example 7.6.12. Consider the matrix $A = \begin{bmatrix} a & 1 & & \\ n & \ddots & \ddots & \\ & \ddots & \ddots & n \\ & & 1 & a \end{bmatrix}$. Find the determinant. (This is the notorious Kac-Sylvester matrix.)

This is seriously annoying, even with all the zero entries. Say if you expand along the last row, then you shall see that you FAIL to set up an induction, because the 1 through n and n through 1 anti-symmetric pattern here.

The magic trick is this. Consider say $n = 5$ for illustration purppose. We have $\begin{bmatrix} a & 1 & & & & \\ 5 & a & 2 & & & \\ & 4 & a & 3 & & \\ & & 3 & a & 4 & \\ & & & 2 & a & 5 \\ & & & & 1 & a \end{bmatrix}$.

Now, we add the i-th row to the $(i-2)$-th row, $(i-4)$-th row and so on, for all i. We do these row operations in the order of increasing i. Effectively, it is as if we are going from A to EA for the matrix $E = \begin{bmatrix} 1 & 0 & 1 & \dots & \cdot \\ & \ddots & \ddots & \ddots & \vdots \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ & & & & 1 \end{bmatrix}$, where the values alternate between 1 and 0 for this upper triangular matrix E.

After all these row shearings, we end up with $EA = \begin{bmatrix} a & 5 & a & 5 & a & 5 \\ 5 & a & 5 & a & 5 & a \\ & 4 & a & 5 & a & 5 \\ & & 3 & a & 5 & a \\ & & & 2 & a & 5 \\ & & & & 1 & a \end{bmatrix}$.

This looks neat, yes?

Next, we do E^{-1} as a column operation, and go from EA to EAE^{-1}. In terms of operations, it means we substract the i-th column from the $(i+2)$-th column, $(i+4)$-th column, and so on, for all i. We do these column operations in the order of increasing i.

This way all the extra a's introduced by shearing will cancel out, we end up with $\begin{bmatrix} a & 5 & & & & \\ 5 & a & & & & \\ & 4 & a & 1 & & \\ & & 3 & a & 2 & \\ & & & 2 & a & 3 \\ & & & & 1 & a \end{bmatrix}$.

Note that this is block lower triangular. So $\det(A_5) = (a^2 - 5^2)\det(A_3) = (a+5)(a-5)\det(A_3)$.

By induction, you can see that $\det(A_n) = (a+n)(a-n)\det(A_{n-2}) = (a+n)(a+n-2)\dots(a-n+2)(a-n)$. For example, for A_5 the determinant is $(a+5)(a+3)(a+1)(a-1)(a-3)(a-5)$.

Note that, essentially, we doing $\det(EAE^{-1})$, and as a linear transformation, EAE^{-1} is merely A after a change of basis! Indeed, for any invertible X, then $\det(XAX^{-1}) = \det(X)\det(A)\det(X)^{-1} = \det(A)$. So just like trace is invariant under a change of basis, determinant is also invariant under a change of basis. And similar matrices must always have the same determinant.

In essence, E^{-1} is a basis on which the linear map A has slightly better behavior. We change basis, and induction appear. ☺

7.6.3 Polynomial Interpretation, Interpolation and the Vandermonde Matrix

One niche use of the determinant is to show that a matrix is invertible. However, bear in mind that MOST of the time it is easier to simply look at the kernel. Nevertheless, the following case is both interesting and useful later.

Example 7.6.13. Two points determines a line. Three points determines a parabola $p(x) = a + bx + cx^2$. In general, you can imagine that $n + 1$ points can determine a polynomial of degree n. Can we prove this?

Consider an unknown parabola $p(x) = a + bx + cx^2$. Suppose we know that $p(1) = 2, p(2) = 3, p(3) = 4$, let us find the unique parabola through them. Then we have a linear system

$$\begin{aligned} a + b + c &= 2 \\ a + 2b + 4c &= 3 \\ a + 3b + 9c &= 4. \end{aligned}$$

Write this in matrix form, and we have

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

So, to find a, b, c, it is enough to solve this system using Gaussian elimination. Note that the matrix here is really nice. As a linear map, it sends "coefficients" of a polynomial to "evaluations" of the polynomial at the given points. So, given evaluations $p(1) = 2, p(2) = 3, p(3) = 4$, to solve the unknown coefficients is the same as doing the inverse of this linear map. ☺

Definition 7.6.14. *The Vandermonde matrix is* $V_{n+1} = \begin{bmatrix} 1 & a_1 & \dots & a_1^n \\ 1 & a_2 & \dots & a_2^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_{n+1} & \dots & a_{n+1}^n \end{bmatrix}$ *for some chosen constants* $a_1, \dots, a_{n+1} \in \mathbb{R}$.

Example 7.6.15. Let V be the space of polynomials of degree at most n. This is an $(n+1)$ dimensional space. Pick any $x_1, \dots, x_{n+1} \in \mathbb{R}$, then we can build a linear map $E : V \to \mathbb{R}^{n+1}$ such that $E(p) = \begin{bmatrix} p(x_1) \\ \vdots \\ p(x_{n+1}) \end{bmatrix}$.

This process is secretly linear. Even though $p(x)$ is probably NOT linear in x, but the expression $p(x)$ is linear in p. What I mean is, given two polynomials p, q, then $(ap + bq)(x) = ap(x) + bq(x)$.

Now, if E is a linear map, what is the matrix for E? Well, pick basis $1, x, x^2, \dots, x^n$ for V, and pick the standard basis for $\mathbb{R}^{n+1}$.

The image for $E(1)$ is $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$. The image $E(x^k)$ is $\begin{bmatrix} x_1^k \\ \vdots \\ x_{n+1}^k \end{bmatrix}$ for each k. So all in all, the matrix for E is $\begin{bmatrix} 1 & x_1 & \dots & x_1^n \\ 1 & x_2 & \dots & x_2^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n+1} & \dots & x_{n+1}^n \end{bmatrix}$.

As you can see, the Vandermonde matrix is precisely the linear map E. ☺

Theorem 7.6.16 (Polynomial Interpolation Theorem). *Given any distinct inputs $x_1, \dots, x_{n+1}$ and corresponding outputs $y_1, \dots, y_{n+1}$, there is a UNIQUE polynomial $p(x)$ of degree at most n such that $p(x_i) = y_i$ for all i.*

Proof. Suppose the unknown polynomial is $p(x) = a_0 + a_1 x + \cdots + a_n x^n$. Then let V be the Vandermonde matrix for $x_1, \dots, x_{n+1}$, we have

$$V \begin{bmatrix} a_0 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_{n+1} \end{bmatrix}.$$

So we have unique solution if and only if V is invertible.

We do this by showing that the determinant is non-zero. Look at the lemma below. $\square$

Proposition 7.6.17. *For the Vandermond matrix*

$$V_{n+1} = \begin{bmatrix} 1 & a_1 & \dots & a_1^n \\ 1 & a_2 & \dots & a_2^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_{n+1} & \dots & a_{n+1}^n \end{bmatrix},$$

its determinant is $\det(V_{n+1}) = \prod_{i<j}(a_j - a_i)$. *In particular, if all a_i are distinct, then V_{n+1} is invertible.*

Proof. We give two proofs.

The first proof is traditional. To find $\det V_{n+1}$, going from RIGHT to LEFT, we smartly replace each column $\boldsymbol{c}_i$ by $\boldsymbol{c}_i - a_{n+1}\boldsymbol{c}_{i-1}$. (Yeah, this is NOT a typo. We meant to write a_{n+1} here.) These are all shearings and they preserve the determinant. (The total column action here is equivalent to multiplying $\begin{bmatrix} 1 & -a_{n+1} & & \\ & \ddots & \ddots & \\ & & \ddots & -a_{n+1} \\ & & & 1 \end{bmatrix}$ to the right of V_{n+1}.)

Then magically, we now only need to calculate $\det \begin{bmatrix} 1 & (a_1 - a_{n+1}) & a_1(a_1 - a_{n+1}) & \dots & a_1^{n-1}(a_1 - a_{n+1}) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & (a_n - a_{n+1}) & a_n(a_n - a_{n+1}) & \dots & a_n^{n-1}(a_n - a_{n+1}) \\ 1 & 0 & 0 & \dots & 0 \end{bmatrix}$.

Now expand along the last row (or consider non-zero terms in the big formula), we see that this is $(-1)^{n+1} \det \begin{bmatrix} (a_1 - a_{n+1}) & a_1(a_1 - a_n) & \dots & a_1^{n-1}(a_1 - a_n) \\ \vdots & \vdots & \ddots & \vdots \\ (a_n - a_{n+1}) & a_n(a_n - a_{n+1}) & \dots & a_n^{n-1}(a_n - a_{n+1}) \end{bmatrix}$.

Take out common factors, we have $(-1)^{n+1}(a_1 - a_{n+1}) \dots (a_n - a_{n+1}) \det \begin{bmatrix} 1 & a_1 & \dots & a_1^{n-1} \\ 1 & a_2 & \dots & a_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \dots & a_n^{n-1} \end{bmatrix} = \prod_{1 \le i \le n}(a_{n+1} - a_i) \det V_n$. Hey! By induction (and the base case is very trivial), we have

$$\det V_{n+1} = [\prod_{1 \le i \le n}(a_{n+1} - a_i)][\prod_{1 \le i < j \le n}(a_j - a_i)] = \prod_{1 \le i < j \le n+1}(a_j - a_i).$$

Well, this is not easy to think of, but it gets the job done. Now let me show you my favorite proof.

Recall that, according to the big formula, $\det(V_n)$ is simply a big polynomial on the variables $a_1, \dots, a_n$. What are its factors?

Note that, if $a_i = a_j$, then the i-th row and j-th row of V_n would coincide, so we would have $\det(V_n) = 0$. This means that the polynomial $a_j - a_i$ is a factor of the polynomial $\det(V_n)$ for all $i \neq j$. (In fact, since we already know the answer, these are ALL the factors. HAHAHA!)

In particular, we see that $\prod_{i<j}(a_j - a_i)$ is a factor of $\det(V_n)$. Now each term in the big formula of $\det(V_n)$ has a total degree of $1 + 2 + \cdots + (n-1) = \frac{1}{2}n(n-1)$, which is EXACTLY the degree of $\prod_{i<j}(a_j - a_i)$. So we have $\det(V_n) = k\prod_{i<j}(a_j - a_i)$.

Now check the coefficient of the term $a_2a_3^2 \dots a_n^{n-1}$ in both sides. $\prod_{i<j}(a_j - a_i)$ can only reach this term by always pick a_j in each factor. $\det(V_n)$ can only reach this term by picking the diagonal entries. So we see that both coefficients are one. Hence $k = 1$ and we are done. (On the product side, this is obvious. On the determinant side, this term can only come from the diagonal.) ☐

Remark 7.6.18. *The second proof above is useful for many similar things. For example, consider the Cauchy matrix* $\begin{bmatrix} \frac{1}{x_1+y_1} & \cdots & \frac{1}{x_1+y_n} \\ \vdots & \ddots & \vdots \\ \frac{1}{x_n+y_1} & \cdots & \frac{1}{x_n+y_n} \end{bmatrix}$. *What is this determinant? Try it yourself. People has worked out the LDU decompositions and inverse of Cauchy matrices as well (see wikipedia and related papers).*

Here's some optional fun fact. A special case of the Cauchy matrix is the Hilbert matrix, where $x_i + 1 = y_i = i$ for all i. If we let V be the space of polynomials of degree at nost $n-1$, with basis $1, x, \dots, x^{n-1}$, and inner product $\langle p(x), q(x)\rangle = \int_0^1 p(x)q(x)\,\mathrm{d}x$, then the $n \times n$ Hilbert matrix is actually the corresponding Gram matrix.

Polynomial interpolation is the major reason we bother with the Vandermonde matrix. (Another reason is the study of eigenspaces in later chapters.) Now that we have its determinant, it is time to find its inverse. How to find $V_n^{-1}\boldsymbol{e}_1$, for example? (I.e., how to find the first column of V_n^{-1}.)

Given any distinct inputs $a_1, \dots, a_n$, we want to find the UNIQUE polynomial $p(x)$ of degree at most $n-1$ such that $p(a_i) = 0$ for all $i \neq 1$ and $p(a_1) = 1$. Then $x - a_i$ for all $i \neq 1$ should be a factor of this polynomial.

Consider the polynomial $\prod_{i\neq 1}(x - a_i)$. It has degree $n-1$, and takes value 0 on $a_2, \dots, a_n$, and it takes value $\prod_{i\neq 1}(a_1 - a_i)$ at a_1. So let $p(x) = \frac{\prod_{i\neq 1}(x-a_i)}{\prod_{i\neq 1}(a_1-a_i)}$, and we are done. (Note the similarity in ideas with the second proof of the Vandermonde determinant! Finding factor polynomials is the essense of this problem. Column operations work, but not as elegant, because it is not the essense of the problem.)

Proposition 7.6.19 (Lagrange Interpolation)**.** *Given any distinct inputs $a_1, \dots, a_{n+1}$ and corresponding outputs $b_1, \dots, b_{n+1}$, then the UNIQUE polynomial $p(x)$ of degree at most n is $p = \sum b_k p_k$, where $p_k(x) = \frac{\prod_{i\neq k}(x-a_i)}{\prod_{i\neq k}(a_k-a_i)}$.*

The coeffients of p_k are the entries of V_{n+1}^{-1}. They are not pretty, and it is usually easier to do abstractly via polynomials.

One can go even further to get the more general Hermite interpolation, though that will require Jordan canonical form, and thus outside of current ability yet.

7.6.4 (Optional) A determinant game

This game is proposed by a Putnam mathematical contest. Suppose Alice and Bob are playing a game. We have an $n \times n$ matrix with unfilled entries. Alice and Bob take turns to fill out the entries. Say if Alice goes first, she can fill out a number in an entry. Then Bob can go, and fill out another number in another entry. The game ends when the matrix is filled.

Now if the game ends with an invertible matrix, then Alice wins. If the game ends with determinant zero, the Bob wins. Suppose Alice goes first, and $n = 2020$, who has the winning strategy?

Proposition 7.6.20. *If n is even and Alice goes first, then Bob always win.*

Proof. If Alice write a number in the first or second column, then Bob write the same number in the same row but in the second or first column, respectively. If Alice write a number outside of the first and second columns, then Bob also fill out an arbitrary entry outside of the first and second columns.

This way, Bob can guarantee that the first two column of the resulting matrix is always the same. So Bob wins. □

Now, this will NOT work if n is odd. In that case, there would be an ODD number of entries outside of the first and second columns. So if Alice keep playing outside of the first two columns, then Bob would eventually be forced to go first in the first and second column as there is nowhere else to play. Thus this strategy would not work.

However, Bob might still win with some other strategies.

Proposition 7.6.21. *If $n = 3$ and Alice goes first, then Bob always win.*

Proof. Note that we can permute the rows and columns at all time without changeing the invertibility of the matrix. So WLOG suppose Alice played in the upper left corner. If Alice fill it by zero, then Bob can fill a zero below. Then Alice must fill the last entry in the first column by a non-zero number, else she would loose by Bob filling out the first column with all zeros. Next Bob can fill out the zero at the center, and Alice has no choice but to block at the center-right entry, else lose by a central row of zeros. Then Bob can fill out the last of the upper 2×2 block, so that this entire block is zero. This 2×2 block of zeros would force the matrix to have rank one, and hence determinant zero. Bob wins.

Obviously Alice is better off playing a non-zero number to start with. Suppose the first step of Alice is non-zero in the upper left corner. Now, by wisdom granted to us via the tic-tac-toe game, Bob's winning strategy is to fill out the center entry by zero, thus we have $\begin{bmatrix} a_1 & & \\ & 0 & \\ & & \end{bmatrix}$. Note that, from here on out, Alice CANNOT allow a row of zeros or a column of zeros. The tic-tac-toe feeling is strong now.

Now Alice play next. Afterwards Bob can keep filling out zeros to force a 2×2 block of zeros, and Alice has no choice but to block rows or columns on the way. This 2×2 block of zeros would force the matrix to have rank one, and hence determinant zero. Specifically, we have the following four possibilities, where the subscript indicates the order of steps. Alice will play $a_1, a_2, \ldots, a_5$ while Bob will only play zeros, $0_1, \ldots, 0_5$.

The four possibilities are (up to taking transpose)

$$\begin{bmatrix} a_1 & a_3 & a_5 \\ a_2 & 0_1 & 0_4 \\ a_4 & 0_2 & 0_3 \end{bmatrix}, \begin{bmatrix} a_1 & a_3 & a_5 \\ a_4 & 0_1 & 0_3 \\ a_2 & 0_2 & 0_4 \end{bmatrix}, \begin{bmatrix} a_1 & 0_4 & 0_3 \\ a_3 & 0_1 & 0_2 \\ a_5 & a_2 & a_4 \end{bmatrix}, \begin{bmatrix} a_1 & 0_3 & 0_4 \\ a_3 & 0_1 & 0_2 \\ a_5 & a_4 & a_2 \end{bmatrix}.$$

□

However, Bob do not always win. For example, if $n = 1$, obviously Alice would win. The generic case when n is odd is still unknown. I have no idea about who might win the $n = 5$ game. (But I think playing only zeros is no longer a viable strategy for Bob.)

What if Bob plays first? Then if n is odd, Bob can just fill out some entry outside of the first and second columns, and then resume the copy-cat strategy to win.

What if n is even and Bob goes first? I have no clue yet. You might want to try the easy case when $n = 2, 4$.

7.7 Exercises

Exercise 7.7.1. *Calculate the following determinants.*

1. $\begin{bmatrix}1 & 2 & 1 & 4\\ 0 & -1 & 2 & 1\\ 0 & 0 & 2 & -1\\ 0 & 0 & 0 & 3\end{bmatrix}$.

2. $\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & 2 & 4 & 8\\ 1 & 3 & 9 & 27\\ 1 & -1 & 1 & -1\end{bmatrix}$. *(This is a Vandermonde matrix for* $1, 2, 3, -1$*. The answer should be* $\pm(3-2)(3-1)(3-(-1))(2-1)(2-(-1))(1-(-1))$*, i.e., plus or minus the product of all possible differences of these values.)*

3. $\begin{bmatrix}1 & 2 & 3 & 4\\ 2 & 3 & 4 & 1\\ 3 & 4 & 1 & 2\\ 4 & 1 & 2 & 3\end{bmatrix}$. *(This is an anti-circulant matrix.)*

4. $\begin{bmatrix}1 & 2 & 3 & 4 & 5\\ 6 & 7 & 8 & 9 & 10\\ 11 & 12 & 0 & 0 & 0\\ 13 & 14 & 0 & 0 & 0\\ 15 & 16 & 0 & 0 & 0\end{bmatrix}$. *(Hint: Don't calculate. Just stare at it until you see the aswer.)*

5. $\begin{bmatrix}0 & \dots & 0 & 1\\ \vdots & \iddots & \iddots & 0\\ 0 & \iddots & \iddots & \vdots\\ 1 & 0 & \dots & 0\end{bmatrix}$. *The matrix is* $n \times n$*.*

Exercise 7.7.2. *Let* A *be a* 3×3 *matrix* $\begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}}\\ \boldsymbol{a}_2^{\mathrm{T}}\\ \boldsymbol{a}_3^{\mathrm{T}}\end{bmatrix}$ *with* $\det(A) = 5$*. Find the determinant of the matrices below.*

1. $2A$.
2. $-A$.
3. A^2.
4. A^{-1}.
5. A^{T}.
6. $\begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}} - \boldsymbol{a}_3^{\mathrm{T}}\\ \boldsymbol{a}_2^{\mathrm{T}} - \boldsymbol{a}_1^{\mathrm{T}}\\ \boldsymbol{a}_3^{\mathrm{T}} - \boldsymbol{a}_2^{\mathrm{T}}\end{bmatrix}$. *(Hint: Do this in two ways if you like. You can do this with row operations, or via decomposition* $\det(EA) = \det(E)\det(A)$*.)*
7. $\begin{bmatrix}\boldsymbol{a}_1^{\mathrm{T}} + \boldsymbol{a}_3^{\mathrm{T}}\\ \boldsymbol{a}_2^{\mathrm{T}} + \boldsymbol{a}_1^{\mathrm{T}}\\ \boldsymbol{a}_3^{\mathrm{T}} + \boldsymbol{a}_2^{\mathrm{T}}\end{bmatrix}$. *(Hint: Do this in two ways if you like. You can do this with row operations, or via decomposition* $\det(EA) = \det(E)\det(A)$*.)*

Exercise 7.7.3. *Let* $A = \begin{bmatrix}\boldsymbol{a}_1 & \dots & \boldsymbol{a}_n\end{bmatrix}$ *be any* $n \times n$ *matrix. Geometrically speaking,* $|\det(A)|$ *is the absolute volumn of the parallelotope* A*, and* $\|\boldsymbol{a}_i\|$ *is the length of the* i*-th edge. As you can imagine, the* *volumn $|\det(A)|$ should be the product $\|\boldsymbol{a}_1\| \dots \|\boldsymbol{a}_n\|$ when all edges are mutually orthogonal. And when the edges are NOT mutualy orthogonal, then intuitively $|\det(A)|$ should be strictely less than the product $\|\boldsymbol{a}_1\| \dots \|\boldsymbol{a}_n\|$.*

This is the famous Hadamard inequality $|\det(A)| \leq \|\boldsymbol{a}_1\| \dots \|\boldsymbol{a}_n\|$. We prove this inequality in this problem.

1. *Suppose A is invertible and we have QR decomposition $A = QR$ where $R = \begin{bmatrix} \boldsymbol{r}_1 & \dots & \boldsymbol{r}_n \end{bmatrix}$ is upper triangular. Show that $\|\boldsymbol{r}_i\| = \|\boldsymbol{a}_i\|$.*

2. *Show that in the set up above, $\det(R) \leq \|\boldsymbol{r}_1\| \dots \|\boldsymbol{r}_n\|$.*

3. *Prove the Hadamard inequality $|\det(A)| \leq \|\boldsymbol{a}_1\| \dots \|\boldsymbol{a}_n\|$.*

Exercise 7.7.4. *Prove or find counter examples.*

1. *We always have $\det(AB - BA) = 0$ for any square matrices A, B.*

2. *We always have $\det(-A) = -\det(A)$ for square matrices A.*

3. *If n is odd and A is $n \times n$ and skew-symmetric, then A is not invertible. (Hint: take anther look at the last sub-problem.)*

4. ☕*Suppose we have LDU decomposition $A = LDU$. Let A_i be the upper left $i \times i$ block of A, and let d_i be the i-th diagonal entry of D. Then $d_i = \frac{\det(A_i)}{\det(A_{i-1})}$. (Hint: also write L, D, U in blocks.)*

Exercise 7.7.5 (Don't do these things)**.** *What's wrong with these arguments?*

1. *For block matrix we have $\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = AD - BC$. (Not part of the HW: What should the correct formula be if A is invertible?)*

2. *We have $\det(\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}) = \det(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}) = \frac{ad-bc}{ad-bc} = 1$.*

3. *If $AB = -BA$, then $\det(A)\det(B) = -\det(B)\det(A)$, so $2\det(A)\det(B) = 0$. So one of A, B is not invertible.*

Exercise 7.7.6 (Block determinant)**.**

1. *Prove that $\det \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = a^2 + b^2$.*

2. *For square $n \times n$ matrices A, B with $AB = BA$, prove that $|\det \begin{bmatrix} A & -B \\ B & A \end{bmatrix}| = |\det(A^2 + B^2)|$. (Hint: check out the multiplication $\begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} A & B \\ -B & A \end{bmatrix}$.)*

3. ☕*In the last subproblem, prove that in fact $\det \begin{bmatrix} A & -B \\ B & A \end{bmatrix} = \det(A^2 + B^2)$. (Hint: what is the coefficient of $(a_{11} \dots a_{nn})^2$ on both sides?)*

Exercise 7.7.7. *Calculate the following determinants.*

1. *A is 3×3 with determinant 5. Find the determinant of the cofactor matrix C for A. (Hint: What is $C^{\mathrm{T}} A$?)*

2. $\begin{bmatrix}1&1&1&1&1\\1&2&0&0&0\\1&0&3&0&0\\1&0&0&4&0\\1&0&0&0&5\end{bmatrix}$. *(You are not required to do the generalization of this, but feel free to try to find the determinant of* $\begin{bmatrix}a_1&1&\dots&1\\1&a_2&&\\\vdots&&\ddots&\\1&&&a_n\end{bmatrix}$. *Also feel free to use two methods: Laplace expansion or low rank perturbation.)*

Exercise 7.7.8 (Leibniz formula)**.** *Use Leibniz formula (the big formula) to help with the following problems.*

1. *Consider* $\det\begin{bmatrix}2x&x&1&2\\1&x&1&-1\\3&2&x&1\\1&1&1&x\end{bmatrix}$. *This is a polynomial in x. What is the coefficient for x^4? What is the coefficient for x^3?*

2. *Suppose we multiply each (i,j) entry of a 4×4 matrix A by j. For each term in the big formula of* $\det(A)$*, how would this term change? How would* $\det(A)$ *change?*

3. *Consider* $A=\begin{bmatrix}a&0&b&0\\0&c&0&d\\e&0&f&0\\0&g&0&h\end{bmatrix}$*, where all letter variables are non-zero. In the big formula of* $\det(A)$*, how many non-zero terms are there? Can you factorize the determinant? (Hint: This is a "hidden" block diagonal matrix.)*

Exercise 7.7.9 (Pascal Matrices)**.** *Let P_n be the $n\times n$ symmetric Pascal's matrix. (E.g., the 4×4 version is* $\begin{bmatrix}1&1&1&1\\1&2&3&4\\1&3&6&10\\1&4&10&20\end{bmatrix}$*.) Similarly, let L_n, U_n be the $n\times n$ lower triangular and upper triangular Pascal's matrix.*

(E.g., the 4×4 version is $L_n=\begin{bmatrix}1&&&\\1&1&&\\1&2&1&\\1&3&3&1\end{bmatrix}$*, and we always have* $U_n=L_n^{\mathrm{T}}$*.)*

For this problem, you may freely use the fact that P_n has LU decomposition $P_n=L_nU_n$.

1. *Find* $\det(P_n)$*.*

2. *Let A_n be obtained by reducing the lower right entry of P_n by 1. Find* $\det(A_n)$*. (Hint: Can you see how cofactor is involved here?)*

Exercise 7.7.10 (Laplace expansion)**.** *Use Laplace expansion or cofactors to help with the following problems.*

1. *A Hessenberg matrix is a matrix that is "almost" triangular, except for an extra diagonal of entries. For example, the matrices* $H_4=\begin{bmatrix}2&1&1&1\\1&2&1&1\\0&1&2&1\\0&0&1&2\end{bmatrix}$ *is Hessenberg. Define the $n\times n$ matrix*

$$H_n = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{bmatrix}.$$ *Show that* $\det(H_n)$ *is the Fibonacci sequence.*

2. *A tridiagonal matrix is a matrix that is simultaneously upper Hessenberg and lower Hessenberg. For examlpe,* $S_4 = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$ *is tridiagonal. Generalize* S_4 *into* $n \times n$ *matrices. Find the inductive formula for* $\det(S_n)$*. How is this related to the Fibonacci sequence? (Hint: to see the relation to the Fibonacci sequence, maybe write out the first several terms and compare?)*

Exercise 7.7.11 (Determinant and inverse matrix)**.** *Let* $f(\begin{bmatrix} a & b \\ c & d \end{bmatrix}) = \ln(ad - bc)$*. Recall that for any multi-variable function* f *and one of its variable* x*, then* $\frac{\partial f}{\partial x}$ *is the derivative of* f *when we hold all other variables constant, and only treat* x *as a variable.*

1. *Find* $\frac{\partial f}{\partial a}, \frac{\partial f}{\partial b}, \frac{\partial f}{\partial c}, \frac{\partial f}{\partial d}$.
2. *Show that* $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} \frac{\partial f}{\partial a} & \frac{\partial f}{\partial b} \\ \frac{\partial f}{\partial c} & \frac{\partial f}{\partial d} \end{bmatrix}^{\mathrm{T}}$.
3. ☕*Is this a coincidence? If not, can you generalize this to higher dimensions? (Hint: What is the derivative* $\frac{\partial \ln(\det(A))}{\partial a_{ij}}$*? Use chain rule and see what happens.)*

Exercise 7.7.12 (Determinant and orthogonal matrices)**.** *Let* Q *be an orthogonal matrix, i.e.,* $Q^{\mathrm{T}} = Q^{-1}$.

1. *Show that* $\det(Q) = \pm 1$.
2. *If* $\det(Q) = -1$*, show that* $\det(I + Q) = 0$*. (Hint:* $I + Q = Q(I + Q^{\mathrm{T}})$*.)*
3. *If* $\det(Q) = -1$*, then show that we can find* $\boldsymbol{n}$ *such that* $Q\boldsymbol{n} = -\boldsymbol{n}$*. So we see that at least a part of* Q *behaves like a reflection. (Hint: Last subproblem shows* $Q + I$ *is not invertible.)*
4. *If* $\det(Q) = 1$ *and it is* $n \times n$ *for an odd number* n*, show that* $\det(I - Q) = 0$.
5. *If* $\det(Q) = 1$*, then show that we can find* $\boldsymbol{n}$ *such that* $Q\boldsymbol{n} = \boldsymbol{n}$*. So we see that the rotation* Q *has an axis of rotation.*
6. *Find an* $n \times n$ *orthogonal matrix* Q*,* $\det(Q) = 1$ *(so it is an rotation),* n *is even, and* $Q\boldsymbol{x} = \boldsymbol{x}$ *only happens when* $\boldsymbol{x} = \mathbf{0}$*. (So there is no axis of rotation.)*

Exercise 7.7.13 (Determinant and diagonally dominant matrices)**.** ☕*An* $n \times n$ *matrix* A *is (row) diagonally dominant if for each row* $\begin{bmatrix} a_{i,1} & \dots & a_{i,n} \end{bmatrix}$*, we have*

$$|a_{i,i}| > \sum_{j \neq i} |a_{i,j}|.$$

So basically it means that, in terms of absolute value, the diagonal entry is so much larger than the other entires combined (in each row).

Suppose A *has positive diagonal entries, and it is (row) diagonally dominant. Prove that* $\det(A) > 0$*. (Hint: Mathematical induction.)*

Exercise 7.7.14 (Special linear integer matrices)**.** *We say a matrix* A *is a special linear integer matrix if all its entries are integers, and* $\det(A) = 1$.

1. *If A is an invertible $n \times n$ matrix with all integer entries, and its inverse also has all integer entries, show that* $\det(A) = \pm 1$.

2. *If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2×2 special linear integer matrix, show that its inverse matrix has integer entries.*

3. *If A is an $n \times n$ special linear integer matrix, show that its inverse matrix has integer entries. (Hint: consider the cofactor matrix of A.)*

Exercise 7.7.15 (2×2 special linear integer matrices)**.** *Note that if p, q are coprime, then $p, p - kq$ are also coprime for all integers k. This obervation gives rise to the Euclidean algorithm, which states that give any two coprime numbers $\begin{bmatrix} p \\ q \end{bmatrix}$, we can keep performing row shearings on them, until we reach $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Search the internet for details online. (This is also taught in some primary school, in the sections regarding greatest common divisors.)*

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ *is a 2×2 matrix with integer entries, and* $\det(A) = 1$.

1. *Show that a, c are coprime.*

2. *Show that A is a composition of elementary shearings with integer entires, i.e., matrices of the form $\begin{bmatrix} 1 & k \\ & 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & \\ k & 1 \end{bmatrix}$ where k is some integer.*

Exercise 7.7.16 (Even and odd permutations)**.** *There are $n!$ permutations on n objects. How many are even and how many are odd?*

1. *Let A be an $n \times n$ matrix whose entries are all 1. Calculate* $\det(A)$.

2. *In the last subproblem, write out the Leibniz formula for* $\det(A)$*, and conclude that the number of even permutations is the same as the number of odd permutations.*

Exercise 7.7.17 (Parallelogram in space)**.** *Let A be a 3×2 matrix, and its two columns are edges for a parallelogram in $\mathbb{R}^3$. Show that the area of this parallelogram is $\sqrt{\det(A^{\mathrm{T}}A)}$. (Hint: consider a QR decomposition $A = QR$ where Q is 3×3 and orthogonal, R is 3×2. Then Q is a rotation/reflection between the parallelogram A and the parallelogram R. Now compare their area, and compare $\sqrt{\det(A^{\mathrm{T}}A)}$ and $\sqrt{\det(R^{\mathrm{T}}R)}$.)*

Chapter 8

Eigenstuff

8.1 Introduction

Eigenvalues first started as a vital piece of dynamics. Think about the following example.

Example 8.1.1. Suppose we are studying in the famous Terribly Happy University (THU for short). Students are generally in two romantic status: single, or in a relationship.

If a student is single, since everyone is so happy, suppose 30% will be in a relationship next month. However, 70% shall remain single. If a student is in a relationship, then suppose 90% shall remain in the relationship next month, while 10% will break up and become single.

$$\begin{tikzcd}
\text{Single this month} \arrow[r, "70\%"] \arrow[dr, "30\%"] & \text{Single next month} \\
\text{In relationship this month} \arrow[r, "90\%"'] \arrow[ur, "10\%"'] & \text{In relationship next month}
\end{tikzcd}.$$

Now as months go by, what would happen in this dynimical system?

Say we started with x students single and y students in a relationship. Write this as a vector $\boldsymbol{x}_0 = \begin{bmatrix} x \\ y \end{bmatrix}$. Then next month, we shall have a distribution of $\boldsymbol{x}_1 = A\boldsymbol{x}_0$, where $A = \begin{bmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{bmatrix}$. Can you see why?

And then the next month, we shall have a distribution of $\boldsymbol{x}_2 = A\boldsymbol{x}_1$. As months go by, we are basically applying A again and again iteratively. So our question is now this: what is the limiting behavior of the sequence $\boldsymbol{x}_0, A\boldsymbol{x}_0, A^2\boldsymbol{x}_0, \dots$?

Say $\boldsymbol{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, i.e., 100% of the students are single initially. Then calculation yields the sequence as (rounding to the forth digit after the decimal point):

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 0.52 \\ 0.48 \end{bmatrix}, \begin{bmatrix} 0.412 \\ 0.588 \end{bmatrix}, \begin{bmatrix} 0.3472 \\ 0.6528 \end{bmatrix}, \begin{bmatrix} 0.3083 \\ 0.6917 \end{bmatrix}, \begin{bmatrix} 0.2850 \\ 0.7150 \end{bmatrix}, \begin{bmatrix} 0.2710 \\ 0.7290 \end{bmatrix}, \begin{bmatrix} 0.2626 \\ 0.7374 \end{bmatrix}, \begin{bmatrix} 0.2576 \\ 0.7424 \end{bmatrix}, \begin{bmatrix} 0.2545 \\ 0.7455 \end{bmatrix}, \begin{bmatrix} 0.2527 \\ 0.7473 \end{bmatrix} \dots.$$

As you can see, the sequence seems to be converging to $\boldsymbol{x}_\infty = \begin{bmatrix} 0.25 \\ 0.75 \end{bmatrix}$. Eventually, 75% of the students are in a relationship, and only 25% of the students will remain single.

Is there a better and more rigorous way to find $\boldsymbol{x}_\infty$? Yes there is! If $\boldsymbol{x}_\infty$ is indeed the stable equilibrium, then we should have $A\boldsymbol{x}_\infty = \boldsymbol{x}_\infty$. In particular, $\boldsymbol{x}_\infty \in \operatorname{Ker}(A - I)$. Now $A - I = \begin{bmatrix} -0.3 & 0.1 \\ 0.3 & -0.1 \end{bmatrix}$, and its kernel is spanned by $\begin{bmatrix}1\\3\end{bmatrix}$, which shows that 25% of the students are single in the equilibrium. We find our limiting behavior right away!

Not only so, we can also find the speed of convergence. Suppose our current distribution is $\begin{bmatrix}0.25+x\\0.75-x\end{bmatrix}$ for some x. Then $A\begin{bmatrix}0.25+x\\0.75-x\end{bmatrix}=\begin{bmatrix}0.25+0.6x\\0.75-0.6x\end{bmatrix}$. As you can see, the deviation is shrunk by a factor of 0.6, wow!

In particular, if we started with $\begin{bmatrix}0.25+x\\0.75-x\end{bmatrix}$, then next month we would have $\begin{bmatrix}0.25+0.6x\\0.75-0.6x\end{bmatrix}$, and next month we would have $\begin{bmatrix}0.25+0.6^2x\\0.75-0.6^2x\end{bmatrix}$. The deviation would shrunk exponentially towards zero.

Let us sum up the phenomina here. First of all, we can observe that

$$\begin{aligned} A\begin{bmatrix}1\\3\end{bmatrix} &= \begin{bmatrix}1\\3\end{bmatrix}, \\ A\begin{bmatrix}1\\-1\end{bmatrix} &= 0.6\begin{bmatrix}1\\-1\end{bmatrix}. \end{aligned}$$

Both of these have significant meanings. The first one describes the eventual equilibrium must be in this direction. The second one tells us how we would converge to this equilibrium: in the direction of $\begin{bmatrix}1\\-1\end{bmatrix}$, and the speed is governed by the value 0.6.

To sum up, the solutions to $A\boldsymbol{x}=\lambda\boldsymbol{x}$ completely describe ALL info we need for this dynamical system. ☺

We'd better give this beautiful info names. They are called eigenstuff.

Definition 8.1.2. *For a matrix A, if $A\boldsymbol{x}=\lambda\boldsymbol{x}$ for some $\lambda\in\mathbb{R}$ and some nonzero vector $\boldsymbol{x}$, then we call λ an* ***eigenvalue*** *of A, and $\boldsymbol{x}$ an* ***eigenvector*** *of A for the eigenvalue λ.*

Why do we require $\boldsymbol{x}$ here to be non-zero? Mainly because if we allow the zero vector, then $A\mathbf{0}=\lambda\mathbf{0}$ for all λ. These solutions are entirely trivial and useless.

Now, a major point of eigenstuff is to help us understand the behavior of $A, A^2, A^3, \ldots$. In some very nice cases, this is very easy to do. For example, if A is diagonal say $\begin{bmatrix}a & \\ & b\end{bmatrix}$, then obviously A^k is done by simply raising the diagonal entries to the corresponding power, i.e., $\begin{bmatrix}a^k & \\ & b^k\end{bmatrix}$. But for a generic A, say $A=\begin{bmatrix}0.7 & 0.1\\0.3 & 0.9\end{bmatrix}$, how can we obtain a formula for calculating A^k? This is where eigenstuff come in to help. They help describe the "dynamics" behind the linear transformation A.

Example 8.1.3. Suppose $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$ is a basis of $\mathbb{R}^n$ and they are also eigenvectors of A, say $A\boldsymbol{x}_i=\lambda_i\boldsymbol{x}_i$. One might call this an ***eigenbasis*** of $\mathbb{R}^n$ for A.

Then in particular we have $A\begin{bmatrix}\boldsymbol{x}_1 & \ldots & \boldsymbol{x}_n\end{bmatrix}=\begin{bmatrix}\lambda_1\boldsymbol{x}_1 & \ldots & \lambda_n\boldsymbol{x}_n\end{bmatrix}=\begin{bmatrix}\boldsymbol{x}_1 & \ldots & \boldsymbol{x}_n\end{bmatrix}\begin{bmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{bmatrix}$.

Let $X=\begin{bmatrix}\boldsymbol{x}_1 & \ldots & \boldsymbol{x}_n\end{bmatrix}, \Lambda=\begin{bmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{bmatrix}$, then we see that $AX=X\Lambda$, and $A=X\Lambda X^{-1}$. Hey!

If you remember, structures such as $X\Lambda X^{-1}$ refers to a change of basis for a linear transformation. In particular, it means that if we change basis from the standard basis to the eigenbasis, then A will change into a diagonal matrix Λ.

This is indeed the case. If we pick $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$ as the new basis, then $A\boldsymbol{x}_i = \lambda_i \boldsymbol{x}_i$ says exactly that A is diagonal in this new basis.

So, how to study A? We first go into this new basis. Now our linear map is represented by a diagonal matrix Λ, and its k-th power is simply $\Lambda^k = \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix}$. Powers of a diagonal matrix is super easy to do! Now we change back to the standard basis, and we see that $A^k = X\Lambda^k X^{-1} = X \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} X^{-1}$.

Alternatively, one might also compute directly via $A^k = X\Lambda X^{-1} X\Lambda X^{-1} \dots X\Lambda X^{-1}$. Note that all the X and X^{-1} in the middle will cancel out, and we have $A^k = X\Lambda^k X^{-1}$.

Alternatively, I also enjoy this lovely diagram below, which make things quite clear.

$$\begin{tikzcd}
\mathbb{R}^n \arrow[r, "\Lambda"] \arrow[d, "X"'] & \mathbb{R}^n \arrow[r, "\Lambda"] \arrow[d, "X"'] & \dots \arrow[r, "\Lambda"] & \mathbb{R}^n \arrow[d, "X"'] \\
\mathbb{R}^n \arrow[r, "A"] & \mathbb{R}^n \arrow[r, "A"] & \dots \arrow[r, "A"] & \mathbb{R}^n
\end{tikzcd}$$

For example, in our previous case of $A = \begin{bmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{bmatrix}$, we see that $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is an eigenvector for the eigenvalue 1, and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector for the eigenvalue 0.6. Since the two linearly independent vectors span $\mathbb{R}^2$, this immediately imply that $\begin{bmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & \\ & 0.6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}^{-1}$. (Calculate and see that this is indeed correct.)

We also immediately obtain the formula

$$\begin{bmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{bmatrix}^k = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & \\ & 0.6 \end{bmatrix}^k \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & \\ & 0.6^k \end{bmatrix} \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{3}{4} & -\frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} + \frac{3}{4}(0.6^k) & \frac{1}{4} - \frac{1}{4}(0.6^k) \\ \frac{3}{4} - \frac{3}{4}(0.6^k) & \frac{3}{4} + \frac{1}{4}(0.6^k) \end{bmatrix}.$$

Thus we have obtained a direct formula for A^k.

Finally, let us calculate $A^\infty = \lim_{k\to\infty} A^k = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{3}{4} & \frac{3}{4} \end{bmatrix}$. What is this? This is the oblique projection to the line in the direction $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ along a kernel direction of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. ☺

This technique is called the similarity diagonalization of A.

Definition 8.1.4. *We say square matrices A, B are **similar** if we can find an invertible X such that $A = XBX^{-1}$. (I.e., A, B only differ by a change of basis.)*

*If A is similar to a diagonal matrix, then we say that A is **diagonalizable**.*

Proposition 8.1.5. *A square matrix A is diagonalizable if and only if eigenvectors of A span the whole domain.*

Proof. If eigenvectors of A span the whole domain, then we can pick some of them to form a basis, and thus A is diagonalizable as shown in the example above. We only need to prove the other direction now.

Suppose $A = XDX^{-1}$ for some invertible $X = \begin{bmatrix} \boldsymbol{x}_1 & \dots & \boldsymbol{x}_n \end{bmatrix}$ and diagonal $D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$. Then $AX = XD$, which means $A\boldsymbol{x}_i = d_i \boldsymbol{x}_i$. Since X is invertible, all these $\boldsymbol{x}_i$ are nonzero, so they are all eigenvectors. Again since X is invertible, they span the whole space. ☐

Now, most of the matrices are diagonalizable. (Exceptions exist but rare.) Then our goal is simple: how to find these eigenvalues and eigenvectors?

Proposition 8.1.6. *λ is an eigenvalue of A if and only if $A-\lambda I$ is not invertible, if and only if $\det(A-\lambda I) = 0$, if and only if $\det(\lambda I - A) = 0$.*

And eigenvectors for the eigenvalue λ are exactly non-zero vectors of $\mathrm{Ker}(A - \lambda I)$.

Proof. Just look at definition. □

Example 8.1.7. Again consider the matrix $A = \begin{bmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{bmatrix}$. If x is an eigenvalue, then we should have $\det(xI - A) = 0$.

Now $\det(xI - A) = \det \begin{bmatrix} x-0.7 & -0.1 \\ -0.3 & x-0.9 \end{bmatrix} = x^2 - 1.6x + 0.6$. So as you can see, x is an eigenvalue if and only if $x^2 - 1.6x + 0.6 = 0$. Solving this gives $x = 1$ or $x = 0.6$. So the only eigenvalues for A are 1 and 0.6. ☺

So the standard procedure goes like this: for our $n \times n$ matrix A, consider $\det(A - xI)$. This is some polynomial in x with degree n. Solve $\det(A - xI) = 0$, then solutions to this polynomial equation are EXACTLY the eigenvalues.

Once we have the eigenvalues, for each eigenvalue λ, we find eigenvectors by solving $\mathrm{Ker}(A - \lambda I)$ using Gaussian elimination.

Definition 8.1.8. *The **characteristic polynomial** of an $n \times n$ square matrix A is $\det(xI - A)$, a polynomial in x of degree n. We usually write this polynomial as $p_A(x)$.*

Note that $\det(xI - A) = (-1)^n \det(A - xI)$. Our ultimate goal is to check if $A - xI$ is invertible or not, so this sign does not matter. We use $\det(xI - A)$ so that x^n will have a positive coefficient. If you choose $\det(A - xI)$ and n is odd, then your x^n will have coefficient -1, which is a bit ugly, but ultimatelty does not matter.

Proposition 8.1.9. *The roots of characteristic polynomial of A are exactly the eigenvalues of A.*

Proof. $\det(xI - A) = 0$ if and only if $A - xI$ is not invertible if and only if $A\boldsymbol{v} = x\boldsymbol{v}$ has a non-zero solution if and only if x is an eigenvalue. □

To illustrate this strategy, here is a nice application.

Example 8.1.10. Consider the Fibonacci sequence, $a_0 = 0, a_1 = 1$, and inductively $a_{n+2} = a_{n+1} + a_n$. So the sequence looks like

$$0, 1, 1, 2, 3, 5, 8, 13, 21, \dots.$$

How to find a formula for a_n?

Well, the progression for a_n seems "linear", but involves two terms. So we consider a vector progression instead. Let $\boldsymbol{x}_n = \begin{bmatrix} a_{n+1} \\ a_n \end{bmatrix}$. Then we have

$$\boldsymbol{x}_{n+1} = \begin{bmatrix} a_{n+2} \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} a_n + a_{n+1} \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \boldsymbol{x}_n.$$

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, then we can clearly see $\boldsymbol{x}_n = A^n \boldsymbol{x}_0$.

Now let us figure out A^n. First, $\det(xI - A) = x^2 - x - 1$. This has two roots, $\phi_1 = \frac{1+\sqrt{5}}{2}$ and $\phi_2 = \frac{1-\sqrt{5}}{2}$. (These are the golden ratios $1.618\dots$ and $-0.618\dots$.)

Now we find eigenvectors. We have $\mathrm{Ker}(A-\phi_1 I)=\mathrm{span}(\begin{bmatrix}\phi_1\\1\end{bmatrix})$ and $\mathrm{Ker}(A-\phi_2 I)=\mathrm{span}(\begin{bmatrix}\phi_2\\1\end{bmatrix})$. It is straightforward to verify that together, they span all of $\mathbb{R}^2$, so A is diagonalizable. Set $X=\begin{bmatrix}\phi_1 & \phi_2\\1 & 1\end{bmatrix}$, and $D=\begin{bmatrix}\phi_1 & \\ & \phi_2\end{bmatrix}$, then we must have $A=XDX^{-1}$.

So computation gives

$$A^n=XD^nX^{-1}=\frac{1}{\phi_1-\phi_2}\begin{bmatrix}\phi_1^{n+1}-\phi_2^{n+1} & -\phi_1^{n+1}\phi_2+\phi_2^{n+1}\phi_1\\ \phi_1^n-\phi_2^n & -\phi_1^n\phi_2+\phi_2^n\phi_1\end{bmatrix}.$$

Note that $\phi_1-\phi_2=\sqrt{5}$ and $\phi_1\phi_2=-1$, so this can be slightly simplified to

$$A^n=\frac{1}{\sqrt{5}}\begin{bmatrix}\phi_1^{n+1}-\phi_2^{n+1} & \phi_1^n-\phi_2^n\\ \phi_1^n-\phi_2^n & \phi_1^{n-1}-\phi_2^{n-1}\end{bmatrix}.$$

Either way, since $\boldsymbol{x}_0=\begin{bmatrix}1\\0\end{bmatrix}$, therefore $\boldsymbol{x}_n=A^n\boldsymbol{x}_0=\frac{1}{\sqrt{5}}\begin{bmatrix}\phi_1^{n+1}-\phi_2^{n+1}\\ \phi_1^n-\phi_2^n\end{bmatrix}$. Since a_n is the second coordinate, we have a formula

$$a_n=\frac{\phi_1^n-\phi_2^n}{\sqrt{5}}.$$

Note that this method can be generalized quite a bit! For example, suppose $a_{n+2}=2a_{n+1}+3a_n+n^2+2n+2^n$. Then we can simply build $\boldsymbol{x}_n=\begin{bmatrix}a_{n+1}\\a_n\\n^2\\n\\1\\2^n\end{bmatrix}$, and we would have

$$\boldsymbol{x}_{n+1}=\begin{bmatrix}2&3&1&2&0&1\\1&0&0&0&0&0\\0&0&1&2&1&0\\0&0&0&1&1&0\\0&0&0&0&1&0\\0&0&0&0&0&2\end{bmatrix}\boldsymbol{x}_n.$$

So we can convert the progression of the sequence into the progression of this matrix, which then can be solved by similar ideas. ☺

Let us also look at an example of non-diagonalizable matrices.

Example 8.1.11. Consider shearing $E=\begin{bmatrix}1&1\\0&1\end{bmatrix}$. The characteristic polynomial is very simple, it is $(x-1)^2$. So the only eigenvalue is 1.

However, $\mathrm{Ker}(E-I)$ is spanned by $\boldsymbol{e}_1$. So the ONLY eigenvectors of E are multiples of $\boldsymbol{e}_1$.

In this case, eigenvectors of E fails to span $\mathbb{R}^2$. E is NOT diagonalizable. In general, all shearings fail to be diagonalizable. ☺

Now, this strategy works. And almost ALL textbook will teach you to use this strategy. However, this strategy is bogus when $n\geq 5$. Why? Because there is no algebraic formula to solve a polynomial of degree 5 or more. (In practice, a 3×3 matrix will already yields a degree three polynomial, and finding those roots will be a headache already.)

In fact, the reality is the opposite. How would a computer nowaday find roots of a polynomial $p(x)$? The computer would build a matrix whose characteristic polynomial is $p(x)$, then use other methods (variants or improvements of the ***iterated QR algorithm***) to approximate eigenvalues, thus finding approximated roots of $p(x)$. Oh the irony.

Nevertheless, as long as we are dealing with square matrices of dimension at most 4, we can use the traditional strategy as desired. Luckily, most problems we face in this class will have $n \leq 4$.

8.2 Intuitions on Eigenstuff

Sometimes, we can tell the eigenstuff by sight. Here are some examples, which shall further help us understand the intuitions behind the eigenstuff.

Example 8.2.1 (Eigenstuff by sight)**.** Sometimes there are some obvious properties among entries, and they in fact indicates eigenstuff right away.

Consider $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. Note how each row adds up to 5? If you apply A to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, you will exactly add up each row of A and get $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$, so this is an eigenvector for the eigenvalue 5. ☺

Proposition 8.2.2. *If each row of a matrix A adds up to the same number λ, then λ is an eigenvalue of A, and $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$ is an eigenvector for this eigenvalue.*

What if each column of A adds up to the same number, say λ? Then we can actually also conclude that A has eigenvalue λ.

Proposition 8.2.3. *A and A^{T} share the same eigenvalues. In fact, they share the same characteristic polynomial.*

Proof. Note that $p_A(x) = \det(xI - A) = \det(xI - A^{\mathrm{T}}) = p_{A^{\mathrm{T}}}(x)$, where the middle equality is true since determinant is invariant under transpose. So A and A^{T} share the same characteristic polynomial, and thus they also share the same eigenvalues. □

Corollary 8.2.4. *If each column of a matrix A adds up to the same number λ, then λ is an eigenvalue of A. (But the eigenvectors are harder to see now.)*

In many applications, such as our dynamical system before, each column of the matrix represents some probability distribution. Therefore, entries in each column will add up to 1. So 1 will always be an eigenvalue for such a matrix. But if the system is evolving via a matrix A and A has eigenvalue 1, then A has a nonzero eigenvector $\boldsymbol{v}$. So $A\boldsymbol{v} = \boldsymbol{v}$, our system must have an equilibrium!

Definition 8.2.5. *A square matrix is a Markov matrix if all entries are non-negative, and each column adds up to 1. (These are exactly the matrices describing some probabilistic evolutions.)*

Proposition 8.2.6. *A Markov matrix must have an eigenvalue 1.*

Proof. Duh. □

For Markov matrices, finding eigenvectors for the eigenvalue 1 is very important. This corresponds to finding the equilibrium of the system. Here is another application of this.

Example 8.2.7. Do you know how google works?

Given a key word, there are many many webpages containing that key word. How can we rank these pages in order, so that the "best" or "most relevant" page comes up on top?

In the pre-google time, search engines do very stupid things, like rank them alphabetically.... And there are also certain search engines that rank pages according to how much these pages pay them, like certain search engine that shall remain unnamed, but you probably have a candidate in mind.

Another way to rank them is to rank according to the number of visits. Surely the most visited website is the "best" result for your search, yes? However, there are some pitfalls. For example, the number of visits are highly volatile. They change every second! It takes a tremendous amount of work and money to keep track of this, and it might also slow down the search process. We want something that is more stable, yet also indicative of the quality of the webpages.

And one day, two young guys named Larry Page and Sergey Brin (the founders of google) came up with a brilliant idea. This idea started google, and its efficiencies, stabilities and speed was far more superior to all other search engines. It immediately beat all competitors and grew into a super company.

So what is this billion dollar idea? This is called the Page rank algorithm. The internet is connected, with webpages linked to one another. So the basic idea is this:

1. If a website is linked to by many many other website, then this website must be important.
2. If some important website links to your website, then this makes your website important.
3. However, if a website is linked to many many many other websites, then it will fail to transfer much importance to each of these individual websites.

Consider the following cases, where we have four websites v_1, v_2, v_3, v_4, v_5, and they are linked to each other as in the graph below. Suppose these websites have importance x_1, x_2, x_3, x_4, x_5. Then v_3 lends weights to both v_1, v_5, so each of v_1, v_5 would get $\frac{1}{2}x_3$ importance from v_3. So on so forth, so we have this graph.

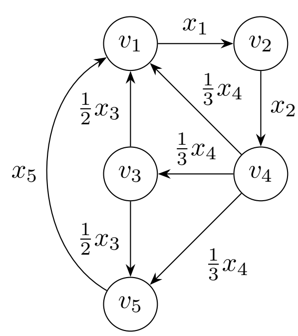


Figure 8.2.1: Graph G of websites

The idea is that your importance equal to the total importance received. For example, the importance of v_1 is x_1, and the total importance received is $\frac{1}{2}x_3 + \frac{1}{3}x_4 + x_5$, so we want $x_1 = \frac{1}{2}x_3 + \frac{1}{3}x_4 + x_5$. Then collectively, we see that our importance $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_5 \end{bmatrix}$ must be solved by this:

$$\begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{3} & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{3} & 0 \end{bmatrix} \boldsymbol{x} = \boldsymbol{x}.$$

So we are looking for an eigenvector $\boldsymbol{x}$ for the eigenvalue 1 of our matrix. Note that this is a Markov matrix, so such an eigenvector must exist. Then the page rank algorithm would simply list the websites from the most important (largest x_i) to the least important (smallest x_i).

In this cases, one solution is $\boldsymbol{x} = \begin{bmatrix} 6 \\ 6 \\ 2 \\ 6 \\ 3 \end{bmatrix}$. So we put v_1, v_2, v_4 up top in the search result, as they are the most important, then we list webpage v_5, and finally we list webpage v_3, which is the least important.

(As you can probably see from the graph, v_4 is probably some "search engine" website, which is important and has link to almost everyone, but lends little weight to each. v_1 is probably some authority website that every one links to. And v_2 is some remote website that is hard to reach, but yet very valuable since v_1 only links to v_2. Here v_3, v_5 are not important. v_5 basically just says "yeah just go check out v_1". v_3 is even less useful, as it links to an unhelpful site v_5 other than v_1.) ☺

Above we have focused on the algebra side of the intuition. Now let us turn to geometry. What does it mean geometrically that $A\boldsymbol{x} = \lambda\boldsymbol{x}$? This means the linear transformation of A would FIX this line spanned by $\boldsymbol{x}$.

Example 8.2.8. Suppose all entries of a 2×2 matrix A are positive. I claim that A has a positive eigenvalue whose eigenvector spans a line through the first-third quadrants.

(And A has another eigenvector which spans a line through the second-forth quadrand, with positive eigenvalue if $\det(A) > 0$, and negative eigenvalue if $\det(A) < 0$, and zero eigenvalue if $\det(A) = 0$. I shall not prove these things in the parenthesis though, which is similar but more tedius. Determinant matters here because they indicate whether your map preserves or reverse orientation.)

How can I see this? Let us say $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

Image in the domain, we have a ray shooting from the origin in the direction of $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Now this ray rotates clockwise, sweeps through the entire first quadrant, and ends up in the direction of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Now these are happening in the domain of A. If we map these to the codomain, then in the codomain my ray will start in the direction of $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then it will sweep counter-clockwise and end up in the direction of $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

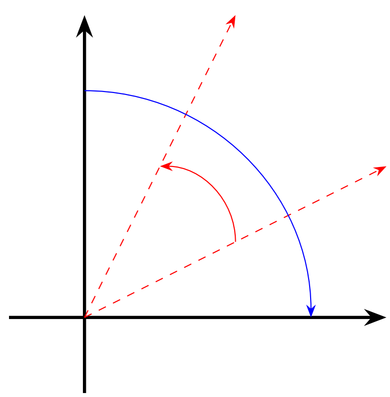


Figure 8.2.2: Sweeping Rays, input = blue and output = red.

Now look at both of these sweeping process in $\mathbb{R}^2$ simultaneously, and we see that somewhere the input ray and the output ray must meet. (This is a variant of the intermediate value theorem in calculus.) This is a line fixed by A, and hence vectors on this line are eigenvectors of A. Since the fixed ray stays in the first quadrant, the eigenvalue must be positive.

Note that $\det(A)$ is negative here. In particular, it is an orientation reversing linear map. As a result, the input clockwise movement becomes a counter-clockwise movement in the output. ☺

I shall mention the following theorem without proof. (We don't use it in this class.) But surely you can imagine how it is true. The proof is basically just a high dimensional version of intermediate value theorem.

Theorem 8.2.9 (Perron-Frobenius Theorem)**.** *If all entries of an $n \times n$ matrix A are positive, then A must have a positive eigenvalue $\lambda > 0$, and an eigenvector $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ for λ such that all $x_i > 0$. (So it is a "positive eigenvector".)*

Finally, let us see some geometric examples.

Example 8.2.10. Consider a reflection in $\mathbb{R}^3$, say about the plane $x+y+z=0$. Note that a normal vector is $\boldsymbol{n} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and thus this reflection is $A = I - 2\frac{\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}} = I - \frac{2}{3}\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}} = \begin{bmatrix} \frac{1}{3} & -\frac{2}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$.

Now, we can in fact find all eigenvectors using only the geometric intuition, without any computation. Obviously our map A will reflect $\boldsymbol{n}$, so $\boldsymbol{n}$ is an eigenvector for the eigenvalue -1. Furthermore, any non-zero vector on the plane $x+y+z=0$ are preserved, so they are all eigenvectors for the eigenvalue 1. Finally, it is geometrically obvious to see that there are no more fixed lines. All other lines will be reflected to a different line. So we are done.

In particular, pick any two vector $\boldsymbol{v}, \boldsymbol{w}$ on the plane $x+y+z=0$, then $\boldsymbol{n}, \boldsymbol{v}, \boldsymbol{w}$ form a basis of $\mathbb{R}^3$ made of eigenvectors of A. As a result, A is diagonalized under this basis into $\begin{bmatrix} -1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$. ☺

Example 8.2.11. Let us now think about rotations R_θ on $\mathbb{R}^2$. Suppose θ is NOT a multiple of π. Then since every line is rotated, there is no fixed line. In particular, R_θ will have NO eigenvalue.

However, there is no need to despair. Consider the matrix $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The characteristic polynomial is x^2+1. So even though it has no REAL root, it has two COMPLEX roots, $\pm\mathrm{i}$.

Consider $R - \mathrm{i}I = \begin{bmatrix} -\mathrm{i} & -1 \\ 1 & -\mathrm{i} \end{bmatrix}$, you can see that the kernel is spanned by $\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix}$. Indeed, you can check that $R\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix} = \mathrm{i}\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix}$.

Similarly, eigenvalues for $-\mathrm{i}$ is $\begin{bmatrix} 1 \\ \mathrm{i} \end{bmatrix}$. (You can get this by imitating the strategy above, or you can simply take complex conjugates on $R\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix} = \mathrm{i}\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix}$.)

Eitherway, $\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix}, \begin{bmatrix} 1 \\ \mathrm{i} \end{bmatrix}$ form a basis for $\mathbb{R}^2$. So we have $R = XDX^{-1}$ where $X = \begin{bmatrix} 1 & 1 \\ -\mathrm{i} & \mathrm{i} \end{bmatrix}, D = \begin{bmatrix} \mathrm{i} & \\ & -\mathrm{i} \end{bmatrix}$. We say that R is NOT diagonalizable over $\mathbb{R}$, but IS diagonalizable over $\mathbb{C}$.

It seems that in certain cases, complex matrices are unavoidable. Previously, we have been completely focused on real matrices. But now, we have to deal with complex matrices. ☺

As such, now is a very good time to take a detour to look at complex linear algebra.

8.3 Complex Numbers

As we can see above, even if we only want to study real matrices, complex nubmers would still pop up as eigenvalues and eigenvectors. Here let us quickly consider what is a complex number, and how would complex linear algebra look like. We assume that the readers have a basic familiarty for complex numbers. Nevertheless, we list the basic definition and calculation rules here.

Definition 8.3.1. *The set of complex numbers is the set* $\mathbb{C} = \{a + b\mathrm{i} : a, b \in \mathbb{R}\}$ *where we define complex addition and complex multiplication as*

$$(a + b\mathrm{i}) + (c + d\mathrm{i}) = (a + c) + (b + d)\mathrm{i},$$

$$(a + b\mathrm{i})(c + d\mathrm{i}) = (ac - bd) + (ad + bc)\mathrm{i}.$$

Here $a + c, b + d, ac - bd, ad + bc$ *are all calculations of real numbers.*

So in short, a complex number is a REAL linear combination of 1 and i, and we require $\mathrm{i}^2 = -1$. You can easily verify that complex addition and multiplication are commutative and associative, and we have law of distribution, and so on. We also have the formula $-(a + b\mathrm{i}) = -a - b\mathrm{i}$, and $(a + b\mathrm{i})^{-1} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}\mathrm{i}$. These allow us to do substractions between complex numbers, and divisions when the divider is non-zero. Everything is verifiably nice.

Here are some final bits of definitions.

Definition 8.3.2. *Given a complex number* $a+b\mathrm{i}$*, we define its real part as* $\mathrm{Re}(a+b\mathrm{i}) = a$*, and its imaginary part as* $\mathrm{Im}(a+b\mathrm{i}) = b$*. (Note that traditionally we use* b*, not* $b\mathrm{i}$*.) We also define its absolute value (or modulus or norm or length or magnitude.... it's alll the same to me) as* $|a+b\mathrm{i}| = \sqrt{a^2 + b^2}$*, and its complex conjugate is* $\overline{a + b\mathrm{i}} = a - b\mathrm{i}$*.*

Now, this traditional definition of complex numbers seems artificial and a bit surreal. What does i even mean? How would such a system exist? Let us observe some interesting comparisons.

	Complex Numbers	Special 2×2 matrices
Elements	$a + b\mathrm{i}$	$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$
Addition	$(a + b\mathrm{i}) + (c + d\mathrm{i}) = (a + c) + (b + d)\mathrm{i}$	$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} a + c & -b - d \\ b + d & a + c \end{bmatrix}$
Multiplication	$(a + b\mathrm{i})(c + d\mathrm{i}) = (ac - bd) + (ad + bc)\mathrm{i}$	$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}\begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} ac - bd & -ad - bc \\ ad + bc & ac - bd \end{bmatrix}$
Real part	$\mathrm{Re}(a + b\mathrm{i}) = a$	$\frac{1}{2}\,\mathrm{trace}\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = a$
Absolute value	$\|a + b\mathrm{i}\|^2 = a^2 + b^2$	$\det\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = a^2 + b^2$
Complex conjugate	$\overline{a + b\mathrm{i}} = a - b\mathrm{i}$	$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$
Motivation	$\mathrm{i}^2 = -1$	$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}^2 = -I.$

The point is this: even though complex numbers might look "unreal", they are actually part of our everyday life. You can simply think of 1 as the identity map I, and think of i as rotation counter-clockwise by 90 degree, then a complex number is simply a linear combination of the two. As far as calculations go, $a + b\mathrm{i}$ is usually easier to work with. But as far as interpretations go, $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ tells you the meaning behind this complex number.

So as we can see, a complex number is basically a linear combination between the identity map and the "rotation by 90" map. In fact, all rotations of $\mathbb{R}^2$ are closely related to complex numbers.

Example 8.3.3. Consider $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. Clearly this corresponds to $\cos\theta + \mathrm{i}\sin\theta$. You can also see that these are precisely the ***unit complex numbers***, i.e., complex numbers with absolute value one. It is in general a very good intuition to think of unit complex numbers as representations of rotations.

Consider $kI = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$. This corresponds to the complex number k, which is purely real. So a purely real number as a complex number would represent the "stretch everything" map, where we stretch everything by a factor of k.

Now consider a generic complex number $a + b\mathrm{i}$. The corresponding linear map is $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$. Now both columns have the same length $\sqrt{a^2+b^2}$. If we divide by this length, then $\begin{bmatrix} \frac{a}{\sqrt{a^2+b^2}} \\ \frac{b}{\sqrt{a^2+b^2}} \end{bmatrix}$ would be a unit vector. Set $\frac{a}{\sqrt{a^2+b^2}} = \cos\theta$ and $\frac{b}{\sqrt{a^2+b^2}} = \sin\theta$ for some theta, we see that $\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} \sqrt{a^2+b^2} & \\ & \sqrt{a^2+b^2} \end{bmatrix} R_\theta$.

In short, a complex number $a + b\mathrm{i}$ as a linear map would mean a rotation map then a "stretch all" map. We have $a + b\mathrm{i} = \sqrt{a^2+b^2}(\cos\theta + \mathrm{i}\sin\theta)$, where the rotation amount θ is determined by the direction of $\begin{bmatrix} a \\ b \end{bmatrix}$, and the stretching factor is determined by the absolute value. ☺

The decomposition $a+b\mathrm{i} = \sqrt{a^2+b^2}(\cos\theta+\mathrm{i}\sin\theta)$ is the famous polar decomposition of complex numbers. It corresponds to decomposing the linear map $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ into a stretching and a rotation. To better write this, we need some extra notational convention.

Definition 8.3.4. *For a complex number z, we define $\mathrm{e}^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$.*

In general, for a square matrix A, we define $\mathrm{e}^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$.

We shall use the fact that e^z and e^A always converge, without any proof. The proof is pure analysis and has nothing to do with linear algebra.

Lemma 8.3.5. $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$, *and* $\mathrm{e}^{\begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix}} = R_\theta$. *For any two complex nubmers $z, w \in \mathbb{C}$, we have* $\mathrm{e}^{z+w} = \mathrm{e}^z\mathrm{e}^w$.

Proof. Direct calculations. Recall that $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$, and $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

Now consider a matrix $A = \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}$. Note that this is a skew-symmetric matrix. We now have

$$\begin{aligned}
\mathrm{e}^A &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix} + \frac{1}{2!}\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}^2 + \frac{1}{3!}\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}^3 + \dots \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix} + \begin{bmatrix} -\frac{a^2}{2!} & 0 \\ 0 & -\frac{a^2}{2!} \end{bmatrix} + \begin{bmatrix} 0 & \frac{a^3}{3!} \\ -\frac{a^3}{3!} & 0 \end{bmatrix}^3 + \dots \\
&= \begin{bmatrix} 1 - \frac{a^2}{2!} + \dots & -a + \frac{a^3}{3!} - \dots \\ a - \frac{a^3}{3!} + \dots & 1 - \frac{a^2}{2!} + \dots \end{bmatrix} \\
&= \begin{bmatrix} \cos(a) & -\sin(a) \\ \sin(a) & \cos(a) \end{bmatrix}.
\end{aligned}$$

□

Corollary 8.3.6 (Euler identity)**.** $\mathrm{e}^{\mathrm{i}\pi} = -1$.

As far as I'm concerned, I think the Euler identity is neither surprising nor mysterious. It simply means that on $\mathbb{R}^2$, rotating things by 180 degree is the same as negating everything.

Corollary 8.3.7 (Polar decomposition)**.** *For any complex number z, there are unique real numbers $r > 0$ and $0 \leq \theta < 2\pi$, such that $z = r\mathrm{e}^{\mathrm{i}\theta}$.*

Remark 8.3.8. *If we think of complex numbers as 2 by 2 matrices as before, then $z = \mathrm{e}^{\mathrm{i}\theta} r$ in fact corresponds to a QR decomposition.*

Now we have understood almost everything about complex numbers. The next property, however, is the most important. It is why we bother.

Theorem 8.3.9 (Fundamental Theorem of Algebra)**.** *Any complex polynomial $p(x)$ of degree n has n roots counting multiplicities. (E.g., we say $(x-a)^3(x-b)^2$ has a root a with multiplicity 3, and a root b with multiplicity 2. So counting multiplicity, the roots are a, a, a, b, b, five in total.)*

Given an $n \times n$ matrix, its characteristic polynomial is some degree n polynomial. The roots of this polynomial are exactly the eigenvalues of your matrix. However, the fundamental theorem of algebra would then reveal that there are exactly n roots (counting multiplicities). Hence an $n \times n$ matrix would always have exactly n eigenvalues (counting algebraic multiplicities, which we shall define later).

Let us see some examples of this. For $n = 2$, we shall study some matrices and see that they all have two (potentially complex) eigenvalues. Further, the complex eigenvalues are indicative of what your matrix is trying to do.

Example 8.3.10. Consider $A = \begin{bmatrix} 3 & 3 \\ 1 & 5 \end{bmatrix}$. Then the characteristic polynomial is $x^2 - 8x + 12$. The roots are exactly 2 and 6, so these are your eigenvalues. $\mathrm{Ker}(A - 2I) = \mathrm{Ker} \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} = \mathrm{span}(\begin{bmatrix} 3 \\ -1 \end{bmatrix})$, and $\mathrm{Ker}(A - 6I)$ is spanned by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. (Note that each row adds up to six.)

In short, we have a diagonalization $A = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & \\ & 6 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}^{-1}$.

Note that both eigenvalues are purely real. What does real complex numbers do? They stretch. Our matrix indeed stretch in the $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$ direction by a factor of 2, and stretch in the $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ direction by a factor of 6. At this stage, you should be able to visualize the geometric action of this linear map on $\mathbb{R}^2$. ☺

Example 8.3.11. Consider $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. Then the characteristic polynomial is $(x - \cos\theta)^2 + \sin^2\theta$. The roots are exactly $\cos\theta \pm \mathrm{i}\sin\theta$, or $\mathrm{e}^{\pm\mathrm{i}\theta}$. As you can see, the matrix is doing rotation, and its eigenvalues are unit complex numbers, which also represent rotations.

Diagonalization here is $R_\theta = \begin{bmatrix} 1 & 1 \\ -\mathrm{i} & \mathrm{i} \end{bmatrix} \begin{bmatrix} \mathrm{e}^{\mathrm{i}\theta} & \\ & \mathrm{e}^{-\mathrm{i}\theta} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\mathrm{i} & \mathrm{i} \end{bmatrix}^{-1}$. ☺

8.4 Complex Linear Algebra

Now moving forward, we shall need to do a lot of complex matrix stuff. In general, most of the things are exactly like before. The most important one is matrix multiplication.

Example 8.4.1. As a simple example, we have $\begin{bmatrix} 1 & \mathrm{i} \\ 1+\mathrm{i} & 1-\mathrm{i} \end{bmatrix} \begin{bmatrix} \mathrm{i} \\ 2+\mathrm{i} \end{bmatrix} = \begin{bmatrix} 1 \times \mathrm{i} + \mathrm{i} \times (2+\mathrm{i}) \\ (1+\mathrm{i}) \times \mathrm{i} + (1-\mathrm{i}) \times (2+\mathrm{i}) \end{bmatrix} = \begin{bmatrix} -1+3\mathrm{i} \\ 2 \end{bmatrix}$. As you can see, such things are exactly as before.

As an optional side note, just like we can think of $a + b\mathrm{i}$ as the real matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, we can similarly think of complex vectors or matrices as "block matrices" with real entries. For example, the same equation above may be written as

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \\ 2 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 3 & -1 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

And you can see that everything works just the same.

In particular, if you seek geometric intuitions on complex vectors, you can think of each complex vector with n coordinates as "a pair of real vectors with $2n$ coordinates". ☺

While doing these, keep in mind of the following: if someting requires only linear structure, but NO inner product strucutre, then all formulas are exactly the same as the real case. These includes:

1. Matrix multiplications.
2. Matrix inversions.
3. Gaussian Elimination.
4. Rank-Nullity Theorem. ($\dim \operatorname{Ran}(L) + \dim \operatorname{Ker}(L) = \dim \operatorname{dom}(L)$.)
5. Trace and determinant formula. (And anything that is similarity-invariant, i.e., invariant under an arbitrary change of basis. Here for example we have $\operatorname{trace}(XAX^{-1}) = \operatorname{trace}(A), \det(XAX^{-1}) = \det(A)$.)
6. Eigenstuff, characteristic polynomial, diagonalizability, etc. (All are similarity-invariant.)

Example 8.4.2. Keep in mind that for trace and determinant, the block matrix analogy does not work, even though all the formula are the same.

For example, consider $A = \begin{bmatrix} 1 & \mathrm{i} \\ 1+\mathrm{i} & 1-\mathrm{i} \end{bmatrix}$. Then the trace is $2-\mathrm{i}$. However, the corresponding real block matrix $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix}$ cannot have a complex trace, since all entries are real. The trace of this real block matrix is in fact always $2\operatorname{Re}(\operatorname{trace}(A))$.

Similarly, $\det(A) = 1(1-\mathrm{i}) - \mathrm{i}(1+\mathrm{i}) = 2-2\mathrm{i}$, while the corresponding real block matrix can only have real determinant since all entries are real. In fact, the real block matrix should always have a determinant of $|\det(A)|^2$.

I shall leave the verification of these correspondences to you. ☺

However, the inner product structure (and therefore transpose) is now vastly different. For the matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, taking transpose would send it to $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$. This corresponds to sending the complex number $a+b\mathrm{i}$ to its complex conjugate $a-b\mathrm{i}$.

Example 8.4.3. Consider the complex matrix $A = \begin{bmatrix} 1 & \mathrm{i} \\ 1+\mathrm{i} & 1-\mathrm{i} \end{bmatrix}$. Its corresponding real block matrix is $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix}$.

Now if we take transpose on the real matrix, we would get $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ -1 & 0 & 1 & 1 \end{bmatrix}$. This corresponds to the complex matrix $\begin{bmatrix} 1 & 1-\mathrm{i} \\ -\mathrm{i} & 1+\mathrm{i} \end{bmatrix}$, which is the transpose then complex conjugate of A.

In summary, when we do transpose to real matrices, the analogous operation on complex matrices should be transpose conjugate. ☺

Definition 8.4.4. *Given a complex matrix A, we define its **adjoint** to be $A^* = \overline{A^{\mathrm{T}}}$, where the bar indicates that we are taking complex conjugates on each entry.*

Definition 8.4.5. *Given $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{C}^n$, we define the dot product between them to be $\boldsymbol{v}^*\boldsymbol{w}$.*

Example 8.4.6. Consider $\boldsymbol{v} = \begin{bmatrix} 1 \\ \mathrm{i} \end{bmatrix}$. Then $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v} = 1 + (-1) = 0$. Clearly this should NOT be the length squared of the vector!

If we have $\boldsymbol{v} = \begin{bmatrix} z \\ w \end{bmatrix}$, then in fact $\boldsymbol{v}^*\boldsymbol{v} = \begin{bmatrix} \overline{z} & \overline{w} \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix} = \overline{z}z + \overline{w}w = |z|^2 + |w|^2$. This now makes much more sense to be the length squared of the vector. ☺

Example 8.4.7. In general, what does it mean for two complex vectors to be perpendicular?

Suppose $\boldsymbol{v} = \begin{bmatrix} 1 \\ \mathrm{i} \end{bmatrix}, \boldsymbol{w} = \begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix}$. You can easily calculate and see that $\boldsymbol{v}^*\boldsymbol{w} = 0$. So these two complex vectors are perpendicular.

In real block form, $\boldsymbol{v}$ corresponds to $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix}$ while $\boldsymbol{w}$ corresponds to $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$. As you can see, ALL FOUR columns are mutually orthogonal. This is what it means to be complex orthogonal. ☺

There are other analogous thing to do. Consider these definitions.

Definition 8.4.8.
1. *A complex matrix A is unitary if $A^* = A^{-1}$. (This is analogous to real orthogonal matrices.)*
2. *A complex matrix A is Hermitian (or self-adjoint as physicists prefer) if $A^* = A$. (This is analogous to real symmetric matrices.)*
3. *A complex matrix A is skew-Hermitian (or skew-adjoint) if $A^* = -A$. (This is analogous to real skew-symmetric matrices.)*

We also have this analogous theorem.

Theorem 8.4.9. *If A is an $m \times n$ complex matrix. Then $\mathrm{Ran}(A)^\perp = \mathrm{Ker}(A^*)$ and $\mathrm{Ker}(A)^\perp = \mathrm{Ran}(A^*)$. Here orthogonality is in the sense of complex dot product.*

8.5 (Optional) Fundamental Theorem of Algebra

Here I present my personal favorite proof, using topology.

Consider the polynomial $p(z) = z^n$. Then $p : \mathbb{C} \to \mathbb{C}$ is a (non-linear) map from the plane to the plane. What is its behavior?

Intuitively, if z represents some rotation, then z^n simply rotate n times the original amount. This is the guiding intuition for all the analysis below.

Suppose in the domain, we are looking at a big circle of radius R, centered at the origin. And we let a point z walk around this circle ONCE, counter-clockwise. What would happen to $p(z)$ in the codomain?

Let $z = R\mathrm{e}^{\mathrm{i}\theta}$, and we are increaing θ from 0 to 2π. Then $p(z) = R^n\mathrm{e}^{\mathrm{i}(n\theta)}$, where θ goes from 0 to 2π. It is not hard to see that $p(z)$ in the codomain would walk around a big circle of radius R^n, centered at the origin, n times! $p(z)$ is moving much faster than z, with n times the angular speed of z.

Now consider a generic polynomial $p(z) = z^n$+lower degree terms. Here we assume the leading coefficient is 1 for simplicity. Now, if I pick a super super large R, then for any z on the big circle of radius R, centered at the origin, it will have a super super large absolute value $|z| = R$. In particular, $|z^n| = R^n$ will be much much larger than any lower degree terms. We would effectively have $p(z) \approx z^n$ for all these z on the big circle of radius R, centered at the origin.

So this big circle of radius R in the domain is approximately mapped to some big circle of radius R^n around the origin n-times. Since this is just an approximation, the actual image of the circle will have some minor perturbations caused by the lower degree terms. But they should be minor if R is super super large.

All in all, if in the domain we pick the big circle of radius R, then its image is some curve winding around the origin n-times, approximately a big circle of radius R^n.

Now, let us gradually shrink R towards the origin. As R is shrinked to the origin, its image in the codomain would shrink towards a single point $p(0)$. Since its image is a curve winding around the origin n-times, when it shrinks to a single point, it shall SWEEP THROUGH the origin n-times.

So, inside the disc of radius R around the origin in the domain, we have n inputs z with $p(z) = 0$.

8.6 Algebraic multiplicity and Schur Decomposition

Important: Without specification, everything in this section is over $\mathbb{C}$. Entries are allowed to be complex, coordinates are allowed to be complex, and so on, unless specifically mentioned.

Just like polynomials can have repeated roots, matrices can have repeated eigenvalues. We use the name multiplicity to refer to the number of times they are repeated. Let us first study algebraic multiplicity.

Given an eigenvalue λ of a matrix A, whose characteristic polynomial is $p_A(x)$, then we know λ is a root of p_A. However, roots of a polynomial have multiplicities. For example, $(x-1)^3(x-2)^4$ has seven roots, 1 with multiplicity 3 and 2 with multiplicity 4.

Definition 8.6.1. *The algebraic multiplicity of an eigenvalue λ of A is the multiplicity of λ as roots of the characteristic polynomial $p_A(x)$. (Later we shall have another concept called geometric multiplicity, which shall be different.)*

Example 8.6.2. Consider $A = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 2 \end{bmatrix}$. Clearly $\det(xI - A) = (x-1)^3(x-2)$.

We say A has eigenvalues $1, 1, 1, 2$ counting algebraic multiplicity. We also say that A has eigenvalues $1, 2$ NOT counting algebraic multiplicity. ☺

Proposition 8.6.3. *If A is $n \times n$, then it has n eigenvalues (counting algebraic multiplicities). Equivalently, if $\lambda_1, \ldots, \lambda_k$ are eigenvalues of A NOT counting algebraic multiplicity, then the sum of algebraic multiplicities is $\sum m_a(\lambda_i) = n$.*

Furthermore, if these eigenvalues are $\lambda_1, \ldots, \lambda_n$ counting algebraic multiplicity, then $p_A(x) = \prod(x - \lambda_i)$.

Proof. This is simply the fundamental theorem of algebra. Each polynomial has at most n roots. □

Here is a nice and useful result: eigenvalues and their algebraic multiplicities are independent of basis.

Proposition 8.6.4. *If $A = XBX^{-1}$ for some invertible X, then $p_A(x) = p_B(x)$. In particular, A, B have the same eigenvalues and the same algebraic multiplicities*

Proof. $\det(xI - A) = \det(xI - XBX^{-1}) = \det(X(xI - B)X^{-1}) = \det(xI - B)$. □

The concept of multiplicities allows us to look at the eigenstuff of a matrix globally, and in fact would produce some amazing results.

Proposition 8.6.5. *Given an $n \times n$ matrix A, let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues (counting algebraic multiplicity). Then $\operatorname{trace}(A) = \sum \lambda_i$ and $\det(A) = \prod \lambda_i$.*

Let us see the 2×2 case before the formal proof.

Example 8.6.6. Consider $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $xI - A = \begin{bmatrix} x-a & -b \\ -c & x-d \end{bmatrix}$. So the characteristic polynomial is $(x-a)(x-d) - bc = x^2 - (a+d)x + ad - bc = x^2 - \operatorname{trace}(A)x + \det(A)$. On the other hand, if the eigenvalues are λ_1, λ_2, then the characteristic polynomial should also be $(x-\lambda_1)(x-\lambda_2) = x^2 - (\lambda_1 + \lambda_2)x + \lambda_1\lambda_2$. Compare the two, and we are done.

Note that this is essentially Vieta's formula, which relates the roots of a polynomial (eigenvalues of your matrix) with the coefficients of a polynomial (calculated from entries of your matrix). ☺

Proof. Say $A = \begin{bmatrix} \boldsymbol{a}_1 & \dots & \boldsymbol{a}_n \end{bmatrix}$. Then $\det(xI - A) = \det(x\boldsymbol{e}_1 - \boldsymbol{a}_1, \dots, x\boldsymbol{e}_n - \boldsymbol{a}_n)$. Now we expand everything using multilinearity.

Note that to get contribution to the term x^{n-1}, we must pick all but one of the $x\boldsymbol{e}_i$. As a result, in the expansion, the coefficient for x^{n-1} is $\det(-\boldsymbol{a}_1, x\boldsymbol{e}_2, \dots, x\boldsymbol{e}_n) + \dots + \det(x\boldsymbol{e}_1, \dots, x\boldsymbol{e}_{n-1}, -\boldsymbol{a}_n) = -a_{11} - \dots - a_{nn} = -\operatorname{trace}(A)$. However, this is also $-\sum \lambda_i$ by Vieta's formula. So we are done.

Similarly, consider the constant term in the expansion of $\det(xI - A) = \det(x\boldsymbol{e}_1 - \boldsymbol{a}_1, \dots, x\boldsymbol{e}_n - \boldsymbol{a}_n)$. Then we must avoid ALL $x\boldsymbol{e}_i$. So we have constant term $\det(-\boldsymbol{a}_1, \dots, -\boldsymbol{a}_n) = (-1)^n \det(A)$. However, this is also $(-1)^n \prod \lambda_i$ by Vieta's formula. So we are done. ☐

Alternative proof. These proofs are just for fun. Suppose $p_A(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$.

Since $p_A(x) = \det(xI - A)$, therefore $p_A(0) = \det(-A) = (-1)^n \det(A)$. On the other hand, $p_A(0) = a_0 = (-1)^n \prod \lambda_i$ by Vieta's formula. Hence we are done.

Now consider $f(t) = \det(I + tA)$. Then $f(-\frac{1}{x}) = \det(I - \frac{1}{x}A) = \frac{1}{x^n}\det(xI - A) = \frac{1}{x^n}p_A(x) = 1 + \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n}$. Therefore, by setting $t = -\frac{1}{x}$, we see that $f(t) = 1 - a_{n-1}t + a_{n-2}t^2 - \dots + (-1)^n a_0 t^n$. So $f'(0) = -a_{n-1} = \sum \lambda_i$ by Vieta's formula. On the other hand, we know $f'(0) = \operatorname{trace}(A)$. So we are done. ☐

Example 8.6.7. Consider $\begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix}$. Since each row adds up to 1, we see that 1 is an eigenvalue. Since the trace is 5, we see that the other eigenvalue is 4. Done. Such techniques are very helpful to speed calculate the eigenvalues of small matrices. ☺

Here is also a handy corollary.

Corollary 8.6.8. *A is invertible iff it has no zero eigenvalue.*

Proof. A is invertible iff $\det(A) \neq 0$ iff $\prod \lambda_i \neq 0$ iff all $\lambda_i \neq 0$. ☐

In fact, by similar methodology (but more complicated computations), we can prove the following. I obmit the proof here since the computations are a bit too messy.

Definition 8.6.9. *A k principal submatrix of A is any $k \times k$ submatrix whose diagonal is on the diagonal of A. (Note that a submatrix does NOT have to be a block. For example, $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ has three principal submatrices $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$, $\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}$ and $\begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}$, whose diagonal is indeed on the diagonal of the original matrix.)*

Theorem 8.6.10. *The coefficient for x^{n-k} in $p_A(x)$ is exactly $(-1)^k S_k$, where S_k is the sum of determinants of all k principal submatrices of A.*

By generalized Vieta's formula, this is also $(-1)^k \sum_{1 \leq i_1 < \dots < i_k \leq n} \lambda_{i_1} \dots \lambda_{i_k}$, sum of all possible products of k of the roots.

So in particular, we have $S_k = \sum_{1 \leq i_1 < \dots < i_k \leq n} \lambda_{i_1} \dots \lambda_{i_k}$.

Remark 8.6.11. *Given an $n \times n$ matrix A, then its only $n \times n$ principal submatrix is A itself. So $S_n = \det(A)$.*

Its 1×1 principal submatrices are the diagonal entries. So $S_1 = \operatorname{trace}(A)$.

You can see that the above theorem is a generalization of the result we have proven.

Let us see some examples both utilizing $\mathrm{trace}(A) = \sum \lambda_i$ and $\det(A) = \prod \lambda_i$, and some examples of the weird theorem above.

Example 8.6.12. Consider $A = \begin{bmatrix} -1 & 1 & 1 \\ -4 & 3 & 2 \\ -4 & 1 & 4 \end{bmatrix}$. What are the eigenvalues? How to diagonalize this matrix?

Well, note that each row adds up to one. So one eigenvalue is 1. We also have $\mathrm{trace}(A) = 6$ and $\det(A) = 6$.

Suppose the eigenvalues are $1, \lambda_2, \lambda_3$, then $\lambda_2 + \lambda_3 = 5$ and $\lambda_2\lambda_3 = 6$. We see that the two unknown eigenvalues must be 2 and 3.

Calculate $\mathrm{Ker}(A - \lambda I)$ with $\lambda = 1, 2, 3$ via Guassian elimination, we have eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for the eigenvalue 1, eigenvector $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ for the eigenvalue 2, and eigenvector $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ for the eigenvalue 3. These three vectors form a basis, so we have found our diagonalization:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 2 & \\ & & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 1 & 2 \end{bmatrix}^{-1}.$$

Now, let us consider the 2×2 principal submatrices. We have

$$S_2 = \det \begin{bmatrix} -1 & 1 \\ -4 & 3 \end{bmatrix} + \det \begin{bmatrix} -1 & 1 \\ -4 & 4 \end{bmatrix} + \det \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = 1 + 0 + 10 = 11.$$

On the other hand, we have

$$\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3 = 1 \times 2 + 2 \times 3 + 1 \times 3 = 11.$$

Indeed, the two are the same. In fact, you can verify that the characteristic polynomial is indeed $p_A(x) = x^3 - 6x^2 + 11x - 6$. The value of S_1 (trace), S_2, S_3 (determinant) corresponds to the three coefficients. ☺

Example 8.6.13. For triangular matrices, this correspondence is even better.

Consider $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. What are the eigenvalues? How to diagonalize this matrix?

Well, note that each row adds up to three. So one eigenvalue is 3. The first column indicates that $A\boldsymbol{e}_1 = \boldsymbol{e}_1$, so one eigenvalue must be 1. Finally, the trace is 6, so the last eigenvalue must be $6 - 1 - 3 = 2$.

Calculate $\mathrm{Ker}(A - \lambda I)$ with $\lambda = 1, 2, 3$ via Guassian elimination, we have eigenvector $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ for the eigenvalue 1, eigenvector $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ for the eigenvalue 2, and eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for the eigenvalue 3. These three vectors form a basis, so we have found our diagonalization:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 2 & \\ & & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^{-1}.$$

Now, let us consider the 2×2 principal submatrices. We have

$$S_2 = \det \begin{bmatrix} 1 & 1 \\ & 2 \end{bmatrix} + \det \begin{bmatrix} 2 & 1 \\ & 3 \end{bmatrix} + \det \begin{bmatrix} 1 & 1 \\ & 3 \end{bmatrix} = 1 \times 2 + 2 \times 3 + 1 \times 3 = 11.$$

On the other hand, we have

$$\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3 = 1 \times 2 + 2 \times 3 + 1 \times 3 = 11.$$

Indeed, the two are the same. You can even see that the three principal submatrices in this case corresponds perfectly with pairs of eigenvalues! This is an amazing thing that should always happen to triangular matrices. ☺

Note that the key here lies on the structure of triangular matrices. Let us extrapolate the result here:

Proposition 8.6.14. *If* $A = \begin{bmatrix} B & \star \\ O & C \end{bmatrix}$*, then the characteristic polynomials have the relation* $p_A(x) = p_B(x)p_C(x)$*.*

Proof. This is a direct corollary of the similar results on the determinants of block triangular matrices. □

Corollary 8.6.15. *If A is triangular, then its eigenvalues (counting algebraic multiplicity) are exactly the diagonal entries.*

As a result, for a triangular matrix such as $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$, it is perfectly clear now why its 2×2 principal submatrices are in perfect correspondence with pairs of eigenvalues: because each 2×2 principal submatrix takes a pair of diagonal entries!

At this stage, it is very clear that triangular matrices are super amazing. Their determinant is easy, their eigenvalues are easy, everything is easy. Don't you wish that all matrices are triangular? Well.... Wish granted!

Theorem 8.6.16 (Schur Decomposition)**.** *For any square matrix A, let its eigenvalues be $\lambda_1, \dots, \lambda_n$ counting algebraic multiplicity, in any prescribed order. Then one can find an invertible matrix X such that $A = XTX^{-1}$ where T is upper triangular, and the diagonal entries of T are exactly $\lambda_1, \dots, \lambda_n$ in the desired order.*

Furthermore, we can take X to be a unitary matrix.

In short, for any matrix, you can change basis and make it triangular. Furthermore, if needed, you can restrict it to an orthonormal change of basis!

Proof. First pick any eigenvector for the eigenvalue λ_1, say $\boldsymbol{x}_1 \neq \mathbf{0}$. WLOG, we can pick $\boldsymbol{x}_1$ to be a unit vector.

Extend $\boldsymbol{x}_1$ into a basis (or orthonormal basis), then these vectors as columns would form a matrix X_1, where the first column of X_1 is $\boldsymbol{x}_1$. (Note that X_1 will be a unitary matrix as well.)

Now consider a change of basis to this new basis X_1, resulting in $X_1^{-1}AX_1$. Note that since $A\boldsymbol{x}_1 = \lambda_1\boldsymbol{x}_1$, the first basis vector is sent to λ_1 times itself. Therefore, after this change of basis, $X_1^{-1}AX_1 = \begin{bmatrix} \lambda_1 & \star \\ \mathbf{0} & A_{n-1} \end{bmatrix}$ for some $(n-1) \times (n-1)$ matrix A_{n-1}.

(Equivalently, observe that $X_1^{-1}AX_1 = X_1^{-1}A\begin{bmatrix} \boldsymbol{x}_1 & \star \end{bmatrix} = X_1^{-1}\begin{bmatrix} \lambda_1\boldsymbol{x}_1 & \star \end{bmatrix} = \begin{bmatrix} \lambda_1\boldsymbol{e}_1 & \star \end{bmatrix}$.)

Either way, now apply mathematical induction on the dimension n of A. The $n = 1$ case is trivial. For generic n, we can assume that the $n-1$ case is already done. Therefore, we can assume that $A_{n-1} = X_{n-1}T_{n-1}X_{n-1}^{-1}$ for some unitary X_{n-1}. Furthermore, since the eigenvalues of A_{n-1} are exactly $\lambda_2, \dots, \lambda_n$, we can make sure that the diagonal entries of T_{n-1} are exactly these values in the desired order by induction hypothesis.

So we have $A = X_1 \begin{bmatrix} 1 & \\ & X_{n-1} \end{bmatrix} \begin{bmatrix} \lambda_1 & \star \\ \mathbf{0} & T_{n-1} \end{bmatrix} \begin{bmatrix} 1 & \\ & X_{n-1}^{-1} \end{bmatrix} X_1^{-1}$. So we are done. □

Remark 8.6.17. *Note that for most of our purpose, it is enough to have $A = XTX^{-1}$ for any invertible X. The fact that we can furthermore choose X to be unitary is not important for our class at the moment. However, in terms of computation stability, this is HUGE. It means this factorization $A = XTX^{-1}$ will NOT magnify errors.*

Currently, if you ask a computer to find roots of a polynomial $p(x)$, it will attempt to first build a matrix A whose characteristic polynomial is $p(x)$, and then perform the Schur decomposition $A = XTX^{-1}$, and then read the diagonal entries of T.

Both diagonalization and triangularization tries to do the same thing: we attempt to find a basis, so that after the change of basis, our matrix looks prettier (and therefore easier to compute with). In this sense, diagonalizations are the better version. Triangular matrices are of course much uglier than diagonal matrices.

However, the advantage is two-fold. First, for diagonalization $A = XDX^{-1}$, the matrix X might not be unitary, and error terms might be magnified severly. For triangularization, this is not a problem since X is unitary. Second, some matrices do NOT have diagonalizations. But ALL matrices have triangularizations. The universality is in itself a huge advantage already.

Example 8.6.18. Suppose A has eigenvalues $\lambda_1, \dots, \lambda_n$ counting algebraic multiplicity, then $A = XTX^{-1}$ where $T = \begin{bmatrix} \lambda_1 & * & * \\ & \ddots & * \\ & & \lambda_n \end{bmatrix}$.

Then $A^2 = XTX^{-1}XTX^{-1} = XT^2X^{-1}$, where $T^2 = \begin{bmatrix} \lambda_1^2 & * & * \\ & \ddots & * \\ & & \lambda_n^2 \end{bmatrix}$ where the star portion might change in ugly manner, but the diagonal portion is still nice. In particular, we see that A^2 has eigenvalues $\lambda_1^2, \dots, \lambda_n^2$ counting algebraic multiplicity. You can imagine that, for A^k, the eigenvalues would be $\lambda_1^k, \dots, \lambda_n^k$ counting algebraic multiplicity.

Now consider $A^2 + A$, which is a polynomial in A. Since $A = XTX^{-1}$ and $A^2 = XT^2X^{-1}$, we can see that $A^2 + A = X(T^2 + T)X^{-1}$ where $T^2 + T = \begin{bmatrix} \lambda_1^2 + \lambda_1 & * & * \\ & \ddots & * \\ & & \lambda_n^2 + \lambda_n \end{bmatrix}$. So the eigenvalues for $A^2 + A$ are $\lambda_1^2 + \lambda_1, \dots, \lambda_n^2 + \lambda_n$ counting algebraic multiplicity. ☺

Proposition 8.6.19. *If A has eigenvalues $\lambda_1, \dots, \lambda_n$, then for any polynomial $p(x)$, the matrix $p(A)$ has eigenvalues $p(\lambda_1), \dots, p(\lambda_n)$. (For example, $A + kI$ would have eigenvalues $\lambda_1 + k, \dots, \lambda_n + k$. And A^2 would have eigenvalues $\lambda_1^2, \dots, \lambda_n^2$.)*

Proof. Write $A = XTX^{-1}$. Then $p(A) = Xp(T)X^{-1}$. Since T is triangular with diagonal entries $\lambda_1, \dots, \lambda_n$, we see that $p(T)$ is still triangular with diagonal entries $p(\lambda_1), \dots, p(\lambda_n)$. So we are done. □

Here is an optional alternative proof without use Schur decomposition. It is a purely algebraic manipulation of the characteristic polynomial.

Optional Proof. Warning: This proof is ugly and not that illuminating. But serious math lovers should know that this is possible.

It is very easy to see that $p(\lambda_i)$ is indeed an eigenvalue for $p(A)$. If $A\boldsymbol{v} = \lambda_i\boldsymbol{v}$, then it is easy to verify that $A^k\boldsymbol{v} = \lambda_i^k\boldsymbol{v}$. Since $p(A)$ is a linear combination of powers of A, we see that $p(A)\boldsymbol{v} = p(\lambda_i)\boldsymbol{v}$. The difficulty here lies in the algebraic multiplicity. To show that, one has to attack the characteristic polynomial.

Consider $\det(\lambda I - p(A))$ for some fixed λ. What is $\lambda I - p(A)$? This is simply another polynomial of A! This is the most vital observation. Let $q(t) = \lambda - p(t)$. By the fundamental theorem of linear algebra, we have $q(t) = k\prod(t - t_i)$ for some complex values $k, t_1, \dots, t_n$. So we have

$$\det(\lambda I - p(A)) = \det(q(A)) = k^n \prod \det(A - t_iI) = (-k)^n \prod_i p_A(t_i).$$

Now use the fact that $p_A(x) = \prod(x - \lambda_j)$, we see that

$$(-k)^n \prod_i p_A(t_i) = (-k)^n \prod_{i,j}(t_i - \lambda_j) = \prod_j (k \prod_i (\lambda_j - t_i)) = \prod_j q(\lambda_j) = \prod_j (\lambda - p(\lambda_j)).$$

This is true for all λ. So $\det(\lambda I - p(A)) = \prod_j (\lambda - p(\lambda_j))$. In particular, the characteristic polynomial has roots $p(\lambda_j)$ as desired. $\square$

By similar arguments (either by Schur decomposition or by characteristic polynomial manipulation), one can see the following:

Proposition 8.6.20. *If A has eigenvalues $\lambda_1, \dots, \lambda_n$, then A^{T} has the same set of eigenvalues. And if A is invertible, then A^{-1} has eigenvalues $\lambda_1^{-1}, \dots, \lambda_n^{-1}$.*

Example 8.6.21. The fact that $p(A)$ has eigenvalues $p(\lambda)$ is very convenient when analyzing eigenvalues of A.

For example, suppose $A^2 = A$, i.e., the (oblique) projection matrices. What are the possible eigenvalues of such a matrix?

We have $A^2 - A = 0$. Therefore, if λ is an eigenvalue of A, then $\lambda^2 - \lambda$ must be an eigenvalue for $A^2 - A = O$, so we have $\lambda^2 - \lambda = 0$. This implies that $\lambda = 0$ or 1.

So all eigenvalues of A are either 0 or 1. Indeed, a projection does two things: kill $\mathrm{Ker}(A)$ and preserve $\mathrm{Ran}(A)$. $\mathrm{Ker}(A)$ is exactly the eigenspace for 0, while $\mathrm{Ran}(A)$ is exactly the eigenspace for 1. ☺

Example 8.6.22. Consider a Householder reflection H. As a reflection, we must have $H^2 = I$. In particular, eigenvalues of H must have $\lambda^2 = 1$. As a result, $\lambda = \pm 1$. Indeed, the eigenspace for 1 is the hyperplane of reflection, while the eigenspace for -1 is the normal direction of the hyperplane, which shall be reflected by H via $H\boldsymbol{n} = -\boldsymbol{n}$. ☺

There is a common theme behind both examples. Any polynomial that kills A would determine the eigenvalues of A. These eigenvalues then in turn determines the behavior of A. So in many cases, if you want to study linear maps with certain behavior, you can just specify some polynomial $p(x)$ and consider all matrices that satisfy this polynomial via $p(A) = 0$. As we have see here, $A^2 = A$ gives (oblique) projections, while $A^2 = I$ would actually give you all (oblique) reflections. If you want the orthogonal versions, you can further require some condition between the matrix and its transpose (or adjoint in case of complex matrices). For example, by requiring $A = A^{\mathrm{T}}$ for real matrices, then $A^2 = A$ would give you orthogonal projection, and $A^2 = I$ would give you orthogonal reflection.

Remark 8.6.23. *This remark is completely optional.*

How to perform a Schur decomposition? If you know all the eigenvalues already, then this is super easy. Just look at the proof and figure it out from there. However, if you do not know the eigenvalues of A, how would you do this?

One famous algorithm is the iterated QR-algorithm. Today computers usually use some variants or improved version of it. The idea is this:

1. *Set $A_0 = A$, and perform QR decomposition $A_0 = Q_0 R_0$.*
2. *Set $A_1 = R_0 Q_0$, and perform QR decomposition $A_1 = Q_1 R_1$.*
3. *Set $A_2 = R_1 Q_1$, and perform QR decomposition $A_2 = Q_2 R_2$.*
4. *....*
5. *R_∞ would probably converge to T in the Schur decomposition $A = XTX^{-1}$.*

(This does not always converge. The precise condition is annoying to state clearly. However, in practice, most of the time it should work.)

What is the idea of this? Well, consider these calculations.

1. $AQ_0 = Q_0R_0Q_0 = Q_0Q_1R_1$.

2. $AQ_0Q_1 = Q_0Q_1R_1Q_1 = Q_0Q_1Q_2R_2$.

3. ....

4. $A(Q_0 \dots Q_k) = (Q_0 \dots Q_{k+1})R_{k+1}$.

Let $X_k = Q_0 \dots Q_k$. Then you see that $AX_k = X_{k+1}R_{k+1}$ where X_k is orthogonal (or unitary in the complex case). Taking limit $k \to \infty$, we see that $AX_\infty = X_\infty R_\infty$, and hence $A = X_\infty R_\infty X_\infty^{-1}$ where X_∞ is unitary and R_∞ is upper triangular. Done.

8.7 Geometric multiplicity and diagonalization

Last section is mostly designed towards eigenvalues. However, to achieve diagonalization, we also need eigenvectors, which are the goal for this section.

Given an eigenvalue λ for a matrix A, then by definition there must be eigenvectors for this eigenvalue, i.e., some non-zero $\boldsymbol{v}$ such that $A\boldsymbol{v} = \lambda\boldsymbol{v}$. In particular, $\mathrm{Ker}(A - \lambda I) \neq \{\mathbf{0}\}$. We can generally think of $\mathrm{Ker}(A - \lambda I)$ as the space of "λ eigenvectors" (with the exception of $\mathbf{0}$ in it).

Definition 8.7.1. *Given an eigenvalue λ for a matrix A, its **eigenspace** is $\mathrm{Ker}(A - \lambda I)$. (This is also exactly made of the zero vector and all eigenvalues for λ.)*

*We say the **geometric multiplicity** of λ is $\dim \mathrm{Ker}(A - \lambda I)$.*

Intuitively, geometric multiplicity measures how many eigenvectors you have for this eigenvalue.

Now, if we want A to be diagonalizable, we want eigenvectors of A to span the whole domain. This means we want the sum space $\sum \mathrm{Ker}(A - \lambda I)$ to be the whole space. In particular, we hope that $\dim(\sum \mathrm{Ker}(A - \lambda I)) = n$.

This sum space is not that easy to study. However, let us show that in fact $\dim(\sum \mathrm{Ker}(A - \lambda I)) = \sum \dim \mathrm{Ker}(A - \lambda I)$. In this sense, we only need to check if the geometric multiplicity adds up to n. The phenomenon here is independent subspaces.

Definition 8.7.2. *For subspaces $W_1, \dots, W_k$ of V, we say they are linearly independent if, for any $\boldsymbol{w}_1 \in W_1, \dots, \boldsymbol{w}_k \in W_k$, then $\sum \boldsymbol{w}_i = \mathbf{0}$ implies all $\boldsymbol{w}_i$ are $\mathbf{0}$.*

Example 8.7.3. Recall that on $\mathbb{R}^2$, the x-axis, the y-axis and the line $x = y$ are pairwise independent, but collectively NOT independent. In particular,

$$\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} + \begin{bmatrix}-1\\-1\end{bmatrix} = \mathbf{0},$$

this violates the linear independence condition. Indeed, in this case $\dim \sum W_i = 2$ while $\sum \dim W_i = 3$.

In contrast, in $\mathbb{R}^3$, the three coordinate axes are independent subspaces. If we have

$$\begin{bmatrix}a\\0\\0\end{bmatrix} + \begin{bmatrix}0\\b\\0\end{bmatrix} + \begin{bmatrix}0\\0\\c\end{bmatrix} = \mathbf{0},$$

then we must have $a = b = c = 0$. Indeed, in this case $\dim \sum W_i = \sum \dim W_i = 3$. ☺

Let us now establish an independence extension lemma for subspaces.

Lemma 8.7.4. *Suppose subspaces $W_1, \dots, W_{k-1}$ of V are linearly independent. Then $W_1, \dots, W_k$ are linearly independent if and only if $(W_1 + \dots + W_{k-1}) \cap W_k = \{\mathbf{0}\}$.*

Proof. Suppose $(W_1 + \cdots + W_{k-1}) \cap W_k = \{\mathbf{0}\}$. Pick any $\boldsymbol{w}_1 \in W_1, \ldots, \boldsymbol{w}_k \in W_k$ and suppose $\sum \boldsymbol{w}_i = \mathbf{0}$.

Then $-\boldsymbol{w}_k = \boldsymbol{w}_1 + \cdots + \boldsymbol{w}_{k-1}$. But the left hand side is in W_k, and the right hand side is in $W_1 + \cdots + W_{k-1}$, so both sides are in the intersection, which must be $\mathbf{0}$. So $\boldsymbol{w}_k = \mathbf{0}$ and $\boldsymbol{w}_1 + \cdots + \boldsymbol{w}_{k-1} = \mathbf{0}$. But since $W_1, \ldots, W_{k-1}$ are linearly independent, therefore all $\boldsymbol{w}_i = \mathbf{0}$.

Conversely, suppose $W_1, \ldots, W_k$ are linearly independent. Pick any $\boldsymbol{w}_k \in W_k \cap (W_1 + \cdots + W_{k-1})$, then $\boldsymbol{w}_k = \boldsymbol{w}_1 + \cdots + \boldsymbol{w}_{k-1}$ where $\boldsymbol{w}_i \in W_i$. But then $\boldsymbol{w}_1 + \cdots + \boldsymbol{w}_{k-1} - \boldsymbol{w}_k = \mathbf{0}$. By linear independence, all $\boldsymbol{w}_i = \mathbf{0}$. So $\boldsymbol{w}_k = \mathbf{0}$. So the only vector in the intersection $(W_1 + \cdots + W_{k-1}) \cap W_k$ is the zero vector. □

Corollary 8.7.5. *Subspaces $W_1, \ldots, W_k$ of V are linearly independent if and only if $\dim \sum W_i = \sum \dim W_i$.*

Proof. If $k = 1$, the statement is trivial. Now for generic k, suppose the statement is true for dimensions less than k.

If $W_1, \ldots, W_k$ are linearly independent, then $(W_1 + \cdots + W_{k-1}) \cap W_k = \{\mathbf{0}\}$. By inclusion-exclusion principle, we have $\dim \sum W_i = \dim W_k + \dim \sum_{i=1}^{k-1} W_i$. Now by induction hypothesis, $\dim \sum_{i=1}^{k-1} W_i = \sum_{i=1}^{k-1} \dim W_i$. So we have $\dim \sum W_i = \sum \dim W_i$.

Conversely, suppose $\dim \sum W_i = \sum \dim W_i$. Then $\sum \dim W_i = \dim \sum W_i \leq \dim W_k + \dim \sum_{i=1}^{k-1} W_i \leq \dim W_k + \sum_{i=1}^{k-1} \dim W_i = \sum \dim W_i$. So all inequalities here are equalities. In particular, we have $W_k \cap (W_1 + \cdots + W_{k-1}) = \{\mathbf{0}\}$, and the fact that $\dim \sum_{i=1}^{k-1} W_i = \sum_{i=1}^{k-1} \dim W_i$. By induction hypothesis, this means that $W_1, \ldots, W_{k-1}$ are linearly independent and $W_k \cap (W_1 + \cdots + W_{k-1}) = \{\mathbf{0}\}$. So $W_1, \ldots, W_k$ of V are linearly independent. □

Next, we shall see that eigenspaces are all independent.

Remark 8.7.6. *Eigenspaces for different eigenvalues clearly have zero intersection. If $\boldsymbol{v} \in \mathrm{Ker}(A - \lambda I) \cap \mathrm{Ker}(A - \mu I)$, then $A\boldsymbol{v} = \lambda \boldsymbol{v}$ and $A\boldsymbol{v} = \mu \boldsymbol{v}$. Therefore $\lambda \boldsymbol{v} = \mu \boldsymbol{v}$. Hence either $\boldsymbol{v} = \mathbf{0}$, or $\lambda = \mu$.*

However, this is merely "pairwise independence", which is not enough for collective independence.

Proposition 8.7.7. *Suppose A has eigenvalues (NOT counting algebraic multiplicity) $\lambda_1, \ldots, \lambda_k$, and let $V_1, \ldots, V_k$ be corresponding eigenspaces. Then these spaces are linearly independent.*

Proof. Pick $\boldsymbol{v}_i \in V_i$ for each i.

Suppose $\sum \boldsymbol{v}_i = \mathbf{0}$. What can we do? Since $A\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$, we can try to repeatedly applying A to this equation. This would yeilds the following system of equations:

$$\begin{aligned} \boldsymbol{v}_1 + \ldots \quad & + \boldsymbol{v}_k = 0 \\ \lambda_1 \boldsymbol{v}_1 + \ldots \quad & + \lambda_k \boldsymbol{v}_k = 0 \\ \lambda_1^2 \boldsymbol{v}_1 + \ldots \quad & + \lambda_k^2 \boldsymbol{v}_k = 0 \\ \vdots \quad & \\ \lambda_1^{k-1} \boldsymbol{v}_1 + \ldots & + \lambda_k^{k-1} \boldsymbol{v}_k = 0. \end{aligned}$$

Writing in terms of matrices we have

$$\begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_k \end{bmatrix} \begin{bmatrix} 1 & \lambda_1 & \ldots & \lambda_1^{k-1} \\ 1 & \lambda_2 & \ldots & \lambda_2^{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_k & \ldots & \lambda_k^{k-1} \end{bmatrix} = \mathbf{0}.$$

Note that the right matrix is a Vandermonde matrix. Since $\lambda_1, \ldots, \lambda_k$ are eigenvalues WIHTOUT counting algebraic multiplicities, they are all distinct. Therefore this Vandermonde matrix is invertible. So $\begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_k \end{bmatrix} = O$. As a result, we have $\boldsymbol{v}_i = 0$ for all i. So these eigenspaces are linearly independent. □

Remark 8.7.8. *The idea behind this proof is called the power method. Recall that we started our study of eigenvalues EXACTLY by trying to study sequences like $\boldsymbol{v}, A\boldsymbol{v}, A^2\boldsymbol{v}, \dots$. Therefore, in many computational applications, people usually compute the sequence $A, A^2, A^3, \dots$ and then extrapolate eigenvalues or eigenvalue behaviors.*

This is also the second major use of Vandermonde matrix.

In some sense, the very motivation of eigensutff come from the study of $A, A^2, A^3, \dots$. It is not surprising that iterations are crucial for many parts of eigenstuff.

We now come to the meat of this section.

Corollary 8.7.9. *An $n \times n$ matrix A is diagonalizable iff its geometric multiplicities $m_g(\lambda)$ add up to n.*

Proof. Note that $\dim(\sum V_i) = \sum \dim V_i$ is the sum of geometric multiplicity. So if the geometric multiplicities add up to n, then $\sum V_i$ is the whole domain. So the whole domain is spanned by eigenvectors, and we have A diagonalizable.

The other direction is simply the reverse of the deductions above. □

Now compare the two multiplicities. $\sum m_a(\lambda_i) = n$ always, while $\sum m_g(\lambda_i)$ might or might not be n. If the eigenvectors fail to span the whole domain, then they shall span some subspace, so we in fact have $\sum m_g(\lambda_i) < n$ in that case. Geometric multiplicities seems to be smaller or equal to algebraic multiplicities. This is indeed the case.

Theorem 8.7.10. *Let λ be an eigenvalue of A. Then $m_g(\lambda) \leq m_a(\lambda)$. I.e., for each eigenvalue, geometric multiplicities are less than or equal to the algebraic multiplicities.*

Proof. Let $V \subseteq \mathbb{R}^n$ be the eigenspace for λ with dimension $m = m_g(\lambda)$. Pick a basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$, and extend this into a basis for the whole space. Then in this new basis, our matrix A changes into $B = X^{-1}AX$ for some invertible X.

Now consider the first m columns of B. Since $A\boldsymbol{v}_i = \lambda\boldsymbol{v}_i$ for all $1 \leq i \leq m$, we see that $B\boldsymbol{e}_i = \lambda\boldsymbol{e}_i$ for all $1 \leq i \leq m$. So $B = \begin{bmatrix} \lambda I_{m\times m} & \star \\ O & B_1 \end{bmatrix}$.

Consider the characteristic polynomial, and use the upper triangular block structure, we see that $\det(xI - B) = \det \begin{bmatrix} (x-\lambda) I_{m\times m} & \star \\ O & xI - B_1 \end{bmatrix} = (x-\lambda)^m p_{B_1}(x)$. Here $p_{B_1}(x)$ is the characteristic polynomial of B_1.

In particular, we see that λ is a root of the characteristic polynomial of B at least m times. Finally, since A, B differs only via a change of basis, they have the same characteristic polynomial, same eigenvalues and same algebraic multiplicity.

As a more computational write-up of the same proof, we have

$$\begin{aligned} AX =& A\begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_m & X_1 \end{bmatrix} \\ =& \begin{bmatrix} \lambda\boldsymbol{v}_1 & \dots & \lambda\boldsymbol{v}_m & Y_1 \end{bmatrix} \\ =& \begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_m & X_1 \end{bmatrix} \begin{bmatrix} \lambda I_{m\times m} & \star \\ O & B_1 \end{bmatrix} \\ =& X \begin{bmatrix} \lambda I_{m\times m} & \star \\ O & B_1 \end{bmatrix}. \end{aligned}$$

So $X^{-1}AX = \begin{bmatrix} \lambda I_{m\times m} & \star \\ O & B_1 \end{bmatrix}$, and hence the characteristic polynomial of A contains factor $(x-\lambda)^m$. □

In some sense, algebraic multiplicity cares only about the result of a triangularization (Schur decomposition). In contrast, geometric multiplicity cares only about the result of a potential diagonalization. In particular, let us now conclude this venture into a criteria for diagonalizability.

Theorem 8.7.11 (Criteria for diagonalizability)**.** *A matrix A is diagonalizable if and only if for each eigenvalue λ, its algebraic multiplicity equals to its geometric multiplicity.*

Proof. Note that $\sum m_g(\lambda) \leq \sum m_a(\lambda) = n$. So the left hand side equal to n iff equality holds iff $m_a = m_g$ for all λ. $\square$

In particular, why does diagonalizability fail? It is because of DEFECTIVE eigenvalues.

Definition 8.7.12. *An eigenvalue is **defective** if its geometric multiplicity is STRICTLY smaller than its algebraic multiplicity.*

Example 8.7.13. Consider an upper triangular matrix. Then the algebraic multiplicity only cares about the diagonal entries. The rest could be whatever. However, if the entries above the diagonal are too bad, your matrix might NOT be diagonalizable. Then you would fail to have enough geometric multiplicity.

The most important example to keep in mind is $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Its only eigenvalue is 1, with algebraic multiplicity 2 and geometric multiplicity 1. (Can you compute this geometric multiplicity yourself?)

In general, we can look at a ***Jordan block***, which is $J_\lambda = \begin{bmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix}$. This is the MOST DEFECTIVE case possible, where the algebraic multiplicity of the only eigenvalue λ is n, while the geometric multiplicity is 1. ☺

Remark 8.7.14. *For matrix that cannot be diagonalized, we can in fact always "Jordanize", which is as close to diagonal as possible. For any square matrix A, we can always find decomposition $A = XJX^{-1}$ where J is block diagonal, and each diagonal block is a Jordan block. This matrix J (also called the Jordan canonical form of A) is unique up to the permutation of these diagonal blocks. In particular, A is diagonalizable if and only if its Jordan canonical form is diagonal.*

We do not prove it here though.

Now that the most defective case is dealt with, let us look at the opposite case. This is a very useful scenario.

Definition 8.7.15. *An eigenvalue is **simple** if its algebraic multiplicity is 1.*

Proposition 8.7.16. *A simple eigenvalue cannot be defective. In particular, if all eigenvaleus of A are simple, then A is diagonalizable.*

In other words, if all n eigenvalues of A are distinct, then A is diagonalizable.

Proof. Note that by definition, an eigenvalue λ must have corresponding non-zero eigenvectors. Its eigenspace has dimension at least 1. So $m_g(\lambda)$ is at least 1.

If the eigenvalue λ is simple, then $1 \leq m_g(\lambda) \leq m_a(\lambda) = 1$. So we must have $m_g = m_a$. $\square$

Corollary 8.7.17. *If a triangular matrix has distinct diagonal entries, then it is diagonalizable.*

Let us see some examples to get a better computational idea on diagonalization. In practice, one computational way to get diagonalization for computers is to first find the Schur triangularization, and then diagonalize using row/column operations.

Example 8.7.18. Consider $\begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$. How can we diagonalize it?

Let us try the shearing $\begin{bmatrix} 1 & k & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -k & \\ & 1 & \\ & & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4+k & 5+6k \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$. (Think in terms of row and column operations to compute faster.) So by setting $k = -4$, we see that our original matrix is similar to $\begin{bmatrix} 1 & 0 & -19 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$.

Next we try $\begin{bmatrix}1 & & \\ & 1 & k \\ & & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -19 \\ 0 & 2 & 6 \\ 0 & 0 & 3\end{bmatrix}\begin{bmatrix}1 & & \\ & 1 & -k \\ & & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & -19 \\ 0 & 2 & 6+k \\ 0 & 0 & 3\end{bmatrix}$. By setting $k = -6$, we are now at $\begin{bmatrix}1 & 0 & -19 \\ 0 & 2 & 0 \\ 0 & 0 & 3\end{bmatrix}$.

Finally, we try $\begin{bmatrix}1 & & k \\ & 1 & \\ & & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -19 \\ 0 & 2 & 0 \\ 0 & 0 & 3\end{bmatrix}\begin{bmatrix}1 & & -k \\ & 1 & \\ & & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & -19+2k \\ 0 & 2 & 0 \\ 0 & 0 & 3\end{bmatrix}$. Set $k = \frac{19}{2}$, and we are done. Indeed A can be diagonalized.

Set $X = \begin{bmatrix}1 & 4 & \\ & 1 & \\ & & 1\end{bmatrix}\begin{bmatrix}1 & & \\ & 1 & 6 \\ & & 1\end{bmatrix}\begin{bmatrix}1 & & -\frac{19}{2} \\ & 1 & \\ & & 1\end{bmatrix} = \begin{bmatrix}1 & 4 & \frac{29}{2} \\ & 1 & 6 \\ & 1 & \end{bmatrix}$. Then the process above shows that $X^{-1}AX = \begin{bmatrix}1 & & \\ & 2 & \\ & & 3\end{bmatrix}$, so $A = X\begin{bmatrix}1 & & \\ & 2 & \\ & & 3\end{bmatrix}X^{-1}$. In particular, A has eigenvalues $1, 2, 3$, and the corresponding eigenspaces are spanned by eigenvectors that are corresponding columns of X, respectively. (Note that since $m_g = m_a = 1$, all eigenspaces are one-dimensional.)

In contrast, recall that $\begin{bmatrix}1 & 1 & \\ & 1 & 1 \\ & & 1\end{bmatrix}$ cannot be diagonalized. If we try shearing, we would typically have $\begin{bmatrix}1 & k & \\ & 1 & \\ & & 1\end{bmatrix}\begin{bmatrix}1 & 1 & \\ & 1 & 1 \\ & & 1\end{bmatrix}\begin{bmatrix}1 & -k & \\ & 1 & \\ & & 1\end{bmatrix} = \begin{bmatrix}1 & 1 & k \\ & 1 & 1 \\ & & 1\end{bmatrix}$. There is no way to kill the $(1,2)$ entry, and the cancellation is precise due to the (algebraically) repeated eigenvalues.

This gives a computational intuition about why algebraically repeated eigenvalues are crucial for defectiveness. Intuitively, for a triangular matrix, to kill the (i,j) entry using row/column operations, we usually need the (i,i) eigenvalue and (j,j) eigenvalue to be different. ☺

Now, let us combine the two multiplicities and have some applications.

Example 8.7.19. Consider again $\begin{bmatrix}1+a_1b_1 & a_1b_2 & \dots & a_1b_n \\ a_2b_1 & 1+a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_nb_1 & a_nb_2 & \dots & 1+a_nb_n\end{bmatrix}$. What is its determinant?

Let us do this by finding the eigenvalues. Then it is enough to find the eigenvalues of $A = \begin{bmatrix}a_1b_1 & a_1b_2 & \dots & a_1b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n\end{bmatrix}$. Note that this matrix has rank at most 1, so $\dim \mathrm{Ker}(A)$ is at least $n-1$. In particular, 0 is an eigenvalue of A with geometric multiplicity at least $n-1$. This in turn implies that 0 also has algebraic multiplicity at least $n-1$.

So, since A has n eigenvalues in total (counting algebraic multiplicity), and we already know that $n-1$ of them are all zero. We just need to find the last one. But since $\mathrm{trace}(A) = \sum a_ib_i$, we see that the last one is simply $\sum a_ib_i$. So A has eigenvalue $0, \dots, 0, \sum a_ib_i$.

Now our goal is to find $\det(I+A)$. Note that $A+I$ has eigenvalues $1, \dots, 1, 1+\sum a_ib_i$, so $\det(I+A) = 1 + \sum a_ib_i$. Done. ☺

Example 8.7.20. Consider again $\begin{bmatrix}1+a_1+b_1 & a_1+b_2 & \dots & a_1+b_n \\ a_2+b_1 & 1+a_2+b_2 & \dots & a_2+b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n+b_1 & a_n+b_2 & \dots & 1+a_n+b_n\end{bmatrix}$. What is its determinant?

Let us do this by finding the eigenvalues. Then it is enough to find the eigenvalues of $A = \begin{bmatrix} a_1+b_1 & a_1+b_2 & \dots & a_1+b_n \\ a_2+b_1 & a_2+b_2 & \dots & a_2+b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n+b_1 & a_n+b_2 & \dots & a_n+b_n \end{bmatrix}$
Note that this matrix has rank at most 2, so $\dim \mathrm{Ker}(A)$ is at least $n-2$. In particular, 0 is an eigenvalue of A with algebraic multiplicity at least $n-2$.

So, since A has n eigenvalues in total (counting algebraic multiplicity), and we already know that $n-2$ of them are all zero. We just need to find the last two. Let them be λ_1, λ_2. How to find them?

From trace we easily have $\lambda_1+\lambda_2 = \mathrm{trace}(A) = \sum a_i + \sum b_i$. This corresponds to the coefficient of the x^{n-1} term in the characteristic polynomial. However, let us look at the coefficient of the x^{n-2} terms. This gives $\sum_{i<j} \lambda_i\lambda_j = S_2$, where S_2 is the sum of all 2×2 principal minors of A. Note that since all but two eigenvalues are zero, the left hand side is in fact simply $\lambda_1\lambda_2$, the product of the only two non-zero eigenvalues. And the right hand side is $\sum_{i<j} \det \begin{bmatrix} a_i+b_i & a_i+b_j \\ a_j+b_i & a_j+b_j \end{bmatrix} = \sum_{i<j}[(a_i+b_i)(a_j+b_j) - (a_i+b_j)(a_j+b_i)] = \sum_{i<j}(a_ib_j + b_ia_j - a_ib_i - a_jb_j)$.

Now our goal is to find $\det(I+A)$. Note that $A+I$ has eigenvalues $1,\dots,1,1+\lambda_1,1+\lambda_2$, so $\det(I+A) = (1+\lambda_1)(1+\lambda_2) = 1+(\lambda_1+\lambda_2)+\lambda_1\lambda_2 = 1+\sum a_i + \sum b_i + \sum_{i<j}(a_ib_j+b_ia_j-a_ib_i-a_jb_j)$. Done.

For aethetic purpose, one might want to further simplify by $\sum_{i<j}(a_ib_j+b_ia_j-a_ib_i-a_jb_j) = \frac{1}{2}\sum_{i,j}(a_ib_j + b_ia_j - a_ib_i - a_jb_j)$, since the cases of $i<j$ and $i>j$ are symmetric, and the terms when $i=j$ are all zero. Then this further simplify to $\frac{1}{2}\sum_{i,j}(a_ib_j+b_ia_j-a_ib_i-a_jb_j) = \frac{1}{2}((\sum a_i)(\sum b_i) + (\sum a_i)(\sum b_i) - n(\sum a_ib_i) - n(\sum a_ib_i)) = (\sum a_i)(\sum b_i) - n\sum a_ib_i$.

Then we have $\det(I+A) = 1+\sum a_i + \sum b_i + (\sum a_i)(\sum b_i) - n\sum a_ib_i$. ☺

8.8 Limit and Conquer

Suppose A is diagonalizable, say $A = XDX^{-1}$. Then we have a very nice formula $A^k = XD^kX^{-1}$ and more generally $p(A) = Xp(D)X^{-1}$, which are all easy to calculate. What if A is NOT diagonalizable? It turned out that we can still calculate $p(A)$ with some formula. Except that derivatives are now involved.

Example 8.8.1. For a diagonal matrix $D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$, then we have $D^k = \begin{bmatrix} d_1^k & & \\ & \ddots & \\ & & d_n^k \end{bmatrix}$. Therefore, for any polynomial $p(x)$, note that $p(D)$ is a linear combination of powers of D. Therefore we can easily verify that in fact $p(D) = \begin{bmatrix} p(d_1) & & \\ & \ddots & \\ & & p(d_n) \end{bmatrix}$.

Now consider a non-diagonalizable matrix, a Jordan block $J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$. Even though this matrix is NOT diagonalizable, consider $J_t = \begin{bmatrix} 2 & 1 \\ 0 & 2+t \end{bmatrix}$. Now for $t \neq 0$, then all eigenvalues of J_t would be simple. So J_t is diagonalizable for $t \neq 0$. In particular, even though J is NOT diagonalizable itself, it is a limit of diagonal matrices.

Now for $t \neq 0$, we have $J_t = \begin{bmatrix} 1 & \frac{1}{t} \\ & 1 \end{bmatrix} \begin{bmatrix} 2 & \\ & 2+t \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{t} \\ & 1 \end{bmatrix}$. As a result, we see that

$$p(J_t) = \begin{bmatrix} 1 & \frac{1}{t} \\ & 1 \end{bmatrix} \begin{bmatrix} p(2) & \\ & p(2+t) \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{t} \\ & 1 \end{bmatrix} = \begin{bmatrix} p(2) & \frac{p(2+t)-p(t)}{t} \\ & p(2+t) \end{bmatrix}.$$

Hey! That looks like a derivative formula.

Now take limit $t \to 0$, we see that $p(J) = \begin{bmatrix} p(2) & p'(2) \\ & p(2) \end{bmatrix}$. This is true for all polynomial $p(x)$. (Note that $\lim p(J_t) = p(\lim J_t) = p(J)$ because p is continuous. Algebraically, continuity means "it commute with the limit operator".) ☺

Remark 8.8.2. *In fact, the argument above extends to all analytic functions. (An analytic function is a function that equal to its own Taylor expansion.)*

Suppose $f(x)$ is a function with Taylor expansion. Then $f(x)$ is a limit of polynomial functions. Hence $f(D)$ for $D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$ is $f(D) = \begin{bmatrix} f(d_1) & & \\ & \ddots & \\ & & f(d_n) \end{bmatrix}$. This gives us a way to define $f(A)$ in general. If $A = XDX^{-1}$, we simply define $f(A) = Xf(D)X^{-1}$, since the X, X^{-1} portion are simple changing the basis.

One can then use the same proof to see that $f(\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}) = \begin{bmatrix} f(2) & f'(2) \\ 0 & f(2) \end{bmatrix}$. For example, $e^{\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}} = \begin{bmatrix} e^2 & e^2 \\ 0 & e^2 \end{bmatrix}$.

This idea of proving easy cases, and then extend by continuity is a very standard strategy, and in fact very useful everywhere. We are doing linear algebra, so everything is linear or at worst polynomial. So everything is continuous. This makes taking limit very easy to impliment. Here are two optional lemmas you might be interested in. The results is more important than the proof. The proof is a standard mathematical technique about tiny perturbations.

Lemma 8.8.3. *For any square matrix A, it is the limit of a sequence of matrices A_n with distinct eigenvalues. (In particular, all A_n are diagonalizable since all eigenvalues are simple.) We may also require that all A_n are invertible.*

Proof. Consider the Schur decomposition $A = QUQ^{-1}$. Say $U = \begin{bmatrix} \lambda_1 & * & * \\ & \ddots & * \\ & & \lambda_n \end{bmatrix}$. We aim to perturb the diagonal entries of U a bit, to get matrices with distinct eigenvalues. Let $A_t = QU_tQ^{-1}$ where $U_t = \begin{bmatrix} \lambda_1 + t & * & * \\ & \ddots & * \\ & & \lambda_n + nt \end{bmatrix}$. Obviously $\lim_{t\to 0} A_t = A$. We only need to show that for small enough non-zero t, all A_t have distinct eigenvalues.

Let g be the smallest non-zero gap between eigenvalues of A. (E.g., if A has eigenvalues $1, 1, 4, 6, 6, 6$, then $g = 2$.) I claim that for all $0 < |t| < \frac{g}{2n}$, A_t has distinct eigenvalues.

Indeed, eigenvalues of A_t must be $\lambda_1 + t, \dots, \lambda_n + nt$. For any $i \neq j$, if $\lambda_i = \lambda_j$, then $\lambda_i + it \neq \lambda_j + jt$ due to $|t| > 0$. If $\lambda_i \neq \lambda_j$, then the gap $|\lambda_i - \lambda_j|$ is non-zero and therefore at least g. So we have

$$|(\lambda_i + it) - (\lambda_j + jt)| \geq |\lambda_i - \lambda_j| - i|t| - j|t| \geq g - i|t| - j|t| > 2nt - nt - nt = 0.$$

Hence $\lambda_i + it \neq \lambda_j + jt$ as well. Either way, A_t has distinct eigenvalues and thus is diagonalizable.

(Intuitively, if g is smallest gap, and $2nt < g$, then for all λ_i, λ_j with a gap, then the total perturbation is $it + jt \leq 2nt < g$, which is not enough to fill in this gap.)

If you want all A_t to be invertible as well, then you can again let g be the smallest non-zero gap, both between eigenvalues of A and between non-zero eigenvalues of A and zero. Then the same proof works in the same way. ☐

So, sometimes to prove some generic theorem about continuous things, we may simply assume that our matrices are diagonalizable or invertible or both, and prove the special case. Then we take limit, and we would have obtained all cases.

As an application, consider the following proof.

Theorem 8.8.4. *Say $n \geq k$. For any $n \times k$ matrix A and $k \times n$ matrix B, then $p_{AB}(x) = x^{n-k} p_{BA}(x)$. In particular, AB and BA have the same eigenvalues and the same algebraic multiplicity, with the exception of the eigenvalue* 0. *AB has $n-k$ more zeros as eigenvalues than BA.*

Note that $\det(I + AB) = \det(I + BA)$ is a very simple corrolary of the fact here. If we add identity, then we see that $I + AB$ has $n - k$ more ones as eigenvalues than $I + BA$, and otherwise they have identical eigenvalues. Taking product, and we are done.

Proof. We prove the theorem by first proving it for invertible A, then proving it for square A, B, and then prove the general case when A, B are rectangular.

A ***is invertible:*** If A is invertible, then $A(BA)A^{-1} = AB$. So AB and BA are similar, and we are done.

A ***is NOT invertible:*** Consider $A + tI$. Since A has finitely many eigenvalues, by choosing really tiny t, we can make sure that eigenvalues of $A + tI$ are ALL non-zero. So A is the limit of invertible matrices A_t. Now since $A_t B$ and BA_t have identical characteristic polynomial, by taking limit, AB and BA will have the same characteristic polynomial.

A, B ***are NOT square:*** We add zero columns to A and obtain $A' = \begin{bmatrix} A & O \end{bmatrix}$, a square matrix, and add zero rows to B and obtain B', a square matrix. Note that $A'B' = AB$ while $B'A' = \begin{bmatrix} BA & \\ & O \end{bmatrix}$.

Since $A'B'$ and $B'A'$ have the same characteristic polynomial, we see that $p_{AB}(x) = x^{n-k} p_{BA}(x)$. □

As you can see, each step is trivial. This is a very fundamental relation between AB and BA, and in turn, between $I + AB$ and $I + BA$. Everything that we've done previously, like $\mathrm{trace}(AB) = \mathrm{trace}(BA)$ or $\det(I + AB) = \det(I + BA)$ are all corollaries of this.

We now end this section with the all powerful Cayley-Hamilton theorem.

Proposition 8.8.5. *The process of sending $n \times n$ matrices A to the complex number $p_A(A)$ is continuous.*

Proof. The coefficients of $p_A(x)$ are $(-1)^k S_k$, which is some polynomials in entries of A. So all coefficients of $p_A(x)$ depends on A continuously. So in the calculation of $p_A(A)$, we are adding or multiplying things that all depends on A continuously. Therefore this is a continuous process. □

Theorem 8.8.6 (Cayley-Hamilton)**.** *We have $p_A(A) = O$ for any square matrix A.*

Proof. If A is diagonalizable, then $A = XDX^{-1}$ where the diagonal entries of D are the eigenvalues. Then $p_A(A) = Xp_A(D)X^{-1}$. However, $p_A(D)$ simply applies p_A to the eigenvalues, which are roots of p_A. So $p_A(D) = O$. So $p_A(A) = O$.

Now suppose A is NOT diagonalizable. Pick a sequence A_n of diagonalizable matrices A_n such that $A = \lim A_t$.

So $p_A(A) = \lim p_{A_n}(A_n) = \lim O = O$. □

There are many theoretical consequence of Cayley-Hamilton. Here is one:

Corollary 8.8.7. *If A is invertible, then there is a polynomial $p(x)$ such that $A^{-1} = p(A)$.*

Proof. Consider $p_A(x)$. Since $\det(A) \neq 0$, the constant term of $p_A(x)$ is non-zero. So $p_A(x) = xq(A) + c$ for some $c \neq 0$ and some polynomial $q(x)$.

Now since $p_A(A) = O$, we see that $Aq(A) + cI = O$. Rearrange terms, and we have $A^{-1} = -\frac{1}{c} q(A)$. Done. □

Here is an even more powerful version.

Corollary 8.8.8. *If $p(A) = O$ for some polynomial $p(x)$ with degree d, then for any polynomial $q(x)$, $q(A) = r(A)$ for some polynomial $r(x)$ with degree strictly less than d.*

Proof. For any polynomial $q(x)$, perform polynomial division, and we have $q(x) = a(x)p(x) + r(x)$ for some polynomial $a(x)$ and remainder $r(x)$. Hence $r(x)$ has degree strictly less than the degree of $p(x)$, and $q(A) = a(A)p(A) + r(A) = r(A)$. □

Since we always have $p_A(A) = O$, we see that for any $n \times n$ matrix A, we only need to consider polynomials of A with degree at most $n - 1$. No higher degree polynomials of A is ever needed.

Example 8.8.9. Suppose $A^2 = A$, show that $I + A$ is invertible and find its inverse.

How to do this? Well, since eigenvalues of A must satisfy $\lambda^2 = \lambda$, we see that $\lambda = 0, 1$. So eigenvalues of A are 0 or 1. Therefore, eigenvalues of $I + A$ are 1 and 2, and in particular $I + A$ is invertible.

Now inverse of $I + A$ must be some polynomial of $I + A$, which is in term some polynomial of A. However, $A^2 = A$, so any polynomial of A must be $a_0 I + a_1 A$.

If $a_0 I + a_1 A$ is the inverse, then we have $I = (I + A)(a_0 I + a_1 A) = a_0 I + (a_0 + 2a_1)A$. So we want $a_0 = 1$ and $a_0 + 2a_1 = 0$, which gives $a_0 = 1$ and $a_1 = -\frac{1}{2}$. So we have $(I + A)^{-1} = I - \frac{1}{2}A$. ☺

Here is a slightly more complicated example.

Example 8.8.10. Suppose $A^3 + A^2 + A + I = O$, show that $2A + I$ is invertible and find its inverse.

How to do this? Well, since eigenvalues of A must satisfy $\lambda^3 + \lambda^2 + \lambda + 1 = 0$, we see that $\lambda = -1, \mathrm{i}, -\mathrm{i}$. So eigenvalues of A are $-1, \mathrm{i}, -\mathrm{i}$. Therefore, eigenvalues of $2A + I$ are $-1, 1 + 2\mathrm{i}, 1 - 2\mathrm{i}$, and in particular $2A + I$ is invertible.

Now inverse of $2A + I$ must be some polynomial of $2A + I$, which is in term some polynomial of A. However, $A^3 + A^2 + A + I = O$, so any polynomial of A must be $a_0 I + a_1 A + a_2 A^2$.

If $a_0 I + a_1 A + a_2 A^2$ is the inverse, then we have

$$I = (2A+I)(a_0 I + a_1 A + a_2 A^2) = a_0 I + (2a_0 + a_1)A + (2a_1 + a_2)A^2 + (2a_2)A^3 = (a_0 - 2a_2)I + (2a_0 + a_1 - 2a_2)A + (2a_1 - a_2)A^2.$$

So we want $a_0 - 2a_2 = 1$, $2a_0 + a_1 - 2a_2 = 0$ and $2a_1 - a_2 = 0$, which gives $a_0 = -\frac{3}{5}$, $a_1 = -\frac{2}{5}$ and $a_2 = -\frac{4}{5}$. So we have $(2A + I)^{-1} = -\frac{3}{5}I - \frac{2}{5}A - \frac{4}{5}A^2$. ☺

So you always have only finitely many coefficients to consider, and can always attempt at something like the example above. In fact, for any function f with converging Taylor expansion, then $f(A) = r(A)$ for some polynomial $r(x)$ with degree strictly less than d.

Example 8.8.11. For $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, we have $\mathrm{e}^A = \begin{bmatrix} e^2 & e^2 \\ 0 & e^2 \end{bmatrix} = \mathrm{e}^2 A - \mathrm{e}^2 I$. ☺

8.9 (Optional) Classification of 2×2 real matrices

Since we were studying real matrices, here are the most important properties of them.

Proposition 8.9.1. *If A is a real square matrix, then any complex eigenvalues would come in conjugate pairs, with same algebraic and geometric multiplicities, and their corresponding eigenvectors are also in conjugate pairs.*

Proof. If $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then $\overline{A\boldsymbol{v}} = \overline{\lambda\boldsymbol{v}}$, and hence $A\overline{\boldsymbol{v}} = \overline{\lambda}\overline{\boldsymbol{v}}$. The rest is easy verifications along similar lines. □

Proposition 8.9.2. *If a real square matrix A has a real eigenvalue, then we can find real eigenvectors for the eigenvalue. Furthermore, real eigenvectors would span the (complex) eigenspace of λ.*

Proof. Note that if $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then $\overline{A\boldsymbol{v}} = \overline{\lambda\boldsymbol{v}}$. Since both A and λ are real, we see that $A\overline{\boldsymbol{v}} = \lambda\overline{\boldsymbol{v}}$. As a result, both $\boldsymbol{v}, \overline{\boldsymbol{v}}$ are eigenvectors for λ. Then $\frac{1}{2}(\boldsymbol{v} + \overline{\boldsymbol{v}})$, $\frac{1}{2\mathrm{i}}(\boldsymbol{v} - \overline{\boldsymbol{v}})$ are real eignevectors for λ and their span would contain $\boldsymbol{v}$.

In particular, not only λ has real eigenvectors, we also see that real eigenvectors are enough to span the eigenspace. □

Another interesting but less important fact happen when n is odd.

Proposition 8.9.3. *If A is an $n \times n$ real square matrix, and n is odd, then A must have a real eigenvalue (and hence with corresponding real eigenvectors).*

Proof. Note that $p_A(x)$ is a polynomial of odd degree. In particular, it is easy to see that $\lim_{x\to\infty} p_A(x) = \infty$ while $\lim_{x\to-\infty} p_A(x) = -\infty$. By intermediate value theorem, somewhere we must be able to find a real λ with $p_A(\lambda) = 0$. □

Corollary 8.9.4. *Any 3-dimensional rotation must have an axis of rotation. (In fact, this is true for all odd-dimensional rotations.)*

Proof. Let this rotation be R, which would be an orthogonal matrix with determinant 1. Then suppose it has an eigenvalue λ. Say $R\boldsymbol{v} = \lambda\boldsymbol{v}$. However, since R is an orthogonal matrix, we have $\|\boldsymbol{v}\| = \|R\boldsymbol{v}\| = |\lambda|\|\boldsymbol{v}\|$, so we see that all eigenvalues must be complex numbers with absolute value 1.

In particular, eigenvalues of R are conjugate pairs of non-real numbers with absolute value 1, and some -1 and some 1. Now since R is a rotation, it preserves orientation, so the product of all its eigenvalues are 1. But the conjugate pairs of complex eigenvalues must multiply to 1, so we see that R must have even numbers of eigenvalue -1.

Now let us count. R has n eigenvalues in total, and n is odd. R must have even numbers of non-real eigenvalues, and even numbers of -1, so it seems that it must have an odd number (hence non-zero number) of eigenvalues 1. In particular, R has eigenvalue 1. And the corresponding real eigenvector must be an axis of rotation. □

Here let us attempt to classify the behavior of all 2×2 real matrices. We finally have enough tools to do so. Throughout this subsection, suppose A is a 2×2.

Example 8.9.5. Suppose A is NOT invertible. If A has rank 0, then obviously the only case is $A = O$. Now suppose A has rank 1.

Then the eigenvalues of A are $0, \lambda$. Suppose $\lambda \neq 0$. Then A has distinct eigenvalues and thus diagonalizable. So $A = XDX^{-1}$ where $D = \begin{bmatrix} \lambda & \\ & 0 \end{bmatrix}$. In particular, $(\frac{1}{\lambda}A)^2 = \frac{1}{\lambda}A$. So A is a multiple of some oblique or orthogonal projection.

As a side note, $\operatorname{trace}(A) = 0 + \lambda = \lambda \neq 0$ in this case.

If $\lambda = 0$, yet $A \neq O$, so we see that A is NOT diagonalizable. By Schur decomposition, we see that $A = QTQ^{-1}$ for some orthogonal matrix Q, where $T = \begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix} = a\boldsymbol{e}_1\boldsymbol{e}_2^{\mathrm{T}}$ for some non-zero a. Also note that $\begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & \\ & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \frac{1}{a} & \\ & 1 \end{bmatrix}$. So all these matrices are similar to $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. The matrix $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ does this: it sends the y-axis to the x-axis, and it sends the x-axis to the origin. It is also funny that $A \neq O$ but $A^2 = O$.

As a side note, $\operatorname{trace}(A) = 0$ in this case.

In summary we have the following classification:

1. If A has rank zero, then $A = O$.
2. If A has rank one and $\operatorname{trace}(A) \neq 0$, then A is a multiple of oblique or orthogonal projection.
3. If A has rank one and $\operatorname{trace}(A) = 0$, then A is similar to the Jordan block $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.

☺

Now we are ready to study the invertible matrices. Note that in this case, A will have two non-zero eigenvectors λ_1, λ_2. And furthermore, we have $\operatorname{trace}(A) = \lambda_1 + \lambda_2$ and $\det(A) = \lambda_1\lambda_2$. In particular, the trace and determinant of A completely determines the eigenvalues!

For the moment, let us assume that $\det(A) = 1$. So A would preserve area. First we study rotation like behavior, which would obviously preserve area.

Example 8.9.6 (Elliptic rotations)**.** If A has non-real eigenvalues, then note that these eigenvalues come in conjugate pairs. So the two eigenvalues must be $\lambda, \overline{\lambda}$. Also note that, by premises, $\lambda\overline{\lambda} = 1$, so λ is a unit complex number.

Now by polar decomposition, we must have $\lambda = \mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ for some θ. Furthermore, since this should NOT be purely real, we have $|\cos\theta| < 1$. For the record, we can see that in this case $|\mathrm{trace}(A)| = |2\cos\theta| < 2$

Now since our eigenvalues are non-real, they are conjugate pairs, and thus they are distinct, so A is diagonalizable. So A is similar to $\begin{bmatrix} \mathrm{e}^{\mathrm{i}\theta} & \\ & \mathrm{e}^{-\mathrm{i}\theta} \end{bmatrix}$. However, the matrix $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is also similar to $\begin{bmatrix} \mathrm{e}^{\mathrm{i}\theta} & \\ & \mathrm{e}^{-\mathrm{i}\theta} \end{bmatrix}$. Hence A is similar to R_θ.

Note that R_θ is a simple rotation. You can think of the plane $\mathbb{R}^2$ as made of concentric circles around the origin, and R_θ simple rotates all these circles by the same amount.

If we change basis and get A, then A would be an "elliptic rotation". You can think of the plane $\mathbb{R}^2$ as made of concentric ellipses around the origin, and A simple rotates all these ellipses simultaneously.

What is the speed of this rotation? It turns out that different points on the ellipse will be rotated with different speed, depending on how close they are to the origin. It is actually a version of Kepler's second law of planetary motion, where the area swept in unit time should be the same. Note the fact that $\det(A) = 1$, so A should preserve area. In particular, the triangle made by $\boldsymbol{v}, A\boldsymbol{v}$ and the triangle made by $A\boldsymbol{v}, A^2\boldsymbol{v}$ should have the same area, and so on. This can help us determine the speed of rotation induced by A. ☺

We are now left with the cases where the eigenvalues are both real. Suppose they are distinct.

Example 8.9.7. Suppose the eigenvalues are distinct. Since their product is 1, they are $\lambda, \frac{1}{\lambda}$ where $\lambda \neq \pm 1$. Note that in this case, we necessarily have $|\mathrm{trace}(A)| > 2$. Let us assume in this case that $\lambda > 0$, so both eigenvalues are positive.

Since the two eigenvalues are distinct, $A = XDX^{-1}$ with $D = \begin{bmatrix} \lambda & \\ & \frac{1}{\lambda} \end{bmatrix}$. So A after a change of basis becomes D. What is the geometric behavior of this D?

Note that $D\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda x \\ \frac{1}{\lambda} y \end{bmatrix}$. In particular, it would fix the product of the coordinates! As you can imagine, it will slide points along hyperbolas such as curves $xy = k$ in $\mathbb{R}^2$. These hyperbolas are "orbits" of D.

After a change of basis, A would become a "hyperbolic rotation". Let $\boldsymbol{v}, \boldsymbol{w}$ be a basis of $\mathbb{R}^2$ made of eigenvectors of A, and consider all hyperbolas with these two lines as asymptotes. Then A simply slide each point along the hyperbola it lies on.

What is the speed of this rotation? Again, since $\det(A) = 1$, Kepler's second law of planetary motion applies. The triangle made by $\boldsymbol{v}, A\boldsymbol{v}$ and the triangle made by $A\boldsymbol{v}, A^2\boldsymbol{v}$ should have the same area, and so on. This can help us determine the speed of rotation induced by A.

What if we started with $\lambda < 0$? Then we can apply the analysis above to $-A$. Hence A is the negation of a hyperbolic rotation. You may think of it as $A = -B$ where B is the hyperbolic rotation. So A would perform a hyperbolic rotation and then negate the outcome vector. ☺

Now we are left with the case where the two eigenvalues are real and identical. We have obvious candidates like $\pm I$. However, there is also another special linear map of this case that preserve area, i.e., shearing.

Example 8.9.8. The two eigenvalues must be $\lambda_1 = \lambda_2 = 1$ or $\lambda_1 = \lambda_2 = -1$. Note that in this case, we have $|\mathrm{trace}(A)| = 2$.

If A is diagonalizable with identical eigenvalues, then $A = XDX^{-1}$ where $D = \pm I$. But then we would have $A = \pm I$. So these are the only possibility in this case.

Suppose A is NOT diagonalizable. Then $A = QTQ^{-1}$ for some triangular T. Suppose $\lambda_1 = \lambda_2 = 1$, then we have $T = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, so A after a change of basis is a shearing transformation.

What if $\lambda_1 = \lambda_2 = -1$? Then again, we simply apply the arguments above to $-A$. So $A = -B$ where B is a shearing transformation. ☺

We have now covered all cases. Let us make a summary.

Proposition 8.9.9. *If* $\det(A) = 1$*, then the geometric behavior of* A *is exactly one of the following, characterized by trace.*

1. *(*$\mathrm{trace}(A) > 2$*) Sliding points along hyperbolas with common asymptotes.*
2. *(*$\mathrm{trace}(A) = 2$*) Either* $A = I$ *or* A *is a shearing along some direction.*
3. *(*$|\mathrm{trace}(A)| < 2$*) Rotations around concentric ellipses of the same eccentricity.*
4. *(*$\mathrm{trace}(A) = -2$*) Either* $A = -I$ *or* A *is a shearing along some direction then negation.*
5. *(*$\mathrm{trace}(A) < -2$*) Sliding points along hyperbolas with common asymptotes, and then negation.*

Note that the case $|\mathrm{trace}(A)| < 2$ is sometimes called elliptic, and the case $|\mathrm{trace}(A)| > 2$ is sometimes called hyperbolic, for obvious reason. A less intuitive name is that the case of $\mathrm{trace}(A) = \pm 2$ is sometimes called parabolic. (This is because the parabola is an intermediate state between the ellipses and the hyperbolas.)

Now what if $\det(A) \neq 1$? We have two scenarios. Let us focus now on the case of positive determinant.

Suppose $\det(A) > 0$. Then let $t = \sqrt{\det(A)}$, we have $\det(\frac{1}{t}A) = 1$. So $\frac{1}{t}A$ is one of the cases above. In particular, $A = tB$ for some B in one of the cases above. A is basically B and then a uniform scaling.

Example 8.9.10. If B is an elliptic rotation and $t > 1$, then A will "spiral out". For each application of A, you shall rotate, and then uniformly scale outward. Repeatedly apply A, and you will see all points spiraling away from the origin. Reversely, if $0 < t < 1$ in this case, then you shall see a spiral in process instead. Note that this is the case of non-real eigenvalues with $\mathrm{trace}(A)^2 < 4\det(A)$ ☺

Example 8.9.11. Suppose B is a hyperbolic sliding (i.e., with positive eigenvalues). Note that this corresponds to the case where A is diagonalizable with distinct eigenvalues. Suppose that both eigenvalues are positive.

Then we have $A = XDX^{-1}$ with $D = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$. If we have $0 < \lambda_1, \lambda_2 < 1$, then A will shrink everything towards the origin. The origin is like a "sink" where everything would flows towards. If $\lambda_1, \lambda_2 > 1$, then A will now push everything away from the origin. The origin is like a "source" where everything pours outwards.

If one eigenvalue is larger than 1, while the other is less than 1, then the picture is similar to a hyperbolic sliding, even though the orbits are not necessarily hyperbolas. The origin is called a "saddle" point, where points would flow in through one eigen-direction, and yet pushed away along another eigen-direction.

The intermediate state, where one eigenvalue is exactly 1, means we have a line of fixed points, and we stretch or shrink along the other direction, depending on whether the other eigenvalue is larger than or less than 1. The geometric behaviors are a bit like the process of opening doors or closing doors.

Finally, the case when A has negative eigenvalues are simply above process and then negation. ☺

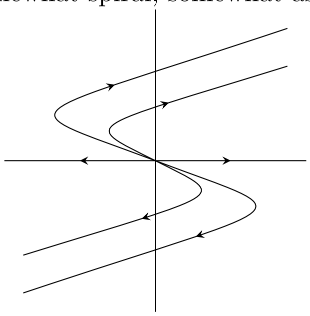
We are again left with the cases of identical real eigenvalues. If $B = \pm I$, then A is also a multiple of identity, and this case is trivial. So let us now look at the case when B is a shearing, or alternatively, when A is similar to some $\begin{bmatrix} \lambda & 1 \\ & \lambda \end{bmatrix}$

Example 8.9.12. Consider the case $A = \begin{bmatrix} \lambda & 1 \\ & \lambda \end{bmatrix}$ where $\lambda = 100$. Then we can think of A as two process: first we perform a tiny shearing $\begin{bmatrix} 1 & \frac{1}{100} \\ & 1 \end{bmatrix}$, and then we scale everything by 100.

First imagine that the imput is a vector in a direction above but close to the negative x-axis. Then since the y-component is tiny, shearing should effect very little of it. Yet the scaling should effect it greatly and move it towards the upper left. As a result, the vector would move towards its upper left.

However, also because $A\begin{bmatrix}x\\y\end{bmatrix}$ is parallel to $\begin{bmatrix}x+\frac{1}{100}y\\y\end{bmatrix}$, the vector is slowly but surely turning clockwise. Eventually, once the vector is turned past the y-axis, then both the shearing and the scaling would push it to the right, and the vector would just go faster and faster right ever since.

This phenomena is called a degenerate source. All points are pushed away, but the direction of push is somewhat spiral, somewhat asymptotic.



In general, suppose A has identical positive eigenvalues but is NOT diagonalizable. When $\det(A) > 1$, its behavior would be some degenerate source as shown. When $\det(A) < 1$, then it would be some degenerate sink, which spirals in. Finally, when $\det(A)$, then this is just a regular shearing.

If A has negative eigenvalues, then A is the above process and then negation. ☺

We have one last, case, the case when $\det(A) < 0$. Note that in this case, $p_A(x)$ is a degree two polynomial with negative constant coefficient, and hence it always have distinct real roots, one positive and one negative. In particular, A MUST be diagonalizable.

Consider $\begin{bmatrix}\lambda_1 & \\ & \lambda_2\end{bmatrix}$ where $\lambda_1 > 0, \lambda_2 < 0$. This is essentially $\begin{bmatrix}1 & \\ & -1\end{bmatrix}$ and then we stretch or shrink a bit in each axis. So all such matrices must be (maybe oblique) reflection and then some stretches.

8.10 Linear Differential Equations

8.10.1 Differential Equations with only One Function

Example 8.10.1. What is the solution to $f'(x) = f(x)$? Consider the differential operator D, which is linear. We are trying to find $Df = f$, which would be eigenvectors of D for eigenvalue 1. ☺

As you can see, taking derivative is a linear operator. To study this operator D, obviously we need to understand its eigenvalues and eigenvectors. Well, guess what? We do.

Proposition 8.10.2. *Let D be the derivative operator. Then any complex number $\lambda \in \mathbb{C}$ is an eigenvalue of D with geometric multiplicity one. Its eigenspace is spanned by $\mathrm{e}^{\lambda x}$.*

Proof. This is a basic calculus class result. □

Before moving on, recall that if $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then it is very easy to verify that $p(A)\boldsymbol{v} = p(\lambda)\boldsymbol{v}$ for all polynomial $p(x)$.

Example 8.10.3. Consider the harmonic osciliator. Say your position is $f(t)$, then we know $mf''(t) = -kf(t)$. However, suppose that we also have frictions proportional to your speed. Then now we have $mf''(t) = -bf'(t) - kf(t)$. Then the differential equation is $mf'' + bf' + kf = 0$ for some positive constants m, b, k. What is the solution?

Let us assume $m = 1, b = 3, k = 2$. Then $f'' + 3f' + 2f = 0$. Let $p(x) = x^2 + 3x + 2$, then naturally $p(D)f = 0$. So we are trying to find eigenvectors for the eigenvalue 0.

Now $p(-1) = p(-2) = 0$. So since e^{-x} and e^{-2x} are eigenvectors of D for the eigenvalues -1 and -2 respectively, it turned out that they are both eigenvectors of $p(D)$ for the eigenvalue $p(-1) = 0 = p(-2)$.

I state without proof here that for any polynomial p of degree d, then any eigenspace of $p(D)$ is complex d dimensional. So the solution to our differential equation is the span of e^{-x} and e^{-2x}.

So our harmonic osciliator with friction in this case behaves like $a\mathrm{e}^{-t} + b\mathrm{e}^{-2t}$ and the limit as $t \to \infty$ is 0. So eventually we stopped moving at the origin. ☺

Proposition 8.10.4. *Let D be the derivative operator. For any polynomial p of degree d, then any eigenspace of $p(D)$ is d dimensional.*

Proof. Check out some odinary differential equation class for this proof. □

With this idea, you should be able to almost solve all differential equations that looks like $p(D)f = 0$. If you see $f''' + 6f'' + 11f' + 6f = 0$? NO PROBLEM. The roots for p are 1,2,3, so all solutions are linear combintions of $\mathrm{e}^x, \mathrm{e}^{2x}, \mathrm{e}^{3x}$. As long as all the roots of p are distinct, we can find all solutions to differential equations this way.

Let us check out another case. This allow us to handle complex eigenvalues with better grace.

Example 8.10.5. Now suppose we have no friction, so we have $f''+f = 0$. Then $p(D)f = 0$ for $p(x) = x^2+1$, and the roots for p are $\pm\mathrm{i}$. The two eigenvectors are $\mathrm{e}^{\mathrm{i}t}$ and $\mathrm{e}^{-\mathrm{i}t}$, and they are both eigenvectors for $p(D)$ for the eigenvalue 0. Any solution must look like $z\mathrm{e}^{\mathrm{i}t} + w\mathrm{e}^{-\mathrm{i}t}$ for some complex numbers z, w. But how can I find all real solutions?

Well, note that the (complex) span of $\mathrm{e}^{\mathrm{i}t}, \mathrm{e}^{-\mathrm{i}t}$ is the same as the (complex) span of $\cos t, \sin t$. So the answer is simply all real linear combinations of $\cos t, \sin t$.

Alternatively, note that $p(D)\mathrm{e}^{\mathrm{i}t} = 0$ means that $p(D)\cos t + \mathrm{i}p(D)\sin t = 0$, so $p(D)\cos t = 0$ and $p(D)\sin t = 0$.

Finally, note that the solutions are all periodic. Indeed, if there is no friction, then the osciliator would simply bounce forever, in a periodic manner. ☺

Techniques here are vaid, as long as $p(D)f = 0$ and the polynomial $p(x)$ have distinct roots. What if $p(x)$ has repeated roots?

Example 8.10.6. Consider any potential solution to $(D^2 - 2D + I)f = 0$. Since the roots of $p(x)$ are both 1, we see that e^x is an eigenvector with eigenvalue 0. However, since our polynomial has degree two, the solution space is suppose to be two dimensional. What could another basis vector be?

This could be $x\mathrm{e}^x$. You can feel free to verify that indeed $D^2 - 2D + I$ would kill this function. This is NOT an eigenvector, but a ***generalized eigenvector***, which we shall discuss next semester. ☺

8.10.2 (Optional) Eigenspace of $p(A)$ and $p(\frac{\mathrm{d}}{\mathrm{d}x})$

In last subsection, we claim that if p has degree n, then $\mathrm{Ker}(p(D))$ would have dimension n for the differential operator D. Why is that? Well, one side of the inequality is easy.

Lemma 8.10.7. $\dim \mathrm{Ker}(AB) \leq \dim \mathrm{Ker}(A) + \dim \mathrm{Ker}(B)$. *Here A, B can be any abstract linear transformation (maybe on infinite dimensional spaces) with finite dimensional kernels.*

Proof. Let B' be the restriction of the map B to the domain $\mathrm{Ker}(AB)$. Since B' is a restriction of B, we automatically have $\mathrm{Ker}(B') \subseteq \mathrm{Ker}(B)$. Also note that for any $\boldsymbol{v}$ in the domain of B', we have $\boldsymbol{v} \in \mathrm{Ker}(AB)$, so $AB\boldsymbol{v} = \boldsymbol{0}$ and hence $B\boldsymbol{v} \in \mathrm{Ker}(A)$. Since $B\boldsymbol{v} = B'\boldsymbol{v}$, we see that $\mathrm{Ran}(B') \subseteq \mathrm{Ker}(A)$.

Now B' has finite dimensional domain, and rank-nullity theorem statess that $\dim \mathrm{Ker}(AB) = \dim \mathrm{Ker}(B') + \dim \mathrm{Ran}(B') \leq \dim \mathrm{Ker}(B) + \dim \mathrm{Ker}(A)$. □

Corollary 8.10.8. $\dim \mathrm{Ker}(A_1 \dots A_k) \leq \sum \dim \mathrm{Ker}(A_i)$. *Here $A_1, \dots, A_k$ can be any abstract linear transformation (maybe on infinite dimensional spaces) with finite dimensional kernels.*

Proof. Mathematical induction. □

Corollary 8.10.9. *For any non-zero polynomial p,* $\dim \mathrm{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x}))$ *is at most the degree of p.*

Proof. Suppose $p(x) = \prod_{i=1}^n (x - \lambda_i)$. Now, $\operatorname{Ker}(\frac{\mathrm{d}}{\mathrm{d}x} - \lambda_i I)$ is the set of all functions f such that $f' = \lambda_i f$, and therefore it is $\{k\mathrm{e}^{\lambda_i x} : k \in \mathbb{R}\}$. So $\dim \operatorname{Ker}(\frac{\mathrm{d}}{\mathrm{d}x} - \lambda_i I) = 1$.

Now $\dim \operatorname{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x})) = \dim \operatorname{Ker}(\prod_{i=1}^n (\frac{\mathrm{d}}{\mathrm{d}x} - \lambda_i I)) \leq \sum_{i=1}^n \dim \operatorname{Ker}(\frac{\mathrm{d}}{\mathrm{d}x} - \lambda_i I) = n$, which is the degree of p. □

Now we wish to show that $\dim \operatorname{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x}))$ is exactly the degree of p. First let us do some easy cases.

Lemma 8.10.10. $\operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x})^n)$ *is made of polynomials of degree* $n-1$. *In particular, indeed we have* $\dim \operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x})^n) = n$.

Proof. $\operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x})^n)$ is made of functions whose n-th derivative is zero. Repeatedly integrate from the zero function, and we have the result. □

Now, suppose we want to investigate $\operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x} - kI)^n)$. How to study $(\frac{\mathrm{d}}{\mathrm{d}x} - kI)^n$? This is an iteration of $\frac{\mathrm{d}}{\mathrm{d}x} - kI$. So maybe we can "diagonalize" it some how?

Lemma 8.10.11. *Pick any non-zero* $g \in \operatorname{Ker}(\frac{\mathrm{d}}{\mathrm{d}x} - kI)$, *say* $g(x) = \mathrm{e}^{kx}$. *Let* M_g *be the map sending any function* h *to* gh. *Then* M_g *is invertible, and* $\frac{\mathrm{d}}{\mathrm{d}x} - kI = M_g \frac{\mathrm{d}}{\mathrm{d}x} M_g^{-1}$.

Proof. Since the function $g(x) = \mathrm{e}^{kx}$ is never zero, therefore $M_g^{-1}(h) = g^{-1}h$ is the desired inverse function.

Now by Leibniz rule (product rule for derivative), we have $(\frac{\mathrm{d}}{\mathrm{d}x} - kI)(gh) = (gh)' - k(gh) = g'h + gh' - kgh = gh' + h(g' - kg) = gh'$. So we have $(\frac{\mathrm{d}}{\mathrm{d}x} - kI)M_g = M_g \frac{\mathrm{d}}{\mathrm{d}x}$. In particular, $\frac{\mathrm{d}}{\mathrm{d}x} - kI = M_g \frac{\mathrm{d}}{\mathrm{d}x} M_g^{-1}$. In particular, we see that $(\frac{\mathrm{d}}{\mathrm{d}x} - kI)^n = M_g (\frac{\mathrm{d}}{\mathrm{d}x})^n M_g^{-1}$. □

Remark 8.10.12. *There is another intuition behind the formula* $\frac{\mathrm{d}}{\mathrm{d}x} - kI = M_g \frac{\mathrm{d}}{\mathrm{d}x} M_g^{-1}$.

For any eigenvector of $\frac{\mathrm{d}}{\mathrm{d}f}$, *say* $\mathrm{e}^{\lambda x}$ *for some* $\lambda \in \mathbb{C}$, *then* $M_g(\mathrm{e}^{\lambda x}) = \mathrm{e}^{(\lambda + k)x}$. *So the map* M_g *will send each eigenvector for* λ *to an eigenvector for* $\lambda + k$. *Consequently, if we use* M_g *as a change of basis, it is simply as if all eigenvalues of* $\frac{\mathrm{d}}{\mathrm{d}x}$ *is dropped by* k, *so* $\frac{\mathrm{d}}{\mathrm{d}x}$ *turns into* $\frac{\mathrm{d}}{\mathrm{d}x} - kI$.

Corollary 8.10.13. $f \in \operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x} - kI)^n)$ *if and only if* $f(x) = q(x)g(x) = q(x)\mathrm{e}^{kx}$ *for some polynomial* $q(x)$ *of degree at most* $n-1$. *In particular,* $\dim \operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x} - kI)^n) = n$.

Proof. Note that $(\frac{\mathrm{d}}{\mathrm{d}x} - kI)^n f = 0$ if and only if $M_g (\frac{\mathrm{d}}{\mathrm{d}x})^n M_g^{-1} f = 0$ if and only if $(\frac{\mathrm{d}}{\mathrm{d}x})^n M_g^{-1} f = 0$ if and only if $M_g^{-1} f$ is a polynomial of degree at most $n-1$. The rest is obvious. □

Now we aim to deal with the cases where $p(x)$ has several distinct roots. Here we need some basic polynoial results.

Definition 8.10.14. *For any two non-zero polynomials* $p(x)$ *and* $q(x)$, *we say* $g(x)$ *is the greatest common factor of them if* $g(x)$ *is a factor of both* $p(x)$ *and* $q(x)$, *and it has the largest degree possible, and its leading coefficient (i.e., the coefficeint for the largest degree term) is one.*

Lemma 8.10.15. *If non-zero polynomials* $p(x)$ *and* $q(x)$ *has greatest common factor* $g(x)$, *then we can find polynomials* $a(x)$ *and* $b(x)$ *such that* $a(x)p(x) + b(x)q(x) = g(x)$.

Proof. Search for the name "Euclidean algorithm". There is a highly similar statements for integers and their greatest common divisors.

For a complete proof, WLOG say $p(x)$ has degree larger than or equal to the degree of $q(x)$. We proceed by mathematical induction on the degree of q. If q has degree zero, then q is a non-zero constant, and the greatest common factor is simply $g(x) = 1$. So set $a(x) = 0$ and set $b(x) = q(x)^{-1}$ and we are done.

Now for generic $q(x)$ with degree at least 1, by polynomial division, $p(x) = q(x)h(x) + r(x)$ for some polynomial $h(x)$ and a polynomial $r(x)$ such that $\deg(r) < \deg(q)$.

Now, if $g(x)$ is the greatest common factor of $p(x)$ and $q(x)$, then $g(x)$ is also the greatest common factor of $p(x)$ and $r(x)$. So by induction hypothesis, we can find polynomials $a'(x), b'(x)$ such that $a'(x)p(x) + b'(x)r(x) = g(x)$.

Then $a'(x)p(x) + b'(x)(p(x) - q(x)h(x)) = g(x)$, and thus set $a(x) = a'(x) + b'(x)$, and set $b(x) = -b'(x)h(x)$, and we are done. □

Proposition 8.10.16. *If $p(x)$ and $q(x)$ has no common root, then* $\operatorname{Ker}(p(A)q(A)) = \operatorname{Ker}(p(A)) + \operatorname{Ker}(q(A))$ *and* $\operatorname{Ker}(p(A)) \cap \operatorname{Ker}(q(A)) = \{\mathbf{0}\}$ *for any abstract linear transformation A (over maybe infinite dimensional spaces).*

Proof. Since $p(x)$ and $q(x)$ has no common root, therefore we can find $a(x), b(x)$ such that $a(x)p(x) + b(x)q(x) = 1$. In particular, for any $\boldsymbol{v}$, we have $\boldsymbol{v} = a(A)p(A)\boldsymbol{v} + b(A)q(A)\boldsymbol{v}$.

Let us show that they have trivial intersection. Suppose $\boldsymbol{v} \in \operatorname{Ker}(p(A)) \cap \operatorname{Ker}(q(A))$. Note that again we have $\boldsymbol{v} = a(A)p(A)\boldsymbol{v} + b(A)q(A)\boldsymbol{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$. So we are done.

Now, if $p(A)q(A)\boldsymbol{v} = \mathbf{0}$, we see that $a(A)p(A)\boldsymbol{v} \in \operatorname{Ker}(q(A))$ and $b(A)q(A)\boldsymbol{v} \in \operatorname{Ker}(p(A))$. So $\boldsymbol{v} = a(A)p(A)\boldsymbol{v} + b(A)q(A)\boldsymbol{v} \in \operatorname{Ker}(p(A)) + \operatorname{Ker}(q(A))$. So $\operatorname{Ker}(p(A)q(A)) \subseteq \operatorname{Ker}(p(A)) + \operatorname{Ker}(q(A))$. On the other hand, obviously $q(A)\boldsymbol{v} = \mathbf{0}$ or $p(A)\boldsymbol{v} = \mathbf{0}$ would both imply $p(A)q(A)\boldsymbol{v} = \mathbf{0}$. So we have $\operatorname{Ker}(p(A)) + \operatorname{Ker}(q(A)) \subseteq \operatorname{Ker}(p(A)q(A))$. We have established that $\operatorname{Ker}(p(A)q(A)) = \operatorname{Ker}(p(A)) + \operatorname{Ker}(q(A))$. □

Theorem 8.10.17. *Suppose $p(x) = \prod(x-\lambda_i)^{m_i}$ is a factorization into distinct roots λ_i with multiplicity m_i. For any abstract linear transformation A (over maybe infinite dimensional spaces), let $V_i = \operatorname{Ker}((A-\lambda_i I)^{m_i})$. Then $V_1, \ldots, V_k$ are linearly independent, and $\operatorname{Ker}(p(A)) = \sum V_i$.*

Proof. Use last proposition and mathematical induction. □

Theorem 8.10.18. *For any non-zero polynomial p, $\dim \operatorname{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x}))$ is exactly the degree of p.*

Proof. Suppose $p(x) = \prod_i (x-\lambda_i)^{m_i}$ is a factorization into distinct roots λ_i with multiplicity m_i.

Let $V_i = \operatorname{Ker}((\frac{\mathrm{d}}{\mathrm{d}x} - \lambda_i I)^{m_i})$, then $V_1, \ldots, V_k$ are all linearly independent, and $\operatorname{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x})) = \sum V_i$.

So $\dim \operatorname{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x})) = \dim \sum V_i = \sum \dim V_i = \sum m_i = \deg(p)$. □

In fact, if $p(x) = \prod_i (x-\lambda_i)^{m_i}$ is a factorization into distinct roots λ_i with multiplicity m_i, then we can now have an exlicit basis for $\operatorname{Ker}(p(\frac{\mathrm{d}}{\mathrm{d}x}))$. It is the collection $\bigcup_i \{\mathrm{e}^{\lambda_i x}, x\mathrm{e}^{\lambda_i x}, \ldots, x^{m_i-1}\mathrm{e}^{\lambda_i x}\}$.

Let me give one last applications.

Theorem 8.10.19. *Fix an integrable function $h(x)$ and a real number k. To find all f such that $f'(x) - kf(x) = h(x)$, the solution is e^{kx} times any anti-derivative of $\mathrm{e}^{-kx}h(x)$.*

Proof. We are looking for the solution to $(\frac{\mathrm{d}}{\mathrm{d}x} - kI)f = h$.

Let $g(x) = \mathrm{e}^{kx}$, and define M_g which sends $f(x)$ to $g(x)f(x)$. By previous results we have $\frac{\mathrm{d}}{\mathrm{d}x} - kI = M_g \frac{\mathrm{d}}{\mathrm{d}x} M_g^{-1}$. So our differential equation is now

$$M_g \frac{\mathrm{d}}{\mathrm{d}x} M_g^{-1} f = h.$$

In particular, we have

$$\frac{\mathrm{d}}{\mathrm{d}x} M_g^{-1} f = M_g^{-1} h.$$

So $M_g^{-1} f$ is simply some anti-derivative of $M_g^{-1} h$. □

8.10.3 Linear Systems of Differential Equations

Example 8.10.20. I take a bottle of milk out of the fridge, and put it into a bowl of hot water to warm it up. Let us say that at time t, the temperature of the milk is $M(t)$ and the temperature of the water is $W(t)$ Then we have $M'(t) = a(W(t) - M(t))$ and $W'(t) = b(M(t) - W(t))$, where a, b are some positive constants to be determined. How to solve this system?

We have two functions, $M(t), W(t)$. We also have the following descriptions: $\begin{cases} M'(t) = -aM(t) + aW(t) \\ W'(t) = bM(t) - bW(t) \end{cases}$

So in fact we have $\begin{bmatrix} M'(t) \\ W'(t) \end{bmatrix} = \begin{bmatrix} -a & a \\ b & -b \end{bmatrix} \begin{bmatrix} M(t) \\ W(t) \end{bmatrix}$. ☺

Example 8.10.21. Say we want to solve $f''' + 6f'' + 11f' + 6f = 0$. Let $g = f'$, and $h = f''$. Then we have $\begin{bmatrix} f' \\ g' \\ h' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} f \\ g \\ h \end{bmatrix}$. So a high order linear differential equation becomes a first order linear system of differential equations.

(Also be mindful of the fact that the matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$ is EXACTLY thetranspose of the companion matrix for the polynomial $x^3 + 6x^2 + 11x + 6$.) ☺

How to solve these systems? We are trying to solve $\boldsymbol{v}' = A\boldsymbol{v}$. If $f' = kf$, then the solution is $\mathrm{e}^{kx} f(0)$. So I can make the following guess: If $\boldsymbol{v}'(t) = A\boldsymbol{v}(t)$, then the solution is probably related to e^{At}, which is a matrix. I claim that the solution should be $\mathrm{e}^{At}\boldsymbol{v}(0)$.

In particular, if the initial conditions are arbitrary, then $\mathrm{e}^{At}\boldsymbol{v}(0)$ could be anything in $\mathrm{Ran}(\mathrm{e}^{At})$, i.e., some linear combinations of columns of e^{At}. So the solution space to $\boldsymbol{v}'(t) = A\boldsymbol{v}(t)$ is simply $\mathrm{Ran}(\mathrm{e}^{At})$.

Proposition 8.10.22. *For any diagonal matrix D, the solution space to $\frac{\mathrm{d}}{\mathrm{d}x}\boldsymbol{v} = D\boldsymbol{v}$ is $\mathrm{Ran}(\mathrm{e}^{Dt})$. In fact, the solution is $\mathrm{e}^{Dt}\boldsymbol{v}(0)$ where $\boldsymbol{v}(0)$ is some arbitrary initial value.*

Proof. Say $D = diag(a, b)$. Then the system reads $f' = af$ and $g' = bg$. So we know the solution is that f is a multiple of e^{at} and g is a multiple of e^{bt}. So $\begin{bmatrix} f \\ g \end{bmatrix}$ is a linear combination of columns of $\begin{bmatrix} \mathrm{e}^{at} & 0 \\ 0 & \mathrm{e}^{bt} \end{bmatrix} = \mathrm{e}^{Dt}$.

Now plut in $t = 0$ to the equation $\boldsymbol{v}(t) = \begin{bmatrix} \mathrm{e}^{at} & 0 \\ 0 & \mathrm{e}^{bt} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$, we see that $\boldsymbol{v}(t) = \begin{bmatrix} \mathrm{e}^{at} & 0 \\ 0 & \mathrm{e}^{bt} \end{bmatrix} \boldsymbol{v}(0)$. □

Proposition 8.10.23. *For a diagonalizable matrix $A = BDB^{-1}$, the solution space to $\boldsymbol{v}'(t) = A\boldsymbol{v}(t)$is $\mathrm{Ran}(\mathrm{e}^{At})$. In fact, the solution is $\mathrm{e}^{At}\boldsymbol{v}(0)$ where $\boldsymbol{v}(0)$ is some arbitrary initial value.*

Proof. Set $\boldsymbol{w}(t) = B^{-1}\boldsymbol{v}(t)$. Then $BD\boldsymbol{w}(t) = (B\boldsymbol{w}(t))' = B\boldsymbol{w}'(t)$. Here B comes out of the derivative because it is constant. Since B is also invertible, we have $D\boldsymbol{w} = \boldsymbol{w}'$. Hence the solution space to $\boldsymbol{w}$ is $\mathrm{Ran}(\mathrm{e}^{Dt})$.

Now $\boldsymbol{v} = B\boldsymbol{w}$. Hence the solution space to $\boldsymbol{v}$ is $\mathrm{Ran}(B\mathrm{e}^{Dt}) = \mathrm{Ran}(B\mathrm{e}^{Dt}B^{-1}) = \mathrm{Ran}(\mathrm{e}^{At})$. Here the first equality is true because B^{-1} is invertible and does not effect the codomain at all. □

The statement is still true for non-diagonalizable A, but we don't prove it here.

Example 8.10.24. In real life, many couples behave in a periodic way. They are very sweet with each other for a while, and they argue and fight for a while, and then they are sweet again, and then they fight again. In short, their romantic relation exhibit a periodic behavior. What are some possible explanations?

Say two person A and B are romantically involved. The love of A for B is $f(t)$, a function of time, and the love of B for A is $g(t)$, a function of time. Now assume that A is a normal person. To a normal person, the more you are loved, the more you love the other. So $f'(t)$ is proportional of $g(t)$. Say $f'(t) = g(t)$. Assume that, unfortunately, B is a very unappreciative person. The more you love B, then more B takes you for granted. Then more you ignore B, the more B is obssessed with you. So $g'(t)$ is proportional to $-f(t)$, say $g'(t) = -f(t)$.

Then $\begin{bmatrix} f' \\ g' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} f \\ g \end{bmatrix}$. So the solution space is $\mathrm{Ran}(\mathrm{e}^{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} t})$.

Now $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = (\frac{1}{-2\mathrm{i}} \begin{bmatrix} 1 & 1 \\ \mathrm{i} & -\mathrm{i} \end{bmatrix}) \begin{bmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{bmatrix} \begin{bmatrix} -\mathrm{i} & -1 \\ -\mathrm{i} & 1 \end{bmatrix}$. So $\mathrm{e}^{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} t}$ is in fact $(\frac{1}{-2\mathrm{i}} \begin{bmatrix} 1 & 1 \\ \mathrm{i} & -\mathrm{i} \end{bmatrix}) \begin{bmatrix} \mathrm{e}^{\mathrm{i}t} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}t} \end{bmatrix} \begin{bmatrix} -\mathrm{i} & -1 \\ -\mathrm{i} & 1 \end{bmatrix}$, which is $\begin{bmatrix} \frac{\mathrm{e}^{\mathrm{i}t} + \mathrm{e}^{-\mathrm{i}t}}{2} & \frac{\mathrm{e}^{\mathrm{i}t} - \mathrm{e}^{-\mathrm{i}t}}{2\mathrm{i}} \\ -\frac{\mathrm{e}^{\mathrm{i}t} - \mathrm{e}^{-\mathrm{i}t}}{2\mathrm{i}} & \frac{\mathrm{e}^{\mathrm{i}t} + \mathrm{e}^{-\mathrm{i}t}}{2} \end{bmatrix} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$.

So there are constants a, b such that $f(t) = a\cos t + b\sin t$ and $g(t) = -a\sin t + b\cos t$. As you can see, they are periodic. ☺

8.10.4 (Optional) Non-linear Romantic Dynamics

Even for non-linear differential equations, linear algebra is still a very powerful tool. To see this, let us first try to have a slightly more realistic dynamic model for romantic relations.

Example 8.10.25. Again let $f(t), g(t)$ be the love the two person A, B have for each other. What is $f'(t)$? Well, we know it should be proportional to $g(t)$, where the proportion is positive if A is appreciative, and negative if A is non-appreciative, and 0 if A love B but does not really care if B loves back or not. So let us say A has an appreciativeness of k_A and B has an appreciativeness of k_B.

On the other hand, our energy is finite. Your love for another person cannot be infinity, because you don't have that much energy. For each person, there might be an ideal amount of love L, and if your love for another person is more than L, you will start to feel tired and emotionally drained. Let us say the ideal amount of love for A and for B are L_A and L_B. Then $f'(t)$ should be proportional to $L_A - f(t)$ and $g'(t)$ should be proportional to $L_B - g(t)$.

So our model is this: $\begin{bmatrix} f'(t) \\ g'(t) \end{bmatrix} = \begin{bmatrix} k_A g(t)(L_A - f(t)) \\ k_B f(t)(L_B - g(t)) \end{bmatrix}$. In a healthy relation, we hope that $\lim f(t) = L_A$ while $\lim g(t) = L_B$. In an indifferent relation, we have $\lim f(t) = \lim g(t) = 0$. In an unhealthy relation, $\lim f(t)$ and $\lim g(t)$ fail to exist because they are both periodic. ☺

In general, a non-linear system is like this: $\begin{bmatrix} f'(t) \\ g'(t) \end{bmatrix} = \begin{bmatrix} h_1(f(t), g(t)) \\ h_2(f(t), g(t)) \end{bmatrix}$. Then in particuylar, given a pairs of values of f and g, we can calculate f' and g'. In short, IF WE ALEADY KNOW where we are, then we know the direction we are moving to. This gives us a vector field.

Example 8.10.26. (Draw pictures of some vector fields. Electromagnetic fields and so on.) ☺

A generic vector field may have many behaviors, but what is important are the ***fixed points*** of this vector field. It could be a sink, a source, or a saddle, or neither. How can we study this behavior?

A fixed point, or an equilibrium, is a point on the fg-plane where if you start there, you never move. So you are looking for (f, g) values that makes $f' = g' = 0$. So this is like solving the values for f and g from $h_1(f, g) = 0$ and $h_2(f, g) = 0$.

Recall that in one-variable calculus, to find a local max or local min of a function $f(x)$, we first find equilibriums by looking at the information in the first derivative, to solve $f'(x) = 0$. Then AT THESE EQUILIBRIUMS, we try to look at their second derivative to classify their behavior. It turned out that we can do the same thing here. This is the Hartman-Grobman Theorem.

We try to solve $f' = g' = 0$ to find all equilibriums. Then at these points, we try to look at their second order derivative to classify their behavior. Recall that $f' = h_1$ and $g' = h_2$, so the second order derivatives for f and g are the derivatives for h_1 and h_2. You will have to look at the matrix $\begin{bmatrix} \frac{\partial h_1}{\partial f} & \frac{\partial h_1}{\partial g} \\ \frac{\partial h_2}{\partial f} & \frac{\partial h_2}{\partial g} \end{bmatrix}$.

Example 8.10.27. Consider a highly simplified case. Say $\boldsymbol{v}'(t) = \begin{bmatrix} h_1(\boldsymbol{v}) \\ h_2(\boldsymbol{v}) \end{bmatrix}$, and for each point $\boldsymbol{v} = \begin{bmatrix} f(t) \\ g(t) \end{bmatrix}$, we have constant $\begin{bmatrix} \frac{\partial h_1}{\partial f} & \frac{\partial h_1}{\partial g} \\ \frac{\partial h_2}{\partial f} & \frac{\partial h_2}{\partial g} \end{bmatrix} = A$. What do we have?

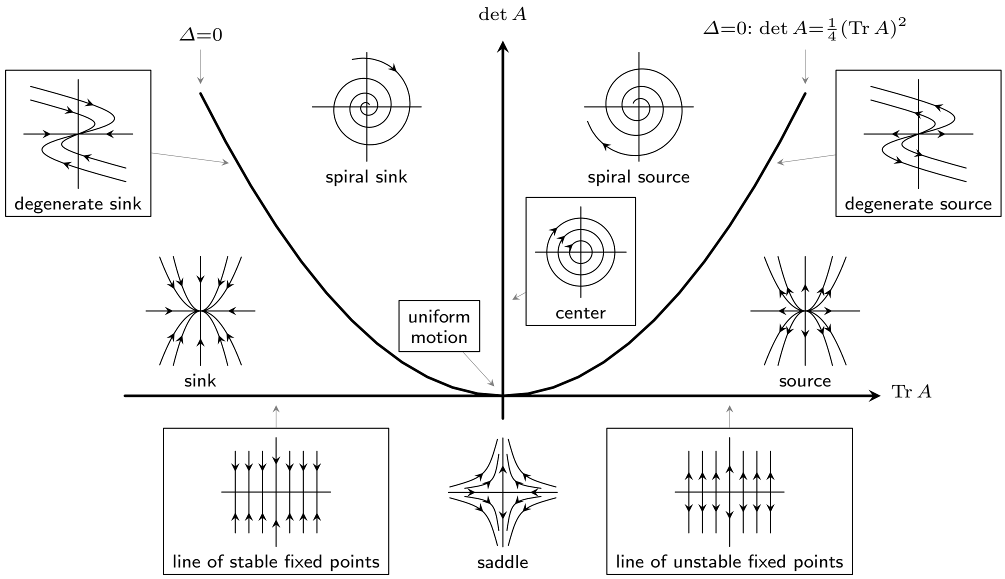
Note that if we have constant $\begin{bmatrix} \frac{\partial h_1}{\partial f} & \frac{\partial h_1}{\partial g} \\ \frac{\partial h_2}{\partial f} & \frac{\partial h_2}{\partial g} \end{bmatrix} = A$, then the vector function $\boldsymbol{h}(\boldsymbol{v}) = \begin{bmatrix} h_1(\boldsymbol{v}) \\ h_2(\boldsymbol{v}) \end{bmatrix}$ will have constant derivative. In particular, it is easy to guess and verify that $\boldsymbol{h}(\boldsymbol{v}) = A\boldsymbol{v}$.

As a result, we see that $\boldsymbol{v}'(t) = \boldsymbol{h}(\boldsymbol{v}) = A\boldsymbol{v}(t)$. So the solution is $\mathrm{e}^{At}\boldsymbol{v}(0)$.

In particular, if the second derivative of your evolution is A, then locally, vectors would flow around your equilibrium in approximately the same way vectors flow according to the linear map e^A.

Now note that e^A is a real 2×2 matrix with positive eigenvalues. By our previous classification, we can obtain the following diagram.

Poincaré Diagram: Classification of Phase Portraits in the $(\det A, \operatorname{Tr} A)$-plane



☺

Example 8.10.28. Equilibriums for our romantic relation system are $(f, g) = (0, 0)$ and $(f, g) = (L_A, L_B)$. The second derivative matrix is $\begin{bmatrix} -k_A g & k_A(L_A - f) \\ k_B(L_B - g) & -k_B f \end{bmatrix}$. So at (L_A, L_B), this second derivative matrix is $\begin{bmatrix} -k_A L_B & 0 \\ 0 & -k_B L_A \end{bmatrix}$. So the two eigenvalues are exactly $-k_A L_B$ and $-k_B L_A$, both are real, and the eigendirctions are horizontal and vertical. Also note that L_A and L_B are by assumption positive.

So if both k_A, k_B are positive, then this is a sink. It attracts everyone around. So two appreciative person are likely to be attracted to their ideal love limits. If k_A, k_B are both negative, this is negative. So two unappreciative person are repelled by their love limits, and they may never work out. Finally, if k_A, k_B have different sign, then this is a saddle. It will attract from the eigendirection for positive k-value and repel from the eigendirection for the negative k-value. It seems to promote periodic behavior.

And at $(0, 0)$, this second derivative matrix is $\begin{bmatrix} 0 & k_A L_A \\ k_B L_B & 0 \end{bmatrix}$. So if k_A, k_B have the same sign, then we have two real eigenvalues of distinct signs, and we have a saddle at the origin. If k_A, k_B have distinct signs, then we have two imaginary eigenvalues and trace zero, and so we tend to rotate around the origin. ☺

Now, in our case we cannot solve for f, g very easily. However, we can find out about their relations!

Example 8.10.29. We have the system

$$\begin{bmatrix} \frac{\mathrm{d}f}{\mathrm{d}t} \\ \frac{\mathrm{d}g}{\mathrm{d}t} \end{bmatrix} = \begin{bmatrix} k_A g(L_A - f) \\ k_B f(L_B - g) \end{bmatrix}.$$

Divide the two equations and we have

$$\frac{\mathrm{d}f}{\mathrm{d}g} = \frac{k_A g(L_A - f)}{k_B f(L_B - g)}.$$

This is separable! So we have

$$\frac{f\,\mathrm{d}f}{k_A(L_A - f)} = \frac{g\,\mathrm{d}g}{k_B(L_B - g)}.$$

Integrate both sides, we have the following result for some constant C

$$\frac{L_A}{k_A}\left(\frac{f(t)}{L_A} + \ln\left|1 - \frac{f(t)}{L_A}\right|\right) = \frac{L_B}{k_B}\left(\frac{g(t)}{L_B} + \ln\left|1 - \frac{g(t)}{L_B}\right|\right) + C.$$

This is the evolution curve! The romantic dynamics must evolve along these curves on the fg-plane. ☺

Example 8.10.30. Let us specifically analyze the case when $k_A > 0$ and $k_B > 0$. Hopefully, we are always in a relation that is mutually appreciative. Then (L_A, L_B) is a sink, while $(0, 0)$ is a saddle, with an attracting eigendirection in the 1st-3rd quadrant and a repelling eigendirection in the 2nd-4th quadrant. Draw the picture to see some flows. As you can see, there seems to be some curve through the origin, above which we will always be attracted to our ideal love limit, and below which we will be repelled to mutual hatred.

This curve has an equation of $\frac{L_A}{k_A}(\frac{f(t)}{L_A}+\ln|1-\frac{f(t)}{L_A}|) = \frac{L_B}{k_B}(\frac{g(t)}{L_B}+\ln|1-\frac{g(t)}{L_B}|)$. I call this the confession line. Suppose you are an appreciative person, and you are secretly in love with another appreciative person. Then should you confess your love and start your romance? Well, if you two are above the confession line, then even if the other person does not love you back, you should still go ahead and confess, because EVERYTHING above the confession line will eventually be attracted to the mutual ideal love limits. However, if you two are below the confession line, then hold it for the moment, because your confession would only result in eventual mutual hatred.

Just as a side note, the curve along which your love will evolve is always $\frac{L_A}{k_A}(\frac{f(t)}{L_A} + \ln|1 - \frac{f(t)}{L_A}|) = \frac{L_B}{k_B}(\frac{g(t)}{L_B} + \ln|1 - \frac{g(t)}{L_B}|) + C$ for some constant C. The direction of your evolution depends on the equilibrium situation.

Take a look at any curve above the confession line. There is a valuable lesson here. Suppose you are trying to persue someone, and you know you two should be above the confession line, then don't give up! Look at the place where your curve intersect with the x or y axis. This is the moment where your love to the other person dropped to the bottum. This is your darkest moment, then moment of despair. But LOOK, this is also exactly the moment where the other person finally starts to like you. This is not chicken soup for the soul. This is mathematics. ☺

Example 8.10.31. What if both k_A and k_B are negative? Then (L_A, L_B) is repelling, and $(0, 0)$ is a saddle, attracting along some direction in the 1st-3rd quadrant, and repelling along some direction of the 2nd-4th quadrant. The solution curves are still $\frac{L_A}{k_A}(\frac{f(t)}{L_A} + \ln|1 - \frac{f(t)}{L_A}|) = \frac{L_B}{k_B}(\frac{g(t)}{L_B} + \ln|1 - \frac{g(t)}{L_B}|) + C$.

Drawing some solution curves, you see that the curves are almost IDENTICAL with the two positive k-value case, except that all directions are negated. And invariabley, it will end up with one person, say A, extremely hate the other, and the other, say B, wierdly reached the ideal love limit, since B loves to be hated. I think it is safe to conclude that two unappreciative person would never be together. ☺

Example 8.10.32. What if we have an appreicative person A and an unappreciative person B? Then (L_A, L_B) is a saddle attracting in the horizontal direction and repelling in the vertical direction. And $(0, 0)$ is a swirling point. The relation would behave in a clockwise and periodic manner. The solution curves still have the same equations though, $\frac{L_A}{k_A}(\frac{f(t)}{L_A} + \ln|1 - \frac{f(t)}{L_A}|) = \frac{L_B}{k_B}(\frac{g(t)}{L_B} + \ln|1 - \frac{g(t)}{L_B}|) + C$.

Let us start from the point where your periodic curve intersect with the negative f-axis. At this moment, the jerk B start to have an interest in A, while A hates B. Then B started to persue A, and after a long persuing process, B eventually mananged to get A's love back. Very shortly after, there will be a sweet spot where the two are almost close to their ideal love limit. They will probably get married at that point. However, immediately after marriage, the jerk B, being the jeriest jerk, starts to lose interest in A. Very shortly after, our curve intersect with the f-axis agiain, marking the point where A completely lose interest in B. This is probably the point where A started cheating on B. Now A is still in love with B, and in fact the love of A for B is at maximum at this moment. So A will desperately try to cling on to B, to win B's love back. But this just annoys B extremely and speed up the process of B hating A. After a very long and mutually painful process, our curve eventually hit negative g-axis, marking the spot where the two no longer love each other. In most cases, they get a divorce and never see each other again.

In some cases some couples manage to stay married. Oh boy, then they will only get to torture themselves over and over agian with a very tiny sweep period and very very long period of mutual hatred. I mean, why bother? And the sweeter the sweet time is, the longer the mutual hatred will be, as can be seen from these orbiting curves. ☺

To sum up, if $f' = h_1(f,g)$ and $g' = h_2(f,g)$, you can find equilibriums by solving $h_1 = h_2 = 0$, and you can classify your equilibriums as sources or sinks or stuff by looking at the eigenstuff of $\begin{bmatrix} \frac{\partial h_1}{\partial f} & \frac{\partial h_1}{\partial g} \\ \frac{\partial h_2}{\partial f} & \frac{\partial h_2}{\partial g} \end{bmatrix}$.

8.11 Spectral Theorem

8.11.1 Spectral Theorem for Normal Matrices

Remark 8.11.1. *(Optional) This is just to explain the name of "spectral theorem".*

Given any system, say a single hydrogen atom, physicists would use a linear operator H to describe its evolution (usually the Hamiltonian operator). Each state of the system is described by some function, and H is a linear map acting on these functions.

If one intends to study potential orbits of the electrons, then we are seeking states that are stable under the evolution, i.e., eigenvectors (or eigenfunctions) of H. If $Hf = \lambda f$, then f is an orbit, and the eigenvalue λ is the corresponding energy state of this orbit.

So when electrons change orbit, it will change energy and therefore emit a light with certain frequence (i.e., certain color). These are called the spectrum of the hydrogen atom.

*As you can see, this ultimately depends on the eigenvalue. As a result, people usually use the name "spectrum" to refer to the set of possible eigenvalues of a matrix or operator, and people usually use the name "spectral theorem" to refer to a big theorem governing the structures of eigenvalues and eigenvectors. The diagonalization $A = XDX^{-1}$ is also sometimes called a **spectral decomposition**.*

In this section we shall study the fundamental eigen structure of normal matrices. First let us see some examples.

Example 8.11.2. Consider $P = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$. It has trace one and determinant zero, so the eigenvalues are 0 and 1 (distinct) and therefore the matrix can be diagonalized as $D = \begin{bmatrix} 1 & \\ & 0 \end{bmatrix}$. Since $D^2 = D$, we see that $P^2 = P$. This is an oblique projection. Indeed, the eigenspace for the eigenvalue 1 is $\mathrm{Ker}(P-I) = \mathrm{Ker}(I-P) = \mathrm{Ran}(P)$ according to properties of projections, and this is spanned by $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. The the eigenspace for the eigenvalue 0 is $\mathrm{Ker}(P)$ obviously spanned by $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. So the two eigenspaces are NOT orthogonal.

But if we have $P = \begin{bmatrix} 0.1 & 0.3 \\ 0.3 & 0.9 \end{bmatrix}$, then since $P^2 = P = P^{\mathrm{T}}$, this is an orthogonal projection. The eigenspace for eigenvalue 1 is $\mathrm{Ker}(P - I) = \mathrm{Ker}(I - P) = \mathrm{Ran}(P)$, which is orthogonal to $\mathrm{Ker}(P)$ the eigenspace for the eigenvalue 0.

This is not a singular occurence. You can also verify this: if $H^2 = I$, it is an (oblique) reflection, and it is an orthogonal projection if and only if $H = H^{\mathrm{T}}$. Note that the "mirror" or the reflection is $\mathrm{Ker}(H - I)$, the set of fixed points, while the direction of reflection is $\mathrm{Ker}(H + I)$, the set of vectors $\boldsymbol{v}$ such that $H\boldsymbol{v} = -\boldsymbol{v}$. So the two eigenspaces are orthogonal if and only if H is symmetric.

But this is not a phenomenon to symmetric matrices alone. Consider $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. We have

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -\mathrm{i} & \mathrm{i} \end{bmatrix} \begin{bmatrix} \mathrm{i} & \\ & -\mathrm{i} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\mathrm{i} & \mathrm{i} \end{bmatrix}^{-1}.$$

As you can see, the eigenspace for i is spanned by $\begin{bmatrix} 1 \\ -\mathrm{i} \end{bmatrix}$ and the eigenspace for $-\mathrm{i}$ is spanned by $\begin{bmatrix} 1 \\ \mathrm{i} \end{bmatrix}$. These two complex vectors are orthogonal as well. ☺

So, what kinds of matrices would have orthogonal eigenspaces? It seems like we need A to have some relation to A^{T}, or in the case of complex matrices, we need A to have some relation to A^*. Now we can have our definition.

As a side note, since we are interested in orthogonal eigenspaces, throughout this section we need inner product structure. We shall always assume that our space is $\mathbb{R}^n$ or $\mathbb{C}^n$ with the usual dot product as the inner product structure.

Definition 8.11.3. *A matrix is normal if $AA^* = A^*A$.*

This includes real symmetric matrices, real skew-symmetric matrices, real orthogonal matrices, complex Hermitian matrices, complex skew-Hermitian matrices and complex unitary matrices. There are also normal matrices that are none of these special cases, such as $\begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix}$.

Now, the following lemma is the key reason why these matrices are so nice: its eigenstructure respect the complex conjugate.

Lemma 8.11.4. *If A is normal and $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then $A^*\boldsymbol{v} = \overline{\lambda}\boldsymbol{v}$.*

Proof. If $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then $(A-\lambda I)\boldsymbol{v} = \mathbf{0}$. Note that we must utilize the inner product structure here. So we rewrite the equation as $\|(A-\lambda I)\boldsymbol{v}\| = 0$, and then $\boldsymbol{v}^*(A-\lambda I)^*(A-\lambda I)\boldsymbol{v} = 0$.

Now comes the use of normality. Since $AA^* = A^*A$, we have a commutativity $(A-\lambda I)^*(A-\lambda I) = A^*A - \overline{\lambda}A - \lambda A^* + \lambda\overline{\lambda}I = AA^* - \overline{\lambda}A - \lambda A^* + \lambda\overline{\lambda}I = (A-\lambda I)(A-\lambda I)^*$.

As a result, $0 = \boldsymbol{v}^*(A-\lambda I)^*(A-\lambda I)\boldsymbol{v} = \boldsymbol{v}^*(A-\lambda I)(A-\lambda I)^*\boldsymbol{v} = \|(A-\lambda I)^*\boldsymbol{v}\| = \|(A^*-\overline{\lambda}I)\boldsymbol{v}\|$. So we see that $(A^*-\overline{\lambda}I)\boldsymbol{v} = \mathbf{0}$, and $A^*\boldsymbol{v} = \overline{\lambda}\boldsymbol{v}$. ☐

Lemma 8.11.5. *If T is a (complex) upper triangular matrix and T is normal, then T is diagonal.*

Proof. Suppose the upper left entry of T is λ, so we have $T = \begin{bmatrix} \lambda & * \\ \mathbf{0} & T' \end{bmatrix}$. Then since T is upper triangular, we have $T\boldsymbol{e}_1 = \lambda\boldsymbol{e}_1$. Then $T^*\boldsymbol{e}_1 = \overline{\lambda}\boldsymbol{e}_1$, and therefore taking adjoint we have $\boldsymbol{e}_1^*T = \lambda\boldsymbol{e}_1^* = \lambda\boldsymbol{e}_1^{\mathrm{T}}$. This implies that the first row of T is $\lambda\boldsymbol{e}_1^{\mathrm{T}}$.

In particular, we have $T = \begin{bmatrix} \lambda & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & T' \end{bmatrix}$.

Now T' is an upper triangular normal matrix of smaller size. So we are done by induction. ☐

We are now ready for the big theorem.

Theorem 8.11.6 (Spectral Theorem for Normal Matrices)**.** *If A is normal, then we can find unitary U and diagonal D such that $A = UDU^* = UDU^{-1}$.*

In particular, A is always diagonalizable, there is an orthonormal basis made of eigenvectors of A, and all geometric multiplicities are equal to corresponding algebraic multiplicities.

Proof. Consider a Schur decomposition $A = UTU^*$. Then $AA^* = A^*A$ implies that $TT^* = T^*T$. But then this means T is triangular and normal, and therefore diagonal. So we are done. ☐

Since A is diagonalizable, the domain $\mathbb{C}^n$ of A can be decomposed as the direct sum of the eigenspaces of A. In fact, these eigenspaces must be mutually orthogonal! This is already evident from the spectral theorem above, but you can also enjoy the following independents proof.

Corollary 8.11.7. *If A is normal, then its eigenspaces are mutually orthogonal.*

Proof. Let us show that eigenspaces are mutually orthogonal. Suppose $A\boldsymbol{v} = \lambda\boldsymbol{v}$ and $A\boldsymbol{w} = \mu\boldsymbol{w}$ where $\lambda \neq \mu$.

Now $\boldsymbol{v}^*(A\boldsymbol{w}) = \mu\boldsymbol{v}^*\boldsymbol{w}$. On the other hand, $(\boldsymbol{v}^*A)\boldsymbol{w} = (A^*\boldsymbol{v})^*\boldsymbol{w} = (\overline{\lambda}\boldsymbol{v})^*\boldsymbol{w} = \lambda\boldsymbol{v}^*\boldsymbol{w}$. Since $\mu \neq \lambda$, we have no choice but to conclude that $\boldsymbol{v} \perp \boldsymbol{w}$. ☐

Remark 8.11.8. *Note that the requirement of being normal is necessary and sufficient for the spectral theorem here. Suppose $A = UDU^*$ for some unitary U. Then $A^* = UD^*U^*$, and $AA^* = UDU^*UDU^* = UDD^*U^*$. Since diagonal matrices commutes with each other, we have $DD^* = D^*D$. So $AA^* = UDD^*U^* = UD^*DU^* = A^*A$. So A is normal.*

Example 8.11.9. Let V be the space of periodic smooth real functions with period 2π. We make it an inner product space via the inner product $\langle f, g\rangle = \int_0^{2\pi} f(x)g(x)\,\mathrm{d}x$, i.e., we integrate the product over a single period.

Now, consider the derivative operator $D = \frac{\mathrm{d}}{\mathrm{d}x}$. (If you are familiar with integration by parts, skip this paragraph.) One of the most important property of the derivative is the product rule or Leibniz rule, i.e., $D(fg) = (Df)g + f(Dg)$. By integration, we would obtain $fg|_a^b = \int_a^b Df(x)g(x)\,\mathrm{d}x + \int_a^b f(x)Dg(x)\,\mathrm{d}x$. This technique is called integration by parts. Namely, if one attempt to do integration of a product $\int_a^b Df(x)g(x)\,\mathrm{d}x$, then one can integrate one factor function while differentiating the other factor function, and have $\int_a^b Df(x)g(x)\,\mathrm{d}x = (fg)|_a^b - \int_a^b f(x)Dg(x)\,\mathrm{d}x$.

Now take integration from 0 to 2π, and note that fg is periodic and thus $(fg)|_0^{2\pi} = 0$. So we have $\langle Df, g\rangle = -\langle f, Dg\rangle$. If you think of $\langle Df, g\rangle$ as $(Df)^{\mathrm{T}}g = f^{\mathrm{T}}D^{\mathrm{T}}g$, and think of the right hand side $-\langle f, Dg\rangle$ as $-f\mathrm{T}Dg$, we see that $D^{\mathrm{T}} = -D$. So the differentiation operator is skew-symmetric.

In practice, many physical phenomena (heat, wave, etc.) are related to the second derivative. Then we have $(D^2)^{\mathrm{T}} = (D^{\mathrm{T}})^2 = (-D)^2 = D^2$, so this would be a symmetrix operator.

In particular, eigenspaces of D^2 are orthogonal. So you immediately see that, for example, $\sin x, \sin(2x), \ldots$ are all orthogonal because they are eigenvectors of D^2 for different eigenvalues. (I.e., for integers $m \neq n$, we must have $\int_0^{2\pi} \sin(nx)\sin(mx)\,\mathrm{d}x = 0$.)

(Optional) It is also fun to do this directly but with the same idea as above, i.e., different eigenvalues lead to $\lambda \boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = \mu \boldsymbol{v}^{\mathrm{T}}\boldsymbol{w}$, which means $\boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = 0$. For integers $m \neq n$, we have

$$\int_0^{2\pi} \sin(nx)\sin(mx)\,\mathrm{d}x = -\frac{1}{n}\cos(nx)\sin(mx)|_0^{2\pi} - \frac{m}{n}\int_0^{2\pi}\cos(nx)\cos(mx)\,\mathrm{d}x = -\frac{m}{n}\int_0^{2\pi}\cos(nx)\cos(mx)\,\mathrm{d}x.$$

But similarly we have

$$\int_0^{2\pi} \sin(nx)\sin(mx)\,\mathrm{d}x = -\frac{1}{m}\sin(nx)\cos(mx)|_0^{2\pi} - \frac{n}{m}\int_0^{2\pi}\cos(nx)\cos(mx)\,\mathrm{d}x = -\frac{n}{m}\int_0^{2\pi}\cos(nx)\cos(mx)\,\mathrm{d}x.$$

So since $m \neq n$, all must be zero. As you can see, the key here is $-m^2 \neq -n^2$, i.e., they are eigenvectors of D^2 for different eigenvalues. ☺

Now let us move on to specific matrices. First we consider real symmetric or complex Hermitian matrices.

Proposition 8.11.10. *If $A = A^*$, then all eigenvalues are real. If A is a real matrix and $A = A^{\mathrm{T}}$, then furthermore, the spectral decomposition $A = QDQ^{-1}$ can be made so that Q, D are both real matrices. (So Q is real orthogonal and D is real diagonal.)*

Proof. For all normal matrices, we know if $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then $A^*\boldsymbol{v} = \overline{\lambda}\boldsymbol{v}$. However, if furthermore $A = A^*$, then in fact we also have $A^*\boldsymbol{v} = A\boldsymbol{v} = \lambda\boldsymbol{v}$. Since an eigenvector $\boldsymbol{v}$ must be non-zero, we must have $\lambda = \overline{\lambda}$.

Now if A is real and symmetric, then immediately we see that A is diagonalizable and all eigenvalues are real. So all eigenspaces are spanned by real vectors, and they are all orthogonal to each other. By picking an orthonormal basis for each eigenspace, together they form an orthonormal basis for $\mathbb{R}^n$ made of eigenvectors. Say this basis is the real orthogonal matrix Q. Then $A = QDQ^{-1}$. □

Remark 8.11.11. *Note that this is a necessary and sufficient condition. Conversely, if $A = QDQ^{-1}$ for unitary Q and real diagonal D, then $A^* = QD^*Q^{-1} = QDQ^{-1} = A$, so A is Hermitian. And if Q is real orthogonal, then A is also real, and thus symmetric.*

Let us consider the geometric implication of this. Most of the matrices are diagonalizable. Therefore, they are scalings along eigendirections.

Example 8.11.12. Consider $A = XDX^{-1}$ where $D = diag(2,3)$ and $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then A would stretch in the x-axis direction by a factor of 2, and stretch in the $x = y$ direction by a factor of 3. (If you are curious, it will repel points along curves $\ln|y| = \frac{\ln 3}{\ln 2}\ln|x-y| + k$, or via parametrization $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} a2^t + b3^t \\ b3^t \end{bmatrix}$ for constants a, b.)

Note that the two direction of stretching are NOT perpendicular. We are stretching along an oblique frame of reference. However, in the case of normal matrices, X would be unitary or in the real case orthogonal. This means we are stretching along some orthogonal frame of reference.

In particular, consider the effect of A on a unit circle. We are stretching the circle in the x-axis direction by a factor of 2, and stretch in the $x = y$ direction by a factor of 3. Since the vectors $\boldsymbol{e}_1, \boldsymbol{u} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are on the original circle, after the action of A, the vectors $2\boldsymbol{e}_1, 3\boldsymbol{u}$ are on the resulting ellipse. You can draw the unique ellipse centered around the origin through points $\pm 2\boldsymbol{e}_1, \pm 3\boldsymbol{u}$. Note that $\boldsymbol{e}_1, \boldsymbol{u}$ are NOT the direction of the major-axis and the minor-axis of the resulting ellipse (because they are oblique).

Consider a symmetric operation $A = QDQ^{-1}$ where Q is orthogonal, and $D = diag(2,3)$. Then we are stretching along an orthogonal fram of reference. In particular, the two eigendirections are EXACTLY the directions of the major-axis and the minor-axis, and the lengths of the resulting ellipse is EXACTLY the eigenvalues of A. (Because that is how much A stretches.)

In short, $A = QDQ^{-1}$ means the geometric action of A is rotation–coordinate stretch–rotation back. ☺

8.11.2 (Optional) Other special cases of spectral theorem

We are mainly concerned with unitary matrices, skew-Hermitian matrices, real orthogonal matrices and real skew-symmetric matrices.

Proposition 8.11.13. *If U is a unitary matrix, then all its eigenvalues are unit complex numbers. In particular, $U = QDQ^{-1}$ where Q is unitary and D is diagonal with diagonal entries $d_k = \mathrm{e}^{\mathrm{i}\theta_k}$.*

Proof. Again note that U is normal, so $U\boldsymbol{v} = \lambda\boldsymbol{v}$ if and only if $U^*\boldsymbol{v} = \overline{\lambda}\boldsymbol{v}$. However, $U^* = U^{-1}$. As a result, we see that $\lambda^{-1} = \overline{\lambda}$, so λ is a unit complex number. □

Remark 8.11.14. *Note that this is also necessary and sufficient. If $D = \begin{bmatrix} \mathrm{e}^{\mathrm{i}\theta_1} & & \\ & \ddots & \\ & & \mathrm{e}^{\mathrm{i}\theta_n} \end{bmatrix}$ and $U = QDQ^*$ for some unitary Q, then $U^{-1} = QD^{-1}Q^* = QD^*Q^* = U^*$. So U is unitary.*

Geometrically, a unitary matrix means up to an orthogonal change of variable, we simply rotate the complex number in each coordinate in their respective complex plane.

For the real case, we have an even better result.

Lemma 8.11.15. *If A is real and normal, and $A(\boldsymbol{v} + \mathrm{i}\boldsymbol{w}) = \lambda(\boldsymbol{v} + \mathrm{i}\boldsymbol{w})$ for real vectors $\boldsymbol{v}, \boldsymbol{w}$ and non-real eigenvalue λ, then we have $\|\boldsymbol{v}\| = \|\boldsymbol{w}\|, \boldsymbol{v} \perp \boldsymbol{w}$.*

Proof. Say $A(\boldsymbol{v} + \mathrm{i}\boldsymbol{w}) = \lambda(\boldsymbol{v} + \mathrm{i}\boldsymbol{w})$ for real vectors $\boldsymbol{v}, \boldsymbol{w}$ and non-real eigenvalue λ. Then since A is real, by taking complex conjugate, we have $A(\boldsymbol{v} - \mathrm{i}\boldsymbol{w}) = \overline{\lambda}(\boldsymbol{v} - \mathrm{i}\boldsymbol{w})$.

However, since eigenspaces for distinct eigenvalues are mutually orthogonal, and $\lambda \neq \overline{\lambda}$, hence $0 = (\boldsymbol{v} + \mathrm{i}\boldsymbol{w})^*(\boldsymbol{v} - \mathrm{i}\boldsymbol{w}) = \|\boldsymbol{v}\|^2 - \|\boldsymbol{w}\|^2 - 2\mathrm{i}(\boldsymbol{w}^*\boldsymbol{v})$. Then the real part and imaginary parts must both be zero, and we see that $\|\boldsymbol{v}\| = \|\boldsymbol{w}\|, \boldsymbol{v} \perp \boldsymbol{w}$. □

Theorem 8.11.16. *If A is real orthogonal, then we have $A = QDQ^{-1}$ where Q is real orthogonal, and D is real block diagonal where each diagonal block is either 1×1 with value ± 1, or it is some 2×2 rotation matrix R_θ.*

Proof. If all eigenvalues of A are ± 1, then since all eigenvalues are real, $A = UDU^*$ for unitary U and real diagonal D. So, taking adjoint, we have $A^* = UD^*U^* = UDU^* = A$. But A is also real, so it is real symmetric. So, we can perform $A = QDQ^{-1}$ with real orthogonal Q and real diagonal D where the diagonal entries are ± 1. So we are done.

Suppose this is not the case. Then A has an eigenvalue $\lambda \neq \pm 1$. Since A is unitary, $|\lambda| = 1$, and hence λ cannot be real. But A is also a real matrix, so complex eigenstuff come in pairs!

Say $A(\boldsymbol{v} + \mathrm{i}\boldsymbol{w}) = \lambda(\boldsymbol{v} + \mathrm{i}\boldsymbol{w})$ for real vectors $\boldsymbol{v}, \boldsymbol{w}$. By previous lemma, we have $\|\boldsymbol{v}\| = \|\boldsymbol{w}\|, \boldsymbol{v} \perp \boldsymbol{w}$. So let us scale $\boldsymbol{v}, \boldsymbol{w}$ simultaneously so that they are now unit vectors.

Since $\boldsymbol{v}, \boldsymbol{w}$ are unit mutually orthogonal vectors, we can complete them into an orthogonal basis for $\mathbb{R}^n$, and as columns they form an orthogonal matrix X. Also since $A(\boldsymbol{v} + \mathrm{i}\boldsymbol{w}) = \lambda(\boldsymbol{v} + \mathrm{i}\boldsymbol{w})$, by setting $\lambda = \cos\theta + \mathrm{i}\sin\theta$, we can deduce that $A\begin{bmatrix}\boldsymbol{v} & \boldsymbol{w}\end{bmatrix} = \begin{bmatrix}\boldsymbol{v} & \boldsymbol{w}\end{bmatrix} R$ for some rotation matrix R. As a result. $AX = X\begin{bmatrix} R & O \\ O & A_1 \end{bmatrix}$. The rest is standard induction. ◻

Consider the implication of this. It gives a complete description of the geometric behavior of an orthogonal matrix A. It means we can decompose the whole space into mutually orthogonal planes which A would rotate independently, and then a subspace of points fixed by A, and finally a subspace of points reflected by A.

However, note that $\begin{bmatrix} -1 & \\ & -1 \end{bmatrix}$ can also be thought of as a rotation. So we have the following result:

Corollary 8.11.17. *If A is orthogonal and $\det(A) = 1$, then A simply rotates mutually orthogonal planes independently. If $\det(A) = -1$, then A would first rotates mutually orthogonal planes independently, and then reflect along a direction orthogonal to all the planes of rotation.*

In particular, an orthogonal matrix is either a rotation, or a rotation plus reflection.

Now let us move on to the skew-Hermitian case. They are intuitively the opposite of being Hermitian.

Proposition 8.11.18. *If A is skew-Hermitian, then all eigenvalues are purely imaginary.*

Proof. Same old same old. Since A is normal, $A\boldsymbol{v} = \lambda\boldsymbol{v}$ if and only if $A^*\boldsymbol{v} = \overline{\lambda}\boldsymbol{v}$. But since $A^* = -A$, we see that $\overline{\lambda} = -\lambda$, so λ has no real part. ◻

Remark 8.11.19. *This is again necessary and sufficient, and the proof is trivial.*

Also note that if A is skew-Hermitian, then all diagonal entries of A must be purely imaginary because $A^* = -A$. In particular, consider $\begin{bmatrix} a\mathrm{i} & \\ & a\mathrm{i} \end{bmatrix}$ for real a. This is NOT a Hermitian matrix, because the diagonal is not real. This is in fact Skew-Hermitian, depsite the apparent look of symmetricity.

If A is a skew-symmetric matrix, then its eigenvalues are precisely pairs of conjugate purely imaginary numbers, so we can have the following fact.

Corollary 8.11.20. *If A is an $n \times n$ real symmetric matrix, and n is odd, then A is NOT invertible.*

Proof. A has an odd number of eigenvalues, but all the non-real eigenvalues come in pairs. So A must have a real eigenvalue. However, the only number that is both real and purely imaginary is 0. So A has eigenvalue 0. ◻

Note that the skew-Hermitian matrix $\begin{bmatrix} a\mathrm{i} & \\ & -a\mathrm{i} \end{bmatrix}$ is similar to the real matrix $\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}$.

Proposition 8.11.21 (Darboux basis)**.** *If A is real skew-symmetric and invertible, then we have $A = Q\begin{bmatrix} O & -D \\ D & O \end{bmatrix} Q^{-1}$ for some real orthogonal Q and invertible diagonal D.*

Proof. Suppose all eigenvalues of A are zero. Then since A is also diagonalizable, we must have $A = O$ and the whole statement is trivial.

Suppose we have some non-zero eigenvalues. Suppose $A(\boldsymbol{v} + \mathrm{i}\boldsymbol{w}) = \lambda(\boldsymbol{v} + \mathrm{i}\boldsymbol{w})$ for real vectors $\boldsymbol{v}, \boldsymbol{w}$ and non-zero λ. Then λ cannot be real. So $\|\boldsymbol{v}\| = \|\boldsymbol{w}\|, \boldsymbol{v} \perp \boldsymbol{w}$. So let us scale $\boldsymbol{v}, \boldsymbol{w}$ simultaneously so that they are now unit vectors.

Since $\boldsymbol{v}, \boldsymbol{w}$ are unit mutually orthogonal vectors, we can complete them into an orthogonal basis for $\mathbb{R}^n$, and as columns they form a real orthogonal matrix X. Also since $A(\boldsymbol{v} + \mathrm{i}\boldsymbol{w}) = \lambda(\boldsymbol{v} + \mathrm{i}\boldsymbol{w})$, by setting $\lambda = a\mathrm{i}$ for a real a, we can deduce that $A \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix} = \begin{bmatrix} \boldsymbol{v} & \boldsymbol{w} \end{bmatrix} R$ for $R = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix}$. As a result. $AX = X \begin{bmatrix} R & O \\ O & A_1 \end{bmatrix}$.

Now by induction, $X^{-1}AX$ is now $\begin{bmatrix} & a & & \\ -a & & & \\ & & & -D \\ & & D & \end{bmatrix}$. Permute the second block row and third block row, while permuting the second block column with the third block column, we see that $P^{-1}X^{-1}AXP = \begin{bmatrix} & & a & \\ & & & -D \\ -a & & & \\ & D & & \end{bmatrix}$, and we are done. Note that a permutation matrix P is real orthogonal. $\square$

Proposition 8.11.22 (Darboux basis)**.** *If A is real skew-symmetric, then we have $A = QDQ^{-1}$ for some real orthogonal Q and block diagonal D, where each block is either zero or of the form* $\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}$.

The most fascinating aspect of a skew symmetric matrix, however, lies in its connection to rotations.

Proposition 8.11.23. *If A is skew-Hermitian, then e^A is unitary. Conversely, if B is unitary, then $B = \mathrm{e}^A$ for some skew-Hermitian A.*

Proof. Say A is skew-Hermitian. Then $A = Q \begin{bmatrix} a_1\mathrm{i} & & \\ & \ddots & \\ & & a_n\mathrm{i} \end{bmatrix} Q^*$. Then $\mathrm{e}^A = Q \begin{bmatrix} \mathrm{e}^{a_1\mathrm{i}} & & \\ & \ddots & \\ & & \mathrm{e}^{a_n\mathrm{i}} \end{bmatrix} Q^*$ will have unit complex eigenvalues, and hence this is unitary.

Conversely, if B is unitary, then $B = Q \begin{bmatrix} \mathrm{e}^{a_1\mathrm{i}} & & \\ & \ddots & \\ & & \mathrm{e}^{a_n\mathrm{i}} \end{bmatrix} Q^*$, and thus the skew-Hermitian $A = Q \begin{bmatrix} a_1\mathrm{i} & & \\ & \ddots & \\ & & a_n\mathrm{i} \end{bmatrix} Q^*$ would yield $\mathrm{e}^A = B$. $\square$

The case is a bit trickier for the real matrices.

Lemma 8.11.24. $\det(\mathrm{e}^A) = \mathrm{e}^{\mathrm{trace}(A)}$

Proof. Let $x_1, ..., x_n$ be eigenvalues for A. Then $\mathrm{e}^{x_1}, \dots, \mathrm{e}^{x_n}$ are the eigenvalues of e^A.

So, $\det(\mathrm{e}^A) = \prod \mathrm{e}^{x_i} = \mathrm{e}^{\sum x_i} = \mathrm{e}^{\mathrm{trace}(A)}$. $\square$

Proposition 8.11.25. *If A is real skew-symmetric, then e^A is orthogonal with determinant 1, i.e., a rotation. Conversely, if B is a real orthogonal matrix with $\det(B) = 1$, i.e., a rotation, then $B = \mathrm{e}^A$ for some skew symmetric A.*

Proof. If A is real, by the formula of exponentiation we see that e^A is also real. Since A is skew-Hermitian, e^A is real and unitary, i.e., real orthogonal. Finally, $\det(\mathrm{e}^A) = \mathrm{e}^{\mathrm{trace}(A)} = \mathrm{e}^0 = 1$, this is because all diagonal entries of A must be zero, courtesy of $A^{\mathrm{T}} = -A$.

Conversely, if B is real orthogonal with $\det(B) = 1$, then QBQ^{T} is block diagonal where each block is either 1×1 with value 1, or some 2×2 rotation matrix. (Note that pairs of -1 on the diagonal is also a rotation matrix, and there is no left out -1 eigenvalues, because $\det(B) = 1$.)

Now each rotation matrix $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is the exponentiation $\mathrm{e}^{\begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix}}$. So we are done. ☐

Example 8.11.26. Sometimes it is much more geometric to consider the logarithm of a rotation, i.e., the skew-symmetric matrices, rather than the rotation matrix. For a quick example, $\mathrm{e}^{\begin{bmatrix} 0 & -\theta \\ \theta & 0 \end{bmatrix}} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, so the skew-symmetric matrix tells you immediately the angle of rotation, whereas the actual rotation matrix looks like a big mess with ugly values coming from sines and cosines.

Consider the skew-symmetric matrix $A = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}$. What is the rotation e^A? Well, let $\boldsymbol{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Then first of all you should notice that $A\boldsymbol{v} = \mathbf{0}$. As a result, $\mathrm{e}^A\boldsymbol{v} = \mathrm{e}^0\boldsymbol{v} = \boldsymbol{v}$. So $\boldsymbol{v}$ is the axis of rotation!

Now how much is this rotation? Consider the eigenvalues of A, which should be $\theta\mathrm{i}, -\theta\mathrm{i}, 0$ for some real θ. Then the eigenvalues of e^A are $\mathrm{e}^{\theta\mathrm{i}}, \mathrm{e}^{-\theta\mathrm{i}}, 1$, so e^A is a rotation by θ. So we just need to find out what θ is.

Now consider the sum of 2×2 principal minors of A, which gives $(\theta\mathrm{i})(-\theta\mathrm{i}) = \det\begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix} + \det\begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} + \det\begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix} = a^2 + b^2 + c^2 = \|\boldsymbol{v}\|^2$. In particular, $\theta = \pm\|\boldsymbol{v}\|$.

So the direction of $\boldsymbol{v}$ gives the axis of rotation, while the length of $\boldsymbol{v}$ gives the amount of rotation. ☺

8.11.3 Definiteness

Here is also an important application.

Definition 8.11.27. *Given a Hermitian matrix A, we say it is* ***positive definite*** *if $\boldsymbol{v}^*A\boldsymbol{v} \geq 0$ with equality if and only if $\boldsymbol{v} = \mathbf{0}$. (Can you see why $\boldsymbol{v}^*A\boldsymbol{v}$ must be real?)*

We can have a bunch of similar definitions.

Definition 8.11.28. *Given a Hermitian matrix A,*

1. *we say it is* ***negative definite*** *if $\boldsymbol{v}^*A\boldsymbol{v} \leq 0$ with equality if and only if $\boldsymbol{v} = \mathbf{0}$.*
2. *we say it is* ***positive semidefinite*** *if $\boldsymbol{v}^*A\boldsymbol{v} \geq 0$.*
3. *we say it is* ***negative semidefinite*** *if $\boldsymbol{v}^*A\boldsymbol{v} \leq 0$.*
4. *we say it is* ***indefinite*** *if $\boldsymbol{v}^*A\boldsymbol{v}$ could be positive for some $\boldsymbol{v}$ and negative for some other $\boldsymbol{v}$.*

The definition is not new in itself. We have already studied positive definite symmetric matrices when we study inner product spaces. However, we now have a new criterion.

Corollary 8.11.29. *A Hermitian matrix is positive definite if and only if all eigenvalues are positive.*

Proof. Suppose A is Hermitian. Note that $\boldsymbol{v}^*A\boldsymbol{v} = \boldsymbol{v}^*QDQ^*\boldsymbol{v} = \boldsymbol{w}^*D\boldsymbol{w}$, here $\boldsymbol{w} = Q^*\boldsymbol{v}$. Now if $\boldsymbol{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$, then $\boldsymbol{w}^*D\boldsymbol{w} = \sum \lambda_i |w_i|^2$.

So if all eigenvalues of A are positive, then $\boldsymbol{v}^*A\boldsymbol{v} = \boldsymbol{w}^*D\boldsymbol{w} = \sum \lambda_i |w_i|^2 \geq 0$, with equality if and only if all $w_i = 0$, if and only if $\boldsymbol{w} = \mathbf{0}$, if and only if $Q^*\boldsymbol{v} = \mathbf{0}$. Note that Q is unitary, and hence invertible, so equality happens if and only if $\boldsymbol{v} = \mathbf{0}$. So A is positive definite.

Conversely, suppose some eigenvalues of A are NOT positive. Say $A\boldsymbol{v} = \lambda\boldsymbol{v}$ for some $\boldsymbol{v} \neq \mathbf{0}$ and $\lambda \leq 0$. Then $\boldsymbol{v}^*A\boldsymbol{v} = \lambda\|\boldsymbol{v}\|^2 \leq 0$, so A is NOT positive definite. ☐

To put this side to side with previous result on positive definiteness, we see that the followings are all equivalent:

1. A is positive definite.
2. All eigenvalues of A are positive.
3. All pivots are positive, i.e., A has LDU decomposition $A = LDL^*$ and the diagonal entries in D are all positive.
4. All leading principal submatrices are positive definite.
5. All leading principal submatrices have positive determinant.
6. All principal submatrices are positive definite.
7. All principal submatrices have positive determinant.
8. We have a lower triangular invertible L with $A = LL^*$.
9. We have $A = BB^*$ for invertible B.

The equivalence between the first three are the most important. The others are not really important. (Except for maybe the last two, as they provide a nice analogy between the positiveness of XX^* and of x^2 for real numbers.)

Similar statements is true for all other (semi)definiteness. For example, the followings are all equivalent:

1. A is positive semi-definite.
2. All eigenvalues of A are non-negative.
3. All pivots are non-negative, i.e., we have LDU decomposition $A = LDL^*$ and the diagonal entries in D are all non-negative.
4. All leading principal submatrices are positive semi-definite.
5. All principal submatrices are positive semi-definite.
6. (Determinants of leading principal matrices are no longer useful. Say $\begin{bmatrix} 0 & \\ & -1 \end{bmatrix}$, then the leading principal minors are all zero (non-negative), yet it is NOT positive semi-definite. It is in fact negative semidefinite, because all pivots are ≤ 0.)
7. All principal matrices have non-negative determinants.
8. We have a lower triangular L with $A = LL^*$. (But we allow L to be non-invertible.)
9. We have $A = BB^*$ for B. (But we allow B to be non-invertible.)

Beware that determinants of leading principal submatrices are no longer useful.

Remark 8.11.30. *(Optional)*

Note that the LDU decompositions for positive semi-definite matrices are a bit tricky. We previously know that when A is INVERTIBLE, then LDU decomposition exists if and only if all leading principal matrices are invertible. Here A could have non-invertible leading principal submatrices. What do we do? We have to start over.

The proof goes like this. Suppose the upper left entry of A is non-zero. Then $A = \begin{bmatrix} a & \boldsymbol{v}^* \\ \boldsymbol{v} & A_1 \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^* \\ \frac{1}{a}\boldsymbol{v} & I \end{bmatrix} \begin{bmatrix} a & \mathbf{0}^* \\ \mathbf{0} & A_1 - \frac{1}{a}\boldsymbol{v}\boldsymbol{v}^* \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{a}\boldsymbol{v}^* \\ \mathbf{0} & I \end{bmatrix}$. Since A is positive semi-definite, therefore $\begin{bmatrix} a & \mathbf{0}^* \\ \mathbf{0} & A_1 - \frac{1}{a}\boldsymbol{v}\boldsymbol{v}^* \end{bmatrix}$ is positive semi-definite, and therefore the lower right block is still positive semi-definite, and therefore if it has LDU decomposition, then we can then deduce a corresponding LDU decomposition for A.

If $a = 0$, then I claim that $\boldsymbol{v} = \mathbf{0}$. Suppose not, let us show that A is NOT positive semi-definite. Pick any $\boldsymbol{w} \in \mathbb{C}^{n-1}$ such that $\boldsymbol{v}^*\boldsymbol{w} = -1$. Then consider $\begin{bmatrix} 1 & t\boldsymbol{w}^* \end{bmatrix} A \begin{bmatrix} 1 \\ t\boldsymbol{w} \end{bmatrix} = -2t + t^2\boldsymbol{w}^*A_1\boldsymbol{w}$. Let t be closer and closer to zero from the positive side, then the t^2 term would be ignorable and this would be negative.

So if $a = 0$ and A is positive semi definite, we in fact have $A = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & A_1 \end{bmatrix}$ and A_1 is positive semi-definite. So again by induction we are done.

For the negative cases, the short answer is that A is negative definite if and only if $-A$ is positive definite, and the same for semi-cases. For example, the followings are all equivalent:

1. A is negative definite.
2. All eigenvalues of A are negative.
3. All pivots are negative. I.e., we have LDU decomposition $A = LDL^*$ and the diagonal entries in D are all negative.
4. The $k \times k$ leading principal submatrix is negative definite if k is odd, and positive definite if k is even.
5. All $k \times k$ principal submatrices are negative definite if k is odd, and positive definite if k is even.
6. The leading principal matrices have alternating determinant signs, i.e., the $k \times k$ leading principal matrix has determinant with sign $(-1)^k$.
7. All $k \times k$ principal matrices have determinant with sign $(-1)^k$.
8. We have a lower triangular invertible L with $A = -LL^*$.
9. We have $A = -BB^*$ for invertible B.

For the negative semidefinite cases, the followings are all equivalent:

1. A is negative semi-definite.
2. All eigenvalues of A are non-positive.
3. All pivots are non-positive. I.e., we have LDU decomposition $A = -LDL^*$ and the diagonal entries in D are all non-positive.
4. All leading principal submatrices are negative semi-definite.
5. All principal submatrices are negative semi-definite.
6. (The determinant of leading principal matrices are no longer useful.)
7. All $k \times k$ principal matrices have non-positive determinant for odd n, and non-negative determinant for even n.
8. We have a lower triangular L with $A = -LL^*$. (But we allow L to be non-invertible.)
9. We have $A = -BB^*$ for B. (But we allow B to be non-invertible.)

Finally, for the indefinite case, the followings are all equivalent:

1. A is indefinite.

2. Some eigenvalues of A are positive and some are negative.

3. Either A has no LDU decomposition, or $A = LDL^*$ where the diagonal entries of D contains both positive and negative values.

In particular, if A is Hermitian but has NO LDU decomposition, then it is indefinite.
Let us see some interesting applications.

Example 8.11.31. Given a twice differentiable function $f : \mathbb{R} \to \mathbb{R}$, how to find a local maximum or minimum? We have the following traditional approach.

1. Solve $f'(x) = 0$ to find critical points, say we find out that $f'(x_0) = 0$.

2. If $f''(x_0) > 0$, then $x = x_0$ is a local minimum.

3. If $f''(x_0) < 0$, then $x = x_0$ is a local maximum.

4. If $f''(x_0) = 0$, then we failed and have no clue what would happen.

Now consider a twice differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$, say $f(x, y) = (x + y + 2)^4 + x^2 + y^2$. Note that its graph would be some surface in $\mathbb{R}^3$. How to find a local maximum or minimum? Note that f has two partial derivatives $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} : \mathbb{R}^2 \to \mathbb{R}$, both are functions with two variables. In our example, we have $\frac{\partial f}{\partial x}(x, y) = 4(x + y)^3 + 2x$ and $\frac{\partial f}{\partial y}(x, y) = 4(x + y)^3 + 2y$.

Then we see that f has four second derivatives, because $\frac{\partial f}{\partial x}$ has two derivatives and $\frac{\partial f}{\partial y}$ has two derivatives. We write these as $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}, \frac{\partial^2 f}{\partial y^2}$. Here the notation $\frac{\partial^2 f}{\partial x \partial y}$ means we take the y-derivative first, and then we take the x-derivative. And the notation $\frac{\partial^2 f}{\partial x^2}$ means we take the x-derivative twice.

Then they form a matrix, the Hessian of f, $H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$. In our case, we have $H(f)(x, y) = \begin{bmatrix} 12(x+y)^2 + 2 & 12(x+y)^2 \\ 12(x+y)^2 & 12(x+y)^2 + 2 \end{bmatrix}$. Hey, this is a real symmetric matrix!

Indeed, your future calculus class would show you that for all real continuously twice-differentiable f, we always have $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$, so $H(f)$ is always symmetric.

Now to find local minimum or local maximum, we do the following approach. (The proof is higher dimensional Taylor expansion, which should be in your future calculus class.)

1. Solve $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ to find critical points, say we find out that one critical point is $x = x_0, y = y_0$.

2. If $H(f)$ at $x = x_0, y = y_0$ is positive definite, then $x = x_0$ is a local minimum.

3. If $H(f)$ at $x = x_0, y = y_0$ is negative definite, then $x = x_0$ is a local maximum.

4. If $H(f)$ at $x = x_0, y = y_0$ is other cases, then we failed and have no clue what would happen.

In our case, solving $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ gives $x = y = 0$, so this is the only critical point. At this point, $H(f) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is positive definite. So we have a local minimum. ☺

8.11.4 (Optional) Some multivariable calculus

These stuff should be covered in any multivariable calculus. However, since they are related to our class, I will put the proof here to be more self-contained.

Theorem 8.11.32. *For any continuously twice differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, let $\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$, so it is a continuously differentiable function $\nabla f : \mathbb{R}^n \to \mathbb{R}^n$. Let Hf be the Hessian of f, so it is a continuous function $Hf : \mathbb{R}^n \to M_{n\times n}$.*

If for some $\boldsymbol{x}_0 \in \mathbb{R}^n$, we have $\nabla f(\boldsymbol{x}_0) = \mathbf{0}$ and $Hf(\boldsymbol{x}_0)$ positive/negative definite, then $\boldsymbol{x}_0$ is a local minimum/maximum of f.

Proof. In lack of calculus materials, this proof is somewhat conceptual and non-rigorous.

Say $n = 2$ and we are looking at a function $f(x, y)$. Then first we increase x by a tiny bit, and our one-variable calculus knowledge tell us that $f(x+\mathrm{d}x, y) = f(x, y) + \mathrm{d}x\frac{\partial f}{\partial x}(x, y) + \frac{1}{2}\mathrm{d}x^2\frac{\partial^2 f}{\partial x^2}(x, y)$ for $\mathrm{d}x \to 0$. (Obviously we are ignoring infinitesimals of degree 3 and above, so we treat things such as $\mathrm{d}x^3$ as zero.)

Now we increase y by a tiny bit. The left hand side is $f(x+\mathrm{d}x, y+\mathrm{d}y)$. On the right hand side, we have three terms.

The first term is $f(x, y+\mathrm{d}y)$ and it can be simplified to $f(x, y) + \mathrm{d}y\frac{\partial f}{\partial y}(x, y) + \frac{1}{2}\mathrm{d}y^2\frac{\partial^2 f}{\partial y^2}(x, y)$.

The second term is $\mathrm{d}x\frac{\partial f}{\partial x}(x, y+\mathrm{d}y)$, and it can be simplified to $\mathrm{d}x\frac{\partial f}{\partial x}(x, y) + \mathrm{d}x\,\mathrm{d}y\frac{\partial^2 f}{\partial y\partial x}(x, y)$.

Finally, the last term is $\frac{1}{2}\mathrm{d}x^2\frac{\partial^2 f}{\partial x^2}(x, y+\mathrm{d}t)$, and it can be simplified to $\frac{1}{2}\mathrm{d}x^2\frac{\partial^2 f}{\partial x^2}(x, y) + \frac{1}{2}\mathrm{d}x^2\,\mathrm{d}y\frac{\partial^3 f}{\partial y\partial x^2}(x, y)$. But the new extra term here has infinitesimal degree 3, so we ignore it. Hence this term is actually unchanged.

All in all, if the input changes from (x, y) to $(x, y) + (\mathrm{d}x, \mathrm{d}y)$, the out put is

$$f(x+\mathrm{d}x, y+\mathrm{d}y) = f(x, y) + \mathrm{d}x\frac{\partial f}{\partial x}(x, y) + \mathrm{d}y\frac{\partial f}{\partial y}(x, y) + \frac{1}{2}\mathrm{d}x^2\frac{\partial^2 f}{\partial x^2}(x, y) + \mathrm{d}x\,\mathrm{d}y\frac{\partial^2 f}{\partial y\partial x}(x, y) + \frac{1}{2}\mathrm{d}y^2\frac{\partial^2 f}{\partial y^2}(x, y).$$

Now use the fact that $\frac{\partial^2 f}{\partial y\partial x} = \frac{\partial^2 f}{\partial x\partial y}$, we can simplify above into the following version of higher dimensional Taylor expansion approximation, assuming $\mathrm{d}\boldsymbol{x} \to \mathbf{0}$.

$$f(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) = f(\boldsymbol{x}) + (\nabla f(\boldsymbol{x}))^{\mathrm{T}}\,\mathrm{d}\boldsymbol{x} + \frac{1}{2}\,\mathrm{d}\boldsymbol{x}^{\mathrm{T}}(Hf(\boldsymbol{x}))\,\mathrm{d}\boldsymbol{x}.$$

So if $\nabla f(\boldsymbol{x}) = 0$, this simplifies to $f(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) = f(\boldsymbol{x}) + \frac{1}{2}\mathrm{d}\boldsymbol{x}^{\mathrm{T}}(Hf(\boldsymbol{x}))\,\mathrm{d}\boldsymbol{x}$. This will always be larger than $f(\boldsymbol{x})$ if $\frac{1}{2}Hf(\boldsymbol{x})$ is positive definite, and this will always be smaller than $f(\boldsymbol{x})$ if $\frac{1}{2}Hf(\boldsymbol{x})$ is negative definite. □

As an extra remark, the idea here also proved $\frac{\partial^2 f}{\partial y\partial x} = \frac{\partial^2 f}{\partial x\partial y}$ by itself. On one hand, we have

$$f(x+\mathrm{d}x, y+\mathrm{d}y) = f(x, y) + \mathrm{d}x\frac{\partial f}{\partial x}(x, y) + \mathrm{d}y\frac{\partial f}{\partial y}(x, y) + \frac{1}{2}\mathrm{d}x^2\frac{\partial^2 f}{\partial x^2}(x, y) + \mathrm{d}x\,\mathrm{d}y\frac{\partial^2 f}{\partial y\partial x}(x, y) + \frac{1}{2}\mathrm{d}y^2\frac{\partial^2 f}{\partial y^2}(x, y).$$

But on the other hand, by symmetry we also have

$$f(x+\mathrm{d}x, y+\mathrm{d}y) = f(x, y) + \mathrm{d}x\frac{\partial f}{\partial x}(x, y) + \mathrm{d}y\frac{\partial f}{\partial y}(x, y) + \frac{1}{2}\mathrm{d}x^2\frac{\partial^2 f}{\partial x^2}(x, y) + \mathrm{d}x\,\mathrm{d}y\frac{\partial^2 f}{\partial x\partial y}(x, y) + \frac{1}{2}\mathrm{d}y^2\frac{\partial^2 f}{\partial y^2}(x, y).$$

Hence we indeed have $\frac{\partial^2 f}{\partial y\partial x} = \frac{\partial^2 f}{\partial x\partial y}$.

Alternatively, consider the following exposition for some more intuitions on this fact.

Example 8.11.33. Given a function on two variables $f(x, y)$, it does not has a single derivative function. Rather, it has partial derivatives. Its first derivatives are $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$. Sometimes in calculus people would write this as a vector $\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$, called the ***gradient*** of f.

What about the second derivatives? Each of the partitial derivative has two further partial derivatives. So we have four second derivatives, i.e., $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$. Here $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}(\frac{\partial f}{\partial y})$. Sometimes in calculus people would write this into a matrix $H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$, called the ***Hessian*** of f. This matrix would record all second order differential information about f.

However, an amazing phenomenon here is the fact that, when the function is continuously twice differentiable, then $H(f)$ is ALWAYS a symmetric matrix!

For example, consider $f(x, y) = x^2 \mathrm{e}^{xy}$. Then $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}(2x\mathrm{e}^{xy} + x^2 y \mathrm{e}^{xy}) = 2x^2\mathrm{e}^{xy} + (x^2\mathrm{e}^{xy} + x^3 y \mathrm{e}^{xy}) = 3x^2\mathrm{e}^{xy} + x^3 y \mathrm{e}^{xy}$, while $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}(x^3\mathrm{e}^{xy}) = 3x^2\mathrm{e}^{xy} + x^3 y \mathrm{e}^{xy}$. Indeed, the mixed derivatives are the same.

We do not provide a full proof here, but merely gives an intuition as to why this happens. Consider the graph of $f(x, y)$, which is some surface in $\mathbb{R}^3$. Suppose I am standing on this surface. Say I extend my arms on the x direction and lay my arm on the surface, then the slope of my arm records the value of $\frac{\partial f}{\partial x}$ at my location.

Now, I keep my arm in the x direction, and I walk in the y direction. Since my arm is laid on the surface, my arm's slope will change up and down depending on how the surface flows. In particular, if I walk in the y direction, the change in my x-directional arm's slope would be $\frac{\partial^2 f}{\partial y \partial x}$.

Say $\frac{\partial^2 f}{\partial y \partial x} > 0$. Then as I walk in the y-direction, my arm would raise, which indicates that in some positive x and positive y direction, there is a bump, which would raise my arm.

Now suppose I lay my arm in the y direction and walk in the x direction. Then the SAME bump in the positive x and positive y direction would again raise my arm. So we have $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} > 0$ because it is caused by the very same bump.

(Note that the assumptions that f is continuously twice differentiable is crutial. It means locally the function can be approximated by degree 2 polynomials, i.e., approximately some paraboloids where there is only one bump or pit. Multiple bumps or pits in the positive x and positive y direction would cause this symmetricity to fail.)

Here is also an alternative intuition. Suppose we start with $f(\boldsymbol{v})$. Consider a super tiny square with vertices $\boldsymbol{a} = \boldsymbol{v}, \boldsymbol{b} = \boldsymbol{v} + \mathrm{d}s\boldsymbol{e}_1, \boldsymbol{c} = \boldsymbol{v} + \mathrm{d}t\boldsymbol{e}_2, \boldsymbol{d} = \boldsymbol{v} + \mathrm{d}s\boldsymbol{e}_1 + \mathrm{d}t\boldsymbol{e}_2$. Now you can also see that one should have $\mathrm{d}s\,\mathrm{d}t\frac{\partial^2 f}{\partial x \partial y} = (f(\boldsymbol{d}) - f(\boldsymbol{b})) - (f(\boldsymbol{c}) - f(\boldsymbol{a})) = f(\boldsymbol{a}) + f(\boldsymbol{d}) - f(\boldsymbol{b}) - f(\boldsymbol{c})$. However, we can also see that $\mathrm{d}s\,\mathrm{d}t\frac{\partial^2 f}{\partial y \partial x} = (f(\boldsymbol{d}) - f(\boldsymbol{c})) - (f(\boldsymbol{b}) - f(\boldsymbol{a})) = f(\boldsymbol{a}) + f(\boldsymbol{d}) - f(\boldsymbol{b}) - f(\boldsymbol{c})$. So the two are the same.

Eitherway, we see that the Hessian of a matrix is always symmetric. From the analysis above, you might also notice that $H(f)$ is crucial in analysing the curvature of surfaces, which is very important in geometry or general relativity, which states that gravity is just curvature in space. ☺

8.12 Singular Value Decomposition

We now go back to the world of the real numbers. No more complex numbers. Futhermore, everything in this section should be about linear maps, not linear transformations. I.e., we think of the domain and codomain as DIFFERENT spaces. So we don't have to change basis simultaneously.

8.12.1 Introduction

Recall that for any linear map $A : \mathbb{R}^n \to \mathbb{R}^m$, we can always change basis so that $RAC = \begin{bmatrix} I_{r\times r} & O \\ O & O \end{bmatrix}$, where r is the rank of A. Now, suppose we only want to perform orthonormal change of basis. Can we still do this?

The Singular Value Decomposition (SVD) is doing exactly this.

Theorem 8.12.1. *For any $m \times n$ real matrix A, we can find $m \times m$ real orthogonal matrix U and $n \times n$ real orthogonal matrix V such that $A = U\Sigma V^{\mathrm{T}}$, where $\Sigma = \begin{bmatrix} D & O \\ O & O \end{bmatrix}$ and D is a real $r \times r$ diagonal matrix with positive diagonal entries, and r is the rank of A.*

In the introduction section, we do not prove this theorem. But first of all, let us get some perspective about why this is useful and important, even for square matrices.

Example 8.12.2. Given a 2×2 matrix A, we know A would map circles in $\mathbb{R}^2$ into ellipses. However, what is the resulting eccentricity fo these ellipses?

We have already seen that when $A = XDX^{-1}$ with $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 2 & \\ & 3 \end{bmatrix}$, then the unit circle would be mapped to an ellipse whose major half-axis has length > 3 and minor half-axis has length < 2. Where is the direction and length of the major half-axis and the minor half-axis?

Suppose now we have a decomposition $A = U\Sigma V^{\mathrm{T}}$, note that in this case, Σ is a genuin diagonal matrix. Now the U, V are both rotations or reflections, and Σ is a stretching along the coordinate axis. So V^{T} would do nothing to the unit circle. Then Σ would stretch the unit circle along coordinate axis to get an ellipse. Finally U rotate/reflect the ellipse. As a result, it is very easy to see that the half-axis lengths are exactly the diagonal entries of Σ. These values are called the singular values of A. As you can see, eigenvalues do NOT define the shape of the resulting ellipse. Only singular values could define their shapes.

A related question is this: suppose I squeeze a ping pong ball in several directions, where will the ping pong ball most likely crack? This is slightly more complicated, but the idea is similar: the forces are not necessarily orthogonal, but their combined action will push the ping pong ball to deform into elliptic shape (ellipsoid). Singluar value decomposition will help us identify the shape of this ellipsoid, and the shape of deformation would tell us where the ping pong ball would crack. ☺

Example 8.12.3. The singular value decomposition is also related to other decompositions.

Suppose we have $A = U\Sigma V^{\mathrm{T}}$ for a square matrix A. Writing this decomposition as $A = (U\Sigma U^{\mathrm{T}})(UV^{\mathrm{T}})$. This is called a polar decomposition, since $U\Sigma U^{\mathrm{T}}$ must have real (and in fact non-negative) eigenvalues, and UV^{T} must have eigenvalues that are unit complex numbers. This is also a matrix analogue of the complex number polar decomposition $z = re^{\mathrm{i}\theta}$. ☺

Now let us finally consider this decomposition from a rank perspective. In many applicaitions of linear algebra, it is veru beneficial to write a rank r matrix as the sum of r matrices of rank one.

Example 8.12.4. Consider the France map. If we use number 1 for blue, 2 for red, and 0 for white, then the France map is something like $M = \begin{bmatrix} 1 & \dots & 1 & 0 & \dots & 0 & 2 & \dots & 2 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 1 & \dots & 1 & 0 & \dots & 0 & 2 & \dots & 2 \end{bmatrix}$. If such a matrix is $m \times n$, then we would need to store mn numbers in the computer.

However, note that such a matrix must have rank one. We must have $M = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ where $\boldsymbol{u} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} 1 & \dots & 1 & 0 & \dots & 0 & 2 & \dots & 2 \end{bmatrix}$. And to store $\boldsymbol{u}, \boldsymbol{v}$, we only need $m + n$ numbers. This way, we saved a lot of memory space in our computer!

Consider the Benin flag. If 3 is green and 4 is yellow, then the 2×3 Benin flag would look like $\begin{bmatrix} 3 & 4 & 4 \\ 3 & 2 & 2 \end{bmatrix}$. In general, the $m \times n$ version of this flag always have rank 2, and we have $\begin{bmatrix} 3 & 4 & 4 \\ 3 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 3 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 4 \\ 0 & 2 & 2 \end{bmatrix}$, the sum of two rank one matrices. Since each rank one matrix must be $\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ for some $\boldsymbol{u}, \boldsymbol{v}$, we see that we only need to store $2(m + n)$ numbers instead of mn numbers.

In general, for an $m \times n$ matrix with rank r, we do not store mn numbers in the computer. Rather, we would only store $r(m+n)$ numbers.

You can also check out the Greece flag, which has rank three. I have not found any rank four flag yet, so please let me know if you find any. Finally, the Chinese flag and the American flag both have infinite rank. ☺

The example above does not involve SVD yet. However, you can think of SVD as the best rank one decomposition.

Consider the nature of SVD for an $m \times n$ matrix $A = U\Sigma V^{\mathrm{T}}$. Suppose $U = \begin{bmatrix}\boldsymbol{u}_1 & \dots & \boldsymbol{u}_m\end{bmatrix}, V = \begin{bmatrix}\boldsymbol{v}_1 & \dots & \boldsymbol{v}_n\end{bmatrix}$, and $\Sigma = \begin{bmatrix}\sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & O\end{bmatrix}$ for nonzero $\sigma_1, \dots, \sigma_r$. Since U, V are invertible, it is easy to see that A has rank exactly r. Furthermore, $A = \sum \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$. So essentially, we are decomposing A into a linear combination of rank one matrices.

Furthremore, consider any two such rank one matrices, say $\sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}, \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^{\mathrm{T}}$ for any $i \neq j$. Then we would have $\boldsymbol{u}_i \perp \boldsymbol{u}_j$ and $\boldsymbol{v}_i \perp \boldsymbol{v}_j$ by construction. So the two rank one matrices are "as orthogonal to each other as possible". So SVD is decomposing A into a linear combination of "mutually orthogonal" rank one matrices!

8.12.2 The fundation of SVD

How can we obtain the singular value decomposition? Again, note that A is NOT necessarily square.

Let us do some backward observation. Suppose $A = U\Sigma V^{\mathrm{T}}$ is possible, with $\Sigma = \begin{bmatrix}\sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & O\end{bmatrix}$. We usually assume that $\sigma_1 \geq \dots \geq \sigma_r > 0$.

A very important observation is that $A^{\mathrm{T}}A = V\Sigma^{\mathrm{T}}\Sigma V^{\mathrm{T}}$. Note that $\Sigma^{\mathrm{T}}\Sigma$ is diagonal and square, and V is orthogonal, so this is in fact the spectral decomposition for the symmetric matrix $A^{\mathrm{T}}A$. So to get V, you can simply find all eigenvectors for $A^{\mathrm{T}}A$ and you are done. Furthermore, we have $\Sigma^{\mathrm{T}}\Sigma = \begin{bmatrix}\sigma_1^2 & & & \\ & \ddots & & \\ & & \sigma_r^2 & \\ & & & O\end{bmatrix}$. So these $\sigma_1, \dots, \sigma_r$ in Σ are simple the square roots of the non-zero eigenvalues of $A^{\mathrm{T}}A$. Fortunately, we know $A^{\mathrm{T}}A$ must always be positive semidefinite, so the square roots are all real.

In short, the spectral decomposition of $A^{\mathrm{T}}A$ will give you both Σ and V. Similarly, the spectral decomposition of AA^{T} will give you both U and Σ. Is this enough to find the whole SVD? Well, there is one more missing ingredient.

Example 8.12.5. Consider $A = \begin{bmatrix}1 & \\ & 1\end{bmatrix}$ and $B = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix}$. Then $AA^{\mathrm{T}} = BB^{\mathrm{T}}$ and $A^{\mathrm{T}}A = B^{\mathrm{T}}B$. However, A, B are different matrices with different SVD.

In particular, in terms of rank decomposition, the SVD of A is $A = \boldsymbol{e}_1\boldsymbol{e}_1^{\mathrm{T}} + \boldsymbol{e}_2\boldsymbol{e}_2^{\mathrm{T}}$, while the SVD for B is $B = \boldsymbol{e}_2\boldsymbol{e}_1^{\mathrm{T}} + \boldsymbol{e}_1\boldsymbol{e}_2^{\mathrm{T}}$. As you can see, the U portion and V portion of the SVD for A and B uses the same columns, but their MATCHING is different.

AA^{T} tells you what columns U should have, and $A^{\mathrm{T}}A$ tells you what columns V should have, but they give no information on how to order these things to match each other. It turns out that the order of columns for U, V are very important. Recall the decomposition $A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$, where the $\boldsymbol{u}_i, \boldsymbol{v}_i$ are in fact paired up. So the order of columns of U and of V must MATCH each other. ☺

A very interesting observation is that we also see that $A^{\mathrm{T}}A$ and AA^{T} has the same positive eigenvalues $\sigma_1^2, \dots, \sigma_r^2$. Can you see why without SVD? (Hint: How are the eigenvalues of AB and BA related? Note that one matrix is $m \times m$ and the other is $n \times n$, they do not have the same amounts of eigenvalues.)

Definition 8.12.6. *A* ***singluar value*** $\sigma > 0$ *for* A *is the square root of a non-zero eigenvalue of* $A^{\mathrm{T}}A$ *or* AA^{T}*.*

Given an $m \times n$ *matrix* A*, we say* $\boldsymbol{v} \in \mathbb{R}^n$ *is a* ***right singular vector*** *if it is an eigenvector for* $A^{\mathrm{T}}A$*. (This is related to the* V *in* $A = U\Sigma V^{\mathrm{T}}$*, hence it is "right".)*

We say it is a singular vector of A *for the singluar value* $\sigma > 0$ *if it is an eigenvector of* $A^{\mathrm{T}}A$ *for the eigenvalue* σ^2*.*

We say $\boldsymbol{u} \in \mathbb{R}^m$ *is a* ***left singular vector*** *if it is an eigenvector of* AA^{T}*. (This is related to the* U *in* $A = U\Sigma V^{\mathrm{T}}$*, hence it is "left".)*

We say it is a singular vector of A *for the singluar value* $\sigma > 0$ *if it is an eigenvector of* AA^{T} *for the eigenvalue* σ^2*.*

How to work out the correspondence between the left and right singular vectors?

Again let us do some backward deduction. Consider $A\boldsymbol{v}_i$. Since all other $\boldsymbol{v}_j$ are orthogonal to $\boldsymbol{v}_j$, via the decomposition $A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$ we see that in fact $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ for $i \leq r$. So actually $\boldsymbol{v}_i$ gives us $\boldsymbol{u}_i$ right away when $i \leq r$. Similarly, we can also see that $\boldsymbol{u}_i^{\mathrm{T}} A = \sigma_i \boldsymbol{v}_i^{\mathrm{T}}$.

Lemma 8.12.7 (Ping Pong of left/right singular vectors)**.** *If* $\boldsymbol{v}$ *is a right singular vector of* A *for the singular value* σ*, then* $A\boldsymbol{v}$ *is a left singular vector of* A *for the singular value* σ*.*

If $\boldsymbol{u}$ *is a left singular vector of* A *for the singular value* σ*, then* $A^{\mathrm{T}}\boldsymbol{u}$ *is a right singular vector of* A *for the singular value* σ*.*

Proof. Suppose $\boldsymbol{v}$ is a right singular vector for the singular value σ. Now $(AA^{\mathrm{T}})(A\boldsymbol{v}) = A(A^{\mathrm{T}}A\boldsymbol{v}) = A(\sigma^2\boldsymbol{v}) = \sigma^2 A\boldsymbol{v}$. Hence $A\boldsymbol{v}$ is a left singular vector of A for the singular value σ. The other statement is proven similarly. □

Lemma 8.12.8. *For a right singular vector* $\boldsymbol{v}$ *for the singular value* σ*,* $\|A\boldsymbol{v}\| = \sigma\|\boldsymbol{v}\|$*. If* $\boldsymbol{v}_1, \boldsymbol{v}_2$ *are orthogonal right singular vectors, then* $A\boldsymbol{v}_1, A\boldsymbol{v}_2$ *are orthogonal left singular vectors.*

Proof. For a right singular vector $\boldsymbol{v}$ for the singular value σ, $\|A\boldsymbol{v}\|^2 = \boldsymbol{v}^{\mathrm{T}}A^{\mathrm{T}}A\boldsymbol{v} = \sigma^2\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}$.

If $\boldsymbol{v}_1, \boldsymbol{v}_2$ are orthogonal right singular vectors, then $\langle A\boldsymbol{v}_1, A\boldsymbol{v}_2\rangle = \boldsymbol{v}_1^{\mathrm{T}}(A^{\mathrm{T}}A\boldsymbol{v}_2) = \sigma^2\boldsymbol{v}_1^{\mathrm{T}}\boldsymbol{v}_2$ for some $\sigma > 0$. Since $\boldsymbol{v}_1, \boldsymbol{v}_2$ are orthogonal, the calculation above gives zero. □

Theorem 8.12.9 (Singular Value Decomposition)**.** *For any real* $m \times n$ *matrix* A *of rank* r*, we can find* $m \times m$ *orthogonal matrix* $U = \begin{bmatrix}\boldsymbol{u}_1 & \dots & \boldsymbol{u}_m\end{bmatrix}$*, and* $n \times n$ *orthogonal matrix* $V = \begin{bmatrix}\boldsymbol{v}_1 & \dots & \boldsymbol{v}_n\end{bmatrix}$*, and "diagonal"* $m \times n$ *matrix* $\Sigma = \begin{bmatrix}\sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & O\end{bmatrix}$ *with positive* $\sigma_1, \dots, \sigma_r$*, such that* $A = U\Sigma V^{\mathrm{T}}$*.*

Here $\boldsymbol{u}_i$ *is a left singular vector for singular value* σ_i *for* $i \leq r$*, and* $\boldsymbol{u}_i$ *is in* $\mathrm{Ker}(A^{\mathrm{T}})$ *if* $i > r$*. Similarly,* $\boldsymbol{v}_i$ *is a right singular vector for singular value* σ_i *for* $i \leq r$*, and* $\boldsymbol{v}_i$ *is in* $\mathrm{Ker}(A)$ *if* $i > r$*.*

Proof. Since eigenvalues of $A^{\mathrm{T}}A$ which are all non-negative, and $A^{\mathrm{T}}A$ have the same rank as A, let us order the eigenvalues from large to small as $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0$. Then we have spectral decomposition $A^{\mathrm{T}}A = VDV^{\mathrm{T}}$ where V is orthogonal and D is diagonal, with diagonal entries $\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0$. And columns of V are right singular vectors by definition.

Let $\boldsymbol{u}_i = \frac{1}{\sigma_i}A\boldsymbol{v}_i$ for all $i \leq r$. Now we have mutually orthogonal unit vectors $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$, we extend this arbitrarily to an orthonormal basis for the codomain $\mathbb{R}^m$. So now we have $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$, and $U = \begin{bmatrix}\boldsymbol{u}_1 & \dots & \boldsymbol{u}_m\end{bmatrix}$ is an orthogonal matrix.

Now $A\begin{bmatrix}\boldsymbol{v}_1 & \dots & \boldsymbol{v}_r & \boldsymbol{v}_{r+1} & \dots & \boldsymbol{v}_n\end{bmatrix} = \begin{bmatrix}\sigma_1\boldsymbol{u}_1 & \dots & \sigma_r\boldsymbol{u}_r & O\end{bmatrix} = U\Sigma$. So $AV = U\Sigma$. So we are done. □

Alternative proof. This proof is a bit more elegant, but lacks some finesse.

Again start by performing any spectral decomposition $A^{\mathrm{T}}A = VDV^{\mathrm{T}}$. Then we see that $(AV)^{\mathrm{T}}(AV) = D$. In particular, columns of AV are mutually orthogonal. Then by a generalized version of QR decomposition, we have $AV = U\Sigma$ where U is orthogonal and Σ is "diagonal" as desired. (We obmit this proof of the generalized QR decomposition. It is strikingly similar to the last proof....) $\square$

Remark 8.12.10. *Note that $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ are all in $\mathrm{Ran}(A)$ since $\boldsymbol{u}_i = \frac{1}{\sigma_i}A\boldsymbol{v}_i$, and $\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ are all in $\mathrm{Ker}(A^{\mathrm{T}})$.*

Given that the two subspaces are orthogonal complements, and given their dimensions, it is not hard to see that $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$ in fact form an orthonormal basis for $\mathrm{Ran}(A)$, while $\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ form an orthonormal basis for $\mathrm{Ker}(A^{\mathrm{T}})$.

The same thing is true for $\boldsymbol{v}_i$ and the subspaces $\mathrm{Ran}(A^{\mathrm{T}}), \mathrm{Ker}(A)$. In particular, SVD means simultaneously finding the BEST orthonormal basis for ALL subspaces related to A.

A common geometric interpretation of SVD is this: $A = U\Sigma V^{\mathrm{T}}$ means A is rotation, then stretch, and the rotation. However, here is another one.

Corollary 8.12.11 (Polar decomposition)**.** *For any square matrix A, we have $A = SQ$ where S is symmetric positive semi-definite and Q is orthogonal.*

Proof. $A = U\Sigma V^{\mathrm{T}} = (U\Sigma U^{\mathrm{T}})(UV^{\mathrm{T}})$. $\square$

This is akin to $z = r\mathrm{e}^{\mathrm{i}\theta}$ for complex numbers, where $r \geq 0$ and $\mathrm{e}^{\mathrm{i}\theta}$ is unit complex. Recall that a symmetric matrix is just a stretch along orthogonal frames. So the geometric action of A is simply a rotation, and the stretch along orthogonal frams (where the stretching factors are the singular values, and the stretching directions are columns of U). In particular, we have the following.

Corollary 8.12.12. *If an $n \times n$ matrix A acts on a unit n-dim sphere, the result is an n-dim ellipsoid whose half-axes have length the same as singular values of A, and the direction of these axes are the same as columns of U for the SVD $A = U\Sigma V^{\mathrm{T}}$.*

In this sence, if we are thinking of A as a linear transformation, then usually U is more prominent in its interpretation.

Example 8.12.13. Consider a Ping Pong ball squeezed by 10 different forces. Where on the ball would crack first?

Well, under these 10 forces, the Ping Pong ball would be "deformed", and the location with the most deformation would be the most likely to crack. The deformed Ping Pong ball could imaginably be some ellipsoid, and the location with most deformation could be determined by the direction of the half-axes of the ellipsoid. As you can imagine, this must be related to SVD some how!

Skipping some details on mechanical analysis involving stress tensor, if the 10 forces are $\boldsymbol{F}_1, \ldots, \boldsymbol{F}_{10}$, then we can set $A = \begin{bmatrix} \frac{\boldsymbol{F}_1}{\sqrt{\|\boldsymbol{F}_1\|}} & \cdots & \frac{\boldsymbol{F}_{10}}{\sqrt{\|\boldsymbol{F}_{10}\|}} \end{bmatrix}$. The SVD for A would yeild these half-axes and the amount of corresponding deformation.

Now instead of a Ping Pong ball under pressure, imagine your head, which is under pressure due to all the classes you are taking. The math class, the science class, the literary class, they each gives you some different pressure. Which class would you most likely fail? I don't know, but I'm sure SVD is involved in this somehow. ☺

We finally end this section with a WARNING: While the singular values and singular vectors are all well defined for A, the decomposition of SVD might NOT be unique. Consider $I = UIU^{\mathrm{T}}$ for any orthogonal matrix U. We see that the U portion is arbitrary.

8.12.3 (Optional) Pseudo Inverse

Given $A = U\Sigma V^{\mathrm{T}}$, where $\Sigma = \begin{bmatrix} D & O \\ O & O \end{bmatrix}$ is an $m \times n$ matrix and D is invertible and diagonal, then we define $\Sigma^{+} = \begin{bmatrix} D^{-1} & O \\ O & O \end{bmatrix}$ which is $n \times m$, and we define $A^{+} = V\Sigma^{+}U^{\mathrm{T}}$ to be the ***pseudo-inverse*** of A.

Proposition 8.12.14. *AA^{+} is the orthogonal projection to* $\mathrm{Ran}(A)$.

Proof. Note that $AA^{+} = U\Sigma\Sigma^{+}U^{\mathrm{T}}$, and $\Sigma\Sigma^{+} = \begin{bmatrix} I_r & O \\ O & O \end{bmatrix}$ is an $m \times m$ square diagonal matrix. Therefore AA^{+} is obviously symmetric, and $(AA^{+})^2 = U(\Sigma\Sigma^{+})^2U^{\mathrm{T}} = U\Sigma\Sigma^{+}U^{\mathrm{T}} = AA^{+}$. So this is an orthogonal projection.

We also see that U represents a change of basis in the product $U\Sigma\Sigma^{+}U^{\mathrm{T}}$, and $\Sigma\Sigma^{+} = \begin{bmatrix} I_r & O \\ O & O \end{bmatrix}$ means a truncation taking only the first r coordinates. So, AA^{+} is a projection to $\mathrm{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r) = \mathrm{Ran}(A)$. □

A very nice aplication is the following. Given a subspace V of $\mathbb{R}^n$, each vector $\boldsymbol{v} \in V$ originally has n coordinates, because it in $\mathbb{R}^n$. However, once we find a basis for V, then $\boldsymbol{v}$ as a vector of V will now have k coordinates under this basis. What is the transition map?

Corollary 8.12.15. *Suppose we have a subspace V in $\mathbb{R}^n$ spanned by linearly independent $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in \mathbb{R}^n$. Then for any $\boldsymbol{w} \in V$, its coordinate under the basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ of V is $A^{+}\boldsymbol{w}$, where $A = \begin{bmatrix} \boldsymbol{v}_1 & \ldots & \boldsymbol{v}_k \end{bmatrix}$.*

Proof. If $\boldsymbol{w} \in V$, then $\boldsymbol{w} = AA^{+}\boldsymbol{w}$, so $\boldsymbol{w}$ is the linear combination of columns of A according to coefficients $A^{+}\boldsymbol{w}$. □

8.12.4 Low Rank Approximation

Consider the rank one decomposition induced by SVD, $A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$. Recall that we order the singular values so that $\sigma_1 \geq \cdots \geq \sigma_r > 0$. Intuitively, you can think of these $\sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$ as components of A. Obviously, $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}}$ is the most important component, and $\sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^{\mathrm{T}}$ is the second most important component, and so on. Finally, the orthogonality between these components means they are recording some independent aspects of A.

Example 8.12.16. Suppose I have a bunch of data to store in the computer. The data are in the form of an $m \times n$ matrix A, say maybe it is a gray-scale picture. Then I would need to store mn numbers in the computer. But I do not have enough data storage to store these many numbers. What should I do?

As we have noted before, if A have a small rank r, then any rank one decomposition $A = \sum \boldsymbol{x}_i \boldsymbol{y}_i^{\mathrm{T}}$ will store A using only $r(m+n)$ numbers. This is super good when r is tiny.

What if A have big rank? Then to store everything, you have no choice but to store all mn numbers. However, maybe we can find a low rank matrix B, such that B is super close to A, and store B instead? The information in A will be lost somewhat, but B should give a good approximation, and it needs much less space to store.

This is the problem of low rank approximation. Fix a small integer k. Given any matrix A, we want to find the best rank k matrix that is "closest" to A. What is the answer?

Well, here is the answer. If the SVD gives $A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$, then the best rank k approximation is $\sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$. (Recall that we order the singular values so that $\sigma_1 \geq \cdots \geq \sigma_r > 0$. So we are simply taking the largest k rank one components of SVD, and we are done.)

Say you are preparing for an exam. There is a lot to memorize, but your memory is finite and probably not enough. What should you do? Ideally, you should perform SVD on all the materials to be memorized, and memorize only the rank one components corresponding to the largest few singular values. This would allow you to most efficiently approximate all the materials given your limited memory capacity. If your teacher tells you some "key points" to memorize in the textbook, those are probably $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}}$, the most important portion to memorize.

Also notice how orthogonality is the key to efficiency. If you memorize two sentences that are essentially saying the same thing, then you wasted some memory space. The orthogonality of SVD guarantees that each $\sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$ contains totally independent informations from others, so there is no waste whatsoever. ☺

Now, we want to find the best low rank matrix that is closest to A. But how to measure closeness? One common way is the Frobenius norm.

Definition 8.12.17. *Given two $m \times n$ real matrices A, B, we can perform their inner product such that $\langle A, B\rangle = \sum a_{ij} b_{ij}$, i.e., this is simply entry-wise "dot product". Then the Frobenius norm of the matrix A is simply $\|A\|_F = \sqrt{\langle A, A\rangle}$.*

Lemma 8.12.18. $\langle A, B\rangle = \operatorname{trace}(A^{\mathrm{T}} B)$*, and* $\|A\|_F^2 = \operatorname{trace}(A^{\mathrm{T}} A)$.

Let us have another application of low rank approximation: lines of best fit.

Example 8.12.19. Consider data points $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \ldots, \begin{bmatrix} x_n \\ y_n \end{bmatrix}$ on the plane $\mathbb{R}^2$. Find the best line to fit the data.

Now previously, we have a least square method, which is trying to find the best k, b to minimize $\sum(y_i - kx_i - b)^2$. In short, we are minimizing the ***vertical distance*** to the line. The least square is used when we are trying to PREDICT y using x, hence vertical distance to the line is more important.

However, sometimes we are not really interested in prediction, and we only want to look at correlation. So I want to minimize the ***orthogonal distance*** to the line. This is called ***the line of best fit***. What should I do then?

First we shift the data so that the average values are $\mathbb{E}(x) = \mathbb{E}(y) = 0$, i.e., my data is centered around the origin. This way, I just need to find the best line through the origin to fit the data.

Consider any line through the origin L. Then given data points $A = \begin{bmatrix} \boldsymbol{p}_1, \ldots, \boldsymbol{p}_n \end{bmatrix}$ which is a $2 \times n$ matrix, we can project each point orthogonally to L, and obtain results $B = \begin{bmatrix} \boldsymbol{q}_1, \ldots, \boldsymbol{q}_n \end{bmatrix}$. Since each point $\boldsymbol{p}_i$ is projected to the point $\boldsymbol{q}_i$ on L, we see that the orthogonal distance from the points in A to the line L is $\sum \|\boldsymbol{p}_i - \boldsymbol{q}_i\|^2 = \operatorname{trace}((A-B)^{\mathrm{T}}(A-B)) = \|A - B\|_F$. So this is simply the Frobenius distance between A and B.

Also note that since all $\boldsymbol{q}_i$ lies on the same line L through the origin, B has parallel columns, and it has rank one. So in short, to find the best line L, I just need to find the best rank one matrix B such that $\|A - B\|_F$ is as small as possible. B should be a best rank one approximation to A.

So, if the SVD gives $A = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$, the best rank one approximation is $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}}$, so this is what B should be. $L = \operatorname{Ran}(B)$ so it is spanned by $\boldsymbol{u}_1$. We have found our line of best fit.

Now suppose the data points are $A = \begin{bmatrix} \boldsymbol{p}_1, \ldots, \boldsymbol{p}_n \end{bmatrix}$ but the points are now in $\mathbb{R}^3$. Again suppose $\mathbb{E}(x) = \mathbb{E}(y) = \mathbb{E}(z) = 0$. How to find the line of best fit? Again you just need the best rank one approximation of A. What if you want to find a plane to best fit the data? Well, you just need the best rank two approximation of A.

So this is why SVD is super important in statistics. ☺

8.12.5 Matrix norms and proofs of low rank approximation

We now provide the proof for low rank approximation. First, let us introduce another matrix norm, which is also super useful, especially in physics

Consider a linear map A acting on the unit circle. This means we have an input $\boldsymbol{u}$, some unit vector, and we are interested in $A\boldsymbol{u}$. Since we know the result is an ellipse, how to find the length of the major half-axis of the ellipse? Essentially, we are asking ourselves how to find $\max\|A\boldsymbol{u}\|$.

Definition 8.12.20. *Given a matrix A, its **operator norm** is $\|A\| = \max_{\boldsymbol{v} \neq \boldsymbol{0}} \frac{\|A\boldsymbol{v}\|}{\|\boldsymbol{v}\|} = \max_{\|\boldsymbol{u}\|=1} \|A\boldsymbol{u}\|$. In particular, this is the maximal output-input length ratio.*

This is called a norm for a reason. For example, here is the triangle inequality:

Lemma 8.12.21. $\|A + B\| \leq \|A\| + \|B\|$.

Proof. For any unit vector $\boldsymbol{u}$, we have $\|(A+B)\boldsymbol{u}\| = \|A\boldsymbol{u}+B\boldsymbol{u}\| \le \|A\boldsymbol{u}\|+\|B\boldsymbol{u}\| \le \|A\|+\|B\|$. Since this is true for all $\|(A+B)\boldsymbol{u}\|$, it is also true for their maximum value $\|A+B\|$. $\square$

The main advantage of this norm over the Frobenius norm is that it can be easily generalized to infinite dimensional settings. This is especially useful in physics.

Now, we are interested in the output-input length ratio $\frac{\|A\boldsymbol{v}\|}{\|\boldsymbol{v}\|}$. If you square this, this is $\frac{\boldsymbol{v}^{\mathrm{T}}A^{\mathrm{T}}A\boldsymbol{v}}{\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}}$. This is called a Rayleigh quotient.

Definition 8.12.22. *Given a symmetric matrix S, a Rayleigh quotient is a quotient $\frac{\boldsymbol{v}^{\mathrm{T}}S\boldsymbol{v}}{\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}}$ for some $\boldsymbol{v}$.*

Proposition 8.12.23. *The Rayleigh quotient of S is always between the largest eigenvalue of S and the smallest eigenvalue of S.*

Proof. $a \le \frac{\boldsymbol{v}^{\mathrm{T}}S\boldsymbol{v}}{\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}} \le b$ for all $\boldsymbol{v}$
if and only if $a\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v} \le \boldsymbol{v}^{\mathrm{T}}S\boldsymbol{v} \le b\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}$ for all $\boldsymbol{v}$,
if and only if $\boldsymbol{v}^{\mathrm{T}}(aI-S)\boldsymbol{v} \le 0 \le \boldsymbol{v}^{\mathrm{T}}(bI-S)\boldsymbol{v}$ for all $\boldsymbol{v}$,
if and only if $aI-S$ is negative semidefinite and $bI-S$ is positive semidefinite,
if and only if all eigenvalues of S are between a and b. $\square$

Now we connect the operator norm to singular values.

Proposition 8.12.24. *The operator norm of A is its largest singular value σ_1.*

Proof. Since $\|A\boldsymbol{v}_1\| = \sigma_1\|\boldsymbol{v}_1\|$, we see that $\|A\| \ge \sigma_1$.
For any $\boldsymbol{v} \ne \boldsymbol{0}$, $\frac{\|A\boldsymbol{v}\|^2}{\|\boldsymbol{v}\|^2} = \frac{\boldsymbol{v}^{\mathrm{T}}A^{\mathrm{T}}A\boldsymbol{v}}{\boldsymbol{v}^{\mathrm{T}}\boldsymbol{v}} \le \lambda_1(A^{\mathrm{T}}A) = \sigma_1^2$. So $\|A\| \le \sigma_1$. $\square$

Alternative proof. First you can prove that $\|UAV\| = \|A\|$, because orthogonal matrices preserve length. Now we do SVD and have $\|A\| = \|U\Sigma V^{\mathrm{T}}\| = \|\Sigma\|$. Now Σ is a very specific matrix and you can simply calculate directly. $\square$

As a notation, we shall now use $\|A\|$ and $\sigma_1(A)$ interchangably, as they refer to the same thing.

Corollary 8.12.25. *Let A acts on an n-dimensional sphere, then the resulting high dimensional ellipsoid has major half-axis length σ_1. (Note that depending on A being injective or not, the resulting ellipsoid might NOT be n dimensional.)*

Remark 8.12.26. *From the ellipsoid result, one can also see the following: If, instead of looking at $\max \frac{\|A\boldsymbol{v}\|}{\|\boldsymbol{v}\|}$, we look at $\max_{\boldsymbol{v}\perp\boldsymbol{v}_1} \frac{\|A\boldsymbol{v}\|}{\|\boldsymbol{v}\|}$ where $\boldsymbol{v}_1$ is the right singular vector for σ_1, then we would get σ_2, i.e. the largest length in the cross section perpendicular to the major half-axis of the ellipsoid. This can be generalized for all singular values.*

Here is a related potential application of Rayleigh quotient.

Example 8.12.27. How to find the maximum and minimum of $\frac{2x^2+4xy+5y^2}{x^2+y^2}$, given that x, y are not both zero?

Well, set $S = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$, then this is a Rayleigh quotient for S. Since S has eigenvalues $1, 6$, this Rayleigh quotient must be between 1 and 6. ☺

Now we go for low rank approximation.

Definition 8.12.28. *Suppose the SVD of A gives $A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$. The **rank** k **truncated SVD** of A for $k \le r$ is $A_k = \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$.*

As we shall see A_k will minimize both the operator norm and the Frobenius norm of $A - A_k$. We do the operator norm first.

Proposition 8.12.29. *Among the set of rank k matrices, A_k is one of the closest to A in terms of operator norm. Specifically, $\|A-B\| \geq \sigma_{k+1}(A)$ for all $m \times n$ rank k matrices B, and $\|A-A_k\|$ reaches this minimal value.*

(Here $\sigma_{k+1}(A)$ refers to the $k+1$-th singular value of A, and if A is rank k, $\sigma_{k+1}(A) = 0$.)

Proof. That A_k gives $\|A - A_k\| = \sigma_{k+1}(A)$ is obvious.

If $k = r$, then the statement is trivial as $A_k = A$. Let us assume $k < r$.

(Before the formal proof process, let us do some analysis. We want to show that $\|A-B\| \geq \sigma_{k+1}(A)$. To achieve this, we need to find a unit vector $\boldsymbol{x}$ such that $\|(A-B)\boldsymbol{x}\| \geq \sigma_{k+1}(A)$. So given fixed B, we want to pick unit vector $\boldsymbol{x}$ to make $A\boldsymbol{x}$ and $B\boldsymbol{x}$ to be as far away as possible. As B has low rank, $B\boldsymbol{x}$ is very likely to be zero, in which case we just want $\|A\boldsymbol{x}\|$ to be as large as possible. This would happen if $\boldsymbol{x}$ is a mixture of the singular vectors of A for large singular values.)

To the formal proof now. Suppose $A = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$, where the singular values are ordered from large to small as usual. Let $V_{k+1} = \begin{bmatrix} \boldsymbol{v}_1 & \dots & \boldsymbol{v}_{k+1} \end{bmatrix}$, so these are singular vectors of A for the $(k+1)$ largest singular values. Obviously $\dim \mathrm{Ran}(V_{k+1}) = k+1$, as all these singular vectors are orthonormal.

On the other hand, $\mathrm{Ker}(B) = n - \mathrm{rank}(B) = n - k$. So by inclusion-exclusion principle for subspaces, $\dim(\mathrm{Ker}(B) \cap \mathrm{Ran}(V_{k+1})) \geq 1$. So we pick $\boldsymbol{x}$ to be a unit vector in $\mathrm{Ker}(B) \cap \mathrm{Ran}(V_{k+1})$.

Now $B\boldsymbol{x} = \boldsymbol{0}$, and $\|(A-B)\boldsymbol{x}\| = \|A\boldsymbol{x}\|$. If $\boldsymbol{x} = a_1\boldsymbol{v}_1 + \cdots + a_{k+1}\boldsymbol{v}_{k+1}$, then $A\boldsymbol{x} = \sigma_1 a_1 \boldsymbol{u}_1 + \cdots + \sigma_{k+1} a_{k+1} \boldsymbol{u}_{k+1}$. Since these $\boldsymbol{u}_i$ are orthonormal, we see that $\|A\boldsymbol{x}\| = \sqrt{\sigma_1^2 a_1^2 + \cdots + \sigma_{k+1}^2 a_{k+1}^2} \geq \sqrt{\sigma_{k+1}^2 a_1^2 + \cdots + \sigma_{k+1}^2 a_{k+1}^2} = \sigma_{k+1}\sqrt{\sum a_i^2}$.

However, since $\boldsymbol{x}$ is a unit vector, and $\boldsymbol{x} = a_1\boldsymbol{v}_1 + \cdots + a_{k+1}\boldsymbol{v}_{k+1}$ for orthonormal $\boldsymbol{v}_i$, therefore $\sum a_i = \|\boldsymbol{x}\|^2 = 1$. So we are done. ☐

Now we move on to Frobenius norm, which is also related to singular values.

Proposition 8.12.30. $\|A\|_F^2 = \sum \sigma_i^2$.

Proof. $\|A\|_F^2 = \mathrm{trace}(A^{\mathrm{T}}A)$, which is the sum of eigenvalues of $A^{\mathrm{T}}A$. ☐

We have already shown that A_k is the best rank k approximation to A in the operator norm. Let us now show that it is also the best rank k approximation to A in the Frobenius norm.

Lemma 8.12.31 (Generalized triangle inequality for the operator norm)**.** *If $A = B+C$, then $\sigma_i(B)+\sigma_j(C) \geq \sigma_{i+j-1}(A)$. (Note that the regular triangle inequality is the case when $i = j = 1$.)*

Proof. Let B_{i-1}, C_{j-i} be the corresponding rank approximation in the operator norm. Then $\sigma_i(B) = \sigma_1(B - B_{i-1}), \sigma_j(C) = \sigma_1(C - C_{j-1})$. So by the regular triangle inequality of operator norm, $\sigma_i(B)+\sigma_j(C) = \sigma_1(B - B_{i-1}) + \sigma_1(C - C_{j-1}) \geq \sigma_1(B + C - B_{i-1} - C_{j-1}) = \sigma_1(A - B_{i-1} - C_{j-1})$.

However, $B_{i-1} + C_{j-1}$ is a rank $i+j-2$ matrix. So by the rank approximation for operator norm, $\sigma_1(A - B_{i-1} - C_{j-1}) \geq \sigma_1(A - A_{i+j-2}) = \sigma_{i+j-1}(A)$. So we are done. ☐

Proposition 8.12.32. *Among the set of rank k matrices, A_k is one of the closest to A in terms of Frobenius norm. Specifically, $\|A - B\|_F \geq \sum_{i=k+1}^n [\sigma_i(A)]^2$ for all $m \times n$ rank k matrices B, and $\|A - A_k\|$ reaches this minimal value.*

(Here $\sigma_i(A)$ refers to the i-th singular value of A, and if A is rank r, $\sigma_i(A) = 0$ for all $i > r$.)

Proof. Consider the decomposition $A = B + (A - B)$. Then $\sigma_i(B) + \sigma_j(A - B) \geq \sigma_{i+j-1}(A)$. Since B has rank k, we see that for $i = k+1$, $\sigma_i(B) = 0$, and hence $\sigma_j(A - B) \geq \sigma_{i+j-1}(A) = \sigma_{j+k}(A)$.

Now $\|A - B\|_F = \sum_{j=1}^n [\sigma_j(A - B)]^2 \geq \sum_{j=1}^n [\sigma_{j+k}(A)]^2 = [\sigma_{k+1}(A)]^2 + \cdots + [\sigma_n(A)]^2$.

Finally, this minimum is reached when $B = A_k$. ☐

In fact, for all norms defined in terms of singular values, then the operator norm would provide a control over that norm.

8.12.6 Principal Component Analysis

Example 8.12.33. Suppose we want to do facial recognition. Given a person's face, we can store all information about this face into a big list of numbers, i.e., a very high diemnsional vector $\boldsymbol{f}$. So each face is now a vector in the "face space", which is just $\mathbb{R}^m$ for some very big m.

Now to build a facial recognition process, we need to first start by collecting data. Say we collected the face of n people for very large n. Now we have $\boldsymbol{f}_1, \ldots, \boldsymbol{f}_n \in \mathbb{R}^m$ for large m, n. These data can fit into a matrix $A = \begin{bmatrix} \boldsymbol{f}_1 & \ldots & \boldsymbol{f}_n \end{bmatrix}$. What do we do next?

Next we want to identifa major features of a face. For example, "nose" is a major feature, "eyes" are also a major feature. To see if two faces are the same, we can compare the nose, compare the eyes, and so on. If enough major features are similar, then we say the two face are the same.

Think about the meaning of this. This means we are decomposing each face $\boldsymbol{f}$ into various features. Since we are doing this for each $\boldsymbol{f}_i$, this means we are decomposing A into a sum of matrices $A = \sum A_i$, where each A_i represent some feature. Furthermore, since "nose" would typically involve less pixels than the whole face, each feature matrix A_i would have a low rank. So we are attempting to decompose A into a sum of low rank matrices. By comparing faces using only the most important features, we are hoping that $A_1 + \cdots + A_k$ for some small k would give a good approximation to the whole matrix A.

You can see now how SVD comes in. The best way to do this is via $A = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$. Here we decomposed A into matrices with the smallest rank possible (rank one), and these "features" are all mutually "independent" (orthogonal to each other), and furthermore, a truncation $\sum_{i=1}^k \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$ is the BEST rank k approximation of A. These features might not have nice interpretations such as "nose", but they are the most efficient way to capture the essense of A while using minimal rank.

So to do facial recognition, we can perform $A = U\varSigma V = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$. Here columns of U are also in $\mathbb{R}^m$, and they are called "eigenfaces" sometimes, since they are eigenvectors of AA^{T}. You can even draw them out. These $\boldsymbol{u}_i$ form an orthonormal basis for the face space $\mathbb{R}^m$, and for each $\boldsymbol{f} \in \mathbb{R}^m$, then $\boldsymbol{f}$ would be some linear combination of them. Since $\boldsymbol{f} = UU^{\mathrm{T}}\boldsymbol{f}$, you can see that the coordinates under the orthonormal basis is just $\boldsymbol{u}_i$. This is akin to breaking down a face into "features", and note that from $\boldsymbol{u}_1$ to $\boldsymbol{u}_m$, we are going in the order of decreasing importance of features.

So fix some small k. Given two faces $\boldsymbol{f}_1, \boldsymbol{f}_2$, you can simply compare the first k coordinates of $U^{\mathrm{T}}\boldsymbol{f}_1, U^{\mathrm{T}}\boldsymbol{f}_2$. If these coordinates are close enough, then all the most important features are close enough, so we can declare them to be the same face.

Note that V here is less important. It records information about how to combine the input faces $\boldsymbol{f}_i$ into features, and it has little consequence in application.

For more on facial recognition using SVD, check out this paper. ☺

Example 8.12.34. I came across this application in the textbook by Gilbert Strang. The information here is written in this paper published in nature.

In short, for each person, we can write a list of number to represent this person's genetic information, i.e., some vector $\boldsymbol{p}$. So if we collect the genetic information of many people, and use these vectors as columns to form a matrix, we get A_0. Then we center the data (substract each column by the average of all columns, and now each row has average zero). Thus we obtain $A = \begin{bmatrix} \boldsymbol{p}_1 & \ldots & \boldsymbol{p}_n \end{bmatrix}$, and perform SVD for $A = U\varSigma V = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}$.

Here the two most principal components to explain A (the genetic variation data set) are $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}}$ and $\sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^{\mathrm{T}}$. Here $\boldsymbol{u}_1, \boldsymbol{u}_2$ would be the most important genetic "features" to explain genetic variation. What are the meanings of $\boldsymbol{v}_1, \boldsymbol{v}_2$?

Note that $\boldsymbol{u}_1^{\mathrm{T}} A = \sigma_1 \boldsymbol{v}_1^{\mathrm{T}}$ by definition of a left singular vector, yet we also have $\boldsymbol{u}_1^{\mathrm{T}} A = \begin{bmatrix} \boldsymbol{u}_1^{\mathrm{T}}\boldsymbol{p}_1 & \ldots & \boldsymbol{u}_1^{\mathrm{T}}\boldsymbol{p}_n \end{bmatrix}$. So the amount of genetic feature $\boldsymbol{u}_1$ contained by the i-th person is exactly the i-th coordinate of $\sigma_1 \boldsymbol{v}_1^{\mathrm{T}}$.

Suppose we ignore ALL other genetic features than the first two. Then the genetic variation of each person would only need two coordinates, and thus each person becomes a point in $\mathbb{R}^2$, and they are columns of $\begin{bmatrix} \sigma_1 \boldsymbol{v}_1^{\mathrm{T}} \\ \sigma_2 \boldsymbol{v}_2^{\mathrm{T}} \end{bmatrix}$. Graph these points in $\mathbb{R}^2$, and we recover something like a map of Europe. (Note that the paper in question only gathered genetic data in Europe.) Check out the paper for a graph of this comparison.

This means that the most important rank 2 factor influencing genetic variations is approximately the geological location of the person. This makes a lot of sense. If you sample the gene of someone in France, with great probability this person will be of French descent, and this person will most likely have the corresponding genetic variations common to French people.

Think about a study about coronavirus policy. Is lockdown an effective method? If you study this by collecting data from different countries, then your result could be misleading. Say one country locked down, while the other remained open, and we observed a difference in the spread of the virus. Does this imply lockdown is effective? Not by itself. If a person's genetic information effects the spreading of the virus, and geography explains the genetic variation, then what you have detected might be genetic variations caused by geography.

Therefore, a rigorous study must try to "correct" the data against this confounding factor. ☺

8.13 Classification of Quadratic surfaces

(This section should have been right next to definiteness of symmetric matrices. However, since it is the most optional, I put it last in case I have no time and need to skip it.)

Previously, given a square matrix A, we think of it as a linear transformation (domain = codomain). In particular, a change of basis would induce a change in matrix into $X^{-1}AX$ for some invertible X. In this section, we try something different. Let us think of A as a bilinear form.

Definition 8.13.1. *A bilinear form is a function $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ such that $f(-, \boldsymbol{v})$ is linear and $f(\boldsymbol{v}, -)$ is also linear. More specifically, we have $f(a\boldsymbol{u} + b\boldsymbol{v}, \boldsymbol{w}) = af(\boldsymbol{u}, \boldsymbol{w}) + bf(\boldsymbol{v}, \boldsymbol{w})$ for the first input, and same for the second input.*

Proposition 8.13.2. *We have $f(\boldsymbol{v}, \boldsymbol{w}) = \boldsymbol{v}^{\mathrm{T}}G\boldsymbol{w}$ for a matrix G whose (i, j) entry is $f(\boldsymbol{e}_1, \boldsymbol{e}_j)$. (So it is a Gram matrix.)*

Proof. Simply use bilinearity to expand $\boldsymbol{v}, \boldsymbol{w}$ in terms of the standard basis, and we are done. □

Proposition 8.13.3. *Suppose we perform a change of basis, so that $\boldsymbol{v}_{new} = X\boldsymbol{v}_{old}$ for invertible X. Then for the bilinear form with Gram matrix A, we have $A_{old} = X^{\mathrm{T}}A_{new}X$.*

Proof. The bottom line is that $\boldsymbol{v}_{new}^{\mathrm{T}}A_{new}\boldsymbol{w}_{new} = \boldsymbol{v}_{old}^{\mathrm{T}}A_{old}\boldsymbol{w}_{old}$. Now substitute in $\boldsymbol{v}_{new} = X\boldsymbol{v}_{old}$, we see that $\boldsymbol{v}_{old}^{\mathrm{T}}X^{\mathrm{T}}A_{new}X\boldsymbol{w}_{old} = \boldsymbol{v}_{old}^{\mathrm{T}}A_{old}\boldsymbol{w}_{old}$ for all $\boldsymbol{v}_{old}, \boldsymbol{w}_{old} \in \mathbb{R}^n$. So we have $A_{old} = X^{\mathrm{T}}A_{new}X$. □

In particular, the change of basis formula for matrices are now different. If the matrix A represent a bilinear form, then $X^{\mathrm{T}}AX$ is the result after a change of basis.

Definition 8.13.4. *Two matrices A, B are said to be* ***congruent*** *if $A = XBX^{\mathrm{T}}$ for some INVERTIBLE X. (Note that we require X to be invertible here, since it should come from some change of basis.)*

Previously, when we study eigenvalues, we would try to find X such that $X^{-1}AX$ is diagonal D, i.e., a decomposition $A = XDX^{-1}$. This is the best basis to study the linear transformation A. Now what is the best basis to study the bilinear form A? We would now try to find X so that $X^{\mathrm{T}}AX$ is as simple as possible.

Thankfully, this is not too bad, when A is symmetric. We have spectral decomposition $A = QDQ^{-1}$ and $Q^{-1} = Q^{\mathrm{T}}$. So by choosing the right basis, we can always assume that A is diagonal. So let us restrict our attention now.

Definition 8.13.5. *A bilinear form is a* ***symmetric bilinear form*** *if $f(\boldsymbol{v}, \boldsymbol{w}) = f(\boldsymbol{w}, \boldsymbol{v})$. (Obviously these are also bilinear forms with symmetric Gram matrix.)*

Remark 8.13.6. *There is also the concept of a* ***quadratic form,*** *which are $q : \mathbb{R}^n \to \mathbb{R}$ such that $q(\boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{x})$ for some symmetric bilinear form f. These corresponds to homogeneous polynomials of degree 2 on n variables.*

The study of quadratic form is pretty much the same as the bilinear form though. If $q(\boldsymbol{x}) = \boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x}$, then by taking transpose on both sides, we also have $q(\boldsymbol{x}) = \boldsymbol{x}^{\mathrm{T}}A^{\mathrm{T}}\boldsymbol{x}$. So $q(\boldsymbol{x}) = \boldsymbol{x}^{\mathrm{T}}G\boldsymbol{x}$ where $G = \frac{1}{2}(A + A^{\mathrm{T}})$ is symmetric. So we can always assume that the bilinear form for the quadratic form q is symmetric.

Note that one can also obtain the symmetric bilinear form via polarization identity $f(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{2}(q(\boldsymbol{x} + \boldsymbol{y}) - q(\boldsymbol{x}) - q(\boldsymbol{y}))$, which can be verified to be bilinear and symmetric, and $q(\boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{x})$.

Proposition 8.13.7. *For any symmetric bilinear form, we can choose a basis so that the Gram matrix is diagonal D where the diagonal entries are $0, 1, -1$.*

Proof. Suppose that under the standard basis, the Gram matrix is symmetric A. By spectral decomposition, $A = QBQ^{\mathrm{T}}$ for diagonal B. So our goal now is to find a decomposition so that $B = XDX^{\mathrm{T}}$, and D is as desired.

Say $B = \begin{bmatrix} b_1 & & \\ & \ddots & \\ & & b_n \end{bmatrix}$. If $b_i \neq 0$, let $x_i = \sqrt{|b_i|}$, and if $b_i = 0$, set $x_i = 1$. Either way, we have $x_i \neq 0$. Let $X = \begin{bmatrix} x_1 & & \\ & \ddots & \\ & & x_n \end{bmatrix}$, and $D = \begin{bmatrix} \mathrm{sign}(b_1) & & \\ & \ddots & \\ & & \mathrm{sign}(b_n) \end{bmatrix}$, where $\mathrm{sign}(x) = 1$ if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$. Then it is easy to verify that $B = XDX^{\mathrm{T}}$ and X is invertible. So we are done. ☐

Corollary 8.13.8. *Any symmetric matrix A is congruent to $\begin{bmatrix} I_a & & \\ & -I_b & \\ & & O \end{bmatrix}$. (We call this the **congruence canonical form of** A.)*

Now, are the integers a, b uniquely determined by A? They are. In fact, they have some very interesting structural meaning.

Note that there are something interesting going on here. Let's say we picked a nice basis and thus our symmetric bilinear form has Gram matrix $G = \begin{bmatrix} I_a & & \\ & -I_b & \\ & & O \end{bmatrix}$. Then consider the subspace $V = \mathrm{span}(\boldsymbol{e}_1, \ldots, \boldsymbol{e}_a)$. For any $\boldsymbol{v} \in V$, you can easily verify that $\boldsymbol{v}^{\mathrm{T}}G\boldsymbol{v} = \|\boldsymbol{v}\|^2 \geq 0$ with equality if and only if $\boldsymbol{v} = \mathbf{0}$. So even though G itself is NOT positive definite, but on the subspace V, it is positive definite.

Definition 8.13.9. *For a symmetrix matrix A, we define its **positive index of inertia** to be n_+, the size of largest subspace on which A can be positive definite. We define its **negative index of inertia** to be n_-, the size of largest subspace on which A can be negative definite.*

Lemma 8.13.10. *If A is positive definite on a subspace V, then A is positive semi-definite on $V + \mathrm{Ker}(A)$.*

Proof. For any $\boldsymbol{u} \in V + \mathrm{Ker}(A)$, suppose $\boldsymbol{u} = \boldsymbol{v} + \boldsymbol{w}$ where $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in \mathrm{Ker}(A)$. (Note that there is a chance of $\boldsymbol{v} = \mathbf{0}$ here, when $\boldsymbol{u}$ is entirely in $\mathrm{Ker}(A)$.)

Then $\boldsymbol{u}^{\mathrm{T}}A\boldsymbol{u} = (\boldsymbol{v} + \boldsymbol{w})^{\mathrm{T}}A(\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{v}^{\mathrm{T}}A\boldsymbol{v} \geq 0$. Here all the $\boldsymbol{w}$ disappear because $A\boldsymbol{w} = \mathbf{0}$ and (by symmetricity of A) $A^{\mathrm{T}}\boldsymbol{w} = \mathbf{0}$. Also note that we merely have $\geq$ rather than $>$, because maybe $\boldsymbol{v} = \mathbf{0}$. ☐

Lemma 8.13.11. *Suppose an $n \times n$ symmetric matrix A has positive index of inertia n_+, negative index of inertial n_-, and kernel of size n_0. Then $n_+ + n_- + n_0 \leq n$.*

Proof. Let V_+ be a subspace of dimension n_+ on which A is positive definite. Then all non-zero $\boldsymbol{v} \in V_+$ will have $\boldsymbol{v}^{\mathrm{T}}A\boldsymbol{v} > 0$, while all non-zero $\boldsymbol{v} \in \mathrm{Ker}(A)$ will have $\boldsymbol{v}^{\mathrm{T}}A\boldsymbol{v} = 0$. In particular, V_+ and $\mathrm{Ker}(A)$ has trivial intersection, and $\dim(V_+ + \mathrm{Ker}(A)) = \dim(V_+) + \dim \mathrm{Ker}(A) = n_+ + n_0$.

Let V_- be a subspace of dimension n_- on which A is negative definite. Then all non-zero $\boldsymbol{v} \in V_+$ will have $\boldsymbol{v}^{\mathrm{T}}A\boldsymbol{v} < 0$. However, in comparison, since A is positive semi-definite on $V_+ + \mathrm{Ker}(A)$, all non-zero $\boldsymbol{v} \in V_+ + \mathrm{Ker}(A)$ will have $\boldsymbol{v}^{\mathrm{T}}A\boldsymbol{v} \geq 0$. In particular, V_- and $V_+ + \mathrm{Ker}(A)$ has trivial intersection, and $\dim(V_- + V_+ + \mathrm{Ker}(A)) = \dim(V_-) + \dim(V_+ + \mathrm{Ker}(A)) = n_- + n_+ + n_0$. But since $V_- + V_+ + \mathrm{Ker}(A) \subseteq \mathbb{R}^n$, we see that $n_+ + n_- + n_0 \leq n$. ☐

Theorem 8.13.12 (Sylvester's Law of Inertia)**.** *Suppose an $n \times n$ symmetric matrix A has positive index of inertia n_+, negative index of inertial n_-, and kernel of size n_0. We have $n_+ + n_- + n_0 = n$, and n_+ is the number of positive eigenvalues of A, and n_- is the number of negative eigenvalues of A.*

In particular, if A, B are congruent, then they have the same positive index of inertia and same negative index of inertial. If A is congruent to $D_{a,b} = \begin{bmatrix} I_a & & \\ & -I_b & \\ & & O \end{bmatrix}$, then we must have $a = n_+$ and $b = n_-$. So the congruence canonical form of A is unique.

Proof. Since A is normal, its eigenspaces are spanning. Let V_+ be the subspace spanned by all eigenspaces of A for positive eigenvalues of A. Then for any vector $\boldsymbol{v} \in V_+$, it is a sum of eigenvectors of A for positive eigenvalues, say $\boldsymbol{v} = \sum \boldsymbol{v}_i$ where each $\boldsymbol{v}_i$ has eigenvalue $\lambda_i > 0$. Then $\boldsymbol{v}^{\mathrm{T}} A \boldsymbol{v} = \sum \lambda_i \|\boldsymbol{v}_i\|^2 \geq 0$, with equality if and only if $\boldsymbol{v}_i$ are all $\mathbf{0}$, if and only if $\boldsymbol{v} = \mathbf{0}$. So A is positive definite on V_+.

Since A is normal, algebraic multiplicity equals geometric multiplicity, so $\dim(V_+)$ is exactly the number of positive eigenvalues of A, say m_+. Since A is positive definite on V_+, and n_+ is by definition the largest possible dimension of such a subspace, we have $n_+ \geq m_+$. Similarly, we also see that $n_- \geq m_-$, where m_- is the number of negative eigenvalues of A. Finally, we have $n_0 = m_0$ where m_0 is the algebraic multiplicity of zero as an eigenvalue of A.

Now since A has n eigenvalues, we have $n = m_+ + m_- + m_0 \leq n_+ + n_- + n_0 \leq n$. So we have equality everywhere. $\square$

In particular, we see that up to a change of basis, a symmetric bilinear form is completely determined by its positive and negative index of inertia. Now we can proceed to classify quadratic curves and quadratic surfaces.

Example 8.13.13 (Classification of quadratic curves)**.** Consider the solution set to some degree two polynomial equation $ax^2 + bxy + dy^2 + ex + fy + c = 0$ in $\mathbb{R}^2$ for constants a, b, c, d, e, f. What is it?

Let $A = \begin{bmatrix} a & \frac{b}{2} \\ \frac{b}{2} & d \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} e \\ f \end{bmatrix}$, then we see that the solution sets are points $\boldsymbol{x}$ such that $\boldsymbol{x}^{\mathrm{T}} A \boldsymbol{x} + \boldsymbol{b}^{\mathrm{T}} \boldsymbol{x} + c = 0$. Suppose for now that A is invertible.

Then we can try to complete the square. Recall that $ax^2 + bx + c$ can be written as $a(x + \frac{b}{2a})^2 + (c - \frac{b^2}{4a})$. So similarly, we change variable with $\boldsymbol{y} = \boldsymbol{x} + \frac{1}{2} A^{-1} \boldsymbol{b}$, and we have $\boldsymbol{y}^{\mathrm{T}} A \boldsymbol{y} = C$ for some constant C. Note that all we did is to translate the solution set around, without changing its shape.

Now we perform a change of basis, and we can assume that A is in congruence canonical form. We have several possibilities.

1. Suppose $n_+ = 2$ and $n_- = 0$, then if $\boldsymbol{y} = \begin{bmatrix} x \\ y \end{bmatrix}$, we are facing $x^2 + y^2 = C$. The solution here is a circle if $C > 0$. Going back to the solution set $\boldsymbol{x}^{\mathrm{T}} A \boldsymbol{x} + \boldsymbol{b}^{\mathrm{T}} \boldsymbol{x} + c = 0$, the change of basis process will make the circle into an ellipse, and then the change of variable process will translate the ellipse around. Anyway, we end up with an ellipse as the solution set. (If $C = 0$, the solution is a single point. If $C < 0$, the solution set is empty.)

2. Suppose $n_+ = 0$ and $n_- = 2$, then if $\boldsymbol{y} = \begin{bmatrix} x \\ y \end{bmatrix}$, we are facing $x^2 + y^2 = -C$. We see that this is the same as above, except that now $C < 0$ gives the ellipse, and $C > 0$ now gives empty solution set.

3. Suppose $n_+ = n_- = 1$. Then if $\boldsymbol{y} = \begin{bmatrix} x \\ y \end{bmatrix}$, we are facing $x^2 - y^2 = C$. We see that this is a hyperbola. Change of basis means the two asymtotes can be in any direction, and the change of variable means we can translate it around.

So all in all, it is either an ellipse, or a hyperbola, or in the degenerate cases, a single point or the empty set.

What if A is NOT invertible? If $A = 0$, then we have a degree one equation $ex + fy + c = 0$, so the solution is a single line, or it is the empty set if $e = f = 0$ while $c = 0$, giving us a contradiction.

We are left with one last case, when A has rank 1. Note that if $\boldsymbol{b} \in \operatorname{Ran}(A)$, say $\boldsymbol{b} = A\boldsymbol{v}$, then we can again change variable $\boldsymbol{y} = \boldsymbol{x} + \frac{1}{2}A^{-1}\boldsymbol{b}$ as before.

1. If $n_+ = 1$ and $n_- = 0$ and $\boldsymbol{b} \in \operatorname{Ran}(A)$. Then by change of variable, we have $\boldsymbol{y}^{\mathrm{T}}A\boldsymbol{y} = C$. By change of basis, we have $x^2 = C$. So we have a pair of parallel lines.
2. If $n_+ = 0$ and $n_- = 1$ and $\boldsymbol{b} \in \operatorname{Ran}(A)$. Then by change of variable, we have $\boldsymbol{y}^{\mathrm{T}}A\boldsymbol{y} = C$. By change of basis, we have $x^2 = -C$. So we have a pair of parallel lines again.
3. If $n_+ = 1$ and $n_- = 0$ and $\boldsymbol{b} \notin \operatorname{Ran}(A)$. Then we change basis directly, and have $\boldsymbol{x}^{\mathrm{T}}D\boldsymbol{x}+(\boldsymbol{b}')^{\mathrm{T}}\boldsymbol{x}+c = 0$, which gives $x^2 + e'x + f'y + c = 0$. Since $\boldsymbol{b} \notin \operatorname{Ran}(A)$, after change of basis, we have $\boldsymbol{b}' \notin \operatorname{Ran}(D)$, so $f' \neq 0$. This gives $y = -\frac{1}{f'}(x^2 + e'x + c)$, so the solution set is a parabola.
4. If $n_+ = 1$ and $n_- = 0$ and $\boldsymbol{b} \notin \operatorname{Ran}(A)$, this is the same as above by changing c into $-c$. So this is a parabola.

Let us now sum up all cases of the solution set to $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} + \boldsymbol{b}^{\mathrm{T}}\boldsymbol{x} + c = 0$. This can be an ellipse, a hyperbola, a parabola, a pair of parallel lines, a single line, a single point or the empty set. Throwing away the degree one cases and the degenerate cases and the contradiction cases, the true degree two cases can only be ellipses, hyperbolas and parabolas. ☺

Example 8.13.14 (Classification of quadric surfaces)**.** We now look at the solution set of degree two polynomials in $\mathbb{R}^3$. Again, we can rearrange so that the equation looks like $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} + \boldsymbol{b}^{\mathrm{T}}\boldsymbol{x} + c = 0$. To save time, let us drop the lower degree cases, degenerate cases and contradictory cases, and only try to classify the rest. This means $A \neq O$ (but A is allowed to be non-invertible), and $\boldsymbol{b}$ and $\operatorname{Ran}(A)$ span the whole $\mathbb{R}^3$. (Otherwise we will be degenerate.)

Suppose A is invertible. Then again we can simplify via translation to $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = C$. If $C = 0$, we are degenerate. So assume $C \neq 0$. By replacing A via $-A$ if needed, we can assume $C > 0$.

1. If $n_+ = 3$, then after change of basis, we have $x^2 + y^2 + z^2 = C$, so this is a sphere. Going back via change of basis and translation, we have an aribtrary ellipsoid.
2. If $n_+ = 2$ and $n_- = 1$, then after change of basis, we have $x^2 + y^2 - z^2 = C$. This is the famous hyperboloid with one sheet. Take the hyperbola $x^2 - z^2 = C > 0$ and rotate around the z-axis, and you can find this shape. It is a connected surface. (Note that there is a single negative sign in the entire equation, which helps you memorize the number of sheets.)
3. If $n_+ = 1$ and $n_- = 2$, then after change of basis, we have $x^2 - y^2 - z^2 = C$. This is the famous hyperboloid with two sheets. Take the hyperbola $x^2 - z^2 = C > 0$ and rotate around the x-axis, and you can find this shape. It is has two connected surfaces as its components. (Note that there are two negative sign in the entire equation, which helps you memorize the number of sheets.)
4. If $n_- = 3$, then A is negative definite. Since we require $C > 0$, $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = C > 0$ is impossible. So there is no solution in this case.
5. (Optional) Note that if $C = 0$, then we are degenerate. In the cases $n_+ = 3$ or $n_+ = 0$, the solution is a single point. In the cases $n_+ = 2$ or $n_+ = 1$, the solution set is a pair of opposite cones.

Since we require $\boldsymbol{b}$ and $\operatorname{Ran}(A)$ to span the whole $\mathbb{R}^3$, the only non-invertible case is $\operatorname{rank}(A) = 2$ and $\boldsymbol{b} \notin \operatorname{Ran}(A)$. By a change of basis, we have $\pm x^2 \pm y^2 + ax + by + cz + d = 0$. In fact by a translation in variable x, y, we are left with $\pm x^2 \pm y^2 + cz + d = 0$. We change variable z so that $-cz$ is now replaced by z, then we have $z = \pm x^2 \pm y^2 + d$, and the signs depends on the index of inertia of A.

1. If $n_+ = 2$, then $z = x^2 + y^2 + d$. This is a paraboloid opening upward. (An elliptic paraboloid after a change of basis.)

2. If $n_+ = n_- = 1$, then $z = x^2 - y^2 + d$. This is a hyperbolic paraboloid.

3. If $n_- = 2$, then $z = -x^2 - y^2 + d$. This is a paraboloid opening downward.(An elliptic paraboloid after a change of basis.)

To sum up, the true quardric surfaces are: ellipsoid, hyperboloid with one sheet, hyperboloid with two sheets, elliptic paraboloid, hyperbolic paraboloid.

If you are interested, the quardric degenerate surfaces are: elliptic cones, elliptic cylinder, hyperbolic cylinder, parabolic cylinder. And then we have lower degree cases line planes, or lower dimensional cases like points or the empty set. ☺

8.14 Exercises

8.14.1 Eigenvalues and eigenvectors

Exercise 8.14.1. *Find all eignevalues and all corresponding eigenvectors for the following matrices. (Including complex eigenvalues and eigenvectors.)*

1. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$.

3. $\begin{bmatrix} 10 & 11 \\ 3 & 18 \end{bmatrix}$.

4. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

5. $\begin{bmatrix} b & -a \\ a & b \end{bmatrix}$, *here* $,b \in \mathbb{R}$.

6. $\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 2 \\ -2 & -2 & 0 \end{bmatrix}$.

Exercise 8.14.2. *Find a formula for the sequence (a_n) under the given conditions. Try to describe the evolutuion using a matrix, and use the eigenvalues and eigenvectors of the matrix to compute the detail.*

1. $a_0 = 0, a_1 = 1$, *and* $a_{n+2} = 3a_{n+1} - 2a_n$.

2. $a_0 = 0$, *and* $a_{n+1} = 2a_n + 1$.

3. $a_0 = 0$, *and* $a_{n+1} = 2a_n + 2^n$.

4. $a_0 = 0, a_1 = 0, a_2 = 1$, *and* $a_{n+3} = 6a_{n+2} - 11a_{n+1} + 6a_n$.

5. $a_0 = 0, b_0 = 1, a_{n+1} = a_n + b_n, b_{n+1} = -2a_n + 4b_n$.

6. ☕$a_0 = 0, a_1 = 1$, *and* $a_{n+2} = 2a_{n+1} - a_n$. *(Hint: the matrix is similar to* $\begin{bmatrix} 1 & k \\ & 1 \end{bmatrix}$, *whose power is also easy to compute.)*

7. ☕$a_0 = 0$, *and* $a_{n+1} = a_n + n^2$. *(Hint: set* $\boldsymbol{x}_n = \begin{bmatrix} a_n \\ n^2 \\ n \\ 1 \end{bmatrix}$. *The matrix is not diagonalizable though, but if you write the matrix as* $I + X$, *then what is* X^4*? What does that tell you about a formula for* $(I + X)^n$*?)*

Exercise 8.14.3 (Formula for 2×2 eigenstuff)**.** *Let* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

1. *Find a formula for the two eigenvalues of* A*, using the quadratic formula (i.e., the solution to* $ax^2 + bx + c = 0$ *is* $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$*).*

2. *If* λ *is an eigenvalue of* A*, show that* $A \begin{bmatrix} b \\ \lambda - a \end{bmatrix} = \lambda \begin{bmatrix} b \\ \lambda - a \end{bmatrix}$, *and* $A \begin{bmatrix} \lambda - d \\ c \end{bmatrix} = \lambda \begin{bmatrix} \lambda - d \\ c \end{bmatrix}$. *(So these are eigenvectors if they are non-zero.)*

3. *If* λ *is an eigenvalue of* A*, and the two vectors above* $\begin{bmatrix} b \\ \lambda - a \end{bmatrix}$ *and* $\begin{bmatrix} \lambda - d \\ c \end{bmatrix}$ *are both zero vectors, find the matrix* A*. (In particular, as long as* A *is not in this case, then the formulas from last subproblem must provide you with at least one eigenvector.)*

4. *(Read only) We have three cases: First,* A *is a multiple of identity. Second,* A *is not a multiple of identity and it has two identical eigenvalues (in this case all eigenspaces are one-dimensional). Finally,* A *has two distinct eigenvalues (in this case all eigenspaces are one-dimensional). In any case, all the eigenvalues and eigenvectors can be found by the formula listed above.*

Exercise 8.14.4 (Companion Matrix)**.** *Given a polynomial* $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$*, we can define the* ***companion matrix*** *to be* $M_p = \begin{bmatrix} 0 & & & & -a_0 \\ 1 & \ddots & & & \vdots \\ & \ddots & \ddots & & \vdots \\ & & 1 & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{bmatrix}$. *Note that this is an* $n \times n$ *square matrix.*

1. *Find the characteristic polynomial and the eigenvalues of* M_p *when* $p(x) = x^2 + 1$.

2. *Find the characteristic polynomial and the eigenvalues of* M_p *when* $p(x) = x^2 - 3x + 2 = (x-1)(x-2)$.

3. *Guess the eigenvalues of* M_p *when* $p(x) = x^3 - 6^2 + 11x - 6 = (x-1)(x-2)(x-3)$.

4. ☕*Can you prove the relation between eigenvalues of* M_p *and roots of* p*? (Hint: expand along the last column when computing* $\det(xI - M_p)$*.)*

5. *(Read Only) For computers, to find all roots of a polynomial* $p(x)$*, one way is to calculate the eigenvalues of* M_p *from other methods (such as Schur decomposition).*

Exercise 8.14.5 (Eigenvalue change)**.** *Suppose* A *has eigenvalues* $1, 1, 2, 3$*. Find the eigenvalues of the following matrices.*

1. $A - I$.

2. A^2.

3. A^{-1}. *(Why is* A *invertible?)*

4. $\overline{A}$.

5. A^*.

6. $A^3 - 6A^2 + 11A - 6I$.

7. $\begin{bmatrix} A & \\ & A \end{bmatrix}$.

8. $\begin{bmatrix} A+I & A \\ & A-I \end{bmatrix}$.

Exercise 8.14.6 (Eigenvector change)**.** *Let $p(x)$ be any polynomial. The intuition here is that any eigenvector of A must be an eigenvector of $p(A)$, but the converse is NOT true! In short, $p(A)$ sometimes could have more eigenvectors than A. Let us verify some details. This problem categorized all possible reasons for this failure.*

1. *Suppose $A\boldsymbol{v} = \lambda\boldsymbol{v}$. Show that $p(A)\boldsymbol{v} = p(\lambda)\boldsymbol{v}$.*

2. *Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $p(x) = (x-1)^2 + 1$. Find all eigenvalues of $p(A)$, and find a vector $\boldsymbol{v}$ which is an eigenvector of $p(A)$ but not an eigenvector of A. (Failure caused by killing non-diagonal portion through $p(x)$.)*

3. *Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $p(x) = (x-1)(x-2) + 1$. Find all eigenvalues of $p(A)$, and find a vector $\boldsymbol{v}$ which is an eigenvector of $p(A)$ but not an eigenvector of A. (Failure caused by merging different eigenvalues into the same eigenvalue through $p(x)$.)*

Exercise 8.14.7. *Find all possible values of the unknown constants given the condition.*

1. $A = \begin{bmatrix} 0 & a \\ 1 & b \end{bmatrix}$ *and the eigenvalues are* $4, 7$.

2. $A = \begin{bmatrix} 0 & 0 & a \\ 1 & 0 & b \\ 0 & 1 & c \end{bmatrix}$ *and the eigenvalues are* $2, 3, 4$.

3. $A = \begin{bmatrix} 2 & -2 & 0 \\ -2 & x & -2 \\ -2 & -2 & 0 \end{bmatrix}, B = \begin{bmatrix} 2 & & \\ & 2 & \\ & & y \end{bmatrix}$, *and A, B have the same characteristic polynomial.*

4. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{bmatrix}$, *and* $\boldsymbol{x} = \begin{bmatrix} 1 \\ b \\ 1 \end{bmatrix}$ *is an eigenvector of A.*

5. $A = \begin{bmatrix} 3 & 2 & -2 \\ -k & -1 & k \\ 4 & 2 & -3 \end{bmatrix}$ *is diagonalizable.*

Exercise 8.14.8. *Let A be a Markov matrix, i.e., all entries are non-negative, and each column has a sum of 1. Let $\boldsymbol{u}$ be the vector whose coordinates are all 1.*

1. *Show that $\boldsymbol{u}^{\mathrm{T}}A = \boldsymbol{u}^{\mathrm{T}}$.*

2. *Let $\boldsymbol{v}$ be a vector whose coordinates add up to 1. Show that $A\boldsymbol{v}$ is also a vector whose coordinates add up to 1, so A sends a "probability distribution vector" to another "probability distribution vector". (Hint: $\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v}$ is the sum of all coordinates of $\boldsymbol{v}$.)*

3. *Show that A^k is a Markov matrix for all natural number k.*

4. *Show that if A is $n \times n$ and Markov, and λ is an eigenvalue of A, then $|\lambda| \leq n$. (Hint: Suppose $A\boldsymbol{x} = \lambda\boldsymbol{x}$, and A is $n \times n$. If the coordinate of $\boldsymbol{x}$ with the largest absolute value is x_i, then the absolute value of the i-th coordinate of $A\boldsymbol{x}$ is at most how much?)*

5. *Show that if A is $n \times n$ and Markov, and λ is an eigenvalue of A, then $|\lambda| \leq 1$. (Hint: Last subproblem should be true for A^k for all natural number k. If the statement in this subproblem is false, pick a super large k and get your contradiction.)*

Exercise 8.14.9 (Sudoku matrix)**.** *Let A be the 4×4 Sudoku matrix. So each row is made of entries $1, 2, 3, 4$, each column is made of entries $1, 2, 3, 4$, and the four 2×2 blocks (upper left, upper right, lower left, lower right) are each made of entries $1, 2, 3, 4$. For example, this is a Sudoku matrix:* $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{bmatrix}$.

There are of course many many Sudoku matrices.

1. *Show that for A has eigenvalue 10, and find all eigenvectors for this eigenvalue.*

2. *If $A\boldsymbol{v} = \lambda\boldsymbol{v}$ where $\lambda \neq 10$, show that $\boldsymbol{v} \perp \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$. (Hint: multiply $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$ to the left on both sides of $A\boldsymbol{v} = \lambda\boldsymbol{v}$.)*

3. *If λ is an eigenvalue of A, show that $|\lambda| \leq 10$. (Hint: $\frac{1}{10}A$ is a Markov matrix. Then look at the last problem about Markov matrix.)*

4. *Find a non-invertible Sudoku matrix.*

5. *Find an invertible Sudoku matrix.*

Exercise 8.14.10 (Haar wavelet Matrix)**.** *Recall that we can define the Haar wavelet basis inductively as this: we set $H_1 = \begin{bmatrix} 1 \end{bmatrix}$, and $H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}$. Then it is easy to see by induction that all entries of H_n are ± 1, and all columns of H_n are mutually orthogonal, and $H_n^{\mathrm{T}} = H_n$. Let us now find its eigenvalues.*

Throughout this problem, n will be some power of 2.

1. *Show that $H_n^2 = nI$. What does this tell you about the eigenvalues of H_n?*

2. *Find the algebraic multiplicity of all eigenvalues. (Hint: consider* $\operatorname{trace}(H_n)$*.)*

3. *Geometrically show that $\frac{1}{\sqrt{2}}H_2$ is a reflection, and find two linearly independent eigenvector this way.*

4. ☕*If $H_n\boldsymbol{v} = \lambda\boldsymbol{v}$, find an eigenvector of H_{2n} for the eigenvalue $\sqrt{2}\lambda$ in terms of $\boldsymbol{v}$, and find an eigenvector of H_{2n} for the eigenvalue $-\sqrt{2}\lambda$ in terms of $\boldsymbol{v}$.*

Exercise 8.14.11 (Real Discrete Fourier Transform)**.** *Suppose A is a 4×4 matrix with diagonal entries 4 and non-diagonal entries -1. Let H_4 be the matrix for the Haar wavelet basis.*

1. *Calculuate AH_4.*

2. *Calculate $H_4^{-1}AH_4$.*

3. *Find* $\det(A), A^{-1}$.

Exercise 8.14.12 (Complex Discrete Fourier Transform)**.** *Consider* $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$. *Let* $F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \mathrm{i} & -1 & -\mathrm{i} \\ 1 & -1 & 1 & -1 \\ 1 & -\mathrm{i} & -1 & \mathrm{i} \end{bmatrix}$.

(This F_4 is the 4×4 Fourier matrix, and it represents a discrete version of Fourier transform. Continuous Fourier transform is useful for the study of periodic functions, whereas here we shall see that discrete Fourier transform is useful for the study of "periodic matrices".)

1. *Calculuate AF_4.*
2. *Find the D, X in the diagonalization $A = XDX^{-1}$. (Hint: the last subproblem.)*
3. *Find* $\det(A), A^{-1}$.

Exercise 8.14.13 (Pavel Grinfeld Problem)**.** ☕*Calculate* $\begin{bmatrix} 110 & 55 & -164 \\ 42 & 21 & -62 \\ 88 & 44 & -131 \end{bmatrix}^{2017}$. *(Hint: Stare at it, until you see two eigenvalues and their eigenvectors. Use trace to find the last eigenvalue.)*

Exercise 8.14.14 (Large power is unstable)**.** *Large power of a large matrix is large? Prove the following problems. Hint: Diagonalization.*

1. *All entries of* $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}^{1024}$ *are larger than* 10^{700}. *(You may use the fact that* $\frac{1}{3}(5^{1024} - 2^{1024}) > 10^{700}$.*)*
2. $\begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix}^{1024} = I_2$
3. $\begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix}^{1024} = \begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix}$.
4. *(Read Only) You can use matrix calculator to verify that all entries of* $\begin{bmatrix} -5 & -6.9 \\ 3 & 4 \end{bmatrix}^{1024}$ *are less than* 10^{-70} *in absolute value. Furthermore, all entries of* $\begin{bmatrix} -5 & -7.1 \\ 3 & 4 \end{bmatrix}^{1024}$ *are larger than* 10^{57} *in absolute value. These things are extremely unstable....*

Exercise 8.14.15 (Eigenvalues are unstable)**.** *Tiny error in the entries could have drastic effect on eigenvalues and diagonalizabilities. Find the eigenvalues of the following matricess, and their algebraic and geometric multiplicity. Are they diagonalizable?*

1. $\begin{bmatrix} 10 & 1 & \\ & 10 & 1 \\ & & 10 \end{bmatrix}$.
2. $\begin{bmatrix} 10 & 1 & \\ & 10.001 & 1 \\ & & 10.002 \end{bmatrix}$. *(Note that diagonalizability could be different for the slightest change in entries.)*
3. $\begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ & & & 0 \end{bmatrix}$.
4. $\begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ 0.0001 & & & 0 \end{bmatrix}$. *(Note how the errors in entries is magnified in the errors in eigenvalues.)*
5. *(Read Only) In general, suppose you pick a random square matrix (all entries are random variables). Then it will have random eigenvalues $\lambda_1, \ldots, \lambda_n$. When randome variables are continuously distributed, the chances that two random variables are equal is zero. So a random matrix A (or anything with unknown tiny error) will have n distinct eigenvalues with probability* 100%. *In particular, a random matrix A is diagonalizable with probability* 100%. *In real life, if you obtain some square matrix with potential tiny errors, then you may simply assume that it is diagonalizable.*

Exercise 8.14.16. *Find all eigenvalues, all algebraic multiplicity, all geometric multiplicity, and a description of all eigenvectors. Finally, find a diagonalization if possible, or find a Schure decomposition if it is not diagonalizable.*

1. $A = \begin{bmatrix} a & b & \dots & b \\ b & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & b \\ b & \dots & b & a \end{bmatrix}$.
2. $A = \begin{bmatrix} 0 & -1 \\ 1 & -2 \end{bmatrix}$.

Exercise 8.14.17. *Are these diagonalizable?*

1. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.
2. $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.
3. $A = \boldsymbol{v}\boldsymbol{w}^{\mathrm{T}}$ *for some* $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$. *Suppose* $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{v} = 0$.
4. $A = \boldsymbol{v}\boldsymbol{w}^{\mathrm{T}}$ *for some* $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$. *Suppose* $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{v} \neq 0$.

Exercise 8.14.18 (Polynomial identity and diagonalization)**.** *Let A be an $n \times n$ complex matrix. In general, if $p(A) = 0$ and $p(x)$ is a polynomial with distinct roots, then A is diagonalizable. Here we prove some special cases.*

1. *Suppose* $A^2 = I$. *Show that* $\mathrm{Ker}(A+I) + \mathrm{Ker}(A-I) = \mathbb{C}^n$, *and A is diagonalizable.*
2. *Suppose* $A^2 = A$. *Show that* $\mathrm{Ker}(A) + \mathrm{Ker}(A-I) = \mathbb{C}^n$, *and A is diagonalizable.*
3. *Suppose* $A^3 = A$. *Find constant* $a, b, c \in \mathbb{C}$ *such that, for all* $\boldsymbol{v} \in \mathbb{C}^n$, *we have* $\boldsymbol{v} = aA(A-I)\boldsymbol{v} + bA(A+I)\boldsymbol{v} + c(A-I)(A+I)\boldsymbol{v}$. *(Maybe you've seen similar techniques in your calculus class.)*
4. *Suppose* $A^3 = A$. *Show that* $\mathrm{Ker}(A+I) + \mathrm{Ker}(A-I) + \mathrm{Ker}(A) = \mathbb{C}^n$, *and A is diagonalizable.*

Exercise 8.14.19. *For the following matrices A, B, find X such that $A = XBX^{-1}$.*

1. $A = \begin{bmatrix} MN & O \\ N & O \end{bmatrix}, B = \begin{bmatrix} O & O \\ N & NM \end{bmatrix}$. *Here M, N might not be square.*
2. $A = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix}, B = \begin{bmatrix} & -\mathrm{i} \\ \mathrm{i} & \end{bmatrix}$. *(The matrices A, B and their diagonalization D are called Pauli matrices and are important in physics. It is crucial in physics that these three matrices do NOT commute, dispite having the same eigenvalues.)*
3. $A = \begin{bmatrix} 1 & 2 & 3 \\ & 1 & 3 \\ & & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & \\ & 1 & 1 \\ & & 1 \end{bmatrix}$. *(Hint: Try change of basis via matrices like* $\begin{bmatrix} 1 & k & \\ & 1 & \\ & & 1 \end{bmatrix}$ *or diagonal matrices.)*
4. $A = \begin{bmatrix} 4 & -2 \\ 3 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & \\ & 1 \end{bmatrix}$.

5. $A = \begin{bmatrix} 4M & -2M \\ 3M & -M \end{bmatrix}, B = \begin{bmatrix} 2M & \\ & M \end{bmatrix}$. *(Hint: It is related to the last sub-problem.)*

6. $A = \begin{bmatrix} M & -N \\ N & M \end{bmatrix}, B = \begin{bmatrix} M + \mathrm{i}N & \\ & M - \mathrm{i}N \end{bmatrix}$. *(Hint: Get hints from the cases when M, N are 1×1.)*

7. $A = \begin{bmatrix} 2I_2 & M \\ & 3I_3 \end{bmatrix}, B = \begin{bmatrix} 2I_2 & \\ & 3I_3 \end{bmatrix}$. *(Hint: Get hints from the cases when all blocks are 1×1.)*

Exercise 8.14.20 (Homogeneous Sylvester's equation)**.** *Suppose $AX = XB$ for some $m \times m$ matrix A and $n \times n$ matrix B and $m \times n$ matrix X.*

1. *Suppose $n = 1$. Show that the $m \times 1$ matrix X must in fact be an eigenvector of A.*
2. *If $\boldsymbol{v} \in \mathrm{Ran}(X)$, show that $A\boldsymbol{v} \in \mathrm{Ran}(X)$. (This means the subspace $\mathrm{Ran}(X)$ is mapped into itself via A. Subspaces such as $\mathrm{Ran}(X)$ is called an **invariant subspace** of A.)*
3. *Show that for any polynomial $p(x)$, we have $p(A)X = Xp(B)$. (Hint: prove it first for powers of A, then note that polynomials are just linear combinations of powers.)*
4. *Suppose A, B has NO common eigenvalue. Show that $AX = XB$ imples $X = O$. (Hint: Why is $p_A(B)$ invertible?)*

Exercise 8.14.21 (Diagonalization via inhomogeneous Sylvester's equation)**.** *Suppose we have a block matrix $\begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ for some $m \times m$ matrix A and $n \times n$ matrix B and $m \times n$ matrix C. We are hoping to block diagonalize this by $\begin{bmatrix} I & X \\ & I \end{bmatrix} \begin{bmatrix} A & C \\ 0 & B \end{bmatrix} \begin{bmatrix} I & -X \\ & I \end{bmatrix} = \begin{bmatrix} A & C - AX + XB \\ & B \end{bmatrix}$.*

In particular, we want to find the solution X to the matrix equation $AX - XB = C$. Then we shall have $\begin{bmatrix} I & X \\ & I \end{bmatrix} \begin{bmatrix} A & C \\ 0 & B \end{bmatrix} \begin{bmatrix} I & -X \\ & I \end{bmatrix} = \begin{bmatrix} A & \\ & B \end{bmatrix}$.

A matrix equation $AX - XB = C$ for constant A, B, C and unknown X is called a Sylvester's equation, and it is extremely important in many applications such as control theory.

1. *Let V be the vector space of $m \times n$ matrices. Show that $L : V \to V$ with $L(X) = AX - XB$ is a linear map.*
2. *Show that, if A, B has no common eigenvalues, then $\mathrm{Ker}(L)$ is trivial. (Hint: Use facts about homogeneous Sylvester's equation in the previous problem.)*
3. *Show that, if A, B has no common eigenvalues, then no matter what C is, the solution X to $AX - XB = C$ exists and is unique. (Consequently, we can always block diagonalize $\begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ using $\begin{bmatrix} I & X \\ & I \end{bmatrix}$.)*

Exercise 8.14.22 (Commuting matrices diagonalization)**.** *We explore some consequences of the identity $AB = BA$. What does it mean for two matrices to commute?*

1. *If $AB = BA$ and A is diagonal with distinct diagonal entries, show that B must be diagonal. (Hint: work an easy example, say $A = \begin{bmatrix} 2 & \\ & 3 \end{bmatrix}$, to get an intuition of why non-diagonal entries of B must be zero.)*
2. *If $AB = BA$ and A has no repeated eigenvalues, then show that one can find an invertible matrix X such that $X^{-1}AX, X^{-1}BX$ are BOTH diagonal. (I.e., A, B are simultaneously diagonalizable. A, B are both diagonal in another basis.)*
3. *If $AB = BA$ and A has no repeated eigenvalues, then show that all eigenvectors of A are eigenvectors of B (for maybe different eigenvalues).*

4. *Find an example of $AB = BA$ where A has no repeated eigenvalues, but B has an eigenvector that is NOT an eigenvector of A. (Simple examples exist. Don't overthink.)*

5. *Find an example of $AB = BA$ where A is diagonalizable, but B is NOT diagonalizable. (Simple examples exist. Don't overthink.) Note how A must have repeated eigenvalues in this case.*

6. *If $A\boldsymbol{v} = \lambda\boldsymbol{v}$, show that $B\boldsymbol{v}$ is also an eigenvector of A for the same eigenvalue λ.*

7. *(No Credit, feel free to give up) $AB = BA$ might not imply simultaneous diagonalizability. However, they can be simultaneously triangularized via some unitary matrix or orthogonal matrix. (Simultaneous Schur decomposition.) Can you see why? (Hint: Start by finding a common eigenvector of A, B, and then it is a straight forward block-matrix induction.)*

Exercise 8.14.23 (Nilpotent matrices)**.** *A matrix A is nilpotent if $A^k = O$ for some positive integer k.*

1. *If A is nilpotent, find all eigenvalues of A counting algebraic multiplicity. Why must A be non-invertible?*

2. *Show that a $n \times n$ matrix A is nilpotent if and only if all eigenvalues of A are zero if and only if $A^n = O$. (Hint: Cayley-Hamilton.)*

3. *For any matrix A, show that $A = B + C$ where B is a normal matrix and C is an nilpotent matrix. (Hint: perform Schur decomposition $A = UTU^{-1}$, and let D be the diagonal matrix taking all the diagonal entries of T. Why is $U(T - D)U^{-1}$ nilpotent?)*

8.14.2 Spectral decompositions

Exercise 8.14.24. *Find the spectral decomposition of the following matrices.*

1. $\begin{bmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{bmatrix}$.

2. $\begin{bmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{bmatrix}$.

3. $\begin{bmatrix} 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 4 \\ 4 & 1 & 0 & 0 \\ 1 & 4 & 0 & 0 \end{bmatrix}$.

4. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

5. *The real symmetric matrix A has eigenvalues $0, 3, 3$, and $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is an eigenvector for 0.*

6. *Each row of the real symmetric matrix A has a sum of 3, and $\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ are in the kernel of A.*

7. $\begin{bmatrix} a & 1 & 1 & -1 \\ 1 & a & -1 & 1 \\ 1 & -1 & a & 1 \\ -1 & 1 & 1 & a \end{bmatrix}$ *and it has an eigenvalue* -3 *with algebraic multiplicity* 1.

8. *The real symmetric matrix* A *satisfies* $A^3 = O$.

9. *The real symmetric matrix* A *satisfies* $A^5 = I$.

10. $a\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}} + b\boldsymbol{v}\boldsymbol{v}^{T}$, *where* $\boldsymbol{u}, \boldsymbol{v}$ *form an orthonormal basis of* $\mathbb{R}^2$.

Exercise 8.14.25. *If* $A = \begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$, *when is this invertible? When is this diagonalizable? When can it be diagonalized through orthogonal matrices?*

Exercise 8.14.26. *When* n *is super large,* $A = \begin{bmatrix} 1 & 10^{-n} \\ 0 & 1+10^{-n} \end{bmatrix}$ *is super close to being diagonal. Calculate the angle between the two independent eigenvectors of* A.

Exercise 8.14.27 (Complex symmetric matrices)**.** *Consider* $A = \begin{bmatrix} \mathrm{i} & 1 \\ 1 & -\mathrm{i} \end{bmatrix}$. *Calculate* A^2. *Is* A *Hermitian? Skew-Hermitian? Unitary? Normal? Diagonalizable?*

Exercise 8.14.28 (Second Derivative)**.** *The second derivative of a function* $f(x)$ *is* $f''(x) = \lim_{\mathrm{d}x \to 0} \frac{f(x+\mathrm{d}x)+f(x-\mathrm{d}x)-2f(x)}{\mathrm{d}x^2}$. *If we approximately imagine that* f *is a vector* $\begin{bmatrix} \vdots \\ f(x+2\,\mathrm{d}x) \\ f(x+\mathrm{d}x) \\ f(x) \\ f(x-\mathrm{d}x) \\ f(x-2\,\mathrm{d}x) \\ \vdots \end{bmatrix}$. *Then you can see that the second derivative operator is related to some matrix like this:*

$$D = \begin{bmatrix} * & 1 & & & & \\ 1 & -2 & 1 & & & \\ & 1 & -2 & \ddots & & \\ & & \ddots & \ddots & 1 & \\ & & & 1 & -2 & 1 \\ & & & & 1 & * \end{bmatrix}.$$

Here the $*$ *portion depends on some boundary conditions.*

1. *Let* $A_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. *Find its spectral decomposition, and write its eigenvectors in the format* $\begin{bmatrix} \sin\theta \\ \sin(2\theta) \\ \sin(3\theta) \end{bmatrix}$.

2. *Let* $B_4 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. *Find its spectral decomposition, and write its eigenvectors in the*

format $\begin{bmatrix} \cos\theta \\ \cos(3\theta) \\ \cos(5\theta) \\ \cos(7\theta) \end{bmatrix}$.

3. *Let V be the space of infinitely differentiable functions. And let $K : V \to V$ be the operator sending $f(x)$ to $f''(x)$. Then for any real number λ, find all eigenvectors of K for the eigenvalue λ. (This is mostly a calculus problem.)*

Exercise 8.14.29 (Heat Distribution on a square)**.** *Imagine a square made of wires. We shall study the heat distribution on this wire. Going around the wire in a counter-clockwise manner, let's say the four vertices currently have temperatures* $\boldsymbol{t} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \end{bmatrix}$.

On the first vertex, if its heat t_1 is higher than the average temperature of its neighbors (i.e., $\frac{t_2+t_4}{2}$), then the heat at the first vertex would go down. If its heat t_1 is lower than the average temperature of its neighbors (i.e., $\frac{t_2+t_4}{2}$), then the heat at the first vertex would go up.

Let us suppose that after a unit time, the heat at the first vertex would go from t_1 to $t_1 - k(t_1 - \frac{t_2+t_4}{2})$ for some constant k. The situation is similar at other vertices.

1. *Write a matrix A such that, if the initial temperature is $\boldsymbol{t}$, $A\boldsymbol{t}$ records the temperature after a unit time.*

2. *Let H be the matrix whose columns are the Haar wavelet basis, calculate AH.*

3. *Find the spectral decomposition of A.*

4. *Find $\lim_{k\to\infty} A^k$. Does this depends on k? Why? What is the eventual heat distribution on the square?*

Exercise 8.14.30 (Statistics and spectral decomposition)**.** *We collect data on n people, and ask about their height, weight and income. These form three vectors $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3 \in \mathbb{R}^n$.*

Let x_1, x_2, x_3 be the average value of these three data respectively. Let $\boldsymbol{u}$ be the vector whose coordinates are all 1. Then we can define the covariance between $\boldsymbol{a}_i$ and $\boldsymbol{a}_j$ as

$$\mathrm{Cov}(\boldsymbol{a}_i, \boldsymbol{a}_j) = \frac{1}{n}(\boldsymbol{a}_i - x_i\boldsymbol{u})^{\mathrm{T}}(\boldsymbol{a}_j - x_j\boldsymbol{u}).$$

We define the covariance matrix as

$$C = \begin{bmatrix} \mathrm{Cov}(\boldsymbol{a}_1, \boldsymbol{a}_1) & \mathrm{Cov}(\boldsymbol{a}_1, \boldsymbol{a}_2) & \mathrm{Cov}(\boldsymbol{a}_1, \boldsymbol{a}_3) \\ \mathrm{Cov}(\boldsymbol{a}_2, \boldsymbol{a}_1) & \mathrm{Cov}(\boldsymbol{a}_2, \boldsymbol{a}_2) & \mathrm{Cov}(\boldsymbol{a}_2, \boldsymbol{a}_3) \\ \mathrm{Cov}(\boldsymbol{a}_3, \boldsymbol{a}_1) & \mathrm{Cov}(\boldsymbol{a}_3, \boldsymbol{a}_2) & \mathrm{Cov}(\boldsymbol{a}_3, \boldsymbol{a}_3) \end{bmatrix}.$$

1. *Let $A = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \boldsymbol{a}_3 \end{bmatrix}$, and let $U = \begin{bmatrix} \boldsymbol{u} & \boldsymbol{u} & \boldsymbol{u} \end{bmatrix}$. Show that there is a constant k such that $kUU^{\mathrm{T}}A = \begin{bmatrix} x_1\boldsymbol{u} & x_2\boldsymbol{u} & x_3\boldsymbol{u} \end{bmatrix}$.*

2. *Express C in terms of A and U, and show that C is a symmetric matrix.*

3. *Suppose C has spectral decomposition $C = Q\Lambda Q^{\mathrm{T}}$. The three columns of AQ now corresponds to three sets of data. Show that these three sets of data has zero covariance pair-wise. (This way we unravelled some entangled data into uncorrelated data. This is a common practice in statistics.)*

Exercise 8.14.31. *When are the following matrices positive definite? Positive semidefinite? Negative definite? Negative semidefinite? Indefinite?*

1. $\begin{bmatrix} c & b \\ b & c \end{bmatrix}$.

2. $\begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix}$.

3. $\begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix}$.

4. $\begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix}$.

5. $\begin{bmatrix} c & -4 & -4 \\ -4 & c & -4 \\ -4 & -4 & c \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix}$.

7. $\begin{bmatrix} t & 3 & 0 \\ 3 & t & 4 \\ 0 & 4 & t \end{bmatrix}$.

8. $\begin{bmatrix} a & a & a \\ a & a+c & a-c \\ a & a-c & a+c \end{bmatrix}$.

Exercise 8.14.32. *For Hermitian matrices A, B, let us write $A \geq B$ if $A - B$ is positive semidefinite.*

1. *Show that if $A \geq B$ and $B \geq C$, then $A \geq C$.*
2. *Show that if $A \geq B$ and $B \geq A$, then $A = B$.*
3. *Show that for any Hermitian A, we can find $a, b \in \mathbb{R}$ such that $aI \leq A \leq bI$.*

Exercise 8.14.33. *Prove the following statements or give counter examples.*

1. *If A is positive definite, then A^{-1} is positive definite.*
2. *If A, B are positive definite, then $A + B$ is positive definite.*
3. *If A, B are positive semidefinite, then $A + B$ is positive semidefinite.*
4. *If A, B are indefinite, then $A + B$ is indefinite.*
5. *If A has full column rank, and B is positive definite, then $A^{\mathrm{T}}BA$ is positive definite.*
6. *If A, B are positive definite, then all eigenvalues of AB are positive.*
7. *If A is positive definite, then A^k is positive definite for all integer k.*
8. *If A is positive definite, B is any matrix, and $A^rB = BA^r$ for some positive integer r, then $AB = BA$.*
9. *If $S = \begin{bmatrix} A & B \\ B^{\mathrm{T}} & C \end{bmatrix}$ for some symmetric matrices A, C and a matrix B. Then S is positive definite if and only if A and $C - B^{\mathrm{T}}A^{-1}B$ are both positive definite.*

Exercise 8.14.34 (Another proof of Hadamard Inequality)**.** *Let A be a positive definite matrix.*

1. *For any $\boldsymbol{y}$, show that we have* $\det \begin{bmatrix} A & \boldsymbol{y} \\ \boldsymbol{y}^{\mathrm{T}} & 0 \end{bmatrix} \leq 0$.

2. *Let a_{ij} be the (i, j) entry of A, and A_{n-1} be the $(n-1)$-th leading principal submatrix, then* $\det(A) \leq a_{nn} \det(A_{n-1})$.

3. *Prove that* $\det(A) \leq a_{11} \dots a_{nn}$.

4. *If $X = \begin{bmatrix} \boldsymbol{x}_1 & \dots & \boldsymbol{x}_n \end{bmatrix}$ is an invertible matrix, note that $X^{\mathrm{T}}X$ is positive definite. Using the above results, prove that*

$$\det(X)^2 \leq \prod_{i=1}^{n} ||\boldsymbol{x}_i||^2.$$

Exercise 8.14.35. *If the (i, j) entry of A is $\frac{1}{i+j}$, we want to prove that it is positive definite.*

1. *Let V be the space of polynomials of degree at most $n-1$, and we define $\langle p(x), q(x) \rangle = \int_0^1 xp(x)q(x)\,\mathrm{d}x$. Show that this is an inner product.*

2. *Investigate the Gram matrix in the above situation for a nice basis and conclude that A is positive definite.*

3. *As an alternative method to above, prove that*

$$\det(A) = (\prod_{1 \leq i,j \leq n} \frac{1}{i+j})(\prod_{1 \leq 1 < j \leq n} (i-j)^2).$$

4. *Use the last subproblem, show that all leading principal submatrices of A have positive determinant.*

5. *Use whatever method, but if the (i, j) entry of H is $\frac{1}{i+j-1}$, show that H is positive definite.*

8.14.3 Singular Value Decomposition

Exercise 8.14.36. *Given that x, y are not both zero, find the maximum value and minimum value of the $\frac{ax^2+bxy+cy^2}{x^2+y^2}$, for the following given constants $a, b, c \in \mathbb{R}$. (Hint: consider the Rayleigh quotient $\frac{\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x}}{\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}}$)*

1. $a = 0, b = 2, c = 0$.

2. $a = 3, b = 2, c = 3$.

3. $a = 1, b = 2, c = 4$.

Exercise 8.14.37. *Find the SVD of the following matrices*

1. $\begin{bmatrix} 3 & 4 & 0 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

3. $\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$.

4. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

5. $\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$.

7. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix}$. *(If you want, you could actually do this by sight alone. No calculation needed.)*

8. $\begin{bmatrix} A & O \\ O & O \end{bmatrix}$ *where we have SVD* $A = U\Sigma V^{\mathrm{T}}$.

9. $\begin{bmatrix} O & A^{\mathrm{T}} \\ A & O \end{bmatrix}$ *where we have SVD* $A = U\Sigma V^{\mathrm{T}}$.

10. *A has QR decomposition $A = QR$ and R has SVD $R = U\Sigma V^{\mathrm{T}}$.*

Exercise 8.14.38. *Let* $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, *then the linear transformation A sends the unit circle to some ellipse. The unit circle is $C = \{\boldsymbol{v} \in \mathbb{R}^2 : ||\boldsymbol{v}|| = 1\}$, so the ellipse is $A(C) = \{A\boldsymbol{v} \in \mathbb{R}^2 : ||\boldsymbol{v}|| = 1\}$*

1. *If $\boldsymbol{w} \in A(C)$, show that $\boldsymbol{w}^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}\boldsymbol{w} = 1$.*
2. *Find the SVD $A = U\Sigma V^{\mathrm{T}}$.*
3. *Now $A(C) = U\Sigma V^{\mathrm{T}}(C)$. Draw the shapes of $C, V^{\mathrm{T}}(C), \Sigma V^{\mathrm{T}}(X), U\Sigma V^{\mathrm{T}}(C)$.*
4. *What is the long half-axis and short half-axis of the ellipse $A(X)$? (I.e., how long and what are their directions?)*

Exercise 8.14.39. ☕*Prove that for a real square matrix A, the absolute values of all eigenvalues are at most the largest singular value of A. (Hint: Recall the definition of the operator norm of A.)*

Exercise 8.14.40. *Consider four points on the plane* $\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \end{bmatrix}, \begin{bmatrix} 4 \\ 20 \end{bmatrix}$. *Find the line $ax+by+c = 0$ that minimize the sum of squared orthogonal distances. (Feel free to use computers to speed up the calculation. Don't forget to center the data first!)*

Exercise 8.14.41 (Singular values are more stable than eigenvalues)**.** *Find the singular values and eigenvalues of the following matrices. (Minor changes in entries could result in huge change in eigenvalues, but not in singular values.)*

1. $\begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ & & & 0 \end{bmatrix}$. *(Since the matrix has rank 3, we say $\sigma_4 = 0$ by convention.)*

2. $\begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ 0.0001 & & & 0 \end{bmatrix}$.

Exercise 8.14.42 (Dihedral angle)**.** *In $\mathbb{R}^3$, we can define the angle between two planes, i.e., the dihedral angle. Let us try to generalize this to any two subspaces of $\mathbb{R}^n$. In any two subspace V, W of $\mathbb{R}^n$, we consider all possible angles between some $\boldsymbol{v} \in V$ and some $\boldsymbol{w} \in W$. Then the smallest possible angle is defined to be the angle between V and W.*

Let A be an ONB for V and B be an ONB for W. Show that if the angle between V, W is θ, then $\cos\theta$ is the largest singular value of $A^{\mathrm{T}}B$. (So this is how we find this angle.)